Introduction to Analytic and Probabilistic Number Theory

THIRD EDITION

Gérald Tenenbaum

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Introduction to Analytic and Probabilistic Number Theory

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Third Edition

Gérald Tenenbaum

Translated by Patrick D. F. Ion

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À Catherine Jablon,

pour la douceur du jour,
ce bouquet de symboles
dont ta conversation
éclaire les secrets.

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Foreword

Arising, as it does, from advanced lectures given in Bordeaux, Paris and Nancy over the past fifteen years (and for which an earlier English version is available from *Cambridge University Press*), this book is a revised, updated, and expanded version of a volume that appeared in 1990 in the *Publications de l'Institut Élie Cartan.* It was written with the purpose of providing young researchers with a self-contained introduction to the analytic methods of number theory, and their elders with a source of references for a number of fundamental questions. Such an undertaking necessarily involves choices. As these were made, they were generally taken on aesthetic grounds—not to forget the categorical imperatives imposed by ignorance.

The double motivation mentioned above has led to a special usage of the traditional subdivision of chapters into text, notes and exercises. Thus the basic text, while restricted as a rule to assertions that are proved in detail, may also contain additional bibliographic comments when providing a useful background upon first reading. Conversely, the notes often give way to statements, and even proofs, of related results which may safely be omitted on first contact. In a parallel way, the exercises serve a double purpose.[1] Whereas some of them are classically designed to facilitate the mastering of previously introduced concepts, some others lead to actual research results, sometimes unpublished, mainly in Part III. We used to believe, naively, that we could avoid an unfortunate current tendency by producing exercises that could be solved without prodigious ingenuity or technical virtuosity. The numerous requests for solutions received after the publication of the first edition have shown that such a goal might be illusory. Result: the reader

[1] Complete solutions to all exercises from this third edition are available as a companion book published by Belin (Paris).

will find in the solution book, written with the collaboration of my colleague Jie Wu, an attempt to make things right. It remains nonetheless true that open questions are exceptional in the formulations of exercises, and that the results aimed at are usually explicit, with the essential steps set out. This part of the work may thus serve, even without making the effort of solving the problems or consulting the solutions, as an informal repository of references.

The writing of this book has been guided by the constant concern of emphasizing methods more than results—a strategy which we believe to be specifically heuristic. This has led to the somewhat artificial subdivision into three parts, respectively devoted to elementary, complex-analytical and probabilistic methods. It will be easy to criticize this taxonomy: is the method of van der Corput, based on the Poisson summation formula, more elementary than the Selberg–Delange method, which employs complex integration? Why qualify as probabilistic the saddle-point method, whose initial step amounts to an inverse Laplace transform? One could multiply the examples of inconsistency with respect to this or that criterion, and it is obvious that the choices have been made on grounds that can be questioned. Thus, we regard as elementary a method that exclusively employs *real variables*, and we choose to view the saddle-point method as probabilistic as much because it is an ever-present tool in probability theory, as for being a specific method implemented to solve problems in probabilistic number theory... One might as well say at the very outset that the classification at work in this book is anything but a Bourbakist choice. Its ambition is limited to the mere wish that it might, at least for a while, shed some light on the path of the neophyte.

Without aiming at complete originality, the text tries to avoid well-trodden paths. We have reconsidered, when it seemed desirable and indeed possible, the exposition of classical results: either by employing new approaches (such as Nair's method for Chebyshev's estimates), or by occasionally introducing technical simplifications that are invisible in the table of contents, but will hopefully be useful to the active reader.

Certain developments, meanwhile, are innovative. This mainly concerns: some uniform results arising from the Selberg–Delange method (Chapter II.5); the version with explicit remainder of the Ikehara–Ingham theorem (§ II.7.5); the study of the sieve function $\Phi(x, y)$ by the saddle-point method (Chapter III.6). The effective form of Ikehara's theorem turns out to be closely related to the Berry–Esseen inequality—an almost conceptual identity which we continue to find fascinating. Besides, a concern for complementarity with respect to the existing literature (and especially the fine book by Elliott) has influenced some of our decisions, such as the choice

of the method of proof for the theorems of Erdős–Wintner, Erdős–Kac, or Halász—see Chapter III.4. This last result corresponds to an extension of Montgomery's method, developed in a way he suggested.

This second edition, like the first, owes much to all those colleagues and friends who helped me clarify and clean up the manuscript. It is a pleasant duty to express my gratitude here to Michel Balazard, Régis de la Bretèche, Gautami Bhomwik, Paul Erdős, Michel Mendès France, Olivier Ramaré, Jean-Luc Rémy, Imre Ruzsa, Patrick Sargos, András Sárközy, Marijke Wijsmuller, and Jie Wu: as long as the list of errata might turn out to be (and experience has shown this is not just a cliché), it would have been a good deal longer without their help. Finally, I would like to warmly thank Daniel Barlet for his friendly and effective involvement in the process of publication of the text by the *Société mathématique de France.*

Nancy, *March 1995* G.T.

Preface to the third edition

While retaining the same structure and the same expository options, we have extensively expanded the contents of this book for its third edition. This meets a three-fold goal: to take recent advances into account, to flesh out the methodological aspect of the exposition, and to provide basic knowledge or useful supplements for university graduate students, in particular for those preparing for higher teaching diplomas.

Updating with the results from the literature is mostly done in the Notes or Exercises. However, such updates may also be done in new subsections, such as § III.6.5 on Kubilius' model. New proofs of previously included statements are also offered, such as for Tauber's theorem (§ II.7.2) or Halász's (§ III.4.3). Finally, as in the case of the Turán–Kubilius inequality and its friable generalization, the influence of recent results led us to substantially modify the exposition.

Numerous new developments have been inserted in order to preserve general consistency. This essentially concerns: section I.4.7, which is devoted to Selberg's sieve in a little known general form; some applications to small gaps between prime numbers given in the Exercises of the same chapter; the description of Ramanujan's method for the maximal order of the divisor function (Exercise 90); the statements of the Kusmin–Landau inequality (I.6.6) and of van der Corput's general theorem (I.6.10); the inclusion of the explicit formulae of the theory of numbers (§§ II.4.4 and II.8.6); a significant expansion of Chapter II.8, devoted to the distribution of prime numbers in arithmetic progressions; the introduction of Jacobsthal's function and of

the proof of Rankin's theorem on large gaps between consecutive primes (§ III.5.6).

Aside from the inclusion in the Exercises of statements following straightforwardly from the main theorems and of synthetic problems, the new items intended for students and future graduate students concern: the Euler–Maclaurin formula (see the exercises of Chapter I.0); an elementary exposition of the Legendre symbol and the theory of quadratic residues (exercises in Chapter I.1); an introduction to the theory of equidistribution modulo 1 (§ I.6.5); a first treatment of Diophantine approximation and a synthetic exposition of continued fractions (Chapter I.7); as well as a *vade mecum* on the theory of Euler's Gamma function (Chapter II.0).

The description sketched above is obviously too succinct to reflect the numerous correlations between developments arising from various motivations. It is also fails to be exhaustive. The text as a whole has been revised, and whole passages have been rewritten. The presentation is further supported by the addition of one hundred and twenty-five new exercises offering, for some important theorems, variations of proofs, or simplified versions, as in the cases of van der Corput's theorem or of the Erdős–Turán inequality. The initial choices of presentation, however, have not been fundamentally modified.

The author wishes to warmly thank all those who have contributed to an attentive and critical rereading of this almost new manuscript, in particular Joseph Basquin, Régis de la Bretèche, Farrell Brumley, Cécile Dartyge, Kevin Ford, Bruno Martin, Michel Mendès France, Aziz Raouj, Jean-Luc Rémy, Olivier Robert, Anne de Roton, Patrick Sargos, and Jie Wu.

Nancy, *November 2007* G.T.

Preface to the English translation

This translation essentially follows the text of the French edition published in 2008, with many corrections and a few updates. It is a pleasure to express here warm thanks to Edward Dunne for his indestructible commitment to making this book available in English, to Patrick Ion, for his careful translation, and to Nicholas Bingham and Matthew de Courcy-Ireland for their invaluable help.

Nancy, *October 2013* G.T.

Notation

The following notation and conventions will be used freely in the text.

Except in explicitly stated or in special cases clear from context, the letter p, with or without subscript, denotes a prime number. We write $\mathbb{P}$ for the set of all primes.

$a|b$ means: a divides b; $p^\nu \| a$ means: $p^\nu | a$ and $p^{\nu+1} \nmid a$; $a|b^\infty$ means: $p|a \Rightarrow p|b$. We also use the notation $[a,b] := \operatorname{lcm}(a,b)$, and $(a,b) := \gcd(a,b)$.

$P^+(n)$ (resp. $P^-(n)$) denotes the largest (resp. the smallest) prime factor of the integer $n > 1$. By convention $P^+(1) = 1$, $P^-(1) = +\infty$.

The lower and upper integer parts, and the fractional part of the real number x are, respectively, denoted by $\lfloor x \rfloor$, $\lceil x \rceil$ and $\langle x \rangle$.

We put $\|x\| := \min_{n\in\mathbb{Z}} |x-n|$, $x^+ := \max(x,0)$ $(x \in \mathbb{R})$ and use the notation $\mathrm{e}(x) := \mathrm{e}^{2\pi i x}$ $(x \in \mathbb{R})$, $\ln^+ x := \max\{0, \ln x\}$ $(x > 0)$. We write $\ln_k$ for the k-fold iterated logarithm. The notation log is reserved for the complex logarithm, taken, if not otherwise specified, in its principal branch.

When the letter s denotes a complex number, we implicitly define real numbers σ and τ by the relation $s = \sigma + i\tau$.

We use interchangeably Landau's notation $f = O(g)$ and Vinogradov's $f \ll g$ both to mean that $|f| \leqslant C|g|$ for a suitable positive constant C, which may be absolute or depend upon various parameters, in which case the dependence may be indicated in a subscript. Moreover, we write $f \asymp g$ to indicate that $f \ll g$ and $g \ll f$ hold simultaneously. We draw the reader's attention to the fact that we have therefore extended the common use of these symbols to complex-valued functions.

We denote the cardinality of a finite set $\mathcal{A}$ either by $\operatorname{card}\mathcal{A}$, or $|\mathcal{A}|$.

We list below page numbers where various notations in the body of the text are introduced.

$b_r(x)$, B_r, $B_r(x)$	5	$\boldsymbol{\delta}\mathcal{A}$	416	σ_a, σ_c	191
$\mathrm{e}(x)$	73	$\delta(n)$	32	$\sigma_k(n)$	30
$\mathbf{d}\mathcal{A}$	415	$\zeta(s)$	19	$\tau(n)$	30
$j(n)$	34	$\zeta(s,y)$	512	$\tau(n,\vartheta)$	240
$k(n)$	64	$\lambda(n)$	64	$\varphi(n)$	30
$N(T)$	243	$\Lambda(n)$	30	$\Phi(x,y)$	70
$N(x,y)$	198	$\mu(n)$	30	$\chi(n)$,	363
$p_j(n)$	460	ν_N	416	$\chi_0(n)$	364
pp	420	$\xi(s)$	242	$\psi(x)$	36
$S(\mathcal{A},\mathcal{P};y)$	69, 91	$\pi(x)$	11	$\psi(x;a,q)$	370
$v_p(n)$	15	$\pi(x;a,q)$	83	$\Psi(x,y)$	511
$\mathbf{1}(n)$	34	$\varrho(u)$	519	$\omega(n)$, $\Omega(n)$	30
				$\Omega_\pm$	111

Part I

Elementary Methods

Some tools from real analysis

0.1. Abel summation

Classically one calls Abel summation, or Abel's transformation, the process of transforming a finite sum of products of two terms by means of partial sums of one of them.

Theorem 0.1 (Abel's transformation). *Let* $\{a_n\}_{n=0}^{\infty}$ *and* $\{b_n\}_{n=0}^{\infty}$ *be complex sequences. For any integers* $N \in \mathbb{Z}$ *and* $M \in \mathbb{N}^*$, *we have*

$$\sum_{N<n\leqslant N+M} a_n b_n = A_{N+M} b_{N+M+1} + \sum_{N<n\leqslant N+M} A_n (b_n - b_{n+1}), \tag{0.1}$$

where $A_n := \sum_{N<m\leqslant n} a_m$ $(n \geqslant 0)$. *In particular, if*

$$\sup_{N<n\leqslant N+M} |A_n| \leqslant A,$$

and if $\{b_n\}_{n=0}^{\infty}$ *is non-negative and non-increasing, then*

$$\left| \sum_{N<n\leqslant N+M} a_n b_n \right| \leqslant A b_{N+1}. \tag{0.2}$$

Proof. The first assertion results from a simple change of variables, putting $a_n = A_n - A_{n-1}$ $(N < n \leqslant N+M)$. When $\{b_n\}_{n=0}^{\infty}$ is non-negative and non-increasing, we deduce from (0.1) that

$$\left| \sum_{N<n\leqslant N+M} a_n b_n \right| \leqslant A b_{N+M+1} + A \sum_{N<n\leqslant N+M} (b_n - b_{n+1}) = A b_{N+1}.$$

□

Corollary 0.2 (Abel's convergence criterion or Abel's rule). *Let $\{a_n\}_{n=0}^{\infty}$ be a sequence of complex numbers and $\{b_n\}_{n=0}^{\infty}$ denote a non-increasing sequence of positive real numbers. Assume that*

$$\lim_{n\to\infty} b_n = 0 \quad \text{and} \quad \sup_{N\geqslant 0}\Big|\sum_{0\leqslant n\leqslant N} a_n\Big| \leqslant A.$$

Then the series $\sum_{n\geqslant 0} a_n b_n$ converges, and we have

$$\Big|\sum_{n>N} a_n b_n\Big| \leqslant 2Ab_{N+1}. \tag{0.3}$$

Proof. This follows immediately from Theorem 0.1. □

Example. The series $\sum_{n\geqslant 1} z^n/n$, whose radius of convergence is 1, converges at any point on the unit circle except at $z = 1$.

In the setting of the Stieltjes integral, Abel summation takes the innocuous form of partial integration. It constitutes a simple but effective tool for handling arithmetic sums. The reader may find the essential information concerning the Stieltjes integral in Chapter 1 of Widder's book (1946).

Theorem 0.3. *Let $\{a_n\}_{n=1}^{\infty}$ be a complex sequence. Put*

$$A(t) := \sum_{n\leqslant t} a_n \qquad (t > 0).$$

Then, for any function $b \in \mathcal{C}^1([1, x])$, we have

$$\sum_{1\leqslant n\leqslant x} a_n b(n) = A(x)b(x) - \int_1^x A(t)\, b'(t)\, \mathrm{d}t.$$

Proof. The quantity to be calculated is

$$\int_{1^-}^{x} b(t)\, \mathrm{d}A(t) = \Big[A(t)\, b(t)\Big]_{1^-}^{x} - \int_1^x b'(t)A(t)\, \mathrm{d}t,$$

by partial integration. This provides the required assertion. □

The same formalism allows us to easily recover other classical results. Let us begin with two important examples.

Theorem 0.4 (Comparison of a sum and an integral). *Let f be a real monotonic function on the interval $[a, b]$, with $a, b \in \mathbb{Z}$. There exists a real number $\vartheta = \vartheta(a, b)$, $0 \leqslant \vartheta \leqslant 1$, such that*

$$\sum_{a<n\leqslant b} f(n) = \int_a^b f(t)\, \mathrm{d}t + \vartheta\big(f(b) - f(a)\big).$$

Proof. Assume, with no loss of generality, that f is continuous at points in $\mathbb{Z}$; introduce the Stieltjes integral of f with respect to the measure $\mathrm{d}\lfloor t \rfloor$. We obtain

$$\sum_{a<n\leqslant b} f(n) - \int_a^b f(t)\,\mathrm{d}t = \int_a^b f(t)\,\mathrm{d}[t] - \int_a^b f(t)\,\mathrm{d}t = -\int_a^b f(t)\,\mathrm{d}\langle t\rangle\,.$$

Integrating this expression by parts gives

$$\Big[-f(t)\,\langle t\rangle\Big]_a^b + \int_a^b \langle t\rangle\;\mathrm{d}f(t) = \int_a^b \langle t\rangle\;\mathrm{d}f(t).$$

Let us suppose, for example, that f is non-decreasing, so that the measure $\mathrm{d}f$ is non-negative. The last integral is then $\vartheta(f(b)-f(a))$ with $0\leqslant\vartheta\leqslant 1$, from which the result follows. □

Corollary 0.5. *For $n\geqslant 1$, we have $\ln n! = n\ln n - n + 1 + \vartheta\ln n$, with $\vartheta=\vartheta_n\in[0,1]$.*

Theorem 0.6 (Second mean value theorem)**.** *Let f be monotonic and let g be an integrable function on the real interval $[a,b]$. Then there exists a real number ξ, $a\leqslant\xi\leqslant b$, such that*

$$\int_a^b f(t)\,g(t)\,\mathrm{d}t = f(a)\int_a^\xi g(t)\,\mathrm{d}t + f(b)\int_\xi^b g(t)\,\mathrm{d}t.$$

Proof. Put $G(t) := \int_a^t g(v)\,\mathrm{d}v$. The left-hand side equals

$$\int_a^b f(t)\,\mathrm{d}G(t) = G(b)\,f(b) - \int_a^b G(t)\,\mathrm{d}f(t),$$

by partial integration. Assume, say, that f is increasing. Then $\mathrm{d}f(t)$ is a positive Stieltjes measure and, since $G(t)$ is continuous, there is a number ξ, $a\leqslant\xi\leqslant b$, such that the last integral equals $G(\xi)\{f(b)-f(a)\}$. The desired result follows by rearranging the terms. □

0.2. The Euler–Maclaurin summation formula

Consider the sequence $\{b_r(x)\}_{r=0}^\infty$ of polynomials defined on $[0,1]$ by the conditions

$$b_0(x)\equiv 1, \tag{0.4}$$

$$b_r'(x)\equiv r\,b_{r-1}(x)\qquad(r\geqslant 1), \tag{0.5}$$

$$\int_0^1 b_r(x)\,\mathrm{d}x = 0\qquad(r\geqslant 1). \tag{0.6}$$

One easily verifies that these assumptions imply the identity

$$\sum_{r\geqslant 0} b_r(x)\,\frac{y^r}{r!} = \frac{y\,\mathrm{e}^{xy}}{\mathrm{e}^y - 1}$$

allowing us to calculate the b_r. We find

$$\begin{aligned} b_0(x) &= 1, & b_3(x) &= x^3 - \tfrac{3}{2}x^2 + \tfrac{1}{2}x,\\ b_1(x) &= x - \tfrac{1}{2}, & b_4(x) &= x^4 - 2x^3 + x^2 - \tfrac{1}{30},\\ b_2(x) &= x^2 - x + \tfrac{1}{6}, & b_5(x) &= x^5 - \tfrac{5}{2}x^4 + \tfrac{5}{3}x^3 - \tfrac{1}{6}x. \end{aligned}$$

One then defines the r-th Bernoulli function $B_r(x)$ as the 1-periodic function which coincides with b_r on $[0,1[$. We set

$$B_r := B_r(0).$$

B_r is the r-th Bernoulli number. It is easy to see that $B_{2r+1} = 0$ for $r \geqslant 1$. The first numerical values are:

r	0	1	2	4	6	8	10	12	14	16
B_r	1	$-\frac{1}{2}$	$\frac{1}{6}$	$-\frac{1}{30}$	$\frac{1}{42}$	$-\frac{1}{30}$	$\frac{5}{66}$	$-\frac{691}{2730}$	$\frac{7}{6}$	$-\frac{3617}{510}$

Let f be a numerical function of class $\mathcal{C}^{k+1}$ on the interval $[a,b]$, with $a,b \in \mathbb{Z}$. Since $B_1(x) = \langle x\rangle - \frac{1}{2}$, we may write

$$\sum_{a<n\leqslant b} f(n) = \int_a^b f(t)\,\mathrm{d}\lfloor t\rfloor = \int_a^b f(t)\,\mathrm{d}t - \int_a^b f(t)\,\mathrm{d}B_1(t).$$

Integrating the last integral by parts yields

$$\begin{aligned} \int_a^b f(t)\,\mathrm{d}B_1(t) &= B_1\cdot(f(b) - f(a)) - \int_a^b B_1(t)f'(t)\,\mathrm{d}t\\ &= B_1\cdot(f(b) - f(a)) - \frac{1}{2!}\int_a^b f'(t)\,\mathrm{d}B_2(t). \end{aligned}$$

Indeed, it can be easily verified that $B_2(t)$ is continuous on $\mathbb{R}$ and is differentiable on $\mathbb{R}\setminus\mathbb{Z}$, where it satisfies $B_2'(t) = 2B_1(t)$. Furthermore, for $r \geqslant 3$, $B_r(t)$ is differentiable on the entire real line, and satisfies

$$B_r'(t) = rB_{r-1}(t).$$

We can thus calculate the integral with respect to $B_2(t)$ by a new partial integration, involving $B_3(t)$. Iterating the process, we obtain in this way the following famous theorem.

Theorem 0.7 (Euler–Maclaurin summation formula). *For any integer $k \geqslant 0$ and any function f of class $\mathcal{C}^{k+1}$ on $[a,b]$ with $a,b \in \mathbb{Z}$, we have*

$$\sum_{a<n\leqslant b} f(n) = \int_a^b f(t)\,\mathrm{d}t + \sum_{0\leqslant r\leqslant k} \frac{(-1)^{r+1}B_{r+1}}{(r+1)!}\,(f^{(r)}(b) - f^{(r)}(a)) + \frac{(-1)^k}{(k+1)!}\int_a^b B_{k+1}(t)\,f^{(k+1)}(t)\,\mathrm{d}t.$$

By way of application, we give an estimate of the partial sums of the harmonic series.

Theorem 0.8. *For $n \geqslant 1$, we have*

$$\sum_{m\leqslant n} \frac{1}{m} = \ln n + \gamma + \frac{1}{2n} - \frac{1}{12n^2} + \frac{\vartheta}{60n^4},$$

where γ is Euler's constant and $\vartheta = \vartheta_n \in [0,1]$.

Proof. Apply Theorem 0.7 with $f(t) = 1/t$, $a = 1$, $b = n$, $k = 3$. We get

$$\sum_{2\leqslant m\leqslant n} \frac{1}{m} = \ln n + \tfrac{1}{2}\Big(\frac{1}{n} - 1\Big) - \tfrac{1}{12}\Big(\frac{1}{n^2} - 1\Big) + \tfrac{1}{120}\Big(\frac{1}{n^4} - 1\Big) - \int_1^n t^{-5}B_4(t)\,\mathrm{d}t.$$

Adding the term corresponding to $m = 1$, and letting n tend to infinity, we obtain

$$\gamma = \tfrac{1}{2} + \tfrac{1}{12} - \tfrac{1}{120} - \int_1^\infty \frac{B_4(t)}{t^5}\,\mathrm{d}t.$$

The result is then implied by the inequality

$$\left|\int_n^\infty \frac{B_4(t)}{t^5}\,\mathrm{d}t\right| \leqslant \frac{1}{120n^4}$$

which may be readily deduced from the fact that $|B_4(t)| \leqslant \frac{1}{30}$ for all t. □

Remark. An immediate generalization of the above calculations provides the formula

$$\gamma = \tfrac{1}{2} + \sum_{2\leqslant r\leqslant k} \frac{B_r}{r} - \int_1^\infty \frac{B_k(t)}{t^{k+1}}\,\mathrm{d}t \qquad (k \geqslant 1)$$

which can be used to calculate γ numerically by subtracting this expression from the expansion of $\sum_{1\leqslant m\leqslant n} 1/m$ and optimizing k as a function of n. We have $\gamma \approx 0.577215664$.

Exercises

1. For $k \in \mathbb{Z}^+$, calculate $\dfrac{1}{2\pi i}\oint_{|z|=\pi(2k+1)} \dfrac{ze^{xz}}{(e^z-1)}\dfrac{dz}{z^{2r+1}}$ and thus establish the Fourier expansion of the Bernoulli functions of even order:

$$B_{2r}(x) = \frac{(-1)^{r-1}2(2r)!}{(2\pi)^{2r}}\sum_{m\geqslant 1}\frac{\cos(2\pi mx)}{m^{2r}} \qquad (r \geqslant 1).$$

Deduce a general formula for $\zeta(2r)$.

2. Establish, by the method of the previous exercise, the Fourier expansion of $B_{2r+1}(x)$, i.e.

$$B_{2r+1}(x) = \frac{(-1)^{r-1}2(2r+1)!}{(2\pi)^{2r+1}}\sum_{m\geqslant 1}\frac{\sin(2\pi mx)}{m^{2r+1}} \qquad (r \geqslant 0),$$

where equality is restricted to $x \in \mathbb{R} \setminus \mathbb{Z}$ when $r = 0$.

Why does this fail to yield a formula for $\zeta(2r+1)$?

3. *Stirling's formula.*

(a) By applying the Euler–Maclaurin formula of order 0 to $f(t) = \ln t$, show that we have

$$n! = n^n e^{-n}\sqrt{An}\,\{1 + O(1/n)\} \qquad (n \geqslant 1)$$

with $\ln A = 2 + 2\int_1^\infty B_1(t)t^{-1}\,dt$.

(b) Show, using integration by parts, that the sequence of Wallis integrals

$$W_n = \int_0^{\pi/2} (\cos t)^n\,dt$$

satisfies the recurrence relation $nW_n = (n-1)W_{n-2}$ for $n \geqslant 2$. Deduce that, as $n \to \infty$, $W_{2n} \sim \pi/\sqrt{2An}$ and $W_{2n+1} \sim \sqrt{A/8n}$.

(c) Prove that $W_n \sim W_{n+1}$ $(n \to \infty)$ and show that $A = 2\pi$.

4. Let $k \in \mathbb{N}$ and $f \in \mathcal{C}^{2k+1}([1,\infty[,\mathbb{R})$ be such that $f^{(2k+1)} \in L^1([1,\infty[)$. Show there is a constant $C_k(f)$, making it explicit, such that, for any integer $n \geqslant 1$,

$$\begin{aligned}\sum_{1\leqslant m\leqslant n} f(m) &= \int_1^n f(x)\,dx + \tfrac{1}{2}f(n) \\ &\quad + \sum_{1\leqslant j\leqslant k}\frac{B_{2j}}{(2j)!}f^{(2j-1)}(n) + C_k(f) + R_{k,n}(f)\end{aligned} \tag{0.7}$$

where $R_{k,n}(f) \to 0$ as $n \to \infty$.

Deduce that, if $f^{(j)} \in L^1([1,\infty[)$ for $j \geqslant 2m+1$, then $C_k(f)$ is independent of k when $k \geqslant m$.

5. *Variations of the $B_r(x)$.* For $m \geqslant 1$, $\varepsilon \in \{0,1\}$, $x \in \mathbb{R}$, put

$$\varphi_{2m+\varepsilon}(x) := (-1)^m\{B_{2m+\varepsilon}(x) - B_{2m+\varepsilon}\}.$$

(a) Show that, for $0 \leqslant x \leqslant 1$, we have

$$\varphi_2(x) = x(1-x), \quad \varphi_3(x) = x(x-\tfrac{1}{2})(1-x).$$

(b) Calculate $\sum_{r\geqslant 0} b_r(\frac{1}{2})y^r/r!$. Deduce that $\varphi_{2m+1}(\frac{1}{2}) = 0$ for $m \geqslant 1$.

(c) Show that, for $m \geqslant 1$, we have

$$\varphi'_{2m+1}(x) = (2m+1)\{\varphi_{2m}(x) - |B_{2m}|\},$$

and

$$\varphi'_{2m+2}(x) = -(2m+2)\varphi_{2m+1}(x).$$

(d) By induction over $m \geqslant 1$, establish that $\varphi_{2m}(x) > 0$ for $0 < x < 1$, and $(x-\frac{1}{2})\varphi_{2m+1}(x) > 0$ for $0 < x < 1$, $x \neq \frac{1}{2}$.

6. *Upper bound for the remainder term.* Let $k \in \mathbb{N}$ and $f \in \mathcal{C}^{2k+2}([1,\infty[,\mathbb{R})$ be such that $f^{(2k+2)} \in L^1[1,\infty[$. Set

$$S_{h,n}(f) := \tfrac{1}{2}f(n) + \sum_{1\leqslant j\leqslant h} \frac{B_{2j}}{(2j)!} f^{(2j-1)}(n) \qquad (0 \leqslant h \leqslant k+1,\ n \in \mathbb{N}^*).$$

(a) Show that, for any integer $n \geqslant 1$,

$$\sum_{1\leqslant m\leqslant n} f(m) = \int_1^n f(x)\,\mathrm{d}x + S_{k,n}(f) + C_k^*(f) + R_{k,n}^*(f) \tag{0.8}$$

where $C_k^*(f)$ is a constant to be made explicit, and where we have put, using the notation φ_r of Exercise 5,

$$R_{k,n}^*(f) := \frac{(-1)^{k+1}}{(2k+2)!}\int_n^\infty f^{(2k+2)}(x)\varphi_{2k+2}(x)\,\mathrm{d}x. \tag{0.9}$$

(b) From now on we make the additional assumption that $f^{(2k)} \in L^1([1,\infty[)$ and that $\lim_{x\to\infty} f^{(2k-1)}(x) = 0$. Show that

$$C_{k-1}^*(f) = C_k^*(f), \qquad R_{k-1,n}^*(f) - R_{k,n}^*(f) = \frac{B_{2k}}{(2k)!}f^{(2k-1)}(n).$$

(c) Provided $f^{(2k+2)}$ and $f^{(2k)}$ are of constant sign, and have the same sign on $[1,\infty[$, show that $R_{k-1,n}^*(f)$ and $R_{k,n}^*(f)$ then have opposite signs. Deduce that

$$|R_{k,n}^*(f)| \leqslant \left|\frac{B_{2k}}{(2k)!}f^{(2k-1)}(n)\right|,$$

that is, in other words, that *the absolute value of the error term in the Euler–Maclaurin formula* (0.8) *does not exceed that of the last term retained.*

7. Show that for any integers $n \geqslant 1$, $k \geqslant 1$, we have

$$\gamma = \sum_{1 \leqslant m \leqslant n} \frac{1}{m} - \ln n - \frac{1}{2n} + \sum_{1 \leqslant j \leqslant k} \frac{B_{2j}}{2jn^{2j}} + \varepsilon_{kn}$$

where γ denotes Euler's constant and where $|\varepsilon_{kn}| \leqslant |B_{2k}|/\{2kn^{2k}\}$. Show that selecting $n = 50$, $k = 7$, allows calculation of γ up to 24 decimal places.

Remark. *Summation to the smallest term.* We know that $|B_{2j}| \sim 2(2j)!/(2\pi)^{2j}$. When n is fixed, the ratio of two consecutive terms in the j-sum is hence, in absolute value, close to $(j/\pi n)^2$. It follows that the best error term obtained by this method corresponds to selecting k near πn. The size of ε_{kn} is then of the order of $\mathrm{e}^{-2\pi n}$. For $n = 10$, we already obtain 27 exact decimal places, provided we know the values of B_{2k} for $k \leqslant 32$.

8. Show that, for any integers $n \geqslant 1$, $k \geqslant 1$, we have

$$\tfrac{1}{6}\pi^2 = \sum_{1 \leqslant m \leqslant n} \frac{1}{m^2} + \frac{1}{n} - \frac{1}{2n^2} + \sum_{1 \leqslant j \leqslant k} \frac{B_{2j}}{n^{2j+1}} + \varepsilon_{kn}$$

with $|\varepsilon_{kn}| \leqslant |B_{2k}|/n^{2k+1}$. Prove that selecting $k = 8$, and $n = 100$ allows the computation of $\pi^2/6$ up to 33 decimal places.

9. By applying the Euler–Maclaurin formula on the interval $[1, N]$ to the function

$$f(x) := \ln\Big(\frac{x\vartheta}{1 - \mathrm{e}^{-x\vartheta}}\Big),$$

and letting N tend to infinity, show that, for any fixed integer $k \geqslant 2$ we have, as ϑ approaches 0 through positive values,

$$\prod_{n \geqslant 1} \Big(\frac{1}{1 - \mathrm{e}^{-n\vartheta}}\Big) = \{1 + O(\vartheta^k)\}\mathrm{e}^{\pi^2/6\vartheta - \vartheta/24}\sqrt{\frac{\vartheta}{2\pi}}.$$

Prime numbers

1.1. Introduction

Addition and multiplication equip the set of natural numbers $\{1, 2, 3, \dots\}$ with a two-fold structure of Abelian semigroup. The first, associated with a total order relation, is generated by the unique number 1. The second, reflecting the partial order of divisibility, has an infinite number of generators: the primes. Although defined since ancient times, this key concept has yet to deliver up all its secrets—far from it. The central place of prime number theory in arithmetic is amply justified by the following result, whose proof we sketch, using Euclid's first theorem, in Exercises 10 to 13.

Theorem 1.1 (Fundamental theorem of arithmetic)). *Each natural number > 1 can be represented in a unique way, up to the order of the factors, as a product of prime numbers.*

Euclid's second theorem asserts the infinitude of the set of prime numbers. It is an immediate consequence of the fundamental theorem of arithmetic: if $p_1 = 2$, $p_2 = 3, \dots,$ p_n are the n smallest prime numbers, then the integer

$$N = 1 + \prod_{1 \leqslant j \leqslant n} p_j$$

is divisible by none of the numbers p_1, $p_2, \dots,$ p_n. Its smallest prime factor is therefore a prime $> p_n$.

One usually denotes by $\pi(x)$ the number of primes not exceeding x so that, for any integer n, we have $\pi(p_n) = n$. Euclid's second theorem expresses the fact that

$$\pi(x) \to \infty \qquad (x \to \infty).$$

For more than 23 centuries mathematicians have been concerned with providing quantitative versions of this qualitative relation. It is one of the goals of this work to describe the various methods they have invented and implemented to achieve this.

The proof given above of Euclid's second theorem is too simple to be ineffective. Indeed, we have

$$p_{n+1} \leqslant 1 + \prod_{1 \leqslant j \leqslant n} p_j$$

from which we deduce by an immediate induction that

$$p_n \leqslant 2^{2^n} \qquad (n \geqslant 1).$$

We obtain in this way the following lower bound.

Theorem 1.2. *We have*

$$\pi(x) > \frac{\ln_2 x}{\ln 2} - \frac{1}{2} \qquad (x \geqslant 2).$$

Proof. Given the upper bound for p_n established above, we may write

$$\pi(x) \geqslant \max\{m \in \mathbb{N} : 2^{2^m} \leqslant x\} = \left\lfloor \frac{\ln(\ln x/\ln 2)}{\ln 2} \right\rfloor \geqslant \frac{\ln_2 x}{\ln 2} - \left(1 + \frac{\ln_2 2}{\ln 2}\right),$$

which implies the stated result. □

The lower bound in Theorem 1.2 is far from optimal. After having been conjectured for more than a century (notably by Legendre and Gauss) the asymptotic formula

$$\pi(x) \sim \frac{x}{\ln x} \qquad (x \to \infty),$$

was proved independently in 1896 by Hadamard (1865–1963) and La Vallée–Poussin (1866–1962). Their methods rested upon complex analysis techniques to be described in Part II. One had to wait until 1949 for the appearance of the first elementary proofs of the prime number theorem, due to Erdős and Selberg. Since this date, many other elementary proofs have been published. That due to Daboussi (1984) is particularly elegant and is based on a principle fundamentally different from the others.[1] It is given a detailed exposition in chap. 4 of Tenenbaum & Mendès France (2000).

[1] Indeed, it does not appeal to Selberg's identity—cf. Exercise 75, p. 65—which plays an essential role in most of the other elementary proofs.

1.2. Chebyshev's estimates

It is to the Russian mathematician Chebyshev that we owe the first significant work on the function $\pi(x)$. In 1852, he proves Bertrand's postulate, according to which each interval $]n, 2n]$, $n \geqslant 1$, contains at least one prime number. He obtains this result by establishing an effective form of the two-sided estimate

$$\{c_1 + o(1)\}\frac{x}{\ln x} \leqslant \pi(x) \leqslant \{c_2 + o(1)\}\frac{x}{\ln x} \qquad (x \to \infty),$$

with $c_1 = \ln(2^{1/2}\,3^{1/3}\,5^{1/5}\,30^{-1/30}) \approx 0.92129$, and $c_2 = \frac{6}{5}\,c_1 \approx 1.10555$.

We shall show by a simple method the following result, which implies a slightly weaker form of Bertrand's postulate: for each $\varepsilon > 0$, there exists an $n_0 = n_0(\varepsilon)$ such that each interval $]n, (2+\varepsilon)n]$, $n \geqslant n_0$, contains at least one prime.

Theorem 1.3. *For $n \geqslant 4$, we have*

$$(\ln 2)\frac{n}{\ln n} \leqslant \pi(n) \leqslant \Big\{\ln 4 + \frac{8\ln_2 n}{\ln n}\Big\}\frac{n}{\ln n}.$$

Proof. The upper bound is an easy consequence of the following classical result.

Theorem 1.4. *For $n \geqslant 1$, we have*

$$\prod_{p \leqslant n} p \leqslant 4^n.$$

Indeed, accepting this result for the moment, we have for all t, $1 < t \leqslant n$,

$$t^{\pi(n)-\pi(t)} \leqslant \prod_{t < p \leqslant n} p \leqslant 4^n$$

whence, by taking logarithms,

$$\pi(n) \leqslant \frac{n \ln 4}{\ln t} + t.$$

The stated result follows by choosing $t = n/(\ln n)^2$—the numerical details being left to the reader. □

Proof of Theorem 1.4. We proceed by induction on the integer n, which we may obviously assume $\geqslant 3$. If n is even, n is not prime, so

$$\prod_{p \leqslant n} p = \prod_{p \leqslant n-1} p \leqslant 4^{n-1} < 4^n.$$

If n is odd, we put $n = 2m + 1$. The argument rests on the integrality of the binomial coefficients of order n. Since $\binom{2m+1}{m} = (2m+1)!/m!(m+1)!$,

we have

$$\Big(\prod_{m+1<p\leqslant 2m+1} p\Big)\Big|\binom{2m+1}{m} \leqslant \tfrac{1}{2}2^{2m+1} = 4^m,$$

where the last inequality arises from the fact that the coefficient $\binom{2m+1}{m}$ equals $\binom{2m+1}{m+1}$ and so appears twice in the binomial expansion of $(1+1)^{2m+1}$. By the induction hypothesis applied to $m+1 < n$, we thus obtain

$$\prod_{p\leqslant n} p = \prod_{p\leqslant m+1} p \prod_{m+1<p\leqslant 2m+1} p \leqslant 4^{m+1}\,4^m = 4^n,$$

which finishes the proof. □

The lower bound in Theorem 1.3 will be obtained by a remarkably simple and efficient method due to Nair (1982a, b). It rests on the inequality

$$\pi(n) \geqslant (\ln d_n)/\ln n \qquad (n \geqslant 2),$$

where d_n denotes the least common multiple of the numbers $1, 2, \ldots, n$. Indeed, if $p^\nu \| d_n$, then there exists an $m \leqslant n$ such that $p^\nu | m$, whence $p^\nu \leqslant n$ and

$$d_n = \prod_{p\leqslant n,\, p^\nu \| d_n} p^\nu \leqslant \prod_{p\leqslant n} n = n^{\pi(n)},$$

which is equivalent to the stated inequality. The desired result then follows from the following theorem.

Theorem 1.5 (Nair). *For $n \geqslant 7$, we have $d_n \geqslant 2^n$.*

Proof. The essential idea introduced by Nair lies in considering the integral

$$I(m,n) = \int_0^1 x^{m-1}(1-x)^{n-m}dx \qquad (1 \leqslant m \leqslant n).$$

On the one hand, the binomial expansion of $(1-x)^{n-m}$ shows that $I(m,n)$ is a rational number with denominator dividing d_n: we have

$$I(m,n) = \sum_{0\leqslant j\leqslant n-m} (-1)^j \binom{n-m}{j} \frac{1}{m+j} \in \frac{1}{d_n}\mathbb{Z}.$$

On the other hand, $I(m,n)$ is "small". Its actual value may even be calculated by noting that, for all y, $0 \leqslant y \leqslant 1$, we have

$$\sum_{1\leqslant m\leqslant n} \binom{n-1}{m-1} y^{m-1} I(m,n) = \int_0^1 (1-x+xy)^{n-1}\,dx = \frac{1}{n}\sum_{1\leqslant m\leqslant n} y^{m-1},$$

whence

$$I(m,n) = 1\Big/ n\binom{n-1}{m-1} = 1\Big/ m\binom{n}{m} \qquad (1 \leqslant m \leqslant n).$$

This shows that $m\binom{n}{m} \mid d_n$ for $1 \leqslant m \leqslant n$, from which we have

$$n\binom{2n}{n} \mid d_{2n} \mid d_{2n+1}, \qquad \text{and} \qquad (n+1)\binom{2n+1}{n} = (2n+1)\binom{2n}{n} \mid d_{2n+1}.$$

Since n and $2n+1$ are relatively prime, it follows that

$$n(2n+1)\binom{2n}{n} \mid d_{2n+1},$$

and finally, since $\binom{2n}{n}$ is the largest of the $(2n+1)$ binomial coefficients appearing in the expansion of $(1+1)^{2n}$,

$$d_{2n+1} \geqslant n\,4^n \qquad (n \geqslant 1).$$

We deduce that

$$d_{2n+1} \geqslant 2{\cdot}4^n = 2^{2n+1} \qquad (n \geqslant 2),$$

and

$$d_{2n+2} \geqslant d_{2n+1} \geqslant 4^{n+1} \qquad (n \geqslant 4),$$

which establishes the stated inequality, $d_n \geqslant 2^n$, for any $n \geqslant 9$. It can be easily checked that it is still satisfied for $n = 7$ and $n = 8$: $d_7 = 420$, $d_8 = 840$. □

1.3. p-adic valuation of $n!$

For each prime number p, the p-adic valuation, denoted by v_p, is defined as the arithmetic function which associates to each integer n the exponent of p in its canonical factorization. The following simple theorem will be useful in the sequel.

Theorem 1.6. *For each prime number p, we have*

$$v_p(n!) = \sum_{k \geqslant 1} \left\lfloor n/p^k \right\rfloor \qquad (n \geqslant 1).$$

Remark. The sum over k is actually finite since the general term vanishes for $k > (\ln n)/\ln p$.

Proof. We have

$$v_p(n!) = \sum_{m \leqslant n} v_p(m) = \sum_{m \leqslant n} \sum_{1 \leqslant k \leqslant v_p(m)} 1 = \sum_{k \geqslant 1} \sum_{m \leqslant n,\, v_p(m) \geqslant k} 1.$$

The inner sum equals the number of integers $m \leqslant n$ which are divisible by p^k. Its value is thus $\lfloor n/p^k \rfloor$, which leads to the desired expression. □

Corollary 1.7. *For each prime p, we have*

$$\frac{n}{p} - 1 < v_p(n!) \leqslant \frac{n}{p} + \frac{n}{p(p-1)} \qquad (n \geqslant 1).$$

This readily follows from Theorem 1.6 and the bounds $x - 1 < \lfloor x \rfloor \leqslant x$, valid for all real numbers x.

1.4. Mertens' first theorem

Certain quantities depending on the primes not exceeding x have an asymptotic behavior which is more easily accessible than that of the function $\pi(x)$ itself. This is certainly the case for the expression evaluated in the following theorem.

Theorem 1.8 (Mertens' first theorem). *For $x \geqslant 2$, we have*

$$\sum_{p \leqslant x} \frac{\ln p}{p} = \ln x + O(1).$$

Moreover, the term $O(1)$ arising in this formula lies in the open interval $]-1-\ln 4, \ln 4[$.

NB. $\ln 4 \approx 1.38629$.

Proof. We evaluate the quantity $\ln(n!)$ in two different ways for $n = \lfloor x \rfloor$.

On the one hand, we have seen in Corollary 0.5 that

$$\ln(n!) = n \ln n - n + 1 + \vartheta_n \ln n$$

with $0 \leqslant \vartheta_n \leqslant 1$.

On the other hand, by the canonical factorization of $n!$ into a product of prime factors, viz.

$$\ln n! = \sum_{p \leqslant n} v_p(n!) \ln p,$$

we deduce from Corollary 1.7 that

$$\ln n! < n \sum_{p \leqslant n} \frac{\ln p}{p} + n \sum_{p \leqslant n} \frac{\ln p}{p(p-1)}$$

and

$$\ln n! > n \sum_{p \leqslant n} \frac{\ln p}{p} - \sum_{p \leqslant n} \ln p.$$

By Theorem 1.4, the last sum over p does not exceed $n \ln 4$. We thus infer

$$n \sum_{p \leqslant n} \frac{\ln p}{p} - n \ln 4 < n \ln n - n + (1 + \ln n) < n \ln n$$

whence

$$\sum_{p \leqslant x} \frac{\ln p}{p} = \sum_{p \leqslant n} \frac{\ln p}{p} \leqslant \ln n + \ln 4 < \ln x + \ln 4.$$

Also

$$\sum_{p\leqslant n}\frac{\ln p}{p(p-1)} < \sum_{m\geqslant 2}\frac{\ln m}{m(m-1)}$$
$$\leqslant \sum_{r\geqslant 1}\sum_{2^{r-1}<m\leqslant 2^r}\frac{r\ln 2}{m(m-1)} = \sum_{r\geqslant 1}\frac{r\ln 2}{2^r} = \ln 4,$$

from which it follows that

$$n\sum_{p\leqslant n}\frac{\ln p}{p} + n\ln 4 > n\ln n - n + 1$$

and finally

$$\sum_{p\leqslant x}\frac{\ln p}{p} > \ln n + \frac{1}{n} - (1+\ln 4) \geqslant \ln x - (1+\ln 4).$$

This concludes the proof. □

1.5. Two new asymptotic formulae

Mertens' first theorem differs conceptually from Chebyshev's estimates inasmuch as it delivers, for a weighted sum over primes, a genuine asymptotic formula. It is, in a sense, a prototype of a class of results of which the acme is the prime number theorem—corresponding to the case when the weighting coefficient equals the constant 1.

We shall see that Theorem 1.8 actually contains other results of the same type. In particular, it allows us to evaluate the expressions

$$\sum_{p\leqslant x}\frac{1}{p} \quad \text{and} \quad \prod_{p\leqslant x}\Big(1-\frac{1}{p}\Big).$$

Let us start by pointing out a close link between these two quantities.

Theorem 1.9. *Set* $c_0 := \sum_p \Big\{\ln\Big(\frac{1}{1-1/p}\Big) - \frac{1}{p}\Big\} \approx 0.315\,718$. *Then we have, for* $x \geqslant 2$,

$$\sum_{p\leqslant x}\frac{1}{p} = \ln\Big\{1\Big/\prod_{p\leqslant x}\Big(1-\frac{1}{p}\Big)\Big\} - c_0 + \frac{\vartheta}{2(x-1)}$$

where $\vartheta = \vartheta(x) \in]0,1[$.

Proof. With the given expression for c_0, we see that the stated formula holds with

$$0 < \vartheta(x) = 2(x-1)\sum_{p>x}\Big\{\ln\Big(\frac{1}{1-1/p}\Big) - \frac{1}{p}\Big\}$$
$$= 2(x-1)\sum_{p>x}\sum_{k\geqslant 2}\frac{1}{kp^k} < \sum_{p>x}\frac{2(x-1)}{2p(p-1)} < \sum_{n>x}\frac{(x-1)}{n(n-1)} = \frac{x-1}{N-1}$$

where $N := 1+\lfloor x\rfloor$ is the smallest integer $> x$. This establishes the required estimate. □

Theorem 1.10. *There is a constant c_1 such that, for $x \geqslant 2$,*

$$\sum_{p\leqslant x}\frac{1}{p} = \ln_2 x + c_1 + O\Big(\frac{1}{\ln x}\Big).$$

In addition, the constant involved in the Landau symbol can be chosen $\leqslant 2(1+\ln 4) < 5$.

Remark. Theorem 1.12 below easily yields a numerical approximation for c_1. We have $c_1 = \gamma - c_0 \approx 0.261497$.

Proof. By Mertens' first theorem, we have, for $t \geqslant 2$,

$$R(t) := \sum_{p\leqslant t}\frac{\ln p}{p} - \ln t = O(1).$$

However, we also have

$$\sum_{p\leqslant x}\frac{1}{p} = \int_{2-}^{x}\frac{1}{\ln t}\,\mathrm{d}\Big\{\sum_{p\leqslant t}\frac{\ln p}{p}\Big\} = \int_2^x\frac{\mathrm{d}t}{t\ln t} + \int_{2-}^{x}\frac{\mathrm{d}R(t)}{\ln t}$$
$$= \ln_2 x - \ln_2 2 + \frac{R(x)}{\ln x} - \frac{R(2-)}{\ln 2} + \int_2^x\frac{R(t)}{t(\ln t)^2}\,\mathrm{d}t,$$

where we have handled the integral involving $R(t)$ by Abel summation.

Let $R := \sup_{t\geqslant 2-}|R(t)|$. We have

$$\Big|\frac{R(x)}{\ln x} - \int_x^\infty\frac{R(t)}{t(\ln t)^2}\,\mathrm{d}t\Big| \leqslant \frac{2R}{\ln x} < \frac{2(1+\ln 4)}{\ln x},$$

by the upper bound in Theorem 1.8. From this we deduce the stated formula with

$$c_1 = -\ln_2 2 + 1 + \int_2^\infty\frac{R(t)}{t(\ln t)^2}\,\mathrm{d}t.$$ □

Theorem 1.11. *With the constants c_0 and c_1 as in Theorems 1.9 and 1.10, we have, for $x \geqslant 2$,*

$$\prod_{p\leqslant x}\Big(1-\frac{1}{p}\Big) = \frac{\mathrm{e}^{-(c_0+c_1)}}{\ln x}\Big\{1+O\Big(\frac{1}{\ln x}\Big)\Big\}.$$

This is an immediate corollary of Theorems 1.9 and 1.10.

1.6. Merten's formula

Mertens' second theorem, known as "Mertens' formula" provides an explicit value for the constant in Theorem 1.11.

Theorem 1.12 (Mertens formula). *With the notation of* § 1.5, *we have* $c_0 + c_1 = \gamma$, *where* γ *denotes Euler's constant. Thus*

$$\prod_{p\leqslant x}\Big(1-\frac{1}{p}\Big) = \frac{\mathrm{e}^{-\gamma}}{\ln x}\Big\{1+O\Big(\frac{1}{\ln x}\Big)\Big\} \qquad (x\geqslant 2).$$

Proof. For $\sigma > 1$, let us put

$$\zeta(\sigma) := \sum_{n\geqslant 1}\frac{1}{n^\sigma}.$$

Comparing the sum to an integral, we easily check that

$$\zeta(1+\sigma) = \frac{1}{\sigma} + O(1) \qquad (\sigma > 0).$$

Moreover, we have

$$\sum_{n\leqslant x}\frac{1}{n^{1+\sigma}} \leqslant \prod_{p\leqslant x}\Big(1-\frac{1}{p^{1+\sigma}}\Big)^{-1} \leqslant \zeta(1+\sigma)$$

since the product over p equals the sum $\sum_{n\geqslant 1}\varepsilon_n/n^{1+\sigma}$ where ε_n equals 1 if all the prime factors of n are $\leqslant x$ and 0 otherwise. Letting x tend to infinity, we obtain Euler's famous formula

$$\zeta(1+\sigma) = \prod_{p}\Big(1-\frac{1}{p^{1+\sigma}}\Big)^{-1}.$$

Now let us consider the function

$$f(\sigma) = \ln\zeta(1+\sigma) - \sum_{p}\frac{1}{p^{1+\sigma}} = \sum_{p}\Big\{\ln\Big(\frac{1}{1-p^{-1-\sigma}}\Big) - \frac{1}{p^{1+\sigma}}\Big\}.$$

Since the general term is positive and bounded above by $1/p(p-1)$, the series $f(\sigma)$ converges uniformly for $\sigma \geqslant 0$ and so its sum is continuous at 0. Hence

$$\lim_{\sigma\to 0} f(\sigma) = f(0) = c_0.$$

We shall transform the two terms of the sum $f(\sigma)$. On the one hand,

$$\ln\zeta(1+\sigma) = \ln\Big\{\frac{1}{\sigma}+O(1)\Big\} = \ln\Big(\frac{1}{\sigma}\Big) + O(\sigma) = \ln\Big(\frac{1}{1-\mathrm{e}^{-\sigma}}\Big) + O(\sigma)$$
$$= \sum_{n\geqslant 1}\frac{\mathrm{e}^{-\sigma n}}{n} + O(\sigma) = \int_0^\infty \mathrm{e}^{-\sigma t}\,\mathrm{d}H(t) + O(\sigma)$$

where we have set

$$H(t) := \sum_{1 \leqslant n \leqslant t} \frac{1}{n}.$$

Integrating the Stieltjes integral by parts then gives

$$\ln \zeta(1+\sigma) = \sigma \int_1^\infty \mathrm{e}^{-\sigma t} H(t)\, \mathrm{d}t + O(\sigma).$$

On the other hand,writing $P(u) := \sum_{p \leqslant u} 1/p$, we have

$$\sum_p \frac{1}{p^{1+\sigma}} = \int_1^\infty \frac{\mathrm{d}P(u)}{u^\sigma} = \sigma \int_1^\infty \frac{P(u)}{u^{1+\sigma}}\, \mathrm{d}u = \sigma \int_0^\infty \mathrm{e}^{-\sigma t} P(\mathrm{e}^t)\, \mathrm{d}t.$$

We hence obtain

$$f(\sigma) = \sigma \int_0^\infty \mathrm{e}^{-\sigma t} \big(H(t) - P(\mathrm{e}^t)\big)\, \mathrm{d}t + O(\sigma).$$

Now we saw in Theorem 0.8 that

$$H(t) = \ln t + \gamma + O(1/t) \qquad (t \geqslant 1),$$

and it follows from Theorem 1.10 that

$$P(\mathrm{e}^t) = \ln t + c_1 + O(1/t) \qquad (t \geqslant 1).$$

Hence, for $0 < \sigma \leqslant \frac{1}{2}$, we have

$$\begin{aligned} f(\sigma) &= \sigma \int_0^\infty \Big\{\gamma - c_1 + O\Big(\frac{1}{t+1}\Big)\Big\} \mathrm{e}^{-\sigma t}\, \mathrm{d}t + O(\sigma) \\ &= \gamma - c_1 + O\Big(\sigma + \sigma \int_0^\infty \mathrm{e}^{-\sigma t} \frac{\mathrm{d}t}{t+1}\Big) = \gamma - c_1 + O\big(\sigma \ln(1/\sigma)\big) \end{aligned}$$

and finally $c_0 = f(0) = \gamma - c_1$, which finishes the proof. □

1.7. Another theorem of Chebyshev

Chebyshev showed that if $\pi(x) \sim cx/\ln x$, then the constant c must equal 1.

Theorem 1.13. *We have*

$$\liminf_{x \to \infty} \frac{\pi(x)}{x/\ln x} \leqslant 1 \leqslant \limsup_{x \to \infty} \frac{\pi(x)}{x/\ln x}.$$

Proof. Since the two inequalities can be handled in analogous ways, let us just establish the left one. Put

$$\ell := \liminf_{x \to \infty} \frac{\pi(x)}{x/\ln x}.$$

For each $\varepsilon > 0$, there exists an $x_0 = x_0(\varepsilon) \geqslant 2$ such that

$$\pi(t) \geqslant (\ell - \varepsilon) \frac{t}{\ln t}, \qquad (t \geqslant x_0(\varepsilon)).$$

This implies, for $x > x_0$,

$$\sum_{p \leqslant x} \frac{1}{p} \geqslant \int_{x_0}^{x} \frac{\mathrm{d}\pi(t)}{t} = \frac{\pi(x)}{x} - \frac{\pi(x_0)}{x_0} + \int_{x_0}^{x} \frac{\pi(t)}{t^2}\,\mathrm{d}t$$

$$\geqslant -1 + (\ell - \varepsilon) \int_{x_0}^{x} \frac{\mathrm{d}t}{t \ln t} \geqslant (\ell - \varepsilon) \ln_2 x + O_\varepsilon(1).$$

By Theorem 1.10, it follows that $\ell - \varepsilon \leqslant 1$ and thus $\ell \leqslant 1$ since ε may be chosen arbitrarily small. □

Notes

§ 1.2. The proof given here of Theorem 1.4 was found independently by Erdős and Kalmár in 1939.

The prime number theorem implies

$$\prod_{p \leqslant n} p \leqslant \mathrm{e}^{(1+\varepsilon)n} \qquad (n \geqslant n_0(\varepsilon))$$

for all $\varepsilon > 0$. It is nonetheless useful to have at one's disposal uniform bounds, like those in Theorem 1.4, which hold without restriction on n. In this spirit Hanson showed in 1972 that

$$\prod_{p \leqslant n} p \leqslant d_n < 3^n \qquad (n \geqslant 1).$$

For numerical bounds on $\pi(x)$ and Chebyshev's functions (cf. § 3.6) see Rosser & Schoenfeld (1962, 1975) and Schoenfeld (1976). For example, we have

$$\frac{x}{\ln x}\Big(1 + \frac{1}{2\ln x}\Big) < \pi(x) < \frac{x}{\ln x}\Big(1 + \frac{3}{2\ln x}\Big) \qquad (x \geqslant 52).$$

The idea exploited by Nair (1982a) in the proof of Theorem 1.5 is not new: see Gelfond (1946). Gorshkov (1956) showed that the prime number theorem cannot be obtained by this method with polynomials in a single variable, and gave some numerical estimates of the best results than can be attained in this way. These bounds were sharpened by Aparicio Bernardo (1981). The generalization given by Nair (1982b) is intrinsically new. It provides very precise two-sided numerical estimates, and might, in principle, lead to a proof of the prime number theorem.

For more classical proofs of Chebyshev's type bounds for $\pi(n)$, see Exercises 16 to 20.

Exercises

10. Prove by induction that each integer > 1 can be written as a product of primes.

11. Consider the following,

(a) Let I be an *ideal* of $\mathbb{Z}$ (i.e. an additive subgroup stable under multiplication by any element of $\mathbb{Z}$). Show that $I = k\mathbb{Z}$, where k is the smallest positive element of I.

(b) Assume $m, n \in \mathbb{Z}$. Show that $m\mathbb{Z} + n\mathbb{Z}$ is an ideal of $\mathbb{Z}$.

(c) Let $d = (m, n)$ be the greatest common divisor of m and n. Prove *Bachet's theorem* (1624), better known, wrongly, as Bézout's theorem:

$$m\mathbb{Z} + n\mathbb{Z} = d\mathbb{Z}.$$

12. *Euclid's first theorem.* Let $a, b \in \mathbb{Z}$ and p be a prime number. Assume that $p \mid ab$ and $p \nmid a$. Show using Bachet's theorem that there exist $u, v \in \mathbb{Z}$ such that $up + va = 1$. Deduce that $p \mid b$.

13. Using Euclid's first theorem, give an inductive proof of the *fundamental theorem of arithmetic*: the decomposition of an integer > 1 into a product of prime numbers is unique, up to the order of the factors.

14. Let p_n be the n-th prime, and $d_n := p_{n+1} - p_n$. Granting the prime number theorem in its minimal form, namely $\pi(x) \sim x/\ln x$ $(x \to \infty)$, show the following assertions.

(a) $p_n \sim n \ln n \quad (n \to \infty)$.

(b) $\displaystyle\sum_{1<n\leqslant x} d_n/\ln n \sim x \quad (x \to \infty)$.

(c) $\displaystyle\liminf_{n\to\infty} (d_n/\ln n) \leqslant 1 \leqslant \limsup_{n\to\infty} (d_n/\ln n)$.

(d) For any $\alpha > 0$ there exists a non-decreasing sequence of integers $\mathcal{A} = \{n_1, n_2, \ldots\}$, such that $p_{n_j} \sim \alpha j$ $(j \to \infty)$.

(e) The set of rational numbers of the form p'/p, where p and p' are primes, is dense in $[0, +\infty[$.

15. Show that the prime number theorem in the strong form

$$\pi(x) = \frac{x}{\ln x} + O\Big(\frac{x}{(\ln x)^2}\Big) \qquad (x \geqslant 2)$$

implies convergence of the series $\sum_{p\in\mathbb{P}} 1/p^{1+i}$.

16. Let $\{a_n\}_{n=1}^{\infty}$ be a sequence of non-negative real numbers. Assume that

$$B(x) := \sum_{n\leqslant x} a_n \lfloor x/n \rfloor = x \ln x + Cx + o(x).$$

Prove *Shapiro's Tauberian theorem (1950): there exist two positive constants* α *and* β *such that:* $\alpha x \leqslant A(x) := \sum_{n \leqslant x} a_n \leqslant \beta x$ $(x > x_0)$.

Make explicit the numerical values obtained for α and β. Examine the case when one only assumes $B(x) = x \ln x + O(x)$. [*Hint: Consider* $A(x) - A(x/2)$ *and use the properties of the function* $u \mapsto \lfloor u \rfloor - 2 \lfloor u/2 \rfloor$.]

17. *von Mangoldt's function.* Define $\Lambda(n) := \ln p$ if $n = p^\nu$, $\Lambda(n) := 0$ if n is not a prime power. Show that

(a) $\sum_{d|n} \Lambda(d) = \ln n$,

(b) $\sum_{d \leqslant x} \Lambda(d) \lfloor x/d \rfloor = x \ln x - x + O(\ln x) \qquad (x \geqslant 2)$.

18. Deduce from the two previous exercises that the function $\psi(x) := \sum_{n \leqslant x} \Lambda(n)$ satisfies

$$\alpha x \leqslant \psi(x) \leqslant \beta x \qquad (x > x_0)$$

for two positive constants α and β, which should be computed.

19. Deduce from Exercise 18 the existence of two positive constants a and b, which should be determined, such that: $ax/\ln x \leqslant \pi(x) \leqslant bx/\ln x$ $(x \geqslant 2)$.[2]

20. *Chebyshev's estimates.* Set $M > 1$, and let h be an arithmetic function such that

(i) $h(1) = 1$, $\sum_{1 \leqslant m \leqslant M} h(m)/m = 0$

(ii) $\chi(x) := \sum_{1 \leqslant m \leqslant M} h(m) \lfloor x/m \rfloor \in [0, 1] \qquad (x \in \mathbb{R})$.

Show that, under the hypotheses of Exercise 16, we have, for large x,

$$(H - \varepsilon)x < A(x) < \Big(\frac{a}{a-1} H + \varepsilon\Big)x,$$

where we have put $H := -\sum_{1 \leqslant m \leqslant M} h(m) \ln m/m$ and where a is defined as the largest integer such that $\chi(x) = 1$ for $1 \leqslant x < a$. To what choice for h does Shapiro's Tauberian theorem correspond? Show that selecting

$$M = 6,\ h(1) = h(6) = 1,\ h(2) = -1,\ h(3) = -2,\ h(4) = h(5) = 0$$

leads to a proof of Bertrand's postulate. Recover Chebyshev's estimates for $\psi(x)$ with his original choice

$$M = 30,\ h(1) = h(30) = 1,\ h(2) = h(3) = h(5) = -1,$$
$$h(m) = 0 \quad (m = 4 \quad \text{or} \quad 6 \leqslant m \leqslant 29).$$

21. Deduce from Bertrand's postulate that there are infinitely many primes whose first digit in base 10 is 1.

22. Show that one may deduce simply from Bertrand's postulate that the number $H_n = \sum_{1 \leqslant m \leqslant n} 1/m$ is not integral for any $n > 1$. Give another proof of this result by considering powers of 2.

[2] For a refinement of Shapiro's Tauberian theorem leading to a proof of the prime number theorem, see Smith (1980).

23. Retain the notations in Exercises 17 and 18.

(a) Show that $\sum_{n\leqslant x}(\ln n)/n = \frac{1}{2}(\ln x)^2 + O(1)$ $(x \geqslant 1)$.

(b) Show that

$$\sum_{n\leqslant x}\frac{\Lambda(n)}{n} = \int_1^x \frac{\psi(t)}{t^2}\,\mathrm{d}t + \frac{\psi(x)}{x} \qquad (x \geqslant 1).$$

(c) From this question on, we assume the prime number theorem in the strong form

$$\psi(x) = x + O\big(x/(\ln 2x)^2\big) \qquad (x \geqslant 1). \tag{1.1}$$

Show the existence of a constant A such that

$$\sum_{n\leqslant x}\frac{\Lambda(n)}{n} = \ln x + A + o(1) \qquad (x \to \infty). \tag{1.2}$$

(d) By using the identity established in Exercise 17(a), show that

$$2\sum_{n\leqslant x}\frac{\ln n}{n} = (\ln x)^2 + 2(A+\gamma)\ln x + o(\ln x) \qquad (x \to \infty).$$

Deduce that $A = -\gamma$.

24. Show that relation (1.2) implies the prime number theorem.

25. Show that the prime number theorem in the strong form (1.1) implies the asymptotic formula

$$\sum_{p\leqslant x}\frac{\ln p}{p-1} = \ln x - \gamma + o(1), \tag{1.3}$$

where the letter p denotes a generic prime number.

Conversely, show that (1.3) implies the prime number theorem.

26. Show that the prime number theorem in the form (1.1) implies the formula

$$\int_1^\infty \frac{t-\psi(t)}{t^2}\,\mathrm{d}t = 1+\gamma.$$

For this, one can use the relation established in Exercise 23(a). Study the converse by introducing, for $0 < \varepsilon < 1$, the quantities $\int_x^{x(1\pm\varepsilon)} \psi(t)\,\mathrm{d}t/t^2$.

27. Is it true that any non-zero natural number can be transformed into a prime number by just changing a single digit in its decimal expansion?

28. Show that $\lim_{x\to\infty}\sum_{\sqrt{x}<p\leqslant x} 1/p = \ln 2$. Deduce from this that, as $x \to \infty$, a positive proportion of the integers $n \leqslant x$ have their largest prime factor $> \sqrt{n}$. Can one sharpen this assertion?

29. The letters p and q denoting prime numbers, show, using Mertens' second theorem, that there exists a constant a such that

$$\sum_{pq \leqslant x} \frac{1}{pq} = (\ln_2 x + a)^2 + o(1) + O(S),$$

where $S := \sum_{p \leqslant x/2} \frac{1}{p} \ln\left(\frac{\ln x}{\ln(x/p)}\right)$. Show that S is bounded as a function of x. Generalize the result obtained to an asymptotic formula for the quantity

$$\sum_{p_1 \cdots p_k \leqslant x} \frac{1}{p_1 \cdots p_k}$$

where the integer k is fixed and $x \to \infty$.

30. Is it possible for the numbers p, $8p-1$ and $8p+1$ to be simultaneously prime? The problem may be reduced modulo m for a suitable value of m.

31. *Mersenne primes.* Show that if $2^k - 1$ is prime then k is prime.

32. Show that there are infinitely many primes of the form $4n+3$. The number $N = 4 \cdot n! - 1$ may be considered.

33. Find those primes p such that $p^2 + 2$ is also prime.

34. Show that a product of integers of the form $4k+1$ is again of that form. Deduce that there are infinitely many primes of the form $4m-1$. Hint: Consider the number $n = 4p_1 \cdots p_r - 1$ where the p_j are the r smallest primes of the form $4m-1$.

35. Show that there are infinitely many prime numbers of the form $6m-1$. Hint: Introduce the number $n = (p_1 \cdots p_r)^2 + 4$ where the p_j are the r smallest primes of the required form.

36. Let $k \in \mathbb{N}^*$. Find an integer N which is followed by k composite integers.

37. *Quadratic residues.* Let p be an odd prime and denote by Q_p the set of *quadratic residues* in $(\mathbb{Z}/p\mathbb{Z})^*$, or in other words the set of those elements a of $(\mathbb{Z}/p\mathbb{Z})^*$ such that the equation $x^2 \equiv a \pmod{p}$ has at least one solution.

(a) Show that, if g is a generator of $(\mathbb{Z}/p\mathbb{Z})^*$, then Q_p coincides with the set of even powers of g. Deduce the value of $|Q_p|$.

(b) In this question, we set about recovering the previous result without making use of the cyclic structure of $(\mathbb{Z}/p\mathbb{Z})^*$.

(i) Show that the equation $x^2 = a$ has 0 or 2 solutions in $(\mathbb{Z}/p\mathbb{Z})^*$.

(ii) Conclude.

38. *Legendre's symbol.* Retain the hypotheses of Exercise 37. For $a \in \mathbb{Z}/p\mathbb{Z}$, put

$$\left(\frac{a}{p}\right) := \begin{cases} 1 & \text{if } a \in Q_p, \\ 0 & \text{if } a \equiv 0 \,(\mathrm{mod}\, p), \\ -1 & \text{if } a \notin Q_p. \end{cases}$$

(a) Show that Q_p is the kernel of the multiplicative homomorphism $x \mapsto x^{(p-1)/2}$ mapping $(\mathbb{Z}/p\mathbb{Z})^*$ into itself.

(b) Deduce that $(a|p) \equiv a^{(p-1)/2} \,(\mathrm{mod}\, p)$ for all a in $\mathbb{Z}/p\mathbb{Z}$.

(c) Under what condition is (-1) a quadratic residue modulo p?

(d) Let $p_1, \ldots, p_k$ be the k smallest primes of the form $4m+1$. By considering the number $N = (p_1 \cdots p_k)^2 + 1$, show that there exist a prime p_{k+1}, distinct from the previous ones and of the same form. Deduce from this that there are infinitely many primes of the form $4m+1$.

39. Let p be an odd prime and let m, n be integers. We extend the definition of Legendre's symbol by putting $(n|p) = (a|p)$ when $n \in \mathbb{Z}$, $n \equiv a \,(\mathrm{mod}\, p)$.

(a) Calculate $S_n := \sum_{0 \leqslant j < p} (j + n \mid p)$.

(b) Calculate $T_{m,n} := \sum_{0 \leqslant j < p} (mj + n \mid p)$.

(c) Calculate $U_n := \sum_{0 \leqslant j < p} (j(j+n) \mid p)$. One may introduce the map $j \mapsto \overline{j}$ which associates to each element j in $(\mathbb{Z}/p\mathbb{Z})^*$ its inverse modulo p.

(d) Using the property established in Exercise 38(b), show that, if $(n, p) = 1$, then $\sum_{0 \leqslant j < p} (j^2 + n \mid p) = -1$.

40. *Euler and Gauss.* Euler showed that $(2|p) = (-1)^{(p^2-1)/8}$, and Gauss established the *law of quadratic reciprocity*:

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{(p-1)(q-1)/4} \qquad (p > 2,\ q > 2,\ p \neq q).$$

We offer here several examples of applications.

(a) Let p be a prime of the form $4m+1$. Show that $m^m \equiv 1 \,(\mathrm{mod}\, p)$.

(b) For which primes p does the equation $x^2 - 3 \equiv 0 \,(\mathrm{mod}\, p)$ have solutions?

(c) For which primes p does the equation have $x^2 - 5 \equiv 0 \,(\mathrm{mod}\, p)$ have solutions?

Arithmetic functions

2.1. Definitions

An *arithmetic function* is a complex-valued function defined on $\mathbb{N}^* = \{1, 2, 3, \ldots\}$. Two classes of arithmetic functions play particularly important roles: additive functions and multiplicative functions. An arithmetic function is called *additive* if it satisfies

$$f(mn) = f(m) + f(n), \quad (\text{if } (m, n) = 1); \tag{2.1}$$

it is said to be *multiplicative* if $f(1) = 1$ and

$$f(mn) = f(m)f(n), \quad (\text{if } (m, n) = 1). \tag{2.2}$$

The condition $f(1) = 1$ is a convenient convention designed to exclude the null function from the class of multiplicative functions.[1]

The main interest of these notions is that an additive or a multiplicative function respects the multiplicative structure of $\mathbb{N}$, in the sense that the image of an integer is the sum or the product of the images of the prime powers appearing in its canonical factorization. One can thus write

$$f(n) = \sum_{p^\nu \| n} f(p^\nu), \quad f(n) = \prod_{p^\nu \| n} f(p^\nu),$$

when f is, respectively, additive or multiplicative.

A function f is called *completely additive* or *completely multiplicative* if the conditions (2.1) or (2.2) hold even when $(m, n) \neq 1$, that is if $f(p^\nu) = \nu f(p)$ or $f(p^\nu) = f(p)^\nu$, respectively. One says that f is *strongly additive* or *strongly multiplicative* if, in addition to (2.1) or (2.2), we have $f(p^\nu) = f(p)$ for all $\nu \geqslant 1$.

[1] However, note Selberg's definition given in § 4.7.

2.2. Examples

The following arithmetic functions are classical and define some fundamental concepts attached to the multiplicative structure of an integer n:

- The counting functions of the total number of prime factors of n, taken with or without multiplicity, thus
$$\Omega(n) := \sum_{p^\nu \| n} \nu, \qquad \omega(n) := \sum_{p^\nu \| n} 1 = \sum_{p \mid n} 1.$$
- The "number of divisors" function and the function "sum of the k-th powers of divisors", traditionally denoted by
$$\tau(n) := \sum_{d \mid n} 1, \qquad \sigma_k(n) := \sum_{d \mid n} d^k \qquad (k \in \mathbb{C}).$$
Usually $\tau(n)$ is referred to as the divisor function and one writes $\sigma_1(n) = \sigma(n)$.
- Euler's totient function, counting the number of invertible residues modulo n, that is
$$\varphi(n) := \sum_{1 \leqslant h \leqslant n,\, (h,n)=1} 1.$$
- The Möbius function, defined by
$$\mu(n) := \begin{cases} (-1)^{\omega(n)}, & \text{if } n \text{ is squarefree,} \\ 0, & \text{otherwise.} \end{cases}$$
- The von Mangoldt function[2]
$$\Lambda(n) := \begin{cases} \ln p, & \text{if } n = p^\nu \text{ for some } \nu \geqslant 1, \\ 0, & \text{if } n \text{ is not a prime power.} \end{cases}$$

It is immediate from their definitions that Ω and ω are additive, the former completely, the latter strongly. The case of the function $\tau(n)$ is less obvious. However, representing the divisors of n as all integers of the form $d = \prod_{p \mid n} p^{\alpha_p}$ where $0 \leqslant \alpha_p \leqslant v_p(n)$ for each prime p, we deduce that

$$\tau(n) = \prod_{p \mid n} (v_p(n) + 1).$$

Thus we can state the following assertion.

Theorem 2.1. *The divisor function is multiplicative. We have*

$$\tau(n) = \prod_{p^\nu \| n} (\nu + 1) \qquad (n \geqslant 1).$$

[2]We also give in § 2.6 a different definition of Λ, which can readily be established to be equivalent to this one.

Euler's totient and the functions σ_k will be examined later. Let us consider now the Möbius function. We have

$$\mu(p^\nu) = \begin{cases} -1 & \text{if } \nu = 1 \\ 0 & \text{if } \nu > 1. \end{cases}$$

and hence immediately obtain that $\mu(n) = \prod_{p^\nu \| n} \mu(p^\nu)$.

Theorem 2.2. *The Möbius function is multiplicative.*

2.3. Formal Dirichlet series

An essential tool in the study of arithmetic functions is the concept of a Dirichlet series, which we shall study in greater detail in Part II. Here we restrict ourselves with identifying certain algebraic properties of arithmetic functions that can be conveniently displayed by employing the notion of a formal Dirichlet series.

Definition 2.3. *Let f be an arithmetic function. The formal Dirichlet series associated to f is the formal series*

$$D(f;s) := \sum_{n \geqslant 1} \frac{f(n)}{n^s}.$$

The sum and the product of two formal Dirichlet series are defined in a natural way by

$$D(f;s) + D(g;s) = \sum_{n \geqslant 1} \frac{f(n) + g(n)}{n^s}, \tag{2.3}$$

and

$$D(f;s)\, D(g;s) = \sum_{n \geqslant 1} \frac{h(n)}{n^s}, \tag{2.4}$$

with

$$h(n) = \sum_{dd'=n} f(d)g(d'). \tag{2.5}$$

This second definition agrees with the the formal calculation

$$\sum_{m \geqslant 1} \frac{f(m)}{m^s} \sum_{k \geqslant 1} \frac{g(k)}{k^s} = \sum_{m \geqslant 1, k \geqslant 1} \frac{f(m)g(k)}{(mk)^s} = \sum_{n \geqslant 1} \frac{1}{n^s} \sum_{km=n} f(m)g(k).$$

One can easily check that the set of formal Dirichlet series equipped with these two operations has the structure of a commutative ring with unity given by the series

$$D(\delta;s) = 1,$$

associated to the arithmetic function

$$\delta(n) := \begin{cases} 1 & (n = 1) \\ 0 & (n > 1). \end{cases}$$

2.4. The ring of arithmetic functions

The correspondence between arithmetic functions and formal Dirichlet series induces on the set of arithmetic functions an addition $(+)$ and a product $(*)$ whose fundamental properties are, respectively,

$$D(f+g;s) = D(f;s) + D(g;s)$$
$$D(f*g;s) = D(f;s)\,D(g;s).$$

Thus we have $(f+g)(n) = f(n) + g(n)$, and $(f*g)(n) = h(n)$ where h is defined by (2.5).

The product $*$ is called *Dirichlet convolution.* These operations give the set $\mathbb{A}$ of arithmetic functions the structure of a commutative ring with unity, isomorphic to that of formal Dirichlet series. Cashwell & Everett (1959) showed that this ring is *factorial*, that is to say an integral domain whose quotient by the group of units satisfies the fundamental theorem of arithmetic.

A necessary and sufficient condition for an element $f \in \mathbb{A}$ to be invertible is that $f(1) \neq 0$. Indeed, under this assumption, the system of equations

$$\sum_{d|n} f(n/d)g(d) = \delta(n) \qquad (n \geqslant 1) \tag{2.6}$$

allows us to calculate $g(n)$ recursively: we have

$$\begin{cases} g(1) = \dfrac{1}{f(1)} \\ g(n) = \dfrac{-1}{f(1)} \displaystyle\sum_{d|n,\, d<n} f(n/d)g(d) \qquad (n > 1). \end{cases} \tag{2.7}$$

Conversely, if $f(1) = 0$, then (2.6) is not solvable for $n = 1$, and f is not invertible.

Theorem 2.4. *The group $\mathbb{G}$ of units in the ring $\mathbb{A}$ of arithmetic functions consists of those arithmetic functions f with $f(1) \neq 0$.*

An element π in $\mathbb{A}$ is prime if it is not a unit and if the relation $\pi = u*v$ implies that either u or v is a unit. One checks easily that the set of prime elements in $\mathbb{A}$ strictly contains the set of functions f such that $f(1) = 0$ and $f(p) \neq 0$ for some prime number p—cf. Exercises 48, 49.

The multiplicative functions are units since they satisfy $f(1) = 1$ by definition. The following result will allow us to show they form a subgroup of $\mathbb{G}$.

Theorem 2.5. *A necessary and sufficient condition for a function f in $\mathbb{A}$ to be multiplicative is that its associated formal series $D(f;s)$ be expandable as an infinite formal product of Euler type, namely*

$$D(f;s) = \prod_p \Big(1 + \sum_{\nu \geqslant 1} \frac{f(p^\nu)}{p^{\nu s}}\Big). \tag{2.8}$$

This is immediate, since equation (2.8) is algebraically equivalent to the conditions

$$f(1) = 1, \quad f(n) = \prod_{p^\nu \| n} f(p^\nu) \qquad (n > 1).$$

Theorem 2.6. *The set $\mathbb{M}$ of multiplicative arithmetic functions is a subgroup of $\mathbb{G}$, the group of units of $\mathbb{A}$.*

Proof. If f and g are in $\mathbb{M}$, formal calculation immediately yields the relation

$$D(f;s)D(g;s) = \prod_p \Big(1 + \sum_{\nu \geqslant 1} \frac{f(p^\nu)}{p^{\nu s}}\Big)\Big(1 + \sum_{\nu \geqslant 1} \frac{g(p^\nu)}{p^{\nu s}}\Big) = \prod_p \Big(1 + \sum_{\nu \geqslant 1} \frac{h(p^\nu)}{p^{\nu s}}\Big)$$

where $h(p^\nu)$ is defined by the formula

$$h(p^\nu) = \sum_{0 \leqslant j \leqslant \nu} f(p^j) g(p^{\nu - j}). \tag{2.9}$$

Since, by definition, $D(f;s)D(g;s) = D(f*g;s)$, it follows that $f*g$ coincides with the multiplicative function determined by (2.9) on prime powers.

It remains to verify that the inverse $\widetilde{f}$ of any function $f \in \mathbb{M}$ is also in $\mathbb{M}$. Equation (2.6) applied with $g = \widetilde{f}$, $n = 1$, and then with $n = p^\nu$ implies immediately that $\widetilde{f}(1) = 1$ and

$$\Big(1 + \sum_{\nu \geqslant 1} \frac{f(p^\nu)}{p^{\nu s}}\Big)\Big(1 + \sum_{\nu \geqslant 1} \frac{\widetilde{f}(p^\nu)}{p^{\nu s}}\Big) = 1$$

for any prime p. Therefore

$$D(f;s) \prod_p \Big(1 + \sum_{\nu \geqslant 1} \frac{\widetilde{f}(p^\nu)}{p^{\nu s}}\Big) = 1 = D(f;s)D(\widetilde{f};s)$$

and relation (2.8) is clearly satisfied by $\widetilde{f}$. By Theorem 2.5, this implies that $\widetilde{f} \in \mathbb{M}$. □

Let $\mathbf{1}$ be the arithmetic function defined by

$$\mathbf{1}(n) = 1 \qquad (n \geqslant 1). \tag{2.10}$$

Then $\mathbf{1}$ is trivially multiplicative, and we have for any $n \geqslant 1$

$$\tau(n) = \sum_{d|n} 1 = \sum_{d|n} \mathbf{1}(d)\mathbf{1}(n/d),$$

whence

$$\tau = \mathbf{1} * \mathbf{1}. \tag{2.11}$$

By Theorem 2.6, this provides a new proof for the multiplicativity of the divisor function τ.

Denote by j the identity function, viz.

$$j(n) = n \qquad (n \geqslant 1). \tag{2.12}$$

Clearly,

$$\sigma = \mathbf{1} * j \tag{2.13}$$

and consequently, we obtain the following result.

Theorem 2.7. *The "sum of divisors" function $\sigma(n)$ is multiplicative.*

Of course, the same holds for the functions

$$\sigma_k(n) = \sum_{d|n} d^k = (\mathbf{1} * j^k)(n)$$

for any real or complex value of the parameter k.

2.5. The Möbius inversion formulae

For any prime number p and any integer $\nu \geqslant 0$, we have

$$(\mathbf{1} * \mu)(p^\nu) = \sum_{j=0}^{\nu} \mu(p^j) = \left\{ \begin{matrix} 1 & \text{if } \nu = 0 \\ 0 & \text{if } \nu \geqslant 1 \end{matrix} \right\} = \delta(p^\nu).$$

Since $\mathbf{1} * \mu$ and δ are multiplicative, these two functions must be equal.

Theorem 2.8. *The Möbius function is the convolution inverse of the function* $\mathbf{1}$. *In other words, we have*

$$\mathbf{1} * \mu = \delta, \tag{2.14}$$

or

$$\sum_{d|n} \mu(d) = \begin{cases} 1 & (n = 1) \\ 0 & (n > 1). \end{cases} \tag{2.15}$$

However trivial it may appear, formula (2.15) is nonetheless rich in applications. In particular, it is the starting point for combinatorial sieve theory—cf. §§ I.4.2–I.4.3.

Before actually using (2.15), we note we could have effectively calculated the convolution inverse of μ by the method described in the course of the proof of Theorem 2.6. In general, the inverse $\widetilde{f}$ of a multiplicative function f is the multiplicative function determined on prime powers by the formal identity

$$\Big(1+\sum_{\nu\geqslant 1}\widetilde{f}(p^\nu)\xi^\nu\Big)\Big(1+\sum_{\nu\geqslant 1}f(p^\nu)\xi^\nu\Big)=1. \tag{2.16}$$

This remark can prove very useful when bounds are needed for $|\widetilde{f}(p^\nu)|$. One applies (2.16) for complex ξ in a common disc of convergence for the two series, and obtains the desired estimate from Cauchy's integral formula.

Theorem 2.9 (First Möbius inversion formula). *Let f and g be arithmetic functions. The following two properties are equivalent:*

$$\text{(i)}\quad g(n)=\sum_{d\mid n}f(d) \qquad (n\geqslant 1)$$

$$\text{(ii)}\quad f(n)=\sum_{d\mid n}g(d)\,\mu(n/d) \qquad (n\geqslant 1).$$

Proof. Condition (i) is equivalent to $g = f * \mathbf{1}$, condition (ii) to $f = g * \mu$. The result therefore follows from (2.14). □

Let us now consider a generalization to functions of a real variable.

Theorem 2.10 (Second Möbius inversion formula). *Let F and G be functions defined on $[1,+\infty[$. The following two conditions are equivalent:*

$$\text{(i)}\quad F(x)=\sum_{n\leqslant x}G(x/n) \qquad (x\geqslant 1)$$

$$\text{(ii)}\quad G(x)=\sum_{n\leqslant x}\mu(n)F(x/n) \qquad (x\geqslant 1).$$

Proof. Let us prove, for instance, the implication (i) ⇒ (ii). The converse goes analogously. For $x \geqslant 1$, we have

$$\sum_{n\leqslant x}\mu(n)F(x/n)=\sum_{n\leqslant x}\mu(n)\sum_{m\leqslant x/n}G(x/mn)=\sum_{k\leqslant x}G(x/k)\sum_{mn=k}\mu(m).$$

By (2.15), the inner sum has value $\delta(k)$. This yields (ii). □

Applying Theorem 2.10 to the case $G(x) \equiv 1$, we obtain

$$\sum_{n \leqslant x} \mu(n) \lfloor x/n \rfloor = 1 \qquad (x \geqslant 1). \tag{2.17}$$

This suggests that

$$\lim_{x \to \infty} \sum_{n \leqslant x} \frac{\mu(n)}{n} = 0. \tag{2.18}$$

We shall see in Section 3.6 that (2.18) is equivalent to the prime number theorem.

2.6. Von Mangoldt's function

Traditionally denoted by Λ, *von Mangoldt's function* is defined as the arithmetic function

$$\Lambda := \mu * \ln. \tag{2.19}$$

For each $n \geqslant 1$, we may thus write

$$\Lambda(n) = \sum_{d|n} \mu(d) \ln(n/d) = \delta(n) \ln n - \sum_{d|n} \mu(d) \ln d = -\sum_{d|n} \mu(d) \ln d$$

that is

$$\Lambda = -\mu \ln * \mathbf{1}. \tag{2.20}$$

For all m, n such that $(m, n) = 1$, it is follows from (2.20) that

$$\begin{aligned} \Lambda(mn) &= -\sum_{d|m} \sum_{t|n} \mu(dt) \ln(dt) = -\sum_{d|m} \mu(d) \sum_{t|n} \mu(t) \{\ln d + \ln t\} \\ &= \sum_{d|m} \mu(d) \{-\delta(n) \ln d + \Lambda(n)\} = \delta(n)\Lambda(m) + \delta(m)\Lambda(n). \end{aligned}$$

Thus $\Lambda(n)$ is zero whenever n is not a prime power. Moreover, it follows from (2.20), that

$$\Lambda(n) = \begin{cases} \ln p & (n = p^\nu,\ \nu \geqslant 1) \\ 0 & (n \neq p^\nu). \end{cases} \tag{2.21}$$

Chebyshev's summatory functions

$$\psi(x) := \sum_{n \leqslant x} \Lambda(n), \tag{2.22}$$

$$\vartheta(x) := \sum_{p \leqslant x} \ln p, \tag{2.23}$$

are important in the analytic theory of prime numbers. We have

$$\psi(x) = \ln\left(\operatorname{lcm}\{n : n \leqslant x\}\right) \geqslant \lfloor x \rfloor \ln 2 \qquad (x \geqslant 7) \tag{2.24}$$

$$\vartheta(x) \leqslant x \ln 4 \qquad (x \geqslant 2) \tag{2.25}$$

where the inequalities result, respectively, from Theorems 1.5 and 1.4. We infer immediately from (2.21) that

$$\psi(x) = \sum_{k \geqslant 1} \vartheta(x^{1/k}) \qquad (x \geqslant 1). \tag{2.26}$$

For each x, the summation over k is finite since the general term vanishes as soon as $2^k > x$.

Theorem 2.11. *For $x \geqslant 2$, we have*

$$\psi(x) = \vartheta(x) + O(\sqrt{x}), \tag{2.27}$$

$$\pi(x) = \frac{\vartheta(x)}{\ln x} + O\left(\frac{x}{(\ln x)^2}\right). \tag{2.28}$$

Proof. Formula (2.27) follows directly from (2.25) and (2.26). To establish (2.28), we evaluate $\vartheta(x)$ by Abel summation:

$$\vartheta(x) = \int_1^x \ln t \, \mathrm{d}\pi(t) = \pi(x) \ln x - \int_1^x \frac{\pi(t)}{t} \, \mathrm{d}t.$$

By Chebyshev's upper bound (Theorem 1.3), the integral is $O(x/\ln x)$, whence the stated formula. □

Corollary 2.12. *Let α, β be constants such that $0 < \alpha < \ln 2$, and $\beta > \ln 4$. For large enough x, we have*

$$\alpha x \leqslant \vartheta(x) \leqslant \psi(x) \leqslant \beta x. \tag{2.29}$$

This follows immediately from Theorems 2.11 and 1.3.

2.7. Euler's totient function

For each $n \geqslant 1$, we have defined $\varphi(n)$, the value at n of Euler's totient function, as the number of invertible residues modulo n. We can hence write

$$\varphi(n) = \sum_{1 \leqslant m \leqslant n} \delta\big((m,n)\big),$$

from which we infer by applying (2.14) that

$$\varphi(n) = \sum_{m \leqslant n} \sum_{d \mid (m,n)} \mu(d) = \sum_{d \mid n} \mu(d) \sum_{\substack{m \leqslant n \\ m \equiv 0 \,(\mathrm{mod}\, d)}} 1 = \sum_{d \mid n} \mu(d) \frac{n}{d}.$$

This relation is equivalent to

$$\varphi = \mu * j. \tag{2.30}$$

In particular, we have for each prime p

$$\varphi(p^\nu) = \mu(1)p^\nu + \mu(p)p^{\nu-1} = p^\nu\Big(1 - \frac{1}{p}\Big).$$

We have therefore shown the following theorem.

Theorem 2.13. *Euler's totient function φ is multiplicative. For each integer $n \geqslant 1$, we have*

$$\varphi(n) = n\prod_{p|n}\Big(1 - \frac{1}{p}\Big). \tag{2.31}$$

Another proof of this result consists in noting that any rational fraction h/n with denominator n can be uniquely written in the form $h/n = a/d$, with $d|n$, $(a,d) = 1$. For any real function F we thus have

$$\sum_{1\leqslant h\leqslant n} F(h/n) = \sum_{d|n}\sum_{\substack{1\leqslant a\leqslant d\\(a,d)=1}} F(a/d). \tag{2.32}$$

Upon applying this identity to the case $F(x) \equiv 1$, it follows that

$$n = \sum_{d|n}\varphi(d). \tag{2.33}$$

In other words: $j = \mathbf{1} * \varphi$, and the desired result follows from (2.14).

The *inclusion–exclusion principle* (cf. Notes) easily provides a third proof of Theorem 2.13.

Notes

§ 2.4. For other properties of the ring of arithmetic functions, see Shapiro (1972).

§ 2.5. *Inclusion–exclusion principle.* A purely combinatorial version of the Möbius inversion formula, in its fundamental form $\mathbf{1} * \mu = \delta$, is known under the name of the inclusion–exclusion principle. It may be formulated in this way:

Let $\mathcal{A}$ be a finite set with cardinality N, and let $\mathcal{P} = \{(1), \dots, (k)\}$ be a set of numbered properties. For each subset I of $\mathcal{P}$, denote by $A(I)$ the number of elements in $\mathcal{A}$ which satisfy at least all the properties in I. Denoting by $S(\mathcal{A}, \mathcal{P})$ the number of elements in $\mathcal{A}$ which do not satisfy any property in $\mathcal{P}$, we have

$$S(\mathcal{A}, \mathcal{P}) = N + \sum_{1 \leqslant s \leqslant k} (-1)^s \sum_{\substack{I \subseteq \mathcal{P} \\ |I| = s}} A(I). \tag{2.34}$$

It is easy to prove the identity (2.34) directly by introducing for each element a of $\mathcal{A}$, the exact number $m(a)$, $0 \leqslant m(a) \leqslant k$, of properties satisfied by a. It is also possible to argue indirectly in the following way. Let $2 = p_1 < p_2 < \cdots < p_k$ be the k smallest prime numbers, put

$$P = \prod_{1 \leqslant j \leqslant k} p_j,$$

and associate to each a in $\mathcal{A}$ the integer $F(a) = \prod_{a \in (j)} p_j$ where the notation $a \in (j)$ means that a satisfies property (j). Then we have

$$S(\mathcal{A}, \mathcal{P}) = \sum_{a \in \mathcal{A}} \delta\big(F(a)\big) = \sum_{a \in \mathcal{A}} \sum_{d \mid F(a)} \mu(d) = \sum_{d \mid P} \mu(d) \sum_{\substack{a \in \mathcal{A} \\ F(a) \equiv 0 \pmod{d}}} 1.$$

If $d = \prod_{(j) \in I} p_j$, we have

$$\mu(d) = (-1)^{|I|}, \qquad \text{and} \qquad \sum_{\substack{a \in \mathcal{A} \\ F(a) \equiv 0 \pmod{d}}} 1 = A(I).$$

This implies (2.34).

Exercises

41. Let $\zeta(s)$ be the Riemann zeta function, which we identify with the formal Dirichlet series associated to the arithmetic function $\mathbf{1}(n) = 1$ $(n \geqslant 1)$. Express in terms of $\zeta(s)$ the Dirichlet series associated with the following arithmetic functions: $\mu(n)$, $\mu(n)^2$, $\varphi(n)$, $\sigma(n)$, $\tau(n)$, $2^{\omega(n)}$, $\Lambda(n)$. Find the convolution relations corresponding to these identities.

42. Show that, for all $n \geqslant 1$, we have $2^{\omega(n)} \leqslant \tau(n) \leqslant 2^{\Omega(n)}$.

43. Show that, for all $n > 1$, we have $6n^2/\pi^2 < \sigma(n)\varphi(n) < n^2$.

44. Show that any integer $n \geqslant 1$ can be uniquely decomposed as $n = qm^2$, where q is squarefree. Denote by $Q(x)$ the number of squarefree integers q not exceeding x. Establish the formulae:

(a) $\lfloor x \rfloor = \sum_{m \leqslant \sqrt{x}} Q(x/m^2) \qquad (x \geqslant 1),$

(b) $Q(x) = \sum_{d \leqslant \sqrt{x}} \mu(d) \lfloor x/d^2 \rfloor \qquad (x \geqslant 1),$

(c) $Q(x) = 6x/\pi^2 + O(\sqrt{x}) \qquad (x \geqslant 1).$

Generalize these considerations to "k-free" integers, that is, integers such that $p|n \Rightarrow p^k \nmid n$.

45. *Ramanujan sums.* Define *Ramanujan sums* $c_n(m)$ by

$$c_n(m) := \sum_{1 \leqslant h \leqslant n,\, (h,n)=1} \mathrm{e}(hm/n) \qquad (m, n \geqslant 1)$$

with the notation $\mathrm{e}(u) := \mathrm{e}^{2\pi i u}$ $(u \in \mathbb{R})$. Show that, for all $m \geqslant 1$, $n \geqslant 1$, we have

(a) $c_n(m) = \sum_{d|(m,n)} \mu(n/d)d,$

(b) $c_n(m) = \mu(t)\varphi(n)/\varphi(t), \quad$ with $\quad t := n/(m,n).$

Deduce that

$$\mu(n) = \sum_{\substack{1 \leqslant h \leqslant n \\ (h,n)=1}} \mathrm{e}(h/n) \qquad (n \geqslant 1).$$

Show that, for each fixed integer m, $c_n(m)$ is a multiplicative function of n. Can you give a direct proof of this result?

46. Let f be a function defined on $\mathbb{R}^+$ and such that

$$\sum_{m \geqslant 1} |f(mx)|\tau(m) < \infty.$$

Write $g(x) = \sum_{m \geqslant 1} f(mx)$. Then $f(x) = \sum_{n \geqslant 1} \mu(n) g(nx)$. Find a converse to this property.

47. Define an arithmetic function $R(n)$ by

$$R(n) := \operatorname{card}\{(d, d') : d \geqslant 1,\ d' \geqslant 1,\ n = [d, d']\}.$$

(a) Show that $R(n)$ is a multiplicative function of n.

(b) Calculate $\sum_{d|n} R(d)$ and deduce an alternative proof of (a).

(c) Show that the formal Dirichlet series $F(s) := \sum_{d,d' \geqslant 1} [d, d']^{-s}$ can be simply expressed as a function of $\zeta(s)$.

(d) Study the generalization $\sum_{d_1,\dots,d_k \geqslant 1} [d_1, \dots, d_k]^{-s}$.

48. Let f be a real-valued arithmetic function. Suppose there exist two prime numbers p and q such that $f(pq)^2 < 4f(p^2)f(q^2)$. Show that f is either a unit or a prime element in the ring $\mathbb{A}_r$ of real arithmetic functions.

49. Let f be a complex-valued arithmetic function such that $f(1) = 0$.

(a) Show that if $f = u * v$, with u and v not units, then for each prime pair (p, q) the solutions to the equation

$$z^2 - f(pq)z + f(p^2)f(q^2) = 0$$

are $u(p)v(q)$ and $u(q)v(p)$.

(b) Denote by $g_i(p, q)$ $(i = 1, 2)$ the two complex numbers of the form

$$g_i(p, q) = f(pq) \pm \sqrt{f(pq)^2 - 4f(p^2)f(q^2)}$$

where a determination of the complex square root has been chosen. Show that, if there exist four prime numbers p, q, r, s such that the 16 determinants of the form

$$\begin{vmatrix} g_i(p, q) & g_k(q, s) \\ g_j(p, r) & g_\ell(r, s) \end{vmatrix} \qquad (i, j, k, \ell = 1, 2)$$

are all non-zero, then f is a prime element in $\mathbb{A}$.

50. *On monotone multiplicative functions.* Let f be a real-valued, non-decreasing multiplicative function. For integers $a \geqslant 3$, $t \geqslant 1$, put $R_t := \sum_{0 \leqslant j \leqslant t} a^j$, $S_t := a^t - \sum_{0 \leqslant j < t} a^j$. Show that $f(R_t) \geqslant f(a)^t \geqslant f(S_t)$ for all t. Deduce that, for any integers a, b, n exceeding 2, we have

$$f(b)^{s-1} \leqslant f(n) \leqslant f(a)^{r+2}$$

with $r := \lfloor \ln n / \ln a \rfloor$, $s := \lfloor \ln n / \ln b \rfloor$. Establish that $f(a)^{1/\ln a} = f(b)^{1/\ln b}$ and so that there exists a constant $k \geqslant 0$ such that $f(n) = n^k$ for all $n \geqslant 1$.[3]

[3] This result is due to Erdős (1946). The simple proof offered here is due to Moser & Lambek (1953).

51. *Stirling numbers of the second kind.* Apply the inclusion–exclusion principle to show that the number $\sigma(n,k)$ of surjections from a set with n elements into a set with k elements[4] evaluates to

$$\sigma(n,k) = \sum_{0 \leqslant j \leqslant k} (-1)^j \binom{k}{j} (k-j)^n.$$

[4]The Stirling number of the second kind $S(n,k)$ is $\sigma(n,k)/k!$ and is traditionally defined as the number of partitions of a set E with n elements into k blocks, that is, as the number of equivalence relations with k classes, on E.

Average orders

3.1. Introduction

We define an *average order* of an arithmetic function f to be any elementary function g of a real variable such that

$$\sum_{n\leqslant x} f(n) \sim \sum_{n\leqslant x} g(n). \tag{3.1}$$

A given function f may possess several average orders. In general one seeks a function g which is simply expressible in terms of elementary functions, and whose asymptotic behavior is easy to determine. Of course, given the choice between several candidates, it is preferable to use the one which minimizes the error term in the asymptotic relation (3.1).

Arithmetic functions often have seemingly very erratic behavior, which is difficult to analyze even with the aid of extensive numerical tables. This is the case, for example, for the function $\tau(n)$ which oscillates between the value 2 (taken at each prime number) and occasional very large values—see Chap. I.5. The averaging procedure often has a substantial regularizing effect, masking atypical values which occur only rarely. Normally, and in particular when f takes only positive values, the average order is the easiest non-trivial piece of information about an arithmetic function to determine.

3.2. Dirichlet's problem and the hyperbola method

We will investigate the average order of the divisor function, $\tau(n)$. A simple interchange of summations gives a first estimate. Indeed we have

$$\sum_{n\leqslant x}\tau(n)=\sum_{n\leqslant x}\sum_{d\mid n}1=\sum_{d\leqslant x}\sum_{\substack{n\leqslant x\\ n\equiv 0\,(\mathrm{mod}\,d)}}1=\sum_{d\leqslant x}\Big\lfloor\frac{x}{d}\Big\rfloor$$
$$=\sum_{d\leqslant x}\Big(\frac{x}{d}+O(1)\Big)=x\ln x+O(x).$$

It is possible to improve this estimate using a general method, due to Dirichlet in its primitive form, concerning the mean value of a convolution product.

Theorem 3.1 (Dirichlet's hyperbola method). *Let f, g be two arithmetic functions, with respective summatory functions F, G. For $1\leqslant y\leqslant x$, we have*

$$\sum_{n\leqslant x}f*g(n)=\sum_{n\leqslant y}g(n)F(x/n)+\sum_{m\leqslant x/y}f(m)G(x/m)-F(x/y)G(y). \tag{3.2}$$

Proof. The left-hand side of (3.2) can be written as

$$\sum_{md\leqslant x}f(m)g(d)=\sum_{md\leqslant x,\,d\leqslant y}f(m)g(d)+\sum_{md\leqslant x,\,d>y}f(m)g(d)$$
$$=\sum_{d\leqslant y}g(d)F(x/d)+\sum_{m\leqslant x/y}f(m)\big\{G(x/m)-G(y)\big\}.$$

This immediately implies the stated result upon expanding the last term. □

As noted above, this result yields us more precise information on the average behavior of $\tau(n)$.

Theorem 3.2. *As x tends to infinity, we have*

$$\sum_{n\leqslant x}\tau(n)=x(\ln x+2\gamma-1)+O(\sqrt{x}) \tag{3.3}$$

where γ denotes Euler's constant.

Proof. Apply (3.2) with $f=g=\mathbf{1}, F(x)=G(x)=\lfloor x\rfloor$, and $y=\sqrt{x}$. We get

$$\sum_{n\leqslant x}\tau(n)=2\sum_{m\leqslant\sqrt{x}}\lfloor x/m\rfloor-\lfloor\sqrt{x}\rfloor^2=2x\sum_{m\leqslant\sqrt{x}}\frac{1}{m}-x+O\big(\sqrt{x}\big). \tag{3.4}$$

By Theorem 0.8, the sum over m has the value $\frac{1}{2}\ln x+\gamma+O\big(1/\sqrt{x}\big)$. This yields (3.3). □

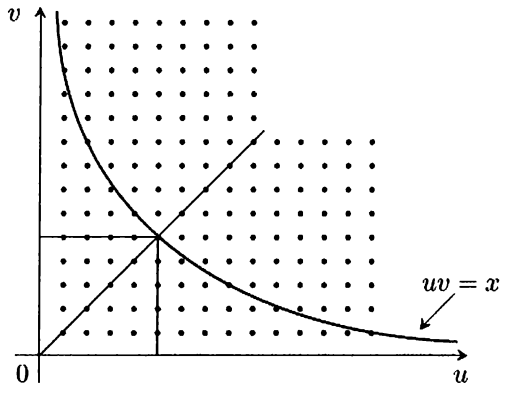

Figure 1. The equilateral hyperbola and the lattice of integral points.

The name *hyperbola method* comes from the following geometrical interpretation of Dirichlet's problem.

In the (u, v)-plane, consider the equilateral hyperbola $uv = x$. Each lattice point (i.e., point with integer coordinates) (a, b) lying between the curve and the coordinate axes (excluded) corresponds to a decomposition of the integer $n = ab \leqslant x$ as a product of two factors. Hence, the left-hand side of (3.3) is exactly the number of lattice points lying under the curve. The calculation carried out in the proof of Theorem 3.2 consists simply in exploiting the symmetry of the figure: the sought-after quantity equals the number of points contained in the square $\left[1, \sqrt{x}\,\right]^2$ augmented by twice the number of points contained in the hatched area.

Let us write

$$\Delta(x) := \sum_{n \leqslant x} \tau(n) - x(\ln x + 2\gamma - 1).$$

We have seen that $\Delta(x) = O(\sqrt{x})$. Voronoï showed in 1903 that

$$\Delta(x) \ll x^{1/3} \ln x$$

and Hardy and Landau proved independently in 1915 that $\Delta(x)$ is not $o(x^{1/4})$. Let α be the infimum of the set of exponents ξ such that

$$\Delta(x) \ll x^{\xi}.$$

The exact value of α remains unknown. It is generally conjectured that $\alpha = 1/4$; the best upper bound known to date is that of Huxley (2003)

$$\alpha \leqslant 131/416 \approx 0.314904,$$

improving on the estimate of Iwaniec & Mozzochi (1988) $\alpha \leqslant 7/22$. We shall show in Chap. I.6 how van der Corput's method (1922) can be applied to prove Voronoï's result in a simple way.

3.3. The sum of divisors function

The following theorem establishes that an average order of the function

$$\sigma(n) = \sum_{d|n} d$$

is $\frac{1}{6}\pi^2 n$. Actually we obtain a more precise result.

Theorem 3.3. *As x tends to infinity, we have*

$$\sum_{n\leqslant x} \sigma(n) = \tfrac{1}{12}\pi^2 x^2 + O(x \ln x). \tag{3.5}$$

Proof. The idea of the proof, which we shall use in several other contexts in this chapter, is very simple: to write the function under consideration as a sum over the divisors of n and then interchange summations. We have here

$$\sum_{n\leqslant x} \sigma(n) = \sum_{md\leqslant x} m = \tfrac{1}{2}\sum_{d\leqslant x} \left\lfloor \frac{x}{d} \right\rfloor \left(\left\lfloor \frac{x}{d} \right\rfloor + 1 \right) = \tfrac{1}{2}\sum_{d\leqslant x} \frac{x^2}{d^2} + O\Big(x \sum_{d\leqslant x} \frac{1}{d}\Big).$$

The stated result follows, given the classical formula

$$\sum_{d\geqslant 1} \frac{1}{d^2} = \frac{\pi^2}{6}\cdot \tag{3.6}$$

□

In Exercise 52 two proofs of (3.6) are suggested.

The best remainder term known to date for the asymptotic formula (3.5) is $O\big(x(\ln x)^{2/3}\big)$. It is due to Walfisz (1963), p. 99.

3.4. Euler's totient function

Starting from the convolution formula

$$\varphi(n) = \sum_{md=n} \mu(d) m, \tag{3.7}$$

the method of Theorem 3.3 can be applied unchanged.

Theorem 3.4. *As x tends to infinity, we have*

$$\sum_{n\leqslant x} \varphi(n) = \frac{3}{\pi^2} x^2 + O(x \ln x). \tag{3.8}$$

Proof. By (3.7), the sum to be evaluated is

$$\sum_{d\leqslant x} \mu(d) \sum_{m\leqslant x/d} m = \tfrac{1}{2}\sum_{d\leqslant x} \mu(d) \left\lfloor \frac{x}{d} \right\rfloor \left(\left\lfloor \frac{x}{d} \right\rfloor + 1 \right) = \tfrac{1}{2} x^2 \sum_{d\leqslant x} \frac{\mu(d)}{d^2} + O\Big(x \sum_{d\leqslant x} \frac{1}{d}\Big).$$

Since the Möbius function is the convolution inverse of **1**, we have formally

$$\Big(\sum_{d\geqslant 1}\frac{\mu(d)}{d^2}\Big)\Big(\sum_{d\geqslant 1}\frac{1}{d^2}\Big)=1.$$

Each of the two series being absolutely convergent, this equality actually holds in the usual sense, and it follows from (3.6) that we have

$$\sum_{d\geqslant 1}\frac{\mu(d)}{d^2}=\frac{6}{\pi^2}. \tag{3.9}$$

This implies the stated formula. □

The best error term currently known for (3.8) is

$$O\Big(x(\ln x)^{2/3}(\ln_2 x)^{4/3}\Big)$$

—cf. Walfisz (1963), p. 144.

For each integer $N\geqslant 1$, the *Farey series of order* N, denoted by $\mathcal{F}_N$, is the ordered sequence of rational fractions in lowest terms with denominators not greater than N. Thus we have

$$\mathcal{F}_1=\{\tfrac{0}{1},\tfrac{1}{1}\},\quad \mathcal{F}_2=\{\tfrac{0}{1},\tfrac{1}{2},\tfrac{1}{1}\},\qquad \mathcal{F}_3=\{\tfrac{0}{1},\tfrac{1}{3},\tfrac{1}{2},\tfrac{2}{3},\tfrac{1}{1}\},\quad \text{etc.}$$

From the definition of of $\varphi(n)$ we immediately note that

$$F(N):=\operatorname{card}\mathcal{F}_N=1+\sum_{n\leqslant N}\varphi(n).$$

Hence it follows from Theorem 3.4 that

$$F(N)\sim\frac{3}{\pi^2}N^2\qquad(N\to\infty).$$

This estimate may be interpreted in the following way: $F(N)-1$ equals the number of rational fractions $m/n, 1\leqslant m\leqslant n\leqslant N$, which are in lowest terms; since there are $\frac{1}{2}N(N+1)$ pairs $\{m,n\}$, it follows that the *probability that a rational fraction in* $]0,1]$ *with denominator not greater than* N *is in lowest terms tends to* $6/\pi^2$ *as* N *tends to infinity.*

This assertion may be generalized in the following way.

Theorem 3.5. *Let* $G(x,y)$ *denote the number of pairs of integers* $\{m,n\}$ *such that* $1\leqslant m\leqslant x$, $1\leqslant n\leqslant y$, $(m,n)=1$. *Then* $G(x,y)\sim(6/\pi^2)xy$ *as* x *and* y *tend to infinity. More precisely, setting* $z:=\min(x,y)$, *we have*

$$G(x,y)=xy\Big\{\frac{6}{\pi^2}+O\Big(\frac{\ln z}{z}\Big)\Big\}\qquad(x,y\geqslant 2).$$

Proof. From (2.14) we have

$$\begin{aligned} G(x,y) &= \sum_{m\leqslant x,\, n\leqslant y} \delta\big((m,n)\big) = \sum_{d\leqslant z} \mu(d) \Big\lfloor \frac{x}{d} \Big\rfloor \Big\lfloor \frac{y}{d} \Big\rfloor \\ &= xy \sum_{d\leqslant z} \frac{\mu(d)}{d^2} + O\Big((x+y) \sum_{d\leqslant z} \frac{1}{d}\Big) \\ &= xy\Big\{ \frac{6}{\pi^2} + O\Big(\frac{1}{z} + \Big(\frac{1}{x} + \frac{1}{y}\Big) \ln z\Big)\Big\}. \end{aligned}$$

□

3.5. The functions ω and Ω

Theorem 3.6. *As x tends to infinity, we have*

$$\sum_{n\leqslant x} \omega(n) = x \ln_2 x + c_1 x + O\Big(\frac{x}{\ln x}\Big) \tag{3.10}$$

where $c_1 \approx 0.261497$ is the constant appearing in Theorem 1.10.

Proof. We have

$$\sum_{n\leqslant x} \omega(n) = \sum_{n\leqslant x} \sum_{p\mid n} 1 = \sum_{p\leqslant x} \lfloor x/p \rfloor = x \sum_{p\leqslant x} 1/p + O(\pi(x)).$$

Formula (3.10) hence follows from Theorem 1.10 and Chebyshev's upper bound (Theorem 1.3). □

The case of the function $\Omega(n) = \sum_{p^\nu \| n} \nu$ is similar but slightly more delicate.

Theorem 3.7. *As x tends to infinity, we have*

$$\sum_{n\leqslant x} \Omega(n) = x \ln_2 x + c_2 x + O\Big(\frac{x}{\ln x}\Big) \tag{3.11}$$

with

$$c_2 := c_1 + \sum_p \frac{1}{p(p-1)} \approx 1.034653. \tag{3.12}$$

Proof. We have

$$A(x) := \sum_{n\leqslant x} \{\Omega(n) - \omega(n)\} = \sum_p \sum_{\nu\geqslant 2} \lfloor x/p^\nu \rfloor .$$

On the one hand, it follows that

$$A(x) \leqslant x \sum_p \sum_{\nu\geqslant 2} \frac{1}{p^\nu} = x \sum_p \frac{1}{p(p-1)},$$

and, on the other,

$$A(x) \geqslant \sum_{p \leqslant \sqrt{x}} \sum_{2 \leqslant \nu \leqslant \ln x / \ln p} \left(\frac{x}{p^\nu} - 1\right) = \sum_{p \leqslant \sqrt{x}} \left\{\frac{x}{p(p-1)} + O\left(\frac{\ln x}{\ln p}\right)\right\}$$
$$= x \sum_p \frac{1}{p(p-1)} + O(\sqrt{x}),$$

by Theorem 1.3. Thus we have

$$A(x) = x(c_2 - c_1) + O(\sqrt{x})$$

and formula (3.11) follows immediately from (3.10). □

3.6. Mean value of the Möbius function and Chebyshev's summatory functions

We saw in § 2.6 that von Mangoldt's function

$$\Lambda = \mu * \ln$$

is closely connected to the characteristic function of the prime numbers, and has a summatory function

$$\psi(x) = \sum_{n \leqslant x} \Lambda(n)$$

satisfying

$$\psi(x) \sim \pi(x) \ln x \qquad (x \to \infty).$$

It is therefore natural to ask if the mean value of $\mu(n)$ has a simple interpretation in terms of the asymptotic behavior of Chebyshev's functions $\pi(x)$, $\vartheta(x)$, $\psi(x)$. The following theorem, due to Landau (1909), completely answers this question.

Theorem 3.8. *The following statements are elementarily equivalent:*

(i) $\psi(x) \sim x \qquad (x \to \infty)$,

(ii) $M(x) := \sum_{n \leqslant x} \mu(n) = o(x) \qquad (x \to \infty)$,

(iii) $\sum_{n \geqslant 1} \mu(n)/n = 0$.

Remark. Statement (iii) means of course that the series occurring on the left converges and has sum zero.

Proof. The implication (iii) ⇒ (ii) results from a simple Abel summation. Indeed, assuming that

$$m(x) := \sum_{n \leqslant x} \mu(n)/n = o(1) \quad (x \to \infty),$$

it follows that

$$M(x) = \int_{1-}^{x} t\,\mathrm{d}m(t) = xm(x) - \int_{1}^{x} m(t)\,\mathrm{d}t = o(x).$$

To prove the converse implication (ii) $\Rightarrow$ (i), we first establish a convolution identity for the function $\Lambda - \mathbf{1}$, the summatory function of which we have to show to be $o(x)$. We have

$$\Lambda - \mathbf{1} = (\ln - \tau) * \mu = (\ln - \tau + 2\gamma\mathbf{1}) * \mu - 2\gamma\delta = f * \mu - 2\gamma\delta,$$

say, where the function f satisfies

$$F(x) := \sum_{n\leqslant x} f(n) = O(\sqrt{x}), \tag{3.13}$$

by Theorem 3.2 and the estimate of Corollary 0.5 for $\sum_{n\leqslant x} \ln n$.

We shall show that

$$H(x) := \sum_{n\leqslant x} f * \mu(n) = o(x) \tag{3.14}$$

using (3.13) and the hyperbola method. By Theorem 3.1, we can write, for each fixed $y > 2$,

$$H(x) = \sum_{n\leqslant x/y} \mu(n)F(x/n) + \sum_{m\leqslant y} f(m)M(x/m) - F(y)M(x/y).$$

Under hypothesis (ii), it therefore follows that

$$\limsup_{x\to\infty} \Big|\frac{H(x)}{x}\Big| \leqslant \limsup_{x\to\infty} \frac{1}{x} \sum_{n\leqslant x/y} \Big|F\Big(\frac{x}{n}\Big)\Big| \ll \limsup_{x\to\infty} \frac{1}{x} \sum_{n\leqslant x/y} \sqrt{\frac{x}{n}} \ll \frac{1}{\sqrt{y}}.$$

Since y may be chosen arbitrarily large, this does indeed imply (3.14).

It remains to establish the implication (i) $\Rightarrow$ (iii). First of all, note that the convolution relation $\mu * \mathbf{1} = \delta$ yields by summation that

$$1 = \sum_{n\leqslant x} \mu(n)\,\lfloor x/n \rfloor = xm(x) + O(x)$$

whence

$$m(x) = O(1). \tag{3.15}$$

The same relation also implies

$$\sum_{md=n} \frac{\mu(m)}{md} = \delta(n)$$

from which

$$\sum_{m\leqslant x} \frac{\mu(m)}{m} \sum_{d\leqslant x/m} \frac{1}{d} = 1 \qquad (x \geqslant 1).$$

Working out the inner sum by means of Theorem 0.8, we obtain

$$1 = \sum_{m\leqslant x} \frac{\mu(m)}{m}\left\{\ln\Big(\frac{x}{m}\Big) + \gamma + O\Big(\frac{m}{x}\Big)\right\} = m(x)(\ln x + \gamma) - G(x) + O(1),$$

where we have written

$$G(x) := \sum_{m\leqslant x} \frac{\mu(m)\ln m}{m}\cdot$$

It is thus enough to establish that we have

$$G(x) = o(\ln x) \qquad (x \to \infty). \tag{3.16}$$

To this end, we use the convolution formula (2.20):

$$\mu \ln = -\Lambda * \mu = (\mathbf{1} - \Lambda) * \mu - \delta.$$

It suffices to consider the case $x \notin \mathbb{N}$. We may therefore write

$$G(x) = \sum_{jk\leqslant x} \Big(\frac{1-\Lambda(j)}{j}\Big)\frac{\mu(k)}{k} - 1 = -1 + \int_{1-}^{x} \frac{m(x/t)}{t}\,\mathrm{d}R(t)$$

where we have put

$$R(t) := \lfloor t\rfloor - \psi(t) \qquad (t \geqslant 1),$$

so that, by hypothesis,

$$R(t) = o(t) \qquad (t \to \infty).$$

By partial summation, it follows that

$$\begin{aligned} G(x) &= -1 + \int_1^x \frac{m(x/t)R(t)}{t^2}\,\mathrm{d}t - \int_{1-}^{x} \frac{R(t)}{t}\,\mathrm{d}m(x/t) \\ &= O(1) + \int_1^x o(1/t)\,\mathrm{d}t - \int_{1-}^{x} o(1)|\,\mathrm{d}m(x/t)|. \end{aligned}$$

The first integral is $o(\ln x)$. On account of the inequality between Stieltjes measures

$$|\,\mathrm{d}m(y)| \leqslant \mathrm{d}\Big\{\sum_{n\leqslant y} 1/n\Big\},$$

we infer that this is also true of the second integral, whence (3.16) and finally (iii). □

Corollary 3.9. *The following assertions are elementarily equivalent to those in Theorem* 3.8*:*

(iv) $\pi(x) \sim x/\ln x \qquad (x \to \infty)$,

(v) $\vartheta(x) \sim x \qquad (x \to \infty)$,

(vi) $\sum_{n\leqslant x} \Lambda(n)/n = \ln x - \gamma + o(1) \qquad (x \to \infty)$.

Proof. The cases of (iv) and (v) follow immediately from Theorem 2.11. The implication (vi) $\Rightarrow$ (i) is proved by mere partial summation, the details of which we leave to the reader. For the converse, we again use the function $f = \ln - \tau + 2\gamma\mathbf{1}$ introduced in the proof of Theorem 3.8. We have

$$\begin{aligned}\sum_{n\leqslant x} \frac{\Lambda(n)-1}{n} &= \sum_{kd\leqslant x} \frac{f(k)}{k}\frac{\mu(d)}{d} - 2\gamma \\ &= \sum_{d\leqslant x/y} \frac{\mu(d)}{d} E\Big(\frac{x}{d}\Big) + \sum_{k\leqslant y} \frac{f(k)}{k} m\Big(\frac{x}{k}\Big) - E(y)\, m\Big(\frac{x}{y}\Big) - 2\gamma\end{aligned} \tag{3.17}$$

with

$$E(z) := \sum_{k\leqslant z} f(k)/k = C + O(1/\sqrt{z}) \qquad (z \geqslant 1).$$

This last relation follows, with a suitable constant C, by partial summation from the estimate (3.13). Letting x, and then y, tend to infinity, we obtain as before, employing (iii), that (3.17) has the value $-2\gamma + o(1)$, giving the required conclusion. □

3.7. Squarefree integers

When an arithmetic function takes only the values 0 and 1, it can be considered as the characteristic function of an integer sequence $\mathcal{A}$. The study of the average order then merges with that of the counting function

$$A(x) := \big|\mathcal{A} \cap [1, x]\big|.$$

The multiplicative function $n \mapsto \mu(n)^2$ is a good example of such a situation. Its summatory function

$$Q(x) := \sum_{n\leqslant x} \mu(n)^2$$

equals the number of squarefree integers not exceeding x.

Theorem 3.10. *As x tends to infinity, we have*

$$Q(x) = \frac{6}{\pi^2}x + O(\sqrt{x}). \tag{3.18}$$

Proof. As in the cases of $\tau(n)$, $\sigma(n)$ or $\varphi(n)$, we want to write $\mu(n)^2$ as a convolution product. In order to achieve this, we decompose each integer n into a canonical product

$$n = qm^2, \quad \mu(q)^2 = 1. \tag{3.19}$$

This decomposition is unique: indeed q equals the product of those prime factors of n with an odd exponent. Hence we have

$$\mu(n)^2 = \delta(m) = \sum_{d|m} \mu(d). \tag{3.20}$$

Noticing that $d \mid m$ is equivalent to $d^2 \mid n$, it follows that

$$Q(x) = \sum_{n\leqslant x} \sum_{d^2|n} \mu(d) = \sum_{d\leqslant\sqrt{x}} \mu(d) \left\lfloor \frac{x}{d^2} \right\rfloor = \frac{6}{\pi^2}x + O\Big(x \sum_{d>\sqrt{x}} \frac{1}{d^2} + \sqrt{x}\Big),$$

by (3.9). This finally shows (3.18). □

Remark. We could have argued in the following slightly different way. For each $m \leqslant \sqrt{x}$, denote by $\mathcal{A}_m$ the set of integers $n \leqslant x$ satisfying (3.19). Then $\{n : n \leqslant x\}$ is the disjoint union of the $\mathcal{A}_m$, and we have

$$|\mathcal{A}_m| = Q(x/m^2),$$

whence

$$\sum_{m\leqslant\sqrt{x}} Q(x/m^2) = \lfloor x \rfloor \qquad (x \geqslant 1).$$

Put $x = y^2$, and apply Möbius's second inversion formula (Theorem 2.10) to the functions $F(y) = \lfloor y^2 \rfloor$ and $G(y) = Q(y^2)$. It follows, as it did before, that

$$Q(x) = \sum_{d\leqslant\sqrt{x}} \mu(d) \lfloor x/d^2 \rfloor \qquad (x \geqslant 1).$$

The following result shows that the prime number theorem allows us to improve the error term in (3.18).

Theorem 3.11. *Under the assumption*

$$M(x) = \sum_{n\leqslant x} \mu(n) = o(x), \tag{3.21}$$

we have

$$Q(x) = \frac{6}{\pi^2}x + o(\sqrt{x}) \qquad (x \to \infty). \tag{3.22}$$

Proof. Relation (3.20) can also be written as

$$\mu^2 = \lambda * \mathbf{1}$$

where λ is the arithmetic function defined by

$$\lambda(n) = \begin{cases} \mu(d) & \text{if } n = d^2, \\ 0 & \text{if } n \text{ is not a square.} \end{cases}$$

By the hyperbola method, we obtain, for $1 \leqslant y \leqslant x$, that

$$\begin{aligned} Q(x) &= \sum_{d \leqslant \sqrt{x/y}} \mu(d) \Big\lfloor \frac{x}{d^2} \Big\rfloor + \sum_{m \leqslant y} M\Big(\sqrt{\frac{x}{m}}\Big) - \lfloor y \rfloor M\Big(\sqrt{\frac{x}{y}}\Big) \\ &= \sum_{d \leqslant \sqrt{x/y}} \mu(d) \Big\{ \frac{x}{d^2} + O(1) \Big\} + o_y\big(\sqrt{x}\big) \\ &= \frac{6x}{\pi^2} - x \int_{\sqrt{x/y}}^{\infty} \frac{\mathrm{d}M(t)}{t^2} + O\Big(\sqrt{\frac{x}{y}}\Big) + o_y\big(\sqrt{x}\big). \end{aligned}$$

Now for all $z \geqslant 1$ we have

$$\int_z^{\infty} \frac{\mathrm{d}M(t)}{t^2} = -\frac{M(z)}{z^2} + 2\int_z^{\infty} \frac{M(t)}{t^3}\,\mathrm{d}t = o\Big(\frac{1}{z}\Big) + \int_z^{\infty} o\Big(\frac{1}{t^2}\Big)\,\mathrm{d}t = o\Big(\frac{1}{z}\Big).$$

We may thus write

$$Q(x) = \frac{6x}{\pi^2} + O\Big(\sqrt{\frac{x}{y}}\Big) + o_y\big(\sqrt{x}\big),$$

as a consequence of which we have, for each fixed $y \geqslant 1$,

$$\limsup_{x \to \infty} |Q(x) - 6x/\pi^2|/\sqrt{x} \ll 1/\sqrt{y}.$$

This implies the required conclusion (3.22), upon letting y tend to infinity. □

3.8. Mean value of a multiplicative function with values in $[0, 1]$

Theorem 3.10 formulates that an average order of the function $\mu(n)^2$ is the constant

$$\frac{6}{\pi^2} = \prod_p \big(1 - 1/p^2\big) = \prod_p (1 - 1/p)\big(1 + \mu(p)^2/p\big).$$

One says that $\mu(n)^2$ possesses a *mean value*, equal to $6/\pi^2$. It is easy to construct arithmetic functions, even with values in $[0, 1]$, which fail to have a mean value. If, for example, we put

$$f(n) = \begin{cases} 1 & \text{if } 2^{2k} < n \leqslant 2^{2k+1} \quad (k = 0, 1, \ldots), \\ 0 & \text{if } 2^{2k+1} < n \leqslant 2^{2k+2} \quad (k = 0, 1, \ldots), \end{cases}$$

we observe that

$$\liminf_{x\to\infty} \frac{1}{x}\sum_{n\leqslant x} f(n) = \tfrac{1}{3}, \qquad \limsup_{x\to\infty} \frac{1}{x}\sum_{n\leqslant x} f(n) = \tfrac{2}{3}.$$

The following theorem settles the question in the case of multiplicative functions.

Theorem 3.12. *Let f be a multiplicative arithmetic function with values in $[0,1]$. Set*

$$M(f) := \prod_p \Big(1 - \frac{1}{p}\Big) \sum_{\nu\geqslant 0} \frac{f(p^\nu)}{p^\nu}$$

where the infinite product is considered to be 0 when it diverges. Then we have

$$\sum_{n\leqslant x} f(n) = x\{M(f) + o(1)\} \qquad (x\to\infty). \tag{3.23}$$

Proof. For each $y > 2$, we introduce the auxiliary completely multiplicative functions α_y and β_y defined by

$$\alpha_y(p) = \begin{cases} 1 & \text{if } p \leqslant y, \\ 0 & \text{if } p > y, \end{cases} \qquad \beta_y(p) = \begin{cases} 0 & \text{if } p \leqslant y, \\ 1 & \text{if } p > y. \end{cases}$$

It is easy to verify that for any multiplicative function f we have

$$f = f\alpha_y * f\beta_y.$$

When f takes values in $[0,1]$, this implies that

$$f \leqslant f_y := f\alpha_y * \beta_y.$$

Moreover we have $\beta_y = \mathbf{1} * h_y$ where $h_y := \beta_y * \mu$ is defined by

$$h_y(p^\nu) = \begin{cases} -1 & \text{if } p \leqslant y \text{ and } \nu = 1, \\ 0 & \text{if } p > y \text{ or } \nu \geqslant 2. \end{cases}$$

In particular, $h_y(n) = 0$ whenever $n > N(y) := \prod_{p\leqslant y} p$. This implies, as x tends to infinity,

$$\frac{1}{x}\sum_{n\leqslant x} \beta_y(n) = \frac{1}{x}\sum_{m\leqslant N(y)} h_y(m)\Big\lfloor \frac{x}{m}\Big\rfloor \to \sum_{m\geqslant 1} \frac{h_y(m)}{m} = \prod_{p\leqslant y}\Big(1 - \frac{1}{p}\Big).$$

Thus, for each fixed y, β_y has a mean value, equal to $M(\beta_y)$. Now we can write

$$\frac{1}{x}\sum_{n\leqslant x} f_y(n) = \sum_{d\leqslant x} \frac{f(d)\alpha_y(d)}{d} \cdot \frac{d}{x} \sum_{m\leqslant x/d} \beta_y(m). \tag{3.24}$$

Since the series of non-negative terms

$$\sum_{d\geqslant 1}\frac{f(d)\alpha_y(d)}{d}=\prod_{p\leqslant y}\sum_{\nu\geqslant 0}\frac{f(p^\nu)}{p^\nu}$$

is convergent, we deduce from (3.24), by the dominated convergence theorem, that f_y has a mean value, equal to

$$\sum_{d\geqslant 1}\frac{f(d)\alpha_y(d)}{d}M(\beta_y)=\prod_{p\leqslant y}\Big(1-\frac{1}{p}\Big)\sum_{\nu\geqslant 0}\frac{f(p^\nu)}{p^\nu}=M(f_y).$$

Hence, for each fixed $y>2$,

$$\frac{1}{x}\sum_{n\leqslant x}f(n)\leqslant M(f_y)+o(1)\qquad(x\to\infty). \tag{3.25}$$

Now $M(f_y)$ is a decreasing function of y. Thus if the infinite product $M(f)$ diverges, we have $M(f_y)\to 0$ as $y\to\infty$. In this case, we conclude that f has mean value zero. If the product $M(f)$ converges, the same holds for the series

$$\sum_p\frac{1-f(p)}{p}\cdot$$

For all integers $n\geqslant 1$, we then have

$$\begin{aligned}f_y(n)-f(n)&=\prod_{p^\nu\|n,\,p\leqslant y}f(p^\nu)-\prod_{p^\nu\|n}f(p^\nu)\leqslant 1-\prod_{p^\nu\|n,\,p>y}f(p^\nu)\\&\leqslant\sum_{p^\nu\|n,\,p>y}\big(1-f(p^\nu)\big)\end{aligned}$$

from which we infer that

$$\begin{aligned}\sum_{n\leqslant x}\{f_y(n)-f(n)\}&\leqslant\sum_{p>y}\sum_{\nu\geqslant 1}\{1-f(p^\nu)\}\,\lfloor x/p^\nu\rfloor\\&\leqslant x\Big\{\sum_{p>y}\frac{1-f(p)}{p}+\frac{1}{p(p-1)}\Big\}=:x\varepsilon(y),\end{aligned}$$

say, where $\varepsilon(y)$ goes to 0 as y tends to infinity. The desired result follows from this estimate in view of (3.25), by making x and y tend successively to infinity. □

Notes

§ 3.2. This presentation of the hyperbola method is due to Diamond (1982).

The original proof by Voronoï is elementary and based on the Euler–Maclaurin summation formula. Two other classical elementary proofs of the inequality $\alpha \leqslant 1/3$ are that of Landau (1912), using the theory of Bessel functions, and that of I. M. Vinogradov (1917), explained in the book by Gelfond & Linnik (1965), which is also based on the Euler–Maclaurin formula. The upper bound $\alpha \leqslant 131/416$ of Huxley (2003) stems from a refinement of the method of Iwaniec & Mozzochi (1988) who derived $\alpha \leqslant 7/22$, improving on a result of Kolesnik (1985), $\alpha \leqslant 139/429$. The numerical improvement is relatively slight ($139/429 \approx 0.324$, $131/416 \approx 0.315$), as in the whole history of this problem, but the ideas involved are important. Kolesnik's method is an extension to several variables of van der Corput's (1922), sketched in Chap. I.6. That of Iwaniec & Mozzochi essentially amounts, by a sophisticated process, to bounding above an average of double exponentials sums. This is a two-dimensional version of the treatment due to Bombieri & Iwaniec (1986), which led to a new upper bound for the Riemann zeta function on the critical line: $|\zeta(\frac{1}{2} + it)| \ll_\varepsilon t^{9/56+\varepsilon}$. The best result known to date in this direction is due to Huxley (2005) who obtains, by elaborating on the same method,

$$|\zeta(\tfrac{1}{2} + it)| \ll_\varepsilon t^{32/205+\varepsilon}.$$

The fact that $\Delta(x)$ is not $o(x^{1/4})$ follows directly from the quadratic mean evaluation by Tong (1956):

$$\int_0^x \Delta(y)^2 \, \mathrm{d}y = \frac{\zeta(3/2)^4}{6\pi^2\zeta(3)} x^{3/2} + O\big(x(\ln x)^5\big).$$

§ 3.4. Sharpening a result of Erdős & Shapiro (1951), Montgomery (1987) shows that

$$\lim \sup_{\inf} \Big\{ \sum_{n \leqslant x} \varphi(n) - \frac{3}{\pi^2} x^2 \Big\} \Big/ x\sqrt{\ln_2 x} \gtrless 0.$$

§ 3.6. The proof of Theorem 3.8 is essentially that proposed by Diamond (1982). The implication (ii) $\Rightarrow$ (i) in Theorem 3.8 is a special case of a general result concerning Möbius inverses of functions of bounded variation—cf. Ellison & Mendès France (1975), th. 3.1.

§ 3.7. Theorem 3.11 is due to Landau (1909). The best error term known at present for formula (3.22) is

$$O\Big(\sqrt{x} \exp\Big\{-c\frac{(\ln x)^{3/5}}{(\ln_2 x)^{1/5}}\Big\}\Big).$$

It is due to Walfisz (1963). Assuming the Riemann hypothesis, S. Graham (1981a), improving on a result of Montgomery & Vaughan (1981), showed that one may derive

$$O_\varepsilon\big(x^{8/25+\varepsilon}\big).$$

§ 3.8. The method of using the functions α_y, β_y, goes back at least to Erdős, in the 1930s. It has been used in a remarkable manner by Daboussi—see in particular 1979, 1984. The case of convergence of the product $M(f)$ can be generalized as follows: *Let f be a complex-valued multiplicative function such that*

$$\sum_{n\geqslant 1} \frac{|f*\mu(n)|}{n} < \infty. \tag{3.26}$$

Then we have

$$\sum_{n\leqslant x} f(n) = x\{M(f) + o(1)\} \qquad (x\to\infty),$$

where the product $M(f)$, defined as in Theorem 3.12, *is absolutely convergent.*

Indeed, putting $h = f*\mu$, we have, for $x \geqslant 1$,

$$\frac{1}{x}\sum_{n\leqslant x} f(n) = \frac{1}{x}\sum_{d\leqslant x} h(d)\left\lfloor \frac{x}{d}\right\rfloor \to \sum_{d\geqslant 1}\frac{h(d)}{d} \qquad (x\to\infty)$$

by the dominated convergence theorem. It is then easy to verify that the hypothesis of absolute convergence implies

$$\sum_{d\geqslant 1}\frac{h(d)}{d} = M(f).$$

We shall see in Theorem II.1.3 that assumption (3.26) is equivalent to

$$\sum_p\sum_{\nu\geqslant 1}\frac{|f(p^\nu) - f(p^{\nu-1})|}{p^\nu} < \infty. \tag{3.27}$$

The case of a zero mean value in Theorem 3.12 can be handled by Theorem III.3.5. Such an approach actually provides an effective upper bound.

Exercises

52. Show that $\sum_{n\geqslant 1} 1/n^2 = \pi^2/6$:

(a) by integrating the Fourier series for $\langle t\rangle$;

(b) by integrating the Taylor series for $(\operatorname{arctanh} x)/x$. [*Calculate the integral by the method of residues.*[1]]

53. Show that $\sum_{n\leqslant x} 2^{\omega(n)} = (6/\pi^2)x\ln x + O(x)$.

54. *A result of M. Nair.*[2]

(a) Show that $2^{\omega(n)} = \sum_{d^2|n} \mu(d)\tau(n/d^2)$ $(n \geqslant 1)$.

(b) Deduce an asymptotic formula for $\sum_{n\leqslant x} 2^{\omega(n)}$ with error term $O(\sqrt{x}\ln x)$.

55. Determine an arithmetic function h such that $h * 2^\omega = \mu^2 2^\omega$.

(a) Show that $\sum_{n>x} |h(n)|/n \ll_\varepsilon x^{-1/2+\varepsilon}$ for all $\varepsilon > 0$.

(b) Deduce that $\sum_{n\leqslant x} \mu(n)^2 2^{\omega(n)} = Cx\ln x + O(x)$ where

$$C := \prod_p (1-1/p)^2(1+2/p).$$

56. Show that

$$\sum_{n\leqslant x} 3^{\omega(n)} = \tfrac{1}{2}Cx(\ln x)^2 + O(x\ln x),$$

where C is the constant defined in Exercise 55. Generalize.

57. (a) Show that $\sum_{n\leqslant x} 2^{\Omega(n)} = \sum_{k\leqslant \ln x/\ln 2} 2^k \sum_{m\leqslant x/2^k} f(m)$ where f is the multiplicative function defined by $f(p^\nu) := 0$ (if $p = 2$), $:= 2^\nu$ (if $p \geqslant 3$).

(b) Writing $f = \tau * h$, where h is some function to be determined, show that $\sum_{n\leqslant x} f(n) = \frac{1}{4}C_0 x\ln x + O(x)$ with $C_0 := \prod_{p>2}\big(1 + 1/p(p-2)\big)$.

(c) Show that $\sum_{n\leqslant x} 2^{\Omega(n)} = (C_0/8\ln 2)x(\ln x)^2 + O(x\ln x)$.

58. *Mean value of* $3^{\Omega(n)}$. In this exercise, let us assume Euler's formula (1.6) for the Riemann function $s \mapsto \zeta(s)$. Recall that $\langle u\rangle$ denotes the fractional part of the real number u.

(a) Set

$$u := \frac{\ln(3/2)}{\ln 2}, \quad v := u + 1 = \frac{\ln 3}{\ln 2}, \quad f(\vartheta) := \sum_{b\geqslant 0} 3^{-ub-\langle \vartheta - vb\rangle} \quad (\vartheta \in \mathbb{R}).$$

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[1] See for example Cartan (1961), pp. 107–109.

[2] Private communication.

Show that

$$\sum_{\substack{a\geqslant 0,\, b\geqslant 0\\ 2^a3^b\leqslant z}} 3^{a+b} = \tfrac{3}{2}z^v f\Big(\frac{\ln z}{\ln 2}\Big) + O(z) \qquad (z \geqslant 1).$$

(b) Let $A(x) := \sum_{n\leqslant x} 3^{\Omega(n)}$. Show that

$$A(x) = \sum_{\substack{m\leqslant x\\ (m,6)=1}} 3^{\Omega(m)} \sum_{\substack{a\geqslant 0,\, b\geqslant 0\\ 2^a3^b\leqslant x/m}} 3^{a+b} \qquad (x \geqslant 1).$$

(c) Let h be the multiplicative function defined by

$$h(p^\nu) := \begin{cases} 0 & \text{if } p = 2 \text{ or } 3 \text{ and } \nu \geqslant 1,\\ 3^\nu & \text{if } p \geqslant 5. \end{cases}$$

Show that, for $c \geqslant 1$, $x \geqslant 1$, we have

$$\sum_{\substack{m\leqslant x\\ (m,6)=1}} \frac{3^{\Omega(m)}}{m^c} = \sum_{m\leqslant x} \frac{h(m)}{m^c} \leqslant \prod_{5\leqslant p\leqslant x} \frac{1}{1-3/p^c}. \tag{3.28}$$

Deduce from this that

$$\sum_{\substack{m\leqslant x\\ (m,6)=1}} \frac{3^{\Omega(m)}}{m} \ll (\ln x)^3 \qquad (x \geqslant 2).$$

(d) Using the inequality $(1-3r)^{-1} \leqslant (1-r)^{-3}(1+8r^2)$ $(0 \leqslant r \leqslant 1/5)$, deduce from (3.28) that there exists an absolute constant A such that

$$\sum_{(m,6)=1} \frac{3^{\Omega(m)}}{m^c} \leqslant A\zeta(c)^3 \qquad (c > 1).$$

(e) Show that, for all $\varepsilon \in]0, v-1[$ and all $x \geqslant 2$, we have

$$\sum_{\substack{m>x\\ (m,6)=1}} \frac{3^{\Omega(m)}}{m^v} \leqslant \sum_{(m,6)=1} \frac{3^{\Omega(m)}}{m^{1+\varepsilon}x^{v-1-\varepsilon}} \leqslant Ax^{1-v+\varepsilon}\zeta(1+\varepsilon)^3.$$

What do we obtain on selecting $\varepsilon := 1/\ln x$ for x sufficiently large?

(f) Show that the series

$$g(\vartheta) := \sum_{\substack{m\geqslant 1\\ (m,6)=1}} \frac{3^{\Omega(m)}}{m^v} f\Big(\vartheta - \frac{\ln m}{\ln 2}\Big)$$

converges for all real ϑ. Using the results of parts (a), (b), (c), (d), prove that

$$A(x) = \tfrac{3}{2}x^v g\Big(\frac{\ln x}{\ln 2}\Big) + O(x(\ln x)^3) \qquad (x \geqslant 2).$$

59. For $r \in \mathbb{N}^*$ and $x > 0$, denote by $S_r(x)$ the number of integer pairs (a, b) such that $1 \leqslant a, b \leqslant x$ and $\gcd(a, b) = r$.

(a) Show that

$$S_1(x) = \frac{6}{\pi^2}x^2 + O(x \ln 2x) \qquad (x \geqslant 1).$$

(b) Give a simple relationship between the functions S_r an S_1.

(c) Deduce that

$$T(x) := \sum_{1 \leqslant a,\, b \leqslant x} \gcd(a, b) = \frac{6}{\pi^2}x^2 \ln x + O(x^2).$$

(d) Using the formula $n = \sum_{d|n} \varphi(d)$, show that

$$T(x) = x^2 \sum_{d \leqslant x} \frac{\varphi(d)}{d^2} + O(x^2).$$

(e) Deduce an asymptotic formula for $\sum_{d \leqslant x} \varphi(d)/d^2$.

60. *Axer's Theorem (1910).* Let $\lambda(x)$ be a bounded function with bounded variation $V_\lambda(I)$ on any bounded interval I, and let $\{a_n\}_{n=1}^{\infty}$ be a real sequence satisfying

$$\text{(i)} \quad \sum_{n \leqslant x} |a_n| = O(x) \qquad (x \geqslant 1)$$

$$\text{(ii)} \quad A(x) := \sum_{n \leqslant x} a_n = o(x) \qquad (x \to \infty).$$

(a) Write the sum $S(x, y) := \sum_{n \leqslant y} a_n \lambda(x/n)$ $(x \geqslant 1,\ y \geqslant 1)$ as a Stieltjes integral. Show that

$$|S(x, x) - S(x, y)| \leqslant o(x) + V_\lambda([1, x/y]) \sup_{y < t \leqslant x} |A(t)| \qquad (1 \leqslant y \leqslant x).$$

(b) By applying the previous result with $y = \varepsilon x$, where ε is an arbitrarily small positive number, show that $S(x, x) = o(x)$ $(x \to \infty)$.

61. *Application of Axer's theorem to equivalent forms of the prime number theorem.* Set

$$M(x) := \sum_{n \leqslant x} \mu(n), \qquad m(x) := \sum_{n \leqslant x} \mu(n)/n.$$

(a) Show that $\sum_{n \leqslant x} \mu(n) \lfloor x/n \rfloor = 1$ $(x \geqslant 1)$.

(b) Applying, for $\{a_n\}_{n=1}^{\infty} = \{\mu(n)\}_{n=1}^{\infty}$, the result proved in Exercise 60 (b) for the function $\lambda(x) = x - \lfloor x \rfloor$, show that, if $M(x) = o(x)$ $(x \to \infty)$, then $\lim_{x\to\infty} m(x) = 0$.

(c) Examine the converse.

62. Assuming the prime number theorem, show that we have

$$\sum_{p\leqslant x} \frac{\ln p}{p-1} = \ln x - \gamma + o(1).$$

63. Let $A(n) := \sum_{p^\nu \| n} \nu p$ be Alladi & Erdős's function (1977, 1979). Assuming the prime number theorem, show that we have

$$\sum_{n\leqslant x} A(n) \sim \frac{\pi^2}{12}\,\frac{x^2}{\ln x} \qquad (x \to +\infty).$$

Make the error term in this relation more precise by invoking a strong form of the prime number theorem—for example Theorem II.4.1.

64. (a) Let f be an additive arithmetic function such that

$$\sum_{p}\sum_{\nu\geqslant 2} \frac{|f(p^\nu)|}{p^\nu} < \infty.$$

Show that

$$\sum_{n\leqslant x} f(n) = \sum_{p\leqslant x} f(p)\left\{ \left\lfloor \frac{x}{p} \right\rfloor - \left\lfloor \frac{x}{p^2} \right\rfloor \right\} + O(x). \tag{3.29}$$

(b) Application. Let f be the additive function defined by $f(p^\nu) = (\nu \ln p)^2$. Give, for the left-hand side of (3.29), an asymptotic formula where the ratio of the error term to the principal term is $\ll 1/\ln x$ as $x \to \infty$.

65. Let $\alpha > 0$ be a fixed real number.

(a) Assuming the prime number theorem in the strong form

$$\pi(x) = x/\ln x + O(x/(\ln x)^2),$$

evaluate, asymptotically as $x \to \infty$, the expression

$$T_\alpha(x) := \sum_{p\leqslant x} p^\alpha$$

with an error term $O(x^{1+\alpha}/(\ln x)^2)$.

(b) Define a strongly additive arithmetic function f_α by $f_\alpha(p) := p^\alpha$. Evaluate the summatory function $S_\alpha(x) := \sum_{n\leqslant x} f_\alpha(n)$ with an error $O(x^{1+\alpha}/(\ln x)^2)$. Dirichlet's hyperbola method may be employed.

66. Let f be a multiplicative arithmetic function such that $f*\mu \geqslant 0$. Show we have for $x \geqslant 1$

$$\sum_{n\leqslant x} f(n) \leqslant x \prod_{p\leqslant x}\Big(1-\frac{1}{p}\Big)\sum_{\nu\geqslant 0}\frac{f(p^\nu)}{p^\nu}.$$

67. Write for $\alpha > 0$, $t \geqslant 1$, $f_\alpha(n) := \big(n/\varphi(n)\big)^\alpha$, and

$$F_x(t) := \big|\{n \leqslant x : n > t\varphi(n)\}\big|.$$

(a) By applying the result of Exercise 66 to f_α with $\alpha = \alpha(t)$ suitably chosen, show that there exists a positive absolute constant c such that

$$F_x(t) \ll x\exp\{-\mathrm{e}^{ct}\} \qquad (x \geqslant 1,\ t \geqslant 1).$$

(b) Show that for fixed $\alpha > 0$, $\varepsilon > 0$, we have

$$\sum_{n\leqslant x}\Big(\frac{n}{\varphi(n)}\Big)^\alpha = x\prod_p\Big(1+\frac{f_\alpha(p)-1}{p}\Big)+O(x^\varepsilon).$$

68. *Farey series, 1.* Let $N \geqslant 1$, $\mathcal{F}_N$ be the Farey series of order N, and a/b, a'/b' be two consecutive elements of $\mathcal{F}_N$ appearing in this order.

(a) Show that the equation $bu - av = 1$ has integer solutions u and v such that $u \geqslant 1$, $N - b < v \leqslant N$. Deduce that $u/v \in \mathcal{F}_N$, $u/v > a/b$.

(b) Show that, if $u/v \neq a'/b'$, then $u/v - a'/b' \geqslant 1/vb'$. Deduce that $u/v - a/b > N/bb'v$. Show that this last inequality does not hold and infer that $a'/b' = u/v$.

(c) Application. Construct the successor to $\frac{4}{9}$ in $\mathcal{F}_{13}$.

69. *Farey series, 2.* Let $\mathcal{F}_N$ be the Farey series of order N and a/b, a''/b'', a'/b' three consecutive elements of $\mathcal{F}_N$ appearing in that order. Show that

$$\frac{a''}{b''} = \frac{a+a'}{b+b'}.$$

70. *Squarefull integers.* Let $S := \{n \geqslant 1 : p|n \Rightarrow p^2|n\}$, $S(x) := |S\cap[1,x]|$.

(a) By estimating $\sum_{n\leqslant x,\, n\in S}\sqrt{x/n}$, show that

$$S(x) \ll \sqrt{x}\ln x \qquad (x \geqslant 2).$$

(b) Show that each integer n in S can be written in a unique way in the form $n = m^3d^2$ where m is squarefree. Deduce that[3]

$$S(x) \sim \frac{\zeta(3/2)}{\zeta(3)}\sqrt{x} \qquad (x \to \infty).$$

[3]See Suryanarayana & Sitaramachandra (1973) for a more precise result.

71. *Core of an integer.* Let $k(n) := \prod_{p|n} p$ be the squarefree *core*[4] of a generic integer n.

(a) Deduce from Theorem 3.12 that $\sum_{n\leqslant x} k(n)/n \sim Cx$ with

$$C := \prod_p \Big(1 - \frac{1}{p(p+1)}\Big).$$

(b) Using the result of Exercise 70(b), show that

$$\sum_{n\leqslant x} \frac{k(n)}{n} = Cx + O(\sqrt{x}).$$

(c) Show that[5]

$$\sum_{n\leqslant x} k(n) = \tfrac{1}{2}Cx^2 + O\big(x^{3/2}\big).$$

72. Show the existence of a positive constant A such that

$$\sum_{n\leqslant x} \omega\big(\tau(n)\big) = Ax + O\Big(\frac{x}{\ln x}\Big)$$

and establish the estimate $\sum_{n\leqslant x} \Omega\big(\tau(n)\big) = x \ln_2 x + O(x)$.[6]

73. Show that the prime number theorem implies

$$\int_1^\infty \frac{\psi(t) - t}{t^2}\,\mathrm{d}t = -1 - \gamma.$$

74. Let $\lambda(n)$ be Liouville's function, defined by $\lambda(n) := (-1)^{\Omega(n)}$, where $\Omega(n)$ denotes the total number of prime factors of n, counted with multiplicities.

(a) Show that $\lambda = \mu * \kappa$, where κ denotes the characteristic function of the squares.

(b) Show that the prime number theorem is elementarily equivalent to the asymptotic formula

$$\sum_{n\leqslant x} \lambda(n) = o(x) \qquad (x \to \infty).$$

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[4] Also called *kernel*, or *radical*.

[5] See also Cohen (1960, 1964).

[6] See Rieger (1972); Heppner (1974).

75. *Selberg's identity (1949).*

Define the arithmetic function $\Lambda_2 := \ln^2 * \mu = \Lambda * \Lambda + \Lambda \ln$.

(a) Show that, for all primes p, q, $p \neq q$, and any integers $\alpha \geqslant 1$, $\beta \geqslant 1$, we have $\Lambda_2(p^\alpha) = (2\alpha - 1)\ln^2 p$, $\Lambda_2(p^\alpha q^\beta) = 2 \ln p \ln q$ and $\Lambda_2(n) = 0$ if $\omega(n) > 2$.

(b) Show that $\sum_{n \leqslant x} \Lambda_2(n) = \sum_{p \leqslant x} \ln^2 p + \sum_{pq \leqslant x} \ln p \ln q + O(x)$.

(c) Show that $\sum_{n \leqslant x} \tau * \mathbf{1}(n) = \frac{1}{2} x \ln^2 x + ax \ln x + bx + O\big(x^{2/3} \ln(2x)\big)$, where a and b are real constants. Deduce that, for suitable constants A and B, the function $h := 2\tau * \mathbf{1} + A\tau + B\mathbf{1} - \ln^2$ has a summatory function $H(x) \ll x^{3/4}$.

(d) Show that $\sum_{n \leqslant x} \Lambda_2(n) = 2x \ln x + O(x)$.[7]

76. Let g be a complex multiplicative function, with values in the unit disk. Put $S(n) := \sum_{d|n} g(d)$. Show the existence of the mean value

$$\lim_{x \to +\infty} \frac{1}{x} \sum_{n \leqslant x} |S(n)/\tau(n)|^2$$

and express it in the form of an Euler product. Deduce that the condition

$$\sum_p \{1 - \Re e\, g(p)\}/p = \infty$$

is necessary and sufficient to ensure that $S(n) = o\big(\tau(n)\big)$ holds for almost all integers n, or, in other words, that there exists a function $\varepsilon(x)$ satisfying $\lim_{x \to \infty} \varepsilon(x) = 0$ and such that

$$\big|\{n \leqslant x : |S(n)| > \varepsilon(x)\tau(n)\}\big| = o(x) \qquad (x \to \infty).$$

[7] Selberg deduces in an elementary way the prime number theorem from this formula. This implication in fact holds in a general setting: see Shapiro (1959).

Sieve methods

4.1. The sieve of Eratosthenes

The inclusion–exclusion principle or the Möbius inversion formula can in theory be used to calculate $\pi(x)$. For sufficiently large x, let us write

$$P = \prod_{p \leqslant \sqrt{x}} p;$$

then an integer n, $\sqrt{x} < n \leqslant x$, is a prime number if, and only if, $(n, P) = 1$. Thus we may write

$$\pi(x) - \pi(\sqrt{x}) + 1 = \sum_{n \leqslant x} \delta((n, P)) = \sum_{d \mid P} \mu(d) \left\lfloor \frac{x}{d} \right\rfloor. \tag{4.1}$$

At this stage, if we content ourselves with estimating $\lfloor x/d \rfloor$ using $x/d + O(1)$, we obtain

$$\pi(x) - \pi(\sqrt{x}) + 1 = x \prod_{p \leqslant \sqrt{x}} \left(1 - \frac{1}{p}\right) + O\left(2^{\pi(\sqrt{x})}\right).$$

By Mertens' formula, the main term in this estimate equals

$$\{2\mathrm{e}^{-\gamma} + o(1)\} x / \ln x,$$

but Chebyshev's estimates show that the error term is larger than any power of x.

This calls for two comments. On the one hand, the exact formula (4.1)—called the sieve formula of Eratosthenes—involves far too many terms for any reasonable practical validity. On the other hand, the estimate for the main term shows *a posteriori*, taking into account the prime number theorem, that the "error terms" created by replacing $\lfloor x/d \rfloor$ by x/d contribute globally

at the same order as the "main terms". This suggests that, even suitably adapted, this method will never allow us to prove the prime number theorem. However, we shall see that it can provide Chebyshev-type estimates in a very general context.

In order to obtain a non-trivial result starting from formula (4.1), it is necessary to introduce a parameter y, $1 \leqslant y \leqslant x$, and to bound $\pi(x)-\pi(y)+1$ from above by the number of integers n not exceeding x and having no prime factor $\leqslant y$. With the same calculations, we obtain

$$\pi(x) \leqslant x \prod_{p \leqslant y} \Big(1 - \frac{1}{p}\Big) + O(2^y) = x \frac{\mathrm{e}^{-\gamma} + o(1)}{\ln y} + O(2^y) \leqslant \{\mathrm{e}^{-\gamma} + o(1)\} \frac{x}{\ln_2 x}$$

for the essentially optimal choice $y = \ln x$.

It was with the aim of improving the efficiency of this method that the Norwegian mathematician Viggo Brun invented the theory of the combinatorial sieve between 1917 and 1924.

4.2. Brun's combinatorial sieve

The sieve of Eratosthenes is based on the identity

$$\mu * \mathbf{1} = \delta.$$

Brun's idea consists in introducing two auxiliary functions μ_1, μ_2, satisfying

$$\mu_1 * \mathbf{1} \leqslant \delta \leqslant \mu_2 * \mathbf{1}, \tag{4.2}$$

and vanishing sufficiently often so that the number of terms in the resulting formula analogous to (4.1) is not prohibitive.

Brun's initial choice, leading to what is customarily referred to in the literature as *Brun's pure sieve*, is the following.

Theorem 4.1 (Brun). *Denote by χ_t the characteristic function of the set of integers n such that $\omega(n) \leqslant t$. Then, for any integer $h \geqslant 0$, the functions defined by*

$$\mu_i(n) := \mu(n)\chi_{2h+2-i}(n) \qquad (i = 1, 2) \tag{4.3}$$

satisfy the inequalities (4.2).

Proof. Since $\mu_i * \mathbf{1}(n)$ only depends on the squarefree kernel of n, it is enough to consider the case when $\mu(n)^2 = 1$. If $\omega(n) = k$, then, for each r, $0 \leqslant r \leqslant k$, it is clear that n has exactly $\binom{k}{r}$ divisors d such that $\omega(d) = r$. We can thus write, for any given $t \geqslant 0$,

$$\mu\chi_t * \mathbf{1}(n) = \sum_{d|n,\, \omega(d) \leqslant t} \mu(d) = \sum_{r \leqslant t} (-1)^r \binom{k}{r} = (-1)^t \binom{k-1}{t},$$

where the last equality is easily obtained by induction on t. □

Corollary 4.2. *Let $\mathcal{A}$ be a finite set of integers, and $\mathcal{P}$ be a set of prime numbers. Write*

$$A_d := \operatorname{card}\{a \in \mathcal{A} : a \equiv 0 \,(\mathrm{mod}\, d)\},$$
$$P(y) := \prod_{p \in \mathcal{P},\, p \leqslant y} p,$$
$$S(\mathcal{A}, \mathcal{P}; y) := \operatorname{card}\{a \in \mathcal{A} : (a, P(y)) = 1\}.$$

Then, for each integer $h \geqslant 0$, we have

$$\sum_{d \mid P(y),\, \omega(d) \leqslant 2h+1} \mu(d) A_d \leqslant S(\mathcal{A}, \mathcal{P}; y) \leqslant \sum_{d \mid P(y),\, \omega(d) \leqslant 2h} \mu(d) A_d. \tag{4.4}$$

Let us see how this result allows us to improve considerably the upper bound for $\pi(x)$ provided by the sieve of Eratosthenes.

We select, in the above corollary, $\mathcal{A} = \{n : n \leqslant x\}$ and $\mathcal{P} = \mathbb{P}$, the set of all prime numbers. Whence

$$\begin{aligned} \pi(x) &\leqslant \sum_{\substack{d \mid P(y) \\ \omega(d) \leqslant 2h}} \mu(d) \left\lfloor \frac{x}{d} \right\rfloor + y = x \sum_{\substack{d \mid P(y) \\ \omega(d) \leqslant 2h}} \frac{\mu(d)}{d} + O\Big(y + \sum_{\substack{d \mid P(y) \\ \omega(d) \leqslant 2h}} 1 \Big) \\ &= x \prod_{p \leqslant y} \Big(1 - \frac{1}{p}\Big) + O\Big(y + \sum_{\substack{d \mid P(y) \\ \omega(d) \leqslant 2h}} 1 + x \sum_{\substack{d \mid P(y) \\ \omega(d) > 2h}} \frac{1}{d} \Big). \end{aligned} \tag{4.5}$$

The second of the three error terms does not exceed y^{2h} since this is an upper bound for all integers d such that $d \mid P(y)$, $\omega(d) \leqslant 2h$. The d-sum arising in the third remainder term is bounded from above, for each value of the parameter $u \geqslant 1$, by

$$\sum_{d \mid P(y)} \frac{u^{\omega(d)-2h}}{d} = u^{-2h} \prod_{p \leqslant y} \Big(1 + \frac{u}{p}\Big) \leqslant \exp\Big\{ -2h \ln u + u \sum_{p \leqslant y} \frac{1}{p} \Big\}.$$

For the optimal choice $u = 2h \big/ \sum_{p \leqslant y} p^{-1}$, we obtain that this quantity is

$$\ll_u (\ln y)^{-v}$$

where $v = u \ln u - u$: this follows from Theorem 1.10. When $u > 5$, we have $v > 3$. It is easy to see that, for y large enough, there is some $u = u(y)$, $5 < u < 6$, such that

$$h := \tfrac{1}{2} u \sum_{p \leqslant y} \frac{1}{p}$$

is an integer. With these choices of the parameters, we have, for sufficiently large x,

$$y^{2h} \leqslant y^{6 \ln_2 y + O(1)} < x^{2/3}$$

whenever

$$y \leqslant x^{1/(10 \ln_2 x)} =: Y(x). \tag{4.6}$$

Collecting all previous estimates, and selecting $y = Y(x)$, we obtain

$$\pi(x) \ll \frac{x \ln_2 x}{\ln x}.$$

Even though inferior to Chebyshev's, this result is remarkable because of the great generality of the argument. The corresponding lower bound for $S(\mathcal{A}, \mathcal{P}; y)$ can be established in the same way. This quantity actually is of intrinsic arithmetic interest: it is equal to the number of integers not exceeding x all of whose prime factors are larger than y. We can thus formulate the following result. Recall that, for each integer $n \geqslant 1$, we denote by $P^-(n)$ the smallest prime factor of n, with the convention $P^-(1) = +\infty$, and we define

$$\Phi(x, y) := \operatorname{card}\{n \leqslant x : P^-(n) > y\}. \tag{4.7}$$

Theorem 4.3. *Under condition* (4.6), *we have*

$$\Phi(x, y) = x \prod_{p \leqslant y} \Big(1 - \frac{1}{p}\Big) \Big\{1 + O\Big(\frac{1}{(\ln y)^2}\Big)\Big\}. \tag{4.8}$$

The choice of the functions μ_1, μ_2 defined by (4.3) is certainly not optimal. At the cost of technical complications, the method can be refined by introducing a partition of $]1, y]$ into subintervals $]y_j, y_{j+1}]$, $0 \leqslant j \leqslant k$, and selecting for $i = 1, 2$,

$$\mu_i(d) = \mu(d) \chi_i^*(d)$$

where χ_i^* is the characteristic function of the set of those integers d having at most $2h_j + 2 - i$ distinct prime factors in $\mathcal{P} \cap]y_j, y]$ for each j, $0 \leqslant j \leqslant k$—see Exercise 86. There are thus two families of parameters to optimize, the y_j and the h_j.

It would take too long to develop here the full theory of the combinatorial sieve, which is still evolving at the present moment. The interested reader will find a pedagogical exposition of the subject in the book by Halberstam & Richert, *Sieve Methods* (1974)—see also the Notes for this chapter.

We confine ourselves to quoting the basic result of the theory, which was obtained with the choice of the $\mu_i(d)$ indicated above. It is this very theorem which is intended when "Brun's method" is invoked in the literature without further qualification.

Theorem 4.4 (Fundamental lemma of the combinatorial sieve). *With the notation of Corollary* 4.2, *assume there exists a multiplicative function* $w \geqslant 0$, *a real number* X, *and positive constants* κ, A, *such that*

$$\text{(a)} \quad A_d =: Xw(d)/d + R_d \qquad (d \mid P(y))$$

$$\text{(b)} \quad \prod_{\eta \leqslant p \leqslant \xi} \Big(1 - \frac{w(p)}{p}\Big)^{-1} < \Big(\frac{\ln \xi}{\ln \eta}\Big)^{\kappa} \Big(1 + \frac{A}{\ln \eta}\Big) \qquad (2 \leqslant \eta \leqslant \xi).$$

Then, uniformly in $\mathcal{A}$, X, y *and* $u \geqslant 1$, *we have*

(4.9)

$$S(\mathcal{A}, \mathcal{P}; y) = X \prod_{p \leqslant y,\, p \in \mathcal{P}} \Big(1 - \frac{w(p)}{p}\Big)\big\{1 + O\big(u^{-u/2}\big)\big\} + O\Big(\sum_{d \leqslant y^u,\, d \mid P(y)} |R_d|\Big).$$

From this result a Chebyshev-type upper bound for $\pi(x)$ may easily deduced, as well as an evaluation of the order of magnitude of $\Phi(x, y)$ for $y \leqslant x^{\delta}$, where δ is some fixed positive real number. We leave the details to the reader.

The best sieve results currently obtained by combinatorial techniques are due to Iwaniec (1980a,b).

4.3. Application to twin primes

Here we illustrate the results of the previous paragraph with Brun's theorem on twin primes.

It is obvious that the difference between two consecutive odd primes must be at least 2. When it is exactly 2, we say that these primes are *twins*, such as $\{3, 5\}, \{5, 7\}, \{11, 13\}, \{17, 19\}, \{29, 31\}$, etc. A famous conjecture asserts the existence of infinitely many twin primes.

Let us write

$$\mathcal{J} := \{p : p + 2 \text{ is prime}\} \qquad \text{and} \qquad J(x) := \big|\mathcal{J} \cap [1, x]\big|.$$

On the basis of an analytic approach, and in agreement with a heuristic probabilistic calculation, Hardy & Littlewood (1922) conjectured that

$$J(x) \sim 2 \prod_{p > 2} \Big(1 - \frac{1}{(p-1)^2}\Big) \frac{x}{(\ln x)^2} \quad (x \to \infty). \tag{4.10}$$

By Brun's pure method, we establish the following result.

Theorem 4.5 (Brun). *As* x *tends to infinity, we have*

$$J(x) \ll x \Big(\frac{\ln_2 x}{\ln x}\Big)^2. \tag{4.11}$$

Corollary 4.6. *We have*

$$\sum_{p\in\mathcal{J}} \frac{1}{p} < \infty. \tag{4.12}$$

Proof. Apply Corollary 4.2 with

$$\mathcal{A} = \{m(m+2) : m \leqslant x\},$$

choosing for $\mathcal{P}$ the set of all prime numbers. For all y, $1 \leqslant y \leqslant x$, we have

$$J(x) \leqslant S(\mathcal{A}, \mathcal{P}; y) + y \leqslant \sum_{d|P(y),\, \omega(d)\leqslant 2h} \mu(d) A_d + y, \tag{4.13}$$

where A_d is the number of integral solutions $m \leqslant x$ to the congruence

$$m(m+2) \equiv 0 \,(\mathrm{mod}\, d). \tag{4.14}$$

This relation is equivalent to

$$m \equiv 0 \,(\mathrm{mod}\, 2^\nu), \quad m \equiv 0 \text{ or } -2 \,(\mathrm{mod}\, p) \quad (p|d,\ p \neq 2), \tag{4.15}$$

where ν is 1 or 0 according to whether d is even or odd. By the Chinese remainder theorem, there are thus $\varrho(d)$ solutions modulo d, where ϱ is the strongly multiplicative function defined by

$$\varrho(2) = 1, \quad \varrho(p) = 2 \qquad (p \geqslant 3). \tag{4.16}$$

Each interval of length d contains $\varrho(d)$ integers m counted in A_d. Therefore we can write

$$A_d = x\frac{\varrho(d)}{d} + O(\varrho(d)) \qquad (\mu(d)^2 = 1). \tag{4.17}$$

Inserting this back into (4.13) and performing a calculation parallel to (4.5), it follows that

$$J(x) \leqslant x \sum_{d|P(y)} \frac{\mu(d)\varrho(d)}{d} + O\Big(y + \sum_{\substack{d|P(y)\\ \omega(d)\leqslant 2h}} \varrho(d) + x \sum_{\substack{d|P(y)\\ \omega(d)>2h}} \frac{\varrho(d)}{d}\Big). \tag{4.18}$$

The main term equals

$$\tfrac{1}{2}x \prod_{2<p\leqslant y} \Big(1 - \frac{2}{p}\Big) \leqslant 2x \prod_{p\leqslant y} \Big(1 - \frac{1}{p}\Big)^2 \sim 2\mathrm{e}^{-2\gamma} x (\ln y)^{-2}.$$

Selecting, as in the application described in § 4.2, $h = c\ln_2 y + O(1)$ for a suitable constant c, and

$$\ln y \sim c' \ln x / \ln_2 x$$

with c' sufficiently small, we check that the error term is of smaller order than $x/(\ln y)^2$. This implies (4.11) and thereby completes the proof of the theorem. The corollary follows at once by Abel summation. □

4.4. The large sieve—analytic form

The large sieve is one of the most powerful tools in analytic number theory. Invented by Linnik in 1941, it has been systematically developed since then, as much from the point of view of its fundamental principle as for its arithmetic applications.

The reader will find complete expositions and an exhaustive bibliography in Bombieri's monograph (1974) and Montgomery's survey (1978a). We will be content here with indicating that the crucial steps in the development of the current theory of the large sieve are due to Rényi (1950), Roth (1965) and Bombieri (1965). The optimal version was obtained by Montgomery & Vaughan (1973, 1974).

It was Davenport & Halberstam (1966) who were the first to isolate the analytic foundations of the large sieve. Let $\{a_n\}_{n=0}^{\infty}$ be a sequence of complex numbers, and for given arbitrary integers $M, N \geqslant 0$, let

$$S(\alpha) := \sum_{M<n\leqslant M+N} a_n \mathrm{e}(\alpha n) \tag{4.19}$$

be a trigonometric polynomial, where we have written $\mathrm{e}(u) := \exp\{2\pi i u\}$ $(u \in \mathbb{R})$. The analytic form of the large sieve is an inequality of the type

$$\sum_{1\leqslant i\leqslant R} |S(\alpha_i)|^2 \leqslant \Delta(N,\delta) \sum_{M<n\leqslant M+N} |a_n|^2 \tag{4.20}$$

which holds for all R-tuples $\{\alpha_1, \dots, \alpha_R\}$ of real numbers which are δ-*well spaced* in the sense that

$$\min_{1\leqslant i<j\leqslant R} \|\alpha_j - \alpha_i\| \geqslant \delta > 0, \tag{4.21}$$

where $\|u\|$ denotes the distance from the real number u to the set of integers. The aim of this section is to prove the following optimal result.

Theorem 4.7 (Montgomery & Vaughan; Selberg). *Under the above conditions, the large sieve inequality* (4.20) *holds with the choice*

$$\Delta(N,\delta) = N + \delta^{-1} - 1. \tag{4.22}$$

The proof we give here, due to Selberg, is presented in the article of Montgomery (1978a). The value of $\Delta(N,\delta)$ obtained by Montgomery & Vaughan in 1974 was only slightly weaker: $\Delta = N + 1/\delta$. Selberg's improvement is unimportant for applications—where an upper bound of the type $\Delta \ll N + 1/\delta$ is usually sufficient—and it is mainly the nature of the argument which motivated our choice.

Let us observe, meanwhile, that the value given in equation (4.22) is attained for certain values of α_i, N and δ. Indeed, for each integer $R \geqslant 1$,

set $\alpha_j = j/R$ $(1 \leqslant j \leqslant R)$, so that $\delta = 1/R$, and let us consider for $N \equiv 1 \,(\mathrm{mod}\, R)$ the case $a_n := \mathbf{1}_{\mathbb{N}}(n/R)$ $(0 \leqslant n < N)$. Then we have

$$\sum_{1\leqslant j\leqslant R} |S(\alpha_j)|^2 = \sum_{1\leqslant j\leqslant R} \Big| \sum_{\substack{0\leqslant n\leqslant N-1 \\ n\equiv 0\,(\mathrm{mod}\, R)}} 1 \Big|^2 = R\Big(\frac{N-1}{R}+1\Big)^2$$

$$= (N-1+R)\Big(1+\frac{N-1}{R}\Big) = \Big(N-1+\frac{1}{\delta}\Big) \sum_{0\leqslant n<N} |a_n|^2.$$

The proof of Theorem 4.7 rests on a general duality principle asserting the equality of the norms of a Banach space operator and its adjoint. We only use this in the case of endomorphisms of $\ell^2(\mathbb{C})$. The duality principle then takes the following simple form.

Lemma 4.8. *Let (c_{nr}) be an $N \times R$ matrix with complex coefficients. The three following assertions concerning the real number D are equivalent:*

(i) $\displaystyle\sum_r \Big|\sum_n c_{nr}x_n\Big|^2 \leqslant D\sum_n |x_n|^2 \qquad (\forall x_n \in \mathbb{C}),$

(ii) $\displaystyle\Big|\sum_{n,r} c_{nr}y_r x_n\Big|^2 \leqslant D\sum_n |x_n|^2 \sum_r |y_r|^2 \qquad (\forall x_n, y_r \in \mathbb{C}),$

(iii) $\displaystyle\sum_n \Big|\sum_r c_{nr}y_r\Big|^2 \leqslant D\sum_r |y_r|^2 \qquad (\forall y_r \in \mathbb{C}).$

Proof. Let us show the equivalence of (i) and (ii). That of (ii) and (iii) follows from this by interchanging the roles of the indices r and n.

(i) $\Rightarrow$ (ii). We have

$$\Big|\sum_{n,r} c_{nr}y_r x_n\Big|^2 = \Big|\sum_r y_r \sum_n c_{nr}x_n\Big|^2$$

$$\leqslant \sum_r |y_r|^2 \sum_r \Big|\sum_n c_{nr}x_n\Big|^2 \leqslant D\sum_n |x_n|^2 \sum_r |y_r|^2$$

where the first upper bound comes from the Cauchy–Schwarz inequality.

(ii) $\Rightarrow$ (i). For each r, define $L_r := \sum_n c_{nr}x_n$ and apply (ii) with $y_r = \overline{L_r}$. We get

$$\Big(\sum_r |L_r|^2\Big)^2 \leqslant D\sum_n |x_n|^2 \sum_r |L_r|^2,$$

that is we have (i). □

In the sequel, we will systematically use notation (4.19). The following statement results immediately from Lemma 4.8.

Lemma 4.9. *Let α_r $(1 \leqslant r \leqslant R)$ be fixed real numbers. The two following assertions, concerning the sequence of real numbers $b_n \geqslant 0$ $(n \in \mathbb{Z})$ such that $b_n > 0$ $(M < n \leqslant M+N)$ and the positive real number B, are equivalent:*

$$\text{(i)} \quad \sum_{1\leqslant r\leqslant R} |S(\alpha_r)|^2 \leqslant B \sum_{M<n\leqslant M+N} |a_n|^2/b_n \qquad (\forall a_n \in \mathbb{C})$$

$$\text{(ii)} \quad \sum_{M<n\leqslant M+N} b_n \Big| \sum_{1\leqslant r\leqslant R} y_r\, \mathrm{e}(n\alpha_r)\Big|^2 \leqslant B \sum_{1\leqslant r\leqslant R} |y_r|^2 \qquad (\forall y_r \in \mathbb{C}).$$

Proof. We use Lemma 4.8 with $c_{nr} = \mathrm{e}(n\alpha_r)\sqrt{b_n}$. Up to replacing a_n by $a_n\sqrt{b_n}$, assertion (i) is equivalent to

$$\sum_{1\leqslant r\leqslant R} \Big| \sum_{M<n\leqslant M+N} a_n\sqrt{b_n}\,\mathrm{e}(\alpha_r n)\Big|^2 \leqslant B \sum_{M<n\leqslant M+N} |a_n|^2 \quad (\forall a_n \in \mathbb{C}).$$

Using the equivalence of (i) and (iii) in Lemma 4.8, we obtain that the above condition can also be written

$$\sum_{M<n\leqslant M+N} \Big| \sum_{1\leqslant r\leqslant R} y_r \mathrm{e}(n\alpha_r)\sqrt{b_n}\Big|^2 \leqslant B \sum_{1\leqslant r\leqslant R} |y_r|^2 \quad (\forall y_r \in \mathbb{C}),$$

which is the required inequality. □

Corollary 4.10. *Let $B(\alpha) := \sum_{n\in\mathbb{Z}} b_n\,\mathrm{e}(n\alpha)$ be a convergent Fourier series such that $b_n \geqslant 0$ $(n \in \mathbb{Z}), b_n > 0$ $(M < n \leqslant M+N)$. Then, for any positive real number B, the inequality*

$$\text{(i)} \quad \sum_{1\leqslant r\leqslant R} |S(\alpha_r)|^2 \leqslant B \sum_{M<n\leqslant M+N} |a_n|^2/b_n \qquad (\forall a_n \in \mathbb{C})$$

is satisfied provided that

$$\text{(ii)} \quad \sum_{1\leqslant r,s\leqslant R} y_r\overline{y_s}\, B(\alpha_r - \alpha_s) \leqslant B \sum_{1\leqslant r\leqslant R} |y_r|^2 \qquad (\forall y_r \in \mathbb{C}).$$

Proof. Expanding $B(\alpha_r - \alpha_s)$ as a series and interchanging summations, we see that (ii) is equivalent to the second inequality of Lemma 4.9 in which the sum over n is extended to $\mathbb{Z}$. Since $b_n \geqslant 0$ for all n, the conclusion is immediate. □

An obvious way to construct functions satisfying the conditions of Corollary 4.10 consists in insisting that

(a) $b_n \geqslant 0$ $(n \in \mathbb{Z})$, $\quad b_n \geqslant 1$ $(M < n \leqslant M+N)$,

(b) $B(\alpha) = 0$ $\quad (\|\alpha\| \geqslant \delta)$,

where δ is defined by (4.21). It is convenient to suppose here that $0 < \delta < \frac{1}{2}$, the case $\delta = \frac{1}{2}$ (which is possible only if $R = 2$) then following by a straightforward limit procedure. When (a) and (b) are realized,

assertion (i) of Corollary 4.10 implies that the large sieve inequality (4.20) holds with

$$\Delta(N,\delta) = B(0). \tag{4.23}$$

The remainder of this section is devoted to making explicit a suitable choice for the sequence $\{b_n\}_{n\in\mathbb{Z}}$ of Fourier coefficients of $B(\alpha)$.

It is natural to try and write b_n as the value at n of a function $b \in L^1(\mathbb{R})$ whose Fourier transform

$$\widehat{b}(\vartheta) := \int_{-\infty}^{+\infty} b(t)\mathrm{e}(-\vartheta t)\,\mathrm{d}t$$

has a support contained in $[-\delta,\delta]$. The Poisson summation formula

$$B(\alpha) := \sum_{n\in\mathbb{Z}} b(n)\mathrm{e}(\alpha n) = \sum_{k\in\mathbb{Z}} \widehat{b}(k-\alpha) \tag{4.24}$$

then guarantees the validity of condition (b).

To verify that the summation formula (4.24) is effectively applicable in this context, we first observe that $\widehat{b} \in L^1(\mathbb{R})$, from which we deduce (cf., for example, Katznelson (1968), p. 126) the integral representation

$$b(t) = \int_{-\delta}^{\delta} \widehat{b}(\vartheta)\mathrm{e}(\vartheta t)\,\mathrm{d}\vartheta. \tag{4.25}$$

In particular, $b(n)$ is the Fourier coefficient of the periodic continuous function $\beta(\alpha) := \sum_{k\in\mathbb{Z}} \widehat{b}(k-\alpha)$. Now, for $N \geqslant 1$, $|\alpha| \leqslant \frac{1}{2}$, we have by (4.25),

$$\begin{aligned}\sum_{|n|\leqslant N} b(n)\mathrm{e}(\alpha n) - \int_{-N-\frac{1}{2}}^{N+\frac{1}{2}} b(t)\mathrm{e}(\alpha t)\,\mathrm{d}t \\ = \int_{-\delta+\alpha}^{\delta+\alpha} \lambda_\alpha(\vartheta)\sin\{(2N+1)\pi\vartheta\}\,\mathrm{d}\vartheta,\end{aligned} \tag{4.26}$$

with

$$\lambda_\alpha(\vartheta) := \widehat{b}(\vartheta-\alpha)\Big\{\frac{1}{\sin(\pi\vartheta)} - \frac{1}{\pi\vartheta}\Big\}.$$

Since $|\pm\delta+\alpha| < 1$ (it is here we use the assumption $\delta < \frac{1}{2}$), we have $\lambda_\alpha \in L^1[-\delta+\alpha,\ \delta+\alpha]$. The Riemann–Lebesgue lemma allows us to conclude that the last ϑ-integral tends to 0 as $N\to\infty$. Since $b \in L^1(\mathbb{R})$, this implies the (symmetric) convergence of the series

$$\sum_{n\in\mathbb{Z}} b(n)\mathrm{e}(\alpha n)$$

to $\widehat{b}(-\alpha) = \beta(\alpha)$. That establishes (4.24) when $|\alpha| \leqslant \frac{1}{2}$, and hence for all α, by periodicity.

We are thus led to investigate an integrable function b such that the quantity

$$B(0) = \widehat{b}(0) = \int_{-\infty}^{+\infty} b(t)\,dt \tag{4.27}$$

is as small as possible within the constraints

$$\begin{cases} b(t) \geqslant 0 & (t \in \mathbb{R}), \\ b(t) \geqslant 1 & (M+1 \leqslant t \leqslant M+N), \\ \widehat{b}(\vartheta) = 0 & (|\vartheta| \geqslant \delta). \end{cases} \tag{4.28}$$

The Fejér kernel provides an easy way to exhibit an initial possibility. For

$$b(t) := C \sum_{\delta(M+1) \leqslant n \leqslant \delta(M+N)} \left(\frac{\sin\left(\frac{1}{2}\pi(n-\delta t)\right)}{\frac{1}{2}\pi(n-\delta t)} \right)^2$$

we have

$$\widehat{b}(\vartheta) = \frac{2C}{\delta}(1 - |\vartheta/\delta|)^+ \sum_{\delta(M+1) \leqslant n \leqslant \delta(M+N)} \mathrm{e}(-n\vartheta/\delta)$$

so that the conditions (4.28) are certainly satisfied with $C = \frac{1}{4}\pi^2$. This gives the inequality

$$\widehat{b}(0) \leqslant \tfrac{1}{2}\pi^2(N - 1 + 1/\delta)$$

which suffices for most applications. Selberg remarked that the following lemma provides an even better choice for $b(t)$.

Lemma 4.11. *Let*

$$F(z) := \left(\frac{\sin \pi z}{\pi}\right)^2 \left\{ \sum_{n \geqslant 0} \frac{1}{(z-n)^2} - \sum_{n \geqslant 1} \frac{1}{(z+n)^2} + \frac{2}{z} \right\}.$$

Then F defines an entire function of z, such that

$$F(z) \ll \mathrm{e}^{2\pi|\Im m\, z|}, \qquad F(x) \geqslant \operatorname{sgn}(x) \quad (x \in \mathbb{R}), \qquad F(0) = 1,$$

and

$$\int_{-\infty}^{+\infty} \{F(x) - \operatorname{sgn}(x)\}\,dx = 1. \tag{4.29}$$

Remark. It plainly follows from (4.29) that $F \notin L^1(\mathbb{R})$. However, we can interpret the upper bound on $F(z)$ as meaning that, in a certain sense, $\widehat{F}(\vartheta) = 0$ for $|\vartheta| \geqslant 1$. Consider for example, for $T > 0$, the function

$F_T(x) := \frac{1}{2}\{F(T+x) + F(T-x)\}$. Then $F_T \in L^1(\mathbb{R})$ for all $T > 0$ and, as shown by Vaaler (1985), we have

$$\widehat{F}_T(\vartheta) = \begin{cases} \dfrac{\sin(2\pi T|\vartheta|)}{\pi} + (1-|\vartheta|)\dfrac{\sin\{\pi\vartheta(2T+1)\}}{\sin(\pi\vartheta)} & \text{if } |\vartheta| \leqslant 1, \\ 0 & \text{if } |\vartheta| > 1. \end{cases}$$

where the Fourier transform is defined by $\widehat{F}_T(\vartheta) := \int_{\mathbb{R}} F_T(x)\mathrm{e}(\vartheta x)\,\mathrm{d}x$.

Proof. The first two assertions are clear: writing $z = x+iy$, then, assuming for instance that $|y| \geqslant 1$, we have $|z \pm n|^2 \geqslant 1 + (|x|-n)^2$ $(n \geqslant 0)$. To establish the third assertion, we first recall Euler's formula

$$\Big(\frac{\sin \pi z}{\pi}\Big)^2 \sum_{n\in\mathbb{Z}} \frac{1}{(z-n)^2} = 1 \qquad (z \in \mathbb{C})$$

and notice that for $x > 0$

$$\sum_{n\geqslant 1} \frac{1}{(x+n)^2} \leqslant \sum_{n\geqslant 1} \int_{x+n-1}^{x+n} \frac{\mathrm{d}u}{u^2} = \frac{1}{x} = \sum_{n\geqslant 0} \int_{x+n}^{x+n+1} \frac{\mathrm{d}u}{u^2} \leqslant \sum_{n\geqslant 0} \frac{1}{(x+n)^2}.$$

This implies, still for $x > 0$,

$$F(x) = \Big(\frac{\sin \pi x}{\pi}\Big)^2 \Big\{ \sum_{n\in\mathbb{Z}} \frac{1}{(x-n)^2} - 2\sum_{n\geqslant 1} \frac{1}{(x+n)^2} + \frac{2}{x} \Big\} \geqslant 1,$$

and for $x < 0$, with $y = -x$,

$$F(x) = \Big(\frac{\sin \pi x}{\pi}\Big)^2 \Big\{ -\sum_{n\in\mathbb{Z}} \frac{1}{(x-n)^2} + 2\sum_{n\geqslant 0} \frac{1}{(y+n)^2} - \frac{2}{y} \Big\} \geqslant -1.$$

Finally

$$F(0) = \lim_{z\to 0} \Big(\frac{\sin \pi z}{\pi z}\Big)^2 = 1 \geqslant 0 = \operatorname{sgn}(0).$$

Let us prove (4.29). We have

$$\begin{aligned} \int_{-\infty}^{+\infty} \{F(x) - \operatorname{sgn}(x)\}\,\mathrm{d}x &= \int_0^{\infty} \{F(x) - 1\}\,\mathrm{d}x + \int_0^{\infty} \{F(-y) + 1\}\,\mathrm{d}y \\ &= \int_0^{\infty} \{F(x) + F(-x)\}\,\mathrm{d}x = 2\int_0^{\infty} \Big(\frac{\sin \pi x}{\pi x}\Big)^2 \mathrm{d}x = 1. \end{aligned}$$

□

Conclusion of the proof of Theorem 4.7. Define

$$b(t) := \tfrac{1}{2}\big\{F\big(\delta(t-M-1)\big) + F\big(\delta(M+N-t)\big)\big\}. \tag{4.30}$$

Then $b(t)$ satisfies the first of conditions (4.28) and relation (4.29) shows that b is integrable on $\mathbb{R}$, together with

$$\int_{-\infty}^{+\infty} b(t)\,\mathrm{d}t = N - 1 + \frac{1}{\delta}. \tag{4.31}$$

This results immediately from the identity

$$b(t) = \mathbf{1}_{[M+1,M+N]}(t) + \tfrac{1}{2}\{F(\delta(t-M-1)) - \operatorname{sgn}(\delta(t-M-1))\} + \tfrac{1}{2}\{F(\delta(M+N-t)) - \operatorname{sgn}(\delta(M+N-t))\},$$

valid for $t \neq M+1, M+N$. In addition, we then have from Lemma 4.11

$$b(z) \ll e^{2\pi\delta|\Im m\, z|} \qquad (z \in \mathbb{C}). \tag{4.32}$$

In particular, b is bounded on $\mathbb{R}$. Since $b \in L^1(\mathbb{R})$, we hence also have $b \in L^2(\mathbb{R})$. The estimate (4.32) then implies, by the Paley–Wiener theorem,[1] that

$$\widehat{b}(\vartheta) = 0 \qquad (|\vartheta| \geqslant \delta).$$

This completes the proof of Theorem 4.7. □

4.5. The large sieve—arithmetic form

Let $\{a_n\}_{n=M+1}^{M+N}$ be a finite sequence of complex numbers, and set

$$S(\alpha) := \sum_{M<n\leqslant M+N} a_n e(n\alpha).$$

Apply Theorem 4.7 to the case where the α_r are all rational numbers of the form $\alpha_r = a/q, (a,q) = 1, q \leqslant Q$. For $r \neq s$, we clearly have

$$\|\alpha_r - \alpha_s\| = \|a/q - a'/q'\| = \|(aq' - a'q)/qq'\| \geqslant 1/Q^2,$$

which shows that the α_r are $1/Q^2$-well spaced. We may thus write

$$\sum_{q\leqslant Q} \sum_{1\leqslant a\leqslant q,\,(a,q)=1} |S(a/q)|^2 \leqslant (N-1+Q^2) \sum_{M<n\leqslant M+N} |a_n|^2. \tag{4.33}$$

The usefulness of this inequality lies in the observation that one may bound the inner sum from below by an explicit function of q related to the numbers, $w(p)$, of classes modulo p, $p|q$, which contain no integer n such that $a_n \neq 0$. More precisely, write, for each prime p,

$$w(p) := \operatorname{card}\{h : 0 \leqslant h < p,\, n \equiv h \,(\mathrm{mod}\, p) \Rightarrow a_n = 0\}, \tag{4.34}$$

and

$$g(q) := \mu(q)^2 \prod_{p|q} \frac{w(p)}{p - w(p)} \tag{4.35}$$

(We may assume that $w(p) < p$ for all p since otherwise $a_n \equiv 0$.) The basis for the arithmetic form of the large sieve is set out in the following result.

[1] See, for example, Katznelson (1968), th. 7.4, p. 176.

Theorem 4.12. *With the previous notation, we have, for all* $q \geqslant 1$,

$$\Big|\sum_{M<n\leqslant M+N} a_n\Big|^2 g(q) \leqslant \sum_{\substack{1\leqslant a\leqslant q\\(a,q)=1}} |S(a/q)|^2. \tag{4.36}$$

Corollary 4.13 (Arithmetic large sieve). *For any finite sequence of complex numbers* $\{a_n\}_{M+1}^{M+N}$ *and any integer* $Q \geqslant 1$, *we have*

$$\Big|\sum_{M<n\leqslant M+N} a_n\Big|^2 \leqslant \frac{N-1+Q^2}{L} \sum_{M<n\leqslant M+N} |a_n|^2 \tag{4.37}$$

with

$$L := \sum_{q\leqslant Q} g(q), \tag{4.38}$$

where $g(q)$ *is defined by* (4.34) *and* (4.35).

Proof of Theorem 4.12. We have to show that, for any sequence $\{a_n\}$, we have

$$|S(0)|^2 g(q) \leqslant \sum_{1\leqslant a\leqslant q,\,(a,q)=1} |S(a/q)|^2. \tag{4.39}$$

Since the definition of the $w(p)$ is unchanged when one replaces a_n by $a_n \mathrm{e}(n\beta)$, relation (4.39) is equivalent to

$$|S(\beta)|^2 g(q) \leqslant \sum_{1\leqslant a\leqslant q,\,(a,q)=1} |S(a/q+\beta)|^2 \quad (\beta \in \mathbb{R}). \tag{4.40}$$

Let us assume (4.40) is satisfied for q and q' with $(q,q')=1$. By the Chinese remainder theorem we may then write

$$\begin{aligned}\sum_{\substack{1\leqslant c\leqslant qq'\\(c,qq')=1}} |S(c/qq')|^2 &= \sum_{\substack{1\leqslant a\leqslant q\\(a,q)=1}} \sum_{\substack{1\leqslant b\leqslant q'\\(b,q')=1}} |S(a/q+b/q')|^2\\ &\geqslant \sum_{1\leqslant a\leqslant q,\,(a,q)=1} |S(a/q)|^2 g(q') \geqslant |S(0)|^2 g(q) g(q').\end{aligned}$$

Since g is multiplicative, it ensues that (4.39) (and consequently (4.40)) holds for qq'. Since $g(q)=0$ when q is not squarefree, we can confine ourselves to establishing (4.39) when q is prime.

For any prime number p, we have

$$|S(0)|^2 + \sum_{1\leqslant a<p} |S(a/p)|^2 = \sum_{0\leqslant a,a'<p} \frac{S(a/p)\overline{S(a'/p)}}{p} \sum_{0\leqslant h<p} \mathrm{e}\big((a-a')h/p\big)$$

$$= \frac{1}{p} \sum_{0\leqslant h<p} \Big| \sum_{0\leqslant a<p} \mathrm{e}(-ah/p) S(a/p) \Big|^2$$

$$= \frac{1}{p} \sum_{0\leqslant h<p} \Big| \sum_n a_n \sum_{0\leqslant a<p} \mathrm{e}\big(a(n-h)/p\big) \Big|^2 = p \sum_{0\leqslant h<p} |S(p,h)|^2, \text{ say,} \tag{4.41}$$

where we have set

$$S(p,h) := \sum_{\substack{M<n\leqslant M+N \\ n\equiv h \,(\mathrm{mod}\, p)}} a_n.$$

Note that, by assumption, $S(p,h)$ vanishes for at least $w(p)$ values of h modulo p. By the Cauchy–Schwarz inequality, we therefore have

$$|S(0)|^2 = \Big| \sum_{0\leqslant h<p} S(p,h) \Big|^2 \leqslant \{p - w(p)\} \sum_{0\leqslant h<p} |S(p,h)|^2$$

from which it follows by (4.41) that

$$\sum_{1\leqslant a<p} \big|S(a/p)\big|^2 = p \sum_{0\leqslant h<p} |S(p,h)|^2 - |S(0)|^2$$

$$\geqslant \left(\frac{p}{p-w(p)} - 1 \right) |S(0)|^2 = g(p)|S(0)|^2.$$

This establishes (4.39) for $q = p$, and thereby completes the proof. □

In view of the identity $S(0) = \sum_{0\leqslant h<p} S(p,h)$, we obtain

$$p \sum_{0\leqslant h<p} \Big| S(p,h) - \frac{1}{p} S(0) \Big|^2 = p \sum_{0\leqslant h<p} |S(p,h)|^2 - |S(0)|^2$$

from which, using (4.41), it follows that

$$p \sum_{0\leqslant h<p} \Big| S(p,h) - \frac{1}{p} S(0) \Big|^2 = \sum_{0\leqslant a<p} |S(a/p)|^2.$$

By (4.33) we hence obtain the following result.

Theorem 4.14. *With the previous notation, we have*

$$\sum_{p\leqslant Q} p \sum_{0\leqslant h<p} \Big| S(p,h) - \frac{1}{p} S(0) \Big|^2 \leqslant (N - 1 + Q^2) \sum_{M<n\leqslant M+N} |a_n|^2. \tag{4.42}$$

Relation (4.42) is a weakened form of the large sieve inequality (4.33), since only the contribution of the numbers q which are prime is estimated. Nonetheless it is a very useful result in applications—cf., Notes. Moreover, it may be extended to congruence classes for composite moduli. Montgomery (1968) showed that, for any squarefree integer q, we have

$$q \sum_{0 \leqslant h < q} \Big| \sum_{d \mid q} \frac{\mu(d)}{d} S(q/d, h) \Big|^2 = \sum_{\substack{1 \leqslant a \leqslant q \\ (a,q)=1}} |S(a/q)|^2.$$

Inserting this in (4.33), we thus obtain that $S(q/d, h)$ is, on average over d dividing q and h in $[0, q-1]$, close to $(d/q)S(0)$.

4.6. Applications of the large sieve

By comparison with Brun's method, the large sieve delivers a remarkably effective upper bound on the number $J(x)$ of prime twins not exceeding x.

Theorem 4.15. *As x tends to infinity, we have*

$$J(x) \leqslant \{8C + o(1)\} x/(\ln x)^2 \tag{4.43}$$

with

$$C := 2 \prod_{p \geqslant 3} \Big(1 - \frac{1}{(p-1)^2}\Big). \tag{4.44}$$

This upper bound is thus asymptotically equal to eight times the value conjectured for $J(x)$—cf., the Notes for references to improvements.

Proof. Let $\varepsilon > 0$. We employ (4.37) with $N = \lfloor x \rfloor$, $Q = x^{1/2-\varepsilon}$, $M = 0$, and $a_n := 1$ if $P^-\big(n(n+2)\big) > Q$, $a_n := 0$ otherwise. We get

$$J(x) - J(\sqrt{x}) \leqslant \{1 + o(1)\} x/L \tag{4.45}$$

where L is defined by (4.38) with $w(2) = 1$, $w(p) = 2$ $(p \geqslant 3)$. We have $g(q) = 2^\omega * h(q)/q$ where h is the multiplicative function defined by

$$h(2) = 0, \quad h(2^\nu) = 2(-1)^{\nu-1} \qquad (\nu \geqslant 2)$$

$$h(p) = \frac{4}{p-2} \quad (p > 2), \quad h(p^\nu) = \frac{2(-1)^{\nu-1}(p+2)}{p-2} \qquad (p > 2,\ \nu \geqslant 2).$$

It is easily checked that the series $\sum_{d \geqslant 1} h(d)/d^\sigma$ converges absolutely for $\sigma > \frac{1}{2}$, whence

$$\sum_{q \leqslant y} g(q) = \sum_{md \leqslant y} \frac{h(d)}{d} \frac{2^{\omega(m)}}{m} \sim \frac{3}{\pi^2} (\ln y)^2 \sum_{d \geqslant 1} \frac{h(d)}{d} \qquad (y \to \infty)$$

where the sum over m is evaluated by partial summation starting with the estimate

$$\sum_{m\leqslant y} 2^{\omega(m)} = \sum_{m\leqslant y} \mathbf{1} * \mu^2(m) \sim \frac{6}{\pi^2} y \ln y.$$

We therefore deduce from (4.45) that we have

$$J(x) \leqslant \frac{\{2C + o(1)\}x}{\big(\ln \sqrt{x}\big)^2}$$

with

$$\begin{aligned} C &= \frac{\pi^2}{6}\Big(\sum_{d\geqslant 1} \frac{h(d)}{d}\Big)^{-1} \\ &= \prod_p (1-p^{-2})^{-1} \tfrac{3}{2} \prod_{p\geqslant 3}\Big(1 + \frac{4}{p(p-2)} - \frac{2(p+2)}{p^2(p-2)(1+p^{-1})}\Big)^{-1} \\ &= 2\prod_{p\geqslant 3}\Big(1 - \frac{1}{(p-1)^2}\Big), \end{aligned}$$

whence the desired conclusion. □

Our second application concerns prime numbers in arithmetic progressions. Let us write

$$\pi(x; a, q) := \operatorname{card}\{p \leqslant x : p \equiv a \,(\mathrm{mod}\, q)\}$$

so that $\pi(x; a, q)$ is possibly unbounded only when a takes on one of the $\varphi(q)$ values of invertible residues modulo q. It is then natural to conjecture that, under suitable assumptions on the relative values of q and x, we have

$$\pi(x; a, q) \sim \frac{\pi(x)}{\varphi(q)} \sim \frac{x}{\varphi(q)\ln x}.$$

We shall show this in Part II for fixed q (Dirichlet's theorem), and even for $q \leqslant (\ln x)^c$ (Siegel–Walfisz theorem). The large sieve allows us to obtain an upper bound of the same type, equally valid for "short intervals".

Theorem 4.16 (Brun–Titchmarsh). *Let x, y be positive numbers and q, a be integers. If $y/q \to \infty$, we have*

$$\pi(x+y; a, q) - \pi(x; a, q) \leqslant \frac{\{2 + o(1)\}y}{\varphi(q)\ln(y/q)}. \tag{4.46}$$

Proof. The left-hand side of (4.46) is at most equal to

$$\sum_n a_n + \pi\big(\sqrt{y/q}\big), \tag{4.47}$$

where $a_n := 1$ if $x < a + nq \leqslant x + y$ and $P^-(a+qn) > \sqrt{y/q}$, and $a_n := 0$ otherwise. The second term in (4.47) is plainly absorbed by the remainder

term in (4.46). With the notation of the large sieve, we have $N \leqslant y/q + 1$ and $w(p) \geqslant 1$ for any prime number p such that $p \leqslant \sqrt{y/q}$ and $p \nmid q$. We therefore obtain that, for all $Q \leqslant \sqrt{y/q}$,

$$\sum_n a_n \leqslant \frac{y/q + Q^2}{L} \tag{4.48}$$

with

$$L := \sum_{m \leqslant Q,\,(m,q)=1} \mu(m)^2 \prod_{p \mid m} \frac{1}{p-1} = \sum_{m \leqslant Q,\,(m,q)=1} \frac{\mu(m)^2}{\varphi(m)}.$$

Noting that each integer $n \leqslant Q$ factorizes uniquely in the form $n = mdt$ with $(m, q) = 1$, $\mu(m)^2 = 1$, $d \mid m^\infty$, $t \mid q^\infty$, we can write

$$\sum_{n \leqslant Q} \frac{1}{n} \leqslant \sum_{m \leqslant Q,\,(m,q)=1} \frac{\mu(m)^2}{m} \sum_{d \mid m^\infty} \frac{1}{d} \sum_{t \mid q^\infty} \frac{1}{t} = \sum_{m \leqslant Q,\,(m,q)=1} \frac{\mu(m)^2}{m} \frac{m}{\varphi(m)} \frac{q}{\varphi(q)}$$

from which we get

$$L \geqslant \frac{\varphi(q)}{q} \ln Q.$$

Inserting this estimate back into (4.48) and selecting $Q = \sqrt{y/q}/\ln(y/q)$, we obtain the stated result. □

4.7. Selberg's sieve

4.7.1. Introduction. Based on an elegant optimization of quadratic forms, the sieve method introduced by Selberg in 1947 has a triple advantage over Brun's sieve, dating from the 1920s: it is simpler in principle, more versatile in its technical implementation, and it provides better estimates, at least in the situations considered in practice.

When the large sieve was discovered and perfected, Selberg's method was merely used for its ability to sieve any sequence and not just the set of integers in an interval—which leads, in certain cases such as that of twin primes, to better constants. From the mid-1970s on, the general idea was that, in the case of an interval, the two techniques are essentially equivalent—see Huxley (1972b), Kobayashi (1973)—but that the large sieve is based on a more fundamental duality principle. This philosophy, in particular, underpins the beautiful book of Elliott (1997).

Subsequently, from the 1980s on, the Rosser–Iwaniec sieve (see the Notes for a precise statement) was, rightly, considered the most efficient in the theory. Besides introducing a great deal of flexibility in the treatment of remainder terms, it allows, in the more propitious cases, to gain asymptotically a constant factor in the main terms of the upper bounds obtained.

The recent and fundamental advances by Goldston, Pintz & Yıldırım (2006) and then Zhang (2014) and Maynard (2014) on small gaps between primes have brought the Selberg sieve out into full light again.

It would take us too far afield to describe these remarkable works. We have chosen to present the Selberg sieve by looking at one of its lesser known aspects: the possibility of sieving by prime powers. This is based on an extension, interesting for its own sake, of the notion of a multiplicative function in one or many variables.

The reader will find, for example, in the book of Halberstam & Richert (1974) or of Greaves (2001), exhaustive expositions about the classical form of Selberg's method, notably the possibility of lower bounds—which we will not mention further here.

4.7.2. Arithmetic functions of several variables. The classic notion of a multiplicative function presented in Chapter I.2 has several drawbacks. One of them is that the structure is not stable if one modifies a finite number of terms in a finite number of factors in the Euler product representing the associated Dirichlet series. Another is that the generalization to several variables is not obvious.

In an article published in 1977, Selberg proposes a new definition which overcomes these two difficulties.

Definition 4.17. *A arithmetic function of r variables $f : (\mathbb{N}^*)^r \to \mathbb{C}$ is said to be multiplicative (in Selberg's sense) if its formal Dirichlet series*

$$F(s_1, \dots, s_r) := \sum_{n_1 \geqslant 1, \dots, n_r \geqslant 1} \frac{f(n_1, \dots, n_r)}{n_1^{s_1} \cdots n_r^{s_r}}$$

is representable as a product

$$F(s_1, \dots, s_r) = \prod_p F_p(p^{-s_1}, \dots, p^{-s_r}) \tag{4.49}$$

where the F_p are power series in r variables

$$F_p(X_1, \dots, X_r) := \sum_{\nu_1 \geqslant 0, \dots, \nu_r \geqslant 0} f_p(\nu_1, \dots, \nu_r) X_1^{\nu_1} \cdots X_r^{\nu_r}$$

such that $F_p(0, \dots, 0) = 1$ except for at most a finite number of values of p.

We say that f is singular if $f(1, \dots, 1) = 0$, regular if $f(1, \dots, 1) \neq 0$, and normal if $f(1, \dots, 1) = 1$.

Just for this section, we adopt the convention that the term multiplicative arithmetic function in one or many variables should be understood in Selberg's sense.

Condition (4.49) is equivalent to

$$f(n_1,\dots,n_r)=\prod_p f_p(v_p(n_1),\dots,v_p(n_r)) \tag{4.50}$$

where v_p denotes the p-adic valuation and the f_p are such that $f_p(0,\dots,0)=1$ except for a finite number of values of p.

In the case of a single variable, the class of ordinary multiplicative functions coincides with that of normal multiplicative functions in Selberg's sense. We have thus obtained a generalization of the notion. If, for example, f is multiplicative and normal, then $n\mapsto f(kn)$ is multiplicative for all $k\in\mathbb{N}^*$.

In the sequel we shall restrict ourselves to arithmetic functions of two variables.

4.7.3. Generalized convolution.

Definition 4.18. *An arithmetic function f of two variables is said to be symmetric if it satisfies $f(m,n)=f(n,m)$ identically. It is said to be lower triangular if $f(m,n)=0$ whenever $n>m$; it is said to be normal lower triangular if it is lower triangular and satisfies $f(m,m)=1$ for all $m\in\mathbb{N}^*$.*

If t is a normal lower triangular arithmetic function, then, for each fixed integer $m\geqslant 1$, the linear system[2]

$$\sum_{n\leqslant k\leqslant m} t(k,n)x_k=\delta_{mn}\qquad(1\leqslant n\leqslant m)$$

is upper triangular, and so can be solved by Cramer's rule. We denote by

$$x_k=t^*(m,k)\quad(1\leqslant k\leqslant m)$$

its solution. We hence have

$$\sum_{n\leqslant k\leqslant m} t^*(m,k)t(k,n)=\delta_{mn}\qquad(1\leqslant n\leqslant m). \tag{4.51}$$

This relation means that, for each integer $q\geqslant 1$, the lower triangular matrices $T_q^*:=\big(t^*(j,k)\big)_{1\leqslant k\leqslant j\leqslant q}$ and $T_q:=\big(t(r,s)\big)_{1\leqslant s\leqslant r\leqslant q}$ are one another's inverses. By transposition, we obtain the dual relation

$$\sum_{n\leqslant k\leqslant m} t(m,k)t^*(k,n)=\delta_{mn}\qquad(1\leqslant n\leqslant m). \tag{4.52}$$

We then say that the normal lower triangular arithmetic functions t and t^* are each other's inverses.

We recover the classical Dirichlet convolution inverse (see §2.4) by putting $t(m,n):=h(m/n)$ (resp. $t^*(m,n):=h^*(m/n)$) where h (resp. h^*)

[2]Here and in the sequel we use Kronecker's notation $\delta_{mn}:=1$ if $m=n$, $\delta_{mn}:=0$ if $m\neq n$.

is a function in one variable, with the convention that the right-hand expression is zero if m/n is not an integer. Then, setting $m = Nn$ and making the change of variables $k = dn$ in (4.52), this formula becomes

$$\sum_{d|N} h^*(d)h(N/d) = \delta_{1N} = \delta(N) \qquad (N \in \mathbb{N}^*).$$

This new framework allows us to state inversion formulae that are more general than those arising from Dirichlet convolution, described in Chapter 1.2.

Proposition 4.19. *Let t be a normal lower triangular arithmetic function of two variables.*

(i) If f and g are arithmetic functions of a single variable connected by the relation

$$f(m) = \sum_{1 \leqslant n \leqslant m} t(m,n)g(n) \qquad (m \in \mathbb{N}^*), \tag{4.53}$$

then

$$g(m) = \sum_{1 \leqslant n \leqslant m} t^*(m,n)f(n) \qquad (m \in \mathbb{N}^*). \tag{4.54}$$

(ii) If f and g are arithmetic functions of a single variable connected by the relation

$$f(n) = \sum_{m \geqslant n} t(m,n)g(m) \qquad (n \in \mathbb{N}^*), \tag{4.55}$$

where the series is assumed to be absolutely convergent for each n, then we have

$$g(n) = \sum_{m \geqslant n} t^*(m,n)f(m) \qquad (n \in \mathbb{N}^*) \tag{4.56}$$

provided that the series is absolutely convergent for each value of n.

Proof. Relation (4.54) results from (4.53) upon inserting the identity

$$f(n) = \sum_{1 \leqslant k \leqslant n} t(n,k)g(k)$$

in the left-hand side and inverting summations. A similar manipulation allows us to deduce (4.56) from (4.55), questions of convergence being taken care of by truncating the series in the right-hand side. We omit the details. □

The case when t is multiplicative is of particular practical interest.

Proposition 4.20. *Let $t : (\mathbb{N}^*)^2 \to \mathbb{C}$ be a normal lower triangular multiplicative function. Then its inverse t^* is also normal lower triangular and multiplicative.*

Proof. The power series $T_p(X,Y) := 1 + \sum_{\substack{0\leqslant\nu\leqslant\mu \\ (\mu,\nu)\neq(0,0)}} t(p^\mu,p^\nu)X^\mu Y^\nu$ are formally invertible. We have, for any prime p,

$$T_p(X,Y)^{-1} = \sum_{j\geqslant 0}(-1)^j \left\{ \sum_{\substack{0\leqslant\nu\leqslant\mu \\ (\mu,\nu)\neq(0,0)}} t(p^\mu,p^\nu)X^\mu Y^\nu \right\}^j$$

$$= 1 + \sum_{\substack{0\leqslant\nu\leqslant\mu \\ (\mu,\nu)\neq(0,0)}} t^*(p^\mu,p^\nu)X^\mu Y^\nu$$

with $t^*(1,1) = 1$ and, for $0 \leqslant \nu \leqslant \mu$, $(\mu,\nu) \neq (0,0)$,

$$t^*(p^\mu,p^\nu) = \sum_{1\leqslant j\leqslant\nu} (-1)^j \sum_{\substack{\mu_1+\cdots+\mu_j=\mu \\ \nu_1+\cdots+\nu_j=\nu}} \prod_{1\leqslant s\leqslant j} t(p^{\mu_s},p^{\nu_s}).$$

The normal lower triangular multiplicative function t^* defined by this formula is the sought after inverse. □

Note that, if t is a normal lower triangular multiplicative function, then $t(m,n)$ vanishes whenever $n \nmid m$. The following result is a very useful special case of Proposition 4.19.

Proposition 4.21. *Let t be a normal lower triangular multiplicative function of two variables.*

(i) *If f and g are arithmetic functions in one variable connected by the relation*

$$f(m) = \sum_{d|m} t(m,d)g(d) \qquad (m \in \mathbb{N}^*), \tag{4.57}$$

then

$$g(m) = \sum_{d|m} t^*(m,d)f(d) \qquad (m \in \mathbb{N}^*). \tag{4.58}$$

(ii) *If f and g are arithmetic functions in one variable connected by the relation*

$$f(n) = \sum_{m\equiv 0 \,(\mathrm{mod}\, n)} t(m,n)g(m) \qquad (n \in \mathbb{N}^*), \tag{4.59}$$

where the series is assumed absolutely convergent for each n, then we have

$$g(n) = \sum_{m\equiv 0 \,(\mathrm{mod}\, n)} t^*(m,n)f(m) \qquad (n \in \mathbb{N}^*) \tag{4.60}$$

provided that the series is absolutely convergent for each value of n.

Let $k \geqslant 1$ be a fixed integer. An example of a pair (t, t^*) of normal lower triangular multiplicative functions is provided by setting

$$t(m,n) := \tau((m/n,k))\mathbf{1}_{\mathbb{N}}(m/n) \qquad (m,n \in \mathbb{N}^*).$$

Putting $\kappa_p := v_p(k)$, we then have, for any prime p,

$$\sum_{0\leqslant\nu\leqslant\mu} t_p(\mu,\nu)X^\mu Y^\nu = \frac{1-X^{\kappa_p+1}}{(1-X)^2(1-XY)}.$$

It follows that

$$t_p^*(\mu,\nu) = \begin{cases} 1 & \text{if } \mu=\nu=0,\\ -1 & \text{if } \mu=1,\ \nu\in\{0,1\},\ \kappa_p=0,\\ 1 & \text{if } \mu=2,\ \nu=1,\ \kappa_p=0,\\ 1 & \text{if } \mu\equiv 0 \text{ or } 2 \,(\mathrm{mod}\,\kappa_p+1),\ \nu=0,\ \kappa_p\geqslant 1,\\ -2 & \text{if } \mu\equiv 1 \,(\mathrm{mod}\,\kappa_p+1),\ \nu=0,\ \kappa_p\geqslant 1,\\ -1 & \text{if } \mu\equiv 1 \text{ or } 3 \,(\mathrm{mod}\,\kappa_p+1),\ \nu=1,\ \kappa_p\geqslant 1,\\ 2 & \text{if } \mu\equiv 2 \,(\mathrm{mod}\,\kappa_p+1),\ \nu=1,\ \kappa_p\geqslant 1,\\ 0 & \text{in any other case.}\end{cases}$$

4.7.4. Quadratic forms. We denote by $\ell_0(\mathbb{R})$ the set of ultimately vanishing real sequences.

Definition 4.22. *We say that an arithmetic function $f : (\mathbb{N}^*)^2 \to \mathbb{R}$ is positive definite if it is symmetric and if the quadratic form*

$$Q(\boldsymbol{x}) := \sum_{m\geqslant 1,\, n\geqslant 1} f(m,n)x_m x_n \tag{4.61}$$

is strictly positive for any non-zero real sequence $\boldsymbol{x} = \{x_n\}_{n=1}^\infty$ in $\ell_0(\mathbb{R})$.

It follows immediately that, for any integer $N \geqslant 1$, the matrix

$$F_N := \big(f(m,n)\big)_{1\leqslant m,n\leqslant N}$$

associated to a positive definite arithmetic function f is symmetric, invertible, and that all its eigenvalues are strictly positive. A classical theorem on the decomposition of a square matrix into triangular factors (see, for example, Gantmacher (1959, 1966), § 2.4) then allows us to write $F_N = TD\,{}^tT$ where T is lower triangular with 1's along the principal diagonal, D is diagonal and ${}^tT$ denotes the transpose of T. Moreover, the matrices D and T are unique under the conditions indicated. This furnishes the canonical representation

$$f(m,n) = \sum_{1\leqslant k\leqslant \min(m,n)} g(k)t(m,k)t(n,k) \tag{4.62}$$

where t is normal lower triangular and g has strictly positive real values.

Whenever f is multiplicative, this formula takes a simpler form.

Proposition 4.23. *Let f be a positive definite multiplicative function of two variables. Then there exist two normal lower triangular multiplicative functions $g : \mathbb{N}^* \to \mathbb{R}^{+*}$ and $t : (\mathbb{N}^*)^2 \to \mathbb{R}$, such that*

$$f(m,n) = \sum_{d|(m,n)} g(d)t(m,d)t(n,d) \qquad (m,n \in \mathbb{N}^*). \tag{4.63}$$

Furthermore, the functions g and t are uniquely determined by this condition.

Proof. For each prime p, the arithmetic function f_p defined by $f_p(\mu,\nu) := f(p^\mu, p^\nu)$ is positive definite on $\mathbb{N}^2$. Therefore there exist multiplicative functions g_p and t_p, respectively of one and two variables, so that t_p is normal lower triangular and we have the identity

$$f(p^\mu, p^\nu) = \sum_{0 \leqslant \kappa \leqslant \min(\mu,\nu)} g_p(\kappa)t_p(\mu,\kappa)t_p(\nu,\kappa). \tag{4.64}$$

Taking the product over p and defining g and t according to (4.50), we obtain (4.63). The result then follows from the uniqueness of the representation given by (4.62). □

We are presently in a position to solve the problem of constrained optimization for quadratic forms. We restrict ourselves to stating the result for the case of multiplicative functions. The reader will be able to reconstruct the general case without difficulty.

Theorem 4.24. *Let f be a positive definite multiplicative function in two variables, $N \in \mathbb{N}^*$, and g, t, be the two multiplicative functions defined by the canonical representation* (4.63). *Then, under the conditions $x_1 = 1$ and $n > N \Rightarrow x_n = 0$, the quadratic form* (4.61) *attains its minimum*

$$Q^* := \Big\{ \sum_{m \leqslant N} t^*(m,1)^2/g(m) \Big\}^{-1} \tag{4.65}$$

for the choice

$$x_n := Q^* \sum_{1 \leqslant m \leqslant N} t^*(m,n)t^*(m,1)/g(m) \qquad (1 \leqslant n \leqslant N). \tag{4.66}$$

Proof. Let us put

$$y_d := \sum_{\substack{1 \leqslant m \leqslant N \\ m \equiv 0 \,(\mathrm{mod}\, d)}} t(m,d)x_m \qquad (d \geqslant 1),$$

so that $y_d = 0$ for $d > N$. Substituting (4.63) in (4.61), we obtain

$$Q(\boldsymbol{x}) = \sum_{1 \leqslant d \leqslant N} g(d)y_d^2 \tag{4.67}$$

and, by Proposition 4.21(ii),

$$x_d = \sum_{m \equiv 0 \,(\mathrm{mod}\, d)} t^*(m,d) y_m \qquad (1 \leqslant d \leqslant N). \tag{4.68}$$

In particular, the condition $x_1 = 1$ reads

$$\sum_{1 \leqslant m \leqslant N} t^*(m,1) y_m = 1.$$

It follows that, for all real λ,

$$\begin{aligned} Q(\boldsymbol{x}) &= \lambda + \sum_{1 \leqslant d \leqslant N} \left\{ \sqrt{g(d)}\, y_d - \lambda \frac{t^*(d,1)}{2\sqrt{g(d)}} \right\}^2 - \tfrac{1}{4}\lambda^2 \sum_{1 \leqslant d \leqslant N} \frac{t^*(d,1)^2}{g(d)} \\ &\geqslant \lambda - \tfrac{1}{4}\lambda^2 \sum_{1 \leqslant d \leqslant N} \frac{t^*(d,1)^2}{g(d)} = \lambda - \frac{\lambda^2}{4Q^*}. \end{aligned}$$

The maximum of the right-hand side is achieved for $\lambda = 2Q^*$. We thus obtain $Q(\boldsymbol{x}) \geqslant Q^*$, with equality when $y_d = \lambda t^*(d,1)/2g(d)$ $(1 \leqslant d \leqslant N)$, whence (4.66) by (4.68). □

4.7.5. The Johnsen–Selberg prime power sieve. For each prime power p^ν $(\nu \geqslant 1)$, let $\mathcal{W}(p^\nu)$ be a set of residues modulo p^ν, identified with the class of its representatives in $\mathbb{Z}$. Let us assume that $\mathcal{W}(p^\mu) \cap \mathcal{W}(p^\nu) = \varnothing$ if $\mu \neq \nu$. We then put

$$\mathcal{W}(d) := \bigcap_{p^\nu \| d} \mathcal{W}(p^\nu), \tag{4.69}$$

so that $n \in \mathcal{W}(d)$ if, and only if, $n \in \mathcal{W}(p^\nu)$ whenever $p^\nu \| d$. By convention, let us also set $\mathcal{W}(1) = \mathbb{Z}$.

Let $\mathcal{A}$ be a finite sequence of integers (not necessarily distinct), let $\mathcal{P}$ be a set of primes and let $z \geqslant 2$ be a real number. We put

$$\mathcal{P}_z := \mathcal{P} \cap [1, z].$$

We seek an upper bound for the quantity

$$S(\mathcal{A}, \mathcal{P}; z) := \big|\{a \in \mathcal{A} : a \notin \mathcal{W}(p^\nu) \ \ (p \in \mathcal{P}_z,\, \nu \geqslant 1)\}\big|.$$

Let us assume we may write

$$\sum_{\substack{a \in \mathcal{A} \\ a \in \mathcal{W}(d)}} 1 = \frac{w(d)}{d} X + r_d \qquad (d \geqslant 1), \tag{4.70}$$

where $X \geqslant 0$ is a quantity independent of d, w is a non-negative multiplicative function, and r_d is, in some appropriate sense, an error term. We can assume with no loss of generality that

$$w(p^\nu) = 0 \qquad (p \notin \mathcal{P}_z,\, \nu \geqslant 1) \tag{4.71}$$

and that

$$\sum_{\nu\geqslant 1}\frac{w(p^\nu)}{p^\nu}<1 \quad (p\in\mathcal{P}_z), \tag{4.72}$$

which, in usual cases of application, is equivalent to the fact that $\bigcup_{\nu\geqslant 1}\mathcal{W}(p^\nu)\neq\mathbb{Z}$ for any prime number p.

We put

$$\vartheta(p^\nu):=1-\sum_{1\leqslant\mu\leqslant\nu}\frac{w(p^\mu)}{p^\mu}>0 \qquad (p\geqslant 2,\nu\geqslant 0). \tag{4.73}$$

Given an arithmetic function f, we also introduce the notation

$$\psi_f(x,y):=\sum_{\substack{n\leqslant x\\ P^+(n)\leqslant y}}\frac{f(n)}{n} \qquad (x\geqslant 1,\ y\geqslant 1).$$

Theorem 4.25. *Under the hypotheses of* (4.70), (4.71) *and* (4.72), *we have, for any integer* $D>1$

$$S(\mathcal{A},\mathcal{P};z)\leqslant\frac{X}{\psi_f(D,z)}+\sum_{\substack{m\leqslant D^2\\ P^+(m)\leqslant z}}3^{\omega(m)}|r_m|, \tag{4.74}$$

where f *is the normal multiplicative function defined by*

$$f(p^\nu):=\frac{p^\nu}{\vartheta(p^\nu)}-\frac{p^\nu}{\vartheta(p^{\nu-1})}=\frac{w(p^\nu)}{\vartheta(p^\nu)\vartheta(p^{\nu-1})} \qquad (\nu\geqslant 1). \tag{4.75}$$

Proof. Let $D>1$. For any real sequence $\{\lambda_d\}_{d\geqslant 1}$ satisfying $\lambda_1=1$ and $\lambda_d=0$ for $d>D$, we have

$$S(\mathcal{A},\mathcal{P};z)\leqslant\sum_{a\in\mathcal{A}}\Big(\sum_{a\in\mathcal{W}(d)}\lambda_d\Big)^2. \tag{4.76}$$

Indeed, the term of index a in the sum in (4.76) is always non-negative (since the λ_d are real) and equals $\lambda_1=1$ if a is counted in $S(\mathcal{A},\mathcal{P};z)$.

Let us introduce the multiplicative function defined by

$$\varepsilon(p^\mu,p^\nu):=\begin{cases}1 & \text{if } \mu=\nu \text{ or } \mu\nu=0,\\ 0 & \text{otherwise.}\end{cases}$$

Our initial disjointness condition for the classes $\mathcal{W}(p^\nu)$ implies that

$$\mathcal{W}(p^\mu)\cap\mathcal{W}(p^\nu)\neq\varnothing$$

is only possible if $\varepsilon(p^\mu,p^\nu)=1$. By multiplicativity, it follows that $\varepsilon(d,d')=1$ whenever $\mathcal{W}(d)\cap\mathcal{W}(d')\neq\varnothing$. Expanding the square in (4.76),

we therefore obtain

$$
\begin{aligned}
S(\mathcal{A},\mathcal{P};z) &\leqslant \sum_{d,d'\leqslant D}\lambda_d\lambda_{d'}\sum_{\substack{a\in\mathcal{A}\\ a\in\mathcal{W}(d)\cap\mathcal{W}(d')}}1\\
&= \sum_{d,d'\leqslant D}\lambda_d\lambda_{d'}\varepsilon(d,d')\sum_{\substack{a\in\mathcal{A}\\ a\in\mathcal{W}([d,d'])}}1 = XQ+R,
\end{aligned}
\tag{4.77}
$$

where we have put

$$
Q := \sum_{d,d'\leqslant D}\lambda_d\lambda_{d'}\varepsilon(d,d')\frac{w([d,d'])}{[d,d']},\quad R := \sum_{d,d'\leqslant D}\lambda_d\lambda_{d'}\varepsilon(d,d')r_{[d,d']}.
\tag{4.78}
$$

We may apply Theorem 4.24 to minimize the quadratic form Q. To this end, we must calculate the functions g, t, and t^* associated to the function $F(m,n) := \varepsilon(m,n)w([m,n])/[m,n]$.

Iterating over the pair (μ,ν), we infer from (4.64) that we have, with the notation (4.73),

$$
g(p^\nu) := \{\vartheta(p^{\nu-1}) - \vartheta(p^\nu)\}\frac{\vartheta(p^\nu)}{\vartheta(p^{\nu-1})}\qquad(\nu\geqslant 1),
$$

$$
t(p^\mu,p^\nu) := \begin{cases}\vartheta(p^{\mu-1})-\vartheta(p^\mu) & \text{if } \nu=0,\\ -\{\vartheta(p^{\mu-1})-\vartheta(p^\mu)\}/\vartheta(p^\nu) & \text{if } 0<\nu<\mu,\\ \delta_{\mu\nu} & \text{if } \mu\leqslant\nu,\end{cases}
$$

from which

$$
t^*(p^\mu,p^\nu) := \begin{cases}-\{\vartheta(p^{\mu-1})-\vartheta(p^\mu)\}/\vartheta(p^{\mu-1}) & \text{if } \nu=0,\\ \{\vartheta(p^{\mu-1})-\vartheta(p^\mu)\}/\vartheta(p^{\mu-1}) & \text{if } 0<\nu<\mu,\\ \delta_{\mu\nu} & \text{if } \mu\leqslant\nu.\end{cases}
$$

The minimum Q^* is reached for the value of $\{\lambda_d\}_{d\geqslant 1}$ given by (4.66), namely

$$
\lambda_d := \frac{\sum_{m\leqslant D}t^*(m,d)t^*(m,1)/g(m)}{\sum_{m\leqslant D}t^*(m,1)^2/g(m)}\quad(d\leqslant D),\quad \lambda_d^* := 0\quad(d>D),
\tag{4.79}
$$

We notice that $t^*(m,1)=0$ if $P^+(m)>z$. We then deduce from (4.65) that

$$
Q^* = \frac{1}{\sum_{\substack{d\leqslant D\\ P^+(d)\leqslant z}}\prod_{p^\nu\| d}\{1/\vartheta(p^\nu)-1/\vartheta(p^{\nu-1})\}} = \frac{1}{\psi_f(D,z)}.
\tag{4.80}
$$

On the other hand, since $\sup_d|\lambda_d^*|=1$, we can write

$$
|R| \leqslant \sum_{\substack{m\leqslant D^2\\ P^+(m)\leqslant z}}\sum_{[d,d']=m}\varepsilon(d,d')|r_m| = \sum_{\substack{m\leqslant D^2\\ P^+(m)\leqslant z}}3^{\omega(m)}|r_m|,
$$

where the inner sum over d,d' has been evaluated by noticing that it is multiplicative in m, and equals exactly 3 whenever m is a prime power. □

When the set $\mathcal{A}$ is an interval of integers, the upper bound (4.74) can take a different form.

Theorem 4.26 (Selberg). *Let $N \in \mathbb{N}^*$ and $\mathcal{A}$ be a set made up of N consecutive integers. If the function $w(d) := |\mathcal{W}(d) \cap [0, d[|$ satisfies* (4.71) *and* (4.72)*, we have, for any integer $D \geqslant 1$,*

$$S(\mathcal{A}, \mathcal{P}; z) \leqslant \frac{N + D^2 - 1}{\psi_f(D, z)}$$

where f is the normal multiplicative function defined by (4.75)*.*

Proof. There exists an integer M such that $\mathcal{A} = \{n \in \mathbb{Z} : M < n \leqslant M+N\}$. Let b be the Beurling function defined by (4.30) with $\delta := 1/D^2$. By the second upper bound in (4.77), we have

$$\begin{aligned} S(\mathcal{A}, \mathcal{P}; z) &\leqslant \sum_{d,\, d' \leqslant D} \lambda_d \lambda_{d'} \varepsilon(d, d') \sum_{\substack{n \in \mathbb{Z} \\ n \in \mathcal{W}([d,d'])}} b(n) \\ &= \sum_{d,\, d' \leqslant D} \lambda_d \lambda_{d'} \varepsilon(d, d') \sum_{\substack{0 \leqslant m < [d,d'] \\ m \in \mathcal{W}([d,d'])}} \sum_{n \equiv m \,(\mathrm{mod}\, [d,d'])} b(n). \end{aligned}$$

The inner sum evaluates to

$$\begin{aligned} &\sum_{n \in \mathbb{Z}} \frac{b(n)}{[d, d']} \sum_{0 \leqslant j < [d,d']} \mathrm{e}\Big(\frac{(n-m)j}{[d, d']}\Big) \\ &\quad = \frac{1}{[d, d']} \sum_{0 \leqslant j < [d,d']} \mathrm{e}\Big(\frac{-mj}{[d, d']}\Big) \sum_{n \in \mathbb{Z}} b(n) \mathrm{e}\Big(\frac{nj}{[d, d']}\Big) = \frac{\widehat{b}(0)}{[d, d']}, \end{aligned}$$

taking account of (4.24). Noticing that $\widehat{b}(0) = N + D^2 - 1$ by (4.31) and summing over m, it follows that, with the notation (4.78),

$$S(\mathcal{A}, \mathcal{P}; z) \leqslant (N + D^2 - 1)Q.$$

The stated result now follows from (4.80). □

We have, for each prime number p,

$$\sum_{\nu \geqslant 0} \frac{f(p^\nu)}{p^\nu} = 1 + \sum_{j \geqslant 1} \frac{1}{\vartheta(p^j)} - \frac{1}{\vartheta(p^{j-1})} = \frac{1}{\lim_{j \to \infty} \vartheta(p^j)} = \Big(1 - \sum_{\nu \geqslant 1} \frac{w(p^\nu)}{p^\nu}\Big)^{-1},$$

whence

$$\lim_{D \to \infty} \frac{1}{\psi_f(D, z)} = W(z) := \prod_{p \leqslant z} \Big(1 - \sum_{\nu \geqslant 1} \frac{w(p^\nu)}{p^\nu}\Big). \tag{4.81}$$

Under some additional hypotheses, hardly limiting in practice, one may give an optimal estimate for the product $\psi_f(D, z)W(z)$. With this in view,

we assume for example that there are some constants, $r \in \mathbb{N}^*$, $\eta \in]0, \frac{1}{2}[$ and $\kappa > 0$ such that

$$\vartheta(p^\nu) \geqslant \eta \quad (p \in \mathcal{P},\ \nu \geqslant 1), \tag{4.82}$$

$$\sum_p \sum_{1 \leqslant \nu \leqslant r} \frac{w(p^\nu)^2 \log p}{p^{2\nu}} + \sum_p \sum_{\nu > r} \frac{w(p^\nu)}{p^{(1-\eta)\nu}} < \infty, \tag{4.83}$$

and

$$\sum_{\substack{y < p \leqslant t \\ 1 \leqslant \nu \leqslant r}} \frac{w(p^\nu) \ln p}{p^\nu} \leqslant \kappa \ln(t/y) + O(1) \qquad (t \geqslant y \geqslant 1). \tag{4.84}$$

The finer asymptotic behavior of the quantity $\psi_f(D, z)W(z)$ is described in terms of continuous solutions to delay differential equations, the prototype of which is the Dickman function, studied in detail in paragraph III.5.4.

We denote by ϱ_κ the continuous solution of the system

$$\begin{cases} \varrho_\kappa(u) = u^{\kappa-1}/\Gamma(\kappa) & \text{if } 0 < u \leqslant 1, \\ u\varrho_\kappa'(u) + (1-\kappa)\varrho_\kappa(u) + \kappa\varrho_\kappa(u-1) = 0 & \text{if } u > 1, \end{cases} \tag{4.85}$$

where Γ is Euler's function. One may thus establish, by evaluating its Laplace transform,[3] that the generalized Dickman function ϱ_κ is the fractional convolution power of order κ of the Dickman function $\varrho := \varrho_1$. For each κ, the function ϱ_κ inherits from ϱ the rapid decay property. We have, for example,

$$\varrho_\kappa(u) = (u \ln u)^{-u} \mathrm{e}^{O_\kappa(u)} \qquad (u \to \infty),$$

and an asymptotic formula is given in the Notes for this chapter.

Let us further put

$$\lambda_\kappa(u) := \mathrm{e}^{-\gamma\kappa} \int_u^\infty \varrho_\kappa(v)\,\mathrm{d}v, \quad j_\kappa(u) = 1 - \lambda_\kappa(u) \qquad (u \geqslant 0), \tag{4.86}$$

where γ denotes Euler's constant. Proceeding as in the case $\kappa = 1$, described in § III.5.4, one may establish that $\lambda_\kappa(0) = j_\kappa(\infty) = 1$—see, for example, Hensley (1986a).

We state without proof the following result, due to Tenenbaum & Wu (2008a).

[3] See Chapter III.5, where the calculation is performed in the case $\kappa = 1$, which is typical of the general situation.

Theorem 4.27. *Let $\kappa > 0$, $\eta \in]0, \frac{1}{2}[$. Under hypotheses* (4.70), (4.71), (4.82), (4.83), (4.84), *there exists a constant B such that, uniformly for $2 \leqslant z \leqslant D^{1/r}$, we have*

$$\frac{1}{\psi_f(D,z)} \leqslant \frac{W(z)}{j_\kappa(v)}\left\{1 + O\left(\frac{\lambda_\kappa^+(v)}{\ln z} e^{Bv(\ln v)/\ln z}\right)\right\} \tag{4.87}$$

where $W(z)$ is defined by (4.81) *and where we have put*

$$v := \min\left\{(\ln D)/\{r \ln z\}, 3(\ln D)/(r\eta \ln_2 D)\right\}, \quad \lambda_\kappa^+(v) := \lambda_\kappa(v) v \ln(1+v).$$

We note that the error term, say R, in (4.87) satisfies the upper bound

$$R \ll v^{-v(1-\varepsilon)}/\ln z \qquad (z > z_0(\varepsilon)).$$

4.8. Sums of two squares in an interval

In the form described in the previous section, Selberg's sieve is perfectly adapted to providing an upper bound for the number of integers in an interval which can be represented as sums of two squares.

Let us begin by characterizing this set. The following theorem was conjectured by Girard in 1632 and established by Fermat in 1654.

The proof we give below rests on the fact that, given an odd prime p, the number -1 is a quadratic residue modulo p—or, in other words, the equation $x^2 + 1 \equiv 0 \pmod p$ is solvable— if, and only if, $p \equiv 1 \pmod 4$.

This classic property can easily be derived from the fact that the group $(\mathbb{Z}/p\mathbb{Z})^*$ of invertible residues modulo p is cyclic.[4] Another approach consists in observing that the multiplicative homomorphism

$$f : (\mathbb{Z}/p\mathbb{Z})^* \to \{-1, 1\}$$

defined by $f(x) = x^{(p-1)/2}$ is identically trivial on the subgroup Q_p of quadratic residues. Since the polynomial equation $f(x) = 1$ has at most $\frac{1}{2}(p-1)$ solutions in the field $\mathbb{Z}/p\mathbb{Z}$, it follows that Q_p coincides with the set of its roots. Thus, $f(a)$ assumes the value 1 if a is a quadratic residue modulo p, and the value -1 otherwise. As a rule, this property is expressed in terms of the *Legendre symbol*:

$$\left(\frac{a}{p}\right) := \begin{cases} 1 & \text{if } a \in Q_p, \\ -1 & \text{if } a \notin Q_p. \end{cases}$$

Theorem 4.28. *For every $a \in (\mathbb{Z}/p\mathbb{Z})^*$, we have*

$$\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod p.$$

[4] See Exercise 232, p. 404.

We are now able to establish the Girard–Fermat theorem in a simple way. Another proof, depending on the theory of continued fractions, is given in Chapter I.7.

Theorem 4.29 (Girard–Fermat). *An odd prime number is the sum of two squares if, and only if, it is congruent to 1 modulo 4.*

Proof. The condition is necessary because a sum of two squares is congruent to 0, 1, or 2 modulo 4. Let us show it is sufficient. Given a prime p such that $p \equiv 1 \pmod 4$, we put $N := \lfloor \sqrt{p} \rfloor$ and consider a solution x of the equation $x^2 \equiv -1 \pmod p$. Amongst the $(N+1)^2 > p$ numbers $u + vx$ with $0 \leqslant u, v \leqslant N$, at least two belong to the same residue class modulo p. By subtraction, we deduce that the congruence $a \equiv bx \pmod p$ has a non-trivial solution in integers a, b such that $\max(|a|, |b|) < \sqrt{p}$. This implies $a^2 \equiv x^2 b^2 \equiv -b^2 \pmod p$, whence $p \mid a^2 + b^2$. Since $0 < a^2 + b^2 < 2p$, we must have $p = a^2 + b^2$. □

Theorem 4.30. *A positive integer n is representable as a sum of two squares if, and only if, for any prime number p such that $p \equiv 3 \pmod 4$, we have $v_p(n) \equiv 0 \pmod 2$.*

Proof. Let h denote the characteristic function of the set of sums of two squares. The identity

$$(a^2 + b^2)(c^2 + d^2) = (ac + bd)^2 + (ad - bc)^2 \tag{4.88}$$

implies that h is super-multiplicative, i.e. satisfies the inequality

$$h(mn) \geqslant h(m)h(n).$$

Taking the Girard–Fermat theorem into account, it follows that any integer satisfying the property stated in the theorem is the sum of two squares.

Conversely, if n has a representation in the form $n = a^2 + b^2$ and if p is an odd prime such that $v_p(n) = 2\nu + 1$ with $\nu \in \mathbb{N}$, let us put $\mu := v_p(a)$, so that $\mu \leqslant \nu$ and hence $v_p(b) = \mu$. Dividing the representation by $p^{2\mu}$, we obtain $\alpha^2 + \beta^2 \equiv 0 \pmod p$ with $p \nmid \alpha\beta$. This implies that -1 is a quadratic residue and therefore $p \equiv 1 \pmod 4$. □

Remark. It results from the above proof that the characteristic function h is in fact multiplicative, defined by

$$h(p^\nu) = \begin{cases} 1 & \text{if } p = 2 \text{ or } p \equiv 1 \pmod 4 \text{ or } 2 \mid \nu, \\ 0 & \text{if } p \equiv 3 \pmod 4 \text{ and } 2 \nmid \nu. \end{cases} \tag{4.89}$$

By standard techniques described in Part II, this allows one to evaluate the summatory function of h. We thus obtain, for example (see Exercise 240, p. 406),

$$\sum_{n\leqslant N} h(n) = \Big\{1 + O\Big(\frac{1}{\ln N}\Big)\Big\}\frac{K_0 N}{\sqrt{\ln N}} \qquad (N \geqslant 2), \tag{4.90}$$

with

$$K_0 := \frac{1}{\sqrt{2}} \prod_{p\equiv 3 \,(\mathrm{mod}\, 4)} \Big(1 - \frac{1}{p^2}\Big)^{-1/2}. \tag{4.91}$$

We can now apply the Johnsen–Selberg sieve to get an upper bound on the number of sums of two squares in an interval. To improve the legibility of the statement, we anticipate, in proving the second part of the following result, the formula

$$\prod_{p>2} \Big(1 - \frac{(-1)^{(p-1)/2}}{p}\Big) = \frac{4}{\pi}. \tag{4.92}$$

This follows, for example, from the prime number theorem for arithmetic progressions in the form given in Theorem II.8.17 and the well-known formula $\sum_{n\geqslant 0}(-1)^n/(2n+1) = \pi/4$.

Theorem 4.31. *Let I be a real interval containing N positive integers. There exists an absolute positive constant K such that the number Z_N of elements of I which can be written as a sum of two squares satisfies*

$$Z_N \leqslant KN \prod_{\substack{p\leqslant N \\ p\equiv 3 \,(\mathrm{mod}\, 4)}} \Big(1 - \frac{1}{p}\Big). \tag{4.93}$$

Furthermore, as N tends to infinity, we have

$$Z_N \leqslant \big\{\pi + o(1)\big\}\frac{K_0 N}{\sqrt{\ln N}}. \tag{4.94}$$

Proof. We apply Theorem 4.26 with the choice

$$\mathcal{P} := \{p \equiv 3 \,(\mathrm{mod}\, 4)\}, \quad \mathcal{W}(p^\nu) := \begin{cases} \{mp^{\nu-1} : p \nmid m\} & \text{if } p \in \mathcal{P},\ 2 \mid \nu, \\ \varnothing & \text{if } p \in \mathcal{P},\ 2 \nmid \nu. \end{cases}$$

Thus, we have

$$w(p^\nu) = \begin{cases} p-1 & \text{if } 2 \mid \nu \text{ and } p \in \mathcal{P}, \\ 0 & \text{if } 2 \nmid \nu \text{ or } p \notin \mathcal{P}. \end{cases}$$

It follows that the hypotheses in Theorem 4.27 are satisfied for $r = 2$: by Theorem 1.10 the value $\kappa = 1$ is trivially admissible—which suffices to

establish (4.93)—and we can in fact select $\kappa = \frac{1}{2}$, anticipating the validity of Theorem II.8.16. We obtain

$$Z_N \leqslant \frac{N+D^2}{j_{1/2}(v)} \prod_{\substack{p\leqslant z\\ p\equiv 3 \,(\mathrm{mod}\, 4)}} \Big(1-\frac{1}{p+1}\Big)\Big\{1+O\Big(\frac{v^{-v/2}}{\ln z}\Big)\Big\},$$

where D is arbitrary and v is defined as indicated in the statement of Theorem 4.27.

Let us choose $D = z^2 := \sqrt{N}/\ln N$, so that $v = 1$ and

$$j_{1/2}(1) = 2\mathrm{e}^{-\gamma/2}/\Gamma(\tfrac{1}{2}) = 2\mathrm{e}^{-\gamma/2}/\sqrt{\pi}.$$

It remains to bound the product, say $W(z)$. We have

$$\begin{aligned} W(z) &= \prod_{\substack{p\leqslant z\\ p\equiv 3 \,(\mathrm{mod}\, 4)}} \Big(1-\frac{1}{p}\Big)\Big(1-\frac{1}{p^2}\Big)^{-1} \\ &= \sqrt{2}\prod_{p\leqslant z}\Big(1-\frac{1}{p}\Big)^{1/2} \prod_{2<p\leqslant z}\Big(1-\frac{(-1)^{(p-1)/2}}{p}\Big)^{-1/2} \prod_{\substack{p\leqslant z\\ p\equiv 3 \,(\mathrm{mod}\, 4)}} \Big(1-\frac{1}{p^2}\Big)^{-1/2} \\ &= \frac{2\sqrt{\pi}\mathrm{e}^{-\gamma/2}K_0+o(1)}{\sqrt{\ln N}}. \end{aligned}$$

□

Remark. The comparison of (4.94) and (4.90) shows that the factor lost by applying the sieve is asymptotically at most π. Since, by the functional equation (4.85), the function $j_{1/2}(v)/\sqrt{v}$ is decreasing for $v \geqslant 1$, it follows that the selection of $v = 1$ made in the preceding proof is optimal for the method used.

Notes

§ 4.2. To estimate the third error term in (4.5), we have appealed to the *parametric method*, presented in detail in chapter 0 of the book by Hall & Tenenbaum (1988). The famous *Rankin's method* (cf. § III.5.1) is another example of this fruitful computational trick, which consists in bounding the characteristic function of a set of integers by a constant multiple of a multiplicative function depending on a parameter to be optimized.

Our application of Brun's method to bounding $\pi(x)$ from above is only a mere illustration of the basic ideas, and should not be considered as a result in itself. Not only, as we observed, does it provides a weaker estimate than Chebyshev's, but one could also object, *stricto sensu*, to a loss of information. Indeed, we appealed to Theorem 1.10, which itself rests on Mertens' first theorem. But this last result immediately yields a Chebyshev-type upper bound for $\pi(x)$ since we have, for large x,

$$\pi(x) - \pi(x/2) \leqslant \frac{x}{\ln x} \sum_{x/2<p\leqslant x} \frac{\ln p}{p} \ll \frac{x}{\ln x}.$$

However, one can easily check that Brun's treatment does not require more than an asymptotic formula for $\sum_{p\leqslant x} 1/p$. This is a much weaker bound than those of Theorems 1.10 or 1.8, and reveals precisely why the scope of this method is so vast.

For other results about $\Phi(x,y)$, see Chapter III.6.

Alladi (1988) has an exposition of some developments of Brun's method for sums of multiplicative functions on certain subsets of $\mathbb{N}$.

The reader interested in the state of the art in the theory of the combinatorial sieve can refer to the survey of Diamond & Halberstam (1985) and the deep articles of Iwaniec (1980a and b, 1981).

A version similar in its principles to that of Brun, but furnishing results of a quality comparable with those of the improvements available in the literature, has been obtained by Hooley (1994) and simplified by Ford and Halberstam (2000), who stated a form that can be deployed "out of the box".

Let us briefly describe the main properties of the coefficients in the Rosser–Iwaniec sieve—see Iwaniec (1980a and b) in the "linear" case, that is to say when the function $w(p)$, representing the number of excluded residue classes modulo p, is bounded by 1 on average, also known as the case of *sieve of dimension* 1. The work of Iwaniec covers the case of any dimension.

Given a set of primes $\mathcal{P}$, we put

$$P(z) := \prod_{p\in\mathcal{P},\, p\leqslant z} p.$$

The Buchstab function $\omega(t)$, vanishing for $0 \leqslant t < 1$, is defined on $[1,\infty[$ as the unique continuous solution of the delay differential equation

$$\big(t\omega(t)\big)' = \omega(t-1) \qquad (t > 2)$$

with the initial condition $t\omega(t) = 1$ for $1 \leqslant t \leqslant 2$. One then defines the sieve functions F and f by

$$F(t) = \mathrm{e}^{\gamma}\{\omega(t) + \varrho(t-1)/t\}, \qquad f(t) = \mathrm{e}^{\gamma}\{\omega(t) - \varrho(t-1)/t\}$$

where γ is Euler's constant and ϱ is the Dickman function. We have $f(t) > 0$ if, and only if, $t > 2$. Moreover (see Corollary III.6.9), we have the asymptotic formulae

$$\begin{matrix} F(t) \\ f(t) \end{matrix} = 1 + O(\varrho(t-1)/t) = 1 + O(t^{-t}) \qquad (t \to \infty).$$

Theorem 4.32 (Iwaniec). *Let $D \geqslant 1$. There exist two sequences $\{\lambda_d^+\}_{d\geqslant 1}$ and $\{\lambda_d^-\}_{d=1}^{\infty}$, vanishing for $d > D$ or $\mu(d) = 0$, satisfying $\lambda_1^{\pm} = 1$, $|\lambda_d^{\pm}| \leqslant 1$, and such that we have*

$$\lambda^- * \mathbf{1} \leqslant \mu * \mathbf{1} \leqslant \lambda^+ * \mathbf{1},$$

and

$$\sum_{d|P(z)} \lambda_d^+ \frac{w(d)}{d} \leqslant \prod_{\substack{p\leqslant z\\ p\in\mathcal{P}}} \Big(1 - \frac{w(p)}{p}\Big)\Big\{F(s) + O\Big(\frac{\mathrm{e}^{\sqrt{L}-s}}{(\ln D)^{1/3}}\Big)\Big\},$$

$$\sum_{d|P(z)} \lambda_d^- \frac{w(d)}{d} \geqslant \prod_{\substack{p\leqslant z\\ p\in\mathcal{P}}} \Big(1 - \frac{w(p)}{p}\Big)\Big\{f(s) + O\Big(\frac{\mathrm{e}^{\sqrt{L}-s}}{(\ln D)^{1/3}}\Big)\Big\}$$

for any $z \in [2, D]$, $D = z^s$, any set $\mathcal{P}$ of prime numbers, and any multiplicative function w satisfying

(i) $0 < w(p) < p \qquad (p \in \mathcal{P})$,

(ii) $$\prod_{u<p\leqslant v,\, p\in\mathcal{P}} \Big(1 - \frac{w(p)}{p}\Big)^{-1} \leqslant \frac{\ln v}{\ln u}\Big(1 + \frac{L}{\ln u}\Big) \qquad (2 \leqslant u \leqslant v \leqslant z).$$

§ 4.4. Although (at least for the optimal form of the result) our proof of the analytic large sieve relies on the theory of analytic functions of a complex variable and on a deep result in harmonic analysis, the large sieve may be considered as an essentially elementary tool. We have seen that the proof of the inequality

$$\Delta(N,\delta) \leqslant \tfrac{1}{2}\pi^2(N - 1 + \delta^{-1})$$

is elementary. We also refer to the proofs, all distinct, of Montgomery (1971) [$\Delta \leqslant N + 2/\delta$], Bombieri (1974) [*idem*], and Elliott (1980) [$\Delta \leqslant N + 1/\delta$].

The function $F(z)$ in Lemma 4.11 was studied by Beurling at the end of the 1930s. For an exhaustive study of the optimization problem (4.27)–(4.28) and numerous applications which have been brought about by the various generalizations of this question, see Graham & Vaaler (1981, 1984), Vaaler (1985).

The large sieve is equally useful for estimating mean values of Dirichlet characters—see § II.8.1. Let us set

$$T(\chi) := \sum_{M<n\leqslant M+N} a_n \chi(n).$$

Gallagher (1967) gave a simple proof of the inequality

$$\sum_{\chi}^{*} |T(\chi)|^2 \leqslant \frac{\varphi(q)}{q} \sum_{1\leqslant a\leqslant q,\,(a,q)=1} \left|S\Big(\frac{a}{q}\Big)\right|^2$$

where the sum is over the primitive characters modulo q, that is, the characters which are not induced by a character modulo d with $d|q$, $d < q$. We then deduce from (4.33) the upper bound

$$\sum_{q\leqslant Q} \frac{q}{\varphi(q)} \sum_{\chi}^{*} |T(\chi)|^2 \leqslant (N - 1 + Q^2) \sum_{M<n\leqslant M+N} |a_n|^2.$$

This inequality plays an essential role in the study of L-functions (cf. § II.8.2) and the distribution of prime numbers in arithmetic progressions. It is, in particular, one of the fundamental ingredients in the proof of the Bombieri–Vinogradov theorem (1965, 1966)—cf. Chapter II.8, Notes.

§ 4.5. Theorem 4.14 has been used for numerous elegant solutions of arithmetic problems. In particular, it underpins the original proof of Daboussi's theorem (Daboussi & Delange, 1974). It is also the starting point of the new method of Hildebrand (1986c, d, e, 1987a) for studying the mean value of multiplicative functions with modulus at most 1. Hildebrand (1986b) also applies this inequality to manufacture a new elementary proof of the prime number theorem. A variant of Theorem 4.14 has been established by Elliott (1979, lemma 4.7), who shows the inequality (see also Theorem III.3.2)

$$\sum_{p\leqslant N} p\Big|S(p,0) - \frac{1}{p}S(0)\Big|^2 \leqslant 16N \sum_{1\leqslant n\leqslant N} |a_n|^2.$$

The result is not directly comparable to (4.42): the summation over p is longer,[5] but here only one congruence class for each prime number p is considered.

[5] Indeed, (4.42) only implies an upper bound of the same order if $Q \ll \sqrt{N}$.

§ 4.6. The best upper bound currently known for $J(x)$ is due to Wu (2004), improving on results by Fouvry & Grupp (1986) and himself (1990). The factor 8 in Theorem 4.15 is replaced by 3.3996. The basic result for this problem provides a factor 4: see Exercise 82. It is due to Bombieri & Davenport (1966)—cf. Halberstam & Richert (1974). Beside the sieve (Selberg's) the proof requires the Bombieri–Vinogradov theorem—cf. Theorem II.8.34.

The inequality (4.46) in fact holds uniformly *without* an error term, i.e.,

$$\pi(x+y;\ell,k)-\pi(x;\ell,k)<\frac{2y}{\varphi(k)\ln(y/k)} \qquad (1\leqslant k<y\leqslant x).$$

This handy result is due to Montgomery & Vaughan (1973).

§ 4.7. The special case of the prime power sieve where $\mathcal{A}$ is an interval was initially considered by Johnsen (1971) under hypotheses which are noticeably more restrictive than those we assumed. These appear in the work of Selberg (1977), who thus extends his own method. Gallagher (1972, 1973/74) provided a simpler proof of Johnsen's estimate. Motohashi (1979) then obtained a new proof, via the large sieve, of Selberg's result, thereby answering a question posed in Selberg's 1977 work.

The final upper bound for $S(\mathcal{A},\mathcal{P};z)$ is made explicit by Selberg when $\mathcal{A}$ is an interval, which corresponds to Theorem 4.26. In Theorem 4.25 we consider the general formulation by adopting the exposition of Tenenbaum & Wu (2008a).

Theorem 4.27 is obtained by observing that, for any non-negative multiplicative function f, we have

$$\psi_f(D,z)\geqslant\psi_{f_r}(D^{1/r},z) \qquad (D\geqslant 1,\ z\geqslant 1) \tag{4.95}$$

where f_r is the multiplicative function defined by

$$f_r(p^\nu):=\begin{cases}\displaystyle\sum_{1\leqslant j\leqslant r}\frac{f(p^j)}{p^{j-1}} & \text{if } \nu=1,\\ 0 & \text{if } 1<\nu\leqslant r\\ f(p^\nu) & \text{if } \nu>r.\end{cases}$$

Under hypotheses (4.82), (4.83) and (4.84), this allows one to apply th. 3.1 of Tenenbaum–Wu to the function f defined by (4.75).

We now provide some information about the functions ϱ_κ and λ_κ occurring above. Denote by $\xi(u)$ the unique non-zero real solution of $\mathrm{e}^\xi=1+u\xi$

if $u > 0$, $u \neq 1$, and set $\xi(1) = \xi(0) = 0$. Define further

$$\begin{aligned} \xi_\kappa(u) &:= \max\{1, \xi(u/\kappa)\}, \\ I(s) &:= \int_0^s \frac{\mathrm{e}^v - 1}{v}\,\mathrm{d}v, \\ \sigma_j(u) &:= \kappa I^{(j)}(\xi_\kappa(u)), \end{aligned} \tag{4.96}$$

so that, by Lemma III.5.11 *infra* and lemma 4.5 of Smida (1991), we have, for any fixed real number $j \geqslant 0$,

$$\begin{aligned} \xi_\kappa(u) &= \ln u + \ln_2 u + O\Big(\frac{\ln_2 u}{\ln u}\Big) \\ \sigma_j(u) &= u\Big\{1 + O\Big(\frac{1}{\ln u}\Big)\Big\} \end{aligned} \qquad (u \to \infty). \tag{4.97}$$

It then follows from theorem 1 of Smida (1991) or from the more general theorem 2 in Hildebrand–Tenenbaum (1993a) that we have

$$\varrho_\kappa(u) = \Big\{1 + O\Big(\frac{1}{u}\Big)\Big\}\frac{\mathrm{e}^{\gamma\kappa - u\xi_\kappa(u) + \sigma_0(u)}}{\sqrt{2\pi\sigma_2(u)}} \quad (u \to \infty), \tag{4.98}$$

where γ denotes Euler's constant.

As noted in the work of Hanrot, Tenenbaum & Wu (2007)—formula (4.12)—we have

$$\lambda_\kappa(u) = \Big\{1 + O\Big(\frac{1}{u}\Big)\Big\}\frac{\mathrm{e}^{-\gamma\kappa}\varrho_\kappa(u)}{\xi_\kappa(u)} \qquad (u > 0). \tag{4.99}$$

For a survey exposition of the work of Goldston, Pintz and Yıldırım, see in particular Soundararajan (2007).

Exercises

77. For $x \geqslant z \geqslant y \geqslant 1$, we denote by $\Psi_0(x,y,z)$ the number of integers $n \leqslant x$ having no prime factor in $]y,z]$.

(a) Use Eratosthenes' sieve to show that

$$\lim_{x\to\infty} x^{-1}\Psi_0(x,y,z) = \prod_{y<p\leqslant z}(1-1/p).$$

(b) Use Brun's pure sieve to show the existence of an absolute positive constant c such that, uniformly for $1 \leqslant y \leqslant z \leqslant \exp\{c\ln x/\ln_2 x\}$, we have

$$\Psi_0(x,y,z) \sim x\prod_{y<p\leqslant z}(1-1/p) \quad (x\to\infty). \tag{4.100}$$

(c) Extend this result using the fundamental lemma of combinatorial sieve theory.

(d) Assuming the prime number theorem, show that (4.100) does not hold uniformly for $1 \leqslant y < z \leqslant \sqrt{x}$.

78. *Primes and quasi-primes of the form* n^2+1.

(a) Show that the number $\varrho(p)$ of solutions to the equation

$$\xi^2 + 1 \equiv 0 \pmod p$$

equals 1 if $p = 2$, 2 if $p \equiv 1 \pmod 4$, and 0 if $p \equiv 3 \pmod 4$.

(b) Deduce that the equation $\xi^2+1 \equiv 0 \pmod d$ has $\varrho(d) := \prod_{p|d}\varrho(p)$ solutions for each squarefree integer d.

(c) Using the fundamental lemma of combinatorial sieve theory, show that the number $S(x)$ of primes $p \leqslant x$ of the form $p = n^2+1$ satisfies

$$S(x) \ll \sqrt{x}\prod_{p\leqslant x,\, p\equiv 1 \pmod 4}\Big(1-\frac{2}{p}\Big).$$

(d) Make the previous calculation more precise by assuming Dirichlet's theorem

$$\sum_{p\leqslant x,\, p\equiv 1 \pmod 4}\frac{1}{p} = \tfrac{1}{2}\ln_2 x + O(1).$$

(e) Under the same assumptions, show there is a positive absolute constant B such that

$$\operatorname{card}\{n \leqslant \sqrt{x} : n^2+1 \in Q(B,x)\} \asymp \sqrt{x}/\ln x$$

where $Q(B,x) := \{m \leqslant x : p|m \Rightarrow p > x^{1/B}\}$.[6]

[6] The elements of this set are usually referred to as "quasi-primes"—see Halberstam & Richert (1974), §2.8. In particular, a quasi-prime only has a bounded number of prime factors. Moreover, $|Q(B,x)| \asymp \pi(x)$.

79. *Almost squares.*

(a) Let p be be a prime > 2. Show that the kernel of the endomorphism of $(\mathbb{Z}/p\mathbb{Z})^* : x \mapsto x^2$ is $\{\pm 1\}$ and deduce that the number of quadratic non-residues modulo p is $(p-1)/2$.

(b) By sieving the set $\mathcal{A} = \{n : n \leqslant x\}$ by the prime numbers $\leqslant \sqrt{x}$ for suitable congruence classes, show that the number S of integers $n \leqslant x$ such that n is a quadratic residue modulo p for all $p \leqslant \sqrt{x}$ satisfies $\lfloor \sqrt{x} \rfloor \leqslant S \leqslant C\sqrt{x}$ where C is an absolute constant to be calculated.

80. Using the large sieve, show that if $\mathcal{A} \subset]M, M+N]$ is a set of integers which, for each prime p, is excluded from $w(p)$ residue classes modulo p with $w(p) \ll 1$, then

$$|\mathcal{A}| \ll N \prod_{p \leqslant N} (1 - w(p)/p).$$

Recover this result using Selberg's sieve.

81. *Majorizing the number of representations in Goldbach's problem.*

Let N be an even integer and let $r(N)$ denote the number of representations of N in the form $N = p + q$ where p and q are primes. According to Goldbach's conjecture, we have $r(N) > 0$ for every even integer $N \geqslant 4$. Probabilistic considerations suggest that we actually have

$$r(N) \sim C_N N/(\ln N)^2 \qquad (N \to \infty)$$

with $C_N := 2\prod_{p>2}(1 - 1/(p-1)^2) \prod_{p|N, p>2}((p-1)/(p-2))$.

(a) For any multiplicative function $f \geqslant 0$ with $\mu(m) = 0 \Rightarrow f(m) = 0$, show that we have

$$\sum_{n \leqslant x} f(n) \leqslant \sum_{d|N} f(d) \sum_{m \leqslant x,\, (m,N)=1} f(m) \qquad (1 \leqslant x \leqslant N).$$

(b) Prove the existence of an absolute constant C such that

$$r(N) \leqslant C \prod_{p|N} \Big(1 + \frac{2}{p}\Big) \frac{N}{(\ln N)^2}.$$

(c) Let h be a multiplicative function satisfying the conditions

$$|h(p)| \leqslant 1 \ (p|N), \quad |h(p)| \leqslant p^{-\delta} \ (p \nmid N), \quad |h(p^\nu)| \ll 1 \ (\nu \geqslant 2),$$

where δ is a positive constant. Show that, for $0 \leqslant \alpha \leqslant 1/\ln_2 N$, we have

$$\sum_{d \geqslant 1} |h(d)| d^{\alpha - 1} \ll \ln_2 N.$$

(d) Applying the above result to a suitable function h, establish the inequality

$$r(N) \leqslant (8 + o(1))C_N \frac{N}{(\ln N)^2},$$

which sharpens that obtained in (b).[7]

82. *Generalized twin primes.*

For $k \in \mathbb{N}^*$, $x \geqslant 2$, let us denote by $J_{2k}(x)$ the number of primes p not exceeding x such that $p+2k$ is also prime. By applying Selberg's sieve to the set $\mathcal{A} := \{n(n+2k) : n \leqslant x\}$, show that, whenever $x \to \infty$ and uniformly in $k \geqslant 1$, we have

$$J_{2k}(x) \leqslant \{8C_2 + o(1)\}\frac{g(k)x}{(\ln x)^2}$$

where $C_2 := 2\prod_{p>2}\{1 - 1/(p-1)^2\}$, $g(k) := \prod_{\substack{p|k \\ p>2}}\{1 - 1/(p-1)\}^{-1}$.

Show how the Bombieri–Vinogradov theorem (Theorem II.8.34) allows us to divide this bound asymptotically by 2. Hint: apply Selberg's sieve to the set $\mathcal{A} := \{p + 2k : p \leqslant x,\ p \in \mathbb{P}\}$.

83. *Small gaps between primes.*

Let $\{p_n\}_{n=1}^{\infty}$ be the sequence of prime numbers and set $d_n := p_{n+1} - p_n$ $(n \geqslant 2)$. We intend to show here that $\alpha := \liminf_n d_n/\ln n < 1$.[8]

For $x \geqslant 2$, $0 < \delta < 1$, we put $I_x := \{n \in \mathbb{N}^* : x < p_n < p_{n+1} \leqslant 2x\}$, and

$$N_x(\delta) := \sum_{\substack{n \in I_x \\ 1-\delta < d_n/\ln x \leqslant 1+\delta}} 1, \quad S(x) := \sum_{n \in I_x} d_n.$$

(a) Assuming $\alpha > 1 - \delta$, show that, as x tends to infinity, we have

$$\{1 + \delta + o(1)\}x - 2\delta N_x(\delta)\ln x \leqslant S(x) \leqslant x.$$

(b) Let g be the arithmetic function defined in Exercise 82. By a convolution argument, show that there is a constant $A > 0$ such that

$$\sum_{k \leqslant y} g(k) = \{A + o(1)\}y \qquad (y \to \infty).$$

(c) Show that, for large x, we have $N_x(\delta) < B\delta x/\ln x$ where B is an absolute constant.

(d) Conclude and provide a numerical upper bound for α.

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[7] This estimate is thus asymptotically equal to eight times the conjectured value for $r(N)$. Employing the Bombieri–Vinogradov theorem—cf. Theorem II.8.34—Bombieri & Davenport (1966) obtain an estimate where the factor 8 is replaced by 4: see Halberstam & Richert—1974, th. 3.11.

[8] This result is due to Erdős (1940). Goldston, Pintz and Yıldırım (2006) established that $\alpha = 0$. Zhang (2014) proved that d_n is bounded on an infinite subsequence of integers n. See also Maynard (2014)

84. *Poisson summation formula.*

Let $f \in L^1(\mathbb{R})$.

(a) Show that the series $\varphi(t) := \sum_{n\in\mathbb{Z}} f(n+t)$ converges for almost all real numbers t.

(b) Assume additionally that f is continuous and of bounded variation on $\mathbb{R}$. Show that Poisson's formula

$$\sum_{n\in\mathbb{Z}} f(n+t) = \lim_{N\to\infty} \sum_{|n|\leqslant N} \widehat{f}(n)\mathrm{e}(tn)$$

holds for all $t \in [0,1[$.

85. *Integers coprime to q.*

For each integer $q \geqslant 1$, set $N_q(x) := |\{n \leqslant x : (n,q) = 1\}|$.

(a) Show that, for each $q \geqslant 1$, we have

$$\lim_{x\to\infty} N_q(x)/x = \varphi(q)/q.$$

(b) For fixed q, show that every integer $n \geqslant 1$ can be written uniquely as $n = hdt$, with $(h,q) = 1$, $d \mid q$, $\mu(d)^2 = 1$, $p \mid t \Rightarrow p \mid d$.

(c) Calculate $\sum_{p|t\Rightarrow p|d} 1/t$.

(d) For $q \geqslant 1$, $Q \geqslant 1$, set $L(q,Q) := \sum_{d|q,\, d\leqslant Q} \mu(d)^2/\varphi(d)$. Show that

$$\ln(Q+1) \leqslant L(q,Q) \prod_{p\leqslant Q,\, p\nmid q} (1-1/p)^{-1}.$$

Deduce that, for $x \geqslant Q \geqslant 1$, $q \geqslant 1$, $P^+(q) \leqslant x$, we have

$$\frac{1}{L(q,Q)} \leqslant \frac{\mathrm{e}^\gamma \varphi(q)\ln x}{q\ln(Q+1)}\Big\{1 + O\Big(\frac{1}{\ln x}\Big)\Big\}.$$

(e) Show that the upper bound

$$N_q(x) \leqslant \Big\{1 + O\Big(\frac{\ln_2 x}{\ln x}\Big)\Big\} 2\mathrm{e}^\gamma \frac{\varphi(q)}{q} x.$$

holds uniformly for $x \geqslant 3$, $q \geqslant 1$, $P^+(q) \leqslant x$.[9]

86. *Functions of the combinatorial sieve.*

(a) Show that for any arithmetic function χ we have

$$\mu\chi * \mathbf{1}(n) = \sum_{d|m/q} \mu(d)\{\chi(d) - \chi(qd)\} \qquad (n \geqslant 1)$$

with $q := P^-(n)$, $m := \prod_{p|n} p$.

(b) Let $\mathcal{P}$ be a set of prime numbers, and y be a real number; define $P(y) := \prod_{p\leqslant y,\, p\in\mathcal{P}} p$. Show that, if χ_1 and χ_2 satisfy the three properties

[9] This bound can be halved: see Exercise 276, p. 471.

(α) $\chi_i(d) = 0$ or $1 \qquad (d \mid P(y))$

(β) $\chi_i(d) = 1 \Rightarrow \chi_i(t) = 1 \qquad (t \mid d)$

(γ) $\chi_i(d)=1,\ \mu(d)=(-1)^{i-1} \Rightarrow \chi_i(pd)=1 \quad (pd|P(y),\ p < P^-(d))$,

then, for n dividing $P(y)$ and $q = P^-(n)$, we have

$$(-1)^i\mu(d)\{\chi_i(d) - \chi_i(qd)\} \geqslant 0 \qquad (d \mid m/q,\ i = 1, 2).$$

Deduce that

$$\mu\chi_1 * \mathbf{1}(n) \leqslant \delta(n) \leqslant \mu\chi_2 * \mathbf{1}(n) \qquad (n \mid P(y)). \tag{4.101}$$

(c) Show that the functions μ_1, μ_2 described after Theorem 4.3 satisfy relation (4.101).

Extremal orders

5.1. Introduction and definitions

Although the average order of an arithmetic function gives an intuitive idea of its overall behavior, it only reflects very crudely the variations of its values. In this chapter we intend to describe, by means of several significant examples, the standard methods leading to individual bounds that are *optimal*, in a sense to be made precise.

Let f be an arithmetic function. Denote by g and h some generic monotone "elementary" functions. From the perspective described above, the information one may gain about f is of one of the following types:

(a) $f(n) = O(g(n)) \qquad (n \geqslant n_0)$

(b) $f(n) = o(g(n)) \qquad (n \to \infty)$

(c) $f(n) = \{1 + o(1)\}g(n) \qquad (n \to \infty)$

(d) $f(n) = g(n) + O(h(n)) \qquad (n \geqslant n_0).$

In the last case it is naturally desirable that $h(n)$ be $o(g(n))$. The optimality of these asymptotic relations can be expressed by estimates of the following forms (where $g(n)$ is assumed to be ultimately strictly positive):

(α) $f(n) = \Omega_+(g(n)) \qquad$ i.e. $\limsup f(n)/g(n) > 0$

(β) $f(n) = \Omega_-(g(n)) \qquad$ i.e. $\liminf f(n)/g(n) < 0$

(γ) $f(n) = \Omega(g(n)) \qquad$ i.e. $\limsup |f(n)|/g(n) > 0.$

The symbol $f(n) = \Omega_\pm(g(n))$ indicates that the relations (α) and (β) hold simultaneously. The use of the letter Ω is traditional; no confusion should arise in practice with the function "number of prime factors".

The following definition introduces an appreciably finer concept than the previous ones for determining the individual order of an arithmetic function.

Definition 5.1. *Let f be an arithmetic function and let g be a non-decreasing function, which is ultimately positive. We say that g is a maximal order (resp. a minimal order) for f if we have*

$$\limsup_{n\to\infty} f(n)/g(n) = 1 \qquad (\textit{resp. } \liminf_{n\to\infty} f(n)/g(n) = 1).$$

Thus the knowledge of the two extremal orders for an arithmetic function f provides, up to a multiplicative factor tending to 1, *the best possible individual bounds for* $f(n)$. We shall investigate such information for the main additive and multiplicative functions presented in Chapter 2.

5.2. The function $\tau(n)$

Theorem 5.2. *Let f be a multiplicative function satisfying*

$$\lim_{p^\nu\to\infty} f(p^\nu) = 0.$$

Then we have $\lim_{n\to\infty} f(n) = 0$.

Corollary 5.3. *For any $\varepsilon > 0$ we have $\tau(n) = O_\varepsilon(n^\varepsilon)$.*

The corollary readily follows from applying Theorem 5.2 to $f(n) = \tau(n)/n^\varepsilon$. Indeed, for $q = p^\nu$, we have

$$f(q) = (\nu+1)/p^{\nu\varepsilon} \leqslant 2(1+\ln q)/q^\varepsilon \to 0 \quad (q\to\infty).$$

Proof of Theorem 5.2. Let q denote a generic prime power. By assumption, for each $\varepsilon > 0$, there exists an integer $Q = Q(\varepsilon)$ such that

$$q > Q \Rightarrow |f(q)| \leqslant \varepsilon.$$

Next, let us consider the following partition of the set of all possible q:

$$\begin{aligned} Q_1 &:= \{q : q \leqslant Q, |f(q)| \leqslant 1\},\\ Q_2 &:= \{q : q \leqslant Q, |f(q)| > 1\},\\ Q_3 &:= \{q : q > Q\}. \end{aligned}$$

Each integer n decomposes uniquely in the form

$$n = n_1n_2n_3 \quad \text{with} \quad n_i := \prod_{q\|n,\, q\in Q_i} q \qquad (i = 1,2,3).$$

It is clear that the n_i are mutually coprime, so

$$f(n) = f(n_1)f(n_2)f(n_3). \tag{5.1}$$

By definition of Q_1, we have $|f(n_1)| \leqslant 1$. Since Q_2 is included in the finite set of those q such that $|f(q)| > 1$, there is a constant A independent of ε such that

$$|f(n_2)| \leqslant A.$$

Moreover, $|f(n_3)| \leqslant \varepsilon$ except for a finite number of integers n. It then follows from (5.1) that

$$\limsup_{n\to\infty} |f(n)| \leqslant A\varepsilon,$$

and hence the desired conclusion. □

The following result shows that Corollary 5.3 is not far from being optimal: ε can only tend to 0 like $c/\ln_2 n$.

Theorem 5.4. *A maximal order for the function* $\ln \tau(n)$ *is* $(\ln 2)(\ln n)/\ln_2 n$.

Proof. For each $\varepsilon > 0$ we have to show that

$$\tau(n) < n^{(\ln 2+\varepsilon)/\ln_2 n} \tag{5.2}$$

for $n \geqslant n_0(\varepsilon)$, and

$$\tau(n) > n^{(\ln 2-\varepsilon)/\ln_2 n} \tag{5.3}$$

for infinitely many integers n.

The upper bound is easy. For each value of the parameter t, $2 \leqslant t \leqslant n$, we may write

$$\begin{aligned}\tau(n) = \prod_{p^\nu \| n} (\nu+1) &\leqslant \prod_{p^\nu \| n,\, p\leqslant t} (\nu+1) \prod_{p^\nu \| n,\, p>t} 2^\nu \\ &\leqslant \Big(1+\frac{\ln n}{\ln 2}\Big)^t \Big(\prod_{p^\nu \| n} p^\nu\Big)^{\ln 2/\ln t} \leqslant \exp\Big\{t(2+\ln_2 n) + \frac{\ln 2\cdot \ln n}{\ln t}\Big\}.\end{aligned}$$

Choosing $t = (\ln n)/(\ln_2 n)^3$, it follows that

$$\tau(n) \leqslant \exp\Big\{\frac{\ln 2\cdot \ln n}{\ln_2 n}\Big(1+O\Big(\frac{\ln_3 n}{\ln_2 n}\Big)\Big)\Big\} \tag{5.4}$$

from which we immediately deduce (5.2).

To establish the lower bound (5.3), it is necessary to find integers with "many" divisors. Natural candidates are the products

$$n_k := \prod_{1\leqslant j\leqslant k} p_j \qquad (k = 1, 2, \ldots) \tag{5.5}$$

where $\{p_j : j \geqslant 1\} = \{2, 3, 5, \ldots\}$ denotes the increasing sequence of all prime numbers. We have

$$\tau(n_k) = 2^k$$

and

$$\ln n_k = \vartheta(p_k) = \sum_{p \leqslant p_k} \ln p \leqslant \pi(p_k) \ln p_k = k \ln p_k.$$

This implies

$$\ln \tau(n_k) \geqslant (\ln 2 \cdot \ln n_k)/\ln p_k. \tag{5.6}$$

The desired inequality (5.3) follows from this by Chebyshev's estimate (2.29) in the form

$$\ln n_k = \vartheta(p_k) \geqslant A p_k \tag{5.7}$$

for a suitable positive constant A. Substituting (5.7) into (5.6), we get

$$\ln \tau(n_k) \geqslant \frac{\ln 2 \cdot \ln n_k}{\ln_2 n_k} \Big\{ 1 + O\Big(\frac{1}{\ln_2 n_k}\Big) \Big\}, \tag{5.8}$$

which implies (5.3). □

Remark. The problem of the minimal order of $\tau(n)$ is trivial: we have $\tau(n) \geqslant 2$ with equality whenever n is prime.

5.3. The functions $\omega(n)$ and $\Omega(n)$

By multiplicativity one easily sees that, for every integer $n \geqslant 1$,

$$2^{\omega(n)} \leqslant \tau(n) \leqslant 2^{\Omega(n)} \leqslant n. \tag{5.9}$$

Each of these inequalities is optimal: the two first are equalities if, and only if, n is squarefree; the third is whenever n is a power of 2.

This allows a rapid evaluation of the maximal orders of $\omega(n)$ and $\Omega(n)$—the minimal orders being trivial.

First, (5.4) and (5.9) imply

$$\omega(n) \leqslant \{1 + o(1)\} \ln n / \ln_2 n \quad (n \to \infty)$$

and relation (5.8) shows that this upper bound is optimal up to the factor $\{1 + o(1)\}$, since the n_k which occur in (5.8) are squarefree.

The situation is even simpler for $\Omega(n)$ since the last inequality of (5.9) is attained for infinitely many integers n. We can thus formulate the following result.

Theorem 5.5. (i) *A maximal order for the function* $\omega(n)$ *is* $\ln n / \ln_2 n$.

(ii) *A maximal order for the function* $\Omega(n)$ *is* $\ln n / \ln 2$.

5.4. Euler's function $\varphi(n)$

Euler's totient function is multiplicative, and—cf. § 2.7—its value at prime powers is given by

$$\varphi(p^\nu) = p^\nu(1 - p^{-1}). \tag{5.10}$$

We thus have

$$\varphi(n) \leqslant n \qquad (n \geqslant 1) \tag{5.11}$$

and, for all $\varepsilon > 0$,

$$\varphi(n) \geqslant n^{1-\varepsilon} \qquad (n \geqslant n_0(\varepsilon)). \tag{5.12}$$

The upper bound (5.11) is trivial; the lower bound (5.12) follows from Theorem 5.2 applied to $f(n) = n^{1-\varepsilon}/\varphi(n)$.

We make these bounds more precise in the following statement.

Theorem 5.6. *A maximal order for $\varphi(n)$ is n. A minimal order is*

$$\mathrm{e}^{-\gamma} n/\ln_2 n$$

where γ denotes Euler's constant.

Proof. The first assertion follows immediately from (5.10). In order to establish the second, we note that we have

$$\varphi(n) = n \prod_{p \mid n} \Big(1 - \frac{1}{p}\Big) \geqslant n \prod_{p \leqslant q} \Big(1 - \frac{1}{p}\Big) \tag{5.13}$$

for any integer q such that $\pi(q) \geqslant \omega(n)$. By Chebyshev's lower estimate (Theorem 1.3) this condition is satisfied as soon as

$$q/\ln q \geqslant (\omega(n)/\ln 2) + 4,$$

and thus, in particular, for $q = \lfloor A\omega(n) \ln\{2 + \omega(n)\} \rfloor$ where A is a suitable positive constant. By Theorem 5.5(i) we have in this case

$$q \ll \ln n$$

whence, by (5.13), applying Mertens' formula,

$$\varphi(n) \geqslant n \frac{\mathrm{e}^{-\gamma}}{\ln q}\Big\{1 + O\Big(\frac{1}{\ln q}\Big)\Big\} \geqslant n \frac{\mathrm{e}^{-\gamma}}{\ln_2 n}\Big\{1 + O\Big(\frac{1}{\ln_2 n}\Big)\Big\}. \tag{5.14}$$

To complete the proof of Theorem 5.6, it remains to show that this lower bound is asymptotically attained on a suitable subsequence. Now for n_k defined by (5.5) we have

$$\frac{\varphi(n_k)}{n_k} = \prod_{p \leqslant p_k} (1 - p^{-1}) = \frac{\mathrm{e}^{-\gamma}}{\ln p_k}\Big\{1 + O\Big(\frac{1}{\ln p_k}\Big)\Big\} = \frac{\mathrm{e}^{-\gamma}}{\ln_2 n_k}\Big\{1 + O\Big(\frac{1}{\ln_2 n_k}\Big)\Big\},$$

by Chebyshev's estimate (2.29) in the form

$$\ln_2 n_k = \ln \vartheta(p_k) = \ln p_k + O(1).$$

This concludes the proof. □

5.5. The functions $\sigma_\kappa(n), \kappa > 0$

In § I.2.2 we defined the arithmetic function

$$\sigma_\kappa(n) = \sum_{d|n} d^\kappa \qquad (\kappa \in \mathbb{R}).$$

This is a multiplicative function ($\sigma_\kappa = j^\kappa * \mathbf{1}$) and the identity $\sigma_\kappa(n) = n^\kappa \sigma_{-\kappa}(n)$ shows that it is sufficient to examine extremal orders for positive values of the parameter—the case $\kappa = 0$ having already been handled, since $\sigma_0 = \tau$.

We have

$$\sigma_\kappa(p^\nu) = \frac{p^{(\nu+1)\kappa} - 1}{p^\kappa - 1} = p^{\nu\kappa} \frac{1 - p^{-(\nu+1)\kappa}}{1 - p^{-\kappa}}, \tag{5.15}$$

so that

$$\sigma_\kappa(n) \geqslant n^\kappa \qquad (n \geqslant 1) \tag{5.16}$$

and, for all $\varepsilon > 0$,

$$\sigma_\kappa(n) \leqslant n^{\kappa(1+\varepsilon)} \qquad (\kappa > 0,\ n \geqslant n_0(\varepsilon)), \tag{5.17}$$

by Theorem 5.2.

When $\kappa \geqslant 1$, the study of $\sigma_\kappa(n)$ can be undertaken in a parallel way to that of $\varphi(n)$, carried out in the previous section. The lower bound (5.16) trivially provides a minimal order for $\sigma_\kappa(n)$ and the upper bound (5.17) can easily be made more precise. Indeed, choosing q as in (5.13), we have

$$\frac{\sigma_\kappa(n)}{n^\kappa} \leqslant \prod_{p \leqslant q} \Big(1 - \frac{1}{p^\kappa}\Big)^{-1} =: \zeta(\kappa, q), \tag{5.18}$$

say. The product converges when $\kappa > 1$, and equals $\mathrm{e}^\gamma \ln q + O(1)$ when $\kappa = 1$. Furthermore, $\ln q \leqslant \ln_2 n + O(1)$. In particular, we can hence write

$$\sigma_\kappa(n) n^{-\kappa} \leqslant \begin{cases} \mathrm{e}^\gamma \ln_2 n + O(1) & \text{if } \kappa = 1, \\ \zeta(\kappa)\Big\{1 + O_\kappa\Big(\dfrac{1}{(\ln n)^\kappa}\Big)\Big\} & \text{if } \kappa > 1. \end{cases} \tag{5.19}$$

These upper bounds are asymptotically achieved for the sequence $\{m_k\}$ defined by

$$m_k := \Big(\prod_{1 \leqslant j \leqslant k} p_j\Big)^{\ell(k)} \qquad (k = 1, 2, \ldots)$$

where $p_1 = 2 < p_2 = 3 < \dots$ denotes the increasing sequence of prime numbers and where we have put $\ell(k) = \lfloor \ln k \rfloor$. Checking this last point is left to the reader.

When $0 < \kappa < 1$, the above considerations are still valid, but the asymptotic behavior of the factor $\zeta(\kappa, q)$ in (5.18) is much more complicated. By Abel summation, the prime number theorem implies that

$$\ln \zeta(\kappa, q) \sim q^{1-\kappa}/(1-\kappa) \ln q \tag{5.20}$$

but even the strongest form does not allow us to infer an elementary asymptotic formula for $\zeta(\kappa, q)$. Another intrinsic difficulty lies in the fact that, because of the fluctuations in the error term in the prime number theorem, the value of q in (5.18) is not determined with any great precision as a function of n. In the case when $\kappa \geqslant 1$, this does not have any important consequences since $\zeta(\kappa, q)$ is a slowly increasing function of $\ln q$. The situation is radically different when $0 < \kappa < 1$.

We can summarize the results obtained so far in the following theorem.

Theorem 5.7. *For $\kappa > 0$, a minimal order of $\sigma_\kappa(n)$ is n^κ. For $\kappa > 1$, a maximal order is $\zeta(\kappa) n^\kappa$. For $\kappa = 1$, a maximal order is $e^\gamma n \ln_2 n$. For $0 < \kappa < 1$, we have*

$$\sigma_\kappa(n) \leqslant n^\kappa \exp\left\{(1 + o(1)) \frac{(\ln n)^{1-\kappa}}{(1-\kappa) \ln_2 n}\right\}$$

and the opposite inequality is satisfied for infinitely many integers n.

Notes

§§ 5.1–5.5. The first systematic study of the maximal order of an arithmetic function is due to Ramanujan (1915), who not only set out to determine the large values of $\tau(n)$ but also to make explicit the structure of those numbers for which these large values are attained. Today, the study of maximal orders of arithmetic functions represents a lively area in the theory of numbers, with powerful methods and original techniques. An excellent survey, supplemented by an exhaustive bibliography, has been written by Nicolas (1988). The reader interested in this subject will find other examples of problems and conjectures in Nicolas (1974/75, 1978, 1983a), Erdős & Nicolas (1981a and b, 1989), Erdős & Tenenbaum (1989b), and Erdős & Sárközy (1994).

The questions connected with extremal orders are often highly non-trivial, particularly in the extent to which they involve the fine structure of integers. In a general way the problem of "large values" naturally induces that of the distribution of those integers at which they are attained. This aspect of the question is often difficult. Even in the case of the famous divisor function $\tau(n)$ and Ramanujan's highly composite numbers, the results attained are incomplete—cf. Nicolas (1988).

§ 5.5. For the asymptotic behavior of $\zeta(\kappa, q)$, see Lemma III.5.16.

Exercises

87. What are the extremal orders of the function "sum of the digits in base r"?

88. Write $\delta_r(n) := \operatorname{card}\{m_1, \dots, m_r : n = [m_1, \dots, m_r]\}$. Calculate $\delta_r * \mathbf{1}$ and show that δ_r is a multiplicative function. Determine the extremal orders of $\ln \delta_r$.

89. We intend here to recover a result of Erdős & Nicolas (1981b) about the extremal orders of the function $F(n) = \varphi(n) + \varphi(n+1)$.

(a) Show that $F(n) \leqslant 3n/2 \quad (n > 1)$.

(b) Denote by $p_1 = 2 < p_2 = 3 < \dots$ the increasing sequence of prime numbers, and put $N_k := 1 + \prod_{1 \leqslant j \leqslant k} p_j$. Show that $p | N_k \Rightarrow p > p_k$ and that $p | (N_k + 1) \Rightarrow p = 2$ or $p > p_k$. Show that $\ln N_k \asymp p_k \asymp k \ln k$. Deduce that, as $k \to \infty$, $\varphi(N_k) \sim N_k$ and $\varphi(N_k + 1) \sim \frac{1}{2} N_k$. What is the maximal order of $F(n)$?

(c) Show that for $n \to \infty$, we have $\varphi(n) \sim n \prod_{p | n,\, p \leqslant \ln n} (1 - 1/p)$.

(d) Deduce from the previous result that

$$F(n) \geqslant \{1 + o(1)\} 2\mathrm{e}^{-\gamma/2} n / \sqrt{\ln_2 n} \quad (n \to \infty). \tag{5.21}$$

[*One may use the inequality* $a + b \geqslant 2\sqrt{ab}$.]

(e) Show there is an integer $r = r(k)$ such that

$$\prod_{1 \leqslant j \leqslant r} \Big(1 - \frac{1}{p_j}\Big) \sim \prod_{r < j \leqslant k} \Big(1 - \frac{1}{p_j}\Big) \quad (k \to \infty).$$

(f) Let M_k be smallest solution of the congruence

$$M \equiv 0 \,(\mathrm{mod}\, p_1 \cdots p_r), \quad M \equiv -1 \,(\mathrm{mod}\, p_{r+1} \cdots p_k).$$

Show that

$$\max\Big(\frac{\varphi(M_k)}{M_k}, \frac{\varphi(M_k + 1)}{M_k + 1}\Big) \leqslant \big(1 + o(1)\big) \prod_{1 \leqslant j \leqslant r} (1 - 1/p_j) \quad (k \to \infty).$$

Deduce that the right-hand side of (5.21) is a minimal order for $F(n)$.

90. *Ramanujan's estimate for the maximal order of* $\tau(n)$.

(a) Let $\varepsilon > 0$. Show that the supremum $\sup_{n \geqslant 1} \tau(n)/n^\varepsilon$ is achieved for $n = N_\varepsilon := \prod_p p^{\nu_p}$, with $\nu_p := \lfloor 1/(p^\varepsilon - 1) \rfloor$.

(b) Writing $x_k := (1 + 1/k)^{1/\varepsilon}$ $(k \geqslant 1)$, $K := \nu_2 = \lfloor 1/(2^\varepsilon - 1) \rfloor$, show that

$$N_\varepsilon = \prod_{1 \leqslant k \leqslant K} \Big(\prod_{x_{k+1} < p \leqslant x_k} p \Big)^k.$$

(c) Using Chebyshev's functions

$$\vartheta(x) := \sum_{p\leqslant x} \ln p, \qquad \pi(x) := \sum_{p\leqslant x} 1 \qquad (x > 0),$$

show that

$$\ln N_\varepsilon = \sum_{1\leqslant k\leqslant K} \vartheta(x_k) = \vartheta(x_1) + O\big(x_1^{3/4}\big),$$
$$\ln \tau(N_\varepsilon) = \sum_{1\leqslant k\leqslant K} \pi(x_k)\ln(1+1/k) = (\ln 2)\pi(x_1) + O\big(x_1^{3/4}\big).$$

(d) Let $R(x)$ be an increasing function such that $\pi(x) - \operatorname{li}(x) \ll R(x)$ and $\vartheta(x) - x \ll R(x)$. We assume that $x^{4/5} \ll R(x) \ll x/(\ln x)^2$. Show that, for every integer $n \geqslant 1$, we have

$$\ln \tau(n) \leqslant (\ln 2)\operatorname{li}(x_1) - \varepsilon x_1 + \varepsilon \ln n + O(R(x_1)).$$

Selecting ε suitably, show that

$$\ln \tau(n) \leqslant (\ln 2)\operatorname{li}(\ln n) + O(R(\ln n)) \qquad (n \geqslant 3), \tag{5.22}$$

and, in particular, that

$$\ln \tau(n) \leqslant \{1 + O(1/\ln_2 n)\}(\ln 2)(\ln n)/\ln_2 n \qquad (n \geqslant 3).$$

(e) Assuming furthermore that $R(2x) \ll R(x)$ for $x \geqslant 1$, show that the inequality (5.22) is an equality for infinitely many integers n.

91. (a) Find a maximal order for the function $n \mapsto \ln \tau(n^2)$.

(b) Let $S := \{s : p|s \Rightarrow p^2|s\}$ be the set of squarefull integers. What is a maximal order for the function $s \mapsto \ln \tau(s)$ $(s \in S)$?

(c) Let $\omega_1(n)$ be the number of prime factors p of n such that $p^2 \nmid n$. Show that

$$\tau(n) \leqslant D(n)^{\lambda + o(1)} 2^{(1-\lambda)\omega_1(n)} \qquad (n \to \infty)$$

with $\lambda := (\ln 3)/\ln 4$ and $D(n) := \max_{m\leqslant n} \tau(m)$.

92. Let $\alpha > 1$. Set $F_\alpha(n) := \sum_{1\leqslant i<\tau(n)} \big((d_{i+1}/d_i) - 1\big)^\alpha$, where $\{d_j\}_{j=1}^{\tau(n)}$ denotes the increasing sequence of divisors of n.

(a) Writing

$$\chi(n;u,v) := \begin{cases} 1 & (\text{ if } d \mid n \Rightarrow d \notin]u,v]) \\ 0 & (\text{ if } \exists d \mid n,\ u < d \leqslant v), \end{cases}$$

show that

$$F_\alpha(n) = \alpha \int_1^n \int_1^v \chi(n;u,v)(v-u)^{\alpha-2}(\alpha v - u)u^{-1-\alpha}\,du\,dv.$$

(b) We define a *decomposition* of the integer n to be any finite sequence $\{n_j : 1 \leqslant j \leqslant k\}$ such that $n_1 = 1$, $n_j | n_{j+1}$ $(1 \leqslant j < k)$, $n_k = n$. We then say that a sequence of real numbers $\{\varepsilon_j : 1 < j \leqslant k\}$ is *admissible* for $\{n_j\}$ if for each $j \in [2, k]$ and each $z \in \left]\sqrt{n_{j-1}}, \sqrt{n_j}\right]$ the interval $]z, (1+\varepsilon_j)z]$ contains at least one divisor of n_j.

Show that

$$F_\alpha(n) \leqslant \alpha \sum_{2 \leqslant j \leqslant k} (1+\varepsilon_j)\varepsilon_j^{\alpha-1} \ln(n_j/n_{j-1})$$

for every decomposition $\{n_j : 1 \leqslant j \leqslant k\}$ of n and each associated admissible sequence $\{\varepsilon_j : 1 < j \leqslant k\}$.[1]

93. We consider the arithmetic function $f(n) := \omega(n)\sigma(n)/n$.

(a) For $k \geqslant 1$, let $r = r(k) := \lfloor \ln k \rfloor$ and $n_k := \prod_{1 \leqslant j \leqslant r} p_j^r \prod_{r < j \leqslant k} p_j$, where p_j denotes the j-th prime number. Assuming the prime number theorem, show that $f(n_k) \sim \mathrm{e}^\gamma \ln n_k$ $(k \to \infty)$.

(b) Show that $\mathrm{e}^\gamma \ln n$ is a maximal order for $f(n)$.

(c) Application. Let $A(n) := \sum_{p^\nu \| n} \nu p$ denote the Alladi–Erdős function (1977, 1979). Write $A^*(n) := \sum_{d|n} A(d)/d$. Find an additive function $g(n)$ such that $A^*(n) = \sigma(n)\{\omega(n) + g(n)\}/n$. Show that $g(p^\nu) \ll 1/p$ and that $g(n) \ll \ln_3 n$. Deduce that $\mathrm{e}^\gamma \ln n$ is a maximal order for $A^*(n)$.[2]

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[1] This remark is the first step in the proof of Erdős's conjecture according to which $\liminf F_\alpha(n) < \infty$ for all $\alpha > 1$. See Vose (1984). Using this point and a rudimentary form of the saddle-point method, Tenenbaum (1987) establishes that $F_\alpha(n!)$, $F_\alpha([1, 2, \ldots, n])$, and $F_\alpha(\prod_{p \leqslant n} p)$ are uniformly bounded for each $\alpha > 1$.

[2] This result sharpens estimates of Sitaramaiah & Subbarao (1993).

The method of van der Corput

6.1. Introduction and prerequisites

In the 1920s, van der Corput developed a method of estimating trigonometric sums which has numerous applications in the theory of numbers. This theory is typically relevant to Dirichlet's divisor problem and to the circle problem—i.e., the evaluation of the number of pairs $(m, n) \in (\mathbb{Z}^+)^2$ such that $m^2 + n^2 \leqslant x$. It is more generally applicable to counting points with integral coordinates within a given planar contour—see Exercise 98.

As early as 1922, van der Corput showed that the remainder term in the Dirichlet divisor problem has order $\ll_\varepsilon x^{33/100+\varepsilon}$. This problem, and others of a similar nature, have subsequently given rise to a prolific literature. A large part of it rests on the ideas of van der Corput—see also the Notes on § 3.2.

In this chapter, our aim consists essentially in describing the principles of this method in its simplest form, leading to Voronoï's theorem, to the estimate $O(x^{1/3})$ for the remainder in the circle problem, and to the exponent $1/6$ in the bound for the zeta function on the critical line (cf. § II.3.4). The reader interested in a deeper study of the method is referred to the work of Titchmarsh (1951), to the original articles of van der Corput (1922–1937), to the works of Kolesnik (1981, 1985), as well as to the developments of Bombieri & Iwaniec (1986), Iwaniec & Mozzochi (1988), Huxley & Watt (1988), Watt (1989), Huxley (1990, 1993a,b, 2005), and Huxley & Kolesnik (1991). The monograph of Graham & Kolesnik (1991) provides a detailed

introduction to the state of the art. That of Huxley (1996), quite thorough, also considers applications.

We shall make use of the Poisson summation formula, which undoubtedly constitutes the most natural approach to the study of trigonometric sums. We define the Fourier transform by

$$\widehat{f}(\vartheta) := \int_{-\infty}^{+\infty} f(t)\mathrm{e}(-\vartheta t)\,\mathrm{d}t \qquad (f \in L^1(\mathbb{R})). \tag{6.1}$$

The following statement provides a precise framework for our usage of Poisson's formula.

Theorem 6.1. *Let $f \in L^1(\mathbb{R})$. Assume that the series*

$$\varphi(t) = \sum_{n\in\mathbb{Z}} f(n+t) \tag{6.2}$$

converges for all t, and that its sum defines a function continuous at 0 and of bounded variation on $[0,1]$. Then we have

$$\lim_{N\to\infty} \sum_{|\nu|\leqslant N} \widehat{f}(\nu) = \sum_{n\in\mathbb{Z}} f(n). \tag{6.3}$$

Proof. Jordan's theorem on Fourier series (see, for example, Titchmarsh (1939), p. 406) asserts that any periodic function of bounded variation over a period is the sum of its Fourier series at each point of continuity. This implies (6.3) since the Fourier coefficient of order ν of φ is $\widehat{f}(\nu)$. □

6.2. Trigonometric integrals

We collect here two results for bounding trigonometric integrals which will later be useful.

Theorem 6.2. *Let $f \in \mathcal{C}^1(]a,b[)$ be such that $f'(t)$ is monotonic, of constant sign on $]a,b[$, and such that $m := \inf_{a<t<b} |f'(t)| > 0$. Then we have*

$$\Big|\int_a^b \mathrm{e}(f(t))\,\mathrm{d}t\Big| \leqslant \frac{1}{\pi m}. \tag{6.4}$$

Proof. Assuming without loss of generality that f' is non-increasing on $]a,b[$, we can write

$$\begin{aligned}\Big|2\pi\int_a^b \mathrm{e}(f(t))\,\mathrm{d}t\Big| &= \Big|\int_a^b \frac{\mathrm{d}\{\mathrm{e}(f(t))\}}{f'(t)}\Big| = \Big|\Big[\frac{\mathrm{e}(f(t))}{f'(t)}\Big]_a^b - \int_a^b \mathrm{e}(f(t))\,\mathrm{d}\Big\{\frac{1}{f'(t)}\Big\}\Big| \\ &\leqslant \frac{1}{|f'(a)|} + \frac{1}{|f'(b)|} + \frac{1}{|f'(b)|} - \frac{1}{|f'(a)|} \leqslant \frac{2}{m}.\end{aligned}$$

□

Theorem 6.3. *Let $f \in \mathcal{C}^2(]a,b[)$ be such that $f''(t)$ has constant sign on $]a,b[$ and set $r := \inf_{a<t<b} |f''(t)| > 0$. Then we have*

$$\left| \int_a^b \mathrm{e}(f(t))\,\mathrm{d}t \right| \leqslant \frac{4}{\sqrt{\pi r}}. \tag{6.5}$$

Proof. Suppose, for example, that $f''(t) \leqslant -r < 0$ for $a < t < b$. Then $f'(t)$ vanishes at most once on $]a,b[$, say at $t = c$. (If $f'(t)$ does not vanish the argument is similar, but simpler.) Now we can write

$$I := \int_a^b \mathrm{e}(f(t))\,\mathrm{d}t = \int_a^{c-\delta} + \int_{c-\delta}^{c+\delta} + \int_{c+\delta}^b = I_1 + I_2 + I_3,$$

say, where the positive parameter δ satisfies $a + \delta \leqslant c \leqslant b - \delta$. We have

$$|f'(t)| = \left| \int_c^t f''(v)\,\mathrm{d}v \right| \geqslant r|t-c| \geqslant r\delta$$

when t is in $[a, c-\delta] \cup [c+\delta, b]$. By Theorem 6.2, it follows that

$$|I_1| + |I_3| \leqslant 2/\pi r\delta.$$

Since, trivially, $|I_2| \leqslant 2\delta$, we get

$$|I| \leqslant 2\delta + 2/\pi r\delta,$$

whence the required result, by selecting $\delta = \sqrt{1/\pi r}$. It is clear that this upper bound remains valid if, with this choice of δ, we have either $c < a + \delta$ or $c > b - \delta$. □

6.3. Trigonometric sums

The original approach of van der Corput rests on explicit approximation of a trigonometric sum by an integral.

Theorem 6.4. *Let $f \in \mathcal{C}^1([a,b])$ be such that $f'(t)$ is monotone on $]a,b[$. Set*

$$\alpha := \inf_{a<t<b} f'(t), \quad \beta := \sup_{a<t<b} f'(t).$$

Then, for each $\varepsilon > 0$, we have

$$\sum_{a<n\leqslant b} \mathrm{e}(f(n)) = \sum_{\alpha-\varepsilon<\nu<\beta+\varepsilon} \int_a^b \mathrm{e}(f(t) - \nu t)\,\mathrm{d}t + O_\varepsilon\big(\ln(\beta - \alpha + 2)\big). \tag{6.6}$$

Proof. The number $\varepsilon > 0$ being fixed, we can assume that

$$-1 \leqslant \alpha - \varepsilon < 0. \tag{6.7}$$

Indeed, formula (6.6) is invariant on replacing $f(t)$ by $f(t) + kt$, for any $k \in \mathbb{Z}$. We can equally well restrict ourselves to the case when a and b are of the form $m + \frac{1}{2}$ with $m \in \mathbb{Z}$: indeed the error involved is $O(1)$ on the left-hand side, and $\ll \ln(\beta - \alpha + 2)$ on the right. This last estimate follows

easily from interchanging summation and integration, using the trivial upper bound $\sum_\nu \mathrm{e}(-\nu t) \ll \min(\beta - \alpha + 2, 1/\|t\|)$, where $\|t\|$ denotes the distance from t to the set of integers. Finally, we suppose that f' is non-increasing on $]a, b[$. Let us set

$$F(t) := \begin{cases} \mathrm{e}(f(t)) & \text{if } a < t \leqslant b, \\ 0 & \text{otherwise.} \end{cases} \tag{6.8}$$

It is immediate that $\varphi(t) := \sum_{n\in\mathbb{Z}} F(n+t)$ is continuous at 0 (because $a, b \notin \mathbb{Z}$) and has bounded variation on $[0,1]$. The Poisson summation formula (6.3) then implies

$$\sum_{a<n\leqslant b} \mathrm{e}(f(n)) = \sum_{|\nu|\leqslant N} \widehat{F}(\nu) + o(1) \quad (N \to \infty)$$

with

$$\widehat{F}(\nu) = \int_a^b \mathrm{e}(f(t) - \nu t)\,\mathrm{d}t.$$

Taking account of (6.7), it remains to show we have

$$\sum_{\substack{|\nu|\leqslant N \\ \nu\notin[0,\beta+\varepsilon[}} \widehat{F}(\nu) \ll_\varepsilon \ln(\beta+2). \tag{6.9}$$

Now

$$2\pi i\widehat{F}(\nu) = \int_a^b \frac{\mathrm{d}\{\mathrm{e}(f(t)-\nu t)\}}{(f'(t)-\nu)} = \left[\frac{\mathrm{e}(f(t)-\nu t)}{f'(t)-\nu}\right]_a^b - \int_a^b \mathrm{e}(f(t)-\nu t)\mathrm{d}\left\{\frac{1}{f'(t)-\nu}\right\}$$
$$= (-1)^\nu \frac{\mathrm{e}(f(b))}{\alpha-\nu} + (-1)^{\nu+1}\frac{\mathrm{e}(f(a))}{\beta-\nu} + O\Big(\frac{1}{\alpha-\nu} - \frac{1}{\beta-\nu}\Big).$$

The contribution to (6.9) of the main term is $O_\varepsilon(1)$. That of the error term is

$$\ll \sum_{\nu\notin[0,\beta+\varepsilon[} \frac{\beta+1}{\nu(\nu-\beta)} = (\beta+1)\sum_{\nu\geqslant 1}\frac{1}{\nu(\nu+\beta)} + (\beta+1)\sum_{\nu>\beta+\varepsilon}\frac{1}{\nu(\nu-\beta)}$$
$$\ll_\varepsilon \sum_{1\leqslant\nu\leqslant\beta+1}\frac{1}{\nu} + \sum_{\beta+\varepsilon<\nu\leqslant 2\beta}\frac{1}{\nu-\beta} + 2(\beta+1)\sum_{\nu>2\beta}\frac{1}{\nu^2} \ll_\varepsilon \ln(\beta+2).$$

This completes the proof. □

When $\|f'(t)\| \geqslant \vartheta$ on $[a,b]$, the sum in (6.6) is restricted to a single value of ν. We thus obtain

$$\sum_{a<n\leqslant b} \mathrm{e}\big(f(n)\big) \ll 1/\vartheta. \tag{6.10}$$

In practice, the condition $\min \|f'(t)\| > 0$ is not often met. The following result, which follows easily from Theorems 6.3 and 6.4, allows us to relax it at the cost of an assumption on the second derivative.

Theorem 6.5 (van der Corput's inequality). *Let $a, b \in \mathbb{R}$ be such that $a < b$, and let $f \in \mathcal{C}^2(]a, b])$ be such that*

$$|f''(t)| \asymp \lambda > 0 \qquad (a < t < b).$$

Then we have

$$\sum_{a<n\leqslant b} \mathrm{e}\big(f(n)\big) \ll (b-a+1)\lambda^{1/2} + \lambda^{-1/2}. \tag{6.11}$$

Proof. We can assume $\lambda \leqslant 1$, since otherwise (6.11) it is trivially satisfied. With the notation of Theorem 6.4, the left-hand side of (6.11) is

$$\begin{aligned} &\ll (\beta-\alpha+1)\max_{\nu}\Big|\int_a^b \mathrm{e}\big(f(t)-\nu t\big)\,\mathrm{d}t\Big| + \ln(\beta-\alpha+2) \\ &\ll (\beta-\alpha+1)\lambda^{-1/2} + \ln(\beta-\alpha+2), \end{aligned}$$

where the second upper bound follows from Theorem 6.3. Now we have

$$\beta - \alpha = \Big|\int_a^b f''(t)\,\mathrm{d}t\Big| \asymp \lambda(b-a).$$

The previous bound is therefore

$$\lambda^{1/2}(b-a) + \lambda^{-1/2} + 1 + \lambda(b-a) \ll \lambda^{1/2}(b-a) + \lambda^{-1/2}. \qquad \square$$

Formula (6.6), which is the basis of van der Corput's method, turns out to be essential for improving on the estimate (6.11)—see the Notes. However, if one is only interested in (6.11), a simpler method may be employed. It rests on a direct proof of the upper bound (6.10), in the form of an optimal inequality.

Lemma 6.6 (Kusmin–Landau). *Let $\{x_n\}_{n=1}^N$ be a finite sequence and let $\vartheta \in]0, \frac{1}{2}]$ be such that*

$$\vartheta < x_2 - x_1 \leqslant \cdots \leqslant x_N - x_{N-1} \leqslant 1 - \vartheta.$$

Then we have

$$\Big|\sum_{1\leqslant n\leqslant N} \mathrm{e}(x_n)\Big| \leqslant \cot(\pi\vartheta/2) \leqslant 2/(\pi\vartheta). \tag{6.12}$$

Proof. For $1 \leqslant n < N$, let us set $y_n := x_{n+1} - x_n$ and

$$c_n := \mathrm{e}(x_n)/\{\mathrm{e}(x_n) - \mathrm{e}(x_{n+1})\} = 1/\{1 - \mathrm{e}(y_n)\} = \tfrac{1}{2}\{1 + i\cot\pi y_n\}.$$

We note at the outset that

$$|c_n| = |1 - c_n| \leqslant \frac{1}{2|\sin\pi y_n|} \leqslant \frac{1}{2\sin\pi\vartheta}.$$

We have

$$\begin{aligned}\sum_{1\leqslant n\leqslant N} \mathrm{e}(x_n) &= \sum_{1\leqslant n<N} c_n\{\mathrm{e}(x_n)-\mathrm{e}(x_{n+1})\}+\mathrm{e}(x_N)\\ &= \sum_{1\leqslant n<N} c_n\mathrm{e}(x_n)-\sum_{2\leqslant n\leqslant N} c_{n-1}\mathrm{e}(x_n)+\mathrm{e}(x_N)\\ &= \sum_{1<n<N}\{c_n-c_{n-1}\}\mathrm{e}(x_n)+c_1\mathrm{e}(x_1)+\{1-c_{N-1}\}\mathrm{e}(x_N).\end{aligned}$$

Observing that $|c_n - c_{n-1}| = \frac{1}{2}\cot(\pi y_{n-1}) - \frac{1}{2}\cot(\pi y_n)$ since y_n is a non-decreasing function of n, it follows that

$$\begin{aligned}\Big|\sum_{1\leqslant n\leqslant N}\mathrm{e}(x_n)\Big| &\leqslant \tfrac{1}{2}\cot(\pi y_1)+\tfrac{1}{2}\cot(\pi y_{N-1})+|c_1|+|1-c_{N-1}|\\ &\leqslant \cot(\pi\vartheta)+\frac{1}{\sin\pi\vartheta}=\frac{1+\cos\pi\vartheta}{\sin\pi\vartheta}=\cot(\pi\vartheta/2).\end{aligned}$$

The second inequality in (6.12) follows from the fact that $\cot x \leqslant 1/x$ for $0 < x < \pi/2$, which results, for example, from the inequality

$$x\cos x - \sin x \leqslant 0 \qquad (0\leqslant x\leqslant \pi/2). \qquad \square$$

Theorem 6.7 (Kusmin–Landau inequality). *Let I be a bounded interval in $\mathbb{R}$, and let $f \in \mathcal{C}^1(I,\mathbb{R})$ be a real function such that f' is monotonic on I and satisfies*

$$\min_{x\in I}\|f'(x)\| \geqslant \lambda > 0. \tag{6.13}$$

Then we have

$$\Big|\sum_{n\in I}\mathrm{e}\big(f(n)\big)\Big| \leqslant \frac{2}{\pi\lambda}.$$

Proof. Up to replacing $f(x)$ by $\pm f(x+a)$, we can assume that $I = [1, N]$ and that f' is non-decreasing. By the assumption on $\min\|f'(x)\|$, there is an integer k such that $k+\lambda \leqslant f'(x) \leqslant k+1-\lambda$ for all x in I. Up to replacing $f(x)$ by $f(x) - kx$, we may hence assume $\lambda \leqslant f'(x) \leqslant 1-\lambda$ for $x \in I$. By the mean value theorem, this implies that the sequence of differences $\{f(n+1)-f(n)\}_{n=1}^{N}$ is non-decreasing and has values in $[\lambda, 1-\lambda]$. We may thus apply Lemma 6.6. $\square$

As stated above, van der Corput's inequality may be deduced from the Kusmin–Landau theorem. Let us derive an effective version. For $N \in \mathbb{N}^*$, $a, b \in \mathbb{Z}$, $b - a = N$, $I :=]a, b]$, $f \in \mathcal{C}^2(I,\mathbb{R})$, $\lambda > 0$, and $C > 0$, such that $\lambda \leqslant |f''(x)| \leqslant C\lambda$ $(x \in I)$, we have

$$\Big|\sum_{n\in I}\mathrm{e}\big(f(n)\big)\Big| \leqslant 3CN\sqrt{\lambda}+6/\sqrt{\lambda}. \tag{6.14}$$

We may assume $\lambda < \frac{1}{9}$ since inequality (6.14) is otherwise trivial. We may also, up to exchanging f and $-f$, assume $f'' > 0$.

Let $[\alpha, \beta]$ be the largest interval with integral endpoints contained in $]f'(a), f'(b)]$, so that $]f'(a), f'(b)] \cap \mathbb{Z} \subset [\alpha, \beta + 1]$. Let $\delta \in]0, \frac{1}{2}]$ be a parameter to be chosen later. For $\nu \in [\alpha, \beta + 1] \cap \mathbb{Z}$, we denote by I_ν the set of real numbers x such that $\nu - \delta < f'(x) \leqslant \nu + \delta$ and by J_ν the set of real numbers x such that $\nu + \delta < f'(x) \leqslant \nu + 1 - \delta$. By the mean value theorem, we have $2\delta \geqslant \lambda |I_\nu|$ for each ν, and thus there are at most $1 + 2\delta/\lambda$ integers n in I_ν. The modulus of the overall contribution of the I_ν to the sum $\sum_{n \in I} \mathrm{e}\big(f(n)\big)$ is hence (trivially) bounded above by

$$(\beta - \alpha + 2)(1 + 2\delta/\lambda) \leqslant (CN\lambda + 2)(1 + 2\delta/\lambda).$$

Furthermore, Theorem 6.7 shows that the contribution of each J_ν cannot exceed $2/\pi\delta$ in modulus. Summing over ν, we therefore obtain that the left-hand side of (6.14) is

$$\leqslant (CN\lambda + 2)(1 + 2\delta/\lambda + 2/\pi\delta).$$

Select $\delta := \sqrt{\lambda/\pi} < \frac{1}{2}$ since $\lambda < \frac{1}{9}$. Then $1 + 2\delta/\lambda + 2/\pi\delta \leqslant 3/\sqrt{\lambda}$. The stated result follows.

Clearly, van der Corput's inequality is only useful provided f'' is small on I. When this is not the case, one may appeal to the following lemma, essentially due to Weyl (1916, 1921) and known as the Weyl–van der Corput inequality.

Lemma 6.8 (Weyl–van der Corput). *For all integers $N \geqslant 1$, $Q \geqslant 1$ and any complex sequence $\{z_n\}_{n=1}^N$, we have*

$$\text{(6.15)} \qquad \Big| \sum_{1 \leqslant n \leqslant N} z_n \Big|^2 \leqslant \Big(1 + \frac{N-1}{Q}\Big) \sum_{|q| < Q} \Big(1 - \frac{|q|}{Q}\Big) \sum_{1 \leqslant n,\, n+q \leqslant N} z_{n+q} \overline{z_n}.$$

Proof. Let us put $z_n = 0$ for $n \notin [1, N]$. With this convention, we have

$$Q \sum_{n \in \mathbb{Z}} z_n = \sum_{1 \leqslant q \leqslant Q} \sum_{n \in \mathbb{Z}} z_{n+q} = \sum_{n \in \mathbb{Z}} \sum_{1 \leqslant q \leqslant Q} z_{n+q}.$$

Only integers n satisfying $1 - Q \leqslant n \leqslant N - 1$ may contribute to the last inner sum. Hence there are at most $N - 1 + Q$ such integers.

By the Cauchy-Schwarz inequality, we obtain

$$\begin{aligned} Q^2\left|\sum_{n\in\mathbb{Z}} z_n\right|^2 &\leqslant (N-1+Q)\sum_{n\in\mathbb{Z}}\left|\sum_{1\leqslant q\leqslant Q} z_{n+q}\right|^2 \\ &\leqslant (N-1+Q)\sum_{1\leqslant q_1\leqslant Q}\sum_{1\leqslant q_2\leqslant Q}\sum_{n\in\mathbb{Z}} z_{n+q_1}\overline{z_{n+q_2}} \\ &\leqslant (N-1+Q)\sum_{1\leqslant q_1\leqslant Q}\sum_{1\leqslant q_2\leqslant Q}\sum_{m\in\mathbb{Z}} z_{m+q_1-q_2}\overline{z_m} \\ &\leqslant (N-1+Q)\sum_{-Q\leqslant q\leqslant Q} r(q)\sum_{m\in\mathbb{Z}} z_{m+q}\overline{z_m} \end{aligned}$$

where $r(q) = \big|\{(q_1,q_2) : 1\leqslant q_1\leqslant Q, 1\leqslant q_2\leqslant Q, q_1-q_2=q\}\big|$. We plainly have

$$r(-q) = r(q) = |\{q_1 : \max(1,q+1)\leqslant q_1\leqslant \min(Q,Q+q)\}| = Q-|q|.$$

This implies (6.15) on dividing by Q^2. □

For a function of class $\mathcal{C}^3$, we establish the following variant of Theorem 6.5—cf. Titchmarsh (1951), § 5.9.

Theorem 6.9. *Let $N\in\mathbb{N}^*$, $a,b\in\mathbb{Z}$, $b-a=N$, $I:=]a,b]$, $f\in\mathcal{C}^3(I,\mathbb{R})$, and $\lambda>0$, be such that*

$$|f'''(t)|\asymp\lambda>0 \qquad (t\in I).$$

Then we have

$$\sum_{n\in I}\mathrm{e}\big(f(n)\big)\ll N\lambda^{1/6}+N^{1/2}\lambda^{-1/6}. \tag{6.16}$$

Proof. Let us apply the Weyl–van der Corput inequality with $z_n := \mathrm{e}\big(f(n)\big)$ $(a<n\leqslant b)$. Considering the case $q=0$ separately, we obtain, under the additional assumption $1\leqslant Q\leqslant N$,

$$\Big|\sum_{n\in I}\mathrm{e}\big(f(n)\big)\Big| \leqslant \frac{2N}{\sqrt{Q}}+2\Big\{\frac{N}{Q}\sum_{1\leqslant q<Q}|S_q(f)|\Big\}^{1/2}, \tag{6.17}$$

where we have put $S_q(f) := \sum_{a<n\leqslant b-q}\mathrm{e}\big(g_q(n)\big)$ with $g_q(x) := f(x+q)-f(x)$. However

$$|g_q''(x)|\asymp q\lambda \qquad (a<x\leqslant b-q).$$

This follows immediately from the Taylor–Lagrange formula applied at order 1 to f''. Using Theorem 6.5 to majorize $S_q(f)$, we get

$$\sum_{a<n\leqslant b}\mathrm{e}\big(f(n)\big)\ll\frac{N}{\sqrt{Q}}+\Big\{\frac{N}{Q}\sum_{1\leqslant q<Q}\Big(N(q\lambda)^{1/2}+(q\lambda)^{-1/2}\Big)\Big\}^{1/2}.$$

This upper bound is

$$\ll NQ^{-1/2} + N(Q\lambda)^{1/4} + N^{1/2}(Q\lambda)^{-1/4}.$$

Provided that $1 \leqslant \lambda^{-1/3} \leqslant N$, we can select $Q = \lfloor \lambda^{-1/3} \rfloor$, so that the first two terms are of the same order of magnitude $N\lambda^{1/6}$. This yields (6.16). We verify easily that the estimate (6.16) remains trivially valid when $\lambda > 1$ or $\lambda < 1/N^3$. □

A suitable iteration provides the following result, the proof of which we omit.

Theorem 6.10 (van der Corput). *Let N and R be integers such that $N \geqslant 1$, $R \geqslant 2$, let $I \subset [N+1, 2N]$ be an interval and let $f \in \mathcal{C}^R(I, \mathbb{R})$. Assume furthermore that there exists a constant $F = F(I)$ such that*

$$FN^{-r} \ll |f^{(r)}(x)| \ll FN^{-r} \qquad (x \in I,\ 1 \leqslant r \leqslant R).$$

Then we have

$$\sum_{n \in I} \mathrm{e}(f(n)) \ll N\{F^u N^{-v} + F^{-1}\}.$$

with $u := 1/(2^R - 2)$, $v := R/(2^R - 2)$.

6.4. Application to Voronoï's theorem

We establish here that the error term in Dirichlet's divisor problem is $O_\varepsilon(x^{1/3+\varepsilon})$. The same technique provides an identical upper bound in the circle problem (cf. Exercise 99) and the estimate $\zeta(1/2 + it) \ll_\varepsilon t^{1/6+\varepsilon}$ (cf. Exercise 95 and § II.3.4).

Theorem 6.11 (Voronoï, 1903). *For $x \geqslant 2$, we have*

$$\sum_{n \leqslant x} \tau(n) = x\big(\ln x + 2\gamma - 1\big) + O\big(x^{1/3} \ln x\big). \tag{6.18}$$

Proof. Let us put $N := \lfloor \sqrt{x} \rfloor$. The hyperbola method—see formula (3.4)—shows that the left-hand side of (6.18) equals

$$2\sum_{d \leqslant N} \Big\lfloor \frac{x}{d} \Big\rfloor - N^2 = 2\sum_{d \leqslant N} \Big(\frac{x}{d} - B_1\Big(\frac{x}{d}\Big)\Big) - N - N^2$$

where $B_1(t) = \langle t \rangle - \frac{1}{2}$ denotes the first Bernoulli function. Estimating $\sum_{d \leqslant N} 1/d$ by Theorem 0.8, we may write the above expression in the form

$$P(x) - 2R(x)$$

with

$$P(x) := 2x\Big(\ln N + \gamma + \frac{1}{2N} + O\Big(\frac{1}{x}\Big)\Big) - N - N^2$$

and

$$R(x) := \sum_{d \leqslant N} B_1(x/d).$$

Putting $N = \sqrt{x} - \vartheta$, with $0 \leqslant \vartheta < 1$, and expanding $P(x)$ up to order 1 as a function of $\vartheta/\sqrt{x}$, we obtain

$$P(x) = x(\ln x + 2\gamma - 1) + O(1).$$

Thus formula (6.18) is equivalent to

$$R(x) \ll x^{1/3} \ln x. \tag{6.19}$$

To establish (6.19), van der Corput's technique consists essentially in expanding $B_1(x/d)$ as a Fourier series, interchanging summations, and estimating the sums in $\sin(2\pi jx/d)$ for fixed j and x, relying on the results of the previous section. The fact that the Fourier series of B_1 is not absolutely convergent induces a technical difficulty which we overcome by replacing $B_1(t)$ with

$$B(t) := \tfrac{1}{2} J \int_{-1/J}^{1/J} B_1(t+u)\,\mathrm{d}u$$

for "large" J. Since B is Lipschitz of order 1, its Fourier series is absolutely convergent. Explicitly, we have

$$B(t) = \sum_{j \geqslant 1} a_j \sin(2\pi jt) \tag{6.20}$$

with

$$a_j = -\frac{J}{2\pi^2 j^2} \sin\Big(\frac{2\pi j}{J}\Big) \qquad (j \geqslant 1). \tag{6.21}$$

Furthermore, the function

$$h(t) := |B(t) - B_1(t)|$$

is also Lipschitz. Indeed we have

$$h(t) = \tfrac{1}{2}(1 - J\|t\|)^+ \qquad (J > 1)$$

where $\|t\|$ denotes the distance from t to the set of integers. An easy calculation shows that we have

$$h(t) = \frac{1}{2J} + \sum_{j \geqslant 1} b_j \cos(2\pi jt) \tag{6.22}$$

with

$$b_j = \frac{J}{\pi^2 j^2} \sin^2\Big(\frac{\pi j}{J}\Big) \qquad (j \geqslant 1). \tag{6.23}$$

In particular, from (6.21) and (6.23) it follows that

$$|a_j| + |b_j| \ll \min(j, J)/j^2 \qquad (j \geqslant 1). \tag{6.24}$$

We are now in position to begin the final phase of the proof. For $M < T \leqslant 2M$, we set

$$R(x; M, T) := \sum_{M<d\leqslant T} B_1(x/d).$$

Obviously, we have

$$\big|R(x; M, T) - \sum_{M<d\leqslant T} B(x/d)\big| \leqslant \sum_{M<d\leqslant T} h(x/d),$$

so that we may write, by (6.20) and (6.22),

$$\begin{aligned} R(x; M, T) = &\sum_{j\geqslant 1} a_j \sum_{M<d\leqslant T} \sin(2\pi jx/d) \\ &+ O\Big(\frac{M}{J} + \sum_{j\geqslant 1} b_j \Big| \sum_{M<d\leqslant T} \cos(2\pi jx/d)\Big|\Big). \end{aligned} \tag{6.25}$$

Now it follows from Theorem 6.5 that, for any real number y,

$$\sum_{M<d\leqslant T} \mathrm{e}(y/d) \ll \Big(\frac{y}{M}\Big)^{1/2} + \Big(\frac{M^3}{y}\Big)^{1/2}. \tag{6.26}$$

Applying this bound with $y = jx$ and inserting it into (6.25), it follows that

$$\begin{aligned} R(x; M, T) &\ll \frac{M}{J} + \sum_{j\geqslant 1} (|a_j| + |b_j|)\Big(\Big(\frac{jx}{M}\Big)^{1/2} + \Big(\frac{M^3}{jx}\Big)^{1/2}\Big) \\ &\ll \frac{M}{J} + \Big(\frac{Jx}{M}\Big)^{1/2} + \Big(\frac{M^3}{x}\Big)^{1/2}, \end{aligned}$$

where the summation over j has been evaluated taking account of (6.24). For the optimal choice $J := Mx^{-1/3}$, we have

$$R(x; M, T) \ll x^{1/3} + \Big(\frac{M^3}{x}\Big)^{1/2}. \tag{6.27}$$

This bound is *a priori* valid only if $J > 1$, that is $M > x^{1/3}$. It is clear that it still holds, trivially, in the complementary case.

Let us set $r_0 := \lfloor \ln N/\ln 2 \rfloor - 1$. We have

$$\begin{aligned} R(x) &= \sum_{-1\leqslant r\leqslant r_0} R(x; 2^r, 2^{r+1}) + R(x; 2^{r_0+1}, N) \\ &\ll \sum_{-1\leqslant r\leqslant r_0} \Big(x^{1/3} + \frac{2^{3r/2}}{\sqrt{x}}\Big) + x^{1/3} \ll r_0 x^{1/3} + \frac{N^{3/2}}{\sqrt{x}} \ll x^{1/3} \ln x. \end{aligned}$$

This establishes (6.19) and completes the proof of Theorem 6.11. □

6.5. Equidistribution modulo 1

6.5.1. Definition, discrepancy, Weyl's criterion. One of the shining applications of bounds on exponential sums in number theory is the study of the distribution of sequences with values in the torus $\mathbb{T} := \mathbb{R}/\mathbb{Z}$, or, what amounts to the same thing, of fractional parts of real sequences.

Definition 6.12. *A real sequence $\{u_n\}_{n=1}^{\infty}$ is said to be equidistributed modulo 1 if, for all α, β such that $0 \leqslant \alpha \leqslant \beta < 1$, we have*

$$\frac{1}{N} \sum_{\substack{1 \leqslant n \leqslant N \\ \alpha < \langle u_n \rangle \leqslant \beta}} 1 = \beta - \alpha + o(1) \qquad (N \to \infty).$$

It is easy to show, from the compactness of the interval $[0,1]$, that a necessary and sufficient condition for the sequence $\{u_n\}_{n=1}^{\infty}$ to be equidistributed modulo 1 is that the *discrepancy*

$$D_N := \sup_I \left| \frac{1}{N} \sum_{1 \leqslant n \leqslant N} \mathbf{1}_I(\langle u_n \rangle) - |I| \right|,$$

where the supremum is taken over the set of subintervals I of $[0,1[$, tends to 0 as $N \to \infty$.

Theorem 6.13 (Weyl's criterion, 1916). *A real sequence $\{u_n\}_{n=1}^{\infty}$ is equidistributed modulo 1 if, and only if, one of the two following equivalent conditions is satisfied:*

(i) We have $\lim_{N\to\infty} \frac{1}{N} \sum_{1 \leqslant n \leqslant N} f(\langle u_n \rangle) = \int_0^1 f(x)\,dx$ for any function f which is Riemann integrable on $[0,1]$.

(ii) We have $\lim_{N\to\infty} \frac{1}{N} \sum_{1 \leqslant n \leqslant N} \mathrm{e}(hu_n) = 0$ for any integer $h \in \mathbb{Z}^*$.

Proof. If $\{u_n\}_{n=1}^{\infty}$ is equidistributed modulo 1, we have (i) for all characteristic functions of intervals, and thus for all step functions. But if f is Riemann integrable on $[0,1]$, there are, for each $\varepsilon > 0$, two step functions g_1 and g_2 such that $g_1 \leqslant f \leqslant g_2$ and $\int_0^1 (g_2 - g_1)\,dx \leqslant \varepsilon$. We deduce that

$$\limsup_{N\to\infty} \frac{1}{N} \sum_{1 \leqslant n \leqslant N} f(\langle u_n \rangle) \leqslant \limsup_{N\to\infty} \frac{1}{N} \sum_{1 \leqslant n \leqslant N} g_2(\langle u_n \rangle) = \int_0^1 g_2(x)\,dx$$
$$\leqslant \int_0^1 f(x)\,dx + \varepsilon$$

and similarly

$$\liminf_{N\to\infty} \frac{1}{N} \sum_{1\leqslant n\leqslant N} f\big(\langle u_n\rangle\big) \geqslant \liminf_{N\to\infty} \frac{1}{N} \sum_{1\leqslant n\leqslant N} g_1\big(\langle u_n\rangle\big) = \int_0^1 g_1(x)\,\mathrm{d}x$$
$$\geqslant \int_0^1 f(x)\,\mathrm{d}x - \varepsilon.$$

Letting ε tend to 0, we indeed obtain (i).

Next we observe that property (i) trivially implies (ii) since

$$\int_0^1 \mathrm{e}(hx)\,\mathrm{d}x = 0$$

for $h \in \mathbb{Z}^*$.

Finally, if (ii) holds, we deduce from the Weierstrass approximation theorem that (i) holds for each continuous 1-periodic function on $[0,1]$. But if I is an interval in $\mathbb{R}/\mathbb{Z}$, there exist 1-periodic functions φ_1 and φ_2 such that $\varphi_1 \leqslant \mathbf{1}_I \leqslant \varphi_2$ and $\int_0^1 \{\varphi_2(t) - \varphi_1(t)\}\,\mathrm{d}t \leqslant \varepsilon$. As above, we deduce that (i) holds for $\mathbf{1}_I$, and therefore that $\{u_n\}_{n=1}^{\infty}$ is equidistributed modulo 1. □

Remark. Clearly enough, the above theorem cannot be extended to Lebesgue integrable functions since $\{u_n : n \geqslant 1\}$ is a Lebesgue null set.

Examples. (a) It immediately follows from the Weyl criterion that the sequence $\{n\vartheta\}_{n=1}^{\infty}$ is equidistributed modulo 1 for any irrational number ϑ: the exponential sums in (ii), which may be computed explicitly since they correspond to geometric series, are bounded.

(b) Let us consider the sequence $\{\ln n\}_{n=1}^{\infty}$. The associated trigonometric sum may be written as

$$S_N = \sum_{n\leqslant N} n^{2\pi ih} = N^{2\pi ih} \sum_{n\leqslant N} \Big(\frac{n}{N}\Big)^{2\pi ih} \sim N^{1+2\pi ih} \int_0^1 x^{2\pi ih}\,\mathrm{d}x = \frac{N^{1+2\pi ih}}{1+2\pi ih},$$

from the theory of the Riemann integral. So S_N/N does not tend to 0, and as a consequence the sequence $\{\ln n\}_{n=1}^{\infty}$ fails to be equidistributed modulo 1. However, we note that it is dense modulo 1 since $\ln n \to \infty$ and $\ln(n+1) - \ln n \to 0$ as $n \to \infty$.

6.5.2. The Erdős–Turán inequality. The Weyl criterion does not provide any information on the discrepancy. Theorem 6.15 *infra*, essentially due to Erdős and Turán, furnishes an explicit connection between the quantities

$$\sigma_N(h) := \frac{1}{N} \sum_{1\leqslant n\leqslant N} \mathrm{e}(hu_n)$$

and the discrepancy D_N.

To clarify expectations, we start with an inequality in the opposite direction.

Proposition 6.14. *We have*

$$|\sigma_N(h)| \leqslant 4|h|D_N \qquad (h \in \mathbb{Z}^*).$$

Proof. Assume $\vartheta \in [0, 2\pi[$ is such that $|\sigma_N(h)| = \sigma_N(h)\mathrm{e}^{i\vartheta}$. Replacing u_n by $u_n + \vartheta/(2\pi h)$ if necessary, which does not alter D_N, we may assume that $\vartheta = 0$. Then let us put

$$Z_N(t) := \frac{1}{N} \sum_{\substack{1 \leqslant n \leqslant N \\ \langle u_n \rangle \leqslant t}} 1,$$

so that

$$\sup_{0 \leqslant t \leqslant 1} |Z_N(t) - t| \leqslant D_N.$$

We can then write

$$\begin{aligned}
|\sigma_N(h)| = \sigma_N(h) &= 1 - \frac{2\pi i h}{N} \sum_{1 \leqslant n \leqslant N} \int_{\langle u_n \rangle}^1 \mathrm{e}(ht)\,\mathrm{d}t \\
&= 1 - 2\pi i h \int_0^1 Z_N(t)\mathrm{e}(ht)\,\mathrm{d}t = 2\pi i h \int_0^1 \big\{t - Z_N(t)\big\}\mathrm{e}(ht)\,\mathrm{d}t \\
&= 2\pi h \int_0^1 \big\{Z_N(t) - t\big\} \sin(2\pi h t)\,\mathrm{d}t \leqslant 2\pi |h| \int_0^1 \big|t - Z_N(t)\big|\,|\sin(2\pi h t)|\,\mathrm{d}t \\
&\leqslant 2\pi |h| D_N \int_0^1 |\sin(2\pi h t)|\,\mathrm{d}t = 4|h|D_N.
\end{aligned}$$

□

Theorem 6.15 (Erdős–Turán, 1948; Rivat–Tenenbaum, 2005). *We have*

$$D_N \leqslant \frac{1}{H+1} + \frac{2}{3} \sum_{1 \leqslant h \leqslant H} \frac{|\sigma_N(h)|}{h} \qquad (H \geqslant 1).$$

We do not prove this result here, but offer in Exercise 107 a simple method leading to a slightly less precise bound of similar nature.

Notes

§§ 6.2–6.3. The setting for the proof of van der Corput's inequality via the Poisson summation formula follows that in chapters 4 and 5 of Titchmarsh (1951).

Van der Corput proved in 1921 the estimate $\ll 1/\vartheta$ in the Kusmin–Landau inequality. Kusmin (1927) established the inequality (6.12) with the bound $1/\vartheta$ and Landau (1928) obtained the optimal value $\cot(\pi\vartheta/2)$.

§ 6.3. Theorem 6.5 represents a relatively rough application of van der Corput's method, insofar as formula (6.6), i.e.,

$$\sum_n F(n) = \sum_{\alpha-\varepsilon<\nu<\beta+\varepsilon} \widehat{F}(\nu) + \text{remainder}, \tag{6.28}$$

is employed in the form

$$\sum_n F(n) \ll \sum_\nu |\widehat{F}(\nu)|.$$

Indeed, the integrals

$$\widehat{F}(\nu) = \int_a^b \mathrm{e}(f(t) - \nu t)\,\mathrm{d}t$$

show an oscillatory asymptotic behavior, which may be brought out by the stationary phase method, essentially due to Laplace. Denoting, in the assumptions of Theorem 6.5, by x_ν the unique solution, if it exists, of the equation

$$f'(x_\nu) = \nu,$$

we obtain (cf. Titchmarsh (1951), chap. 4)

$$\widehat{F}(\nu) = \mathrm{e}^{\pm i\pi/4}|f''(x_\nu)|^{-1/2}\mathrm{e}(f(x_\nu) - \nu x_\nu) + \text{remainder}$$

where the error term can be explicitly bounded as a function of parameters λ, μ such that

$$|f'(t)| \asymp \lambda, \quad |f'''(t)| \ll \mu \qquad (a < t \leqslant b).$$

Within an acceptable level of approximation, one may thus consider the right-hand side of (6.28) as a new trigonometric sum, weighted by the factors $|f''(x_\nu)|^{-1/2}$. A non-trivial treatment of this sum allows an improvement of the final estimate. Of course, a new application of the Poisson summation formula will only lead to recovering the initial sum, but we can always hope for a substantial gain by inserting a simple transformation, resting on the Cauchy–Schwarz inequality, such as that in Lemma 6.8, known as the Weyl–van der Corput transformation—cf. Weyl (1916, 1921) and van der Corput (1922). One can thus alternate the applications of this formula

(say, procedure A) and the Poisson summation formula (say, procedure B) according to the scheme

$$A^{r_1}BA^{r_2}\dots BA^{r_n} \qquad (r_i \geqslant 0,\ 1 \leqslant i \leqslant n).$$

Selecting the r_i then poses a difficult optimization problem. That subject was formalized by van der Corput in the 1920s then simplified by Phillips (1933). This is what one today calls the *theory of exponent pairs*—cf. Ivić (1985), chap. 2, and Graham & Kolesnik (1991).

A modern exposition, intended for specialists or students wanting to deepen their knowledge of the subject, of methods for estimating exponential sums and their applications, is given in the book of Huxley (1996).

Lemma 6.8 due to Weyl–van der Corput can be generalized to inequalities involving only correlations

$$\gamma_N(q) := \frac{1}{N} \sum_{1\leqslant n,\, n+q\leqslant N} z_{n+q}\overline{z_n}$$

for *certain* values of q: see for example Montgomery—1994, chap. 2, lemma 1. This type of extension can be used to strengthen the theorem of van der Corput (1931) according to which *a sequence* $\{u_n\}_{n=1}^{\infty}$ *is equidistributed modulo 1 if, and only if, the sequence* $\{u_{n+h} - u_n\}_{n=1}^{\infty}$ *is equidistributed for each* $h \in \mathbb{N}^*$. In this way we have derived, for example, that we may restrict, in this statement to those values of h which are squares, or of the form $p+1$, or indeed $p-1$, with $p \in \mathbb{P}$, while it is insufficient to only consider h in $\mathbb{P}$. These results were initially due to Kamae & Mendès France (1978).

§ 6.4. The theorem of van der Corput (Theorem 6.10) depends on an integer parameter R. In practice, the phase function f is often of the type $f(x) = Ax^{\alpha} + u(x)$, where $\alpha \in \mathbb{R}$—the case $\alpha = 0$ being considered in the form $f(x) = F\ln x + u(x)$—and where u is a regular function whose successive derivatives can be neglected relative to those of f. The choice of R is then made from those values of r for which $f^{(r)}$ is of a "small" order of magnitude. Thus, in the proof of Voronoï's theorem (Theorem 6.11), van der Corput's result is used with $R = 2$.

We shall see in Chapter II.3 (Corollary II.3.7) how the choice $R = 3$ can be used to obtain the fundamental bound for the Riemann zeta function on the critical line.

Another important application concerns the sequence $\{\gamma_n\}_{n=1}^{\infty}$ of zeros of $\zeta(s)$ on the critical line.[1] Let β be the infimum of the set of real numbers ξ such that $\gamma_{n+1} - \gamma_n \ll \gamma_n{}^{\xi}$. We know[2] that $\gamma_n \sim 2\pi n/\ln n$ and it is conjectured that $\beta = 0$. A refinement of Hardy's method described in the

[1] See § II.3.7.

[2] See the notes on § II.3.7, p. 251.

Notes to § II.4.3, p. 272, introduces some exponential sums relevant to Theorem 6.10 with $R = 4$. As Karatsuba (1981) showed, the bound $\beta \leqslant 5/32$ can be obtained in this way. This result has only been improved a little, in spite of considerable effort brought to bear on it, notably by Huxley (1996, th. 21.4.4).

Theorem 6.10 with $R = 5$ appears in the recent treatment of Piatetski–Shapiro's problem on the distribution of prime numbers of the form $\lfloor n^c \rfloor$ with $c \in \mathbb{R}^+ \setminus \mathbb{N}$: see Rivat & Sargos (2001), Rivat & Wu (2001).

Finally, we remark that arbitrarily large values of R in Theorem 6.10 lead to a non-trivial bound for $\zeta(1 + it)$; however, such results are largely overstridden by those provided by Vinogradov's method, developed during the 1930s.

§ 6.5. Erdős and Turán initially showed the existence of two absolute constants c_1 and c_2 such that

$$D_N \leqslant \frac{c_1}{H+1} + c_2 \sum_{1 \leqslant h \leqslant H} \frac{|\sigma_N(h)|}{h} \qquad (H \geqslant 1).$$

The numerical values have been improved subsequently. It is conjectured that one can choose $c_1 = 1$ and $c_2 = 2/\pi$. Rivat & Tenenbaum (2005) have shown that $c_1 = 1$ and $c_2 = 0.6528 < 2/3$ are acceptable. We have $2/\pi \approx 0.6366$.

Exercises

94. Define $\tau(n,\vartheta) := \sum_{d|n} d^{i\vartheta}$ $(\vartheta \in \mathbb{R})$. Making use of Theorem 6.4, show that[3]

$$\frac{1}{x}\Big|\sum_{n\leqslant x}\tau(n,\vartheta)\Big| \ll \ln(2+|\vartheta|) + \frac{1}{|\vartheta|} \qquad (\vartheta \neq 0,\ x > 1+|\vartheta|).$$

95. Show that we have $\sum_{n\leqslant t} n^{-(1/2)-it} \ll t^{1/6}\ln t$ $(t \geqslant 2)$. [*Use, in suitable regions, Theorems* 6.5 *and* 6.9.]

96. Find an upper bound for $\sum_{n\leqslant x}\exp\{in^\alpha\}$ for $0 < \alpha < 2$.

97. *Sums of fractional parts.*

(a) Show that

$$\langle x\rangle = \tfrac{1}{2} + \int_0^1 \{y - \mathbf{1}_{[\langle x\rangle,1]}(y)\}\,\mathrm{d}y \qquad (x \in \mathbb{R}).$$

(b) Let I be an interval of length $N \in \mathbb{N}^*$, let λ and C denote positive real numbers, and let $f \in \mathcal{C}^2(I)$ satisfy $\lambda \leqslant |f''(t)| \leqslant C\lambda$ for $t \in I$. Prove that

$$\sum_{n\in I}\langle f(n)\rangle = \tfrac{1}{2}N + O\big(N\lambda^{1/3} + \lambda^{-1/2}\big)$$

where the implicit constant may depend on C.

98. *Sums of integer parts.* Use the Euler–Maclaurin formula at order 0, and the second mean value theorem, to show that, if $I =]a,b]$ is an interval with integer endpoints and if f satisfies the assumptions of Exercise 97 on I with $\lambda \leqslant 1$, we have

$$\sum_{n\in I}\lfloor f(n)\rfloor = \int_I f(t)\,\mathrm{d}t - \tfrac{1}{2}N + \tfrac{1}{2}\,[f(t)]_I + O\big(N\lambda^{1/3} + \lambda^{-1/2}\big).$$

99. *The circle problem, or the mean value for the number of representations of sums of two squares.* By applying the result of Exercise 98 to the function $t \mapsto \sqrt{x-t^2}$ for suitably selected intervals, show that[4]

$$\sum_{\substack{m,n\in\mathbb{Z}\\ m^2+n^2\leqslant x}} 1 = \pi x + O(x^{1/3}) \qquad (x \geqslant 1).$$

[3] See also Theorems II.3.5 and II.3.9.

[4] The best known estimate for the error term is due to Huxley (2003): $\ll x^{131/416+\varepsilon}$.

100. Show that for all $\lambda \in \mathbb{R} \setminus \{0\}$ we have

$$\sum_{n \leqslant x} \exp\{i\lambda n \ln n\} = o(x) \qquad (x \to \infty).$$

101. Let f be a continuous and differentiable map from $[1, +\infty[$ into $\mathbb{R}^+$ such that $f'(x)$ is non-increasing, $f'(x) \to 0$ $(x \to +\infty)$, and $xf'(x) \to +\infty$ $(x \to +\infty)$.

(a) Making use of either Theorem 6.4, or of the Euler–Maclaurin formula at order 0, show that

$$\sum_{1 \leqslant n \leqslant N} \mathrm{e}\big(f(n)\big) = o(N) \qquad (x \to +\infty). \tag{6.29}$$

(b) Integrating $\int_1^N \mathrm{e}\big(f(t)\big)\,\mathrm{d}t$ by parts, show that (6.29) does not hold if $xf'(x)$ tends to a finite limit.

102. Show that a real sequence is equidistributed modulo 1 if, and only if, its discrepancy tends to 0.

103. Let $\vartheta \in [0, 1[$. For $x \in \mathbb{R}$, express $\langle x - \vartheta \rangle - \langle x \rangle + \vartheta$ as a function of $\langle x \rangle$. Deduce that a real sequence $\{u_n\}_{n=1}^{\infty}$ is equidistributed modulo 1 if, and only if, we have

$$\lim_{N \to \infty} \frac{1}{N} \sum_{n \leqslant N} B_1(u_n - \vartheta) = 0$$

for all $\vartheta \in [0, 1[$.

104. Let $\{u_n\}_{n=1}^{\infty}$ be the sequence

$$\tfrac{0}{1}, \tfrac{1}{1}, \tfrac{0}{2}, \tfrac{1}{2}, \tfrac{2}{2}, \tfrac{0}{3}, \tfrac{1}{3}, \tfrac{2}{3}, \tfrac{3}{3}, \ldots.$$

Show that $\{u_n\}_{n=1}^{\infty}$ is equidistributed modulo 1: (a) directly, (b) by applying the Weyl criterion.

105. (a) Let ϑ be an irrational number and let $\{x_n\}_{n=1}^{\infty}$ be a real sequence such that $x_{n+1} - x_n \to \vartheta$. Show that $\{x_n\}_{n=1}^{\infty}$ is equidistributed modulo 1.

(b) Show that the sequence $\{x_n\}_{n=1}^{\infty}$ defined by $x_n = \sqrt{2}n + \sqrt{n}\sin(n^{1/3})$ is equidistributed modulo 1.

106. *Fejér's criterion.* Let $\{x_n\}_{n=1}^{\infty}$ be a real sequence such that

$$x_{n+1} - x_n \searrow 0 \quad \text{and} \quad n(x_{n+1} - x_n) \to \infty$$

Using the Kusmin–Landau inequality (Theorem 6.7), show that $\{x_n\}_{n=1}^{\infty}$ is equidistributed modulo 1.

107. *Erdős–Turán inequality.* Denote by

$$F_H(t) := \sum_{|h|\leqslant H} \Big(1 - \frac{|h|}{H}\Big) \cos(2\pi ht) = \frac{1}{H}\Big(\frac{\sin \pi Ht}{\sin \pi t}\Big)^2$$

the Fejér kernel of order $H \in \mathbb{N}^*$. Define the Fourier coefficient of index $h \in \mathbb{Z}$ of a function $f \in L^1(\mathbb{T})$ by

$$c_h(f) := \int_{\mathbb{T}} \mathrm{e}(-hx) f(x)\, \mathrm{d}x,$$

with $\mathrm{e}(u) := \exp(2\pi iu)$. For $f, g \in L^1(\mathbb{T})$ define the convolution product

$$f * g(x) := \int_{\mathbb{T}} f(t) g(x-t)\, \mathrm{d}t.$$

Finally, denote by $\langle x \rangle$ the fractional part of a real number x.

(a) Let $f \in L^1(\mathbb{T})$ be a real-valued function satisfying, for a suitable integer $H \geqslant 4$,

$$c_h(f) = 0 \quad (|h| \leqslant H). \tag{6.30}$$

Show that $f * F_H = 0$. Deduce that, for all $x \in \mathbb{T}$, we have

$$f(x)(1-I) + \int_{|t|\leqslant 2/H} \{f(x-t\pm 2/H) - f(x)\} F_H(t)\, \mathrm{d}t + \int_{2/H<|t|\leqslant 1/2} f(x-t\pm 2/H) F_H(t)\, \mathrm{d}t = 0,$$

where $I := \int_{\|t\|>2/H} F_H(t)\, \mathrm{d}t$.

(b) Show that $I \leqslant \frac{1}{4}$.

(c) Assume that f is bounded on $\mathbb{T}$ and set

$$K := \sup_{\substack{0\leqslant y\leqslant 1/H \\ x\in\mathbb{T}}} \{f(x+y) - f(x)\}.$$

Show that, under the assumption (6.30), we have

$$\|f\|_\infty \leqslant 4K(1-I)/(1-2I) \leqslant 6K.$$

(d) Show that, without assuming (6.30), we have

$$\|f\|_\infty \leqslant 6K + 13 \sum_{|h|\leqslant H} |c_h(f)|.$$

(e) Let $\{u_n\}_{n=1}^N$ be a finite sequence of real numbers. Define

$$F(x) := \frac{1}{N} \sum_{1\leqslant n\leqslant N} \mathbf{1}_{[0,x]}(\langle u_n \rangle).$$

By applying the previous results to the function

$$f(x) := \langle x \rangle - \tfrac{1}{2} + c_0(F) - F(x),$$

show that we have, for every integer $H \geqslant 1$,

$$\sup_{I \subset \mathbb{T}} \Big| \frac{1}{N} \sum_{1 \leqslant n \leqslant N} \mathbf{1}_I(\langle u_n \rangle) - |I| \Big| \leqslant \frac{12}{H} + 9 \sum_{0 < h \leqslant H} \frac{|\sigma_N(h)|}{h},$$

where $\sigma_N(h) := (1/N) \sum_{1 \leqslant n \leqslant N} \mathrm{e}(hu_n)$.

108. Show that the validity of a general inequality of Erdős–Turán type

$$D_N \leqslant \frac{c_1}{H+1} + c_2 \sum_{1 \leqslant h \leqslant H} \frac{|\sigma_N(h)|}{h}$$

implies $c_1 \geqslant 1$, $c_2 \geqslant 2/\pi$. Hint: consider the special sequences $x_n := n/N$ and $x_n := n/(2N)$ $(1 \leqslant n \leqslant N)$.

109. Show that the sequence $\{\sqrt{n}(\ln n)^2\}_{n=1}^{\infty}$ is equidistributed modulo 1 and give an explicit upper bound for the discrepancy.

110. Show that, for all $\alpha \in]0,1[$, the sequence $x_n = n^\alpha$ is equidistributed modulo 1. Give an estimate for the discrepancy.

111. Show that, for $1 < \alpha < 2$, the sequence $\{n^\alpha\}_{n=1}^{\infty}$ is equidistributed modulo 1 and give an estimate for the discrepancy.

112. Let I be an interval containing N integers, and let $f \in \mathcal{C}^2(I, \mathbb{R})$, $\lambda > 0$, and $C > 1$ be such that $\lambda \leqslant |f''(x)| \leqslant C\lambda$ for $x \in I$. Show that the discrepancy D_N of the sequence $\{f(n)\}_{n \in I}$ satisfies $D_N \ll \lambda^{1/3} + N^{-1}\lambda^{-1/2}$ where the implicit constant may depend on C.

113. *With the k-th derivatives.* Let k be an integer $\geqslant 3$, let I be an interval containing N integers and let $f \in \mathcal{C}^k(I, \mathbb{R})$. Suppose there exist $\lambda > 0$ and $C > 1$ such that $\lambda \leqslant |f^{(k)}(x)| \leqslant C\lambda$ for $x \in I$. Show by induction on k that

$$\Big| \sum_{n \in I} \mathrm{e}(f(n)) \Big| \ll N\lambda^{\alpha_k} + N^{1-\alpha_k}\lambda^{-\alpha_k}$$

with $\alpha_k := 1/\{2^k - 2\}$.

114. Show that, for all non-integral $\alpha > 0$, the sequence $\{n^\alpha\}_{n=1}^{\infty}$ is equidistributed modulo 1.

115. *Squarefree numbers in short intervals.*

(a) Deduce from the result in Exercise 44(c) (p. 40) the existence of an absolute constant $c > 0$ such that, for large x and $y \geqslant c\sqrt{x}$, the interval $]x - y, x]$ contains at least one squarefree integer.

(b) Let R be an integer $\geqslant 2$, let N be an integer $\geqslant 1$ and let I be an interval contained in $[N+1, 2N]$. Show that if $f \in \mathcal{C}^R(I, \mathbb{R})$ and if there exists $F > 0$ such that

$$\left|f^{(r)}(x)\right| \asymp FN^{-r} \qquad (1 \leqslant r \leqslant R,\ x \in I)$$

then

$$\sum_{n \in I} B_1(f(n)) \ll (FN^{-R})^{1/(2^R-1)} N + F^{-1}N.$$

(c) Assume $1 \leqslant y \leqslant x/2$. Set $S := \sum_{x-y<n\leqslant x} \mu^2(n)$. Show that $S = S_1 + O(S_2)$, with

$$S_1 := \sum_{\substack{x-y<d^2m\leqslant x \\ d\leqslant\sqrt{y}}} \mu(d); \qquad S_2 := \sum_{\substack{x-y<d^2m\leqslant x \\ d>\sqrt{y}}} 1.$$

(d) Show that

$$S_1 = \frac{6}{\pi^2} y + O(\sqrt{y}).$$

(e) Assume $\sqrt{y} \leqslant D \leqslant \sqrt{x}$. Write $S_2 = S_3 + S_4$, with

$$S_3 := \sum_{\substack{x-y<d^2m\leqslant x \\ \sqrt{y}<d\leqslant D}} 1; \qquad S_4 := \sum_{\substack{x-y<d^2m\leqslant x \\ d>D}} 1.$$

Show that $S_3 = -R_3 + O(\sqrt{y})$, with

$$R_3 := \sum_{d\leqslant D} \left\{ B_1\left(\frac{x}{d^2}\right) - B_1\left(\frac{x-y}{d^2}\right) \right\}.$$

(f) Show that $R_3 \ll x^{1/7} D^{2/7} + x^{-1} D^3$.

(g) Show that $S_4 \ll xD^{-2} + yD^{-1}$.

(h) Show that $S = 6y/\pi^2 + O\big(x^{1/4} + yx^{-3/8} + \sqrt{y}\big)$.

(i) Show that, for sufficiently large x and $y \geqslant Cx^{1/4}$, where $C > 0$ is a suitable constant, the interval $]x-y, x]$ contains at least one squarefree integer.

Diophantine approximation

7.1. From Dirichlet to Roth

Diophantine approximation is that branch of number theory which is concerned with the solubility of inequalities involving integral parameters. The basic result of the theory was obtained by Dirichlet in 1842. Its proof is essentially based on the pigeonhole principle.

Notation. Recall that we denote by $\lfloor \vartheta \rfloor$ the integer part of a real number ϑ, by $\lceil \vartheta \rceil$ the smallest integer at least equal to ϑ, by $\langle \vartheta \rangle$ its fractional part, and by $\|\vartheta\|$ the distance from ϑ to the set of integers.

We have $\|\vartheta\| = \min(\langle \vartheta \rangle, 1 - \langle \vartheta \rangle) \in [0, \frac{1}{2}]$.

Theorem 7.1 (Dirichlet). *Let $\vartheta \in \mathbb{R}$. We have*

$$\min_{1 \leqslant q \leqslant Q} \|q\vartheta\| \leqslant \frac{1}{Q+1} \qquad (Q \in \mathbb{N}^*). \tag{7.1}$$

Proof. Set $Q \in \mathbb{N}^*$. We may assume that

$$\max_{1 \leqslant m \leqslant Q} \langle m\vartheta \rangle < Q/(Q+1)$$

since (7.1) is otherwise trivially satisfied. The $Q+1$ numbers $\langle m\vartheta \rangle$ $(0 \leqslant m \leqslant Q)$ are thus in $[0, Q/(Q+1)[$. Therefore two of them, say $\langle u\vartheta \rangle$ and $\langle v\vartheta \rangle$, have a difference in $[0, 1/(Q+1)[$. It follows that, for suitable integers a and b,

$$0 \leqslant (u\vartheta - a) - (v\vartheta - b) < 1/(Q+1),$$

whence $\|q\vartheta\| < 1/(Q+1)$ with $q := |u - v| \leqslant Q$. □

Remark. Equality holds in (7.1) if, and only if, ϑ is a rational number of the form $\vartheta = a/(Q+1)$ with $(a, Q+1) = 1$.

Corollary 7.2. *Let $\vartheta \in \mathbb{R}$. The following three assertions are equivalent:*

(i) $\vartheta \in \mathbb{Q}$;

(ii) *there is a constant $c = c(\vartheta) > 0$ such that*

$$\|q\vartheta\| \neq 0 \Rightarrow \|q\vartheta\| \geqslant c; \tag{7.2}$$

(iii) *the double inequality*

$$0 < q\|q\vartheta\| < 1 \tag{7.3}$$

has only a finite number of solutions in integers q.

Proof. (i)⇒(ii). If $\vartheta = a/b \in \mathbb{Q}$, then $\|q\vartheta\| \in \{j/b : 0 \leqslant j \leqslant b/2\}$, and so $\|q\vartheta\| \neq 0$ implies $\|q\vartheta\| \geqslant 1/b$.

The implication (ii)⇒(iii) is trivial.

(iii)⇒(i). If $\vartheta \in \mathbb{R} \setminus \mathbb{Q}$ and if $q_1, \ldots, q_n$ are solutions of (7.3), Dirichlet's theorem implies the existence of an integer q such that

$$\|q\vartheta\| \leqslant \tfrac{1}{2} \min_{1 \leqslant j \leqslant n} \|q_j\vartheta\|$$

and $q\|q\vartheta\| < 1$. We must have $q \neq q_j$ for $1 \leqslant j \leqslant n$. □

Example. It is easy to deduce from criterion (ii) that the number e is irrational. The details are left to the reader.

Liouville showed in 1844 that a result analogous to (7.2) holds more generally for all algebraic numbers, that is to say for those real or complex numbers which are roots of polynomials with integer coefficients. We say an algebraic number has degree d if d is the least degree of a polynomial in $\mathbb{Z}[X]$ of which it is a root. It may be shown that such a polynomial is unique, up to requiring its leading coefficient to be positive and the coefficients to be mutually coprime. It is then referred to as the minimal polynomial.

Theorem 7.3 (Liouville). *Let ϑ be an real algebraic number with degree $d \geqslant 1$. Then there is a constant $c = c(\vartheta) > 0$ such that we have*

$$\|q\vartheta\| \neq 0 \Rightarrow \|q\vartheta\| > c/q^{d-1} \qquad (q \in \mathbb{N}^*).$$

Proof. We may assume ϑ real and $d \geqslant 2$. Let F be the minimal polynomial of ϑ. By the mean value theorem, for any rational p/q, there exists ξ between ϑ and p/q such that

$$F(\vartheta) - F(p/q) = (\vartheta - p/q)F'(\xi).$$

We have $F(\vartheta) = 0$ and $F(p/q) \neq 0$ since F is irreducible. Therefore $q^d F(p/q)$ is an integer and so $|F(p/q)| \geqslant 1/q^d$. Now we may assume

$|\vartheta - p/q| \leqslant \frac{1}{2}$, since otherwise p is not the nearest integer to $q\vartheta$. We thus have $|F'(\xi)| \leqslant M(\vartheta) := \sup_{|z-\vartheta|\leqslant 1/2} |F'(z)|$, whence

$$|\vartheta - p/q| \geqslant 1/(M(\vartheta)q^d). \qquad \square$$

The above theorem allows an easy construction of transcendental numbers, that is, numbers which are not algebraic.

Corollary 7.4. *For any integer $a > 1$ the number*

$$\vartheta := \sum_{k\geqslant 0} a^{-k!}$$

is transcendental.

Proof. Let us put $p_n/q_n := \sum_{0\leqslant k\leqslant n} a^{-k!}$. We have $q_n | a^{n!}$ and

$$|\vartheta - p_n/q_n| \leqslant 1/q_n^{n+1} + 1/q_n^{(n+1)(n+2)} + \cdots < 2/q_n^{n+1} = o(1/q_n^d)$$

for any given $d > 1$. $\square$

Remark. Cantor showed the existence of transcendental numbers by observing that the set of algebraic numbers is countable: to each algebraic number one may injectively associate the finite set of those integers which are the coefficients of its minimal polynomial. Since the set $\mathbb{R}$ is not countable (Cantor's diagonal argument, see Exercises 116 and 117), it must contain some transcendental numbers. From Lebesgue's theory, we even see that almost all numbers are transcendental.

Liouville's theorem has given rise to a very rich posterity: Thue showed in 1909 that the exponent $d - 1$ may be replaced by any $\delta > d/2$; Siegel improved this to $\delta > 2\sqrt{d} - 1$ in 1921; Dyson and Gelfond & Linnik independently showed in 1947–1948 that one may indeed choose $\delta > \sqrt{2d} - 1$; and finally, in 1955, K. F. Roth showed that any $\delta > 1$ is admissible, which earned him the Fields Medal (1958). This result is clearly final. The proof is very complicated and, as was actually the case for all the improvements upon Liouville's theorem, ineffective: given ϑ and $\delta > 1$, it provides no way of explicitly calculating the value of the constant $c := \inf_{q\geqslant 1} q^\delta \|q\vartheta\|$.

7.2. Best approximations, continued fractions

Let $\vartheta \in \mathbb{R}$. We shall define a sequence of rational fractions approximating ϑ in the most economical manner possible. To start the process, we set

$$z(\vartheta) := \begin{cases} 0 & \text{if } \langle\vartheta\rangle \leqslant \frac{1}{2} \\ 1 & \text{if } \frac{1}{2} < \langle\vartheta\rangle < 1, \end{cases}$$

and

$$p_0/q_0 := \lfloor \vartheta \rfloor /1 \qquad \text{if} \quad z(\vartheta) = 0,$$
$$p_0/q_0 := \lfloor \vartheta \rfloor /1, \quad p_1/q_1 := (1 + \lfloor \vartheta \rfloor)/1 \qquad \text{if} \quad z(\vartheta) = 1.$$

Finally, we introduce the set of best denominators of ϑ

$$\mathcal{D}^+(\vartheta) := \{q \geqslant 2 : \|q\vartheta\| < \min_{1 \leqslant m < q} \|m\vartheta\|\}.$$

Proposition 7.5. *$\mathcal{D}^+(\vartheta)$ is finite if, and only if, $\vartheta \in \mathbb{Q}$.*

Proof. If $\vartheta \in \mathbb{Q}$, write $\vartheta = a/b$, so $\|m\vartheta\| \in \{j/b : 0 \leqslant j \leqslant b/2\}$. Therefore $|\mathcal{D}^+(\vartheta)| \leqslant 1 + \lfloor b/2 \rfloor$ in this case.

If $\vartheta \in \mathbb{R} \setminus \mathbb{Q}$, then, for each $q \in \mathcal{D}^+(\vartheta)$, Dirichlet's approximation theorem implies the existence of at least one integer $q' \leqslant Q := 1 + \lfloor 1/\|q\vartheta\| \rfloor$ such that

$$\|q'\vartheta\| \leqslant 1/Q < \|q\vartheta\|.$$

The smallest among the integers q' such that $\|q'\vartheta\| < \|q\vartheta\|$ belongs to $\mathcal{D}^+(\vartheta)$ and is larger than q. □

Definition 7.6. *For $\vartheta \in \mathbb{R}$, we denote by $\{q_j\}_{j>z(\vartheta)}$ the finite or infinite sequence of elements of $\mathcal{D}^+(\vartheta)$ arranged in increasing order. We denote by $\mathcal{D}(\vartheta)$ the sequence obtained by adjoining to $\{q_j\}_{j>z(\vartheta)}$ the sequence of one or two elements $\{q_j\}_{0 \leqslant j \leqslant z(\vartheta)}$.*

Proposition 7.7. *For all $\vartheta \in \mathbb{R}$, the sequence $\{q_j\}_{j>z(\vartheta)}$ is strictly increasing. If $k > z(\vartheta)$ and $\|q_k\vartheta\| \neq 0$, then q_k has a successor in $\mathcal{D}^+(\vartheta)$ and we have*

$$\|q_k\vartheta\| \leqslant 1/q_{k+1}.$$

Proof. If $q_k \in \mathcal{D}^+(\vartheta)$ and $q_k\vartheta \notin \mathbb{Z}$, Dirichlet's approximation theorem implies the existence of an integer q such that $\|q\vartheta\| < \|q_k\vartheta\|$. Furthermore, by the definition of q_{k+1}, we have

$$\|q_k\vartheta\| \leqslant \min_{q < q_{k+1}} \|q\vartheta\|$$

and the right-hand side is $\leqslant 1/q_{k+1}$ by Theorem 7.1. □

Definition 7.8. *Let $\vartheta \in \mathbb{R}$. For $q_k \in \mathcal{D}^+(\vartheta)$, we denote by p_k the integer closest to $q_k\vartheta$. The sequence $\{p_k/q_k\}_{k=0}^{\infty}$ is called the sequence of* convergents *of ϑ.*

Notation. In the sequel, we put

$$\vartheta_k := q_k\vartheta - p_k. \tag{7.4}$$

We initially use this notation for $k \geqslant 0$, but we shall extend it later to $k \geqslant -2$.

Proposition 7.9. *If $\vartheta \in \mathbb{Q}$, we have $\vartheta = p_k/q_k$ for $k := |\mathcal{D}(\vartheta)| - 1$. If $\vartheta \in \mathbb{R} \setminus \mathbb{Q}$, we have*

$$\lim_{k\to\infty} p_k/q_k = \vartheta.$$

Proof. This is an immediate consequence of Proposition 7.7. □

Remark. Observe that, if ϑ is rational, say $\vartheta = p/q$, then $\mathcal{D}(\vartheta)$ is finite, i.e. $\mathcal{D}(\vartheta) = \{q_0, \dots, q_k\}$, and thus $q_k = q$ for $k = |\mathcal{D}(\vartheta)| - 1$.

Lemma 7.10. *Let $\vartheta \in \mathbb{R}$. If ϑ_k and ϑ_{k+1} are non-zero, they have opposite signs.*

Proof. This is clear when $k = 0$, $z(\vartheta) = 1$: we then have $\vartheta_0 \geqslant 0 > \vartheta_1$.

Put $q := q_{k+1} - q_k$, $p := p_{k+1} - p_k$, so that

$$q\vartheta - p = \vartheta_{k+1} - \vartheta_k.$$

We have $q < q_{k+1}$, and thus $|q\vartheta - p| \geqslant \|q\vartheta\| \geqslant |\vartheta_k|$ if $k > z(\vartheta)$. This does imply $\vartheta_k\vartheta_{k+1} < 0$ in this circumstance.

The cases $k = z(\vartheta) = 0$ and $k = z(\vartheta) = 1$ remain to be examined. For the latter, we have $\langle\vartheta\rangle \leqslant \frac{1}{2}$, therefore $m := \lfloor 1/\langle\vartheta\rangle \rfloor \geqslant 2$. Moreover,

$$\langle\vartheta\rangle \leqslant j\langle\vartheta\rangle \leqslant 1 - \langle\vartheta\rangle \quad (1 \leqslant j < m), \qquad 1 - \langle\vartheta\rangle < m\langle\vartheta\rangle \leqslant 1.$$

We then have $q_1 = m$ and $\vartheta_1 = m\vartheta - p_1 = m\langle\vartheta\rangle - 1 \leqslant 0$. The remaining case can be treated similarly. □

Theorem 7.11. *Let $\vartheta \in \mathbb{R}$. If $1 \leqslant k < |\mathcal{D}(\vartheta)|$, we have*

$$q_k p_{k-1} - q_{k-1}p_k = (-1)^k. \tag{7.5}$$

Proof. For $k = 1$, $z(\vartheta) = 1$, we have $q_0 = q_1 = 1$, $p_0 = \lfloor\vartheta\rfloor$, $p_1 = p_0 + 1$, and thus

$$q_1p_0 - q_0p_1 = p_0 - p_1 = -1.$$

For $k = 1$, $z(\vartheta) = 0$, we have $q_0 = 1$, $p_0 = \lfloor\vartheta\rfloor$ and, from the calculation carried out during the proof of Lemma 7.10, $q_1 := \lfloor 1/\langle\vartheta\rangle\rfloor$, $p_1 = q_1p_0 + 1$.[1] This again implies (7.5) for $k = 1$.

Let us now consider the case $k \geqslant 2$, so that $k > z(\vartheta)$. We first remark that $q_kp_{k-1} - q_{k-1}p_k \neq 0$: otherwise, we would have $p_{k-1}/q_{k-1} = p_k/q_k$, and therefore

$$|\vartheta_{k-1}| = (q_{k-1}/q_k)|\vartheta_k| < |\vartheta_k|,$$

which would contradict the definition of q_k. Next we observe that

$$q_kp_{k-1} - q_{k-1}p_k = q_{k-1}(q_k\vartheta - p_k) - q_k(q_{k-1}\vartheta - p_{k-1}) = q_{k-1}\vartheta_k - q_k\vartheta_{k-1},$$

[1] We have $1 - \langle\vartheta\rangle < q_1\langle\vartheta\rangle \leqslant 1$, or $1 - \langle\vartheta\rangle < q_1(\vartheta - p_0) \leqslant 1$, and therefore

$$q_1p_0 + 1 - \langle\vartheta\rangle < q_1\vartheta \leqslant q_1p_0 + 1.$$

whence

$$|q_k p_{k-1} - q_{k-1} p_k| \leqslant q_{k-1} \|q_k \vartheta\| + q_k \|q_{k-1} \vartheta\| \leqslant q_{k-1}/q_{k+1} + 1 < 2,$$

by Proposition 7.7. We must therefore have

$$q_k p_{k-1} - q_{k-1} p_k = \pm 1.$$

Since we verified relation (7.5) for $k = 1$, it is sufficient to establish that the sign of the expression alternates. However, for $k < |\mathcal{D}(\vartheta)| - 1$, we have

$$(q_k p_{k-1} - q_{k-1} p_k)(q_{k+1} p_k - q_k p_{k+1}) = (q_{k-1}\vartheta_k - q_k \vartheta_{k-1})(q_k \vartheta_{k+1} - q_{k+1}\vartheta_k)$$

and from Lemma 7.10 this last product can be expanded as a sum of four negative terms. □

Corollary 7.12. *The convergents are in lowest terms: we have* $(p_k, q_k) = 1$ *for all* $k \geqslant 0$.

Proof. This is an immediate consequence of Bachet's theorem. □

Theorem 7.13. *Let* $\vartheta \in \mathbb{R}$. *Set* $(p_{-2}, q_{-2}) := (0, 1)$, $(p_{-1}, q_{-1}) := (1, 0)$ *and*

$$a_k := \lfloor -\vartheta_{k-2}/\vartheta_{k-1} \rfloor \quad (k \geqslant 0) \tag{7.6}$$

where ϑ_k *is defined by* (7.4). *Then, assuming* $\vartheta_{k-1} \neq 0$, $k \geqslant 0$, *we have*

$$\begin{cases} p_k = a_k p_{k-1} + p_{k-2}, \\ q_k = a_k q_{k-1} + q_{k-2}. \end{cases} \tag{7.7}$$

Proof. Since the determinant

$$\begin{vmatrix} p_{k-1} & p_{k-2} \\ q_{k-1} & q_{k-2} \end{vmatrix}$$

equals $(-1)^k$ for $k \geqslant 0$, there exist integers a_k and b_k such that

$$\begin{cases} p_k = a_k p_{k-1} + b_k p_{k-2}, \\ q_k = a_k q_{k-1} + b_k q_{k-2}. \end{cases}$$

Elimination immediately yields

$$b_k = \frac{q_{k-1} p_k - q_k p_{k-1}}{q_{k-1} p_{k-2} - q_{k-2} p_{k-1}} = 1,$$

by Theorem 7.11. Furthermore, we check that (7.7) is satisfied for $k = 0$ in view of the definition of (p_0, q_0). For $k \geqslant 1$, we have

$$\begin{aligned} \frac{\vartheta_k}{\vartheta_{k-1}} &= \frac{q_k \vartheta - p_k}{\vartheta_{k-1}} = \frac{(a_k q_{k-1} + q_{k-2})\vartheta - (a_k p_{k-1} + p_{k-2})}{\vartheta_{k-1}} \\ &= a_k + \frac{\vartheta_{k-2}}{\vartheta_{k-1}} \in \,]-1, 0], \end{aligned} \tag{7.8}$$

from Lemma 7.10 and the definition of the ϑ_k. This implies that relation (7.7) does indeed hold for $k \geqslant 1$ when the a_k are defined by (7.6). □

Definition 7.14. *Let $\vartheta \in \mathbb{R}$. The numbers a_k defined by* (7.6) *are called* incomplete quotients *of the continued fraction expansion of ϑ. The numbers $\alpha_k := -\vartheta_{k-2}/\vartheta_{k-1}$ are called the* complete quotients *of the expansion of ϑ.*

The name complete quotients is justified by the relation

$$\vartheta = \frac{\alpha_{k+1}p_k + p_{k-1}}{\alpha_{k+1}q_k + q_{k-1}} \qquad (k \geqslant 0).$$

Theorem 7.15. *Let $\vartheta \in \mathbb{R}$. We have $\alpha_0 = \vartheta$, $\alpha_{k+1} = 1/\langle \alpha_k \rangle$ $(k \geqslant 0)$. For $k < |\mathcal{D}(\vartheta)|$, we have*

$$\frac{p_k}{q_k} = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{\ddots + \cfrac{1}{a_k}}}}. \tag{7.9}$$

Proof. Relation (7.8) can also be written

$$1/\alpha_{k+1} = \alpha_k - a_k = \langle \alpha_k \rangle .$$

This shows the first part of the statement. The second is an immediate consequence of the following result, where we employ the notation

$$[a_0, a_1, \ldots, a_k]$$

for the right-hand side of (7.9) and extend it to all finite sequences $\{a_j\}_{j=0}^{k}$ such that $a_j > 0$ for $j \geqslant 1$. □

Proposition 7.16. *For $k \in \mathbb{N}$, $a_0 \in \mathbb{R}$, $a_1 > 0$, ..., $a_{k-1} > 0$ and any real number $x > 0$ we have*

$$[a_0, a_1, \ldots, a_{k-1}, x] = \frac{xp_{k-1} + p_{k-2}}{xq_{k-1} + q_{k-2}}$$

where the p_j, q_j are defined by $(p_{-2}, q_{-2}) := (0, 1)$, $(p_{-1}, q_{-1}) := (1, 0)$ and the recurrence relation (7.7).

Proof. We argue by induction on k. Formula (7.9) is obvious if $k = 0$ or 1. Assuming the property holds for k, we have

$$\begin{aligned}
[a_0, \ldots, a_k, x] &= [a_0, \ldots, a_k + 1/x] \\
&= \frac{(a_k + 1/x)p_{k-1} + p_{k-2}}{(a_k + 1/x)q_{k-1} + q_{k-2}} \\
&= \frac{(a_k p_{k-1} + p_{k-2})x + p_{k-1}}{(a_k q_{k-1} + q_{k-2})x + q_{k-1}} = \frac{p_k x + p_{k-1}}{q_k x + q_{k-1}}.
\end{aligned}$$

□

Algorithmic computation of incomplete quotients is quite straightforward since we only have to iterate the operation $x \mapsto 1/\langle x \rangle$. For example, let us consider the case of the number π. We find successively

$$\begin{aligned}
&\alpha_0 = \pi \approx 3.141592653589793, && a_0 = 3, && \frac{p_0}{q_0} = \frac{3}{1},\\
&\alpha_1 = \frac{1}{\pi - 3} \approx 7.062513305931045, && a_1 = 7, && \frac{p_1}{q_1} = \frac{7\cdot 3+1}{7\cdot 1+0} = \frac{22}{7},\\
&\alpha_2 = \frac{1}{\alpha_1 - 7} \approx 15.996594406685719, && a_2 = 15, && \frac{p_2}{q_2} = \frac{15\cdot 22+3}{15\cdot 7+1} = \frac{333}{106},\\
&\alpha_3 = \frac{1}{\alpha_2 - 15} \approx 1.003417231013372, && a_3 = 1, && \frac{p_3}{q_3} = \frac{1\cdot 333+22}{1\cdot 106+7} = \frac{355}{113}.
\end{aligned}$$

We have

$$355/113 \approx 3.141592920353982.$$

In four steps we have thus obtained a fraction which is exact to six decimal places.

The following theorem furnishes, for any rational number, the partial quotients of the continued fraction expansion in terms of the quotients and remainders in the Euclidean algorithm. Given a rational number r/s with $s \geqslant 1$, we write the Euclidean algorithm for the greatest common divisor in the form

$$\begin{aligned}
r &= a_0 s + r_1, \quad 0 \leqslant r_1 < s,\\
s &= a_1 r_1 + r_2, \quad 0 \leqslant r_2 < r_1,
\end{aligned}$$

$$\begin{aligned}
r_{N-2} &= a_{N-1} r_{N-1} + r_N, \quad 0 \leqslant r_N < r_{N-1},\\
r_{N-1} &= a_N r_N,
\end{aligned}$$

where $r_N = (r, s)$.

Theorem 7.17. *Let $r/s \in \mathbb{Q}^*$, with $s \geqslant 1$, $(r, s) = 1$. With the previous notation, we have*

$$\frac{r}{s} = [a_0, \ldots, a_N].$$

Proof. Set $s = r_0$, $r = r_{-1}$. We show by downward induction over $j \in [0, N]$ that

$$\frac{r_{j-1}}{r_j} = [a_j, \ldots, a_N].$$

The result is obvious for $j = N$. Assuming it is true for $j+1$, we can write

$$\begin{aligned}[a_j, \dots, a_N] &= [a_j, a_{j+1}, \dots, a_N] \\ &= [a_j, [a_{j+1}, \dots, a_N]] = [a_j, r_j/r_{j+1}] \\ &= a_j + r_{j+1}/r_j = (a_j r_j + r_{j+1})/r_j = r_{j-1}/r_j.\end{aligned}$$

□

7.3. Properties of the continued fraction expansion

Let us begin with a simple remark about finite continued fractions.

Proposition 7.18. *Let $\vartheta \in \mathbb{R}$. For $k < |\mathcal{D}(\vartheta)|$, we have*

$$[a_0, a_1, \dots, a_k] = a_0 + \sum_{0 \leqslant j < k} \frac{(-1)^j}{q_j q_{j+1}}. \tag{7.10}$$

Proof. This is an immediate consequence of the formula (7.5) in the form

$$\frac{p_k}{q_k} = \frac{p_{k-1}}{q_{k-1}} + \frac{(-1)^{k-1}}{q_{k-1}q_k}.$$

□

When $a_k \geqslant 2$ we have

$$[a_0, a_1, \dots, a_k] = [a_0, a_1, \dots, a_k - 1, 1].$$

So we may choose arbitrarily the parity of the length of the continued fraction expansion of a rational number.

Theorem 7.19. *With the convention that $a_k \geqslant 2$ when $\vartheta \in \mathbb{Q}$ and $k = |\mathcal{D}(\vartheta)| - 1$, the continued fraction expansion is unique; in other words, for each real number ϑ, the integers a_j $(0 \leqslant j \leqslant k)$ are uniquely determined by the formula* (7.9).

Furthermore, for every infinite integer sequence $\{a_j\}_{j=0}^{\infty}$ such that $a_j \geqslant 1$ for $j \geqslant 1$, the sequence

$$[a_0, a_1, \dots, a_k] \qquad (k \geqslant 0)$$

converges.

Proof. We have $a_0 = \lfloor p_0/q_0 \rfloor$, $a_1 = \lfloor 1/\langle p_0/q_0 \rangle \rfloor$, etc. This shows the first part of the assertion. The second results from formula (7.10). □

The following result shows that the growth of the convergents is always at least exponential. We denote by $\varphi := \frac{1}{2}(1+\sqrt{5})$ the golden ratio. Recall that the Fibonacci sequence is defined by $F_0 = 0$, $F_1 = 1$ and

$$F_{k+1} = F_k + F_{k-1} \qquad (k \geqslant 1).$$

Theorem 7.20. *Let $\{F_k\}_{k=0}^{\infty}$ be the Fibonacci sequence. For all $\vartheta \in \mathbb{R}$, we have*

$$q_k \geqslant F_k = \frac{1}{\sqrt{5}}\{\varphi^k - (-1)^k/\varphi^k\} \qquad (k \geqslant 1)$$

with equality when $\vartheta = \varphi$.

Proof. We have $q_k = a_k q_{k-1} + q_{k-2} \geqslant q_{k-1} + q_{k-2}$. This implies the inequality $q_k \geqslant F_k$. We have equality when $a_k = 1$ for all k, which corresponds to $\vartheta = 1 + 1/\vartheta$ and $\vartheta > 0$, hence $\vartheta = \varphi$. □

Theorem 7.21. *Let $\vartheta \in \mathbb{R}$. If $\|q_k\vartheta\| \neq 0$, we have*

$$\vartheta - \frac{p_k}{q_k} = \frac{(-1)^k}{q_k(\alpha_{k+1}q_k + q_{k-1})} \qquad (k \geqslant 0). \tag{7.11}$$

In particular

$$\frac{1}{q_k(q_{k+1} + q_k)} < \left|\vartheta - \frac{p_k}{q_k}\right| \leqslant \frac{1}{q_k q_{k+1}}. \tag{7.12}$$

Proof. From the formula $\alpha_{k+1} = -\vartheta_{k-1}/\vartheta_k$, we infer

$$\vartheta = \frac{\alpha_{k+1}p_k + p_{k-1}}{\alpha_{k+1}q_k + q_{k-1}}.$$

Whence

$$\vartheta - \frac{p_k}{q_k} = \frac{q_k(\alpha_{k+1}p_k + p_{k-1})}{q_k(\alpha_{k+1}q_k + q_{k-1})} - \frac{p_k(\alpha_{k+1}q_k + q_{k-1})}{q_k(\alpha_{k+1}q_k + q_{k-1})} = \frac{q_k p_{k-1} - q_{k-1}p_k}{q_k(\alpha_{k+1}q_k + q_{k-1})}.$$

This implies (7.11), by (7.5). We further deduce immediately the two-sided bound (7.12) since

$$q_{k+1} = a_{k+1}q_k + q_{k-1} \leqslant \alpha_{k+1}q_k + q_{k-1} < (a_{k+1} + 1)q_k + q_{k-1} = q_{k+1} + q_k.$$

□

Corollary 7.22. *Let $\vartheta \in \mathbb{R} \setminus \mathbb{Q}$. We have*

$$1/(2 + \limsup_{k\to\infty} a_k) \leqslant \liminf_{q\to\infty} q\|q\vartheta\| \leqslant 1/\limsup_{k\to\infty} a_k.$$

Proof. We clearly have $\liminf_{q\to\infty} q\|q\vartheta\| \leqslant \liminf_{k\to\infty} q_k\|q_k\vartheta\|$. However, for $q_k \leqslant q < q_{k+1}$, we have $q\|q\vartheta\| \geqslant q_k\|q_k\vartheta\|$, by definition of q_k. Thus

$$\liminf_{q\to\infty} q\|q\vartheta\| = \liminf_{k\to\infty} q_k\|q_k\vartheta\|$$

and, since $0 < q_{k-1}/q_k \leqslant 1$, the required result is an immediate consequence of (7.12). □

Theorem 7.23 (Lagrange's criterion). *Let $\vartheta \in \mathbb{R} \setminus \mathbb{Q}$. A necessary and sufficient condition that p/q be a convergent of ϑ is that there exist integers $p' < p$, $q' < q$, and a real number $\alpha > 1$ such that $qp' - pq' = \pm 1$ and*

$$\vartheta = \frac{\alpha p + p'}{\alpha q + q'}. \tag{7.13}$$

Proof. The necessary condition is immediate: it suffices to select $(p', q') = (p_{k-1}, q_{k-1})$ and then $\alpha = \alpha_{k+1}$ if $(p, q) = (p_k, q_k)$. To show that it is sufficient, assume (7.13) with (p, q), (p', q') as indicated. Then expand p/q as a continued fraction $[a_0^*, \cdots, a_k^*]$, choosing the parity of the length k so that

$$q_k^* p_{k-1}^* - q_{k-1}^* p_k^* = qp' - pq'.$$

Since p' (resp. q') is determined modulo p (resp. q) and since $p' < p$ (resp. $q' < q$), this implies $(p', q') = (p_{k-1}^*, q_{k-1}^*)$. We deduce immediately by substitution in (7.13) that we have $\vartheta = [a_0^*, \cdots, a_k^*, \alpha]$ and thus by uniqueness of the continued fraction expansion, that $\alpha = \alpha_{k+1} = -\vartheta_k/\vartheta_{k-1}$. It follows that p_k/q_k is indeed the convergent of ϑ with index k. □

Corollary 7.24. *Let $\vartheta \in \mathbb{R} \setminus \mathbb{Q}$. If $|\vartheta - p/q| < 1/(2q^2)$, then p/q is a convergent of ϑ.*

Proof. Let $[a_0, a_1, \ldots, a_k]$ be the continued fraction expansion of p/q chosen so that $p/q - p_{k-1}/q_{k-1}$ has the same sign as $p/q - \vartheta$. Define α by

$$\vartheta = \frac{\alpha p_k + p_{k-1}}{\alpha q_k + q_{k-1}}.$$

We have

$$\vartheta - \frac{p}{q} = \frac{\alpha p_k + p_{k-1}}{\alpha q_k + q_{k-1}} - \frac{p_k}{q_k} = \frac{\operatorname{sgn}(\vartheta - p/q)}{q_k(\alpha q_k + q_{k-1})}$$

whence

$$|\vartheta - p/q| = \frac{1}{q_k(\alpha q_k + q_{k-1})} < \frac{1}{2q_k^2}$$

and therefore $\alpha > 2 - q_{k-1}/q_k > 1$. The desired conclusion then follows from Lagrange's criterion. □

We end this section with an arithmetic application: the proof by Hermite (1848) of the Girard–Fermat theorem (Theorem 4.29).

Theorem 7.25 (Girard–Fermat). *Let p be a prime congruent to 1 modulo 4. Then there exist integers r and s such that $p = r^2 + s^2$.*

Proof. We know from the formula for the Legendre symbol that -1 is a quadratic residue modulo p. So let a be an integer in $[1, p-1]$ such that $a^2 \equiv -1 \pmod{p}$. Let us expand a/p as a continued fraction and denote

the convergents by $p_0/q_0, \ldots, p_N/q_N$. There is a unique index k such that $q_k < \sqrt{p} < q_{k+1}$. We infer from Proposition 7.7 that

$$\left|\frac{a}{p} - \frac{p_k}{q_k}\right| \leqslant \frac{1}{q_k q_{k+1}},$$

and so $|aq_k - pp_k| \leqslant p/q_{k+1} < \sqrt{p}$.

Now set $r = q_k$, $s = aq_k - pp_k$. We have

$$r^2 + s^2 \equiv q_k^2 + a^2 q_k^2 \equiv (1 + a^2) q_k^2 \equiv 0 \,(\mathrm{mod}\, p).$$

Furthermore,

$$0 < r^2 + s^2 = q_k^2 + s^2 < 2p.$$

It follows that $r^2 + s^2 = p$. □

Remark. The above proof provides a powerful practical algorithm for finding the numbers r and s.

7.4. Continued fraction expansion of quadratic irrationals

Definition 7.26. *Two irrational numbers ϑ and ϑ' are said to be equivalent for the continued fraction expansion if their expansions coincide from a certain rank on, i.e., if, for suitable integers m and n, we have*

$$\vartheta = [a_0, \ldots, a_m, c_0, c_1, \ldots], \quad \vartheta' = [b_0, \ldots, b_n, c_0, c_1, \ldots].$$

Theorem 7.27. *A necessary and sufficient condition for two irrational numbers ϑ and ϑ' to be equivalent for the continued fraction expansion is that there exist integers a, b, c, d such that $ad - bc = \pm 1$ and*

$$\vartheta' = \frac{a\vartheta + b}{c\vartheta + d}. \tag{7.14}$$

Proof. If ϑ and ϑ' are equivalent, we have, writing $\sigma := [c_0, c_1, \ldots]$,

$$\vartheta = \frac{p_m \sigma + p_{m-1}}{q_m \sigma + q_{m-1}},$$

and similarly

$$\vartheta' = \frac{p'_n \sigma + p'_{n-1}}{q'_n \sigma + q'_{n-1}},$$

where the p_k/q_k (resp. p'_k/q'_k) are the convergents of $[a_0, \ldots, a_m]$ (resp. of $[b_0, \ldots, b_n]$). The required formula follows since unimodular linear fractional transformations form a group.

Conversely, if equation (7.14) is satisfied, we may assume, up to changing all the signs of a, b, c, d, that $c\vartheta + d > 0$. Then, we expand ϑ as a continued fraction to rank k. We obtain

$$\vartheta = \frac{\alpha_{k+1} p_k + p_{k-1}}{\alpha_{k+1} q_k + q_{k-1}}.$$

Put $p := ap_k + bq_k$, $p' := ap_{k-1} + bq_{k-1}$, $q = cp_k + dq_k$, $q' := cp_{k-1} + dq_{k-1}$. We have

$$\vartheta' = \frac{a\vartheta + b}{c\vartheta + d} = \frac{p\alpha_{k+1} + p'}{q\alpha_{k+1} + q'}.$$

We check by calculation that $pq' - qp' = \pm 1$. Moreover,

$$q = c(p_k - \vartheta q_k) + q_k(c\vartheta + d) = q_k(c\vartheta + d) - c\vartheta_k \sim q_k(c\vartheta + d) \qquad (k \to \infty),$$

so we have $q > 0$ for sufficiently large k. Similarly, we get that $p > 0$ for large k. Since $\alpha_{k+1} > 1$ (as this is a complete quotient of ϑ), we deduce from Lagrange's criterion that p'/q' and p/q are consecutive convergents of ϑ', say p'_{n-1}/q'_{n-1} and p'_n/q'_n. Therefore there is an integer n such that, with an obvious notation,

$$\vartheta' = \frac{p'_n\alpha'_{n+1} + p'_{n-1}}{q'_n\alpha'_{n+1} + q'_{n-1}}.$$

We have $\alpha_{k+1} = \alpha'_{n+1} = [c_0, c_1, \ldots]$ and

$$\vartheta = [a_0, \ldots, a_k, c_0, c_1, \ldots], \qquad \vartheta' = [b_0, \ldots, b_k, c_0, c_1, \ldots].$$

□

A continued fraction is said to be periodic (or ultimately periodic) if there exist integers m and h such that $a_{k+h} = a_k$ for $k \geqslant m$. When this holds, we write

$$\vartheta = [a_0, \ldots, a_{m-1}, \overline{a_m, \ldots, a_{m+h-1}}] \tag{7.15}$$

for the continued fraction expansion of the corresponding real number.

Theorem 7.28. *Let $\vartheta \in \mathbb{R}$. Then ϑ is a quadratic irrational number if, and only if, its continued fraction expansion is ultimately periodic.*

Proof. Let us first assume that the continued fraction of ϑ is periodic, so that $\alpha_{k+h} = \alpha_k$ for $k \geqslant m$. It follows that

$$\begin{aligned}\vartheta = [a_0, \ldots, a_{m-1}, \alpha_m] &= \frac{p_{m-1}\alpha_m + p_{m-2}}{q_{m-1}\alpha_m + q_{m-2}} \\ = [a_0, \ldots, a_{m+h-1}, \alpha_m] &= \frac{p_{m+h-1}\alpha_m + p_{m+h-2}}{q_{m+h-1}\alpha_m + q_{m+h-2}}.\end{aligned}$$

Therefore α_m is the solution of a quadratic equation with leading coefficient

$$q_{m+h-1}p_{m-1} - p_{m+h-1}q_{m-1} \neq 0$$

since two convergents with distinct indices are distinct.[2] So α_m is quadratic, and thus also ϑ by Theorem 7.27.

[2]Otherwise we would have $|\vartheta_m| = (q_m/q_{m+h-1})|\vartheta_{m+h-1}| < |\vartheta_{m+h-1}|$, which would contradict the definition of q_{m+h-1}.

It remains to show that the condition is necessary. Let ϑ be quadratic irrational, root of the equation $a\vartheta^2 + b\vartheta + c = 0$ with $a \neq 0$. For $k \geqslant 0$, we have

$$\vartheta = \frac{p_k\alpha_{k+1} + p_{k-1}}{q_k\alpha_{k+1} + q_{k-1}},$$

hence α_{k+1} is a solution of the quadratic equation

$$A_k\alpha_{k+1}^2 + B_k\alpha_{k+1} + C_k = 0$$

with

$$\begin{cases} A_k := ap_k^2 + bp_kq_k + cq_k^2, \\ B_k := 2ap_kp_{k-1} + b(q_kp_{k-1} + p_kq_{k-1}) + 2cq_kq_{k-1}, \\ C_k := ap_{k-1}^2 + bp_{k-1}q_{k-1} + cq_{k-1}^2. \end{cases}$$

We shall show that the triplet (A_k, B_k, C_k) can only assume a finite number of values. This will imply that, for suitable h and k with $h > 1$, α_{k+1} and α_{k+h} are roots of the same equation and are actually equal since at least one of these equations must be repeated infinitely many times. Hence the expansion will indeed be shown to be periodic.

A routine calculation furnishes

$$B_k^2 - 4A_kC_k = (b^2 - 4ac)(q_kp_{k-1} - p_kq_{k-1})^2 = b^2 - 4ac,$$

so that $B_k^2 = b^2 - 4ac + 4A_kC_k$. Since $C_k = A_{k-1}$, it is enough to show that A_k can only assume a finite number of values. However

$$\begin{aligned} A_k &= a(q_k\vartheta - \vartheta_k)^2 + b(q_k\vartheta - \vartheta_k)q_k + cq_k^2 \\ &= q_k^2(a\vartheta^2 + b\vartheta + c) - q_k\vartheta_k(2a\vartheta + b) + a\vartheta_k^2 = -q_k\vartheta_k(2a\vartheta + b) + a\vartheta_k^2. \end{aligned}$$

Since $|q_k\vartheta_k| \leqslant 1$ by Dirichlet's theorem, we obtain

$$|A_k| \leqslant |2a\vartheta + b| + |a|,$$

which implies the desired property. □

Notes

§ 7.1. In 1872, Cantor met Dedekind in Interlaken (Switzerland). Their discussions gave birth to modern set theory. At the end of 1873, Cantor realized that $\mathbb{Q}$ is equipotent[3] to $\mathbb{N}$ and Dedekind established that this also holds for $\overline{\mathbb{Q}}$, the set of algebraic numbers. Whence Cantor's idea to show that $\mathbb{R}$ is not equipotent to $\mathbb{N}$ and thus obtain a new proof of Liouville's theorem (Theorem 7.3) asserting the existence of transcendental numbers. In letters addressed to Dedekind (7 and 9 December 1873), Cantor fully achieves his goal, thereby demonstrating the existence of several categories of infinity. This is the starting point of topology, of abstract set theory, and of the theory of order types.

Cantor's ideas revolutionized the very foundations of mathematics and led to new types of proofs. The Cantor–Bernstein theorem asserts that if two sets E and F are such that there is an injection from each one into the other, then E and F are equipotent.

Charles Hermite showed in 1873 that the number e, the base of the natural logarithm, is a transcendental number. It was already known that e is neither rational nor quadratic irrational, since its continued fraction expansion, known to Euler, is:

$$\mathrm{e} = [2, 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, \ldots].$$

A proof of this formula may be found, for example, in the book by D. P. Parent (1978).

Hermite's method depended on the approximation of analytic functions by rational functions. Hermite thought that his method could be adapted to establish the transcendence of π. His intuition was confirmed by Lindemann in 1882. This resolved, in the negative, the famous Greek problem of squaring the circle.

[3]Two sets are said to be equipotent if there is a bijection from one onto the other.

Exercises

116. *Cantor's diagonal argument.*

(a) Let E be the set of those real numbers x in $[0,1]$ whose triadic representation $x = \sum_{n\geqslant 1} \varepsilon_n/3^n$ only includes digits ε_n equal to 0 or 1. Show that the triadic representation of an element of E is unique.

(b) Arguing by contradiction and using the representation described in (a), show that E, and thus *a fortiori* $[0,1]$, is not countable.

117. *Cantor–Mendès France's diagonal argument.*[4]

(a) Show that the set of algebraic numbers is countable.

(b) Let $\{x_n\}_{n\geqslant 1}$ denote the sequence of all algebraic numbers in the interval $[0,1]$, dyadic rational numbers[5] appearing twice. We then write the expansion in base 2

$$x_n = \sum_{k\geqslant 1} e_{nk}/2^k \qquad (n \geqslant 1)$$

with the convention that, when x_n is dyadic rational, one of the two expansions is finite (i.e. $e_{kn} = 0$ ultimately vanishes) while the other is such that $e_{kn} = 1$ for all large k. Show that to each x_n we can associate at least one x_m such that $e_{mk} \neq e_{nk}$ for all $k \geqslant 1$.

(c) Show that the diagonal number $x := \sum_{k\geqslant 1} e_{kk}/2^k$ is transcendental.

118. Is the number $\vartheta := \sum_{n\geqslant 0} 1/2^{3^n}$ transcendental?

119. *First theorem of Hurwitz.*

Let $\vartheta \in \mathbb{R} \setminus \mathbb{Q}$ and let $\{p_k/q_k\}_{k=0}^{\infty}$ denote the sequence of convergents of ϑ. Show that

$$|\vartheta - p_k/q_k| + |\vartheta - p_{k+1}/q_{k+1}| = 1/q_k q_{k+1} \qquad (k \geqslant 0).$$

Deduce that the inequality $\|q_j\vartheta\| < 1/(2q_j)$ holds for $j = k$ or $j = k+1$.

120. *Second theorem of Hurwitz.*

Let $\vartheta \in \mathbb{R} \setminus \mathbb{Q}$ and let $\{p_k/q_k\}_{k=0}^{\infty}$ denote the sequence of convergents of ϑ. Put $t_k := q_k/q_{k-1}$.

(a) Show that $t_{k+1} \geqslant 1 + 1/t_k$ $(k \geqslant 0)$.

(b) Show that if the inequality

$$|\vartheta - p_j/q_j| \leqslant 1/(\sqrt{5}q_j^2) \tag{7.16}$$

does not hold for $j = k-1$ and $j = k$, then $t_k^2 + 1 < \sqrt{5}t_k$.

(c) Let $k \geqslant 1$. Show that (7.16) holds for $j = k-1$, $j = k$ or $j = k+1$.

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[4] See Brlek, Mendès France, Robson & Rubey (2004), and Mendès France (2007).

[5] A dyadic rational number is a rational number of the form $a/2^b$, with $a \in \mathbb{Z}$, $b \in \mathbb{N}$.

121. *Markov's constant.*

The Markov constant of an irrational number ϑ is the finite or infinite quantity

$$\gamma(\vartheta) := 1/\liminf_{q\to\infty}(q\|q\vartheta\|).$$

(a) Show that $\gamma(\vartheta) = 1/\liminf_{k\to\infty} q_k\|q_k\vartheta\|$.

(b) Show that $\gamma(\vartheta) = \infty$ if, and only if, $\limsup_{k\to\infty} a_k = \infty$, where a_k denotes the k-th partial quotient of ϑ.

(c) What can be said about the Markov constant of quadratic irrationals?

(d) Let $\varphi := \frac{1}{2}(1+\sqrt{5})$ denote the golden ratio. Compute $\gamma(\varphi)$ and deduce that the second theorem of Hurwitz is optimal.

122. Given an irrational real number ϑ, let us denote by $\{p_k/q_k\}_{k=0}^{\infty}$ the sequence of convergents of its continued fraction expansion, by $\{\alpha_k\}_{k=0}^{\infty}$ the sequence of its complete quotients and by $\{a_k\}_{k=0}^{\infty}$ that of its incomplete quotients.

(a) Put $s_k := q_{k-1}/q_k$ $(k \geqslant 1)$. Show that the Markov constant $\gamma(\vartheta)$ satisfies

$$\gamma(\vartheta) = \limsup_{k\to\infty}(\alpha_{k+1} + s_k).$$

(b) Show that, if $a_k \geqslant 3$ for infinitely many values of k, then $\gamma(\vartheta) \geqslant 3$.

(c) Show that, if $a_k = 1$ from a certain rank on, then, on the one hand, we have $\alpha_k = \varphi := \frac{1}{2}(1+\sqrt{5})$ for sufficiently large k, and, on the other hand, there is a constant $\lambda > 0$ such that $q_k \sim \lambda\varphi^k$ as $k \to \infty$. Deduce that $\gamma(\vartheta) = \sqrt{5}$.

(d) Show that, if $a_k = 2$ from a certain rank on, then $\alpha_k = 1+\sqrt{2}$ for large enough k, $\lim_{k\to\infty} s_k = \sqrt{2}-1$ and $\gamma(\vartheta) = \sqrt{8}$.

(e) Show that, if a_k equals 1 or 2 for large enough k, both values being assumed infinitely many times, then we have $a_k = 1$ and $a_{k+1} = 2$ for infinitely many values of k. Noticing that $\alpha_{k+2} < 3$ implies

$$\alpha_{k+1} = a_{k+1} + 1/\alpha_{k+2} \geqslant 7/3,$$

show that $\gamma(\vartheta) \geqslant 17/6 > \sqrt{8}$.

(f) Establish the following theorem. *For any real number ϑ, we have $\gamma(\vartheta) \geqslant \sqrt{5}$ with equality if, and only if, $a_k = 1$ for large enough k. If one excludes those real numbers equivalent to the golden ratio φ, we have $\gamma(\vartheta) \geqslant \sqrt{8}$ with equality if, and only if, $a_k = 2$ for sufficiently large k, i.e., ϑ is equivalent to $\sqrt{2}$.*

123. Show that if ϑ and ϑ' have the same incomplete quotients up to rank k, then $|\vartheta - \vartheta'| \leqslant 10/\varphi^{2k-1}$ where $\varphi := \frac{1}{2}(1+\sqrt{5})$.

124. Let ϑ be an irrational number whose incomplete quotients do not exceed some integer a. Show that $q_n \leqslant b^n$ with $b := \frac{1}{2}(a + \sqrt{a^2+4})$.

125. Find the irrational number ϑ whose continued fraction expansion is $[8, \overline{1,1,5,5,1,1,16}]$, in the notation introduced in (7.15).

126. Given an irrational real number ϑ, consider the power series

$$F_\vartheta(z) := \sum_{n \geqslant 1} \frac{z^n}{\|n\vartheta\|}.$$

Denote by $R(\vartheta)$ the radius of convergence of F_ϑ and by $\{p_k/q_k\}_{k\geqslant 0}^{\infty}$ the sequence of convergents of the continued fraction expansion of ϑ.

(a) Show that $R(\vartheta) \leqslant 1$.

(b) Show that, for any integer $k \geqslant 0$ and every integer n satisfying $q_k \leqslant n < q_{k+1}$, we have $\|q_k\vartheta\|^{1/q_k} \leqslant \|n\vartheta\|^{1/n}$. Deduce that

$$R(\vartheta) = \liminf_{k\to\infty} \|q_k\vartheta\|^{1/q_k}.$$

(c) Show that, for any integer $k \geqslant 0$, we have

$$\frac{1}{2q_{k+1}} < \|q_k\vartheta\| < \frac{1}{q_{k+1}}.$$

(d) Let $\varphi := \frac{1}{2}(1+\sqrt{5})$ denote the golden ratio. Show that $R(\varphi) = 1$.

(e) Calculate $R(\vartheta)$ when ϑ is an arbitrary irrational algebraic number.

(f) Show that, for every integer $k \geqslant 0$, we have

$$q_k a_{k+1} \leqslant q_{k+1} \leqslant 2q_k a_{k+1}.$$

Making use of question (c), deduce that, for any irrational number ϑ, we have

$$R(\vartheta) = \liminf_{k\to\infty} a_{k+1}^{-1/q_k}.$$

(g) Show that the map $R : \mathbb{R} \smallsetminus \mathbb{Q} \to [0,1]$, which to ϑ associates $R(\vartheta)$, is surjective.

127. *Good approximations.* Given a real number ϑ, we say that a rational number p/q, with $p \in \mathbb{Z}$, $q \in \mathbb{N}^*$, is a *good approximation* to ϑ if

$$|\vartheta - p/q| = \min_{\substack{(u,v)\in\mathbb{Z}\times\mathbb{N}^* \\ 1\leqslant v\leqslant q}} |\vartheta - u/v|.$$

(a) Let $\vartheta \in \mathbb{R}$. Show that every convergent of ϑ is a good approximation to ϑ.

(b) By considering $\vartheta = \frac{1}{3}$ and $p/q = \frac{1}{2}$, show that a good approximation to a real number is not necessarily a convergent.

128. *Secondary convergents.* Let $\vartheta \in \mathbb{R}$, denote by $\{p_k/q_k\}_{k\geqslant 0}$ the sequence, finite or infinite, of convergents of ϑ, and let $\{a_k\}_{k\geqslant 0}$ be that of its partial quotients. For $k \geqslant 1$, $h \geqslant 0$, put $p_{k,h} := p_{k-1} + hp_k$, $q_{k,h} := q_{k-1} + hq_k$. We then say that a rational number p/q is a *secondary convergent* of ϑ if there are integers k and h such that $k \geqslant 1$, $1 \leqslant h < a_{k+1}$, and $p/q = p_{k,h}/q_{k,h}$.

(a) Determine the continued fraction expansion of $\vartheta := 1+\sqrt{2}$ and show that $\frac{17}{7}$ is a secondary convergent of ϑ.

(b) Show that $\frac{17}{7}$ is not a good approximation to $1+\sqrt{2}$. Hint: consider the convergent p_2/q_2 of ϑ and use the inequalities $\frac{7}{5} < \sqrt{2} < \frac{99}{70} < \frac{10}{5}$.

129. *Good approximations and secondary convergents.*

(a) Show that if the rational numbers with positive denominators a/b, r/s and c/d satisfy $a/b < r/s < c/d$ and $bc - ad = 1$, then $s > \max(b, d)$. Start by proving that $r/s - a/b \geqslant 1/bs$.

(b) Let ϑ be an irrational real number, let $\{p_k/q_k\}_{k\geqslant 0}$ be the sequence of convergents of ϑ and let $\{a_k\}_{k\geqslant 0}$ denote the sequence of its partial quotients. Calculate $p_{k,h+1}q_{k,h} - q_{k,h+1}p_{k,h}$ for all values of the integer parameters $k \geqslant 1$, $h \geqslant 0$. Deduce from this that, for any integer $k \geqslant 1$, the finite sequence $\{p_{k,h}/q_{k,h}\}_{h=0}^{a_{k+1}}$ is strictly increasing if k is odd, and strictly decreasing if k is even.

(c) Let p/q be a good approximation to ϑ such that $q \geqslant q_1$. Show that p/q is a convergent or a secondary convergent of ϑ. Hint: Assuming, for example, that $\vartheta < p/q$, show there exists some $k \geqslant 1$ such that

$$\vartheta < \frac{p_{k+1}}{q_{k+1}} < \frac{p}{q} \leqslant \frac{p_{k-1}}{q_{k-1}},$$

and use the results established in questions (b) and (a).

130. Show that if ϑ, ϑ', ϑ'' are three irrational numbers such that $\vartheta < \vartheta' < \vartheta''$, and if ϑ and ϑ'' have the same convergents up to rank k, then ϑ' also has the same convergents up to rank k.

131. Let ϑ be irrational and let p/q be a rational number such that $|q\vartheta - p| < 1/q$.

(a) Show that there are $n \in \mathbb{N}$, $a_0 \in \mathbb{Z}$, $a_1 \in \mathbb{N}^*, \ldots, a_n \in \mathbb{N}^*$ such that

$$\frac{p}{q} = [a_0, \ldots, a_n] \quad \text{and} \quad (-1)^n\Big(\vartheta - \frac{p}{q}\Big) > 0.$$

(b) Denote by $\{p_j/q_j\}_{j=0}^{n}$ the sequence of convergents of p/q and define α by the equation

$$\vartheta = \frac{\alpha p_n + p_{n-1}}{\alpha q_n + q_{n-1}},$$

with the usual convention that $(p_{-1}, q_{-1}) = (1, 0)$.

Show that $|q\vartheta - p| = 1/(\alpha q_n + q_{n-1})$ and infer that $\alpha > 0$.

(c) Show that, if $\alpha > 1$, then p/q is a convergent of ϑ.

(d) Show that, if $\alpha \leqslant 1$, then $n \geqslant 1$ and $\vartheta = [a_0, a_1, \cdots, a_n + 1/\alpha]$. Deduce that p/q is not a convergent of ϑ.

(e) Show that p/q is a convergent of ϑ if, and only if, we have

$$|q\vartheta - p| < \frac{1}{q_n + q_{n-1}}.$$

132. Let ϑ be an irrational real number and let p/q denote a rational number such that

$$|\vartheta^2 q^2 - p^2| < \vartheta.$$

Show that p/q is a convergent of ϑ. To this end, use the criterion established in Exercise 131(e) and distinguish between the cases $n = 0$, $n = 1$ and $n \geqslant 2$.

133. *Quadratic irrationals in normal form.*

(a) Show that any quadratic irrational real number ϑ may be written in the form $\vartheta = (u + \sqrt{d})/v$ where d is a non-square positive integer, and u, v are integers such that $v \neq 0$, $v|(d - u^2)$. We call such a representation a normal form of ϑ.

(b) Let $\vartheta = (u + \sqrt{d})/v$ be a quadratic irrational in normal form. We denote by $\{\alpha_n\}_{n=0}^{\infty}$ the sequence of its complete quotients and by $[a_0, a_1, \ldots]$ its continued fraction expansion. We further define two sequences $\{u_n\}_{n=0}^{\infty}$ and $\{v_n\}_{n=0}^{\infty}$ by the formulae

$$\begin{aligned} u_0 &= u, \quad v_0 = v \\ u_{n+1} &= a_n v_n - u_n \qquad (n \geqslant 0), \\ v_{n+1} &= (d - u_{n+1}^2)/v_n \qquad (n \geqslant 0). \end{aligned} \tag{7.17}$$

Show that, for any integer $n \geqslant 0$, we have

$$u_n \in \mathbb{Z},\ v_n \in \mathbb{Z}^*, \quad v_n|(d - u_n^2), \quad \alpha_n = (u_n + \sqrt{d})/v_n.$$

134. *Characterization of purely periodic expansions.*

Let $\vartheta = (u + \sqrt{d})/v$ be a real quadratic irrational number in normal form. Put $\vartheta' = (u - \sqrt{d})/v$. Let $\{\alpha_n\}_{n=0}^{\infty}$ be the sequence of complete quotients of ϑ and, for each $n \geqslant 0$, denote by α_n' the conjugate of α_n. The purpose of this exercise is to show that the condition

$$\vartheta > 1, \quad \vartheta' \in]-1, 0[\tag{7.18}$$

is necessary and sufficient for the continued fraction expansion of ϑ to be purely periodic, i.e., of the form $[\overline{a_0, a_1, \ldots, a_{h-1}}]$.

(a) Show that, if ϑ satisfies (7.18), then $1/\alpha_{n+1}' = \alpha_n' - a_n$ and $\alpha_n' \in]-1, 0[$ for all integers $n \geqslant 0$. Deduce that $a_n = \lfloor -1/\alpha_{n+1}' \rfloor$ $(n \geqslant 0)$ and thus that, if the relation $\alpha_j = \alpha_{j+h}$ holds for two integers $j \geqslant 0$, $h \geqslant 1$, we must have $\alpha_0 = \alpha_h$, and $\vartheta = [\overline{a_0, a_1, \ldots, a_{h-1}}]$.

(b) Conversely, show that if ϑ is a real quadratic irrational number whose continued fraction expansion is purely periodic, with period length $h \geqslant 1$, then $\vartheta > 1$ and ϑ is a root of the polynomial

$$f(x) := q_{h-1}x^2 + (q_{h-2} - p_{h-1})x - p_{h-2}.$$

Show that $f(-1) > 0$, $f(0) < 0$, and infer that $\vartheta' \in]-1, 0[$.

135. *Expansion of square roots.*

Let d be a non-square positive integer. Put $m_d := \lfloor \sqrt{d} \rfloor$.

(a) Show that $\vartheta := m_d + \sqrt{d}$ satisfies condition (7.18), and so has a purely periodic continued fraction expansion.

(b) Let h be the period length in the expansion of ϑ and let $\{\alpha_n\}_{n=0}^{\infty}$ denote the sequence of its complete quotients. Show that $\alpha_n = \alpha_0$ if, and only if, $n \equiv 0 \,(\mathrm{mod}\, h)$. Deduce that, with the notation of Exercise 133, we have $v_n = 1$ if, and only if, $n \equiv 0 \,(\mathrm{mod}\, h)$.

(c) Show that $\vartheta = [a_0, \overline{a_1, \ldots, a_0}]$ with $a_0 = 2m_d$. Deduce that the continued fraction expansion of $\sqrt{d}$ has the form

$$\sqrt{d} = \left[m_d, \overline{a_1, \ldots, a_{h-1}, 2m_d}\right].$$

136. *Solution to the Pell equation.*

Let d be a non-square positive integer. The Diophantine equation

$$x^2 - dy^2 = 1 \tag{7.19}$$

is called the Pell equation.

Denote by u_n and v_n the integers constructed from $u_0 = 0$, $v_0 = 1$ according to the rule (7.17), with $a_n := \left\lfloor (u_n + \sqrt{d})/v_n \right\rfloor$. Write $\{p_n/q_n\}_{n=0}^{\infty}$ for the sequence of convergents of $\sqrt{d}$ and let h designate the period length of its continued fraction expansion.

(a) Writing

$$\sqrt{d} = \frac{\alpha_{n+1}p_n + p_{n-1}}{\alpha_{n+1}q_n + q_{n-1}}$$

and appealing to the formulae established in Exercise 133, show that

$$p_n^2 - dq_n^2 = (-1)^{n+1}v_{n+1} \qquad (n \geqslant 0).$$

(b) Show that, for each integer $j \geqslant 0$, the pair (p_{2jh-1}, q_{2jh-1}) is a solution of (7.19).

(c) Show that any solution (x, y) of (7.19) with $x > 0$, $y > 0$, is such that x/y is a convergent of $\sqrt{d}$.

137. (a) Show that, for every interval I in $\mathbb{R}$ containing N integers, we have

$$\left|\sum_{n\in I} \mathrm{e}(n\vartheta)\right| \leqslant \min\Big(N, \frac{1}{2\|\vartheta\|}\Big),$$

where, by convention, the right-hand side equals N if $\vartheta \in \mathbb{Z}$. This convention will be employed systematically in the sequel.

(b) Show that, for any $\vartheta \in \mathbb{R} \setminus \mathbb{Q}$, we have

$$S_N(\vartheta) := \sum_{1\leqslant n\leqslant N} \mathrm{e}(n^2\vartheta) = o(N).$$

(c) Determine the set E of those real numbers ϑ such that the sequence $\{n^2\vartheta\}_{n=1}^{\infty}$ is equidistributed modulo 1.

(d) By applying the Weyl–van der Corput lemma (Lemma 6.8) with a suitable selection of the free parameter, show that, for any integer $N \geqslant 1$ and any real number ϑ, we have

$$|S_N(\vartheta)|^2 \leqslant 2 \sum_{0\leqslant m\leqslant 2N} \min\Big(2N, \frac{1}{\|m\vartheta\|}\Big).$$

(e) Show that, for any integer $N \geqslant 1$ and any real number ϑ, there exist integers $a_N \in \mathbb{Z}$, $q_N \in \mathbb{N}^*$ such that $(a_N, q_N) = 1$, $q_N \leqslant 4N$, and

$$\left|\vartheta - \frac{a_N}{q_N}\right| \leqslant \frac{1}{4Nq_N}.$$

Deduce that $\|m\vartheta\| \geqslant \frac{1}{2}\|a_N m/q_N\|$ $(1 \leqslant m \leqslant 2N)$. Hint: For each relevant value of m, introduce the unique integer $b_m \in [0, q_N/2]$ such that $a_N m \equiv \pm b_m \pmod{q_N}$ and distinguish between the cases $b_m = 0$ and $b_m \geqslant 1$.

(f) Show that, retaining the notation introduced in the previous questions, we have, for any integer $N \geqslant 1$ and any real number ϑ,

$$|S_N(\vartheta)|^2 \leqslant 4\Big(1 + \frac{2N}{q_N}\Big)\Big\{N + q_N \sum_{1\leqslant j\leqslant q_N/2} \frac{1}{j}\Big\}.$$

Deduce the existence of an absolute constant C such that

$$|S_N(\vartheta)| \leqslant C\Big\{\frac{N}{\sqrt{q_N}} + \sqrt{N\ln(2q_N)}\Big\}.$$

Part II

Complex Analysis Methods

The Euler Gamma function

0.1. Definitions

The original definition of the function dates back to 1729, in a letter from Euler to Goldbach:

$$\Gamma(s) := \frac{1}{s} \prod_{n \geqslant 1} \frac{(1+1/n)^s}{1+s/n} \qquad (s \neq 0, -1, -2, \ldots). \tag{0.1}$$

It is easy to see that the product converges uniformly on any compact subset of $\mathbb{C} \setminus (-\mathbb{N}^*)$ and thus defines a meromorphic function on $\mathbb{C}$.

Euler quickly deduces the integral formula

$$\Gamma(s) = \int_0^\infty t^{s-1} \mathrm{e}^{-t} \, \mathrm{d}t \qquad (\sigma > 0). \tag{0.2}$$

Here and in the sequel, we implicitly define the real and the imaginary parts of a complex number s by

$$s = \sigma + i\tau.$$

Theorem 0.1 (Euler). *Let*

$$\Gamma_n(s) := \int_0^n \left(1 - \frac{t}{n}\right)^n t^{s-1} \, \mathrm{d}t \qquad (\sigma > 0).$$

Then we have

$$\Gamma_n(s) = \frac{n^s n!}{s(s+1)\cdots(s+n)}, \tag{0.3}$$

and

$$\Gamma(s) = \lim_{n \to \infty} \Gamma_n(s) = \int_0^\infty t^{s-1} \mathrm{e}^{-t} \, \mathrm{d}t \qquad (\sigma > 0).$$

Proof. The change of variables $u = t/n$ allows us to write

$$\Gamma_n(s) = n^s \int_0^1 (1-u)^n u^{s-1}\,\mathrm{d}u.$$

We can then calculate the integral by successive partial integrations:

$$\begin{aligned}\int_0^1 (1-u)^n u^{s-1}\,\mathrm{d}u &= \left[\frac{u^s(1-u)^n}{s}\right]_0^1 + \frac{n}{s}\int_0^1 (1-u)^{n-1}u^s\,\mathrm{d}u \\ &= \cdots = \frac{n!}{s(s+1)\cdots(s+n-1)}\int_0^1 u^{s+n-1}\,\mathrm{d}u \\ &= \frac{n!}{s(s+1)\cdots(s+n)}.\end{aligned}$$

We conclude by noting that, on the one hand,

$$\begin{aligned}\Gamma_n(s) &= \frac{n^s n!}{s(s+1)\cdots(s+n)} = \frac{1}{s}(1+1/n)^{-s}\prod_{1\leqslant j\leqslant n}\frac{(1+1/j)^s}{1+s/j} \\ &= \Gamma(s) + o(1) \qquad (n\to\infty),\end{aligned}$$

and, on the other, that

$$\lim_{n\to\infty}\Gamma_n(s) = \int_0^\infty t^{s-1}\mathrm{e}^{-t}\,\mathrm{d}t.$$

This last formula can be derived elementarily,[1] and we omit the details. □

The following result shows that the meromorphic continuation can easily be recovered by taking (0.2) as the definition of Γ.

Theorem 0.2 (Functional equation). *We have*

$$\Gamma(s+1) = s\Gamma(s) \qquad (\sigma > 0).$$

Proof. An integration by parts immediately yields the result. □

Corollary 0.3. *For all $n \in \mathbb{N}$, we have $\Gamma(n+1) = n!$.*

The next property leads to another definition of Γ.

Theorem 0.4. *The Γ function is logarithmically convex on $\mathbb{R}^{+*}$.*

Proof. It suffices to apply the Cauchy–Schwarz inequality

$$\Gamma'(x)^2 = \left(\int_0^\infty \mathrm{e}^{-u}(\ln u)u^{x-1}\,\mathrm{d}u\right)^2 \leqslant \Gamma(x)\Gamma''(x).$$ □

[1] The historical perspective prevents us from employing here Lebesgue's dominated convergence theorem; however we can use the fact that $(1-t/n)^n \leqslant \mathrm{e}^{-t}$ for $0 \leqslant t \leqslant n$.

Emil Artin (1898–1962) showed that the functional equation and the logarithmic convexity property characterize the Γ function. We shall later see a remarkable application of this result.

Theorem 0.5 (Artin). *Let $\Phi :]0,\infty[\to]0,\infty[$ be a differentiable function such that $\ln\Phi$ is convex and $x\Phi(x) = \Phi(x+1)$ for all $x > 0$. Then we have $\Phi(x) = \Phi(1)\Gamma(x)$ for all $x > 0$.*

Proof. $H := \Phi/\Gamma$ is 1-periodic and $H(1) = \Phi(1)$. Since Φ'/Φ and Γ'/Γ are non-decreasing, we have for $x \geqslant 0$, $n \geqslant 1$,

$$\frac{\Phi'(n)}{\Phi(n)} - \frac{\Gamma'(n+1)}{\Gamma(n+1)} \leqslant \frac{H'(n+x)}{H(n+x)} = \frac{\Phi'(n+x)}{\Phi(n+x)} - \frac{\Gamma'(n+x)}{\Gamma(n+x)} \leqslant \frac{\Phi'(n+1)}{\Phi(n+1)} - \frac{\Gamma'(n)}{\Gamma(n)}.$$

However, Φ and Γ both satisfy the functional equation

$$\frac{f'(x+1)}{f(x+1)} = \frac{f'(x)}{f(x)} + \frac{1}{x},$$

thus

$$\frac{\Phi'(n)}{\Phi(n)} - \frac{\Gamma'(n+1)}{\Gamma(n+1)} = \frac{H'(n)}{H(n)} - \frac{1}{n}, \quad \frac{\Phi'(n+1)}{\Phi(n+1)} - \frac{\Gamma'(n)}{\Gamma(n)} = \frac{H'(n)}{H(n)} + \frac{1}{n}.$$

It follows that

$$\frac{H'(1)}{H(1)} - \frac{1}{n} = \frac{H'(n)}{H(n)} - \frac{1}{n} \leqslant \frac{H'(x)}{H(x)} \leqslant \frac{H'(n)}{H(n)} + \frac{1}{n} = \frac{H'(1)}{H(1)} + \frac{1}{n}.$$

Letting n tend to infinity, we obtain the existence of a constant k such that $H'(x)/H(x) = k$ for all $x > 0$, whence $H(x) = H(1)\mathrm{e}^{kx}$. Since H is periodic we must have $k = 0$. □

The Bohr–Mollerup theorem, proposed in Exercise 138, avoids the differentiability hypothesis in Artin's theorem. Another characterization is established in Exercise 139.

0.2. The Weierstrass product formula

Theorem 0.6 (Weierstrass). *For $\sigma > 0$, we have*

$$\frac{1}{\Gamma(s)} = s\mathrm{e}^{\gamma s}\prod_{j\geqslant 1}\Big(1+\frac{s}{j}\Big)\mathrm{e}^{-s/j}, \tag{0.4}$$

where γ denotes Euler's constant.

Furthermore, formula (0.4) *defines an analytic continuation of $1/\Gamma(s)$ as an entire function.*

Proof. Set $H_n := \sum_{1\leqslant j\leqslant n} 1/j$ $(n \geqslant 1)$, so that

$$H_n = \ln n + \gamma + o(1) \qquad (n \to \infty).$$

By (0.3), we have

$$\Gamma_n(s) = \frac{n^s}{s} \prod_{1\leqslant j\leqslant n} \frac{\mathrm{e}^{s/j}\mathrm{e}^{-s/j}}{1+s/j} = \frac{\mathrm{e}^{s(\ln n - H_n)}}{s} \prod_{1\leqslant j\leqslant n} \frac{\mathrm{e}^{s/j}}{1+s/j}.$$

We derive the stated formula by letting n tend to infinity, since the general term of the product is $1 + O(1/j^2)$ uniformly on any compact subset of $\mathbb{C} \setminus (-\mathbb{N})$. □

Corollary 0.7. *We have* $\gamma = -\Gamma'(1)$.

Proof. On taking a logarithmic derivative, formula (0.4) yields

$$\frac{-\Gamma'(s)}{\Gamma(s)} = \gamma + \frac{1}{s} + \sum_{j\geqslant 1} \frac{1}{s+j} - \frac{1}{j}. \tag{0.5}$$

For $s = 1$ the series is telescopic and sums to -1. □

0.3. The Beta function

The Beta function, or Euler's integral of the first kind, is defined by

$$B(x,y) := \int_0^1 t^{x-1}(1-t)^{y-1}\,\mathrm{d}t \qquad (x>0,\, y>0).$$

Theorem 0.8. *We have*

$$B(x,y) = \Gamma(x)\Gamma(y)/\Gamma(x+y) \qquad (x>0,\, y>0). \tag{0.6}$$

First proof: change of variables. We have

$$\Gamma(x)\Gamma(y) = \int_0^\infty t^{x-1}\mathrm{e}^{-t}\,\mathrm{d}t \int_0^\infty u^{y-1}\mathrm{e}^{-u}\,\mathrm{d}u.$$

From the change of variables $u = tv$ and Fubini's theorem, we obtain

$$\begin{aligned}
\Gamma(x)\Gamma(y) &= \int_0^\infty t^{x-1}\mathrm{e}^{-t}\,\mathrm{d}t \int_0^\infty t^y v^{y-1}\mathrm{e}^{-vt}\,\mathrm{d}v \\
&= \int_0^\infty v^{y-1}\,\mathrm{d}v \int_0^\infty t^{x+y-1}\mathrm{e}^{-(v+1)t}\,\mathrm{d}t = \int_0^\infty \frac{v^{y-1}}{(1+v)^{x+y}}\Gamma(x+y)\,\mathrm{d}v \\
&= \Gamma(x+y)\int_0^\infty \Big(\frac{v}{1+v}\Big)^{y-1}\Big(\frac{1}{1+v}\Big)^{x-1}\frac{\mathrm{d}v}{(1+v)^2} \\
&= \Gamma(x+y)\int_0^1 w^{y-1}(1-w)^{x-1}\,\mathrm{d}w = \Gamma(x+y)B(x,y).
\end{aligned}$$

□

Second proof: via Artin's theorem. Let $y > 0$ be fixed and put $f(x) := \Gamma(x+y)B(x,y)/\Gamma(y)$. We thus have to show that $f(x) = \Gamma(x)$. We have $f(1) = yB(1,y) = 1$ and

$$f(x+1) = \frac{(x+y)\Gamma(x+y)}{\Gamma(y)} B(x+1,y).$$

Since

$$\begin{aligned} B(x+1,y) &= \int_0^1 t^x (1-t)^{y-1}\,\mathrm{d}t = \int_0^1 \Big(\frac{t}{1-t}\Big)^x (1-t)^{x+y-1}\,\mathrm{d}t \\ &= \Big[\frac{(1-t)^{x+y}}{-(x+y)}\Big(\frac{t}{1-t}\Big)^x\Big]_0^1 + \frac{x}{x+y}\int_0^1 (1-t)^{x+y}\Big(\frac{t}{1-t}\Big)^{x-1}\frac{\mathrm{d}t}{(1-t)^2} \\ &= \frac{x}{x+y} B(x,y), \end{aligned}$$

we deduce from this that

$$f(x+1) = xf(x).$$

Finally, for $1/p + 1/q = 1$, we can write, by Hölder's inequality,

$$\begin{aligned} B(x/p + z/q, y) &= \int_0^1 t^{(x-1)/p + (z-1)/q}(1-t)^{(y-1)/p + (y-1)/q}\,\mathrm{d}t \\ &\leqslant B(x,y)^{1/p} B(z,y)^{1/q}, \end{aligned}$$

which establishes that $x \mapsto B(x,y)$, and therefore also $x \mapsto f(x)$, is log-convex.

Artin's theorem then allows us to deduce from the above that $f(x) = \Gamma(x)$. □

Corollary 0.9. *We have for $x > 0$, $y > 0$,*

$$\Gamma(x)\Gamma(y)/\Gamma(x+y) = 2\int_0^{\pi/2} (\sin\vartheta)^{2x-1}(\cos\vartheta)^{2y-1}\,\mathrm{d}\vartheta.$$

In particular $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.

Proof. It suffices to make the change of variables $t = (\sin\vartheta)^2$ in (0.6). □

We also note that $\Gamma(\frac{1}{2}) = \int_0^\infty t^{-1/2}\mathrm{e}^{-t}\,\mathrm{d}t = 2\int_0^\infty \mathrm{e}^{-u^2}\,\mathrm{d}u$, whence

$$\int_{\mathbb{R}} \mathrm{e}^{-u^2}\,\mathrm{d}u = \sqrt{\pi}. \tag{0.7}$$

Corollary 0.10 (Real Stirling's formula). *We have*

$$\Gamma(x+1) \sim x^x \mathrm{e}^{-x}\sqrt{2\pi x} \qquad (x \to \infty). \tag{0.8}$$

Proof. We employ here a classical method of asymptotic analysis, consisting in emphasizing, through a suitable change of variable, the dominant domain in an integral depending on a parameter.

The function $t^x e^{-t}$, whose integral over $\mathbb{R}^+$ equals $\Gamma(x+1)$, is maximal at $t = x$. The change of variables $t = x(1+u)$ furnishes

$$\begin{aligned}\Gamma(x+1) &= \int_{-1}^{\infty} \{x(1+u)\}^x e^{-x(1+u)} x \, du \\ &= x^{x+1} e^{-x} \int_{-1}^{\infty} \{(1+u)e^{-u}\}^x \, du \\ &= x^x e^{-x} \sqrt{2x} \int_{-\sqrt{x/2}}^{\infty} e^{-v^2 h\left(v\sqrt{2/x}\right)} \, dv,\end{aligned} \tag{0.9}$$

where we set $u = v\sqrt{2/x}$ and

$$h(w) := \frac{2}{w^2}\{w - \ln(1+w)\},$$

so that $h(0) = 1$. Write

$$g_x(v) := \mathbf{1}_{[-\sqrt{x/2},\infty[}(v) e^{-v^2 h\left(v\sqrt{2/x}\right)}.$$

For any fixed real v, we have

$$\lim_{x\to\infty} g_x(v) = e^{-v^2}.$$

Furthermore, noticing that

$$h(w) = \frac{2}{w^2}\int_0^w \frac{t\,dt}{1+t} = 2\int_0^1 \frac{z\,dz}{1+zw},$$

we see that $h(w) \geqslant 1$ for $w \leqslant 0$ and that $h(w)$ is non-increasing for $w \geqslant 0$, whence

$$h\left(v\sqrt{2/x}\right) \geqslant h\left(v\sqrt{2}\right) \geqslant \frac{v\sqrt{2} - \ln(1+v\sqrt{2})}{v^2} \qquad (x \geqslant 1).$$

It follows that

$$g_x(v) \leqslant e^{-v^2}\mathbf{1}_{]-\infty,0[}(v) + (1+v\sqrt{2})e^{-v\sqrt{2}}\mathbf{1}_{[0,\infty[}(v).$$

This allows us to apply Lebesgue's theorem, to obtain

$$\lim_{x\to\infty}\int_{\mathbb{R}} g_x(v)\,dv = \int_{\mathbb{R}} e^{-v^2}\,dv = \sqrt{\pi},$$

in view of (0.7). We deduce the desired formula by substitution in (0.9). □

Corollary 0.11 (Legendre duplication formula). *For $x > 0$, we have*

$$\Gamma\Big(\frac{x}{2}\Big)\Gamma\Big(\frac{x+1}{2}\Big) = \sqrt{\pi}2^{1-x}\Gamma(x). \tag{0.10}$$

Proof. Consider the function $f(x) := 2^{x-1}\pi^{-1/2}\Gamma\big(\frac{1}{2}x\big)\Gamma\big(\frac{1}{2}(x+1)\big)$. We have $f(1) = 1$ from (0.7) and

$$\begin{aligned} f(x+1) &= 2^x\pi^{-1/2}\Gamma\big(\tfrac{1}{2}(x+1)\big)\Gamma\big(\tfrac{1}{2}x+1\big) \\ &= 2^x\pi^{-1/2}\Gamma\big(\tfrac{1}{2}(x+1)\big)\tfrac{1}{2}x\Gamma\big(\tfrac{1}{2}x\big) = xf(x). \end{aligned}$$

Finally, f is logarithmically convex since

$$\ln f(x) = (x-1)\ln 2 - \tfrac{1}{2}\ln\pi + \ln\Big\{\Gamma\big(\tfrac{1}{2}(x+1)\big)\Gamma\big(\tfrac{1}{2}x\big)\Big\}.$$

By Artin's theorem, we deduce that $f(x) = \Gamma(x)$. □

A generalization of this result to the product $\prod_{j=0}^{n-1}\Gamma\left((x+j)/n\right)$ $(n \geqslant 2)$ may be found in Exercise 147.

0.4. Complex Stirling's formula

Recall the definitions of the Bernoulli numbers and functions from § I.0.2.

Theorem 0.12 (Complex Stirling's formula). *For any $s \in \mathbb{C}\setminus\mathbb{R}^-$, we have*

$$\log\Gamma(s) = (s-\tfrac{1}{2})\log s - s + \tfrac{1}{2}\ln(2\pi) - \int_0^\infty B_1(t)\frac{\mathrm{d}t}{s+t}, \tag{0.11}$$

where, on the right-hand side, the complex logarithm is understood as its principal branch.

Remarks. (i) The Weierstrass product formula shows that $\Gamma(s)$ does not vanish and that its only poles are at negative integers.

(ii) The integral is uniformly $O(1/s)$ on every sector $s \neq 0$, $|\arg(s)| \leqslant \pi - \delta$, where δ is an arbitrary positive constant. Indeed, we have for every integer $R \geqslant 1$

$$\int_0^\infty B_1(t)\frac{\mathrm{d}t}{s+t} = \sum_{1\leqslant r\leqslant R}\frac{(-1)^r B_{r+1}}{r(r+1)s^r} + \frac{(-1)^{R+1}}{R+1}\int_0^\infty \frac{B_{R+1}(t)}{(t+s)^{R+1}}\,\mathrm{d}t.$$

Proof. Applying the Euler–Maclaurin summation formula at order 0 to the function $f(t) := \log(t+s)$ for $s \in \mathbb{C}\setminus\mathbb{R}^-$ with $a = 0$ and $b = N$, we obtain

$$\begin{aligned} \sum_{1\leqslant n\leqslant N}\log(n+s) &= \int_0^N \log(t+s)\,\mathrm{d}t + \tfrac{1}{2}\big\{\log(N+s) - \log s\big\} + \int_0^N \frac{B_1(t)\,\mathrm{d}t}{t+s} \\ &= (s+N)\log(s+N) - (s+N) - s\log s + s + \tfrac{1}{2}\log(s+N) \\ &\quad - \tfrac{1}{2}\log s + \int_0^N \frac{B_1(t)\,\mathrm{d}t}{t+s} \\ &= (s+N+\tfrac{1}{2})\log(s+N) - N - (s+\tfrac{1}{2})\log s + \int_0^N \frac{B_1(t)\,\mathrm{d}t}{t+s}. \end{aligned}$$

Subtract the classic Stirling's formula for $\ln N!$ in the form

$$\ln N! = (N+\tfrac{1}{2})\ln N - N + \tfrac{1}{2}\ln(2\pi) + o(1) \qquad (N \to \infty).$$

We obtain

$$\begin{aligned}
&\sum_{1\leqslant n\leqslant N} \log\Big(1+\frac{s}{n}\Big) \\
&\quad = (N+\tfrac{1}{2})\log\Big(1+\frac{s}{N}\Big) + s\log(s+N) - (s+\tfrac{1}{2})\log s \\
&\qquad - \tfrac{1}{2}\ln(2\pi) + \int_0^N \frac{B_1(t)\,\mathrm{d}t}{t+s} + o(1) \\
&\quad = s\big(1+\ln N\big) - (s+\tfrac{1}{2})\log s - \tfrac{1}{2}\ln(2\pi) + \int_0^\infty \frac{B_1(t)\,\mathrm{d}t}{t+s} + o(1).
\end{aligned}$$

Setting $H_N := \sum_{1\leqslant n\leqslant N} 1/n$, it follows, for example for $s \in \mathbb{R}^{+*}$, that

$$\begin{aligned}
&\ln\Big\{ s\mathrm{e}^{\gamma s} \prod_{1\leqslant n\leqslant N} \Big(1+\frac{s}{n}\Big)\mathrm{e}^{-s/n} \Big\} \\
&\quad = \ln s + s(\gamma - H_N + 1 + \ln N) - (s+\tfrac{1}{2})\ln s - \tfrac{1}{2}\ln(2\pi) + \int_0^\infty \frac{B_1(t)\,\mathrm{d}t}{t+s} + o(1) \\
&\quad = -(s-\tfrac{1}{2})\ln s + s - \tfrac{1}{2}\ln(2\pi) + \int_0^\infty \frac{B_1(t)\,\mathrm{d}t}{t+s} + o(1).
\end{aligned}$$

Letting $N \to \infty$, we obtain the stated result, first for $s \in \mathbb{R}^+$, then for all s by analytic continuation. □

Corollary 0.13 (Behavior in vertical strips). *Let σ_1, σ_2 be real numbers such that $\sigma_1 < \sigma_2$. We have uniformly for $\sigma_1 \leqslant \sigma \leqslant \sigma_2$, $|\tau| \to \infty$,*

$$\Gamma(s) = \Big\{1 + O\Big(\frac{1}{\tau}\Big)\Big\}\sqrt{2\pi}|\tau|^{\sigma-\frac{1}{2}}\mathrm{e}^{-\pi|\tau|/2}\mathrm{e}^{ih_\sigma(\tau)} \tag{0.12}$$

where $h_\sigma(\tau) := \tau\ln|\tau| - \tau + \frac{1}{2}\pi(\sigma - \frac{1}{2})\operatorname{sgn}(\tau)$.

Proof. Observe that $\arg s = \operatorname{sgn}(\tau)\frac{1}{2}\pi - \sigma/\tau + O(1/\tau^2) \in]-\pi, \pi[$, and so we have $\log s = \ln|s| + i\arg s$, with

$$\ln|s| = \tfrac{1}{2}\ln(\sigma^2+\tau^2) = \ln|\tau| + O(1/\tau^2).$$

Integration by parts yields

$$\int_0^\infty \frac{B_1(t)\,\mathrm{d}t}{t+s} = O\Big(\frac{1}{|s|}\Big).$$

It follows that

$$\begin{aligned}
(s-\tfrac{1}{2})\log s &= (\sigma - \tfrac{1}{2} + i\tau)\Big(\ln|\tau| + i\Big\{\tfrac{1}{2}\pi\operatorname{sgn}\tau - \frac{\sigma}{\tau}\Big\}\Big) + O\Big(\frac{1}{\tau^2}\Big) \\
&= (\sigma-\tfrac{1}{2})\ln|\tau| - \tfrac{1}{2}\pi|\tau| + \sigma + ih_\sigma(\tau) + O(1/\tau).
\end{aligned}$$

□

Corollary 0.14 (Mellin inversion formula). *For $x > 0$, we have*

$$\mathrm{e}^{-x} = \frac{1}{2\pi i}\int_{\sigma-i\infty}^{\sigma+i\infty} \Gamma(s)x^{-s}\,\mathrm{d}s \qquad (\sigma > 0). \tag{0.13}$$

Proof. For $\sigma > 0$, we have

$$\Gamma(\sigma - i\tau) = \int_0^\infty \mathrm{e}^{-x}x^{\sigma-i\tau}\frac{\mathrm{d}x}{x} = \int_{\mathbb{R}} \mathrm{e}^{-\mathrm{e}^u+\sigma u}\mathrm{e}^{-i\tau u}\,\mathrm{d}u = \widehat{f_\sigma}(\tau),$$

where $f_\sigma(u) := \exp\{-\mathrm{e}^u + \sigma u\}$. Since $\widehat{f_\sigma} \in \mathcal{L}^1(\mathbb{R})$ by Corollary 0.13,[2] the Fourier inversion formula is applicable. We obtain

$$\begin{aligned} f_\sigma(u)\mathrm{e}^{-\sigma u} &= \frac{\mathrm{e}^{-\sigma u}}{2\pi}\int_{\mathbb{R}} \widehat{f_\sigma}(\tau)\mathrm{e}^{i\tau u}\,\mathrm{d}\tau \\ &= \frac{1}{2\pi}\int_{\mathbb{R}} \Gamma(\sigma - i\tau)\mathrm{e}^{-(\sigma-i\tau)u}\,\mathrm{d}\tau = \frac{1}{2\pi}\int_{\mathbb{R}} \Gamma(\sigma + i\tau)\mathrm{e}^{-(\sigma+i\tau)u}\,\mathrm{d}\tau. \end{aligned}$$

The stated result follows by putting $x = \mathrm{e}^u$. □

Corollary 0.15 (Reflection formula). *For $s \in \mathbb{C} \setminus \mathbb{Z}$, we have*

$$\Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin(\pi s)}. \tag{0.14}$$

Proof. Consider the meromorphic function

$$f(s) := \frac{1}{\Gamma(s)\Gamma(1-s)\sin\pi s} = \frac{s(1-s)\mathrm{e}^\gamma}{\sin\pi s}\prod_{j\geqslant 1}\left(1 + \frac{1}{j} + \frac{s(1-s)}{j^2}\right)\mathrm{e}^{-1/j},$$

where the expansion follows from Weierstrass's formula. Since all zeros of $\sin\pi s$ are offset by those in the infinite product,[3] f is entire. Moreover

$$\begin{aligned} f(s+1) &= \frac{-1}{\Gamma(s+1)\Gamma(-s)\sin\pi s} = \frac{1}{\Gamma(s)(-s)\Gamma(-s)\sin\pi s} \\ &= \frac{1}{\Gamma(s)\Gamma(1-s)\sin\pi s} = f(s), \end{aligned}$$

so f is 1-periodic. For $\sigma \in [0,1]$ and $|\tau| \to \infty$, we have by (0.12)

$$\frac{1}{|f(s)|} \sim |\tau|^{\sigma-\frac12}\mathrm{e}^{-\pi|\tau|/2}\sqrt{2\pi}|\tau|^{\frac12-\sigma}\mathrm{e}^{-\pi|\tau|/2}\sqrt{2\pi}|\sin\pi s| \sim 2\pi\mathrm{e}^{-\pi|\tau|}|\sin\pi s|.$$

However, $\sin\pi s = \sin\pi\sigma\cosh\pi\tau + i\sinh\pi\tau\cos\pi\sigma$, hence

$$|\sin\pi s|^2 = \sin^2\pi\sigma\cosh^2\pi\tau + \sinh^2\pi\tau\cos^2\pi\sigma \sim \tfrac14\mathrm{e}^{2\pi|\tau|}.$$

---

[2] The assumption $\sigma > 0$ guarantees that $\Gamma(\sigma + i\tau)$ is well-defined for all $\tau \in \mathbb{R}$.

[3] Note that $1 + 1/j + s(1-s)/j^2 = (s+j)(j+1-s)/j^2$.

It follows that $\lim_{|\tau|\to\infty} |f(s)| = 1/\pi$, thus f is bounded in the vertical strip $0 \leqslant \sigma \leqslant 1$. By periodicity it is therefore bounded in $\mathbb{C}$, so that, by Liouville's theorem, it must be constant. Since we also have

$$f(s) = \frac{1}{\Gamma(s)\Gamma(1-s)\sin \pi s} = \frac{s}{\Gamma(1+s)\Gamma(1-s)\sin \pi s} \to \frac{1}{\pi} \qquad (s \to 0),$$

we conclude that

$$f(s) = 1/\pi \qquad (s \in \mathbb{C}). \qquad \square$$

Two alternative proofs of the reflection formula are proposed in Exercises 150 and 151.

Remark. When the line of integration in (0.13) is moved to the left, the integral on the shifted line is

$$\ll \int_{\mathbb{R}} |s|^{\sigma-\frac{1}{2}} \mathrm{e}^{-\pi|\tau|/2} x^{-\sigma}\, \mathrm{d}\tau.$$

For sufficiently large $|\sigma|$, we have $|s|/x > 1$, so, by Lebesgue's theorem, this integral tends to 0 as $\sigma \to -\infty$. Now, it follows from (0.14) that, as $s \to 0$,

$$\Gamma(s-n) = \frac{\pi}{\Gamma(n+1-s)\sin\{\pi(s-n)\}} \sim \frac{(-1)^n}{n!s},$$

so the residue of $\Gamma(s)$ at $s = -n$ equals $(-1)^n/n!$. The residue theorem therefore allows us to recover the power series expansion

$$\mathrm{e}^{-x} = \sum_{n\geqslant 0} \frac{(-1)^n}{n!} x^n.$$

Corollary 0.16 (Euler). *For all $z \in \mathbb{C}$, we have*

$$\frac{\sin \pi z}{\pi z} = \prod_{n\geqslant 1} \Big(1 - \frac{z^2}{n^2}\Big). \tag{0.15}$$

Proof. This is an immediate consequence of the reflection and Weierstrass' formulae since

$$\begin{aligned} \frac{\sin \pi z}{\pi z} &= \frac{1}{z\Gamma(z)\Gamma(1-z)} = \frac{-1}{z^2\Gamma(z)\Gamma(-z)} \\ &= \mathrm{e}^{\gamma z} \prod_{n\geqslant 1} \Big(1 + \frac{z}{n}\Big)\mathrm{e}^{-z/n} \mathrm{e}^{-\gamma z} \prod_{n\geqslant 1} \Big(1 + \frac{-z}{n}\Big)\mathrm{e}^{z/n} = \prod_{n\geqslant 1} \Big(1 - \frac{z^2}{n^2}\Big). \ \square \end{aligned}$$

By logarithmic differentiation, Euler's formula furnishes

$$\pi \cot \pi z = \frac{1}{z} + \sum_{n\geqslant 1} \frac{2z}{z^2 - n^2} \qquad (z \in \mathbb{C} \setminus \mathbb{Z}), \tag{0.16}$$

from which we get, by letting z tend to 0, that

$$\sum_{n\geqslant 1} \frac{1}{n^2} = \tfrac{1}{6}\pi^2. \tag{0.17}$$

0.5. Hankel's formula

As we saw above, the Weierstrass product formula implies that $1/\Gamma$ is an entire function. Hankel's formula provides us with an integral representation. For each value of the positive parameter r, we designate as *Hankel contour* the path consisting of the circle $|s| = r$ excluding the point $s = -r$, and of the half-line $]-\infty, -r]$ covered twice, with respective arguments $+\pi$ and $-\pi$.

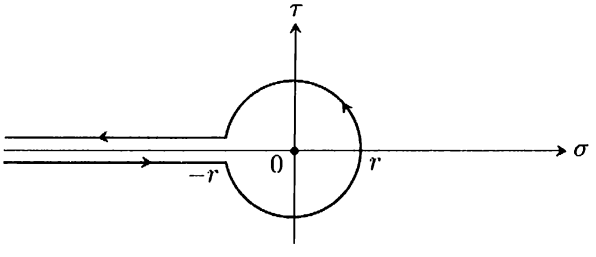

Figure 1. Hankel's contour.

Theorem 0.17 (Hankel's formula). *Let $\mathcal{H}$ be a Hankel contour. For any complex number z, we have*

$$\frac{1}{\Gamma(z)} = \frac{1}{2\pi i}\int_{\mathcal{H}} s^{-z}\mathrm{e}^{s}\,\mathrm{d}s. \tag{0.18}$$

Proof. The integral is absolutely convergent for each z and uniformly convergent on each compact set. It thus defines an entire function of z which, by the residue theorem, is independent of r since the only singularity of the integrand occurs at $s = 0$. When $\Re\mathrm{e}\, z < 1$, the integral round the circular part $|s| = r$ of the Hankel contour tends to 0 with r, while the integral over the double half-line tends to

$$\frac{1}{2\pi i}\int_0^\infty (\mathrm{e}^{i\pi z} - \mathrm{e}^{-i\pi z})\frac{\mathrm{d}\sigma}{\sigma^z\mathrm{e}^\sigma} = \frac{\sin \pi z}{\pi}\int_0^\infty \frac{\mathrm{d}\sigma}{\sigma^z\mathrm{e}^\sigma} = \frac{\sin \pi z}{\pi}\Gamma(1-z) = \frac{1}{\Gamma(z)}\cdot$$

This establishes (0.18) when $\Re\mathrm{e}\, z < 1$, and thus for all z, by analytic continuation. □

It is often useful to have at one's disposal an effective version of Hankel's formula, where the integral is over a contour of finite length.

Corollary 0.18. *For each $X > 1$, let $\mathcal{H}(X)$ denote the part of the Hankel contour situated in the half-plane $\sigma > -X$. Then we have, uniformly for $z \in \mathbb{C}$,*

$$\frac{1}{2\pi i}\int_{\mathcal{H}(X)} s^{-z}\mathrm{e}^s\,\mathrm{d}s = \frac{1}{\Gamma(z)} + O\big(47^{|z|}\,\Gamma(1+|z|)\,\mathrm{e}^{-\frac{1}{2}X}\big). \tag{0.19}$$

Proof. For $s = \sigma\mathrm{e}^{\pm i\pi}, \sigma > 1$, we have $|s^{-z}\mathrm{e}^s| \leqslant (\mathrm{e}^\pi\sigma)^{|z|}\mathrm{e}^{-\sigma}$. The difference between the left-hand side of (0.19) and the integral (0.18) is thus

$$\ll \mathrm{e}^{\pi|z|}\int_X^\infty \sigma^{|z|}\mathrm{e}^{-\sigma}\,\mathrm{d}\sigma \leqslant \mathrm{e}^{\pi|z|-\frac{1}{2}X}\int_0^\infty \sigma^{|z|}\mathrm{e}^{-\frac{1}{2}\sigma}\,\mathrm{d}\sigma.$$

The change of variable $\sigma = 2t$ provides the stated upper bound since $2\mathrm{e}^\pi < 47$. □

Exercises

138. *The Bohr–Mollerup theorem.* Let $\varphi : \mathbb{R} \to \mathbb{R}$ be a convex function such that $\varphi(1) = 0$, $\varphi(x+1) = \varphi(x) + \ln x$ $(x > 0)$.

(a) Show that, for any integer $n \geqslant 1$, the function

$$\psi_n(x) := \{\varphi(n+1+x) - \varphi(n+1)\}/x$$

is non-decreasing on $[-1, 0[\cup]0, 1]$.

(b) Deduce that

$$0 \leqslant \varphi(x) - \ln\Big(n!\, n^x / \prod_{0 \leqslant j \leqslant n} (j+x)\Big) \leqslant 1/n \qquad (0 < x \leqslant 1,\ n \geqslant 1).$$

(c) Show that φ is uniquely determined and satisfies Gauss' formula

$$\mathrm{e}^{\varphi(x)} = \lim_{n\to\infty} n!\, n^x \prod_{0 \leqslant j \leqslant n} (j+x)^{-1} \qquad (x > 0).$$

139. *Another characterization of* Γ.

(a) Show that $\Gamma'(x)/\Gamma(x) < \ln x$ $(x > 0)$. Deduce that $x \mapsto \mathrm{e}^x x^{-x}\Gamma(x)$ is non-increasing for $x > 0$.

(b) Show that Γ is uniquely determined by the decreasingness property in (a) and the conditions $\Gamma(1) = 1$, $\Gamma(x+1) = x\Gamma(x)$.

140. *Gauss' proof of Euler's formula.*

(a) Show that, for every integer $n \geqslant 0$ and every complex number s such that $\sigma > 0$,

$$\frac{\Gamma'(s)}{\Gamma(s)} = -\sum_{0 \leqslant j < n} \frac{1}{s+j} + \frac{\Gamma'(s+n)}{\Gamma(s+n)}.$$

(b) Show that $\Gamma'(s)/\Gamma(s) = \ln s + O(1/s)$ for $s \in \mathbb{R}$, $s > 1$.

(c) Deduce Euler's formula $\Gamma(s) = \lim_{n\to\infty} n^s\, n! \prod_{0 \leqslant j \leqslant n} 1/(s+j)$.

141. Let $\{a_j\}_{j=1}^k$ and $\{b_j\}_{j=1}^k$ be two finite complex sequences such that

$$b_j \notin \mathbb{Z} \quad (1 \leqslant j \leqslant k), \qquad \sum_{1 \leqslant j \leqslant k} a_j = \sum_{1 \leqslant j \leqslant k} b_j.$$

Show that

$$\prod_{n \geqslant 1} \prod_{1 \leqslant j \leqslant k} \frac{n - a_j}{n - b_j} = \prod_{1 \leqslant j \leqslant k} \frac{\Gamma(1-b_j)}{\Gamma(1-a_j)}.$$

142. Let $k \geqslant 2$ and put $\varepsilon_k = \mathrm{e}^{2\pi i/k}$. Using the result of Exercise 139, show that

$$\prod_{n \geqslant 1} \Big(1 - \frac{x}{n^k}\Big) = \prod_{0 \leqslant j < k} \Gamma\big(1 - \varepsilon_k^j x^{1/k}\big)^{-1} \quad (x > 0).$$

143. Show that $\displaystyle\int_{-1}^{1}(1+t)^{x-1}(1-t)^{y-1}\,\mathrm{d}t = 2^{x+y-1}\frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$ $(x>0,\ y>0)$.

144. *Dirichlet's integral.*

Show that, for $f\in\mathcal{C}[0,1]$, $\alpha_j>0$ $(1\leqslant j\leqslant n)$, $\beta=\sum_{j=1}^{n}\alpha_j$, we have

$$\int_{\sum_{j=1}^{n}t_j\leqslant 1} f\Big(\sum_{1\leqslant j\leqslant n} t_j\Big)\prod_{1\leqslant j\leqslant n}\frac{\mathrm{d}t_j}{t_j^{1-\alpha_j}} = \frac{\prod_{j=1}^{n}\Gamma(\alpha_j)}{\Gamma(\beta)}\int_0^1 f(u)u^{\beta-1}\,\mathrm{d}u.$$

145. Compute the double integral $\int_\Delta x^\alpha y^\beta\,\mathrm{d}x\,\mathrm{d}y$, where α and β are positive parameters and Δ is the domain $x\geqslant 0$, $y\geqslant 0$, $x^\mu+y^\nu\leqslant 1$ $(\mu>0,\ \nu>0)$.

146. Calculate $\int_z^{z+1}\log\Gamma(w)\,\mathrm{d}w$ for all $z\in\mathbb{C}\setminus\mathbb{R}^-$.

147. *The Legendre–Gauss formula.*

(a) Using Legendre's formula, show that $\int_0^1\ln\Gamma(x)\,\mathrm{d}x=\frac{1}{2}\ln(2\pi)$.

(b) Show that for any integer $n\geqslant 2$ we have

$$\prod_{0\leqslant j<n}\Gamma\Big(\frac{x+j}{n}\Big) = (2\pi)^{\frac{1}{2}(n-1)}n^{\frac{1}{2}-x}\Gamma(x)\qquad(x>0).$$

148. (a) By directly using the definition of $\Gamma(s)$ in integral form, show that

$$-\Gamma'(1) = \ln x - \frac{1}{x}\int_0^\infty(\ln t)\mathrm{e}^{-t/x}\,\mathrm{d}t\qquad(x>1).$$

(b) Employ the approximation $\ln t=\sum_{n\leqslant t}1/n-\gamma+O(1/t)$ $(t>1)$ to recover the formula

$$\Gamma'(1)=-\gamma.$$

149. *Formulae of Gauss and Dirichlet.*

(a) Show that $\displaystyle\gamma=\int_0^1\frac{1-\mathrm{e}^{-t}}{t}\,\mathrm{d}t-\int_1^\infty\frac{\mathrm{e}^{-t}}{t}\,\mathrm{d}t$ and then that

$$\gamma=\int_0^\infty\Big(\frac{1}{1-\mathrm{e}^{-t}}-\frac{1}{t}\Big)\mathrm{e}^{-t}\,\mathrm{d}t.$$

(b) Show that for $\Re\mathrm{e}\,z>0$ we have $\displaystyle\frac{\Gamma'(z)}{\Gamma(z)}=-\gamma-\frac{1}{z}-\sum_{n\geqslant 1}\frac{z}{n(z+n)}$.

Using (a) and writing $1/(z+n)=\int_0^\infty\mathrm{e}^{-t(z+n)}\,\mathrm{d}t$ $(\Re\mathrm{e}\,z>0)$, show Gauss' formula

$$\frac{\Gamma'(z)}{\Gamma(z)}=\int_0^\infty\Big(\frac{\mathrm{e}^{-t}}{t}-\frac{\mathrm{e}^{-zt}}{1-\mathrm{e}^{-t}}\Big)\,\mathrm{d}t\qquad(\Re\mathrm{e}\,z>0).$$

(c) Show that $\lim_{\varepsilon\to 0}\int_\varepsilon^\infty \Big(\frac{1}{(1+x)^2\ln(1+x)} - \frac{\mathrm{e}^{-x}}{x}\Big)\,\mathrm{d}x = 0$. Using (b), deduce Dirichlet's formula

$$\frac{\Gamma'(z)}{\Gamma(z)} = \int_0^\infty \big(\mathrm{e}^{-x} - (1+x)^{-z}\big)\frac{\mathrm{d}x}{x} \qquad (\Re\mathrm{e}\, z > 0).$$

150. *Another proof of the reflection formula.*

(a) For $\alpha \in]0,1[$, show the existence of the integral

$$I(\alpha) := \int_0^\infty t^{\alpha-1}\,\mathrm{d}t/(1+t)$$

and compute its value. Hint: use the method of residues, or treat first the case $\alpha = p/q \in \mathbb{Q}$ by the change of variable $t = u^q$, and conclude by a continuity argument.

(b) Deduce from (a) that the reflection formula $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$ holds for $z \in \mathbb{C}\setminus\mathbb{Z}$.

151. (a) Let $T_N(\xi)$ be the polynomial determined by the equation

$$T_N(\sin x) = \sin\{(2N+1)x\}.$$

Show that

$$T_N(\xi) = (2N+1)\xi \prod_{1\leqslant k\leqslant N}\Big(1 - \xi^2/\sin^2\frac{\pi k}{2N+1}\Big).$$

(b) Choose $\xi = \sin\{\pi x/(2N+1)\}$ and let N tend to infinity, carefully justifying the limit process, to obtain Euler's formula

$$\sin(\pi x) = \pi x \prod_{k\geqslant 1}\big(1 - x^2/k^2\big).$$

(c) From the Weierstrass product formula, deduce the reflection formula for $\Gamma(x)$.

152. (a) Prove the formulae

$$\int_0^\infty t^{z-1}\cos t\,\mathrm{d}t = \Gamma(z)\cos\big(\tfrac{1}{2}\pi z\big) \qquad (0 < \Re\mathrm{e}\, z < 1),$$

$$\int_0^\infty t^{z-1}\sin t\,\mathrm{d}t = \Gamma(z)\sin\big(\tfrac{1}{2}\pi z\big) \qquad (-1 < \Re\mathrm{e}\, z < 1).$$

(b) Calculate the values of the Fresnel integrals

$$I = \int_0^\infty \cos(t^2)\,\mathrm{d}t, \quad J = \int_0^\infty \sin(t^2)\,\mathrm{d}t.$$

153. *The volume of the unit ball in $\mathbb{R}^n$ and radial functions.*

A function $f : \mathbb{R}^n \to \mathbb{R}$ is said to be radial if there exists a further function $g : \mathbb{R}^+ \to \mathbb{R}$ such that $f(\boldsymbol{x}) = g(\|\boldsymbol{x}\|)$, where $\|\boldsymbol{x}\|$ denotes the Euclidean norm of $\boldsymbol{x} \in \mathbb{R}^n$. Denote by μ the measure on $\mathbb{R}^+$ which is the image of Lebesgue measure on $\mathbb{R}^n$ under the map $\boldsymbol{x} \mapsto \|\boldsymbol{x}\|$, so that $\int_0^\infty g(t)\,\mathrm{d}\mu(t) = \int_{\mathbb{R}^n} g(\|\boldsymbol{x}\|)\,\mathrm{d}\boldsymbol{x}$ for any function $g : \mathbb{R}^+ \to \mathbb{R}$ such that the right-hand side makes sense.

(a) Show that, for all $y \in \mathbb{R}^+$, we have

$$\mu([0,y]) = y^n V_n$$

with $V_n = \mu([0,1])$. Deduce that $\mathrm{d}\mu(y) = nV_n y^{n-1}\,\mathrm{d}y$ on $[0,\infty[$.

(b) Selecting a suitable radial function f, show that

$$V_n = \pi^{n/2}/\Gamma(1+n/2).$$

(c) Show that $\displaystyle\int_{\mathbb{R}^n} \bigl(1+\|\boldsymbol{x}\|^2\bigr)^{-\frac{1}{2}(n+1)}\,\mathrm{d}\boldsymbol{x} = \pi^{\frac{1}{2}(n+1)}/\Gamma\bigl(\tfrac{1}{2}(n+1)\bigr)$.

154. (a) Show that, for all $\sigma \geqslant \frac{1}{2}$, there is a constant $M_0(\sigma)$ such that

$$|\Gamma(x+iy)| \leqslant M_0(\sigma)\mathrm{e}^{-|y|} \qquad (\tfrac{1}{2} \leqslant x \leqslant \sigma,\ y \in \mathbb{R}).$$

(b) Show that for $z = \frac{1}{2} + iy$, $s = \sigma + i\tau$, $\sigma \geqslant 1$, we have

$$\Gamma(z-k)\Gamma(s-z+k) = (-1)^k\Gamma(z)\Gamma(s-z)\prod_{1\leqslant j\leqslant k}\Bigl(1+\frac{s-1}{j-z}\Bigr).$$

Using the bound $|1+w| \leqslant \exp\bigl\{\Re\mathrm{e}\, w + O(|w|^2)\bigr\}$ $(w \in \mathbb{C})$, deduce that, for each s satisfying $\sigma = \Re\mathrm{e}\, s \geqslant 1$, there is a constant $M(s) > 0$ such that

$$|\Gamma(z-k)\Gamma(s-z+k)| \leqslant M(s)k^{\sigma-1}\mathrm{e}^{-|y|} \qquad (k \geqslant 1,\ z = \tfrac{1}{2}+iy,\ y \in \mathbb{R}).$$

(c) Define for $s \in \mathbb{C}$, $\Re\mathrm{e}\, s \geqslant 1$, $u \in]0,1[$, $k \geqslant 0$,

$$I_k(s,u) := \frac{1}{2\pi i}\int_{\frac{1}{2}-k-i\infty}^{\frac{1}{2}-k+i\infty} \Gamma(z)\Gamma(s-z)u^{-z}\,\mathrm{d}z.$$

Show that this integral is absolutely convergent and tends to 0 as k tends to infinity.

(d) Show that, for $v \in \mathbb{C}$, $|v| < 1$, we have

$$\Gamma(s)(1+v)^{-s} = \sum_{k\geqslant 0}(-1)^k\frac{\Gamma(s+k)}{k!}v^k.$$

(e) Show that $I_0(s,u) = (1+u)^{-s}\Gamma(s)$ for $\Re\mathrm{e}\, s \geqslant 1$, $u \in]0,1[$.

(f) Show that, for $a, b \in [1, \infty[$, $\Re e\, s \geqslant 1$, we have

$$\frac{\Gamma(s)}{(a+b)^s} = \frac{1}{2\pi i} \int_{\frac{1}{2}-i\infty}^{\frac{1}{2}+i\infty} \frac{\Gamma(z)\Gamma(s-z)}{a^z b^{s-z}}\, \mathrm{d}z.$$

[Distinguish between the cases $a < b$, $a = b$ and $a > b$.]

(g) Show that there exists a constant σ_0, to be calculated (the optimal value is not required), such that, for $\sigma > \sigma_0$, we have

$$\Gamma(s) \sum_{m \geqslant 1} \sum_{n \geqslant 1} (m^3 + n^2)^{-s} = \frac{1}{2\pi i} \int_{\frac{1}{2}-i\infty}^{\frac{1}{2}+i\infty} \zeta(3z)\zeta(2s-2z)\Gamma(z)\Gamma(s-z)\, \mathrm{d}z$$

where ζ denotes the Riemann zeta function.

Generating functions: Dirichlet series

1.1. Convergent Dirichlet series

Let f be an arithmetic function. If the power series

$$S(z) := \sum_{n \geqslant 1} f(n) z^n$$

converges in some neighborhood of 0, several classical theorems allow us to relate the analytic behavior of the sum to the values of the coefficients—for example the relation $S^{(n)}(0) = n! f(n)$ or Cauchy's formulae.

Similarly, the analytic behavior of a convergent Dirichlet series

$$F(s) := \sum_{n \geqslant 1} \frac{f(n)}{n^s} \tag{1.1}$$

correlates closely with the asymptotic nature of the sequence $f(n)$. One of the principal aims of Part II is to develop methods that allow us to make these connections explicit.

We let the letter s denote a complex variable. The real numbers σ and τ are then implicitly defined by the relation

$$s = \sigma + i\tau.$$

Definition 1.1. *Let f be an arithmetic function. The Dirichlet series associated to f is the function of a complex variable $F(s)$ defined by* (1.1) *for those points s where the series converges.*

The properties of formal Dirichlet series studied in Chapter I.2 suggest the interest for number theory of the notion of a convergent Dirichlet series. We start with a simple, but fundamental, result concerning Dirichlet convolution.

Theorem 1.2. *Let f, g and h be arithmetic functions, with respective Dirichlet series F, G and H. Assume that*

$$h = f * g. \tag{1.2}$$

Then the series $H(s)$ converges in any domain where both series $F(s)$ and $G(s)$ converge absolutely, and in such circumstances we have

$$H(s) = F(s)G(s). \tag{1.3}$$

Proof. If F and G are absolutely convergent at the point s, then, for each $x \geqslant 1$, we have

$$\sum_{n \leqslant x} \left| \frac{h(n)}{n^s} \right| \leqslant \sum_{md \leqslant x} \left| \frac{f(m)g(d)}{m^s d^s} \right| \leqslant \sum_{m \leqslant x} \left| \frac{f(m)}{m^s} \right| \sum_{d \leqslant x} \left| \frac{g(d)}{d^s} \right|.$$

This implies the absolute convergence of $H(s)$. Relation (1.3) then follows from the associated formal identity—cf. § I.2.4. □

Remark. The conclusion of Theorem 1.2 fails if we consider conditional convergence instead of absolute convergence. It is relatively easy to construct functions f, g and h satisfying (1.2) and such that, at certain points s, the series $F(s)$ and $G(s)$ converge while $H(s)$ diverges—cf. Exercise 211; see also the Notes.

1.2. Dirichlet series of multiplicative functions

We saw in Theorem I.2.5 that formal Dirichlet series of multiplicative functions have the characteristic property of being representable as formal Euler products. The following theorem provides a sufficient condition for such an algebraic identity to have an analytic meaning.

Theorem 1.3. *Let f be a multiplicative function and let s be a complex number. Under the hypothesis*

$$\sum_{p} \sum_{\nu \geqslant 1} \left| \frac{f(p^\nu)}{p^{\nu s}} \right| < \infty, \tag{1.4}$$

the Dirichlet series (1.1) is absolutely convergent, and we have

$$F(s) = \prod_{p} \sum_{\nu \geqslant 0} \frac{f(p^\nu)}{p^{\nu s}}. \tag{1.5}$$

Remarks. (i) If the series (1.1) is absolutely convergent, the same holds for (1.4). Thus the absolute convergence of (1.1) is actually equivalent to that of (1.4).

(ii) When f is completely (resp. strongly) multiplicative the right-hand side of (1.5) takes the simpler form

$$\prod_p \Big(1 - \frac{f(p)}{p^s}\Big)^{-1}, \qquad \Big(\text{resp.} \quad \prod_p \Big(1 + \frac{f(p)}{p^s - 1}\Big)\Big).$$

Proof. Observe first that condition (1.4) implies the convergence of the infinite product $M := \prod_p \big(1 + \sum_{\nu \geqslant 1} |f(p^\nu)/p^{\nu s}|\big)$. As a result, we may write, for all $x \geqslant 1$,

$$\sum_{n \leqslant x} \Big|\frac{f(n)}{n^s}\Big| \leqslant \sum_{P^+(n) \leqslant x} \Big|\frac{f(n)}{n^s}\Big| = \prod_{p \leqslant x} \Big(1 + \sum_{\nu \geqslant 1} \Big|\frac{f(p^\nu)}{p^{\nu s}}\Big|\Big) \leqslant M.$$

This shows that the series $F(s)$ is absolutely convergent. The inequality

$$\Big|\sum_{n \geqslant 1} \frac{f(n)}{n^s} - \prod_{p \leqslant x} \sum_{\nu \geqslant 0} \frac{f(p^\nu)}{p^{\nu s}}\Big| = \Big|\sum_{P^+(n) > x} \frac{f(n)}{n^s}\Big| \leqslant \sum_{n > x} \Big|\frac{f(n)}{n^s}\Big|$$

then implies (1.5), upon letting x tend to infinity. □

Proposition 1.4 (Euler's formula). *For $\sigma > 1$, we have*

$$\zeta(s) = \prod_p \Big(1 - \frac{1}{p^s}\Big)^{-1}. \tag{1.6}$$

This simple formula highlights the explicit connection between Riemann's zeta function and the sequence of primes. This is one of the most important tools in the analytic theory of numbers.

1.3. Fundamental analytic properties of Dirichlet series

Let $n \mapsto a_n$ be an arithmetic function. Set

$$A(t) := \sum_{n \leqslant \mathrm{e}^t} a_n.$$

Then the Dirichlet series associated to (a_n) can also be written in the form

$$F(s) := \sum_{n \geqslant 1} \frac{a_n}{n^s} = \int_{0-}^{\infty} \mathrm{e}^{-ts} \, \mathrm{d}A(t).$$

This integral is called the *Laplace–Stieltjes transform* of the function $A(t)$. Most of the fundamental theorems concerning Dirichlet series can be generalized to this framework, where the Stieltjes integral formalism reveals itself to be a handy technical tool.

Let $\mathcal{V}$ be the class of functions defined on $\mathbb{R}$ and of bounded variation on every finite interval. Although we primarily aim to study Dirichlet series, we consider, whenever possible, the Laplace–Stieltjes transforms of functions in $\mathcal{V}$—with more concern for clarity than generality.

Theorem 1.5. *Let A be a function in $\mathcal{V}$ and let*

$$F(s) := \int_{0-}^{+\infty} \mathrm{e}^{-st}\,\mathrm{d}A(t) \tag{1.7}$$

be its Laplace–Stieltjes transform.

(i) If the integral (1.7) *converges for $s = s_0 = \sigma_0 + i\tau_0$, then it converges for all s such that $\sigma > \sigma_0$, and the convergence is uniform in every sector*

$$S(\vartheta) := \{s \in \mathbb{C} : |\arg(s - s_0)| \leqslant \vartheta\} \qquad (0 \leqslant \vartheta < \pi/2).$$

(ii) If the integral (1.7) *converges absolutely for $s = s_0$, then it converges absolutely and uniformly in the closed half-plane $\sigma \geqslant \sigma_0$.*

(iii) The function $F(s)$ is holomorphic in any open domain where (1.7) *converges, and we have*

$$F^{(k)}(s) = \int_{0-}^{\infty} (-t)^k \mathrm{e}^{-st}\,\mathrm{d}A(t) \qquad (k = 0, 1, 2, \ldots). \tag{1.8}$$

Proof. (i) Let $\vartheta \in [0, \pi/2[$. The sector $S(\vartheta)$ is the set of complex numbers s such that

$$|s - s_0| \leqslant \frac{\sigma - \sigma_0}{\cos\vartheta}.$$

For each $\varepsilon > 0$, we show the existence of a real number $x_0 = x_0(\varepsilon, \vartheta)$ such that, whenever $y \geqslant x \geqslant x_0$, we have

$$\Big| \int_x^y \mathrm{e}^{-ts}\,\mathrm{d}A(t) \Big| \leqslant \varepsilon \qquad (s \in S(\vartheta)).$$

Let us write

$$g(u) := \int_{0-}^{u} \mathrm{e}^{-ts_0}\,\mathrm{d}A(t) \qquad (u \geqslant 0).$$

By hypothesis, there exists some $x_0 = x_0(\varepsilon, \vartheta)$ such that, for $v \geqslant u \geqslant x_0$, we have

$$|g(v) - g(u)| \leqslant \tfrac{1}{2}\varepsilon\cos\vartheta.$$

Now, for $y \geqslant x \geqslant x_0$ and $s \in S(\vartheta) \setminus \{s_0\}$, we can write

$$\begin{aligned}\Big|\int_x^y \mathrm{e}^{-ts}\,\mathrm{d}A(t)\Big| &= \Big|\int_x^y \mathrm{e}^{-u(s-s_0)}\,\mathrm{d}\{g(u)-g(x)\}\Big| \\ &= \Big|\mathrm{e}^{-y(s-s_0)}\{g(y)-g(x)\}+(s-s_0)\int_x^y \mathrm{e}^{-u(s-s_0)}\{g(u)-g(x)\}\,\mathrm{d}u\Big| \\ &\leqslant \tfrac{1}{2}\varepsilon\cos\vartheta+|s-s_0|\tfrac{1}{2}\varepsilon\cos\vartheta\int_x^y \mathrm{e}^{-u(\sigma-\sigma_0)}\,\mathrm{d}u \\ &\leqslant \tfrac{1}{2}\varepsilon\cos\vartheta\Big(1+\frac{|s-s_0|}{\sigma-\sigma_0}\Big) \leqslant \tfrac{1}{2}\varepsilon(\cos\vartheta+1) \leqslant \varepsilon.\end{aligned}$$

This furnishes the required conclusion.

Assertion (ii) is merely a reformulation of the trivial upper bound

$$\int_x^y |\mathrm{e}^{-ts}|\,|\mathrm{d}A(t)| \leqslant \int_x^y |\mathrm{e}^{-ts_0}|\,|\mathrm{d}A(t)|,$$

valid for $y \geqslant x$ and $\sigma \geqslant \sigma_0$.

Let us show (iii). The uniform convergence of the exponential series allows us to write, for all $x \geqslant 0$,

$$\int_{0-}^{x} \mathrm{e}^{-ts}\,\mathrm{d}A(t) = \sum_{n\geqslant 0}\frac{s^n}{n!}\int_{0-}^{x}(-t)^n\,\mathrm{d}A(t).$$

The left-hand side of this equality is thus an entire function of s, whose k-th derivative is

$$\sum_{n\geqslant k}\frac{s^{n-k}}{(n-k)!}\int_{0-}^{x}(-t)^n\,\mathrm{d}A(t) = \int_{0-}^{x}(-t)^k\mathrm{e}^{-ts}\,\mathrm{d}A(t).$$

Assertion (iii) follows from the foregoing, by Weierstrass's theorem on the analyticity of limits of sequences of analytic functions converging uniformly on compact subsets—see for example Cartan (1961), chap. V, th. 1. □

Consider an integral $F(s)$ of the type (1.7). From parts (i) and (ii) of the above theorem, the set of abscissas σ of those points s where the integral converges (resp. converges absolutely) is a half-line with origin σ_c (resp. σ_a). One defines respectively σ_c and σ_a as the *abscissa of convergence* and the *abscissa of absolute convergence* of the Stieltjes integral $F(s)$. By convention, we allow these numbers to take the values $\pm\infty$.

It is easy to construct examples of Dirichlet series converging everywhere, or diverging everywhere, on the line of convergence $\sigma = \sigma_c$. The following result shows that, when it exists, the sum of the series can be evaluated on the line of convergence by analytic continuation.

Theorem 1.6. *Suppose that the integral $F(s)$ defined by (1.7) for $\sigma > \sigma_c$ has an analytic continuation $\widetilde{F}(s)$ for certain points s with real part $\sigma = \sigma_c$. We then have*

$$\widetilde{F}(s) = \int_{0-}^{+\infty} \mathrm{e}^{-ts}\, \mathrm{d}A(t)$$

at any point where the integral converges.

Proof. We employ uniformity, as established in Theorem 1.5(i), in the form

$$F(s) = \lim_{\delta \to 0+} F(s+\delta) \qquad (\sigma = \sigma_c).$$

Since $\widetilde{F}$ is analytic, this implies

$$\widetilde{F}(s) = \lim_{\delta \to 0+} \widetilde{F}(s+\delta) = \lim_{\delta \to 0+} F(s+\delta) = F(s),$$

as asserted. □

As an application of Theorem 1.6, let us consider the Riemann zeta function. For $\sigma > 1$, we have

$$\zeta(s) = \int_{1-}^{\infty} \frac{\mathrm{d}\lfloor t \rfloor}{t^s} = s \int_1^{\infty} \lfloor t \rfloor \frac{\mathrm{d}t}{t^{s+1}} = \frac{s}{s-1} - s \int_1^{\infty} \frac{\langle t \rangle}{t^{s+1}}\, \mathrm{d}t. \tag{1.9}$$

Since this last integral is absolutely convergent for $\sigma > 0$, this shows that $\zeta(s)$ can be continued to a meromorphic function for $\sigma > 0$ having for its only singularity a simple pole at $s = 1$. From Theorem 1.6, we then obtain that, under the hypothesis that the series converges, we necessarily have

$$\sum_{n \geqslant 1} \frac{\mu(n)}{n} = \zeta^{-1}(1) = 0. \tag{1.10}$$

Thus, taking account of Theorem I.3.8, it follows that the prime number theorem is equivalent to the *convergence* of the series (1.10).

The following theorem demonstrates that σ_c and σ_a cannot be chosen independently when $F(s)$ is a Dirichlet series.

Theorem 1.7. *Let $F(s)$ be a Dirichlet series, defined by (1.1). Then we have*

$$\sigma_c \leqslant \sigma_a \leqslant \sigma_c + 1. \tag{1.11}$$

Proof. Let $\varepsilon > 0$. The convergence of the series $\sum_{n \geqslant 1} f(n)/n^{\sigma_c + \varepsilon}$ implies the bound

$$f(n) \ll_\varepsilon n^{\sigma_c + \varepsilon}$$

and, consequently, the absolute convergence of the series (1.1) at

$$s = \sigma_c + 1 + 2\varepsilon.$$

Thus we have $\sigma_a \leqslant \sigma_c + 1 + 2\varepsilon$, whence the result on passing to the limit. □

It is easy to establish that the two-sided bound (1.11) is optimal. For the series

$$G(s) := \sum_{n\geqslant 1} \frac{(-1)^n}{n^s} = (2^{1-s}-1)\zeta(s) \tag{1.12}$$

we have $\sigma_c = 0$ (summing an alternating series) and, of course, $\sigma_a = 1$.

A noteworthy consequence of Theorem 1.7 is the uniqueness of the Dirichlet series expansion.

Theorem 1.8. *Let $F(s) = \sum_{n\geqslant 1} a_n/n^s$ be a Dirichlet series which vanishes identically for sufficiently large σ. Then $a_n = 0$ for $n \geqslant 1$.*

Proof. We argue by contradiction, and consequently assume $\{a_n\}_{n=1}^{\infty}$ is not identically zero. Let m be the smallest integer such that $a_m \neq 0$. For sufficiently large σ, we have

$$0 = F(s) = \frac{a_m}{m^s}\{1 + G(s)\} \tag{1.13}$$

with

$$G(s) := \sum_{k\geqslant 1} \frac{a_{m+k}}{a_m}\Big(\frac{m}{m+k}\Big)^s.$$

By Theorem 1.7, this series converges absolutely for $\sigma > \sigma_c + 1$. For $\sigma > \sigma_1 > \sigma_c + 1$, we then have

$$|G(s)| \leqslant \frac{(1+1/m)^{\sigma_1-\sigma} m^{\sigma_1}}{|a_m|} \sum_{k\geqslant 1} \frac{|a_{m+k}|}{(m+k)^{\sigma_1}} \ll_{\sigma_1} (1+1/m)^{-\sigma}.$$

Therefore $G(s)$ tends to 0 when $\sigma \to +\infty$, which contradicts (1.13). □

By analogy with power series, one might be tempted to believe that a Dirichlet series necessarily possesses a singularity on the line of convergence. This is not the case. The series (1.12) furnishes in this respect an excellent counter-example. We shall indeed see in Chapter II.3 that the Riemann zeta function can be continued to a meromorphic function on all of $\mathbb{C}$ with only a single, simple, pole at $s = 1$. This implies that $G(s)$ can be continued to an entire function. One can also show directly that $G(s)$ has a regular analytic continuation for $\sigma > -1$: see Exercise 156.

The following theorem, usually referred to as *Landau's theorem* (cf. Notes), describes a situation when the line of convergence must contain a singularity.

Theorem 1.9 (Phragmén–Landau). *Let A be a function in $\mathcal{V}$ and let $F(s)$ be its Laplace–Stieltjes transform, defined by* (1.7). *If A is non-decreasing, then the point $s = \sigma_c$ is a singularity of $F(s)$.*

Corollary 1.10. *The abscissa of convergence is a singularity of the sum of any Dirichlet series with non-negative coefficients.*

Proof of Theorem 1.9. We argue by contradiction and assume that F has an holomorphic continuation in a suitable neighborhood of $s = \sigma_c$. There then exist two numbers $\sigma > \sigma_c$ and $r > \sigma - \sigma_c$, such that the Taylor series of F at σ, namely

$$F(s) = \sum_{k\geqslant 0} \frac{1}{k!} F^{(k)}(\sigma)(s-\sigma)^k,$$

converges in the disk $|s - \sigma| < r$. By (1.8), we may then write, under this condition,

$$\begin{aligned} F(s) &= \sum_{k\geqslant 0} \frac{1}{k!}(s-\sigma)^k \int_{0-}^{\infty} (-t)^k \mathrm{e}^{-\sigma t}\, \mathrm{d}A(t) \\ &= \sum_{k\geqslant 0} \frac{1}{k!} \int_{0-}^{\infty} t^k (\sigma - s)^k \mathrm{e}^{-\sigma t}\, \mathrm{d}A(t). \end{aligned}$$

When s is real with $\sigma - r < s < \sigma$, we may interchange summation and integration since the integrand and the measure $\mathrm{d}A(t)$ are both non-negative. It follows that

$$\begin{aligned} F(s) &= \int_{0-}^{\infty} \sum_{k\geqslant 0} \frac{1}{k!} t^k (\sigma - s)^k \mathrm{e}^{-\sigma t}\, \mathrm{d}A(t) \\ &= \int_{0-}^{\infty} \mathrm{e}^{t(\sigma - s)} \mathrm{e}^{-\sigma t}\, \mathrm{d}A(t) = \int_{0-}^{\infty} \mathrm{e}^{-st}\, \mathrm{d}A(t). \end{aligned}$$

Therefore the Laplace–Stieltjes integral converges at s, which leads to a contradiction by selecting $s < \sigma_c$. This concludes the proof. □

The Phragmén–Landau theorem assumes special significance in practice inasmuch as it underlies most *oscillation theorems*. Here, we confine ourselves to the following two results, which are typical of its use.

Theorem 1.11. *Let $A : [1, +\infty[\to \mathbb{R}$ be a locally bounded measurable function. If the integral*

$$H(s) := \int_1^{\infty} \frac{A(t)}{t^{s+1}}\, \mathrm{d}t \tag{1.14}$$

has a finite abscissa of convergence σ_c, and if $H(s)$ has a regular continuation at the point $s = \sigma_c$, then for each $\varepsilon > 0$ we have

$$A(x) = \Omega_{\pm}(x^{\sigma_c - \varepsilon}). \tag{1.15}$$

Proof. Assume, for example, the existence of a constant $K = K(\varepsilon)$ such that, for all sufficiently large t, we have

$$A(t) \leqslant K t^{\sigma_c - \varepsilon}.$$

Up to adjusting the value of K, we may assume this inequality is satisfied for all $t \geqslant 1$. So we can write

$$H(s) - \frac{K}{s-\sigma_c+\varepsilon} = \int_1^\infty \frac{A(t) - Kt^{\sigma_c-\varepsilon}}{t^{s+1}}\,\mathrm{d}t = -\int_0^\infty B(u)\mathrm{e}^{-su}\,\mathrm{d}u$$

with

$$B(u) := K\mathrm{e}^{(\sigma_c-\varepsilon)u} - A(\mathrm{e}^u) \geqslant 0.$$

The abscissa of convergence of the last integral is still σ_c, since

$$\int_1^\infty t^{\sigma_c-\varepsilon-s-1}\,\mathrm{d}t$$

is absolutely convergent for $\sigma > \sigma_c - \varepsilon$. By the Phragmén–Landau theorem, the point $s = \sigma_c$ must be a singularity of $H(s) - K/(s-\sigma_c+\varepsilon)$, and hence $H(s)$ is not holomorphic at σ_c. □

Theorem 1.12. *Let $F(s) = \sum_{n\geqslant 1} a_n/n^s$ be a Dirichlet series with real coefficients, having a finite abscissa of convergence. Assume there is a real number $\sigma_0 > 0$ such $F(s)$ has an analytic continuation which is regular at all points of the half-line $[\sigma_0, +\infty[$ and has a pole on the vertical line $\sigma = \sigma_0$. Then we have*

$$\sum_{n\leqslant x} a_n = \Omega_\pm(x^{\sigma_0}).$$

Proof. Write $A(t) := \sum_{n\leqslant t} a_n$ and denote by σ_c the abscissa of convergence of the integral $H(s)$ defined by (1.14). By partial summation, we have, for sufficiently large σ,

$$F(s) = sH(s)$$

so that $H(s)$ can be holomorphically continued on the half-line $[\sigma_0, +\infty[$. We can assume there exists some constant K such that $A(t) + Kt^{\sigma_0}$ has constant sign, say positive, for large t. Then there is a constant C such that

$$B(t) := A(t) + Kt^{\sigma_0} + C \geqslant 0 \qquad (t \geqslant 1).$$

Since the integral

$$L(s) := \int_1^\infty \frac{B(t)}{t^{s+1}}\,\mathrm{d}t = H(s) + \frac{K}{s-\sigma_0} + \frac{C}{s},$$

is holomorphic on $]\sigma_0, +\infty[$, it must be convergent for $\sigma > \sigma_0$ by the Phragmén–Landau theorem. If $s_0 = \sigma_0 + i\tau_0$ is a pole of $F(s)$ with principal part $\lambda(s-s_0)^{-m}$, we have, on the one hand,

$$|L(s_0+\delta)| \leqslant L(\sigma_0+\delta) \qquad (\delta > 0)$$

and, on the other,

$$L(\sigma_0+\delta) \sim \frac{K}{\delta}, \qquad L(s_0+\delta) \sim \frac{\lambda}{s_0\delta^m} \quad (\delta \to 0+).$$

This implies $|K| \geqslant |\lambda/s_0|$. In particular, $A(t) + Kt^{\sigma_0}$ does not have a constant sign at infinity if

$$|K| < |\lambda/s_0|. \qquad \square$$

In Exercise 160, we exhibit three examples of classical oscillation theorems.

1.4. Abscissa of convergence and mean value

Let A be a function in $\mathcal{V}$ and let

$$F(s) = \int_{0-}^{\infty} \mathrm{e}^{-ts}\, \mathrm{d}A(t) \tag{1.16}$$

be its Laplace–Stieltjes transform. In this section, our aim is to give an explicit formula for the abscissa of convergence of (1.16) in terms of the asymptotic behavior of A. It is clear that the value of A in any bounded neighborhood of the origin does not affect the value of σ_c. Hence, without loss of generality, we can make the simplifying assumption that

$$A(0\pm) = 0. \tag{1.17}$$

Explicit calculation of σ_c, analogous to Cauchy's formula for the radius of convergence of a power series, is the concern of Theorem 1.14 *infra*. The proof rests on the following result.

Theorem 1.13. *Let σ_c be the abscissa of convergence of the integral* (1.16).

(i) *If we have $A(x) \ll \mathrm{e}^{\delta x}$ for some real δ, then $\sigma_c \leqslant \delta$.*

(ii) *If the integral* (1.16) *converges for $s = s_0$ with $\sigma_0 > 0$, then*

$$A(x) = o(\mathrm{e}^{\sigma_0 x}) \qquad (x \to \infty).$$

(iii) *If the integral* (1.16) *converges for $s = s_0$ with $\sigma_0 < 0$, then there exists a real number α such that*

$$A(x) = \alpha + o(\mathrm{e}^{\sigma_0 x}) \qquad (x \to \infty).$$

Proof. (i). For all $x > 0$, we have

$$\int_0^x \mathrm{e}^{-st}\, \mathrm{d}A(t) = A(x)\mathrm{e}^{-sx} + s\int_0^x \mathrm{e}^{-st} A(t)\, \mathrm{d}t.$$

The assumption concerning the rate of increase of $A(t)$ then implies the convergence of the integral (1.16) for all s such that $\sigma > \delta$. Hence $\sigma_c \leqslant \delta$.

(ii). By hypothesis,

$$B(x) := \int_0^x \mathrm{e}^{-s_0 t}\, \mathrm{d}A(t) = F(s_0) + o(1) \qquad (x \to \infty). \tag{1.18}$$

We can then write

$$A(x) = \int_0^x \mathrm{e}^{s_0 t}\,\mathrm{d}B(t) = \mathrm{e}^{s_0 x}B(x) - s_0\int_0^x \mathrm{e}^{s_0 t}B(t)\,\mathrm{d}t$$
$$= s_0\int_0^x \{B(x) - B(t)\}\mathrm{e}^{s_0 t}\,\mathrm{d}t + B(x).$$

The expected result then follows from (1.18).

(iii). By Theorem 1.5(i), we can assert that $F(s)$ converges for $s = 0$. Setting $\alpha := F(0)$, we have, with the notation (1.18),

$$\alpha - A(x) = \int_x^\infty \mathrm{e}^{s_0 t}\,\mathrm{d}B(t) = -\mathrm{e}^{s_0 x}B(x) - s_0\int_x^\infty \mathrm{e}^{s_0 t}B(t)\,\mathrm{d}t$$
$$= s_0\int_x^\infty \{B(x) - B(t)\}\mathrm{e}^{s_0 t}\,\mathrm{d}t = s_0\int_x^\infty o(\mathrm{e}^{\sigma_0 t})\,\mathrm{d}t = o(\mathrm{e}^{\sigma_0 x}).$$

□

Remark. Assertions (ii) and (iii) of Theorem 1.13 do not hold for $\sigma_0 = 0$. A counter-example to the first assertion is provided by the function

$$A(x) = \begin{cases} 0 & (0 \leqslant x \leqslant 1) \\ 1 & (x > 1). \end{cases}$$

The integral (1.16) then converges for $s_0 = 0$, but we do not have $A(x) = o(1)$ at infinity. For the second statement, consider

$$A(x) = \begin{cases} 0 & (0 \leqslant x \leqslant 1) \\ 2\sqrt{x} & (x > 1). \end{cases}$$

Clearly we have

$$F(i) = 2\mathrm{e}^{-i} + \int_1^\infty t^{-1/2}\mathrm{e}^{-it}\,\mathrm{d}t.$$

The integral is well-known to be convergent; however $A(x)$ does not tend to a finite limit as $x \to \infty$.

Theorem 1.14. *Put* $\kappa := \limsup_{x\to\infty} x^{-1}\ln|A(x)|$.

(i) *If* $\kappa \neq 0$, *then we have* $\sigma_c = \kappa$.

(ii) *If* $\kappa = 0$, *then either* $A(x)$ *does not tend to a finite limit as* $x \to \infty$, *and we have* $\sigma_c = 0$, *or there exists some real number* α *such that* $A(x) \to \alpha$ *as* $x \to \infty$, *and we have*

$$\sigma_c = \limsup_{x\to\infty} x^{-1}\ln|A(x) - \alpha| \leqslant 0.$$

Proof. For each fixed $\varepsilon > 0$, we have $A(x) \ll_\varepsilon \mathrm{e}^{(\kappa+\varepsilon)x}$. Assertion (i) of Theorem 1.13 then implies that in all cases

$$\sigma_c \leqslant \kappa. \tag{1.19}$$

Suppose, initially, that $\kappa > 0$. Then the integral $F(s)$ diverges when $0 < \sigma < \kappa$: otherwise one would have, by Theorem 1.13(ii), that $A(x) = o(\mathrm{e}^{\sigma x})$, contradicting the definition of κ. Thus $\sigma_c \geqslant \kappa$, and the required equality follows.

If $\kappa < 0$, then $A(x) \to 0$ as $x \to \infty$; assertion (iii) of Theorem 1.13 then implies

$$A(x) = o(\mathrm{e}^{\sigma x}) \qquad (x \to \infty) \tag{1.20}$$

for all $\sigma > \sigma_c$. But, by definition of κ, no $\sigma < \kappa$ satisfies (1.20). Hence $\sigma_c \geqslant \kappa$, and once again $\sigma_c = \kappa$.

Let us now consider the case $\kappa = 0$. If $A(x)$ does not have a limit at infinity, the integral $F(s)$ diverges at $s = 0$, so that $\sigma_c \geqslant 0$ and hence the required conclusion still follows from (1.19). If $A(x) = \alpha + o(1)$, we need to show that $\sigma_c = \xi$, where ξ is the infimum of the real numbers σ_1 such that

$$A(x) = \alpha + o(\mathrm{e}^{\sigma_1 x}). \tag{1.21}$$

From Theorem 1.13(iii), we have $\sigma_c \geqslant \xi$. Integration by parts shows trivially, by (1.21), that $F(s)$ converges for $\sigma > \xi$, whence $\sigma_c \leqslant \xi$. □

1.5. An arithmetic application: the core of an integer

The abscissa of convergence of a Dirichlet series with an Euler product is often easy to calculate. By using Theorems 1.13 and 1.14, we may sometimes deduce non-trivial information on the associated summatory functions.

Consider, for example, the case of the *core*[1] of an integer n, (i.e., the greatest squarefree divisor of n), which we denote by

$$k(n) := \prod_{p \mid n} p. \tag{1.22}$$

Theorem 1.13 readily provides information about the distribution function

$$N(x, y) := \mathrm{card}\{n \leqslant x : k(n) \leqslant y\}.$$

Theorem 1.15. *For each $\varepsilon > 0$, and uniformly for $1 \leqslant y \leqslant x$, we have*

$$N(x, y) \ll_\varepsilon y x^\varepsilon. \tag{1.23}$$

Proof. The function $k(n)$ is multiplicative. Since the series of positive terms

$$\sum_p \sum_{\nu \geqslant 1} \frac{1}{k(p^\nu) p^{\nu\varepsilon}} = \sum_p \frac{1}{p(p^\varepsilon - 1)} \qquad (\varepsilon > 0),$$

[1] Also called *kernel*, or *radical*.

converges, we deduce from Theorem 1.3 that the abscissa of convergence of

$$F(s) := \sum_{n\geqslant 1} \frac{1}{k(n)n^s}$$

is $\sigma_c = 0$. By Theorem 1.13, it follows that, for all $\varepsilon > 0$,

$$\sum_{n\leqslant x} \frac{1}{k(n)} \ll_\varepsilon x^\varepsilon. \tag{1.24}$$

Estimate (1.23) is now immediate from this bound, since

$$N(x,y) \leqslant \sum_{n\leqslant x} \frac{y}{k(n)}. \qquad \square$$

In Exercise 165, we outline a lower bound of the form

$$N(x,y) \gg_m y\{\ln(2x/y)\}^m,$$

valid for all $m > 0$ and $x \geqslant y \geqslant y_0(m)$. (For the best results known to date about $N(x,y)$, see the Notes.)

The idea of comparing the summatory function of an arithmetic function to an Euler product underpins *Rankin's method* (cf. § III.5.1). We show below how to apply this simple method to refine (1.23).

Theorem 1.16. *Uniformly for $x \geqslant y \geqslant 2$, we have*

$$N(x,y) \ll y(\ln y)\mathrm{e}^{\sqrt{8\ln(x/y)}}.$$

Proof. If the integer n is counted in $N(x,y)$, we have, for each ε with $0 \leqslant \varepsilon \leqslant 1$,

$$1 \leqslant \Big(\frac{x}{n}\Big)^\varepsilon \Big(\frac{y}{k(n)}\Big)^{1-\varepsilon}.$$

Write $v := \ln(x/y)$. We may assume that $v \geqslant 2$ since the conclusion is otherwise trivial. It follows that

$$\begin{aligned} N(x,y) &\leqslant y\mathrm{e}^{\varepsilon v} \sum_{\substack{n\leqslant x\\ p\mid n\Rightarrow p\leqslant y}} \frac{1}{n^\varepsilon k(n)^{1-\varepsilon}} \leqslant y\mathrm{e}^{\varepsilon v} \prod_{p\leqslant y} \Big(1+\frac{1}{p}+\frac{1}{p(p^\varepsilon-1)}\Big) \\ &\leqslant y\exp\Big\{\varepsilon v + \frac{K}{\varepsilon} + \sum_{p\leqslant y}\frac{1}{p}\Big\} \end{aligned}$$

with $K := \sum_p 1/(p\ln p) \leqslant 2$. Choosing $\varepsilon = \sqrt{2/v}$, and estimating the last sum over p by Theorem I.1.10, we obtain the stated result. $\square$

1.6. Order of magnitude in vertical strips

It is evident that a Dirichlet series $F(s)$ is bounded in any closed half-plane contained in its domain of absolute convergence. However $|F(s)|$ may take large values as $\sigma \to \sigma_a$ and $|\tau| \to \infty$. The following theorem provides an important example.

Theorem 1.17. *For all real $T > 0$, there exists some real number $\tau > T$ such that*

$$\sup_{\sigma>1} |\zeta(\sigma + i\tau)| \geqslant \tfrac{1}{10} \ln_2(3+\tau). \tag{1.25}$$

In order to establish this result, we appeal to a classical variation, concerning simultaneous approximation modulo 1 of N real numbers, of Dirichlet's approximation theorem, Theorem I.7.1. We recall the statement in the lemma below. As is conventional, we put, for $x \in \mathbb{R}$,

$$\|x\| := \min_{n\in\mathbb{Z}} |x-n|.$$

Lemma 1.18 (Dirichlet). *Let $\alpha_1, \alpha_2, \dots, \alpha_N$ be real numbers and let D be an integer $\geqslant 1$. For any integer $Q \geqslant 2$, there exists some integer q, $D \leqslant q \leqslant D{\cdot}Q^N$, such that*

$$\max_{1\leqslant j\leqslant N} \|q\alpha_j\| \leqslant 1/Q. \tag{1.26}$$

Proof. Consider the partition

$$\bigcup_{0\leqslant j_1,\dots,j_N<Q} \prod_{h=1}^{N} \Big[\frac{j_h}{Q}, \frac{j_h+1}{Q}\Big[$$

of the unit cube $[0,1[^N$ into Q^N subcubes. By the pigeonhole principle, at least two of the Q^N+1 points

$$(mD\alpha_1, mD\alpha_2, \dots, mD\alpha_N) \qquad (0 \leqslant m \leqslant Q^N)$$

belong, modulo 1, to the same subcube. If m and m' are the corresponding indices, (1.26) is satisfied with $q = |m'-m|D \in [D, D{\cdot}Q^N]$. □

Proof of Theorem 1.17. For any $\sigma > 1$, and any integer $N \geqslant 1$, we can write

$$\Re e\, \zeta(s) \geqslant \sum_{n\leqslant N} \frac{\cos(\tau \ln n)}{n^\sigma} - \sum_{n>N} \frac{1}{n^\sigma}. \tag{1.27}$$

Apply the lemma with $Q := 6$, $D := Q^N$, and $\alpha_n := \frac{1}{2\pi} \ln n$ $(1 \leqslant n \leqslant N)$. Then there exists some real number τ, $6^N \leqslant \tau \leqslant 6^{2N}$, such that

$$\min_{1\leqslant n\leqslant N} \cos(\tau \ln n) \geqslant \cos(\pi/3) = \tfrac{1}{2}.$$

Hence we deduce from (1.27) that

$$\Re e\,\zeta(s) \geqslant \tfrac{1}{2}\sum_{n\leqslant N}\frac{1}{n^\sigma} - \sum_{n>N}\frac{1}{n^\sigma} = \tfrac{1}{2}\zeta(\sigma) - \tfrac{3}{2}\sum_{n>N}\frac{1}{n^\sigma} \geqslant \frac{1-3N^{1-\sigma}}{2(\sigma-1)}$$

where the last inequality follows from the usual comparison estimates between series and integrals. For $\sigma = 1 + 4/\ln N$, we obtain

$$\Re e\,\zeta(s) > \tfrac{1}{9}\ln N \geqslant \tfrac{1}{10}\ln_2(3+\tau)$$

if $N \geqslant N_0$. The extra condition $\tau > T$ is satisfied by choosing $N > \ln T/\ln 6$. This completes the proof. □

The following theorem shows that, in its domain of convergence, a Dirichlet series necessarily satisfies certain bounds.

Theorem 1.19. *Let $F(s) = \sum_{n\geqslant 1} a_n/n^s$ be a Dirichlet series with abscissa of convergence σ_c. Let $\sigma_0 > \sigma_c$ and $\varepsilon > 0$. Then, uniformly for $\sigma_0 \leqslant \sigma \leqslant \sigma_c + 1$, we have*

$$F(s) \ll |\tau|^{1-(\sigma-\sigma_c)+\varepsilon} \qquad (|\tau| \geqslant 1). \tag{1.28}$$

Proof. We may obviously suppose we have $0 < \varepsilon < \sigma_0 - \sigma_c$. Put

$$A(t) := \sum_{n\leqslant \mathrm{e}^t} \frac{a_n}{n^{\sigma_c+\varepsilon}}$$

so that $A(t) = F(\sigma_c+\varepsilon) + o(1)$ as $t \to \infty$. Furthermore

$$\begin{aligned} |F(s)| &= \Big|\sum_{n\leqslant N}\frac{a_n}{n^s} + \int_{\ln N}^{\infty} \mathrm{e}^{-t(s-\sigma_c-\varepsilon)}\,\mathrm{d}A(t)\Big| \\ &\leqslant \sum_{n\leqslant N}\frac{|a_n|}{n^\sigma} + \frac{|A(\ln N)|}{N^{\sigma-\sigma_c-\varepsilon}} + |s-\sigma_c-\varepsilon|\int_{\ln N}^{\infty}|A(t)|\mathrm{e}^{-t(\sigma-\sigma_c-\varepsilon)}\,\mathrm{d}t. \end{aligned}$$

Since $|a_n| \ll_\varepsilon n^{\sigma_c+\varepsilon}$, we deduce that

$$|F(s)| \ll_\varepsilon N^{1-(\sigma-\sigma_c)+\varepsilon} + |s|N^{-(\sigma-\sigma_c)+\varepsilon}.$$

The stated result follows from this estimate by choosing $N = 1 + \lfloor|\tau|\rfloor$. □

The problem of determining whether a given function, analytic in a certain half-plane, is or is not representable as a Dirichlet series is difficult in general. Theorem 1.19 shows that functions satisfying a relation of the type

$$F(s) \ll |\tau|^A \qquad (|\tau| \geqslant 1) \tag{1.29}$$

for some $A > 0$, play a special role.[2] If F satisfies (1.29) in a domain $\mathcal{D}$, we say that F has *finite order* in $\mathcal{D}$. A Dirichlet series has finite order in any closed half-plane included in its domain of convergence. If the sum of the series can be analytically continued, it can happen that the continuation still has finite order in a larger domain. For example, let us consider the series $G(s) := \sum_{n\geqslant 1}(-1)^n/n^s$, which provides an analytic continuation of $\zeta(s)$ to the domain $\sigma > 0$, $s \neq 1$, with the formula

$$\zeta(s) = G(s)/(2^{1-s} - 1).$$

By Theorem 1.19, we have for $0 < \sigma \leqslant 1$, $|s-1| \gg 1$, that

$$\zeta(s) \ll G(s) \ll \tau^{1-\sigma+\varepsilon}$$

and so the continuation of $\zeta(s)$ has finite order for $0 < \sigma \leqslant 1$.

For any function F of finite order in a domain $\mathcal{D}$, we let $\mu(\sigma) = \mu_F(\sigma)$ denote the infimum of the set of real numbers ξ such that

$$F(s) \ll_{\sigma,\xi} |\tau|^{\xi} \qquad (s \in \mathcal{D},\ |\tau| \geqslant 1).$$

Theorem 1.20. *Let $F(s)$ be a function of finite order in a vertical strip $\sigma_1 \leqslant \sigma \leqslant \sigma_2$. Then the function $\mu(\sigma)$ is convex in this interval. In particular, it is continuous for $\sigma_1 < \sigma < \sigma_2$.*

Proof. It is an immediate consequence of the classical Phragmén–Lindelöf theorem[3] that the upper bounds

$$(\forall \varepsilon > 0) \qquad F(s) \ll_{\varepsilon} \mathrm{e}^{\varepsilon|\tau|} \qquad (\sigma_1 \leqslant \sigma \leqslant \sigma_2)$$

and

$$F(\sigma_1 + i\tau) \ll |\tau|^{k_1}, \quad F(\sigma_2 + i\tau) \ll |\tau|^{k_2} \qquad (|\tau| \geqslant 1)$$

imply

$$F(\sigma + i\tau) \ll |\tau|^{k(\sigma)} \qquad (\sigma_1 \leqslant \sigma \leqslant \sigma_2,\ |\tau| \geqslant 1)$$

where $k(\sigma)$ is the linear function taking the values k_1 and k_2, at σ_1 and σ_2, respectively. We hence deduce that

$$\mu(\sigma) \leqslant \frac{(\sigma_2 - \sigma)\mu(\sigma_1) + (\sigma - \sigma_1)\mu(\sigma_2)}{\sigma_2 - \sigma_1} \qquad (\sigma_1 \leqslant \sigma \leqslant \sigma_2). \tag{1.30}$$

□

Remark. The Phragmén–Lindelöf theorem actually implies that, for any $\varepsilon > 0$, we have

$$F(s) \ll_{\varepsilon,\sigma_1,\sigma_2} |\tau|^{k(\sigma)+\varepsilon}$$

in the domain $\sigma_1 \leqslant \sigma \leqslant \sigma_2$, $|\tau| \geqslant 1$. We shall later have occasion to use this local uniformity in σ.

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[2] In certain circumstances, it may be useful to replace the condition $|\tau| \geqslant 1$ in (1.29) by $|\tau| \geqslant \tau_0(F)$ for some arbitrarily large constant $\tau_0(F)$. All results presented in the sequel remain valid in this setting.

[3] See, for example, Titchmarsh (1939), § 5.65, or Valiron (1955), § 242.

Theorem 1.21. *For any Dirichlet series $F(s)$, we have*

$$\mu(\sigma) = 0 \qquad (\sigma > \sigma_a). \tag{1.31}$$

Moreover $\mu(\sigma)$ is a non-increasing function of σ in any region where F has finite order.

Proof. For $\sigma > \sigma_a$, $F(s)$ is bounded, so $\mu(\sigma) \leqslant 0$. Furthermore, if a_m is the first non-zero coefficient, we have

$$|F(s)| \geqslant \frac{|a_m|}{m^\sigma} - \sum_{n \geqslant m+1} \frac{|a_n|}{n^\sigma} > 0$$

provided that σ is sufficiently large. Since this lower bound is independent of τ, we have $\mu(\sigma) \geqslant 0$, and consequently $\mu(\sigma) = 0$. Now applying (1.30) with σ and σ_2 sufficiently large (so that $\mu(\sigma) = \mu(\sigma_2) = 0$), we obtain $\mu(\sigma_1) \geqslant 0$ for $\sigma_1 > \sigma_a$, thereby establishing (1.31). The second assertion is also a consequence of (1.30), selecting $\sigma_2 > \sigma_a$: since $\mu(\sigma_2) = 0$, we can write

$$\mu(\sigma) \leqslant \frac{\sigma_2 - \sigma}{\sigma_2 - \sigma_1}\mu(\sigma_1) \leqslant \mu(\sigma_1) \qquad (\sigma_1 \leqslant \sigma)$$

with strict inequality $\mu(\sigma) < \mu(\sigma_1)$ if $\mu(\sigma_1) \neq 0$ and $\sigma > \sigma_1$ □

Corollary 1.22. *Let $F(s)$ be a Dirichlet series of finite order for $\sigma > \sigma_0$ with $\sigma_0 < \sigma_a$. Then $\mu(\sigma_a) = 0$.*

Proof. This follows immediately from (1.31) and the last assertion of Theorem 1.20. □

Notes

§ 1.1. It is easy to extend Theorem 1.2 in the following manner: *If the series $F(s)$ converges, and if the series $G(s)$ converges absolutely, then the series $H(s)$ converges and we have $H(s) = F(s)G(s)$.*

Indeed, let us define, for fixed s, $A(x) := \sum_{m\leqslant x} f(m)m^{-s}$. Then, for any $\varepsilon > 0$, there exists some $y = y(\varepsilon)$ such that

$$|A(z) - F(s)| < \varepsilon \qquad (z > y).$$

Hence, for sufficiently large x, we can write,

$$\begin{aligned}\sum_{n\leqslant x} \frac{h(n)}{n^s} &= \sum_{md\leqslant x} \frac{f(m)g(d)}{(md)^s} = \sum_{d\leqslant x} \frac{g(d)}{d^s} A\Big(\frac{x}{d}\Big) \\ &= \sum_{d\leqslant x/y} \frac{g(d)}{d^s}\{F(s) + O(\varepsilon)\} + O\Big(\sum_{x/y<d\leqslant x} \Big|\frac{g(d)}{d^s}\Big|\Big).\end{aligned}$$

We obtain the result stated above by letting x tend to infinity, and then ε tend to 0.

Using Cesàro summability for Dirichlet series, it can also be shown that *the equality $H(s) = F(s)G(s)$ remains valid in any domain of convergence common to the three series F, G, H*—cf. Landau (1909), pp. 762, 904, or Hardy & Riesz (1915), p. 64.

As shown by the example in Exercise 155, it is not in general possible to deduce the convergence of $H(s) = \sum_{n\geqslant 1}(f * g)(n)/n^s$ from that of $F(s)$ and $G(s)$. By Theorem 1.7, if $F(s_0)$ and $G(s_0)$ both converge, then for any $\varepsilon > 0$, $F(s_0 + 1 + \varepsilon)$ and $G(s_0 + 1 + \varepsilon)$ converge absolutely, so $H(s)$ is (absolutely) convergent for $\sigma > \sigma_0 + 1$. Without much effort this result can be made considerably more precise—cf. Landau (1909) pp. 759 and ff., or Hardy & Riesz (1915), p. 67.

Theorem 1.23 (Stieltjes, 1887). *If $F(s)$ and $G(s)$ converge for $s = s_0$, then $H(s) = F(s)G(s)$ converges at $s = s_0 + \frac{1}{2}$.*

Proof. We may assume, without loss of generality, that $s_0 = 0$. Moreover, Abel summation readily shows that the hypotheses imply

$$\sum_{n>y} f(n)n^{-1/2} = o(y^{-1/2}), \quad \sum_{n>y} g(n)n^{-1/2} = o(y^{-1/2}) \quad (y \to \infty). \tag{1.32}$$

Now we can write

$$\sum_{n\leqslant x} h(n)n^{-1/2} = \sum_{m\leqslant\sqrt{x}} f(m)m^{-1/2} \sum_{d\leqslant\sqrt{x}} g(d)d^{-1/2} + R_1 + R_2 \tag{1.33}$$

where we have defined

$$R_1 := \sum_{d \leqslant \sqrt{x}} g(d) d^{-1/2} \sum_{\sqrt{x} < m \leqslant x/d} f(m) m^{-1/2}$$

and where R_2 is the analogous quantity obtained by interchanging the roles of f and g. By (1.32), we see that

$$R_1 = \sum_{d \leqslant \sqrt{x}} g(d) d^{-1/2} \cdot o\big(x^{-1/4}\big) \ll \sum_{d \leqslant \sqrt{x}} d^{-1/2} \cdot o\big(x^{-1/4}\big) = o(1).$$

Similarly, we obtain that $R_2 = o(1)$. By hypothesis, the first term on the right of (1.33) tends to $F(\frac{1}{2})G(\frac{1}{2})$ as $x \to \infty$. This completes the proof. □

Stieltjes' theorem was refined by Landau (1909, vol. § 216, *Satz* 24). Delange & Tenenbaum (1992) made the result more precise and generalized it.

Theorem 1.24 (Delange & Tenenbaum, 1992). *Let $k \geqslant 2$ and let*

$$F_j(s) := \sum_{n \geqslant 1} a_j(n)/n^s \quad (1 \leqslant j \leqslant k)$$

be Dirichlet series. We assume that, for each j, the series $F_j(s)$ converges for $s = \sigma_j + i\tau_j$, and that

$$\max_j \sigma_j - \min_j \sigma_j < 1/(k-1).$$

Then the product series $F(s) := \sum_{n \geqslant 1} a(n)/n^s$ where $a := a_1 * \cdots * a_k$ converges on the line $\sigma = (\sigma_1 + \cdots + \sigma_k)/k + 1 - 1/k$, and the convergence is uniform on any bounded segment of this line.

Using Bohr's counter-example mentioned below, it can easily be shown that the number $1 - 1/k$ appearing in this statement is optimal.

For an exhaustive study of the convergence of a product of Dirichlet series depending on that of its terms, see Kahane & Queffélec (1997).

Theorems of this type can have non-trivial applications. Consider for example the case of $F(s) = G(s) = \sum_{n \geqslant 1} (-1)^n/n^s$. Writing each integer in the canonical form $n = 2^\nu m$, $2 \nmid m$, we see that

$$h(n) = \begin{cases} \tau(m) & (\nu = 0) \\ (\nu - 3)\tau(m) & (\nu \geqslant 1). \end{cases}$$

Theorem 1.23 then immediately implies, for any $\varepsilon > 0$,

$$\sum_{n \leqslant x} h(n) \ll_\varepsilon x^{1/2+\varepsilon}. \tag{1.34}$$

Indeed, this results, by Theorem 1.13, from the fact that the series associated to $h(n)$ converges for $\sigma > \frac{1}{2}$.

A deeper analysis allows us to elucidate this example further. By the identity

$$\sum_{m\leqslant x,\, 2\nmid m} \tau(m) = T(x) - 2T\big(\tfrac{1}{2}x\big) + T\big(\tfrac{1}{4}x\big)$$

where

$$T(x) := \sum_{n\leqslant x} \tau(n),$$

we can see, after an elementary calculation, that

$$\sum_{n\leqslant x} h(n) = \Delta(x) - 4\Delta\big(\tfrac{1}{2}x\big) + 4\Delta\big(\tfrac{1}{4}x\big) \qquad (x > 0)$$

where we have set

$$\Delta(x) := T(x) - x(\ln x + 2\gamma - 1).$$

Voronoï's theorem (Theorem I.6.11) thus allows us to replace the exponent $1/2$ in (1.34) by $1/3$, and it is natural to conjecture that the abscissa of convergence for the series $H(s) = F(s)^2$ is $\sigma_c = 1/4$.

In Exercise 155, we give a simple example of a Dirichlet series $F(s)$ which converges everywhere on the line of convergence $\sigma = \sigma_c$ but such that the series $F(s)^2$ diverges everywhere on the same line. If $\sigma_c(F)$ and $\sigma_c(F^2)$ are the respective abscissas of convergence of $F(s)$ and $F(s)^2$, Theorem 1.23 implies

$$\sigma_c(F^2) \leqslant \sigma_c(F) + \tfrac{1}{2}. \tag{1.35}$$

Disproving a conjecture of Cahen (1894), Landau (1909, p. 773) showed that one may have

$$\sigma_c(F^2) > \sigma_c(F). \tag{1.36}$$

His argument consists in employing the estimate

$$\zeta(s) = \Omega(|\tau|^{\frac{1}{2}-\sigma}) \qquad (0 < \sigma < \tfrac{1}{2}), \tag{1.37}$$

which we shall see in § 3.4 to be a straightforward consequence of the functional equation for the zeta function $\zeta(s)$. The by now familiar relation

$$G(s) := \sum_{n\geqslant 1} \frac{(-1)^n}{n^s} = (2^{1-s} - 1)\zeta(s) \qquad (\sigma > 1)$$

implies that (1.37) is equally valid for $G(s)$, although $\sigma_c(G) = 0$. This implies

$$G(s)^4 = \Omega(|\tau|^{2-4\sigma}) \qquad (0 < \sigma < \tfrac{1}{2}),$$

so that $\sigma_c(G^4) \geqslant \frac{1}{4}$, by Theorem 1.19. Therefore (1.36) holds either for $F = G$, or for $F = G^2$.

Actually one can even have equality in (1.35). Bohr (1910) provided an example of a Dirichlet series such that $\sigma_c = 0, \sigma_a = 1$, and

$$F(s) = \Omega_\varepsilon(|\tau|^{1-\sigma-\varepsilon}) \qquad (\varepsilon > 0,\ 0 < \sigma < 1). \tag{1.38}$$

Theorem 1.19 readily implies that, for this series, $\sigma_c(F^2) \geqslant \frac{1}{2}$, from which equality follows by (1.35).

Bohr's construction is ingenious but relatively easy. Let us consider a sequence $\{\tau_k\}_{k=1}^{\infty}$ of real numbers tending to infinity sufficiently fast ($\tau_k := \exp 2^k$ suffices), and let $\{\delta_k\}_{k=1}^{\infty}$ be any sequence such that $\delta_k \to 0$ and $\tau_k^{\delta_k} \to +\infty$. Now let the sequence $\{a_n\}_{n=1}^{\infty}$ be defined by the formula

$$A_n := \sum_{m \leqslant n} a_m = \begin{cases} 0 & (\tau_k^{1/2} < n \leqslant \tau_k^{1+\delta_k}) \\ n^{i\tau_k} & (\tau_k^{1+\delta_k} < n \leqslant \tau_k^2) \\ 1 & (\tau_k^2 < n \leqslant \tau_{k+1}). \end{cases}$$

Then the series $F(s) := \sum_{m \geqslant 1} a_m/m^s$ satisfies (1.38). Indeed, we first check that $\sigma_c = 0$ since A_n does not tend to a limit at infinity (whence $\sigma_c \geqslant 0$) and Abel's rule shows that $F(\sigma)$ converges for all $\sigma > 0$ (hence $\sigma_c \leqslant 0$). By Theorem 1.7, we have $\sigma_a \leqslant 1$, and by Theorem 1.19, $\mu(\sigma) \leqslant 1 - \sigma$ for $0 \leqslant \sigma \leqslant 1$. We now prove that for all σ, $0 < \sigma < 1$, we have

$$F(\sigma + i\tau_k) \sim (i/\sigma)\tau_k^{1-\sigma(1+\delta_k)} \qquad (k \to \infty),$$

which plainly establishes (1.38) and shows that

$$\mu(\sigma) = 1 - \sigma \qquad (0 \leqslant \sigma \leqslant 1). \tag{1.39}$$

Setting $s_k = \sigma + i\tau_k$, we have

$$F(s_k) = \sum_{n \geqslant 1} A_n\{n^{-s_k} - (n+1)^{-s_k}\} = \sum_{n \leqslant \sqrt{\tau_k}} + \sum_{n > \tau_k^{1+\delta_k}} (\cdots)$$

since $A_n = 0$ when $\sqrt{\tau_k} < n \leqslant \tau_k^{1+\delta_k}$. The first of the two sums above is clearly

$$\ll \sum_{n \leqslant \sqrt{\tau_k}} n^{-\sigma} \ll_\sigma \tau_k^{(1-\sigma)/2}.$$

The second equals

$$\sum_{n > \tau_k^{1+\delta_k}} \left\{ s_k A_n n^{-1-s_k} + O(s_k^2 n^{-2-s_k}) \right\}$$

$$= s_k \sum_{\tau_k^{1+\delta_k} < n \leqslant \tau_k^2} n^{-1-\sigma} + O\left(\sum_{n > \tau_k^2} \tau_k n^{-1-\sigma} + \sum_{n > \tau_k^{1+\delta_k}} \tau_k^2 n^{-2-\sigma} \right)$$

$$= s_k\left\{ \sigma^{-1}\tau_k^{-\sigma(1+\delta_k)} + O_\sigma(\tau_k^{-2\sigma}) \right\} + O_\sigma\left(\tau_k^{1-\delta_k-\sigma(1+\delta_k)}\right) \sim \frac{i}{\sigma}\tau_k^{1-\sigma(1+\delta_k)}.$$

§ 1.3. For a historical clarification of the decisive contribution made by Phragmén to the so-called "Landau's theorem", the reader is invited to consult the detailed article of Dress (1983–84). This also contains an expository treatment and a thorough survey of oscillation theorems available in the literature. See also Kaczorowski & Pintz (1986–7).

The Phragmén–Landau theorem can be stated in the following form: *Let f be a non-negative arithmetic function; if the function $F(s)$ defined by $F(s) = \sum_{n\geqslant 1} f(n)/n^s$ for $\sigma > \sigma_0$ has a regular analytic continuation to a domain containing the line $\sigma = \sigma_0$, then the series converges for $\sigma = \sigma_0$.*

Ingham (1935) showed that in this statement *one can replace the condition $f \geqslant 0$ by $|f| \leqslant M$ when $\sigma_0 = 1$*. This allows one to deduce the prime number theorem directly from the fact that $\zeta(s) \neq 0$ on the line $\sigma = 1$ (cf. § 3.7). Indeed we obtain the convergence of $\sum_{n\geqslant 1} \mu(n)/n$, from which the desired conclusion follows by Theorem I.3.8.

Ingham's proof rests on considerations from Fourier analysis. D. J. Newman (1980) gave a much simpler proof of his theorem by using an ingenious method of contour integration. Korevaar (1982), and then Zagier (1997), reduced the prerequisites even more in an elegant and ultra-short presentation of a variant of Newman's proof: see II.7.31.

The proof given here of Theorem 1.12 provides a lower bound for the implicit constant in the $\Omega_\pm$ notation. For further information of this type see Dress (1983–84) or Grosswald (1972).

§ 1.5. The best results known to date concerning $N(x, y)$ are due to Robert & Tenenbaum (2013), and significantly improve those of Squalli (1985) in a thesis supervised by the author. Let F be the continuous function on $\mathbb{R}^+$, differentiable on $\mathbb{R}^+ \setminus \{\ln m : m \in \mathbb{N}^*\}$, defined for $v \geqslant 0$ by

$$F(v) := \frac{6}{\pi^2} \sum_{m\geqslant 1} \frac{\min(1, \mathrm{e}^v/m)}{\prod_{p|m}(p+1)}.$$

Robert & Tenenbaum establish the existence of a sequence of polynomials Q_j $(j \geqslant 1)$, with $\deg Q_j \leqslant j$, such that, for each $N \geqslant 1$, we have

$$F(v) = \exp\left\{\sqrt{\frac{8v}{\ln v}}\left(1 + \sum_{1\leqslant j\leqslant N} \frac{Q_j(\ln_2 v)}{(\ln v)^j} + O_N\Big(\Big(\frac{\ln_2 v}{\ln v}\Big)^{N+1}\Big)\right)\right\} \quad (v \geqslant 3).$$

In particular, $Q_1(t) = 1 + \frac{1}{2}t$. Writing systematically

$$v := \ln(x/y) \qquad (1 \leqslant y \leqslant x),$$

they show in particular that

$$N(x, y) \ll yF(v) \qquad (x \geqslant y \geqslant 2)$$

and that, with the notations

$$\mathcal{Y}_x := \mathrm{e}^{\frac{1}{4}\sqrt{2\ln x}(\ln_2 x)^{3/2}} \quad \text{and} \quad M_x := \sqrt{2\ln x \ln_2 x}\,\ln_3 x,$$

we have

$$N(x,y) \sim yF(v) \Leftrightarrow y > \mathcal{Y}_x \mathrm{e}^{-3M_x/8}\mathrm{e}^{\psi_x\sqrt{\ln x \ln_2 x}} \qquad (\psi_x \to \infty).$$

The same article contains much information, too long to mention here *in extenso*, on the local and global behavior of $N(x,y)$. We restrict here to a brief description of some of it.

Rankin's upper bound for $N(x,y)$ can be written

$$N(x,y) \leqslant \frac{6y}{\pi^2}\mathrm{e}^{g(\sigma_v)}$$

where $g(\sigma) := \sum_p \ln\Big(1 + \dfrac{1-p^{\sigma-1}}{p(p^\sigma-1)}\Big)$ $(\sigma > 0)$ and σ_v is defined by the equation $g'(\sigma_v)+v = 0$. We then have the generalized asymptotic expansion[4]

$$\sigma_v \approx \sqrt{\frac{2}{v\ln v}}\Big\{1 + \sum_{k\geqslant 0}\frac{P_k(\ln_2 v)}{(\ln v)^k}\Big\} \qquad (v \to \infty)$$

where P_k is a polynomial of degree $\leqslant k$. Thus

$$P_1(X) := \tfrac{1}{2}(X - \ln 2), \quad P_2(X) = \tfrac{3}{8}X^2 - (\tfrac{3}{4}\ln 2 + \tfrac{1}{2})X + \tfrac{1}{2}\ln 2 + \tfrac{3}{8}(\ln 2)^2 + \tfrac{2}{3}\pi^2.$$

Robert & Tenenbaum establish the following six assertions:

(i) $\forall c \in]0,1[\ \exists b = b_c :\ N(x,y) = yF(v)^{c+o(1)} \Leftrightarrow y = \mathrm{e}^{\{b+o(1)\}\sqrt{\ln x \ln_2 x}}$ where the quantity b_c is defined as the solution of an explicit equation depending on c;

(ii) $N(x,y) = yF(v)^{o(1)} \Leftrightarrow (\ln y)/\sqrt{\ln x \ln_2 x} \to 0$;

(iii) $N(x,y) \sim yF(v)\mathrm{e}^{-\{1+o(1)\}\mathcal{K}}$ $(x \geqslant y \geqslant 2)$ where $\mathcal{K} = \mathcal{K}(x,y)$ is an explicit quantity satisfying $\mathcal{K} \asymp v\sigma_v/y^{\sigma_v}$;

(iv) $N(2x,y) \sim N(x,y) \Leftrightarrow y = o(x)$;

(v) $N(x,2y) \sim 2N(x,y) \Leftrightarrow \ln y > (\ln x)^{1/2+o(1)}$;

(vi) $(\forall b > 1) \qquad N(x,2y) \sim 2^b N(x,y) \Leftrightarrow \ln y = (\ln x)^{1/(b+1)+o(1)}$.

These results allow a precise formulation of the *abc*-conjecture of Masser & Oesterlé—cf. Masser (1985)—which is one of the most famous open problems in mathematics to date. One of its equivalent formulations consists in stipulating that, for any $\varepsilon > 0$, there exists a constant M_ε such that the conditions $a \in \mathbb{N}^*$, $b \in \mathbb{N}^*$, $(a,b) = 1$ imply $c := a + b \leqslant M_\varepsilon k(abc)^{1+\varepsilon}$. This provides an essential restriction on the multiplicative structure of a product whose terms are connected by an additive relation. Its numerous

[4] In the sense that, truncating the series at any order, we obtain an asymptotic formula with an error term whose order does not exceed that of the first neglected term.

consequences, notably pertaining to Fermat's Last Theorem, have been extensively described in the literature.

The heuristic hypothesis underlying the *abc*-conjecture consists in imagining that the numbers $k(a)$, $k(b)$ and $k(c)$ essentially behave like independent random variables when the sizes of the parameters have a fixed order of magnitude. Since the estimates described above for $N(x,y)$ essentially provide a description of the distributions of these random variables, they naturally allow a precise formulation of the conjecture. Robert, Stewart & Tenenbaum (2014) thus obtain the following version by a probabilistic argument of Borel–Cantelli type.

Conjecture. *Assume $\varepsilon > 0$. There exists a positive constant A such that, for every triplet (a,b,c) of positive integers satisfying $(a,b) = 1$, $c = a+b$, $k = k(abc)$, we have*

$$c \leqslant AkF\big(\tfrac{2}{3}\ln k\big)^{3-(\ln 4-\varepsilon)/\ln_2(k+2)}.$$

Moreover the inequality $c > kF\big(\frac{2}{3}\ln k\big)^{3-(\ln 4+\varepsilon)/\ln_2(k+2)}$ holds for infinitely many such triplets.

For an application of Theorem 1.16, see Exercise 276, p. 471.

§ 1.6. Theorem 1.19, which states that

$$\mu(\sigma) \leqslant 1-(\sigma-\sigma_c) \qquad (\sigma_c \leqslant \sigma \leqslant \sigma_c+1),$$

is optimal, as shown by Bohr's counter-example mentioned above—cf. Hardy & Riesz (1915), p. 19.

We shall see in § 2.2 (Theorem 2.8) that Theorem 1.21 has a partial converse: *if $F(s)$ is holomorphic and satisfies $\mu(\sigma) = 0$ for $\sigma > \sigma_0$, then $\sigma_c \leqslant \sigma_0$.*

Thus, if $F(s) := 1+\sum_{n\geqslant 2} f(n)/n^s$ converges for $\sigma > \sigma_0$, does not vanish in this half-plane, and satisfies there the lower bound $|F(s)| \gg_\varepsilon (1+|\tau|)^{-\varepsilon}$ for any $\varepsilon > 0$, then the series $G(s) = F(s)^{-1}$ converges for all $\sigma > \sigma_0$. A theorem of Landau (1933) shows that the growth condition is actually superfluous.

Exercises

155. Show that the Dirichlet series $F(s) = \sum_{n\geqslant 1}(-1)^n(\ln 2n)^{-2}n^{-s}$ converges at every point on the line of convergence $\sigma = 0$. Define

$$F(s)^2 = \sum_{n\geqslant 1}\frac{h(n)}{n^s} =: H(s).$$

Show that $h(n)$ is unbounded, and deduce that $H(s)$ diverges everywhere on the line $\sigma = 0$.

156. Write $A(t) := \sum_{n\leqslant t}(-1)^n n \qquad (t \geqslant 0)$.

(a) Show that $A(N) = (-1)^N \left\lfloor \frac{1}{2}(N+1)\right\rfloor$ for any integer $N \geqslant 1$.

(b) Calculate the abscissa of convergence of $G(s) := \int_{1^-}^{+\infty} t^{-s-1}\,\mathrm{d}A(t)$.

(c) Show that, for $\sigma > 0$, we have

$$G(s) = (s+1)\int_1^\infty \frac{A(t)}{t^{s+2}}\,\mathrm{d}t. \tag{1.40}$$

(d) Show that the series $\sum_{n\geqslant 1} A(n)\int_n^{n+1}\mathrm{d}t/t^{1+\varepsilon}$ converges for each $\varepsilon > 0$.

(e) Deduce that (1.40) defines a regular analytic continuation of $G(s)$ into the half-plane $\sigma > -1$.

157. Let $F(s) = \sum_{n\geqslant 1} a_n/n^s$ be a Dirichlet series with a finite abscissa of convergence σ_c.

(a) Show that the series $\varphi(z) = \sum_{n\geqslant 1} a_n \mathrm{e}^{-nz}$ is uniformly convergent for $\Re\mathrm{e}\, z \geqslant \varepsilon > 0$.

(b) Define $H := \{s : \sigma > \max(0, \sigma_c)\}$. Show that for any $s \in H$, there exists a constant $C(\sigma)$ such that $|\varphi(t)| \leqslant C(\sigma)t^{-\sigma}$ $(t > 0)$. Deduce that the domain of absolute convergence of the integral $I(s) = \int_0^\infty t^{s-1}\varphi(t)\,\mathrm{d}t$ contains the half-plane H.

(c) Show that for $s \in H$, we have $F(s)\Gamma(s) = I(s)$.

158. Let $F(s) = \sum_{n\geqslant 1} a_n/n^s$ be a Dirichlet series. Assume there exists $z \in \mathbb{C}$, with $\Re\mathrm{e}\, z > -1$, such that

$$A(x) := \sum_{1\leqslant n\leqslant x} a_n \sim x(\ln x)^z \qquad (x \to \infty).$$

Show that $F(s) = \Gamma(z+1)/(s-1)^{z+1} + o\big(1/(\sigma-1)^{\Re\mathrm{e}\, z+1}\big)$ as $s \to 1$ while remaining in the half-plane $\sigma > 1$.

159. Retain the notation of Exercise 158, but now assume that $A(x) \ll 1$. Show that $F(s)$ is holomorphic in the half-plane $\sigma > 0$ and that, if $F(s)$ has a pole on the line $\sigma = 0$, it must be simple.

160. *Three oscillation results.* In this exercise, we assume the following weak consequence of Theorems 3.3 and 3.19 below: *The function $\zeta(s)$ has at least one zero in the closed half-plane $\sigma \geqslant \frac{1}{2}$ and the zeros in the strip $0 < \sigma < 1$ are placed symmetrically about the axis $\sigma = \frac{1}{2}$.*

(a) Using the properties of the series $G(s) := \sum_{n\geqslant 1}(-1)^n/n^s$, show that $\zeta(s)$ can be continued to a meromorphic function for $\sigma > 0$, which has as its only singularity a pole at $s = 1$ and does not vanish for $s \in]0, +\infty[$.

(b) By considering the series $\zeta(s) + \zeta'(s)/\zeta(s)$, show that

$$\psi(x) = x + \Omega_{\pm}(x^{\frac{1}{2}}).$$

(c) By considering the series $1/\zeta(s)$, show that

$$M(x) := \sum_{n\leqslant x} \mu(n) = \Omega_{\pm}(x^{\frac{1}{2}}).$$

(d) By considering the series $\zeta(s)/\zeta(2s)$, show that

$$Q(x) := \sum_{n\leqslant x} \mu(n)^2 = \frac{6}{\pi^2}x + \Omega_{\pm}(x^{\frac{1}{4}}).$$

161. (a) Let f be the additive function defined by

$$f(p^\nu) = \sqrt{p} \qquad (p \in \mathbb{P}, \nu \in \mathbb{N}^*).$$

Determine the order of magnitude of the summatory function

$$F(x) := \sum_{n\leqslant x} f(n).$$

(b) Consider the same question, but now assuming the function f is multiplicative while leaving unchanged the definition of the numbers $f(p^\nu)$. Hint: compare $f(n)$ to $\sqrt{n}$ and apply either Theorem I.3.10, or the inequality $\mu(n)^2 \geqslant 1 - \sum_{p^2|n} 1$.

(c) Determine the abscissas of convergence of the Dirichlet series associated to the functions defined in (a) and (b).

162. (a) Show that for $s \in]1, \infty[$, $x > 0$, $N := \lfloor x \rfloor$, we have

$$\frac{1}{(s-1)(N+1)^{s-1}} < \sum_{j>x} \frac{1}{j^s} < \frac{1}{(s-1)N^{s-1}},$$

where, if $N = 0$, the upper bound is interpreted as $\zeta(s) < \infty$.

(b) Show that $1/(s-1) < \zeta(s) < s/(s-1)$ for all $s \in]1, \infty[$.

(c) Establish that the equation $\zeta(s) = 2$ has a unique root ϱ on the half-line $]1, \infty[$. Prove that $\frac{3}{2} < \varrho < 2$.

(d) For $k \in \mathbb{N}^*$, $n \in \mathbb{N}^*$, denote by $t_k(n)$ the number of solutions of the equation $d_1 d_2 \cdots d_k = n$ where the unknowns $d_1, \dots, d_k$ are integers $\geqslant 2$. Let $T_k(s) := \sum_{n\geqslant 1} t_k(n)/n^s$ denote the Dirichlet series associated to the arithmetic function t_k. Express $T_1(s)$ as a function of $\zeta(s)$ in a suitable sub-domain of the complex plane, to be described explicitly.

(e) Show that $t_k = t_{k-1} * t_1$ for $k \geqslant 2$. Deduce a simple expression for $T_k(s)$. Determine the abscissa of convergence of this series.

(f) Show that, for each fixed integer n, we have $t_k(n) = 0$ for all sufficiently large k .

(g) Set $f(n) := \sum_{k\geqslant 1} t_k(n)$. Show that, for $\Re e\, s > \varrho$, we have

$$F(s) := \sum_{n\geqslant 1} \frac{f(n)}{n^s} = \frac{1}{2-\zeta(s)} - 1.$$

Deduce the value of the abscissa of convergence for the Dirichlet series $F(s)$.

(h) Show that $f * t_1 = f - t_1$. Deduce that, writing $A(x)$ for the summatory function of the function f, we have, for all $x \geqslant 1$,

$$A(x) = \lfloor x \rfloor - 1 + \sum_{2\leqslant j\leqslant x} A(x/j).$$

(i) Put $C := (\varrho - 1)2^{\varrho-1}$ where ϱ is the real number introduced in (c). Show that $A(x) \leqslant Cx^\varrho$ for all $x \geqslant 1$. Hint: argue by induction over the unique integer h such that $2^h \leqslant x < 2^{h+1}$, and appeal to the lower bound established in (a).

163. Let $P(m,n) := m^2 + n^3 + mn$.

(a) Show that $m^2 + n^3 \leqslant P(m,n) \leqslant 2(m^2+n^3)$ for $m \geqslant 1$, $n \geqslant 1$.

(b) Show that we have

$$\sum_{n\geqslant 1} \frac{1}{(n+a)^\alpha} = \int_1^\infty \frac{1}{(t+a)^\alpha}\,\mathrm{d}t + O\Big(\frac{1}{a^\alpha}\Big) \asymp \frac{a^{1-\alpha}}{\alpha-1}$$

uniformly for $a \geqslant 1$ and $1 < \alpha \leqslant a$.[5]

(c) What is the abscissa of convergence of $\sum_{n\geqslant 1}\sum_{m\geqslant 1} P(n,m)^{-s}$?

(d) What estimates can be deduced for

$$A(x) := |\{n, m \geqslant 1 : P(n,m) \leqslant x\}|?$$

(e) Using a direct argument, evaluate the order of magnitude of $A(x)$ up to a multiplicative constant. Recover in this way the result of (c).

[5] Recall that $f \asymp g$ means $f \ll g \ll f$.

164. Put $a_n = 1$ if the base 10 expansion of n does not contain the digit 6, and $a_n = 0$ otherwise. Show that there exists a real number $\delta > 0$, to be calculated, such that

$$\sum_{n\leqslant x} a_n \asymp x^{1-\delta} \qquad (x \geqslant 1).$$

What can be deduced about the abscissa of convergence of the associated Dirichlet series?

165. *A lower bound for the number of integers with a small core.*

(a) Let $m \in \mathbb{N}^*$. Show that the number of integral solutions $\nu_1 \geqslant 0$, $\nu_2 \geqslant 0, \dots, \nu_m \geqslant 0$ of the inequality

$$\sum_{1\leqslant i\leqslant m} \nu_i \leqslant N$$

equals $\binom{N+m}{m}$. [*Use the Taylor series of* $(1-x)^{-m-1}$.]

(b) Let $p_1 < p_2 < \cdots < p_m$ be distinct prime numbers. Show that each integer n of the form

$$n = p_1^{\nu_1} \cdots p_m^{\nu_m} r, \quad \text{with} \quad \mu(p_1 \cdots p_m r)^2 = 1,$$

satisfies $k(n) \leqslant r p_1 \cdots p_m$.

(c) Deduce from (a) and (b) that, for any integer $m \geqslant 1$, we have

$$N(x,y) \gg_m \big(\ln(2x/y)\big)^m \sum_{r\leqslant y/p_1\cdots p_m} \mu(p_1\cdots p_m r)^2.$$

(d) Establish the identity $\mu(n)^2 = \sum_{d^2|n} \mu(d)$, and deduce that, uniformly for $M \geqslant 1$, $\mu(M)^2 = 1$, $y \geqslant 1$, we have

$$\sum_{r\leqslant y} \mu(Mr)^2 = \frac{6}{\pi^2} \prod_{p|M} \Big(1+\frac{1}{p}\Big)^{-1} y + O\big(2^{\omega(M)}\sqrt{y}\big).$$

(e) Prove that, for $m \geqslant 1$ and $x \geqslant y \geqslant p_1 \cdots p_m$, we have

$$N(x,y) \gg_m y\big\{\ln(2x/y)\big\}^m.$$

166. Let f be a multiplicative function. Show that the Dirichlet series associated to $n \mapsto f(kn)$ can, for each integer $k \geqslant 1$, be expanded as an infinite product of Eulerian type. Applications: for $\sigma > 1$, we have

$$\text{(a)} \quad \sum_{n\geqslant 1} \frac{\tau(kn)}{n^s} = \zeta(s)^2 \prod_{p^\nu \| k} (\nu + 1 - \nu p^{-s}),$$

$$\text{(b)} \quad \sum_{n\geqslant 1} \frac{1}{\tau(kn) n^s} = \prod_{p\nmid k} \Big\{p^s \log\Big(\frac{1}{1-p^{-s}}\Big)\Big\} \prod_{p^\nu \| k} \sum_{j\geqslant 0} \frac{1}{(j+\nu+1)p^{js}}.$$

167. Let $\alpha \in]0,1[$ and $\vartheta \neq 0$ be fixed real numbers.

(a) Show that $A(x) := \sum_{n \leqslant x} \mathrm{e}(\vartheta n^\alpha) \ll x^{1-\alpha}$.

(b) Show that, uniformly for $x \geqslant 1$, $0 < \varepsilon < 1$, we have

$$A(x + \varepsilon x^{1-\alpha}) - A(x) = \mathrm{e}(\vartheta x^\alpha)\varepsilon x^{1-\alpha} + O(1 + \varepsilon^2 x^{1-\alpha}).$$

(c) Show that the abscissa of convergence of the Dirichlet series

$$F(s) := \sum_{n \geqslant 1} \frac{\mathrm{e}(\vartheta n^\alpha)}{n^s}$$

is $\sigma_c = 1 - \alpha$.

168. (a) Let $\vartheta \in \mathbb{R}$ and let α, β be real numbers in $]0,1[$. Show that, if there exists an integer m such that $0 < \alpha \leqslant |\vartheta - m| \leqslant \beta < 1$, then $\|\vartheta\| \geqslant \min(\alpha, 1-\beta)$.

(b) Show that if ϑ is a real number and n, p, q, Q, v are integers such that $0 < n < Q$, $0 < q < Q$, $(p,q) = 1$, $|\vartheta - p/q| \leqslant 1/(qQ)$, $1 < v \leqslant q/2$, $np \equiv \pm v \pmod q$, then $\|n\vartheta\| \geqslant (v-1)/q$.

(c) Let ϑ be a fixed irrational number and let $\{p_k/q_k\}_{k=0}^\infty$ be the sequence of its convergents. Here, we aim at determining bounds on the quantity

$$A_\vartheta(x) := \sum_{1 \leqslant n \leqslant x} \frac{1}{\|n\vartheta\|}$$

for $x \geqslant 1$. In all the sequel, we consider x as given and implicitly define an integer $t \geqslant 0$ by the inequalities $q_t \leqslant x < q_{t+1}$.

(i) Show that $1/(2q_tq_{t+1}) < |\vartheta - p_t/q_t| < 1/(q_tq_{t+1})$. Deduce a two-sided estimate for $\|\vartheta q_t\|$.

(ii) Show that $A_\vartheta(x) > q_{t+1}$.

(iii) Show that $\|n\vartheta\| > 1/(2q_{t+1})$ for any integer n in $[1,x]$.

(iv) Using the inequality established in (b), show that, for all integers $v \in [2, \frac{1}{2}q_t]$, $n \in [1,x]$, such that $np_t \equiv \pm v \pmod{q_t}$, we have

$$\|n\vartheta\| \geqslant (v-1)/q_t.$$

(v) Show that, for any real number $\nu \geqslant 1$, we have

$$1/\nu \leqslant \int_{\nu-1/2}^{\nu+1/2} \mathrm{d}w/w,$$

and deduce that

$$\sum_{1 \leqslant n \leqslant z} \frac{1}{n} \leqslant \ln(2z+1) \qquad (z > 0).$$

(vi) Infer from the three previous questions that, for $m \in \mathbb{N}$, we have

$$\sum_{\substack{1\leqslant n\leqslant x\\ mq_t<n\leqslant(m+1)q_t}} \frac{1}{\|n\vartheta\|} < 6q_{t+1} + 2\sum_{1<v\leqslant q_t/2} \frac{q_t}{v-1} \leqslant 6q_{t+1} + 2q_t \ln q_t.$$

(vii) Show that

$$A_\vartheta(x) < 4x\Big\{3\frac{q_{t+1}}{q_t} + \ln q_t\Big\} \qquad (x \geqslant 1).$$

(d) Assume $\delta \geqslant 1$. Let $\{a_k\}_{k=0}^{\infty}$ be the integer sequence defined by $a_0 = 1$ and the inequalities $q_k^{\delta-1} - \frac{1}{2} < a_{k+1} \leqslant q_k^{\delta-1} + \frac{1}{2}$, where q_k is the denominator of the finite continued fraction $[a_0, a_1, \dots, a_k]$. Note that $a_k \geqslant 1$ for all k.

(i) Show that $\frac{1}{2}q_k^\delta \leqslant q_{k+1} \leqslant \frac{5}{2}q_k^\delta$ for all $k \geqslant 1$.

(ii) Define $\vartheta := [a_0, a_1, \dots]$. Deduce from (c-vii) that

$$A_\vartheta(x) < 30\,x^\delta + 4x \ln x \qquad (x \geqslant 1),$$

and show, using the result of question (c-ii), that $A_\vartheta(x) > \frac{1}{2}x^\delta$ for infinitely many integral x.

(iii) Determine the abscissa of convergence of the Dirichlet series

$$F(s) := \sum_{n\geqslant 1} \frac{1}{\|n\vartheta\| n^s}.$$

Summation formulae

2.1. Perron formulae

Most of the theorems in the previous chapter were intended to obtain analytic properties of Dirichlet series given the asymptotic behavior of their coefficients, or of their summatory functions. We are now going to proceed in the opposite direction. For the applications we have in mind, the ultimate aim is to understand arithmetic functions, so Dirichlet series are here to be regarded more as an indispensable tool than as an intrinsic subject of study.

In this context, the Perron formulae play the fundamental role of an inversion principle, analogous to that of the Cauchy formula in the theory of power series.

Let

$$F(s) := \sum_{n \geqslant 1} \frac{a_n}{n^s} \tag{2.1}$$

be a Dirichlet series with abscissa of convergence σ_c and abscissa of absolute convergence σ_a. Extending the definition of the function $n \mapsto a_n$ by putting $a_x = 0$ if $x \in \mathbb{R} \setminus \mathbb{N}^*$, we introduce the *normalized summatory function*

$$A^*(x) := \sum_{n<x} a_n + \tfrac{1}{2} a_x \qquad (x \geqslant 0). \tag{2.2}$$

Theorem 2.1 (Perron's formula). *Let $\kappa > \max(0, \sigma_c)$. We have*

$$A^*(x) = \frac{1}{2\pi i} \int_{\kappa - i\infty}^{\kappa + i\infty} F(s) x^s \frac{\mathrm{d}s}{s} \qquad (x > 0), \tag{2.3}$$

where the integral is conditionally convergent for $x \in \mathbb{R} \setminus \mathbb{N}$ and converges in the sense of Cauchy's principal value when $x \in \mathbb{N}$.

The proof rests on the following lemma, which amounts to an explicit calculation of the Laplace inversion formula for the function

$$h(x) := \begin{cases} 1 & (x > 1) \\ \frac{1}{2} & (x = 1) \\ 0 & (0 < x < 1). \end{cases} \tag{2.4}$$

Lemma 2.2. *For any positive κ, T, T', we have*

$$\text{(i)} \quad \Big|h(x) - \frac{1}{2\pi i}\int_{\kappa-iT'}^{\kappa+iT} x^s \frac{\mathrm{d}s}{s}\Big| \leqslant \frac{x^\kappa}{2\pi|\ln x|}\Big(\frac{1}{T} + \frac{1}{T'}\Big) \quad (x \neq 1),$$

$$\text{(ii)} \quad \Big|h(1) - \frac{1}{2\pi i}\int_{\kappa-iT}^{\kappa+iT} \frac{\mathrm{d}s}{s}\Big| \leqslant \frac{\kappa}{T+\kappa}.$$

Provisionally assuming this lemma, let us see how we can deduce formula (2.3). First suppose that $\kappa > \sigma_a$. Then the series $F(s)$ is absolutely and uniformly convergent for $\sigma = \kappa$, hence

$$\frac{1}{2\pi i}\int_{\kappa-iT'}^{\kappa+iT} F(s)\frac{x^s}{s}\,\mathrm{d}s = \frac{1}{2\pi i}\sum_{n\geqslant 1} a_n \int_{\kappa-iT'}^{\kappa+iT} \Big(\frac{x}{n}\Big)^s \frac{\mathrm{d}s}{s}.$$

Applying part (i) of the lemma, it follows that, for $x \in \mathbb{R} \setminus \mathbb{N}$,

$$\Big|\frac{1}{2\pi i}\int_{\kappa-iT'}^{\kappa+iT} F(s)\frac{x^s}{s}\,\mathrm{d}s - A^*(x)\Big| \leqslant \frac{x^\kappa}{2\pi}\Big(\frac{1}{T} + \frac{1}{T'}\Big)\sum_{n\geqslant 1} \frac{|a_n|}{|n|^\kappa\,|\ln(x/n)|}. \tag{2.5}$$

Since the factor $|\ln(x/n)|$ is bounded from below independently of n, we obtain the first assertion of the theorem by allowing T and T' to tend independently to infinity. The second assertion is proved in much the same way: it suffices to take $T = T'$, and to replace on the right-hand side of (2.5) the (infinite) term corresponding to $n = x$ by $\kappa|a_x|/(T+\kappa)$.

Now suppose that $\sigma_c < \kappa \leqslant \sigma_a$. We have $\kappa + 1 > \sigma_a$ by Theorem 1.7. Consider the integral

$$I := \int_{\mathcal{R}} F(s)\frac{x^s}{s}\,\mathrm{d}s$$

where $\mathcal{R}$ is the rectangle formed by the lines $\sigma = \kappa$, $\tau = T$, $\sigma = \kappa + 1$, $\tau = -T'$. By Theorem 1.19 we have

$$F(s)x^s s^{-1} \ll \tau^{-(\sigma-\sigma_c)+\varepsilon}x^\sigma \quad (s \in \mathcal{R},\ |\tau| \geqslant 1)$$

so the contribution to I from the horizontal segments of $\mathcal{R}$ tends to 0 as T and T' tend to infinity. Since $F(s)$ is analytic for $\sigma > \sigma_c$, the residue theorem shows that $I = 0$, whence if T, $T' \to +\infty$,

$$\int_{\kappa-iT'}^{\kappa+iT} F(s)x^s\,\frac{\mathrm{d}s}{s} = \int_{\kappa+1-iT'}^{\kappa+1+iT} F(s)x^s\,\frac{\mathrm{d}s}{s} + o(1).$$

This completes the proof of Theorem 2.1.

Proof of Lemma 2.2. Consider first the case $x > 1$. Let k be a sufficiently large integer and let $\mathcal{R}_k$ denote the rectangle with vertices $\kappa - iT'$, $\kappa + iT$, $\kappa - k + iT$, $\kappa - k - iT'$. By the residue theorem, we may write

$$\frac{1}{2\pi i}\int_{\mathcal{R}_k} x^s \frac{\mathrm{d}s}{s} = 1 = h(x).$$

Now we have the following upper bounds

$$\Big|\int_{\kappa+iT}^{\kappa-k+iT} x^s \frac{\mathrm{d}s}{s}\Big| \leqslant \frac{x^\kappa}{T|\ln x|}, \quad \Big|\int_{\kappa-k-iT'}^{\kappa-iT'} x^s \frac{\mathrm{d}s}{s}\Big| \leqslant \frac{x^\kappa}{T'|\ln x|},$$
$$\Big|\int_{\kappa-k+iT}^{\kappa-k-iT'} x^s \frac{\mathrm{d}s}{s}\Big| \leqslant \frac{x^{\kappa-k}}{k-\kappa}(T+T').$$

We deduce the stated result by letting k tend to infinity.

The case $0 < x < 1$ can be dealt with in a symmetric way, applying the same argument with k replaced by $-k$. We omit the details.

When $x = 1$, we simply note that we have

$$\frac{1}{2\pi i}\int_{\kappa-iT}^{\kappa+iT} \frac{\mathrm{d}s}{s} = \frac{1}{2\pi}\big(\arg(\kappa+iT) - \arg(\kappa-iT)\big) = \frac{1}{\pi}\arctan(T/\kappa).$$

The stated upper bound is now immediate from the following inequalities, valid for all $y > 0$

$$0 \leqslant \frac{\pi}{2} - \arctan y = \int_y^\infty \frac{\mathrm{d}t}{1+t^2} \leqslant \frac{2}{1+y}.$$

This concludes the proof of the lemma. □

In practice, it is useful to have at hand effective versions of Perron's formula, i.e., explicit bounds on the contribution from the domain $|\tau| \geqslant T$ to the integral (2.3).

Theorem 2.3 (First effective Perron formula). *For $\kappa > \max(0, \sigma_a)$, $T \geqslant 1$, and $x \geqslant 1$, we have*

$$A(x) = \frac{1}{2\pi i}\int_{\kappa-iT}^{\kappa+iT} F(s)x^s \frac{\mathrm{d}s}{s} + O\Big(x^\kappa \sum_{n\geqslant 1} \frac{|a_n|}{n^\kappa(1+T|\ln(x/n)|)}\Big). \tag{2.6}$$

Proof. It suffices to show that, for any fixed $\kappa > 0$, we have, uniformly for $y > 0, T > 0$

$$\Big|h(y) - \frac{1}{2\pi i}\int_{\kappa-iT}^{\kappa+iT} y^s \frac{\mathrm{d}s}{s}\Big| \ll y^\kappa/\big(1+T|\ln y|\big). \tag{2.7}$$

Indeed, applying this estimate with $y = x/n$ and summing over $n \geqslant 1$ (after multiplication by a_n) we obtain precisely the stated formula.

When $T|\ln y| > 1$, the upper bound (2.7) follows from Lemma 2.2 (i). Otherwise, we can write

$$\int_{\kappa-iT}^{\kappa+iT} y^s \frac{\mathrm{d}s}{s} = y^\kappa \int_{\kappa-iT}^{\kappa+iT} \frac{\mathrm{d}s}{s} + y^\kappa \int_{\kappa-iT}^{\kappa+iT} (y^{i\tau} - 1) \frac{\mathrm{d}s}{s}.$$

The second integral is

$$\ll \int_0^T |(\tau \ln y)/s| \, \mathrm{d}\tau \leqslant T|\ln y| \leqslant 1$$

and consequently, by Lemma 2.2 (ii), we see that the left-hand side of (2.7) is $\ll y^\kappa$. This concludes the proof. □

Corollary 2.4 (Second effective Perron formula). *Let $F(s) := \sum_{n\geqslant 1} a_n/n^s$ be a Dirichlet series with a finite abscissa of absolute convergence σ_a. Suppose there exists a real number $\alpha \geqslant 0$ such that*

$$\text{(i)} \qquad \sum_{n\geqslant 1} |a_n| n^{-\sigma} \ll (\sigma - \sigma_a)^{-\alpha} \qquad (\sigma_a < \sigma \leqslant \sigma_a + 1),$$

and that B is a non-decreasing function satisfying

$$\text{(ii)} \qquad |a_n| \leqslant B(n) \qquad (n \geqslant 1).$$

Then, for $x \geqslant 2$, $T \geqslant 2$, $\sigma \leqslant \sigma_a$, $\kappa := \sigma_a - \sigma + 1/\ln x$, we have

$$\sum_{n\leqslant x} \frac{a_n}{n^s} = \frac{1}{2\pi i} \int_{\kappa-iT}^{\kappa+iT} F(s+w) x^w \frac{\mathrm{d}w}{w} + O\Big(x^{\sigma_a-\sigma} \frac{(\ln x)^\alpha}{T} + \frac{B(2x)}{x^\sigma}\Big(1 + x\frac{\ln T}{T}\Big)\Big). \tag{2.8}$$

Proof. Apply formula (2.6) to the series $\sum b_n/n^w$ with $b_n := a_n/n^s$. The contribution of integers n which do not belong to $[\frac{1}{2}x, 2x]$ is

$$\ll x^\kappa T^{-1} \sum_{n\geqslant 1} |a_n| n^{-\kappa-\sigma} \ll x^{\sigma_a-\sigma} T^{-1} (\ln x)^\alpha.$$

When $\frac{1}{2}x \leqslant n \leqslant 2x$, we write $n = N + h$ where N is the nearest integer to x. We have $|\ln(x/n)| \gg |h|/x$. This leads to the following estimate for the extra contribution:

$$x^{-\sigma} \sum_{x/2\leqslant n\leqslant 2x} \frac{|a_n|}{1 + T|\ln(x/n)|} \ll \frac{B(2x)}{x^\sigma} \sum_{0\leqslant h\leqslant x+1} \frac{1}{1 + Th/x}$$
$$\ll \frac{B(2x)}{x^\sigma}\Big\{1 + \sum_{1\leqslant h\leqslant x/T} 1 + \sum_{x/T<h\leqslant x+1} \frac{x}{Th}\Big\} \ll \frac{B(2x)}{x^\sigma}\Big\{1 + x\frac{\ln T}{T}\Big\}. \square$$

In certain circumstances, it is useful to work with an absolutely convergent Perron integral. This necessitates a suitable smoothing of the function $A(x)$. The following theorem provides two examples.

Theorem 2.5. *For $\kappa > \max(0, \sigma_c)$ and $x \geqslant 1$, we have*

$$\sum_{n \leqslant x} a_n \ln(x/n) = \frac{1}{2\pi i} \int_{\kappa - i\infty}^{\kappa + i\infty} F(s) x^s \frac{\mathrm{d}s}{s^2}, \tag{2.9}$$

$$\int_0^x A(t)\,\mathrm{d}t = \frac{1}{2\pi i} \int_{\kappa - i\infty}^{\kappa + i\infty} F(s) x^{s+1} \frac{\mathrm{d}s}{s(s+1)}. \tag{2.10}$$

Proof. For $w \geqslant 0$ and $x \in \mathbb{R}^+ \setminus \mathbb{N}$, the Perron formula (2.3) implies

$$\sum_{n \leqslant x} a_n n^w = \frac{1}{2\pi i} \int_{\kappa - i\infty}^{\kappa + i\infty} F(s) x^{s+w} \frac{\mathrm{d}s}{s+w}$$

when $\kappa > \max(0, \sigma_c)$. Thus

$$\sum_{n \leqslant x} a_n (x^w - n^w) = \frac{1}{2\pi i} \int_{\kappa - i\infty}^{\kappa + i\infty} F(s) x^{s+w} \frac{w}{s(s+w)}\,\mathrm{d}s \tag{2.11}$$

and it is clear that this formula remains true also when x is an integer. By Theorem 1.19, for $\sigma = \kappa$, we have

$$F(s) \ll 1 + |\tau|^{1-(\kappa-\sigma_c)+\varepsilon}, \tag{2.12}$$

so both the integral, and its derivative with respect to w, are absolutely and uniformly convergent. The conditions for differentiation under the integral sign are therefore satisfied, and we obtain (2.9) by taking the derivatives at $w = 0$ of each side of (2.11). Formula (2.10) corresponds to the case $w = 1$ of (2.11), whose left-hand side then equals

$$\int_0^x A(t)\,\mathrm{d}t. \qquad \square$$

Formula (2.7) can sometimes be advantageously replaced by a more precise formula at the cost of altering the integrand. For example, we have the following result.

Lemma 2.6. *For any $\delta > 0$, there exist complex constants $a = a(\delta)$ and $b = b(\delta)$ such that, putting*

$$w_\delta(s) := \frac{1}{s} + \frac{a}{s+1} + \frac{b}{s+2} \quad (s \in \mathbb{C}), \quad g_\delta(y) := h(y)\{1 + a/y + b/y^2\} \quad (y > 0),$$

we have, uniformly for $y > 0$, $\kappa > 0$,

$$\frac{1}{2\pi i} \int_{\kappa - i\delta}^{\kappa + i\delta} w_\delta(s) y^s\,\mathrm{d}s = g_\delta(y) + O\Big(\frac{y^\kappa}{1 + (\ln y)^2} + \kappa y^\kappa\Big). \tag{2.13}$$

Proof. We first note that (2.7) implies

$$\frac{1}{2\pi i}\int_{\kappa-i\delta}^{\kappa+i\delta} w_\delta(s)y^s\,\mathrm{d}s = g(y) + O(y^\kappa)$$

if $|\ln y| \leqslant 1$. Then, an integration by parts, furnishes, when $|\ln y| > 1$,

$$\int_{\kappa-i\delta}^{\kappa+i\delta} w_\delta(s)y^s\,\mathrm{d}s = -\int_{\kappa-i\delta}^{\kappa+i\delta} w'_\delta(s)\frac{y^s}{\ln y}\,\mathrm{d}s + O\Big(\frac{y^\kappa}{|\ln y|}\{|w_\delta(\kappa+i\delta)| + |w_\delta(\kappa-i\delta)|\}\Big). \tag{2.14}$$

By the mean value theorem, the error term in (2.14) is $O(\kappa y^\kappa/|\ln y|)$ if a and b are chosen so that $w_\delta(\pm i\delta) = 0$. Now, by the residue theorem, we have, still assuming $|\ln y| > 1$,

$$-\frac{1}{2\pi i}\int_{\kappa-i\delta}^{\kappa+i\delta} w'_\delta(s)\frac{y^s}{\ln y}\,\mathrm{d}s = g(y) + O\Big(\frac{y^\kappa}{(\ln y)^2}\Big)$$

by moving the line of integration to the left or to the right towards infinity, according to whether $y > \mathrm{e}$ or $y < \mathrm{e}$. □

From the lemma above, we deduce a new variant of Perron's formula.

Theorem 2.7. *Let $F(s) := \sum_{n\geqslant 1} a_n/n^s$ be a Dirichlet series, convergent for $\sigma > 1$ and whose coefficients satisfy an upper bound of the type*

$$\sum_{n\leqslant x} |a_n| \ll xb(x) \qquad (x \geqslant 2) \tag{2.15}$$

where b is non-decreasing. Then we have, for $\kappa = \kappa_x := 1/\ln x$,

$$\sum_{n\leqslant x}\frac{a_n}{n} = \frac{1}{2\pi i}\int_{\kappa_x-i\delta}^{\kappa_x+i\delta} F(s+1)w_\delta(s)x^s\,\mathrm{d}s + O\big(b^*(x)\big) \qquad (x \geqslant 2), \tag{2.16}$$

where

$$b^*(x) := \kappa_x\int_1^\infty \frac{b(t)}{t^{1+\kappa_x}}\,\mathrm{d}t.$$

Proof. We observe at the outset that

$$b(x) \leqslant b\big(\sqrt{x}\big) \leqslant 2\kappa_x\int_1^{\sqrt{x}}\frac{b(t)}{t}\,\mathrm{d}t \leqslant 2\sqrt{\mathrm{e}}\,b^*(x). \tag{2.17}$$

By Lemma 2.6, the error term in (2.16) is $\ll \sum_{1\leqslant j\leqslant 4}|R_j|$ with

$$R_j := x^{-j}\sum_{n\leqslant x} a_n n^{j-1} \qquad (j=1,2)$$

$$R_3 := \sum_{n\geqslant 1}\frac{|a_n|}{n^{1+\kappa}\{1+(\ln x/n)^2\}}$$

$$R_4 := \frac{1}{\ln x}\sum_{n\geqslant 1}\frac{|a_n|}{n^{1+\kappa}}.$$

We have trivially $R_2 \ll R_1 \ll b(x)$ and $R_4 \ll b^*(x)$ by Abel summation. To estimate R_3, we first remark, by (2.17), that, uniformly for $x \geqslant 2$ and $k\in\mathbb{Z}$, $1/x \leqslant 2^k \leqslant \sqrt{x}$, we have

$$\sum_{x/2^{k+1}<n\leqslant x/2^k}\frac{|a_n|}{n\{1+(\ln x/n)^2\}} \ll \frac{b^*(x)}{1+k^2} \tag{2.18}$$

and we consider separately the contributions to R_3 from the summation domains $n\notin[\sqrt{x},x^2]$ and $\sqrt{x}\leqslant n\leqslant x^2$. Appealing to the estimate $R_4 \ll b^*(x)$ for the first and to (2.18) for the second, we indeed obtain $R_3 \ll b^*(x)$. □

2.2. Applications: two convergence theorems

Perron's formula can be used to derive the following result, which is a particular case of an important theorem of Schnee–Landau—see Landau (1909, § 238) and the Notes.

Theorem 2.8. *Let $F(s) := \sum_{n\geqslant 1} a_n/n^s$ be a Dirichlet series with finite abscissa of convergence σ_c. If σ_0 is a real number such that $F(s)$ has a regular analytic continuation satisfying $\mu(\sigma)=0$ for $\sigma>\sigma_0$, then we have*

$$\sigma_c \leqslant \sigma_0.$$

Proof. Without loss of generality, the value of σ_0 may be arbitrarily fixed. We let $\sigma_0<0$ and show that $F(s)$ converges at $s=0$.

Let δ be some real number satisfying $0<\delta<-\sigma_0$. We apply Theorem 2.3 with fixed $\kappa>\sigma_a+1$, and $x\in\frac12+\mathbb{N}$. For each integer n we have $|\ln(x/n)|\geqslant \ln\big((n+\frac12)/n\big)\gg 1/n$, so that the remainder term in (2.6) is

$$\ll x^\kappa\sum_{n\geqslant 1}\frac{|a_n|}{n^\kappa(1+T/n)} \ll \frac{x^\kappa}{T}\sum_{n\geqslant 1}|a_n|n^{1-\kappa} \ll \frac{x^\kappa}{T}.$$

Next, we deform the contour of integration $[\kappa-iT,\kappa+iT]$ into the polygonal line passing through the points $-\delta-iT, -\delta+iT$. This alters the

integral by the value of the residue at the pole $s = 0$, that is to say $F(0)$. We can then write

$$\sum_{n\leqslant x} a_n = F(0) + R_1 + R_2 + O(x^\kappa/T) \tag{2.19}$$

where R_1 and R_2 denote the respective contributions to the integral

$$\frac{1}{2\pi i}\int F(s)x^s\,\frac{\mathrm{d}s}{s}$$

from the horizontal segments and from the vertical segment of the new contour. Since, by assumption, $\mu(\sigma) = 0$ pour $\sigma \geqslant -\delta$, we immediately have, for all $\varepsilon > 0$,

$$R_1 \ll_\varepsilon x^\kappa T^{\varepsilon-1}, \qquad R_2 \ll_\varepsilon x^{-\delta}T^\varepsilon.$$

(Note that we have used here the fact that the bound $F(s) \ll T^\varepsilon$ ($|\tau| \leqslant T$) is uniform for $-\delta \leqslant \sigma \leqslant \kappa$—see the remark following the proof of Theorem 1.20.)

Substituting in (2.19) and choosing $T := x^{\kappa+\delta}$, $\varepsilon := \delta/2(\kappa+\delta)$, we obtain the convergence of the series $\sum a_n$ in the "explicit" form

$$\sum_{n\leqslant x} a_n = F(0) + O(x^{-\delta/2}).$$

This concludes the proof. □

Similarly, we can use Theorem 2.7 to establish the following theorem due to Riesz (1909).

Theorem 2.9. *Let $F(s) := \sum_{n\geqslant 1} a_n/n^s$ be a Dirichlet series convergent for $\sigma > 1$ and having a regular analytic continuation to a neighborhood of $s = 1$. Assume furthermore that*

$$A(x) := \sum_{n\leqslant x} a_n = o(x) \qquad (x \to \infty). \tag{2.20}$$

Then $F(s)$ converges at $s = 1$.

Proof. We start by noticing that, if $\lim_{n\to\infty} a_n = 0$, then by (2.16) we have

$$\sum_{n\leqslant x} \frac{a_n}{n} = \frac{1}{2\pi i}\int_{\kappa-i\delta}^{\kappa+i\delta} F(s+1)w_\delta(s)x^s\,\mathrm{d}s + o(1).$$

The stated result then follows from the residue theorem by choosing δ to be sufficiently small, so that the square with vertices $\pm\delta \pm i\delta$ is included in the domain of holomorphy of $F(1+s)$, and by moving, for sufficiently large x, the integration abscissa to $-\delta$. Indeed, the integrals over the horizontal segments are $\ll 1/\ln x$ and that on the displaced segment is $\ll 1/x^\delta$.

To handle the general case, we introduce the series

$$G(s) := \sum_{n\geqslant 1} A_n/n^{s+1}$$

with $A_n := A(n)$. By Abel summation, we obtain, for all $N \in \mathbb{N}^*$, $s \in \mathbb{C}$,

$$\sum_{1\leqslant n\leqslant N} \frac{a_n}{n^s} - s\sum_{1\leqslant n\leqslant N} \frac{A_n}{n^{s+1}} = \sum_{1\leqslant n\leqslant N} A_n\varphi_n(s) + \frac{A_N}{(N+1)^s} \tag{2.21}$$

where $\varphi_n(s) := n^{-s} - (n+1)^{-s} - sn^{-s-1}$ $(s \in \mathbb{C})$. Moreover, by the mean value theorem,

$$|\varphi_n(s)| \leqslant |s(s+1)|n^{-\sigma-2} \qquad (\sigma > 0).$$

This implies that $F(s) - sG(s)$ has a regular analytic continuation to the half-plane $\sigma > 0$ and thus that $G(s)$ can be holomorphically continued to a neighborhood of $s = 1$. Applying to G the result of the first part of this proof, we obtain that the series $\sum_{n\geqslant 1} A_n/n^2$ is convergent and sums to $G(1)$. Using (2.21) with $s = 1$ and (2.20), we finally obtain that the series $\sum_{n\geqslant 1} a_n/n$ converges and sums to $F(1)$. □

2.3. The mean value formula

The following result is the analogue, for Dirichlet series, of Parseval's theorem concerning trigonometric series.

Theorem 2.10. *Let $F(s) := \sum_{n\geqslant 1} a_n/n^s$ and $G(s) := \sum_{n\geqslant 1} b_n/n^s$ be two Dirichlet series with respective abscissas of absolute convergence σ_1 and σ_2. For $\alpha > \sigma_1$, $\beta > \sigma_2$, we have*

$$\lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} F(\alpha+i\tau)\,G(\beta-i\tau)\,\mathrm{d}\tau = \sum_{n\geqslant 1} \frac{a_nb_n}{n^{\alpha+\beta}}. \tag{2.22}$$

Proof. We have

$$F(\alpha+i\tau)\,G(\beta-i\tau) = \sum_{n\geqslant 1} \frac{a_nb_n}{n^{\alpha+\beta}} + \sum_{m\neq n} \frac{a_mb_n}{m^\alpha n^\beta}(n/m)^{i\tau},$$

the series being absolutely and uniformly convergent in τ. We can therefore integrate term by term. This gives

$$\frac{1}{2T}\int_{-T}^{T} F(\alpha+i\tau)\,G(\beta-i\tau)\,\mathrm{d}\tau = \sum_{n\geqslant 1} \frac{a_nb_n}{n^{\alpha+\beta}} + \sum_{m\neq n} \frac{a_mb_n}{m^\alpha n^\beta}\left(\frac{\sin(T\ln(n/m))}{T\ln(n/m)}\right).$$

The factor involving T is uniformly bounded with respect to T, m, n and tends to 0 as $T \to \infty$. The stated result then follows from the theorem of dominated convergence. □

Corollary 2.11. *For $\sigma > \sigma_a$, we have*

$$\lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |F(s)|^2 \, d\tau = \sum_{n\geqslant 1} \frac{|a_n|^2}{n^{2\sigma}}. \tag{2.23}$$

Corollary 2.12. *For $\sigma > \sigma_a$, we have*

$$\lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} F(s) n^s \, d\tau = a_n. \tag{2.24}$$

The formulae (2.23) and (2.24) provide two new instant proofs of the uniqueness theorem for the representation of a function as a Dirichlet series (Theorem 1.8).

The question of the range of validity of relation (2.23) is, in general, rather difficult. It can be shown (see Titchmarsh (1939), § 9.5) that the set of points s where $F(s)$ is holomorphic, of finite order, and satisfies (2.23) is a half-plane with boundary the vertical line $\sigma = \sigma_m$, with

$$\sigma_m \geqslant \max\left(\sigma_c, \sigma_a - \tfrac{1}{2}\right).$$

It is, however, possible that, even when $F(s)$ is not holomorphic for $\sigma \geqslant \sigma_0$, the left-hand side of (2.23) converges for $\sigma = \sigma_0$. In the case of the zeta function, Titchmarsh (1951, § 7.2) showed that

$$\lim_{T\to\infty} \frac{1}{T} \int_{1}^{T} |\zeta(s)|^2 \, d\tau = \zeta(2\sigma) \quad (\sigma > \tfrac{1}{2}) \tag{2.25}$$

and hence that (2.23) holds for $F = \zeta$ whenever $\sigma > \frac{1}{2}$, $\sigma \neq 1$.

Notes

§ 2.1. The idea for Lemma 2.6 goes back to Landau (1910), who proves Theorem 2.9 in this way. It allows a refined use of Perron's formula when the integral is taken over a short vertical segment, which sometimes turns out to be essential. A variant of a lemma due to Tenenbaum & Wu (2003) may be formulated as follows.

Lemma 2.13. *For any given real number $T > 0$, there exist real constants b_j $(1 \leqslant j \leqslant 5)$ such that, writing*

$$w_T(s) := \frac{1}{s} + \sum_{1\leqslant j\leqslant 5} \frac{b_j}{s+j}, \quad g_T(x) := h(x)\Big\{1 + \sum_{1\leqslant j\leqslant 5} \frac{b_j}{x^j}\Big\} \quad (x > 0),$$

we have, on the one hand

$$\sum_{1\leqslant j\leqslant 5} b_j/j = 0, \tag{2.26}$$

and, on the other, uniformly for $x > 0$, $\kappa > 0$,

$$\frac{1}{2\pi i}\int_{\kappa-iT}^{\kappa+iT} w_T(s)x^s\,\mathrm{d}s = g_T(x) + O\left(\frac{x^\kappa}{1+(\ln x)^2} + \frac{\kappa^2 x^\kappa}{1+|\ln x|}\right). \tag{2.27}$$

§ 2.2. The Schnee–Landau theorem can be stated thus: *If $a_n \ll_\varepsilon n^\varepsilon$ for any $\varepsilon > 0$*[1] *and if $F(s) = \sum_{n\geqslant 1} a_n/n^s$ has, for some σ_0, a regular continuation in $\sigma > \sigma_0$ satisfying $\mu(\sigma) \leqslant \alpha$ in this half-plane, then*

$$\sigma_c \leqslant \min\left(\frac{\sigma_0+\alpha}{1+\alpha},\ \sigma_0+\alpha\right).$$

For a proof, see Landau (1909) pp. 853 et seq.

When $\alpha = 0$, the Schnee–Landau theorem implies the convergence of $\sum_{n\geqslant 1} a_n n^{-s-\kappa}$ for $\kappa > \sigma_a$, $\sigma > \sigma_0 - \kappa$. The condition $a_n \ll_\varepsilon n^\varepsilon$ is therefore unnecessary, and one obtains Theorem 2.8.

Theorem 2.8 is proved in Landau (1909), p. 848. Another proof appears in Hardy & Riesz (1915), th. 50. See also Titchmarsh (1951), th. 3.13.

§ 2.3. Carlson (1922) showed that, if $F(s)$ is holomorphic and of finite order for $\sigma \geqslant \sigma_1$ and if $\int_{-T}^{T} |F(s)|^2\,\mathrm{d}\tau \ll T$ for $\sigma = \sigma_1$, then formula (2.23) holds for $\sigma > \sigma_1$. See, for example, Titchmarsh (1939), § 9.51.

Exercises

169. Show that, for $\kappa > \max(0, \sigma_c)$, we have

$$\sum_{n \leqslant x} a_n \left(\ln \frac{x}{n}\right)^k = \frac{k!}{2\pi i} \int_{\kappa - i\infty}^{\kappa + i\infty} F(s) x^s \frac{\mathrm{d}s}{s^{k+1}}$$

where $k \geqslant 1$ is an arbitrary positive integer.

170. Let $w \in L^1(\mathbb{R})$ be a function such that the Fourier transform

$$\widehat{w}(\tau) := \int_{-\infty}^{+\infty} \mathrm{e}^{-it\tau} w(t)\, \mathrm{d}t$$

is itself integrable on $\mathbb{R}$. Show that, for $\kappa > \sigma_a$, we have

$$\sum_{n \geqslant 1} a_n \left(\frac{x}{n}\right)^\kappa w\left(\ln \frac{x}{n}\right) = \frac{1}{2\pi i} \int_{\kappa - i\infty}^{\kappa + i\infty} F(s) x^s \widehat{w}(\tau)\, \mathrm{d}s.$$

171. By applying the conclusion of Exercise 170 to the function

$$w(t) := \left(\frac{\sin(t/2\varepsilon)}{t/2\varepsilon}\right)^2,$$

show that, if $a_n \geqslant 0$ for all n, then, for any $\varepsilon > 0$, we have

$$\sum_{|\ln(x/n)| \leqslant \varepsilon} a_n \leqslant C\varepsilon \mathrm{e}^{\kappa\varepsilon} \int_{\kappa - iT}^{\kappa + iT} |F(s) x^s|\, |\mathrm{d}s| \qquad (\kappa > \sigma_a)$$

with $C = 1/\{2 \sin \frac{1}{2}\}^2$, and $T = 1/\varepsilon$.

172. Let $F(s) := \sum a_n/n^s$ be a Dirichlet series such that $\lim_{\tau \to \infty} F(\sigma + i\tau)$ exists for at least one value of $\sigma > \sigma_a$. Denote this limit by λ.

(a) Show that $|\lambda|^2 = \sum_{n \geqslant 1} |a_n|^2/n^{2\sigma}$.

(b) Show that $\lambda = a_1$. [*Use Corollary* 2.12.]

(c) Deduce that $F(s)$ is constant.

173. Let $F(s) = \sum_{n \geqslant 1} a_n/n^s$ be a Dirichlet series converging for $\sigma = \Re e\, s > 1$. Assume that F has a regular analytic continuation $\widetilde{F}$ to a neighborhood of $s = 1$.

(a) Is it true that we necessarily have $\widetilde{F}(1) = \lim_{\sigma \to 1+} F(\sigma)$?

(b) Suppose, in addition, that we have $\sum_{m \leqslant n} a_m = o(n)$ $(n \to \infty)$. Show that the series $\sum_{n \geqslant 1} a_n/n$ converges and sums to $\widetilde{F}(1)$.

(c) Assume $\sigma \in]1, \infty[$ and $N \in \mathbb{N}^*$.

(i) Evaluate $\sum_{1 \leqslant n \leqslant N} 1/n^\sigma$ by applying the Euler–Maclaurin formula at order 0.

(ii) Deduce an expression for $\zeta(\sigma) = \sum_{n\geqslant 1} 1/n^{\sigma}$ valid when $\sigma > 1$ and an evaluation of $\sum_{1\leqslant n\leqslant N} 1/n$. Hint: in the second case, bring Euler's constant γ into play.

(d) Show that $\zeta(s)$ may be meromorphically continued to a neighborhood of $s = 1$ and give its Laurent expansion to order 1 at $s = 1$. Consider the same question for $\zeta'(s)$.

(e) From the prime number theorem in Chebyshev's form $\psi(x) \sim x$ $(x \to \infty)$ and from the result of (b) deduce the existence and compute the value of

$$K := \lim_{x\to\infty} \left\{ \sum_{n\leqslant x} \frac{\Lambda(n)}{n} - \ln x \right\}$$

where Λ denotes the von Mangoldt function. Hint: consider the Dirichlet series $\zeta(s) + \zeta'(s)/\zeta(s)$.

The Riemann zeta function

3.1. Introduction

The zeta function, in terms of which a wide variety of Dirichlet series may be expressed, occupies a central role in arithmetic, significantly because of its close correlation with prime numbers via Euler's formula—cf. § 1.2. In connection with the Perron formulae proved in Chapter 2, the analytic study of the zeta function is hence of major interest.

In general, evaluating an integral of the type

$$\int_{-T}^{T} F(s)x^s s^{-1}\,\mathrm{d}\tau,$$

where F is defined by a Dirichlet series, can be greatly simplified, as we have seen in § 2.2, by introducing a closed contour and the use of the residue theorem. Since $F(s)$ has no singularity in the half-plane $\sigma > \sigma_c$ (cf. Theorem 1.5), it is natural to use this approach in circumstances when the contour can be selected so as to intersect the complementary half-plane $\sigma \leqslant \sigma_c$. This requires the existence of an analytic continuation of $F(s)$. We shall see this condition is realized in the case of $\zeta(s)$—and, consequently, for series expressed in terms of it.

From this perspective, the study of the analytic properties of $\zeta(s)$ (appropriately identified with its continuation) constitutes an indispensable investment for a number theorist. In this chapter, we merely establish the basic results of the theory—and so already invite the reader interested in more detailed or more sophisticated expositions to consult the classic works

of Titchmarsh (1951), as updated by Heath-Brown (1986), and of Edwards (1974), whose historical approach is enlightening, or the comprehensive survey of Ivić (1985).

3.2. Analytic continuation

Theorem 3.1. *The function $\zeta(s)$ can be analytically continued to a meromorphic function on the entire complex plane $\mathbb{C}$, its only singularity being a simple pole at $s = 1$, with residue 1.*

Proof. For $\sigma > 1$, we have

$$G(s) := \sum_{n\geqslant 1} \frac{(-1)^n}{n^s} = \sum_{m\geqslant 1} \frac{1}{(2m)^s} - \left(\sum_{n\geqslant 1} \frac{1}{n^s} - \sum_{m\geqslant 1} \frac{1}{(2m)^s} \right) = (2^{1-s} - 1)\zeta(s).$$

Now $G(s)$ converges for $\sigma > 0$, and we have

$$G(1) = -\ln 2 \neq 0, \qquad G(s_k) = 0 \quad \big(s_k := 1 + 2\pi i k/\ln 2,\ k \neq 0\big).$$

The second formula follows from the estimate

$$\sum_{n\leqslant x} \frac{(-1)^n}{n^{s_k}} = 2^{1-s_k} \sum_{n\leqslant x/2} \frac{1}{n^{s_k}} - \sum_{n\leqslant x} \frac{1}{n^{s_k}} = - \sum_{x/2<n\leqslant x} \frac{1}{n^{s_k}} \ll \frac{1}{x}$$

obtained by partial summation.[1] We thus see that the stated property holds for the half-plane $\sigma > 0$. The same formula can also be obtained from the partial integration formula

$$\zeta(s) = \int_{1-}^{\infty} \frac{\mathrm{d}\lfloor t \rfloor}{t^s} = \int_1^{\infty} \frac{\mathrm{d}t}{t^s} - \int_{1-}^{\infty} \frac{\mathrm{d}\langle t \rangle}{t^s} = \frac{s}{s-1} - s\int_1^{\infty} \langle t \rangle \frac{\mathrm{d}t}{t^{s+1}},$$

initially valid for $\sigma > 1$, which provides an alternative explicit form of the continuation for $\sigma > 0$.

The stated result could be derived by successive integrations, starting from one or the other of the preceding formulae—using, for example, in the second case, the Euler–Maclaurin formula, cf. Exercise 183. We employ another method due to Riemann, which provides additional information useful in establishing the functional equation.

The starting point is the formula

$$\Gamma(s)n^{-s} = \int_0^{\infty} t^{s-1}\mathrm{e}^{-nt}\,\mathrm{d}t \qquad (\sigma > 0). \tag{3.1}$$

Summing over $n \geqslant 1$, we have, for $\sigma > 1$,

$$\Gamma(s)\zeta(s) = \sum_{n\geqslant 1} \int_0^{\infty} t^{s-1}\mathrm{e}^{-nt}\,\mathrm{d}t = \int_0^{\infty} \frac{t^{s-1}}{\mathrm{e}^t - 1}\,\mathrm{d}t.$$

[1] See, e.g., (3.18) below.

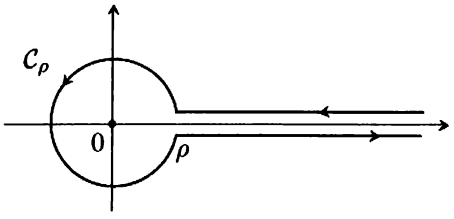

Figure 1. The Hankel contour $\mathcal{C}_\varrho$.

We obtain an analytic continuation of this integral by replacing the half-line of integration by the *Hankel contour* $\mathcal{C}_\varrho$ where ϱ is a real parameter, $0 < \varrho < 2\pi$.

$\mathcal{C}_\varrho$ is composed of three parts: the real half-line $[\varrho, +\infty[$ covered from right to left with argument $0+$, the circle $|z| = \varrho$, traced in the positive sense and from which the point $z = \varrho$ is excluded, and the half-line $[\varrho, +\infty[$ covered from left to right with argument $2\pi-$.

Since the function $z \mapsto z^{s-1}(\mathrm{e}^z - 1)^{-1}$ is holomorphic in the horizontal strip $|\Im m\, z| < 2\pi$ excluding the half-line $[0, +\infty[$, the integral

$$I(s) := \int_{\mathcal{C}_\varrho} z^{s-1} \frac{\mathrm{d}z}{\mathrm{e}^z - 1}$$

is independent of ϱ in $]0, 2\pi[$. It is absolutely convergent for $s \in \mathbb{C}$ and uniformly convergent on any compact subset. Thus it defines an entire function of s. We have

$$I(s) = \oint_{|z|=\varrho} z^{s-1} \frac{\mathrm{d}z}{\mathrm{e}^z - 1} + (\mathrm{e}^{2\pi i s} - 1) \int_\varrho^\infty t^{s-1} \frac{\mathrm{d}t}{\mathrm{e}^t - 1}. \tag{3.2}$$

Taking into account the upper bound

$$|z^{s-1}/(\mathrm{e}^z - 1)| \ll_s \varrho^{\sigma-2} \qquad (|z| = \varrho \leqslant \pi),$$

and letting ϱ tend to 0, we obtain the formula

$$I(s) = (\mathrm{e}^{2\pi i s} - 1)\Gamma(s)\zeta(s) \qquad (\sigma > 1).$$

The reflection formula $\Gamma(s)\Gamma(1-s) = \pi/\sin(\pi s)$ therefore implies

$$\zeta(s) = \frac{\mathrm{e}^{-i\pi s}}{2\pi i}\Gamma(1-s)I(s). \tag{3.3}$$

This last formula, initially valid for $\sigma > 1$, explicitly gives the analytic continuation of $\zeta(s)$ to the whole complex plane. When $\sigma \leqslant 0$, the factor $\Gamma(1-s)$ is holomorphic, hence $\zeta(s)$ has no singularity other than that already noted at $s = 1$. This concludes the proof. □

Formula (3.2) readily gives the value of $\zeta(s)$ at negative integers.

Theorem 3.2. *Letting B_n denote the nth Bernoulli number, we have*

$$\zeta(-n) = (-1)^n \frac{B_{n+1}}{n+1} \qquad (n \geqslant 0). \tag{3.4}$$

In particular, $\zeta(-2n) = 0$ for every $n \geqslant 1$.

Proof. The Bernoulli numbers were defined in Chapter I.0 by the Laurent expansion

$$\frac{1}{e^z - 1} = \sum_{m \geqslant 0} \frac{B_m}{m!} z^{m-1}.$$

Hence, by (3.2),

$$I(-n) = \oint_{|z|=\varrho} (e^z - 1)^{-1} z^{-n-1} \, dz = 2\pi i \frac{B_{n+1}}{(n+1)!}.$$

We obtain the stated formula by substituting in (3.3). □

3.3. Functional equation

Theorem 3.3. *For each $s \neq 1$, we have*

$$\zeta(s) = 2^s \pi^{s-1} \sin\left(\tfrac{1}{2}\pi s\right) \Gamma(1-s) \zeta(1-s). \tag{3.5}$$

Remark. The functional equation takes the more symmetric form

$$\Phi(s) = \Phi(1-s) \qquad (s \neq 0, 1) \tag{3.6}$$

with the notation

$$\Phi(s) := \pi^{-s/2} \Gamma(s/2) \zeta(s). \tag{3.7}$$

This follows immediately from (3.5), the reflection formula, and the duplication formula (0.10) for the Γ function.

Proof. For $k \geqslant 1$, let $\mathcal{H}_k$ be the Hankel contour with parameter $\varrho_k := (2k+1)\pi$. Then $|e^z - 1|^{-1}$ remains bounded for z on $\mathcal{H}_k$, and we have

$$\left| \int_{\mathcal{H}_k} z^{s-1} (e^z - 1)^{-1} \, dz \right| \ll_s k^\sigma. \tag{3.8}$$

Let ϱ satisfy $0 < \varrho < 2\pi$. The contour $\mathcal{C}_\varrho - \mathcal{H}_k$ encircles, clockwise, the poles $z = 2n\pi i$ $(n = \pm 1, \pm 2, \dots, \pm k)$. The residue theorem then implies

$$\begin{aligned} I(s) &= \int_{\mathcal{C}_\varrho} z^{s-1} (e^z - 1)^{-1} \, dz = \int_{\mathcal{H}_k} z^{s-1} (e^z - 1)^{-1} \, dz - 2\pi i \sum_{1 \leqslant |n| \leqslant k} (2n\pi i)^{s-1} \\ &= \int_{\mathcal{H}_k} z^{s-1} (e^z - 1)^{-1} \, dz - (2\pi i)^s (1 - e^{i\pi s}) \sum_{1 \leqslant n \leqslant k} n^{s-1}. \end{aligned}$$

Using (3.8) and letting k tend to infinity, we obtain for each s in the half-plane $\sigma < 0$ that

$$I(s) = (2\pi i)^s(e^{i\pi s} - 1)\zeta(1-s).$$

Substituting in (3.3), this gives equation (3.5) for $\sigma < 0$. By analytic continuation, it persists for all s. □

An immediate consequence of (3.4) and (3.5) is the following formula for $\zeta(2n)$, which can also be obtained as a special case of the Fourier series expansion for Bernoulli functions—cf. Exercise 1, p. 8.

Theorem 3.4. *We have*

$$\zeta(2n) = (-1)^{n-1}2^{2n-1}\frac{B_{2n}}{(2n)!}\pi^{2n} \qquad (n \geqslant 1). \tag{3.9}$$

3.4. Approximations and bounds in the critical strip

Let us write

$$\chi(s) := 2(2\pi)^{s-1}\Gamma(1-s)\sin\left(\tfrac{1}{2}\pi s\right), \tag{3.10}$$

so that the functional equation (3.5) can be written as

$$\zeta(s) = \chi(s)\zeta(1-s). \tag{3.11}$$

The asymptotic behavior of $\chi(s)$ when σ is fixed and $|\tau| \to \infty$ can easily be determined in terms of that of $\Gamma(s)$—cf. Corollary 0.13, derived from the complex Stirling formula stated in Theorem 0.12. We obtain

$$|\chi(s)| \sim \left(\frac{|\tau|}{2\pi}\right)^{\frac{1}{2}-\sigma} \qquad (|\tau| \to +\infty). \tag{3.12}$$

Thus, it follows from (3.11) that the order of magnitude of $\zeta(s)$ on vertical lines is completely determined when $\sigma < 0$. With the notation of § 1.6, we deduce in particular that, for the zeta function,

$$\mu(\sigma) = \begin{cases} \frac{1}{2} - \sigma & (\sigma \leqslant 0), \\ 0 & (\sigma \geqslant 1). \end{cases} \tag{3.13}$$

The exact value of $\mu(\sigma)$ in the *critical strip* $0 < \sigma < 1$ is a deep problem which is still unsolved. Given (3.13), the simplest hypothesis consists in assuming that the graph of $\mu(\sigma)$ is formed by two half-lines, viz.

$$\mu(\sigma) = \max(\tfrac{1}{2} - \sigma, 0). \tag{3.14}$$

This formula is known as *Lindelöf's hypothesis*. Taking into account the convexity properties of $\mu(\sigma)$ (cf. Theorem 1.20), it is equivalent to $\mu(\frac{1}{2}) = 0$.

The equations $\mu(0) = \frac{1}{2}$ and $\mu(1) = 0$ immediately imply, by convexity, that $\mu(\sigma) \leqslant \frac{1}{2}(1-\sigma)$ for $0 \leqslant \sigma \leqslant 1$. The following result allows us to improve this estimate.

Theorem 3.5. *Assume $\sigma_0 > 0$, $0 < \delta < 1$. Uniformly for $\sigma \geqslant \sigma_0$, $x \geqslant 1$, $0 < |\tau| \leqslant (1-\delta)2\pi x$, we have that*

$$\zeta(s) = \sum_{n \leqslant x} \frac{1}{n^s} - \frac{x^{1-s}}{1-s} + O\Big(\frac{1}{x^\sigma}\Big). \tag{3.15}$$

The proof depends on an easy lemma.

Lemma 3.6. *For $\sigma > 0$ and $N \in \mathbb{Z}^+$, we have*

$$\zeta(s) = \sum_{n \leqslant N} \frac{1}{n^s} - \frac{N^{1-s}}{1-s} - s\int_N^\infty \frac{\langle t\rangle}{t^{s+1}}\,\mathrm{d}t. \tag{3.16}$$

Proof. For $\sigma > 1$, we have

$$\begin{aligned} \zeta(s) - \sum_{n \leqslant N} \frac{1}{n^s} &= \int_N^\infty \frac{\mathrm{d}\lfloor t\rfloor}{t^s} = -\frac{N^{1-s}}{1-s} - \int_N^\infty \frac{\mathrm{d}\langle t\rangle}{t^s} \\ &= -\frac{N^{1-s}}{1-s} - s\int_N^\infty \frac{\langle t\rangle}{t^{s+1}}\,\mathrm{d}t. \end{aligned} \tag{3.17}$$

By analytic continuation, this formula remains true for $\sigma > 0$. □

Proof of Theorem 3.5. For $|\tau| \leqslant (1-\delta)2\pi x$ and all $y \geqslant x$, we have

$$\sum_{x < n \leqslant y} n^{-i\tau} = \int_x^y t^{-i\tau}\,\mathrm{d}t + O(1), \tag{3.18}$$

applying Theorem I.6.4 to the function $f(t) := -(\tau/2\pi)\ln t$, which satisfies $|f'(t)| \leqslant 1-\delta$ for $x \leqslant t \leqslant y$. Hence we may write, for $N \geqslant x$,

$$\sum_{x < n \leqslant N} \frac{1}{n^s} = \int_x^N \frac{\mathrm{d}y}{y^s} + \int_x^N \frac{\mathrm{d}\{O(1)\}}{y^\sigma} = \frac{N^{1-s} - x^{1-s}}{1-s} + O\Big(\frac{1}{x^\sigma}\Big).$$

Substituting in (3.16) and letting N tend to infinity, we obtain precisely (3.15). □

Corollary 3.7. *We have*

$$\zeta\big(\tfrac{1}{2} + i\tau\big) \ll \tau^{1/6}\ln|\tau| \qquad (|\tau| \to \infty). \tag{3.19}$$

Proof. We may assume that $\tau > 0$. Theorems I.6.5 and I.6.9 applied to $f(t) := -(\tau/2\pi)\ln t$ readily provide the estimate

$$\sum_{a < n \leqslant b} n^{-i\tau} \ll \min\Big(\tau^{1/2} + a\tau^{-1/2}, a^{1/2}\tau^{1/6} + a\tau^{-1/6}\Big),$$

valid uniformly for $\tau > 0$, $a < b \leqslant 2a$. Under the additional assumption $a \ll \tau$, we infer by partial summation that

$$\sum_{a<n\leqslant b} n^{-1/2-i\tau} \ll \min\left((\tau/a)^{1/2} + (a/\tau)^{1/2},\ \tau^{1/6} + a^{1/2}\tau^{-1/6}\right)$$

$$\ll \min\left(\tau^{1/6}, (\tau/a)^{1/2}\right).$$

For $r \leqslant \ln x/\ln 2$, let us choose $a := 2^r$, $b := \min(2^{r+1}, x)$ and sum all corresponding estimates. It follows that

$$\sum_{n\leqslant x} n^{-1/2-i\tau} \ll \tau^{1/6}\ln\tau \qquad (x \ll \tau).$$

Substituting $x = \tau$ in (3.15), we obtain the required bound (3.19). □

The convexity of the function $\mu(\sigma)$ allows us to deduce from the above result a new bound in the critical strip.

Theorem 3.8. *The zeta function satisfies the inequalities*

$$\mu(\sigma) \leqslant \begin{cases} \frac{1}{3}(1-\sigma) & \left(\frac{1}{2} \leqslant \sigma \leqslant 1\right), \\ \frac{1}{6}(3-4\sigma) & \left(0 \leqslant \sigma \leqslant \frac{1}{2}\right). \end{cases} \tag{3.20}$$

Proof. This follows immediately from the estimates $\mu(0) = \frac{1}{2}$, $\mu(\frac{1}{2}) \leqslant \frac{1}{6}$, $\mu(1) = 0$. □

The bound (3.20) provides no new information on the line $\sigma = 1$. However, using Theorem 3.5 directly, the estimate $\zeta(s) \ll |\tau|^{\varepsilon}$ $(\sigma = 1)$ can be considerably improved.

Theorem 3.9. *For $0 < \alpha < 1$, we have*

$$|\zeta(s)| \leqslant \frac{3|\tau|^{1-\alpha}}{2\alpha(1-\alpha)} \qquad (\sigma \geqslant \alpha,\ |\tau| \geqslant 1). \tag{3.21}$$

In particular, for each positive constant c, we have

$$\zeta(s) \ll \ln|\tau| \qquad (|\tau| \geqslant 2,\ \sigma \geqslant 1 - c/\ln|\tau|). \tag{3.22}$$

Remark. The explicit bound (3.21) is clearly of theoretical interest only when $\alpha = \alpha(\tau) \to 1$. Otherwise, relation (3.20) provides a superior result.

Proof. By (3.16), we can write

$$|\zeta(s)| \leqslant \sum_{n\leqslant N} n^{-\sigma} + N^{1-\sigma}|\tau|^{-1} + \tfrac{1}{2}N^{-\sigma} + \tfrac{1}{2}|s|\sigma^{-1}N^{-\sigma}.$$

Moreover, we plainly have

$$\sum_{n\leqslant N} \frac{1}{n^{\sigma}} \leqslant 1 + \int_1^N \frac{\mathrm{d}t}{t^{\sigma}} < \frac{N^{1-\sigma}}{1-\sigma} \qquad (0 < \sigma < 1).$$

Under the assumption $\sigma \geqslant \alpha$, it then follows that

$$|\zeta(s)| \leqslant N^{1-\alpha}\left\{\frac{1}{1-\alpha}+\frac{1}{|\tau|}+\frac{2\alpha+|\tau|}{2\alpha N}\right\}.$$

Choosing $N = \lfloor |\tau| \rfloor$, we obtain

$$|\zeta(s)| \leqslant \frac{N^{1-\alpha}}{\alpha(1-\alpha)}\{\alpha+\alpha(1-\alpha)+(1+\alpha)(1-\alpha)\}.$$

The maximum of the function in curly brackets is attained for $\alpha = \frac{1}{2}$ and equals $\frac{3}{2}$. □

Corollary 3.10. *For any positive constant c and any integer $k \geqslant 0$ we have*

$$\zeta^{(k)}(s) \ll (\ln|\tau|)^{k+1} \quad (|\tau| \geqslant 2,\ \sigma \geqslant 1 - c/\ln|\tau|). \tag{3.23}$$

Proof. We may plainly suppose $|\tau| \geqslant 3$. For some suitable constant c_0, the disk $|z-s| \leqslant r := c_0/\ln|\tau|$ is contained in a domain where (3.22) applies. The desired bound then follows from Cauchy's formula

$$\zeta^{(k)}(s) = \frac{k!}{2\pi i}\oint_{|z|=r}\zeta(s+z)\frac{\mathrm{d}z}{z^{k+1}}.$$ □

3.5. Initial localization of zeros

The convergence of the Euler product $\zeta(s) = \prod_p(1-p^{-s})^{-1}$ guarantees that $\zeta(s)$ does not vanish for $\sigma > 1$. We shall show in this section that this property extends to the closed half-plane $\sigma \geqslant 1$. The following result, simple but cunning, enables us to perform the limit process. It was introduced by Mertens (1898), thereby formalizing Hadamard's argument. Although essentially equivalent, that of La Vallée–Poussin drew upon deeper properties.

Theorem 3.11 (Mertens). *Let $F(s) := \sum_{n\geqslant 1} a_n/n^s$ be a Dirichlet series with non-negative coefficients and with abscissa of convergence σ_c. Then we have*

$$3F(\sigma) + 4\Re\mathrm{e}\,F(\sigma+i\tau) + \Re\mathrm{e}\,F(\sigma+2i\tau) \geqslant 0 \quad (\sigma > \sigma_c). \tag{3.24}$$

Proof. Write $V(\vartheta) := 3 + 4\cos\vartheta + \cos 2\vartheta$ $(\vartheta \in \mathbb{R})$. A simple verification shows that $V(\vartheta) = 2(1+\cos\vartheta)^2 \geqslant 0$. Now the left-hand side of (3.24) equals

$$\sum_{n\geqslant 1}\frac{a_n V(\tau\ln n)}{n^\sigma}.$$

This implies the stated conclusion. □

Corollary 3.12. *We have*

$$\zeta(\sigma)^3|\zeta(\sigma+i\tau)|^4|\zeta(\sigma+2i\tau)| \geqslant 1 \quad (\sigma > 1). \tag{3.25}$$

Proof. It suffices to apply the previous theorem to the series

$$F(s) = \log \zeta(s) = -\sum_p \log(1 - p^{-s}) = \sum_{n \geqslant 2} \frac{\Lambda(n)}{n^s \ln n}$$

which converges for $\sigma > 1$. □

We are now in a position to prove the following important result.

Theorem 3.13. *The function $\zeta(s)$ has no zero in the half-plane $\sigma \geqslant 1$.*

Proof. Let us argue by contradiction and assume that $\zeta(1+i\tau_0) = 0$. Then $\tau_0 \neq 0$ and $\zeta(s)$ is holomorphic in some neighborhood of $1 + i\tau_0$. Therefore

$$\zeta(\sigma + i\tau_0) \ll \sigma - 1 \qquad (\sigma > 1).$$

On the other hand

$$\zeta(\sigma) \ll 1/(\sigma - 1), \quad \zeta(\sigma + 2i\tau_0) \ll 1 \quad (\sigma > 1),$$

since $s = 1$ is a simple pole and $\zeta(s)$ is holomorphic everywhere except at $s = 1$. It follows that

$$\zeta(\sigma)^3 |\zeta(\sigma + i\tau_0)|^4 |\zeta(\sigma + 2i\tau_0)| \ll \sigma - 1 \quad (\sigma > 1)$$

which contradicts (3.25) as $\sigma \to 1+$. □

When combined with the functional equation (3.5), Theorem 3.13 clearly implies the following corollary.

Corollary 3.14. *In the half-plane $\sigma \leqslant 0$, the function $\zeta(s)$ has zeros only at the points $-2n$ $(n = 1, 2, \ldots)$. These are simple zeros.*

The zeros at even negative integers are called the *trivial zeros* of the zeta function. We shall see in § 3.7 that $\zeta(s)$ also has other zeros, located in the critical strip $0 < \sigma < 1$. The formula

$$\sum_{n \geqslant 1} \frac{(-1)^n}{n^\sigma} = (2^{1-\sigma} - 1)\zeta(\sigma) \quad (0 < \sigma < 1)$$

shows that *these non-trivial zeros are not real.* The functional equation and the fact that $\zeta(s)$ is real for real s imply that they are distributed symmetrically with respect to the line $\sigma = \frac{1}{2}$ and the real axis $\tau = 0$.

Another, particularly elegant, proof of Theorem 3.13 is due to Ingham (1930). It rests on Ramanujan's identity

$$\frac{\zeta(s)^2 \zeta(s + i\vartheta)\zeta(s - i\vartheta)}{\zeta(2s)} = \sum_{n \geqslant 1} \frac{|\tau(n, \vartheta)|^2}{n^s} \tag{3.26}$$

which holds for $\sigma > 1$, $\vartheta \in \mathbb{R}$, with

$$\tau(n, \vartheta) := \sum_{d|n} d^{i\vartheta}. \tag{3.27}$$

Formula (3.26) may be easily checked by identifying the factors in the Euler products on both sides—see also the Notes. Let us consider the abscissa of convergence σ_c of (3.26). We have $\sigma_c \leqslant 1$ since $|\tau(n, \vartheta)| \leqslant \tau(n)$. By Theorem 1.5, the sum of the series is holomorphic for $\sigma > \sigma_c$. Since the coefficients are non-negative, the point $s = \sigma_c$ must be a singularity. Now, if $s = 1 + i\vartheta$ is a zero of $\zeta(s)$, so is $1 - i\vartheta$ (since $\zeta(\bar{s}) = \overline{\zeta(s)}$) and these two zeros compensate for the double pole of $\zeta(s)^2$ at $s = 1$. But $\zeta(s)$ has no other singularity than $s = 1$, hence it follows from the above that (3.26) is holomorphic *on the real axis* down to the first zero of $\zeta(2s)$, that is to say $s = -1$. We would then have $\sigma_c = -1$, which is absurd since $|\tau(n, \vartheta)|$ does not tend to 0.

3.6. Lemmas from complex analysis

The following classical results will be useful in our further investigation of the zeta function.

Theorem 3.15 (Jensen's formula). *Let $F(s)$ be a holomorphic function for $|s| \leqslant R$, with $F(0) = 1$. For $0 < r \leqslant R$, let $n(r)$ denote the number of zeros (counted with multiplicities) of F in the disk $|s| \leqslant r$. Then we have*

$$\int_0^R \frac{n(r)}{r}\,\mathrm{d}r = \frac{1}{2\pi}\int_0^{2\pi} \ln|F(R\mathrm{e}^{i\vartheta})|\,\mathrm{d}\vartheta. \tag{3.28}$$

Proof. If $F(s)$ is non-zero for $|s| = r$, we have

$$n(r) = \frac{1}{2\pi i}\oint_{|s|=r} \frac{F'(s)}{F(s)}\,\mathrm{d}s = \frac{1}{2\pi}\int_0^{2\pi} \frac{F'(r\mathrm{e}^{i\vartheta})}{F(r\mathrm{e}^{i\vartheta})} r\mathrm{e}^{i\vartheta}\,\mathrm{d}\vartheta. \tag{3.29}$$

Formally, equality (3.28) is then obtained by dividing by r, integrating over $[0, R]$, and taking the real parts of the two sides. It remains to show that the critical values of r corresponding to the moduli of the zeros do not interfere with this process.

Let $0 < r_1 < r_2 < \cdots < r_k \leqslant R$ denote the finite sequence of these exceptional moduli. Set $r_0 := 0$, $r_{k+1} := R$. We can integrate (3.29) over each interval

$$\big[(1+\varepsilon)r_j, (1-\varepsilon)r_{j+1}\big] \qquad (j = 0, \ldots, k),$$

and sum the equalities thus obtained. To establish (3.28), it therefore suffices to show that we can allow ε to tend 0, or, in other words, that for each $j \geqslant 1$

we have

$$\lim_{\varepsilon\to 0}\int_0^{2\pi} \ln\left|\frac{F\big((1+\varepsilon)r_j e^{i\vartheta}\big)}{F\big((1-\varepsilon)r_j e^{i\vartheta}\big)}\right| \mathrm{d}\vartheta = 0. \tag{3.30}$$

Since F has only a finite number of zeros on the circle $|s| = r_j$, it is clear that we may without loss of generality restrict to the case when $s = r_j$ is the only zero, say with multiplicity $m \geqslant 1$, of $F(s)$ on this circle. Then

$$\ln|F(re^{i\vartheta})| = m\ln\left|1 - \frac{r}{r_j}e^{i\vartheta}\right| + H(r,\vartheta)$$

for $|r - r_j| \leqslant \varepsilon r_j$, $0 \leqslant \vartheta \leqslant 2\pi$, where $H(r,\vartheta)$ is a continuous function of r and ϑ. Therefore (3.30) is equivalent to

$$\lim_{\varepsilon\to 0}\int_{-\pi}^{\pi} \ln\left|\frac{1-(1+\varepsilon)e^{i\vartheta}}{1-(1-\varepsilon)e^{i\vartheta}}\right| \mathrm{d}\vartheta = 0. \tag{3.31}$$

We have

$$\left|\frac{1-(1+\varepsilon)e^{i\vartheta}}{1-(1-\varepsilon)e^{i\vartheta}}\right|^2 = \frac{\varepsilon^2 + 4(1+\varepsilon)\sin^2\frac{1}{2}\vartheta}{\varepsilon^2 + 4(1-\varepsilon)\sin^2\frac{1}{2}\vartheta} = 1 + O(\varepsilon),$$

so the integrand of (3.31) is uniformly $O(\varepsilon)$ for $|\vartheta| \leqslant \pi$. This implies (3.31). □

Theorem 3.16 (Real part lemma, Borel–Carathéodory). *Let $F(s)$ be a holomorphic function for $|s| \leqslant R$, with $F(0) = 0$. Assume*

$$\max_{|s|=R} \Re e\, F(s) \leqslant A.$$

Then we have

$$|F^{(k)}(s)| \leqslant \frac{2ARk!}{(R-|s|)^{k+1}} \qquad (k \geqslant 0,\ |s| < R). \tag{3.32}$$

Proof. Consider the Taylor series of F at $s = 0$,

$$F(s) = \sum_{n\geqslant 1} a_n s^n \qquad (|s| \leqslant R).$$

If ϑ_n denotes, for each n, the argument of a_n, we can write

$$\Re e\, F(Re^{i\vartheta}) = \sum_{n\geqslant 1} |a_n| R^n \cos(n\vartheta + \vartheta_n),$$

since the series is absolutely and uniformly convergent for $0 \leqslant \vartheta \leqslant 2\pi$. But the fact that F is holomorphic implies that the function $\Re e\, F(s)$ has the mean value property, namely

$$\int_0^{2\pi} \Re e\, F(Re^{i\vartheta})\, \mathrm{d}\vartheta = 0.$$

For each integer $n \geqslant 1$, it then follows that

$$\pi |a_n| R^n = \int_0^{2\pi} \bigl(1 + \cos(n\vartheta + \vartheta_n)\bigr) \Re e\, F(Re^{i\vartheta})\, d\vartheta \leqslant 2\pi A.$$

We obtain (3.32) by multiplying by $\binom{n}{k}(|s|/R)^n$ and summing over $n \geqslant 1$. □

Another useful consequence of Theorem 3.16 is contained in the following statement.

Corollary 3.17. *Let $F(s)$ be a holomorphic function in the disk $|s| \leqslant R$, with $F(0) = 1$, and such that $|F(s)| \leqslant M$ for $|s| = R$. Let Z denote the finite sequence of zeros ϱ of F, counted with multiplicity, in the disk $|s| \leqslant \frac{1}{2}R$. For any complex number s with $|s| < \frac{1}{2}R$, we have*

$$\left| \frac{F'(s)}{F(s)} - \sum_{\varrho \in Z} \frac{1}{s - \varrho} \right| \leqslant \frac{4R \ln M}{(R - 2|s|)^2}. \tag{3.33}$$

Proof. Let $G(s) := F(s) \prod_{\varrho \in Z} (1 - s/\varrho)^{-1}$. We have $|G(s)| \leqslant M$ for $|s| = R$ since each factor in the product over ϱ has modulus at most 1. The function $H(s) := \log G(s)$ hence satisfies the conditions of Theorem 3.16, with $A = \ln M$, in the disk $|s| \leqslant \frac{1}{2}R$. The stated result follows then from (3.32) with $k = 1$. □

3.7. Global distribution of zeros

In the next section, we shall develop, in the special case of the zeta function, the general factorization method of Hadamard for entire functions of finite order. Here we establish the preliminary results.

The function $(s-1)\zeta(s)$ is entire, but it is more convenient to consider instead

$$\xi(s) := s(s-1)\pi^{-s/2}\Gamma\bigl(\tfrac{1}{2}s\bigr)\zeta(s) = s(s-1)\Phi(s) \tag{3.34}$$

which, in addition, satisfies the functional equation

$$\xi(s) = \xi(1-s) \qquad (s \in \mathbb{C}). \tag{3.35}$$

The regularity of ξ for $\sigma \geqslant \frac{1}{2}$ follows from that of ζ, since the pole at $s = 1$ is compensated for by the factor $(s-1)$. By (3.35), $\xi(s)$ is therefore holomorphic in the whole complex plane. We note that $\xi(0) = \xi(1) = 1$.

From Theorem 3.13, $\xi(s) \neq 0$ for $\sigma \geqslant 1$. By (3.35), it follows that $\xi(s)$ does not vanish for $\sigma \leqslant 0$. Therefore all the zeros of $\xi(s)$ are contained in the critical strip $0 < \sigma < 1$. In this strip, $\xi(s)$ and $\zeta(s)$ have the same zeros. The situation is different in the half-plane $\sigma \leqslant 0$, since the trivial zeros of $\zeta(s)$ are compensated by the poles of $\Gamma\bigl(\frac{1}{2}s\bigr)$.

Traditionally, a generic zero of ξ (i.e. non-trivial zero of ζ) is denoted by $\varrho = \beta + i\gamma$. We set accordingly

$$N(T) := \sum_{\varrho\,:\,0\leqslant\gamma\leqslant T} 1.$$

Here and in the sequel, we adopt the convention that all zeros ϱ are counted with multiplicity.

In this section, our aim is to establish an asymptotic formula for $N(T)$ as $T \to \infty$.

With this in mind, it is clearly worthwhile to know the order of growth of $\xi(s)$. By Theorem 3.8, we have

$$\zeta(s) \ll \tau \qquad (\sigma \geqslant 0,\ |\tau| \geqslant 1). \tag{3.36}$$

Moreover, integrating by parts the last term in the complex Stirling formula (0.11), we obtain

$$\log\Gamma(s) = (s-\tfrac{1}{2})\log s - s + \tfrac{1}{2}\ln(2\pi) + \frac{1}{12s} - \tfrac{1}{2}\int_0^\infty \frac{B_2(t)}{(t+s)^2}\,\mathrm{d}t \tag{3.37}$$

where $B_2(t) = \langle t\rangle^2 - \langle t\rangle + \frac{1}{6}$ denotes the second Bernoulli function. This implies that

$$\ln|\xi(s)| \ll |s|\ln|s| \qquad (|s| \to \infty) \tag{3.38}$$

for $\sigma \geqslant \frac{1}{2}$, and hence in the whole plane, by (3.35).

We shall use an analogous upper bound to prove the following crucial result.

Theorem 3.18. *For $T \geqslant 2$, we have*

$$N(T+1) - N(T) \ll \ln T. \tag{3.39}$$

Proof. We apply Jensen's formula (3.28) to the function

$$F(s) := \xi(s+2+iT)/\xi(2+iT)$$

with $R = 3$.

In the first instance, for all r with $\sqrt{5} \leqslant r \leqslant 3$, the disk $|s| \leqslant r$ contains the rectangle with vertices $-2, -2+i, -1, -1+i$, whence

$$n(r) \geqslant N(T+1) - N(T).$$

Next, from (3.36) and (3.37), we readily deduce the estimate

$$\ln|F(s)| \leqslant O(\ln T) \qquad (|s| \leqslant 3).$$

Jensen's formula then implies that

$$\{N(T+1) - N(T)\}\ln\left(3/\sqrt{5}\right) \leqslant \int_{\sqrt{5}}^{3} \frac{n(r)}{r}\,\mathrm{d}r \ll \ln T,$$

and (3.39) follows. □

We are now in a position to establish the following fundamental theorem.

Theorem 3.19. *As T tends to infinity, we have*

$$N(T) = \frac{T}{2\pi}\ln\frac{T}{2\pi} - \frac{T}{2\pi} + O(\ln T). \tag{3.40}$$

Remark. This implies, in particular, that $\zeta(s)$ has infinitely many non-trivial zeros.

Proof. Without loss of generality, we may assume that T is not the ordinate of a zero of $\xi(s)$. Let $\mathcal{R}$ be the positively oriented rectangle with vertices $2 \pm iT, -1 \pm iT$. By Cauchy's argument principle, we have

$$2N(T) = \frac{1}{2\pi i}\int_{\mathcal{R}} \frac{\xi'(s)}{\xi(s)}\,\mathrm{d}s = \frac{1}{2\pi}\Im m \int_{\mathcal{R}} \frac{\xi'(s)}{\xi(s)}\,\mathrm{d}s.$$

Since $\mathcal{R}$ is symmetric with respect to the axes $\sigma = \frac{1}{2}$ and $\tau = 0$, and $\xi'(s)/\xi(s)$ respects this symmetry because of the functional equation, we deduce that

$$N(T) = \frac{1}{\pi}\Im m \int_{\mathcal{L}} \frac{\xi'(s)}{\xi(s)}\,\mathrm{d}s = \frac{1}{\pi}\big[\arg \xi(s)\big]_{\mathcal{L}} \tag{3.41}$$

where $\mathcal{L}$ is the polygonal line $\big[2, 2+iT, \frac{1}{2}+iT\big]$. Now, by (3.34), we have

$$\xi(s) = 2(s-1)\pi^{-s/2}\Gamma(\tfrac{1}{2}s+1)\zeta(s) \tag{3.42}$$

and

$$\begin{aligned} [\arg(s-1)]_{\mathcal{L}} &= \arg(iT - \tfrac{1}{2}) = \tfrac{1}{2}\pi + O\Big(\frac{1}{T}\Big), \\ \big[\arg \pi^{-s/2}\big]_{\mathcal{L}} &= -\tfrac{1}{2}T\ln\pi. \end{aligned} \tag{3.43}$$

Moreover, by the complex Stirling formula (II.0.12),

$$\begin{aligned} \big[\arg\Gamma(\tfrac{1}{2}s+1)\big]_{\mathcal{L}} &= \Im m \log\Gamma(\tfrac{1}{2}iT + \tfrac{5}{4}) \\ &= \Im m \Big\{(\tfrac{1}{2}iT + \tfrac{3}{4})\log(\tfrac{1}{2}iT + \tfrac{5}{4}) - \tfrac{1}{2}iT - \tfrac{5}{4} + \tfrac{1}{2}\ln 2\pi + O\Big(\frac{1}{T}\Big)\Big\} \\ &= \tfrac{1}{2}T\ln\tfrac{1}{2}T - \tfrac{1}{2}T + \tfrac{3}{8}\pi + O\Big(\frac{1}{T}\Big). \end{aligned} \tag{3.44}$$

Thus, it only remains to prove that

$$\Im m \int_{\mathcal{L}} \frac{\zeta'(s)}{\zeta(s)}\,\mathrm{d}s \ll \ln T. \tag{3.45}$$

Since the vertical segment of $\mathcal{L}$ lies in the half-plane of absolute convergence, we can integrate termwise. Hence

$$\left|\int_2^{2+iT} \frac{\zeta'(s)}{\zeta(s)}\,\mathrm{d}s\right| \leqslant 2\sum_{n\geqslant 2} \frac{\Lambda(n)}{n^2\ln n} < \infty.$$

In order to deal with the horizontal segment, we apply Corollary 3.17 to $F(s) = \zeta(2+iT+s)/\zeta(2+iT)$ with $R = 4$ and $r = 3/2$. The numerator is of the order of a power of T and the denominator satisfies

$$|\zeta(2+iT)| \geqslant 1 - \sum_{n\geqslant 2} \frac{1}{n^2} = 2 - \tfrac{1}{6}\pi^2 > 0.$$

Letting Z denote the finite sequence of zeros of $\zeta(s)$ (counted with multiplicity) in the disk $|s-2-iT| \leqslant \frac{3}{2}$, we therefore obtain that, for each s in the horizontal segment $\left[\frac{1}{2}+iT, 2+iT\right]$,

$$\frac{\zeta'(s)}{\zeta(s)} - \sum_{\varrho\in Z} \frac{1}{s-\varrho} \ll \ln T. \tag{3.46}$$

By Theorem 3.18, we have $\sum_{\varrho\in Z} 1 \ll \ln T$ and, for each ϱ in Z, we have

$$\left| \Im m \int_{2+iT}^{\frac{1}{2}+iT} \frac{\mathrm{d}s}{s-\varrho} \right| \leqslant \pi.$$

This implies (3.45), and completes the proof of Theorem 3.19. □

3.8. Expansion as a Hadamard product

We are now in a position to make explicit, for the entire function $\xi(s)$—and consequently for $\zeta(s)$—, the Hadamard expansion in terms of its zeros. As will be clear from the proof, only Theorem 3.18, which locally bounds the number of non-trivial zeros, will be needed.

Theorem 3.20 (Hadamard product formula). *For suitable constants a, b, we have*

$$\xi(s) = \mathrm{e}^{as} \prod_{\varrho} (1 - s/\varrho)\mathrm{e}^{s/\varrho} \qquad (s\in\mathbb{C}), \tag{3.47}$$

$$\zeta(s) = \frac{\mathrm{e}^{bs}}{2(s-1)} \Gamma\left(\tfrac{1}{2}s+1\right)^{-1} \prod_{\varrho} (1 - s/\varrho)\mathrm{e}^{s/\varrho} \qquad (s\neq 1). \tag{3.48}$$

Remark. In Exercise 175 we outline a proof of the formula

$$b = \ln(2\pi) - \tfrac{1}{2}\gamma - 1 \approx 0.54927, \tag{3.49}$$

which implies $a = \frac{1}{2}\ln(4\pi) - \frac{1}{2}\gamma - 1 \approx -0.02310$.

Proof. By (3.39), we have

$$\sum_{|\gamma|\leqslant T} \frac{1}{|\varrho|^2} \leqslant 2\int_0^T \frac{\mathrm{d}N(t)}{t^2} \ll \sum_{k\leqslant T} \frac{\ln(2k)}{k^2} \ll 1.$$

This implies the convergence of the infinite product

$$P(s) := \prod_{\varrho} (1 - s/\varrho)\mathrm{e}^{s/\varrho} \qquad (s \in \mathbb{C})$$

whose general term is, for each s, of the form $1 + O(1/|\varrho|^2)$. Therefore $P(s)$ is an entire function of s with the same zeros as $\xi(s)$ and, since $P(0) = \xi(0) = 1$, we can find a holomorphic branch $F(s)$ of $\log\big(\xi(s)/P(s)\big)$ taking the value 0 at $s = 0$. We shall see that we have

$$\max_{0 \leqslant \vartheta \leqslant 2\pi} \Re\mathrm{e}\, F(R\mathrm{e}^{i\vartheta}) \ll R(\ln R)^2 \tag{3.50}$$

for a sequence of values of R tending to infinity. By the Borel–Carathéodory theorem, it follows from (3.50) that F is a linear function of s, which is precisely the required result.

We shall prove that (3.50) holds for all R satisfying

$$\min_{\varrho} \big|R - |\varrho|\big| \geqslant 1/(C \ln R) \tag{3.51}$$

where C is a suitable positive constant. Each interval $[k, k+2]$ with $k \geqslant 2$ contains at least one such R. Indeed, if C is large enough, we have

$$\mathrm{card}\{\varrho : k \leqslant |\varrho| \leqslant k+2\} \leqslant 2\Big(N(k+2) - N(k-1)\Big) < J := \lfloor C \ln k \rfloor .$$

By the pigeonhole principle, there is at least one interval

$$[k + 2j/J, k + (2j+2)/J] \quad (0 \leqslant j < J)$$

containing no $|\varrho|$. We can thus choose $R := k + (2j+1)/J$.

Consider some R satisfying (3.51). For $|s| = R$ we have

$$\begin{aligned} \ln|P(s)| &= \sum_{\varrho} \ln\big|(1 - s/\varrho)\mathrm{e}^{s/\varrho}\big| \\ &\geqslant -\Bigg\{ \sum_{|\varrho| \leqslant R/2} \frac{R}{|\varrho|} + \sum_{R/2 < |\varrho| \leqslant 2R} \Big(2 + \ln(2CR\ln R)\Big) + \sum_{|\varrho| > 2R} \frac{R^2}{|\varrho|^2} \Bigg\}, \end{aligned}$$

where we have used the inequalities

$$\ln|(1-z)\mathrm{e}^z| \geqslant \begin{cases} -|z| & (|z| \geqslant 2) \\ -2 + \ln|1-z| & (\tfrac{1}{2} \leqslant |z| < 2) \\ -|z|^2 & (|z| < \tfrac{1}{2}). \end{cases}$$

Using (3.39), we deduce from the above lower bound that

$$\ln(1/|P(s)|) \ll R(\ln R)^2.$$

We omit the details, which are easy. Now, by (3.38), we also have

$$\ln|\xi(s)| \ll R \ln R.$$

This implies (3.50) and so completes the proof. □

We conclude this section with a remark. The bound

$$N(T+1) - N(T) \ll \ln T$$

sufficed to prove the product formula (3.47) for $\xi(s)$. Now this formula easily implies that $N(T) \to \infty$ with T. Indeed, otherwise $\xi(s)$ would have the form

$$\xi(s) = \mathrm{e}^{cs} Q(s)$$

where c is a constant and $Q(s)$ is a polynomial in s. The functional equation $\xi(s) = \xi(1-s)$ would then imply that $c = 0$, but Stirling's formula for positive real s prevents ξ from having polynomial growth. Titchmarsh (1951, § 9.1) provides a simple argument allowing one to deduce merely from the functional equation the lower bound

$$N(T+A) - N(T) \gg 1$$

for some suitable absolute constant A.

3.9. Zero-free regions

It is obvious that the proof of Theorem 3.13, leading to the assertion $\zeta(s) \neq 0$ for $\sigma \geqslant 1$, could yield an explicit region with no zero of $\zeta(s)$ and yet intersecting the half-plane $\sigma < 1$. We provide the details in § 4.2, for the purpose of information only. Indeed, the results of the previous sections, and notably the product formula, enable us to obtain rather easily a much wider region.

Theorem 3.21. *There exists a positive constant c such that $\zeta(s)$ has no zero in the region of the complex plane defined by the inequality*

$$\sigma \geqslant 1 - c/\ln(2 + |\tau|). \tag{3.52}$$

Proof. The Dirichlet series

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n \geqslant 1} \frac{\Lambda(n)}{n^s}$$

has non-negative coefficients. By Theorem 3.11, it hence follows that, for all $\sigma > 1$ and real γ, we have

$$-3\frac{\zeta'(\sigma)}{\zeta(\sigma)} - 4\Re\mathrm{e}\,\frac{\zeta'(\sigma+i\gamma)}{\zeta(\sigma+i\gamma)} - \Re\mathrm{e}\,\frac{\zeta'(\sigma+2i\gamma)}{\zeta(\sigma+2i\gamma)} \geqslant 0. \tag{3.53}$$

We shall arrive at the stated result by bounding each of the three terms on the left-hand side from above when γ is the ordinate of a zero $\varrho = \beta + i\gamma$ of $\zeta(s)$.

In the first place, we have

$$-\frac{\zeta'(\sigma)}{\zeta(\sigma)} = \frac{1}{\sigma - 1} + O(1).$$

Then, taking the logarithmic derivative of the product formula (3.48), we obtain, for $s \neq 1$, $s \neq \varrho$,

$$-\frac{\zeta'(s)}{\zeta(s)} = -b + \frac{1}{s-1} + \frac{\Gamma'(\frac{1}{2}s+1)}{2\Gamma(\frac{1}{2}s+1)} - \sum_{\varrho}\Big(\frac{1}{\varrho} + \frac{1}{s-\varrho}\Big). \tag{3.54}$$

Estimating the term involving Γ by the complex Stirling formula (0.11), and noting that, for $\sigma > 1$, the numbers ϱ and $(s-\varrho)$ have positive real parts, we have

$$-\Re e\,\frac{\zeta'(s)}{\zeta(s)} \leqslant O(\ln|\tau|) \qquad (\sigma > 1,\ |\tau| \geqslant 2),$$

and also, taking into account the contribution of the zero $\beta + i\gamma$,

$$-\Re e\,\frac{\zeta'(\sigma+i\gamma)}{\zeta(\sigma+i\gamma)} \leqslant O(\ln|\gamma|) - \frac{1}{\sigma-\beta}.$$

Substituting these estimates into (3.53), we deduce the existence of a positive constant c_1 such that for $|\gamma| \geqslant 2$ we have

$$\frac{3}{\sigma-1} - \frac{4}{\sigma-\beta} \geqslant -c_1 \ln|\gamma|$$

from which we get

$$1-\beta \geqslant \frac{1 - c_1(\sigma-1)\ln|\gamma|}{(3/(\sigma-1)) + c_1\ln|\gamma|}.$$

Choosing $\sigma = 1 + 1/(2c_1\ln|\gamma|)$, it follows that

$$1-\beta \geqslant c_2/\ln|\gamma|$$

with $c_2 = 1/14c_1$. This implies the stated result, since (3.52) holds trivially for $|\tau| < 2$ with sufficiently small c. □

3.10. Bounds for ζ'/ζ, $1/\zeta$ and $\log\zeta$

Theorem 3.21 determines a region in which the three functions in the title of this section are all holomorphic. It is naturally desirable to have explicit upper bounds available in the same domain.

Theorem 3.22. *There exists a positive constant c such that, for $|\tau| \geqslant 3$ and $\sigma \geqslant 1 - c/\ln|\tau|$, we have*

$$\zeta'(s)/\zeta(s) \ll \ln|\tau|, \tag{3.55}$$

$$1/\zeta(s) \ll \ln|\tau|, \tag{3.56}$$

$$|\log\zeta(s)| \leqslant \ln_2|\tau| + O(1). \tag{3.57}$$

Proof. We can assume $\tau > 0$. By Theorem 3.21, there is a constant c, $0 < c < \frac{1}{16}$, such that each non-trivial zero $\varrho = \beta + i\gamma$ of $\zeta(s)$ satisfies

$$\beta < 1 - 8c/\ln(|\gamma| + 2).$$

We shall see that this implies that

$$\min_{\varrho} \Re e \left\{ \frac{1}{\varrho} + \frac{1}{s-\varrho} \right\} \geqslant 0 \qquad (\tau \geqslant 4,\ \sigma \geqslant 1 - 4c/\ln\tau). \tag{3.58}$$

When $|s-\varrho| > \frac{1}{2}|\varrho|$, the quantity to be minimized can be rewritten, setting $\vartheta := 2|s-\varrho|/|\varrho| \geqslant 1$, in the form

$$\frac{\beta}{|\varrho|^2} + \frac{\sigma-\beta}{|s-\varrho|^2} = \frac{\vartheta^2\beta/4 + \sigma - \beta}{|s-\varrho|^2} \geqslant \frac{\sigma - 3/4}{|s-\varrho|^2} > 0.$$

When $|s-\varrho| \leqslant \frac{1}{2}|\varrho|$, we have $|\tau-\gamma| \leqslant \frac{1}{2}(|\gamma|+1)$, so that $|\gamma| \leqslant 2\tau + 2$, and hence

$$\beta < 1 - 8c/\ln(2\tau+4) \leqslant 1 - 4c/\ln\tau \leqslant \sigma.$$

This implies (3.58).

Substituting in (3.54), it follows that

$$-\Re e \frac{\zeta'(s)}{\zeta(s)} \leqslant K \ln\tau \qquad (\tau \geqslant 4,\ \sigma \geqslant 1 - 4c/\ln\tau) \tag{3.59}$$

where K is some suitable absolute constant.

We are now in a position to establish the upper bound (3.55). It is clear that we can assume τ to be sufficiently large. Let then $s = \sigma + i\tau$ be a fixed complex number such that $\tau \geqslant 5, \sigma \geqslant 1 - c/\ln\tau$. Write

$$\eta := c/\ln\tau, \qquad s_0 := 1 + \eta + i\tau.$$

Then for each w in the disk $|w| \leqslant 4\eta$ the point $s_0 + w = \sigma' + i\tau'$ satisfies $\tau' \geqslant 4$, $\sigma' \geqslant 1 - 4c/\ln\tau'$, and as a result, by (3.59), we may write

$$-\Re e \frac{\zeta'(s_0+w)}{\zeta(s_0+w)} \leqslant 2K\ln\tau.$$

This implies that the function

$$F(w) := \frac{\zeta'(s_0)}{\zeta(s_0)} - \frac{\zeta'(s_0+w)}{\zeta(s_0+w)}$$

satisfies, in the disk $|w| \leqslant 4\eta$, the hypotheses of the Borel–Carathéodory theorem with $A := 2K\ln\tau + |\zeta'(s_0)/\zeta(s_0)|$. Since $|s - s_0| \leqslant 2\eta$, Theorem 3.16 implies

$$\left|\frac{\zeta'(s)}{\zeta(s)}\right| \leqslant 4K\ln\tau + 3\left|\frac{\zeta'(s_0)}{\zeta(s_0)}\right|.$$

The upper bound (3.55) follows from this inequality since

$$\left|\frac{\zeta'(s_0)}{\zeta(s_0)}\right| \leqslant \sum_{n\geqslant 1} \frac{\Lambda(n)}{n^{1+\eta}} = \frac{1}{\eta} + O(1) \ll \ln \tau.$$

The estimate (3.56) is immediate from (3.57). To establish this last inequality, we use (3.55) in the form

$$\log\left(\frac{\zeta(s)}{\zeta(s_0)}\right) = \int_{s_0}^{s} \frac{\zeta'(w)}{\zeta(w)}\,\mathrm{d}w \ll |s - s_0| \ln \tau \ll 1. \tag{3.60}$$

This implies the required result upon noting that

$$\begin{aligned} |\log \zeta(s_0)| &= \left|\sum_{n\geqslant 2} \frac{\Lambda(n)}{n^{s_0} \ln n}\right| \leqslant \sum_{n\geqslant 2} \frac{\Lambda(n)}{n^{1+\eta} \ln n} = \ln \zeta(1+\eta) \\ &= \ln(1/\eta) + O(1) = \ln_2 \tau + O(1). \end{aligned}$$

□

Notes

§§ 3.2–3.3. In the proofs of the fundamental Theorems 3.1 and 3.2, we followed the original method of Riemann (1859). Titchmarsh (1951) gives several other proofs for the analytic continuation and the functional equation of $\zeta(s)$. See in particular Exercises 183 and 184.

§ 3.4. Theorem 3.5 is due to Hardy & Littlewood (1921). In most applications, the fact that the sum over n contains at least $C|\tau|$ terms is prohibitive. Another result of Hardy & Littlewood (1923, 1929) allows one to reduce this number of terms to $O(\sqrt{|\tau|}\,)$. This is the famous *approximate functional equation*:

Theorem 3.23. *For each $\delta > 0$, and uniformly for $0 \leqslant \sigma \leqslant 1$, $x > \delta$, $y > \delta$, $\tau = 2\pi xy$, we have*

$$\zeta(s) = \sum_{n \leqslant x} n^{-s} + \chi(s) \sum_{n \leqslant y} n^{s-1} + O\big(x^{-\sigma} + \tau^{\frac{1}{2}-\sigma} y^{\sigma-1}\big)$$

where $\chi(s)$ is defined by (3.10).

For a proof of this result, its extension to powers of $\zeta(s)$ and numerous notes on these questions, see chapter 4 of Ivić (1985) and chapter IV of Titchmarsh–Heath-Brown (1986).

Improving on results of Bombieri & Iwaniec (1986) and of his own (1993b), Huxley showed in 2005 that the exponent $\frac{1}{6}$ in the bound for $\zeta\big(\frac{1}{2} + i\tau\big)$ can be replaced by $\frac{32}{205} + \varepsilon$ for any $\varepsilon > 0$.

§ 3.5. Formula (3.26) is a special case of Ramanujan's formula

$$\frac{\zeta(s)\zeta(s-a)\zeta(s-b)\zeta(s-a-b)}{\zeta(2s-a-b)} = \sum_{n \geqslant 1} \frac{\sigma_a(n)\sigma_b(n)}{n^s},$$

which holds for $\sigma > 1 + \max\{0, \Re e\, a, \Re e\, b, \Re e(a+b)\}$. Titchmarsh (1951) gives a proof of this result using the Euler product. An alternative approach consists in noting that the left-hand side is the series associated with the arithmetic function

$$\begin{aligned}\sum_{h^2 k\ell m | n} h^{a+b} \mu(h) k^a \ell^b m^{a+b} &= \sum_{KLm|n} K^a L^b m^{a+b} \sum_{h|(K,L)} \mu(h) \\ &= \sum_{\substack{KLm|n \\ (K,L)=1}} K^a L^b m^{a+b} = \sum_{d|n} \sum_{[\kappa,\lambda]=d} \kappa^a \lambda^b \\ &= \sum_{\kappa,\lambda|n} \kappa^a \lambda^b = \sigma_a(n)\sigma_b(n).\end{aligned}$$

§ 3.7. Theorem 3.19, conjectured by Riemann in his memoir of 1859, was proved by von Mangoldt in 1895. Let $S(T) = (1/\pi)\arg\zeta(\frac{1}{2} + iT)$, where the argument is defined by continuity along the polygonal line $\mathcal{L}$ joining the points 2, $2 + iT$, $\frac{1}{2} + iT$. By (3.41), (3.42), (3.43) and (3.44), we have—cf. Titchmarsh (1951, § 9.2)—

$$N(T) = \frac{T}{2\pi}\ln\frac{T}{2\pi} - \frac{T}{2\pi} + \frac{7}{8} + S(T) + O\Big(\frac{1}{T}\Big). \tag{3.61}$$

Littlewood (1924) establishes the bound

$$\int_0^T S(\tau)\,\mathrm{d}\tau \ll \ln T,$$

which shows that, at least on average, the term $\frac{7}{8}$ in (3.61) is significant. However, Selberg proved in 1946 that

$$S(T) = \Omega_{\pm}\Big(\frac{(\ln T)^{1/3}}{(\ln_2 T)^{7/3}}\Big).$$

The estimate $S(T) \ll \ln T$ proved in Theorem 3.19 remains the best-known result to date.

An immediate consequence of Theorem 3.19 is that, if $\{\gamma_n\}_{n=1}^{\infty}$ denotes the non-decreasing sequence of ordinates of the zeros of $\xi(s)$ in the upper half-plane, then

$$\gamma_n \sim 2\pi n/\ln n \quad (n \to \infty). \tag{3.62}$$

Littlewood (1924) showed that $\gamma_{n+1} - \gamma_n \to 0$.

§ 3.9. Note that, in general, an upper bound for $\zeta(s)$ on the line $\sigma = 1$ provides, by Corollary 3.17, an upper bound for $-\Re e\big(\zeta'(s)/\zeta(s)\big)$ in terms of the abscissae of the zeros of $\zeta(s)$ near to the line $\sigma = 1$, which allows one to obtain a zero-free region by an argument analogous to that used in the proof of Theorem 3.21. Thus, it can be stated that, conceptually, it is equivalent to bound $|\zeta(1 + i\tau)|$ and to determine a zero-free region for $\zeta(s)$.

Korobov (1958) and Vinogradov (1958) established the upper bound

$$\zeta(s) \ll \big(1 + \tau^{A(1-\sigma)^{3/2}}\big)(\ln\tau)^{2/3} \qquad (\sigma \geqslant 0,\ \tau \geqslant 2) \tag{3.63}$$

from which follows the best zero-free region for $\zeta(s)$ presently known, namely

$$\sigma \geqslant 1 - C(\ln\tau)^{-2/3}(\ln_2\tau)^{-1/3} \qquad (\tau \geqslant 3). \tag{3.64}$$

For a detailed proof of (3.63) and the deduction of (3.64), see Ivić (1985, chap. 6).

§ 3.10. Knowing the zero-free region (3.64) for $\zeta(s)$ easily implies improvements on the estimates of Theorem 3.22. One obtains

$$\zeta'(s)/\zeta(s) \ll (\ln \tau)^{2/3}(\ln_2 \tau)^{1/3},$$

$$1/\zeta(s) \ll (\ln \tau)^{2/3}(\ln_2 \tau)^{1/3},$$

$$|\log \zeta(s)| \leqslant \tfrac{2}{3} \ln_2 \tau + \tfrac{1}{3} \ln_3 \tau + O(1),$$

for s satisfying (3.64).

Exercises

174. Show that we have $\zeta(s) = 1/(s-1) + \gamma + O(s-1)$ in a neighborhood of $s = 1$, where γ denotes Euler's constant.

175. *Computation of the constant b in Theorem* 3.20.

(a) Show that $\Gamma'(1) = -\gamma$ by using any of the following: the infinite product expansion for $\Gamma(z)$, the complex Stirling formula, or the definition of $\Gamma(z)$ in integral form.

(b) Show that $\zeta'(0) = -\frac{1}{2}\ln(2\pi)$ by considering, in a neighborhood of $s = 1$, a suitable Taylor approximation of each side of the functional equation for $\zeta(s)$.

(c) Show that the constant b such that

$$\zeta(s) = \frac{e^{bs}}{2(s-1)\Gamma\left(\frac{1}{2}s+1\right)} \prod_{\varrho} (1 - s/\varrho)e^{s/\varrho}$$

has the value $b = \ln(2\pi) - 1 - \frac{1}{2}\gamma$. To this end, one may calculate the logarithmic derivative of $\zeta(s)$ at $s = 0$.

176. Define $R(n) := |\{(d, d') : [d, d'] = n\}|$ $(n \geqslant 1)$. Using the result of Exercise 47(c) (p. 41), determine the abscissa of convergence of the associated Dirichlet series.

177. Let $\tau_k(n)$ denote the number of solutions in integers $m_1, m_2, \dots, m_k \geqslant 1$ of the equation $n = m_1 m_2 \dots m_k$.

(a) Show that $\sum_{n\geqslant 1} \tau_k(n)/n^s = \zeta(s)^k$.

(b) Show that, for each $k \geqslant 1$, there exists a $\delta_k > 0$, to be calculated explicitly, such that

$$\sum_{n\leqslant x} \tau_k(n) = xP_{k-1}(\ln x) + O_\varepsilon(x^{1-\delta_k+\varepsilon})$$

where P_{k-1} denotes a polynomial of degree $k-1$, whose coefficients should be calculated for $k = 1, 2, 3$. What value of δ_k can be obtained under the Lindelöf hypothesis?

178. Show that, for fixed $\tau \in \mathbb{R}$ and $s = 1 + i\tau$, we have

$$\sum_{n\leqslant x} \frac{\mu(n)}{n^s} = \frac{1}{2\pi i} \int_{\kappa-iT}^{\kappa+iT} \frac{x^w}{w\zeta(s+w)}\, dw + O\Big(\frac{\ln x}{T} + \frac{1}{x}\Big)$$

uniformly for $x \geqslant 2$, $T \geqslant 1$, with $\kappa := 1/\ln x$. Using the evaluations in § 3.10, deduce the estimate

$$\sum_{n\leqslant x} \frac{\mu(n)}{n^s} = \frac{1}{\zeta(s)} + O_s\Big(e^{-c\sqrt{\ln x}}\Big) \qquad (\sigma = 1,\ x > 1).$$

179. Use the effective Perron formula and some estimates from § 3.10 to show that

$$\sum_{n\leqslant x}\frac{\Lambda(n)}{n^{1+i\tau}}=\begin{cases}-\dfrac{\zeta'(1+i\tau)}{\zeta(1+i\tau)}-\dfrac{x^{-i\tau}}{i\tau}+O_\tau\bigl(\mathrm{e}^{-c\sqrt{\ln x}}\bigr) & (\tau\neq 0)\\ \ln x-\gamma+O\bigl(\mathrm{e}^{-c\sqrt{\ln x}}\bigr) & (\tau=0).\end{cases}$$

Deduce that the following expressions hold for $\sigma=1$, $\tau\neq 0$:

$$\log\zeta(s)=\sum_{n\geqslant 2}\frac{\Lambda(n)n^s}{\ln n},\qquad \zeta(s)=\prod_p\Bigl(1-\frac{1}{p^s}\Bigr)^{-1}.$$

180. Let $\tau^*(n)$ be the number of odd divisors of an integer n. Determine the Dirichlet series associated to $\tau^*(n)$ and use complex integration to deduce the asymptotic estimate

$$T^*(x):=\sum_{n\leqslant x}\tau^*(n)=\tfrac{1}{2}x\bigl\{\ln(2x)+2\gamma-1\bigr\}+O_\varepsilon(x^{\frac{1}{2}+\varepsilon}). \tag{3.65}$$

Set $T(x):=\sum_{n\leqslant x}\tau(n)$. Show that $T^*(x)=T(x)-T(\frac{1}{2}x)$ and deduce an improved error term in (3.65).

181. Let z be a complex number of modulus $\leqslant 1$. Using Euler products, show that the Dirichlet series $F(s,z):=\sum_{n\geqslant 1}z^{\Omega(n)}/n^s$ converges for $\sigma>1$ and has an analytic continuation, given by the convergent infinite product

$$F(s,z)=\prod_{k\geqslant 1}\zeta(ks)^{a_k(z)},$$

with $a_k(z):=k^{-1}\sum_{d|k}\mu(d)z^{k/d}$, in any simply connected open subset of the half-plane $\sigma>0$ excluding all values of s for which ks $(k=1,2,\ldots)$ is either a zero or a pole of the zeta function.

182. Show the following asymptotic relations are equivalent to Lindelöf's hypothesis:[2]

(a) $\displaystyle\int_1^T|\zeta(\tfrac{1}{2}+i\tau)|^{2k}\,\mathrm{d}\tau\ll_{\varepsilon,k}T^{1+\varepsilon}\qquad(\varepsilon>0,\ k=1,2,\ldots);$

(b) $\displaystyle\int_1^T|\zeta(\sigma+i\tau)|^{2k}\,\mathrm{d}\tau\ll_{\sigma,\varepsilon,k}T^{1+\varepsilon}\qquad(\sigma>\tfrac{1}{2},\ \varepsilon>0,\ k=1,2,\ldots);$

(c) $\displaystyle\frac{1}{T}\int_1^T|\zeta(\sigma+i\tau)|^{2k}\,\mathrm{d}\tau\sim\sum_{n\geqslant 1}\frac{\tau_k(n)^2}{n^{2\sigma}}\qquad(\sigma>\tfrac{1}{2},\ k=1,2,\ldots).$

[2] Cf. Titchmarsh (1951), chap. XIII.

183. *Analytic continuation by the Euler–Maclaurin formula.*

Applying the Euler–Maclaurin formula to $f(t) := t^{-s}$, show that, with the notation of Chapter I.0 and for $s > 1$, we have

$$\zeta(s) = \frac{s}{s-1} + \sum_{0 \leqslant r \leqslant k} \frac{B_{r+1}}{r+1} \binom{s+r-1}{r} - \binom{s+k}{k+1} \int_1^\infty B_{k+1}(t) t^{-s-k-1} \, \mathrm{d}t.$$

Deduce an alternative proof of Theorems 3.1 and 3.2.

184. *The functional equation by Jacobi's theta function.*

Let $\vartheta(x) := \sum_{n \in \mathbb{Z}} \mathrm{e}^{-\pi n^2 x}$ $(x > 0)$.

(a) Applying Poisson's summation formula to $f(t) := \mathrm{e}^{-\pi t^2 x}$, show that

$$\vartheta(1/x) = \sqrt{x}\, \vartheta(x) \qquad (x > 0).$$

(b) Make the change of variables $x = \pi n^2 y$ in the integral

$$\Gamma(s/2) = \int_0^\infty \mathrm{e}^{-x} x^{s/2-1} \, \mathrm{d}x$$

and deduce that we have, for $\sigma > 1$,

$$\zeta(s)\Gamma(s/2)\pi^{-s/2} = \int_0^\infty \psi(x) x^{s/2-1} \, \mathrm{d}x, \tag{3.66}$$

where we put $\psi(x) := \frac{1}{2}\big(\vartheta(x) - 1\big)$.

(c) Show that the right-hand side of (3.66) also equals

$$\frac{1}{s(s-1)} + \int_1^\infty \big(x^{-(s+1)/2} + x^{s/2-1}\big)\psi(x) \, \mathrm{d}x,$$

where the last integral defines an entire function of s which is invariant under the transformation $s \mapsto 1 - s$. Deduce an alternative proof of Theorems 3.3 and 3.4.

185. Prove Theorem 3.5 without appealing to van der Corput's theory by employing the following strategy to obtain formula (3.18): apply the Euler–Maclaurin formula of order 0 to $f(t) := t^{-i\tau}$, expand $B_1(t)$ as a Fourier series, integrate termwise, and bound each of the resulting integrals by using the second mean value theorem.

186. *Hurwitz's zeta function.* For $\alpha \in \,]0, 1]$, let $\zeta(s; \alpha) := \sum_{n=0}^\infty (n + \alpha)^{-s}$ denote the Hurwitz zeta function. Show that $\zeta(s; \alpha)$ may be continued to a meromorphic function in the whole complex plane having, as its only singularity, a simple pole with residue 1 at $s = 1$, and satisfying, for $\sigma < 0$, the "functional equation"

$$\zeta(s; \alpha) = \Gamma(1-s) \sum_{n \in \mathbb{Z} \setminus \{0\}} \frac{\mathrm{e}(n\alpha)}{(2n\pi i)^{1-s}} = \frac{2\Gamma(1-s)}{(2\pi)^{1-s}} \sum_{n \geqslant 1} \frac{\sin \pi(\frac{1}{2}s + 2n\alpha)}{n^{1-s}}$$

where the complex logarithm is defined as the principal branch.

187. Assuming that all zeros of $\zeta(s)$ are simple and that there exists, for each $\varepsilon > 0$, a sequence $\{T_\nu\}_{\nu=1}^{\infty}$ such that $\min_{-1\leqslant\sigma\leqslant 2}|\zeta(\sigma \pm iT_\nu)| \gg T_\nu^{-\varepsilon}$, show that, for $x \in]1,\infty] \setminus \mathbb{N}$, we have

$$M(x) := \sum_{n\leqslant x}\mu(n) = \sum_{\varrho}\frac{x^\varrho}{\varrho\zeta'(\varrho)} - 2 - \sum_{n\geqslant 1}\Big(\frac{2\pi}{x}\Big)^{2n}\frac{(-1)^n}{(2n)!\,n\zeta(2n+1)},$$

where the sum over ϱ is defined as the limit of $\sum_{|\Im m\,\varrho|\leqslant T_\nu}$ as $\nu\to\infty$.

188. Show that, for any fixed $k > 0$ and all $x > 1$, we have

$$\sum_{n\geqslant 2}(-1)^n\zeta(n)\frac{x^n}{n!} = x\ln x + (2\gamma-1)x + \tfrac{1}{2} + O_k(1/x^k).$$

Hint: consider the integral $\dfrac{1}{2\pi i}\displaystyle\int_{a-i\infty}^{a+i\infty}\frac{\pi\zeta(s)x^s}{\Gamma(s+1)\sin\pi s}\,ds$ with $a \in]0,1[$.

189. *A natural boundary for a natural Dirichlet series.*[3]

Let $P(s)$ be the Dirichlet series defined as $P(s) := \sum_{p\in\mathbb{P}} 1/p^s$ for $\sigma > 1$.

(a) Express $P(s)$ as a function of $\log\zeta(s)$ in the half-plane $\sigma > 1$.

(b) Show that $P'(s)$ may be continued in the half-plane $\sigma > 0$ to a meromorphic function whose poles are of order 1.

(c) For $\delta > 0$, $t > 0$, denote by $Q(\delta,t)$ the square defined by the inequalities $0 < \sigma < \delta$, $t < \tau < t+\delta$. Show that to each pair of positive real numbers (δ,t) one can associate a number $q_0(\delta,t) > 0$ such that, for each prime $q \geqslant q_0(\delta,t)$, there exists at least one zero $\varrho_q = \sigma_q + i\tau_q$ of $\zeta(s)$ satisfying $\frac{1}{2} \leqslant \sigma_q < 1$ and $qt < \tau_q < q(t+\delta)$.

(d) For any zero ϱ of $\zeta(s)$, let $\nu(\varrho)$ denote its order, i.e. the largest integer ν such that one may write, in a neighborhood of $s = \varrho$, $\zeta(s) = (s-\varrho)^\nu H(s)$ where H is holomorphic at $s = \varrho$. Show that, for a suitable absolute constant A, we have $1 \leqslant \nu(\varrho_q) \leqslant A\ln q(t+\delta)$. Deduce that $q_0(\delta,t)$ may be chosen such that $q \nmid \nu(\varrho_q)$.

(e) Show that ϱ_q/q is a possible pole of $P'(s)$ whose coefficient is exactly

$$c_q = \sum_{1\leqslant m\leqslant 2q-1}\frac{\mu(m)}{m}\nu\Big(\frac{m\varrho_q}{q}\Big).$$

(f) Deduce that $c_q \neq 0$, and hence that ϱ_q/q is a singularity of $P(s)$.

(g) Show that the line $\sigma = 0$ is a natural boundary for the function $P(s)$—in other words, that $P(s)$ cannot be analytically continued to a region intersecting the half-plane $\sigma < 0$—by considering its Taylor series at a point with positive abscissa.

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[3] See Landau & Walfisz (1919).

190. Define the arithmetic function $f = \mu^2 * j$, where μ denotes the Möbius function and j is the identity function.

(a) Show that f is multiplicative and satisfies $f(n) \geqslant n$ for every integer n.

(b) Put $a_m := \big|\{n \geqslant 1 : f(n) = m\}\big|$. Show that, for $\sigma > 1$, we have

$$\sum_{m\geqslant 1} \frac{a_m}{m^s} = \sum_{n\geqslant 1} \frac{1}{f(n)^s}.$$

(c) Write down, wherever it is valid, the Euler product expansion of the Dirichlet series

$$F(s;w) := \sum_{n\geqslant 1} \frac{\{n/f(n)\}^w}{n^s}$$

where w is a complex parameter.

(d) Put $F(s) := F(s;s)$. Show that, for $\sigma > 1$, we have $F(s) = \zeta(s)G(s)$, with

$$G(s) := \prod_p \big(1 + (p+1)^{-s} - p^{-s}\big).$$

(e) Show that, for each prime $p \geqslant 2$, we have

$$\big|(p+1)^{-s} - p^{-s}\big| \leqslant \min\big\{2p^{-\sigma},\, |s|p^{-\sigma-1}\big\} \qquad (\sigma \geqslant 0).$$

(f) Show that, for $\sigma \geqslant 1 - 1/\ln(3+|\tau|)$, we have

$$\sum_{p\leqslant 3+|\tau|} \frac{1}{p^\sigma} \leqslant \mathrm{e}\ln_2(3+|\tau|) + O(1).$$

(g) Deduce from (e) and (f) that $G(s)$ can be holomorphically continued to the domain $\sigma \geqslant 1 - 1/\ln(3+|\tau|)$, where it satisfies the bound

$$G(s) \ll \big(\ln(3+|\tau|)\big)^{2\mathrm{e}}.$$

(h) Set $\sigma(\tau) := 1 - 1/\ln(3+|\tau|)$. Show that

$$\int_{-\infty}^{+\infty} x^{1+\sigma(\tau)} \Big|\zeta\big(\sigma(\tau)+i\tau\big)G\big(\sigma(\tau)+i\tau\big)\Big| \frac{\mathrm{d}\tau}{\sigma(\tau)^2+\tau^2} \ll x^2 \exp\Big\{-\tfrac{1}{2}\sqrt{\ln x}\Big\}.$$

Hint: for $T := \exp\big\{\sqrt{\ln x}\big\}$, estimate separately the contributions from the domains $|\tau| \leqslant T$ and $|\tau| > T$.

(i) Define $A(t) := \sum_{n\leqslant t} a_n$. Show that

$$\int_1^x A(t)\,\mathrm{d}t = \tfrac{1}{2}G(1)x^2 + O\Big(x^2\mathrm{e}^{-\frac{1}{2}\sqrt{\ln x}}\Big).$$

Using the fact that $A(t)$ is non-decreasing, deduce that

$$A(x) = G(1)x + O\Big(x\mathrm{e}^{-\frac{1}{4}\sqrt{\ln x}}\Big).$$

191. Let f be the arithmetic function defined by

$$f(n) = \sum_{d^3|n} \mu(d)^2 d^2.$$

Denote by $F(s) := \sum_{n\geqslant 1} f(n)/n^s$ the associated Dirichlet series and by $S(x) := \sum_{n\leqslant x} f(n)$ the corresponding summatory function.

(a) From the result obtained in Exercise 44, deduce an asymptotic formula for $S(x)$.

(b) Show that f is multiplicative. One may either argue directly or by employing a convolution argument, observing that the characteristic function of the cubes is multiplicative.

(c) Calculate $f(p^\nu)$ for any prime p and any integer $\nu \geqslant 1$. Find, in its convergence domain, the sum of the series $\sum_{\nu\geqslant 0} f(p^\nu)/p^{\nu s}$, and express it in the form of a rational function of p^{-s}.

(d) Show that $F(s)$ converges absolutely for $\sigma > 1$.

(e) Using the expansions of $\zeta(s)$ and $F(s)$ as Euler products, show that $F(s)$ can be given a simple expression in terms of $\zeta(s)$, $\zeta(3s-2)$ and $\zeta(6s-4)$.

(f) Show that $F(s)$ has a pole at $s = 1$, and determine its order. Find the polar part of the Laurent expansion of $F(s)$ in the neighborhood of $s = 1$. [Use the result of Exercise 174.] Express the coefficients using γ, π and $\kappa := -\zeta'(2)/\zeta(2)$.

(g) Determine the abscissa of convergence of $F(s)$.

(h) Calculate the order of the pole of $\zeta'(s)$ at $s = 1$ and its corresponding Laurent expansion.

(i) Show that $G(\sigma) := \sum_{n\geqslant 1}(-1)^n n^{-\sigma}$ is non-zero for $\sigma > 0$. Deduce that $\zeta(s)$ has no zero on the real half-line $]0, \infty[$.

(j) Find real numbers a and b such that the Dirichlet series

$$H(s) := F(s) + a\zeta'(s) - b\zeta(s)$$

has a holomorphic continuation at all points of the real half-line $]\frac{1}{3}, \infty[$.

(k) Assuming that $F(s)$ has a pole on the line $\sigma = \frac{3}{4}$, give an oscillation result for the quantity

$$R(x) := S(x) - ax \ln x + (a - b)x.$$

What is the meaning of this hypothesis for the zeros of the zeta function?

192. (a) Show that each natural number $n \geqslant 1$ can be written uniquely in the form $n = m\ell^2$ with $\mu(m)^2 = 1$. Then set $f(n) := \ell$. Is this function multiplicative?

(b) Show that, for any prime p, the power series $\sum_{\nu\geqslant 0} f(p^\nu)z^\nu$ converges in the open disk $|z| < 1/\sqrt{p}$. Calculate its sum. Deduce that the series $F(s) := \sum_{n\geqslant 1} f(n)/n^s$ is absolutely convergent for $\sigma > 1$, and can be simply expressed as a function of $\zeta(s)$, $\zeta(2s)$ and $\zeta(2s-1)$.

(c) Show that $F(s)$ has a pole at $s = 1$. What is the abscissa of convergence of $F(s)$?

(d) Define $Q(x) := \sum_{n\leqslant x} \mu(n)^2$ and $S(x) := \sum_{n\leqslant x} f(n)$. Establish the formula $S(x) = \sum_{\ell\leqslant\sqrt{x}} \ell Q(x/\ell^2)$ $(x \geqslant 1)$ and find an asymptotic formula for $S(x)$. Recover in this way the result obtained in (c).

(e) Determine the order and the principal part of the pole of $F(s)$ at $s = 1$.

(f) Determine the order and the principal part of the pole of $\zeta'(s)$ at $s = 1$.

(g) Find real numbers a and b—to be written in terms of the Euler constant γ and the number $\delta := \zeta'(2)$—such that the series

$$F(s) - a\zeta(s) + b\zeta'(s)$$

has a holomorphic continuation at all points of the half-line $\left[\frac{1}{4}, +\infty\right[$. Assuming $F(s)$ has at least one non-real pole on the line $\sigma = \frac{1}{4}$, deduce an oscillation theorem for the quantity $S(x) - ax - b(x\ln x - x)$.

193. *An entire relative of the zeta function.* Let $\vartheta \in]0,1[$ and $r := \|\vartheta\|$. Consider the Dirichlet series

$$F_\vartheta(s) := \sum_{n\geqslant 1} \frac{\mathrm{e}(n\vartheta)}{n^s}.$$

Put $I_\vartheta(s) := \int_{\mathcal{H}} z^{s-1}\,\mathrm{d}z/\{\mathrm{e}^{z-2\pi i\vartheta}-1\}$ where $\mathcal{H}$ is the Hankel contour consisting of the real half-line $[r, +\infty[$ covered from right to left with argument $0+$, the circle $|z| = r$ excluding the point $z = r$ traced in the positive sense, and the half-line $[r, +\infty[$ covered from left to right, with argument $2\pi-$.

(a) Show that $I_\vartheta(s)$ is an entire function of s.

(b) Show that $I_\vartheta(1) = 0$.

(c) Show that, for $\sigma = \Re\mathrm{e}\, s > 1$, we have $F_\vartheta(s)(\mathrm{e}^{2\pi i s} - 1)\Gamma(s) = I_\vartheta(s)$.

(d) Deduce that $F_\vartheta(s)$ may be holomorphically continued to an entire function.

The prime number theorem and the Riemann hypothesis

4.1. The prime number theorem

The information gained in the previous chapter about the function $\zeta(s)$ allows us easily to obtain the prime number theorem in its classical form.

Theorem 4.1. *There exists a strictly positive constant c such that we have, as x tends to infinity,*

$$\psi(x) = x + O\Big(x\mathrm{e}^{-c\sqrt{\ln x}}\Big), \tag{4.1}$$

$$\pi(x) = \mathrm{li}(x) + O\Big(x\mathrm{e}^{-c\sqrt{\ln x}}\Big). \tag{4.2}$$

Proof. The second formula readily follows from the first by partial summation. Let us prove (4.1). Since $\Lambda(n) \leqslant \ln n$ for all n in $\mathbb{N}^*$ and

$$|\zeta'(1+\sigma)/\zeta(1+\sigma)| \ll 1/\sigma \qquad (\sigma > 0),$$

the second effective Perron formula[1] allows us to write, for $x \geqslant 2$, $T \geqslant 2$,

$$\psi(x) = -\frac{1}{2\pi i}\int_{\kappa-iT}^{\kappa+iT} \frac{\zeta'(s)}{\zeta(s)}\frac{x^s}{s}\,\mathrm{d}s + O\left(\ln x\Big(1+\frac{x\ln T}{T}\Big)\right) \tag{4.3}$$

with $\kappa := 1 + 1/\ln x$. By Theorem 3.21, there exists some positive constant c_0 such that the point $s = 1$ is the only singularity of the integrand in the

[1] See § 2.1.

rectangle $|\tau| \leqslant T$, $1 - c_0/\ln T \leqslant \sigma \leqslant \kappa$. Since the residue at $s = 1$ equals x, we can write

$$-\frac{1}{2\pi i}\int_{\kappa-iT}^{\kappa+iT} \frac{\zeta'(s)}{\zeta(s)}\frac{x^s}{s}\,\mathrm{d}s = x - \frac{1}{2\pi i}\int_{\mathcal{G}} \frac{\zeta'(s)}{\zeta(s)}\frac{x^s}{s}\,\mathrm{d}s \tag{4.4}$$

where $\mathcal{G}$ is the polygonal line joining the points $\kappa - iT$, $1 - c_0/\ln T - iT$, $1 - c_0/\ln T + iT$, $\kappa + iT$. The upper bound from Theorem 3.22, namely

$$\frac{\zeta'(s)}{\zeta(s)} \ll \ln T \qquad (s \in \mathcal{G}),$$

then implies that the contribution to the integral of the horizontal segments of $\mathcal{G}$ is $\ll x(\ln T)/T$ whereas that of the vertical segment is

$$\ll x \exp\Big\{-c_0\frac{\ln x}{\ln T}\Big\}(\ln T)^2.$$

Choosing $T := \exp\sqrt{c_0 \ln x}$, we see that the integral over $\mathcal{G}$ is of the order of the remainder term of (4.1) for any constant $c < \sqrt{c_0}$. Substituting (4.4) into (4.3) then implies the desired result. □

4.2. Minimal hypotheses

It is conceptually interesting to investigate what are, in the theory of the zeta function, the minimal pieces of information which lead to a proof of the prime number theorem.

The fundamental (but easy) inequality

$$\zeta(\sigma)^3|\zeta(\sigma + i\tau)|^4\,|\zeta(\sigma + 2i\tau)| \geqslant 1 \qquad (\sigma > 1), \tag{4.5}$$

essentially due to Mertens (§ 3.5), led us to show that $\zeta(s)$ is non-zero for $\sigma = 1$. Actually, this is almost sufficient on its own to yield the prime number theorem. Indeed, using the elementary upper bounds (cf. § 3.4)

$$\zeta^{(k)}(s) \ll (\ln|\tau|)^{k+1} \qquad (|\tau| \geqslant 2,\ \sigma \geqslant 1 - c/\ln|\tau|) \tag{4.6}$$

for $k = 0$ and 1, we can employ (4.5) to determine effectively a zero-free region of $\zeta(s)$ which intersects the half-plane $\sigma < 1$.

Let $s = \sigma + i\tau$, $0 < \eta < c/\ln|\tau|$, $s_0 = 1 + \eta + i\tau$; then, for $\sigma \geqslant 1 - \eta$, we have

$$|\zeta(s) - \zeta(s_0)| = \Big|\int_{s_0}^{s} \zeta'(w)\,\mathrm{d}w\Big| \leqslant C_0\eta(\ln|\tau|)^2.$$

Now (4.5) and (4.6) imply

$$|\zeta(s_0)|^4 \geqslant C_1\eta^3(\ln|\tau|)^{-1},$$

from which it follows that

$$|\zeta(s)| \geqslant C_1^{1/4}\eta^{3/4}(\ln|\tau|)^{-1/4} - C_0\eta(\ln|\tau|)^2.$$

Selecting η optimally, we obtain

$$|\zeta(s)| \gg 1/(\ln|\tau|)^7 \qquad \big(|\tau| \geqslant 2,\ \sigma \geqslant 1 - C_2/(\ln|\tau|)^9\big), \tag{4.7}$$

where C_2 is some suitable positive constant.

We leave to the reader the easy task of checking that the proof of Theorem 4.1 given in the previous section can be adapted to these weakened assumptions. The sole modification needed resides in the choice of the parameter T, for which the optimal value becomes

$$T := \exp\{c_3(\ln x)^{1/10}\}.$$

We thus obtain

$$\psi(x) = x + O\Big(x\exp\{-c(\ln x)^{1/10}\}\Big).$$

It is remarkable that *one can equally well arrive at the prime number theorem* (but in its weakest form, i.e. $\psi(x) \sim x$) *using only the properties of* $\zeta(s)$ *in the half-plane* $\sigma \geqslant 1$.

Indeed, let us apply Perron's formula (2.10) to the Dirichlet series

$$F(s) = -\frac{\zeta'(s)}{\zeta(s)} - \zeta(s) = \sum_{n\geqslant 1} \frac{\Lambda(n)-1}{n^s}.$$

We obtain for each $\kappa > 1$

$$\int_0^x \big(\psi(t) - \lfloor t\rfloor\big)\,\mathrm{d}t = \frac{x^{\kappa+1}}{2\pi}\int_{-\infty}^{+\infty} \frac{F(\kappa+i\tau)x^{i\tau}}{(\kappa+i\tau)(\kappa+1+i\tau)}\,\mathrm{d}\tau. \tag{4.8}$$

Estimates (4.6) and (4.7) show that

$$F(\kappa+i\tau) \ll (\ln(2+|\tau|))^9 \qquad (\kappa \geqslant 1). \tag{4.9}$$

Hence the theorem of dominated convergence applies to the integral occurring on the right-hand side of (4.8). It follows that

$$\int_0^x \big(\psi(t) - \lfloor t\rfloor\big)\,\mathrm{d}t = x^2 J(x)$$

with

$$J(x) := \frac{1}{2\pi}\int_{-\infty}^{+\infty} \frac{F(1+i\tau)x^{i\tau}}{(1+i\tau)(2+i\tau)}\,\mathrm{d}\tau.$$

Thus $J(x)$ is the Fourier transform at the point $(-\ln x)$ of an integrable function. By the Riemann–Lebesgue lemma, we then deduce that $J(x) = o(1)$ $(x \to \infty)$, whence

$$\int_0^x \psi(t)\,\mathrm{d}t = \tfrac{1}{2}x^2 + x^2\varepsilon(x)$$

with $\varepsilon(x) = J(x) + O(1/x) = o(1)$.

Let us define $\eta(x) := \max_{x/2\leqslant y\leqslant 3x/2} |\varepsilon(y)|$. For all h, $0 < h \leqslant \frac{1}{2}x$, we can write

$$x - \frac{h}{2} - \frac{2}{h}\eta(x)x^2 \leqslant \frac{1}{h}\int_{x-h}^{x} \psi(t)\,\mathrm{d}t \leqslant \psi(x)$$
$$\leqslant \frac{1}{h}\int_{x}^{x+h} \psi(t)\,\mathrm{d}t \leqslant x + \frac{h}{2} + \frac{4}{h}x^2\eta(x).$$

Choosing $h := x\sqrt{\eta(x)}$, we obtain

$$\psi(x) = x\{1 + O(\sqrt{\eta(x)})\} = x\{1 + o(1)\}. \tag{4.10}$$

Thus the prime number theorem follows easily from the fact that $\zeta(s)$ does not vanish for $\sigma = 1$ and from any "reasonable" upper bound of $F(s)$ for $\sigma \geqslant 1$. Indeed, it would be possible to replace, without additional difficulty, the estimate (4.9) in the above proof by

$$F(s) \ll 1 + |\tau|^{1-\delta} \qquad (\sigma \geqslant 1)$$

We shall see in Chapter II.7 that the *Wiener–Ikehara Tauberian theorem* (generalized in the form given by Theorem 7.13) allows us to prove (4.10) assuming only that $\zeta(1+i\tau) \neq 0$ without needing *any* upper bound on $F(s)$.

The absence of zeros of $\zeta(s)$ on the line $\sigma = 1$ thus constitutes a *minimal* hypothesis for the prime number theorem, of which it is also an easy implication.

Indeed, let us assume that $\psi(x) \sim x$ $(x \to \infty)$ and that $s_0 = 1 + i\tau$ is a zero of order $m \geqslant 1$ of $\zeta(s)$. In these circumstances, we have

$$\lim_{\sigma\to 1+} (\sigma - 1)\frac{\zeta'(\sigma + i\tau)}{\zeta(\sigma + i\tau)} = m \tag{4.11}$$

whereas the following formula, proved by partial integration,

$$-\frac{\zeta'(s)}{\zeta(s)} = \frac{s}{s-1} + s\int_1^\infty \frac{\psi(t) - t}{t^{s+1}}\,\mathrm{d}t$$

implies that

$$(\sigma - 1)\left|\frac{\zeta'(\sigma + i\tau)}{\zeta(\sigma + i\tau)}\right| \leqslant \left|\frac{\sigma + i\tau}{\tau}\right|(\sigma - 1) + |\sigma + i\tau|(\sigma - 1)\int_1^\infty \frac{o(t)}{t^{\sigma+1}}\,\mathrm{d}t$$
$$= o(1) \qquad (\sigma \to 1+).$$

This contradicts (4.11).

4.3. The Riemann hypothesis

Riemann conjectured in 1859 that all non-trivial zeros of $\zeta(s)$ lie on the line $\sigma = \frac{1}{2}$. Having stated that the number of *complex* roots of the equation

$$\zeta(\tfrac{1}{2} + i\tau) = 0 \qquad (0 < \Re e\,\tau \leqslant T) \tag{4.12}$$

is "*approximately equal to* $(T/2\pi)\ln(T/2\pi) - (T/2\pi)$" (cf. Theorem 3.19), he writes: "*indeed, one encounters, between these limits, a number of real roots approximately equal to this, and it is very probable that all the roots are real*". However, a note has been found in Riemann's personal papers, specifying that he has "*not yet completed the proof*" of the first point—cf. Riemann (1859) pp. 168–169 and 175.

Neither of Riemann's two claims has yet been proved or disproved. In 1914, Hardy established that $\zeta(s)$ *has infinitely many zeros on the critical line* $\sigma = \frac{1}{2}$; and, in 1942, Selberg showed that a positive proportion of the zeros is definitely located on the critical line, in other words that, for some $c > 0$,

$$N_0(T) := \operatorname{card}\{\tau : 0 \leqslant \tau \leqslant T,\ \zeta\left(\tfrac{1}{2} + i\tau\right) = 0\} \geqslant cN(T) \quad (T \to \infty). \tag{4.13}$$

Today explicit values of c are available, following the work of Levinson (1974) who showed that for T sufficiently large one can take $c = 0.342$.

The statement that all the roots of (4.12) are real—in other words that $N_0(T) = N(T)$—is known under the name of the *Riemann hypothesis*. It is one of the most famous conjectures in mathematics. It has profound implications throughout analytic number theory. In this section we shall develop two of them.

Theorem 4.2. *The Riemann hypothesis implies that of Lindelöf. More precisely, if all non-trivial zeros of* $\zeta(s)$ *have real parts equal to* $\frac{1}{2}$*, then, for any* $\varepsilon > 0$*, we have*

$$\log\zeta(s) \ll_\varepsilon (\ln|\tau|)^{2-2\sigma+\varepsilon} \qquad (\tfrac{1}{2} < \sigma \leqslant 1,\ |\tau| \geqslant 2). \tag{4.14}$$

The proof uses the following classical result.

Lemma 4.3 (Hadamard's three circles lemma)**.** *Let* $F(s)$ *be a holomorphic function in the annulus* $R_1 \leqslant |s| \leqslant R_2$*. Then the function*

$$M(r) := \max_{|s|=r} |F(s)| \qquad (R_1 \leqslant r \leqslant R_2)$$

is a log-convex function of $\ln r$ *in the interval* $R_1 \leqslant r \leqslant R_2$*.*

Proof of lemma. For $R_1 < r < R_2$, the maximum modulus principle applied to $s^m F(s)^n$ immediately implies that,

$$r^m M(r)^n \leqslant \frac{R_1^m M(R_1)^n}{r - R_1} + \frac{R_2^m M(R_2)^n}{R_2 - r}.$$

Let α be such that $R_1^\alpha M(R_1) = R_2^\alpha M(R_2)$. In the above inequality, let m and n tend to infinity in such a way that $m/n \to \alpha$. We then obtain

$$r^\alpha M(r) \leqslant R_1^\alpha M(R_1),$$

so that, on replacing α by its actual value,

$$\ln M(r) \leqslant \frac{\ln(R_2/r)}{\ln(R_2/R_1)} \ln M(R_1) + \frac{\ln(r/R_1)}{\ln(R_2/R_1)} \ln M(R_2). \tag{4.15}$$

This establishes the desired property. □

Proof of Theorem 4.2. Observe first that the Riemann hypothesis implies that, for sufficiently large τ, the function

$$F(w) := \log\left(\frac{\zeta(s_0 + w)}{\zeta(s_0)}\right) \qquad (s_0 := 2 + i\tau)$$

is holomorphic in the disk $|w| < \frac{3}{2}$. Since it follows from Theorem 3.9 (for example) that, for $0 \leqslant \delta \leqslant 1$,

$$\Re e\, F(w) \leqslant O(\ln \tau) \qquad (|w| \leqslant \tfrac{3}{2} - \delta),$$

we readily deduce from the Borel-Carathéodory lemma (Theorem 3.16) that we have

$$\log \zeta(s) \ll_\sigma \ln \tau \qquad (\sigma > \tfrac{1}{2},\ \tau \geqslant 2). \tag{4.16}$$

Then, let $s := \sigma + i\tau$ with $\sigma > \frac{1}{2}$, $\tau \geqslant \tau_0$. We consider two parameters σ_1 and ε such that $1 < \sigma_1 \leqslant \tau$, $0 < \varepsilon < \min\{\sigma_1 - 1, \sigma - \frac{1}{2}\}$. Applying the three circles lemma to $\log \zeta(\sigma_1 + i\tau + w)$ for

$$R_1 := \sigma_1 - 1 - \varepsilon, \quad r := \sigma_1 - \sigma, \quad R_2 := \sigma_1 - \tfrac{1}{2} - \varepsilon$$

(so that, on these three circles, the point $\sigma_1 + i\tau + w$ passes respectively through $1 + \varepsilon + i\tau$, s and $\frac{1}{2} + \varepsilon + i\tau$), we obtain

$$|\log \zeta(s)| \leqslant M_1^{1-\alpha} M_2^{\alpha}$$

with

$$\alpha := \frac{\ln(r/R_1)}{\ln(R_2/R_1)} = \frac{\ln\left(\dfrac{\sigma_1 - \sigma}{\sigma_1 - 1 - \varepsilon}\right)}{\ln\left(\dfrac{\sigma_1 - \frac{1}{2} - \varepsilon}{\sigma_1 - 1 - \varepsilon}\right)} = 2(1 - \sigma + \varepsilon) + O\Big(\frac{1}{\sigma_1 - 1 + \varepsilon}\Big)$$

and

$$M_1 \leqslant \sup_{\sigma \geqslant 1+\varepsilon} |\log \zeta(\sigma + i\tau)| \ll_\varepsilon 1, \qquad M_2 \ll_\varepsilon \ln \tau,$$

where the last estimate follows from (4.16). Selecting, for instance,

$$\sigma_1 = 1 + \ln \tau,$$

we obtain the stated result. □

Theorem 4.4. *Let Θ be the infimum of those real numbers ξ such that*

$$\psi(x) - x \ll x^{\xi} \qquad (x \geqslant 2).$$

Then we have

$$\Theta = \sup_{\varrho} \beta \tag{4.17}$$

where the supremum is taken over non-trivial zeros $\varrho = \beta + i\gamma$ of $\zeta(s)$.

Corollary 4.5. *The Riemann hypothesis is equivalent to*

$$(\forall \varepsilon > 0) \qquad \psi(x) = x + O_{\varepsilon}(x^{1/2+\varepsilon}).$$

Proof of Theorem. Write $R(x) := \psi(x) - x$. For $\sigma > 1$, we have

$$-\frac{\zeta'(s)}{\zeta(s)} = \int_{1-}^{\infty} \frac{\mathrm{d}\psi(t)}{t^s} = \frac{1}{s-1} + \int_{1-}^{\infty} \frac{\mathrm{d}R(t)}{t^s} = \frac{1}{s-1} + 1 + s\int_{1}^{\infty} \frac{R(t)}{t^{s+1}}\,\mathrm{d}t.$$

By definition of Θ, we have $R(t) \ll_{\varepsilon} t^{\Theta+\varepsilon}$, so the formula above defines an analytic continuation of $\zeta'(s)/\zeta(s)$ to a meromorphic function for $\sigma > \Theta$ with $s = 1$ as its sole singularity. In particular, this implies that $\zeta(s)$ does not vanish for $\sigma > \Theta$.

Conversely, if $\zeta(s)$ does not vanish for $\sigma > \Theta$, the upper bound

$$\ln|\zeta(s)| \leqslant O(\ln|\tau|) \qquad (\sigma > \tfrac{1}{2},\ |\tau| \geqslant 2)$$

and the Borel-Carathéodory theorem imply[2]

$$\log\zeta(s) \ll_{\sigma} \ln|\tau| \qquad (\sigma > \Theta,\ |\tau| \geqslant 2). \tag{4.18}$$

Cauchy's formula applied with a circle of radius $\delta < \sigma - \Theta$ now yields

$$\frac{\zeta'(s)}{\zeta(s)} \ll_{\sigma} \ln|\tau| \qquad (\sigma > \Theta,\ |\tau| \geqslant 2). \tag{4.19}$$

By the effective Perron formula (4.3) and the residue theorem in the form (4.4)—choosing now $\mathcal{G}$ to be the polygonal line joining $\kappa - iT$, $\Theta + \delta \pm iT$, $\kappa + iT$—, we arrive at

$$\psi(x) - x \ll \frac{x\ln T}{T} + x^{\Theta+\delta}(\ln T)^2 + (\ln x)\left(1 + \frac{x\ln T}{T}\right).$$

Selecting $T := x$, this upper bound is $O\big(x^{\Theta+2\delta}\big)$. Since δ is arbitrarily small, this yields the desired conclusion. □

[2]We omit the details, the argument being identical with that leading to the estimate (4.16).

4.4. Explicit formula for $\psi(x)$

Mentioned by Riemann and established by von Mangoldt in 1895, the *explicit formula* for $\psi(x)$ provides a complete description of the distribution of primes in terms of the zeros of the zeta function.

We define

$$\langle\langle x\rangle\rangle := \min_{p\in\mathbb{P},\nu\geqslant 1,\, p^\nu\neq x} |x-p^\nu|.$$

Theorem 4.6. *For $x \geqslant 2$, we have*

$$\psi_0(x) := \tfrac{1}{2}\{\psi(x)+\psi(x-)\} = x - {\sum_{\varrho}}' \frac{x^\varrho}{\varrho} - \ln(2\pi) - \tfrac{1}{2}\ln\Big(1-\frac{1}{x^2}\Big), \tag{4.20}$$

where the apostrophe indicates that, in the sum over non-trivial zeros ϱ of $\zeta(s)$, the zeros ϱ and $\overline{\varrho}$ should be taken into account simultaneously.

Furthermore, for $x \geqslant 2$, $T \geqslant 2$, we have

$$\psi_0(x) = x - \sum_{|\gamma|\leqslant T} \frac{x^\varrho}{\varrho} - \ln(2\pi) + \tfrac{1}{2}\ln\Big(\frac{1}{1-1/x^2}\Big) + R(x,T), \tag{4.21}$$

with

$$R(x,T) \ll \frac{x\ln^2(xT)}{T} + \frac{x\ln x}{x+T\langle\langle x\rangle\rangle}. \tag{4.22}$$

Remarks. (i) For any fixed $x > 2$, we have $\lim_{T\to\infty} R(x,T) = 0$. Moreover, if x is a half-integer, then $\langle\langle x\rangle\rangle \geqslant \frac{1}{2}$, hence

$$R(x,T) \ll \frac{x(\ln xT)^2}{T}. \tag{4.23}$$

(ii) Recall that

$$\zeta'(0)/\zeta(0) = \ln(2\pi). \tag{4.24}$$

We prove Theorem 4.6 in several stages. It is enough to establish (4.21). The natural starting point is the effective Perron formula (Theorem 2.3). Extending the function Λ to be 0 on $\mathbb{R}\setminus\mathbb{N}^*$, and appealing to the inequality obtained in Lemma 2.2(ii) to handle the contribution to the integral of the term with rank x when $\Lambda(x)\neq 0$, we obtain, for $\kappa > 1$, $T > 0$, $x > 0$,

$$|\psi_0(x) - J_\kappa(x,T)| < 2\sum_{\substack{n\geqslant 1\\ n\neq x}} \frac{\Lambda(n)(x/n)^\kappa}{1+T|\ln(x/n)|} + \frac{\kappa\Lambda(x)}{T}, \tag{4.25}$$

with

$$J_\kappa(x,T) := \frac{1}{2i\pi}\int_{\kappa-iT}^{\kappa+iT} \frac{-\zeta'(s)x^s}{s\zeta(s)}\,\mathrm{d}s.$$

The first step consists in verifying that the right-hand side of (4.25) is indeed compatible with (4.21).

Lemma 4.7. *For $x \geqslant 2$, $\kappa := 1 + 1/\ln x$, $T > 0$, we have*

$$|\psi_0(x) - J_\kappa(x,T)| \ll \frac{x(\ln x)^2}{T} + \frac{x \ln x}{x + T\langle\langle x\rangle\rangle}.$$

Proof. The contribution to the right-hand side of (4.25) of those integers n such that $n \leqslant 3x/4$ or $n \geqslant 5x/4$ is

$$\ll \frac{x^\kappa}{T} \sum_{n \geqslant 1} \frac{\Lambda(n)}{n^\kappa} = \frac{-x^\kappa \zeta'(\kappa)}{T\zeta(\kappa)} \ll \frac{x \ln x}{T}.$$

Set $N_1 := \max_{p^\nu < x} p^\nu$. The contribution, say S_1, of the integers n such that $3x/4 < n < x$ vanishes if $N_1 \leqslant 3x/4$. Otherwise, we observe that

$$\ln \frac{x}{n} \geqslant \ln \frac{N_1}{n} = \ln\left(\frac{1}{1 - v/N_1}\right) \geqslant \frac{v}{N_1} \qquad (n = N_1 - v,\ 1 \leqslant v \leqslant x/4),$$

and

$$\ln \frac{x}{N_1} \geqslant \frac{x - N_1}{x} \geqslant \frac{\langle\langle x\rangle\rangle}{x}.$$

It follows that

$$S_1 \ll \sum_{1 \leqslant v < x/4} \frac{\Lambda(N_1 - v)N_1}{Tv} + \frac{x\Lambda(N_1)}{x + T\langle\langle x\rangle\rangle} \ll \frac{x(\ln x)^2}{T} + \frac{x \ln x}{x + T\langle\langle x\rangle\rangle}.$$

The contribution of those integers n such that $x < n < 5x/4$ is handled similarly, by introducing $N_2 := \min_{p^\nu > x} p^\nu$. We omit the details. □

It remains to evaluate $J_\kappa(x,T)$. To this end, we start by noting that, at the cost of shifting T by a value not exceeding 1, we can assume that

$$\inf_\varrho |\gamma - T| \gg 1/\ln T, \tag{4.26}$$

where the implicit constant is absolute: this results directly from the bound

$$N(T+1) - N(T) \ll \ln T \tag{4.27}$$

obtained in Theorem 3.18.

Under the assumption (4.26), let us consider a sufficiently large odd integer U and apply the residue theorem for the rectangle $\mathcal{R}(T,U)$ with vertices $\kappa \pm iT$, $-U \pm iT$, whose left boundary hence avoids the trivial zeros of $\zeta(s)$. Taking (4.24) into account, we obtain that

$$\frac{-1}{2i\pi} \int_{\mathcal{R}(T,U)} \frac{\zeta'(s)x^s}{\zeta(s)s}\,\mathrm{d}s = x - \sum_{|\gamma| < T} \frac{x^\varrho}{\varrho} - \ln(2\pi) + \sum_{0 < 2m < U} \frac{1}{2mx^{2m}}.$$

The integral $J_\kappa(x,T)$ coincides with the contribution to the integral of the right vertical side of $\mathcal{R}(T,U)$. Handling of the three complementary contributions requires bounding $\zeta'(s)/\zeta(s)$.

Lemma 4.8. (i) *Under the assumption* (4.26), *we have*

$$\frac{\zeta'(s)}{\zeta(s)} \ll (\ln T)^2 \qquad (-1 \leqslant \sigma \leqslant 2)$$

(ii) *Under the assumption* $\min_{m\in\mathbb{N}} |s+2m| \geqslant \frac{1}{2}$, *we have*

$$\left|\zeta'(s)/\zeta(s)\right| \ll \ln(2\,|s|) \qquad (\sigma \leqslant -1).$$

Proof. (i) The functional equation allows us to restrict to the case $\sigma \geqslant \frac{1}{2}$. By (3.46), we have

$$\frac{\zeta'(s)}{\zeta(s)} = \sum_{|\gamma-T|\leqslant 3/2} \frac{1}{s-\varrho} + O(\ln T).$$

However, each term in the sum is $\ll \ln T$ and the number of terms is $\ll \ln T$: this is sufficient to imply the stated estimate.

(ii) Logarithmic differentiation of the functional equation (3.5) yields

$$\frac{\zeta'(s)}{\zeta(s)} = \ln 2\pi + \frac{\pi}{2}\cot\Big(\frac{\pi s}{2}\Big) - \frac{\Gamma'(1-s)}{\Gamma(1-s)} - \frac{\zeta'(1-s)}{\zeta(1-s)}.$$

If $\min_{m\in\mathbb{N}} |s+2m| \geqslant \frac{1}{2}$, the term $\cot(\pi s/2)$ is bounded. The term involving Γ is $\ll \ln(2\,|s|)$ for $\sigma \leqslant -1$, using (0.5) or the complex Stirling formula (0.11). Finally, the last term is bounded since $\Re e\,(1-s) \geqslant 2$. □

We are now able to complete the proof of Theorem 4.6. Denote by $R^* := R^*(x,T)$ the left-hand side of (4.22) and by $\mathcal{L}$ the polygonal line

$$[\kappa - iT, -U - iT, -U + iT, \kappa + iT];$$

it results from the above that, under the assumption (4.26), we have

$$\psi_0(x) = x - \sum_{|\gamma|\leqslant T} \frac{x^\varrho}{\varrho} - \ln(2\pi) + \sum_{0<m<U/2} \frac{x^{2m}}{2m} + O(R^* + |E|),$$

with

$$E := \int_{\mathcal{L}} \frac{\zeta'(s)x^s}{s\zeta(s)}\,\mathrm{d}s.$$

By Lemma 4.8, we then have, provided, for example, $U > T^2$,

$$E \ll \int_{-U}^{-1} \frac{x^\sigma \ln(T+|\sigma|)}{T+|\sigma|}\,\mathrm{d}\sigma + \int_{-1}^{\kappa} \frac{x^\sigma (\ln T)^2}{T}\,\mathrm{d}\sigma + \int_{-T}^{T} \frac{\ln U}{Ux^U}\,\mathrm{d}\tau \ll \frac{x(\ln T)^2}{T\ln x}.$$

We may therefore let U tend to infinity, noting that

$$\sum_{m\geqslant 1} \frac{1}{2mx^{2m}} = \tfrac{1}{2}\ln\Big(\frac{1}{1-1/x^2}\Big).$$

This gives the stated result under condition (4.26). However, this limitation can be relaxed. Indeed, if T varies by a bounded amount, the sum over ϱ

is changed by at most $O(\ln T)$ terms, which are all $\ll x/T$. The whole sum thus changes by at most $\ll x(\ln T)/T$. This last quantity is absorbed by the error term.

We may deduce from the explicit formula an improvement upon Theorem 4.4.

Theorem 4.9. *Let* $\Theta := \sup_\varrho \beta$. *We have*

$$\psi(x) = x + O\big(x^\Theta (\ln x)^2\big), \quad \pi(x) = \operatorname{li} x + O(x^\Theta \ln x).$$

Proof. The estimate

$$\sum_{|\gamma| \leqslant T} \frac{1}{|\varrho|} \ll (\log T)^2$$

follows immediately from (4.27). We thus obtain the formula concerning $\psi(x)$ by taking $T = x$ in (4.22). The estimate involving $\pi(x)$ results from this by partial summation. □

Notes

§ 4.1. The best error term known for the prime number theorem is

$$O\big(x \exp\{-c(\ln x)^{3/5}(\ln_2 x)^{-1/5}\}\big).$$

It stems from the zero-free region of Korobov–Vinogradov—cf. Chapter II.3, Notes. For a proof of this result see Ellison & Mendès France (1975) or Ivić (1985). To date, elementary methods have only been able to yield a weaker bound, namely

$$|\psi(x) - x| \ll x \exp\big\{-\tfrac{1}{40}(\ln x)^{1/6}\big\}$$

due to Srinivasan & Sampath (1988). On this topic, one may consult the remarkable survey of Diamond (1982).

§ 4.2. The proof given here of the prime number theorem, using only the properties of $\zeta(s)$ for $\sigma \geqslant 1$, essentially follows that of Widder (1971, chap. 4).

§ 4.3. Titchmarsh (1951) gives several proofs of Hardy's theorem and also establishes that of Selberg. Here we limit ourselves to briefly indicating how to prove that

$$N_0(T) \gg \ln T \qquad (T \to \infty).$$

The basic idea simply consists of making use of the fact that, if $Z(\tau)$ is a real continuous function and if

$$\int_T^{2T} Z(\tau)\,\mathrm{d}\tau = o\Big(\int_T^{2T} |Z(\tau)|\,\mathrm{d}\tau\Big) \qquad (T \to \infty), \tag{4.28}$$

then $Z(\tau)$ certainly vanishes, for sufficiently large T, on the interval $[T, 2T]$.

Titchmarsh's choice is

$$Z(\tau) := \chi\big(\tfrac{1}{2} + i\tau\big)^{-1/2}\zeta\big(\tfrac{1}{2} + i\tau\big)$$

where $\chi(s)$ denotes the "elementary" factor of the functional equation for $\zeta(s)$—see formula (3.10). The equation $\zeta(s) = \chi(s)\zeta(1-s)$ shows immediately that

$$\chi(s)\chi(1-s) = 1$$

(which can also be seen by noting that $\chi(s) = \pi^{s-\frac{1}{2}}\Gamma\big(\frac{1}{2}(1-s)\big)/\Gamma\big(\frac{1}{2}s\big)$), and hence that $\chi\big(\frac{1}{2} + i\tau\big)$ has modulus 1. Therefore

$$Z(\tau) = \chi\big(\tfrac{1}{2} + i\tau\big)^{-1/2}\zeta\big(\tfrac{1}{2} - i\tau\big)\chi\big(\tfrac{1}{2} + i\tau\big) = \overline{Z(\tau)},$$

that is, the function $Z(\tau)$ is real when τ is real.

To estimate the left-hand side of (4.28), we use the residue theorem in the form

$$\int_{\mathcal{R}} \chi(s)^{-1/2}\zeta(s)\,\mathrm{d}s = 0$$

where $\mathcal{R}$ is the rectangle with vertices $\frac{1}{2}+iT$, $\frac{1}{2}+2iT$, $\frac{5}{4}+2iT$, $\frac{5}{4}+iT$. The integral on the left-hand side of $\mathcal{R}$ has the value

$$i\int_T^{2T} Z(\tau)\,\mathrm{d}\tau$$

and we bound this quantity by estimating the integrals on the other three sides.

The upper bounds of § 3.4

$$\chi(s)^{-1} \ll \tau^{\sigma-1/2}, \qquad \zeta(s) \ll_\varepsilon \tau^{(1-\sigma)/2+\varepsilon} \qquad (0 \leqslant \sigma \leqslant 1)$$

immediately imply that

$$\chi(s)^{-1/2}\zeta(s) \ll_\varepsilon \begin{cases} \tau^{(1/4)+\varepsilon} & (\frac{1}{2} \leqslant \sigma \leqslant 1) \\ \tau^{(3/8)+\varepsilon} & (1 \leqslant \sigma \leqslant \frac{5}{4}) \end{cases}.$$

The integrals over the horizontal segments of $\mathcal{R}$ are therefore $\ll T^{(3/8)+\varepsilon}$.

In order to estimate the contribution of the vertical segment

$$\Big[\tfrac{5}{4}+iT, \tfrac{5}{4}+2iT\Big],$$

we evaluate $\chi(s)$ by the complex Stirling formula as given in Corollary 0.13. We find this contribution is

$$i\int_T^{2T} \Big(\frac{\tau}{2\pi}\Big)^{(3/8)+i\tau} \mathrm{e}^{-i\tau/2-i\pi/8}\Big\{1+O\Big(\frac{1}{\tau}\Big)\Big\}\zeta\big(\tfrac{5}{4}+i\tau\big)\,\mathrm{d}\tau.$$

We can treat the remainder term trivially. It gives a quantity $\ll T^{3/8}$. Replacing $\zeta(\frac{5}{4}+i\tau)$ by the series and interchanging summations, we see that the main term has the value

$$i\mathrm{e}^{-i\pi/8}\sum_{n\geqslant 1} n^{-5/4}\int_T^{2T}\Big(\frac{\tau}{2\pi}\Big)^{3/8}\mathrm{e}^{iF_n(\tau)}\,\mathrm{d}\tau \tag{4.29}$$

with $F_n(\tau) := \tau\big(\ln(\tau/2\pi) - \frac{1}{2} + \ln n\big)$. We have

$$F_n''(\tau) \asymp 1/T \qquad (T \leqslant \tau \leqslant 2T)$$

from which, by Theorem I.6.3,

$$\int_T^{\tau} \mathrm{e}^{iF_n(t)}\,\mathrm{d}t \ll T^{1/2} \qquad (T \leqslant \tau \leqslant 2T).$$

An integration by parts then allows us to deduce from the above that the expression (4.29) is $\ll T^{7/8}$, and so finally

$$\int_T^{2T} Z(\tau)\,\mathrm{d}\tau \ll_\varepsilon T^{7/8}. \tag{4.30}$$

Now let us estimate the right-hand side of (4.28). We have

$$\int_T^{2T} |Z(\tau)|\,\mathrm{d}\tau = \int_T^{2T} \left|\zeta\left(\tfrac{1}{2}+i\tau\right)\right| \mathrm{d}\tau \geqslant \left| \int_T^{2T} \zeta\left(\tfrac{1}{2}+i\tau\right) \mathrm{d}\tau \right|.$$

By the residue theorem, we can write

$$i\int_T^{2T} \zeta\left(\tfrac{1}{2}+i\tau\right) \mathrm{d}\tau = \int_{\mathcal{G}} \zeta(s)\,\mathrm{d}s$$

where $\mathcal{G}$ is the polygonal line joining $\frac{1}{2}+iT$, $2+iT$, $2+2iT$, $\frac{1}{2}+2iT$. Since $\zeta(s) \ll_\varepsilon T^{(1/4)+\varepsilon}$ on $\mathcal{G}$, the contribution of the horizontal segments is $\ll_\varepsilon T^{(1/4)+\varepsilon}$. That of the vertical segment has the value

$$i\sum_{n\geqslant 1} \frac{1}{n^2} \int_T^{2T} n^{-i\tau}\,\mathrm{d}\tau = i\Big\{T + O\Big(\sum_{n\geqslant 2} \frac{1}{n^2 \ln n}\Big)\Big\}.$$

We thus have

$$\int_T^{2T} |Z(\tau)|\,\mathrm{d}\tau \gg T$$

which establishes (4.28).

A trivial modification of the argument presented above would give a lower bound of the type

$$N_0(T) \gg T^{(1/4)-\varepsilon}$$

—see Blanchard (1969, chap. IV.6).

The best numerical constant obtained so far by refining the work of Levinson—see (4.13)—is due to Conrey (1989) and exceeds 2/5.

In the same order of ideas, Theorem 4.9, Pintz (1984) established a precise link between the size of the remainder term $R(x) = \psi(x) - x$ and the localization of the zeros of $\zeta(s)$: we have

$$\ln\{x/R(x)\} \sim \min_{\varrho}\{(1-\beta)\ln x + \ln|\gamma|\} \qquad (x \to \infty).$$

§ 4.4. For other proofs of the explicit formulae, see for example Ellison & Mendès France (1975), § 5.5, or chap. XIV of Titchmarsh (1951), or chap. 12 of Ivić (1985).

Ingham (1990) shows that a slight modification of the estimate for $R(x,T)$ allows extending the validity of formula (4.20) to $1 < x < 2$. He also provides a variant which holds for $0 < x < 1$.

Exercises

194. Let k be an integer $\geqslant 1$.

(a) Show that the complex numbers h_ν determined by the identity

$$1+\sum_{\nu\geqslant 1} h_\nu z^\nu = (1-z)^k\Big(1+\frac{kz}{1-z}\Big)$$

satisfy an upper bound of the type $|h_\nu| \leqslant C(k)$ $(\nu = 1, 2, \ldots)$.

(b) Show that we have $\sum_{n\geqslant 1} k^{\omega(n)}/n^s = H_k(s)\zeta(s)^k$ where $H_k(s)$ is an absolutely convergent Dirichlet series for $\sigma > \frac{1}{2}$.

(c) Deduce the formula

$$\sum_{n\leqslant x} k^{\omega(n)} = xP_{k-1}(\ln x) + O_\varepsilon(x^{1-\delta_k+\varepsilon})$$

where P_{k-1} is a polynomial of degree $k-1$ whose leading coefficient should be determined, and where δ_k is some positive constant to be made explicit.

(d) Calculate δ_k assuming the Riemann hypothesis.

195. *Bateman's theorem (1972).*

Let φ denote Euler's totient function.

(a) Show that the number $a_n := \operatorname{card}\{m : m \geqslant 1,\ \varphi(m) = n\}$ is finite for each integer $n \geqslant 1$.

(b) For $\sigma > 1$, define $G(s) := \prod_p(1 + (p-1)^{-s} - p^{-s})$. Establish the relation $\sum_{n\geqslant 1} a_n/n^s = \zeta(s)G(s)$ $(\sigma > 1)$.

(c) Show that the infinite product defining $G(s)$ is absolutely convergent for $\sigma > 0$.

(d) Prove that $|(p-1)^{-s} - p^{-s}| \leqslant \min\big(2(p-1)^{-\sigma},\ |s|(p-1)^{-\sigma-1}\big)$ $(p \geqslant 2,\ \sigma > 0)$ and deduce the existence of an absolute constant A such that

$$G(s) \ll (\ln|\tau|)^A \qquad (|\tau| \geqslant 2,\ \sigma \geqslant 1 - 1/\ln 2|\tau|).$$

(e) Set $\Phi(x) := \operatorname{card}\{m : m \geqslant 1,\ \varphi(m) \leqslant x\}$. Show that $\Phi(x)$ is the summatory function for a_n and establish the asymptotic formula

$$\int_0^x \Phi(t)\,\mathrm{d}t = \tfrac{1}{2}\alpha x^2 + O\Big(x^2\mathrm{e}^{-c_0\sqrt{\ln x}}\Big) \qquad (x \to \infty)$$

with $\alpha := G(1) = \zeta(2)\zeta(3)/\zeta(6)$ and suitable constant $c_0 > 0$.

(f) Using the monotonicity of Φ show—by an argument analogous to that leading to (4.10)—that we have, for suitable fixed $c > 0$,

$$\Phi(x) = \alpha x + O\Big(x\mathrm{e}^{-c\sqrt{\ln x}}\Big).$$

(g) Show that $a_n \ll n\mathrm{e}^{-c\sqrt{\ln n}}$ $(n \geqslant 1)$.

196. Show that the Riemann hypothesis implies the convergence of the series $\sum \mu(n)/n^s$ for any s in the half-plane $\sigma > \frac{1}{2}$. [*Hint: use Theorem* 4.2 *and Schnee–Landau's Theorem* 2.8.]

197. *The prime number theorem via the explicit formula.* Show that, for a suitable positive constant c_1, we have

$$\sum_{|\gamma|\leqslant T} \left|\frac{x^\varrho}{\varrho}\right| \ll (\ln T)^2 x^{1-c_1/\ln T} \qquad (x \geqslant 2,\ T \geqslant 2)$$

and deduce from this a new proof of the prime number theorem.

198. Put $\tau(n,\vartheta) := \sum_{d|n} d^{i\vartheta}$. Using Ramanujan's identity (3.26), show that, uniformly for $x \geqslant 2$, $|\vartheta| \leqslant 1$, $\vartheta \neq 0$, we have

$$\sum_{n\leqslant x} |\tau(n,\vartheta)|^2 = \frac{6x}{\pi^2}\Big\{|\zeta(1+i\vartheta)|^2 \ln x + O\big(1/|\vartheta|^3\big)\Big\}.$$

199. Consider the arithmetic function

$$T(n) := \big|\{(d,d') : d|n,\ d'|n,\ |\ln(d'/d)| \leqslant 1\}\big|.$$

Using the weight function $w(t) := \dfrac{1}{2\pi}\Big(\dfrac{\sin(t/2)}{t/2}\Big)^2$, with Fourier transform

$$\widehat{w}(\vartheta) = \big(1-|\vartheta|\big)^+,$$

show that

$$w(1)\int_{-1}^{1} |\tau(n,\vartheta)|^2\,\mathrm{d}\vartheta \leqslant T(n) \leqslant \frac{1}{2\pi w(1)}\int_{-1}^{1} |\tau(n,\vartheta)|^2\,\mathrm{d}\vartheta.$$

Using the result of Exercise 198, deduce that

$$\sum_{n\leqslant x} T(n) \asymp x(\ln x)^2 \qquad (x \geqslant 2).$$

The Selberg–Delange method

5.1. Complex powers of $\zeta(s)$

The analytic study of the Riemann zeta function enabled us to estimate the summatory functions of certain arithmetic functions whose Dirichlet series could be simply expressed in terms of $\zeta(s)$. Thus the asymptotic formulae

$$\psi(x) := \sum_{n \leqslant x} \Lambda(n) = x + O\Big(x\mathrm{e}^{-c\sqrt{\ln x}}\Big),$$

$$M(x) := \sum_{n \leqslant x} \mu(n) = O\Big(x\mathrm{e}^{-c\sqrt{\ln x}}\Big),$$

$$T_k(x) := \sum_{n \leqslant x} \tau_k(n) = xP_{k-1}(\ln x) + O_k\Big(x^{1-\delta_k}\Big)$$

(where P_{k-1} is a polynomial of degree $k-1$, and δ_k a positive constant), are straightforward consequences of properties established for $\zeta(s)$ in Chapter II.3: they were obtained by applying the Perron formulae to the meromorphic functions

$$-\zeta'(s)/\zeta(s), \quad 1/\zeta(s), \quad \zeta(s)^k.$$

A double extension of this method is considered in this chapter. On the one hand, we want to be able to deal with Dirichlet series having singularities *which are not poles*. On the other hand, we want to exhibit some kind of *stability* for the phenomenon, in the sense of an invariance in the nature of the asymptotic formulae obtained for two Dirichlet series whose ratio is a sufficiently regular analytic function. In this direction, we will consider

series admitting a representation of the type

$$F(s) = G(s;z)\zeta(s)^z \qquad (z \in \mathbb{C}) \tag{5.1}$$

for $\sigma > 1$, where $G(s;z)$ satisfies certain conditions to which we will return later.

Before developing this theory, which is essentially due to Selberg (1954) and Delange (1959, 1971), let us take a deeper look at the analytic nature of $\zeta(s)^z$ when z is a fixed complex number.

Defining the generalized binomial coefficient by

$$\binom{w}{\nu} := \frac{1}{\nu!} \prod_{0 \leqslant j < \nu} (w-j) \qquad (w \in \mathbb{C},\ \nu \in \mathbb{N}),$$

we can write for $|\xi| < 1$, $z \in \mathbb{C}$,

$$(1-\xi)^{-z} = \sum_{\nu \geqslant 0} \binom{z+\nu-1}{\nu} \xi^\nu.$$

When z is a negative integer, this formula reduces to the classical binomial formula. For $\sigma > 1$, we deduce that

$$\zeta(s)^z = \prod_p (1-p^{-s})^{-z} = \prod_p \Big\{ 1 + \sum_{\nu \geqslant 1} \binom{z+\nu-1}{\nu} p^{-\nu s} \Big\}, \tag{5.2}$$

where the infinite product is absolutely convergent. By Theorem 1.3, it follows that $\zeta(s)^z$ is representable in the half-plane $\sigma > 1$ as the Dirichlet series of a multiplicative function, say $\tau_z(n)$, defined by

$$\tau_z(p^\nu) := \binom{z+\nu-1}{\nu} \qquad (p \in \mathbb{P},\ \nu \geqslant 0). \tag{5.3}$$

This definition generalizes that of the functions $\tau_k(n)$, corresponding to the case when $z = k$ is a positive integer. In these circumstances, τ_k is the k-th convolution power of $\mathbf{1}$, and so $\tau_k(n)$ can be interpreted combinatorially as the number of decompositions of n into a product of k factors, viz.

$$\tau_k(n) = \sum_{d_1 d_2 \cdots d_k = n} 1. \tag{5.4}$$

Of course, $\tau_2 = \tau$.

We shall see below that the function

$$Z(s;z) := \{(s-1)\zeta(s)\}^z / s \tag{5.5}$$

plays a special role. It is defined in the intersection of $\mathbb{C}^*$ with any simply connected domain which does not contain a zero of $\zeta(s)$. We shall always

suppose that this domain includes the real half-line $[1, +\infty[$. We can then choose the holomorphic branch of $\log\{(s-1)\zeta(s)\}$ so that

$$Z(1; z) = 1.$$

Theorem 5.1. *The function $Z(s; z)$ is holomorphic in the disk $|s-1| < 1$. It can be represented there by the Taylor series*

$$Z(s; z) = \sum_{j \geqslant 0} \frac{1}{j!} \gamma_j(z)(s-1)^j \tag{5.6}$$

where the coefficients $\gamma_j(z)$ are entire functions of z, satisfying, for all $A > 0$ and $\varepsilon > 0$, the upper bound

$$\frac{1}{j!}\gamma_j(z) \ll_{A,\varepsilon} (1+\varepsilon)^j \qquad (|z| \leqslant A). \tag{5.7}$$

Proof. All the assertions in the statement of the theorem follow immediately from the fact that $\zeta(s)$ does not vanish for $|s-1| < 1$, via Cauchy's formula

$$\frac{1}{j!}\gamma_j(z) = \frac{1}{2\pi i} \oint_{|s-1|=r} \frac{Z(s; z)}{(s-1)^{j+1}}\, \mathrm{d}s.$$

It is indeed well known—see for example Titchmarsh (1951), ch. XV—that the non-trivial zero with smallest modulus in the half-plane $\tau \geqslant 0$ is

$$\varrho = \tfrac{1}{2} + i\,14.13472\ldots.$$

For the sake of completeness, we provide a quick proof of the fact that $\zeta(s) \neq 0$ for $|s-1| < 1$. After integration by parts, formula (3.17) gives

$$\zeta(s) = \frac{1}{s-1} + \tfrac{1}{2} + \tfrac{1}{12}s - \tfrac{1}{2}s(s+1)\int_1^\infty \frac{B_2(t)}{t^{s+2}}\, \mathrm{d}t.$$

Suppose $\zeta(s)$ has a zero $\varrho = \beta + i\gamma$ such that $|\varrho - 1| < 1$. Then, by symmetry, we may assume that $\beta \geqslant \frac{1}{2}$. Since $|B_2(t)| \leqslant B_2 = \frac{1}{6}$, we obtain, by setting $s = \varrho$ in the formula above, that

$$1 + \tfrac{1}{2}(\varrho - 1)\Big\{1 + \tfrac{1}{6}\varrho - \tfrac{1}{6}\vartheta\varrho(\varrho+1)\int_1^\infty \frac{\mathrm{d}t}{t^{5/2}}\Big\} = 0,$$

with $|\vartheta| \leqslant 1$; whence

$$1 \leqslant \tfrac{1}{2}|\varrho - 1|\{1 + \tfrac{1}{6}|\varrho| + \tfrac{1}{9}|\varrho(\varrho+1)|\} < \tfrac{1}{2}\{1 + \tfrac{1}{3} + \tfrac{2}{3}\} = 1,$$

which is the expected contradiction. □

By Theorem 3.21, there is a strictly positive absolute constant c such that $\zeta(s)$ does not vanish in the region defined by the inequality

$$\sigma \geqslant 1 - c/(1 + \ln^+ |\tau|). \tag{5.8}$$

Up to the end of this chapter, *we let $\mathcal{D}$ denote the simply connected domain obtained by deleting the real segment* $[1-c,1]$ *from the region* (5.8). We then have the analytic continuation

$$\zeta(s)^z = sZ(s;z)(s-1)^{-z} \qquad (s \in \mathcal{D}). \tag{5.9}$$

Moreover, the upper bound $|\log\zeta(s)| \leqslant \ln_2|\tau| + O(1)$ from Theorem 3.22 shows that, for each constant $A > 0$, we have

$$\zeta(s)^z \ll_A \{1+\ln^+|\tau|\}^A \qquad (|z| \leqslant A,\ s \in \mathcal{D},\ |s-1| \gg 1). \tag{5.10}$$

5.2. The main result

We are now in a position to establish a general theorem providing, for Dirichlet series sufficiently close to a complex power of $\zeta(s)$, an evaluation of the summatory function of the coefficients comparable in quality to that of the prime number theorem.

The vast field of application for this result and the complete uniformity of its statement justify the explicit presence of rather numerous parameters. We have attempted to clarify the situation by introducing the following terminology.

Let $z \in \mathbb{C}$, $c_0 > 0$, $0 < \delta \leqslant 1$, $M > 0$. We say that a *Dirichlet series* $F(s)$ *has the property* $\mathcal{P}(z; c_0, \delta, M)$ if the Dirichlet series

$$G(s;z) := F(s)\zeta(s)^{-z}$$

may be continued as a holomorphic function for $\sigma \geqslant 1 - c_0/(1+\ln^+|\tau|)$, and, in this domain, satisfies the bound

$$|G(s;z)| \leqslant M(1+|\tau|)^{1-\delta}. \tag{5.11}$$

If $F(s) = \sum_{n\geqslant 1} a_n/n^s$ has property $\mathcal{P}(z; c_0, \delta, M)$, and if there exists a sequence of non-negative real numbers $\{b_n\}_{n=1}^{\infty}$ such that $|a_n| \leqslant b_n$ $(n = 1, 2, \ldots)$, and the series

$$\sum_{n\geqslant 1} \frac{b_n}{n^s}$$

satisfies $\mathcal{P}(w; c_0, \delta, M)$ for some complex number w, we shall say that $F(s)$ *has type* $\mathcal{T}(z, w; c_0, \delta, M)$. It is worthwhile to bear in mind that a series with non-negative coefficients having property $\mathcal{P}(z; c_0, \delta, M)$ is trivially of type $\mathcal{T}(z, z; c_0, \delta, M)$.

In the domain where $G(s;z)$ is holomorphic, we set

$$G^{(k)}(s;z) := \frac{\partial^k}{\partial s^k} G(s;z) \tag{5.12}$$

and

$$\lambda_k(z) := \frac{1}{\Gamma(z-k)} \sum_{h+j=k} \frac{1}{h!j!} G^{(h)}(1;z)\gamma_j(z) \tag{5.13}$$

where the $\gamma_j(z)$ are the entire functions appearing in Theorem 5.1. Defining the sequence $\{\mu_k(z)\}_{k=0}^{\infty}$ by the Taylor expansion at the origin

$$\frac{s^z F(s+1)}{s+1} = \sum_{k\geqslant 0} \mu_k(z) s^k \qquad \big(|s| < \min(1, c_0)\big), \tag{5.14}$$

we then have

$$\lambda_k(z) = \frac{\mu_k(z)}{\Gamma(z-k)} \qquad (k \in \mathbb{N}). \tag{5.15}$$

For the sake of brevity, we shall set in the sequel, for $x > 1$, $a > 0$, $b > 0$,

$$R_N(x;a,b) := \mathrm{e}^{-a\sqrt{\ln x}} + \Big(\frac{bN+1}{\ln x}\Big)^{N+1}. \tag{5.16}$$

Theorem 5.2. *Let $F(s) := \sum_{n\geqslant 1} a_n/n^s$ be a Dirichlet series of type $\mathcal{T}(z, w; c_0, \delta, M)$. For $x \geqslant 3$, $N \geqslant 0$, $A > 0$, $|z| \leqslant A$, $|w| \leqslant A$, we have*

$$\sum_{n\leqslant x} a_n = x(\ln x)^{z-1}\Big\{ \sum_{0\leqslant k\leqslant N} \frac{\lambda_k(z)}{(\ln x)^k} + O\Big(M R_N(x)\Big)\Big\} \tag{5.17}$$

with $R_N(x) := R_N(x; c_1, c_2)$. The positive constants c_1, c_2 and the implicit constant in the Landau symbol depend at most on c_0, δ and A.

We shall see in the next chapter (§ 6.2) how one can take advantage of the uniformity in M in formula (5.17). Uniformity in N is equally significant. Consider, for example, the case when $z \in \mathbb{Z}$. Formula (5.15) then shows that $\lambda_k(z) = 0$ whenever $k \geqslant z$. We can hence choose N so as to minimize the error term in (5.17). For the value $N := \lfloor (\ln x)/ec_2 \rfloor$, we obtain

$$\sum_{n\leqslant x} a_n = x(\ln x)^{z-1}\Big\{ P\Big(\frac{1}{\ln x}\Big) + O\Big(M\mathrm{e}^{-c_1\sqrt{\ln x}}\Big)\Big\}$$

where P is a polynomial of degree at most $z-1$.

Exercise 195 (p. 275) involved a direct study in a situation of this type. The corresponding result is actually an immediate consequence of Theorem 5.2. For the choice $a_n := \mathrm{card}\{m \geqslant 1 : \varphi(m) = n\}$ we indeed have

$$F(s) = \sum_{m\geqslant 1} \frac{1}{\varphi(m)^s} = \zeta(s)G(s)$$

with

$$G(s) = \prod_p \Big(1 + \frac{1}{(p-1)^s} - \frac{1}{p^s}\Big) \qquad (\sigma > 0).$$

The easy upper bound

$$G(s) \ll (\ln|\tau|)^{O(1)} \qquad (|\tau| \geqslant \tau_0,\ \sigma \geqslant 1 - 1/\ln|\tau|)$$

shows that $F(s)$ is of type $\mathcal{T}(1,1;1,\delta,M(\delta))$ for any fixed $\delta < 1$, where $M(\delta)$ only depends on δ. Since

$$\lambda_0(1) = G(1) = \zeta(2)\zeta(3)/\zeta(6) \approx 1.9436$$

is the only non-zero number among the $\lambda_k(1)$, we obtain the following statement.

Theorem 5.3 (Bateman, 1972). *There exists a positive constant c such that, for $x \geqslant 1$, we have*

$$\operatorname{card}\{n \geqslant 1 : \varphi(n) \leqslant x\} = \frac{\zeta(2)\zeta(3)}{\zeta(6)}x + O\Big(x\mathrm{e}^{-c\sqrt{\ln x}}\Big). \tag{5.18}$$

Of course, Theorem 5.2 provides an analogous result when $\varphi(n)$ is replaced by $\sigma(n)$ or, more generally, by any positive multiplicative function f such that

$$f(p^\nu) = p^\nu + O(p^{\nu-\delta}) \qquad (\nu \geqslant 1)$$

with $\delta > 0$.

5.3. Proof of Theorem 5.2

Let c be a positive constant such that $c < c_0$ and such that $\zeta(s)$ has no zero in the region

$$\sigma \geqslant 1 - c/(1 + \ln^+|\tau|).$$

Then $F(s)$ is continuable to a holomorphic function in the domain $\mathcal{D}$ defined in §5.1 and we have, by (5.10) and (5.11),

$$F(s) \ll_A M(1+\ln^+|\tau|)^A(1+|\tau|)^{1-\delta} \ll_{A,\delta} M(1+|\tau|)^{1-\delta/2} \tag{5.19}$$

uniformly for $s \in \mathcal{D}$, $|s-1| \gg 1$, $|z| \leqslant A$.

Let $A(x) := \sum_{n\leqslant x} a_n$. Then Perron's formula (2.10) allows us to write

$$\int_0^x A(t)\,\mathrm{d}t = \frac{1}{2\pi i}\int_{\kappa-i\infty}^{\kappa+i\infty} F(s)x^{s+1}\frac{\mathrm{d}s}{s(s+1)}$$

with $\kappa := 1 + 1/\ln x$. Let $T > 1$ be a parameter whose value will be determined later. The residue theorem allows us to deform the segment of integration $[\kappa - iT, \kappa + iT]$ into some path joining the endpoints and contained entirely in $\mathcal{D}$. We choose the path symmetrically with respect to the real axis. Its upper part is made up of: a truncated Hankel contour Γ surrounding the point $s = 1$ with radius $r = 1/(2\ln x)$ and linear part joining $1 - r$ to $1 - \frac{1}{2}c$; the curve

$$\sigma = \sigma(\tau) := 1 - \tfrac{1}{2}c/(1 + \ln^+\tau)$$

for $0 \leqslant \tau \leqslant T$; and the horizontal segment $[\sigma(T) + iT, \kappa + iT]$.

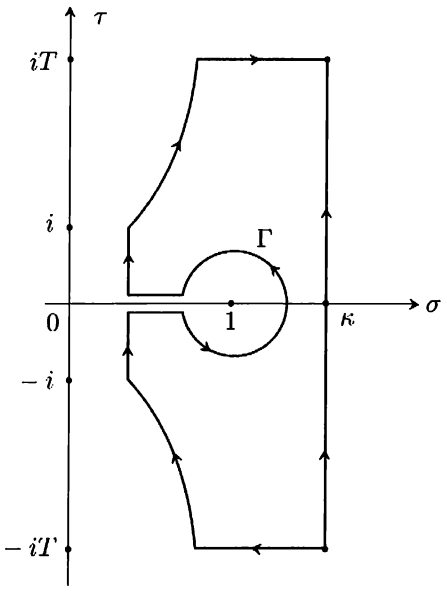

Figure 1. The integration path associated to $F(s) = \zeta(s)^z G(s;z)$.

We shall see that the main contribution arises from the integral over the truncated Hankel contour Γ.

Appealing to (5.19), we see immediately that the contribution from the vertical half-lines $[\kappa \pm iT, \kappa \pm i\infty[$ is $\ll_{A,\delta} Mx^2T^{-\delta/2}$.

This upper bound is equally valid for the contributions of the horizontal segments $[\sigma(T) \pm iT, \kappa \pm iT]$. Finally, that of the arcs $\sigma = \sigma(\tau)$ is

$$\ll_{A,\delta} Mx^{1+\sigma(T)} \int_0^T (1+\tau)^{-1-\delta/2}\,\mathrm{d}\tau \ll_{A,\delta} Mx^{1+\sigma(T)}.$$

Selecting $T = \exp\sqrt{(c/\delta)\ln x}$, for $x \geqslant x_0$, it then follows that

$$\int_0^x A(t)\,\mathrm{d}t = \Phi(x) + O(Mx^2\mathrm{e}^{-c_3\sqrt{\ln x}}) \tag{5.20}$$

with

$$\Phi(x) := \frac{1}{2\pi i}\int_\Gamma F(s)x^{s+1}\frac{\mathrm{d}s}{s(s+1)}. \tag{5.21}$$

Here, and for the rest of the proof, we make the convention that all constants, implicit or explicit, depend at most on c_0, δ and A.

It remains to study the main term $\Phi(x)$ of (5.20). Clearly it is an infinitely differentiable function of x on $\mathbb{R}^+$, and in particular we have

$$\Phi'(x) = \frac{1}{2\pi i}\int_\Gamma F(s)x^s\frac{\mathrm{d}s}{s}, \qquad \Phi''(x) = \frac{1}{2\pi i}\int_\Gamma F(s)x^{s-1}\,\mathrm{d}s.$$

For $s \in \mathcal{D}$, we can write

$$F(s) = sG(s;z)Z(s;z)(s-1)^{-z} \tag{5.22}$$

with the result that, by (5.6), (5.7) and (5.11),

$$F(s) \ll M|s-1|^{-A} \qquad (s \in \Gamma).$$

Since $r = 1/(2\ln x)$, it follows that

$$\Phi''(x) \ll M(\ln x)^A. \tag{5.23}$$

When $s \in \Gamma$, we have by (5.14)

$$G(s;z)Z(s;z) = \frac{(s-1)^z F(s)}{s} = \sum_{k\geqslant 0} \mu_k(z)(s-1)^k$$

with

$$\mu_k(z) := \frac{1}{k!} \sum_{h+j=k} \binom{k}{j} G^{(h)}(1;z)\gamma_j(z) = \Gamma(z-k)\lambda_k(z) \qquad (k \in \mathbb{N}).$$

In addition, since $G(s;z)Z(s;z)$ is holomorphic and $O(M)$ in the disk $|s-1| \leqslant c$, the Cauchy formulae imply that

$$\mu_k(z) \ll Mc^{-k} \qquad (k \geqslant 0). \tag{5.24}$$

Observing that Γ is contained in the disk $|s-1| \leqslant \frac{1}{2}c$, we can write for $s \in \Gamma$, $N \geqslant 0$,

$$G(s;z)Z(s;z) = \sum_{0\leqslant k\leqslant N} \mu_k(z)(s-1)^k + O\Big(M\big(|s-1|/c\big)^{N+1}\Big).$$

Therefore

$$\Phi'(x) = \sum_{0\leqslant k\leqslant N} \mu_k(z) \frac{1}{2\pi i} \int_\Gamma x^s (s-1)^{k-z}\,\mathrm{d}s + O\big(Mc^{-N}R(x)\big), \tag{5.25}$$

with

$$\begin{aligned} R(x) &:= \int_\Gamma \big|x^s(s-1)^{N+1-z}\big|\,|\mathrm{d}s| \\ &\ll \int_{1-c/2}^{1-r} (1-\sigma)^{N+1-\Re\mathrm{e}\,z} x^\sigma\,\mathrm{d}\sigma + x^{1+r} r^{N+2-\Re\mathrm{e}\,z} \\ &\ll x(\ln x)^{\Re\mathrm{e}\,z-N-2}\Big\{\int_{\frac{1}{2}}^{+\infty} t^{N+1-\Re\mathrm{e}\,z}\mathrm{e}^{-t}\,\mathrm{d}t + 2^{-N}\Big\} \\ &\ll x(\ln x)^{\Re\mathrm{e}\,z-N-2}\Gamma(N+A+2) \ll x(\ln x)^{z-1}\Big(\frac{c_4 N+1}{\ln x}\Big)^{N+1}. \end{aligned}$$

Using the change of variable $w = (s-1)\ln x$, we can equally well write, with the notation of Corollary 0.18,

$$\frac{1}{2\pi i}\int_\Gamma x^s(s-1)^{k-z}\,\mathrm{d}s = \frac{x}{2\pi i}(\ln x)^{z-1-k}\int_{\mathcal{H}\left(\frac{1}{2}c\ln x\right)} w^{k-z}\mathrm{e}^w\,\mathrm{d}w$$
$$= x(\ln x)^{z-1-k}\Big\{\frac{1}{\Gamma(z-k)} + O\big((c_5k+1)^k x^{-c/4}\big)\Big\}.$$

The main term of (5.25) thus has the value

$$x(\ln x)^{z-1}\Big\{\sum_{0\leqslant k\leqslant N}\frac{\lambda_k(z)}{(\ln x)^k} + O(E_N)\Big\}$$

with

$$E_N := \frac{1}{x^{c/4}}\sum_{0\leqslant k\leqslant N}|\mu_k(z)|\Big(\frac{c_5k+1}{\ln x}\Big)^k \ll \frac{M}{x^{c/4}}c_6^N\sum_{0\leqslant k\leqslant N}k!\Big(\frac{5}{c\ln x}\Big)^k$$
$$\leqslant \frac{M}{x^{c/4}}\Big(\frac{5c_6}{c\ln x}\Big)^N\sum_{0\leqslant k\leqslant N}\frac{N!}{(N-k)!}\Big(\frac{c\ln x}{5}\Big)^{N-k}$$
$$\leqslant \frac{M}{x^{c/20}}N!\Big(\frac{5c_6}{c\ln x}\Big)^N \leqslant M\Big(\frac{c_7N+1}{\ln x}\Big)^{N+1}.$$

Substituting in (5.25), it follows that

$$\Phi'(x) = x(\ln x)^{z-1}\left\{\sum_{0\leqslant k\leqslant N}\frac{\lambda_k(z)}{(\ln x)^k} + O\left(M\Big(\frac{c_8N+1}{\ln x}\Big)^{N+1}\right)\right\}. \tag{5.26}$$

We shall show, using (5.20) and (5.23), that $\Phi'(x)$ is a suitable approximation to $A(x)$. To this end, let us take a parameter h, $0 < h < \frac{1}{2}x$, and apply (5.20) for both x and $x+h$. Subtracting these estimates, we obtain

$$\int_x^{x+h} A(t)\,\mathrm{d}t = \Phi(x+h) - \Phi(x) + O\big(Mx^2\mathrm{e}^{-c_3\sqrt{\ln x}}\big) \tag{5.27}$$

while (5.23) implies that

$$\Phi(x+h) - \Phi(x) = h\Phi'(x) + h^2\int_0^1(1-t)\Phi''(x+th)\,\mathrm{d}t$$
$$= h\Phi'(x) + O\big(Mh^2(\ln x)^A\big). \tag{5.28}$$

We can thus write

$$A(x) = \Phi'(x) + O\Big(\frac{Mx^2}{h}\mathrm{e}^{-c_3\sqrt{\ln x}} + Mh(\ln x)^A + \frac{L}{h}\Big) \tag{5.29}$$

with

$$L := \int_x^{x+h}|A(t) - A(x)|\,\mathrm{d}t.$$

It is at this stage that we use the hypothesis that the series associated to b_n has property $\mathcal{P}(w; c_0, \delta, M)$. Writing $B(t) := \sum_{n \leqslant t} b_n$, we have

$$L \leqslant \int_x^{x+h} \big(B(t) - B(x)\big)\,\mathrm{d}t \leqslant \int_x^{x+h} B(t)\,\mathrm{d}t - \int_{x-h}^{x} B(t)\,\mathrm{d}t. \tag{5.30}$$

Now, our assumptions on the sequence $\{b_n\}_{n=1}^{\infty}$ imply the existence of an infinitely differentiable function Φ_1 satisfying (5.28) and such that (5.27) is valid on replacing $A(t)$ by $B(t)$ and Φ by Φ_1. By (5.30) this implies

$$L \ll Mx^2 \mathrm{e}^{-c_9\sqrt{\ln x}} + Mh^2(\ln x)^A.$$

In other words, up to modifying the value of c_3, we can suppress the term L/h in the remainder of (5.29). Thus, choosing $h := x\exp\{-(c_3/2)\sqrt{\ln x}\}$, it follows that

$$A(x) = \Phi'(x) + O\big(Mx\mathrm{e}^{-c_{10}\sqrt{\ln x}}\big). \tag{5.31}$$

The conclusion of Theorem 5.2 follows by combining (5.26) and (5.31).

5.4. A variant of the main theorem

We are now going to establish a result of the same type as Theorem 5.2, but in which the assumptions on the analytic continuation of the series

$$G(s; z) := \zeta(s)^{-z}F(s) =: \sum_{n \geqslant 1} \frac{g_z(n)}{n^s} \tag{5.32}$$

are replaced by convergence conditions on the successive derivatives (possibly fractional) at $s = 1$. We recall notation (5.16).

Theorem 5.4. *Let $F(s) := \sum_{n\geqslant 1} a_n/n^s$ be a Dirichlet series converging for $\sigma > 1$. Suppose there exists a complex number z and an integer $N \geqslant 0$ such that*

$$H_N(z) := \sum_{n \geqslant 1} |g_z(n)| \frac{(\ln 3n)^{N+1+w}}{n} < \infty \tag{5.33}$$

where $w := \max(1 - \Re\mathrm{e}\, z, 0)$. Then we have, for $|z| \leqslant A$,

$$\sum_{n \leqslant x} a_n = x(\ln x)^{z-1}\Big\{ \sum_{0 \leqslant k \leqslant N} \frac{\lambda_k(z)}{(\ln x)^k} + O_A\big(H_N(z)R_N(x)\big)\Big\} \tag{5.34}$$

with $R_N(x) := R_N(x; c_1, c_2)$ where c_1, c_2 are positive constants depending at most on A, and where the coefficients $\lambda_k(z)$ are defined by (5.13) *for $0 \leqslant k \leqslant N$.*

Remark. Condition (5.33) is certainly fulfilled if the derivatives of order $\leqslant N + 1 + \lceil w \rceil$ of the Dirichlet series $G(s; z)$ are absolutely convergent for $s = 1$.

Proof. To start with, let us apply Theorem 5.2 to the Dirichlet series $\zeta(s)^z$. This is possible since the upper bound

$$|\tau_z(n)| \leqslant \tau_{|z|}(n) \qquad (n = 1, 2, \dots)$$

guarantees that $\zeta(s)^z$ is of type $\mathcal{T}(z, |z|; c_0, 1, 1)$ for some absolute positive constant c_0. Setting

$$T_z(x) := \sum_{n \leqslant x} \tau_z(n), \tag{5.35}$$

we obtain

$$T_z(x) = x(\ln x)^{z-1} \Big\{ \sum_{0 \leqslant k \leqslant N} \frac{\gamma_k(z)}{k!\,\Gamma(z-k)(\ln x)^k} + O_A\big(R_N(x; c_3, c_4)\big) \Big\} \tag{5.36}$$

where the $\gamma_k(z)$ are the entire functions from Theorem 5.1, and where c_3, c_4 are positive constants only depending on A.

Now, the convolution formula $a_n = \tau_z * g_z(n)$ $(n = 1, 2, \dots)$ allows us to write

$$A(x) = \sum_{mn \leqslant x} \tau_z(m) g_z(n) = \sum_{n \leqslant x} g_z(n) T_z(x/n).$$

Since $T_z(x) = 1$ for $1 \leqslant x < 2$, we further obtain

$$A(x) = \sum_{n \leqslant x/2} g_z(n) T_z(x/n) + \sum_{x/2 < n \leqslant x} g_z(n). \tag{5.37}$$

The stated result follows from this formula, evaluating $T_z(x/n)$ by (5.36). At several stages, we shall employ the following estimate, valid for $y \geqslant 1$ and $0 \leqslant h \leqslant N + 1 + w$:

$$\sum_{n \geqslant y} \frac{|g_z(n)|(\ln n)^h}{n} \leqslant \frac{H_N(z)}{(\ln 3y)^{N+1+w-h}}. \tag{5.38}$$

This results immediately from the definition of $H_N(z)$ and the trivial bound

$$(\ln n)^h \leqslant \Big(\frac{\ln 3n}{\ln 3y}\Big)^{N+1+w} (\ln 3y)^h \qquad (n \geqslant y).$$

Applied with $h = 0$, $y = x/2$, inequality (5.38) implies that

$$\Big| \sum_{x/2 < n \leqslant x} g_z(n) \Big| \leqslant x \sum_{n > x/2} \frac{|g_z(n)|}{n} \leqslant \frac{H_N(z) x}{(\ln x)^{N+1+w}}.$$

On the other hand, formula (5.36) with $N = 0$ gives

$$T_z(x) \ll_A x(\ln x)^{z-1} \qquad (x \geqslant 2),$$

so that, for $x \geqslant 4$, we may write

$$\sum_{\sqrt{x}<n\leqslant x/2} g_z(n)T_z\Big(\frac{x}{n}\Big) \ll_A x \sum_{\sqrt{x}<n\leqslant x/2} \frac{|g_z(n)|}{n}\Big(\ln\frac{x}{n}\Big)^{\Re e\, z-1}$$
$$\ll_A x(\ln x)^{z-1+w} \sum_{\sqrt{x}<n\leqslant x/2} \frac{|g_z(n)|}{n} \ll_A 2^N H_N(z)x(\ln x)^{z-N-2}$$

where the final bound results from (5.38) with $h = 0$, $y = \sqrt{x}$. Using these estimates in (5.37), it follows that

$$A(x) = \sum_{n\leqslant\sqrt{x}} g_z(n)T_z\Big(\frac{x}{n}\Big) + O_A\big(2^N H_N(z)x(\ln x)^{z-N-2}\big). \tag{5.39}$$

Formula (5.36) clearly implies the existence of positive constants $c_5 = c_5(A)$ and $c_6 = c_6(A)$ such that, for $n \leqslant \sqrt{x}$, we have

$$T_z\Big(\frac{x}{n}\Big) = \frac{x}{n}\Big\{\sum_{0\leqslant k\leqslant N} \frac{\gamma_k(z)}{k!\,\Gamma(z-k)}\Big(\ln\frac{x}{n}\Big)^{z-k-1} + O_A\big(R_N(x;c_5,c_6)\big)\Big\}. \tag{5.40}$$

The desired estimate for $A(x)$ will be obtained by expanding, in the previous formula, the terms $(\ln x - \ln n)^{z-k-1}$ by means of the generalized binomial formula, and then substituting in (5.39).

For $|\xi| \leqslant \frac12$, $0 \leqslant k \leqslant N$, we have

$$(1-\xi)^{z-k-1} = \sum_{0\leqslant h\leqslant N-k} \binom{z-k-1}{h}(-\xi)^h + O_A\big(3^N|\xi|^{N-k+1}\big).$$

This readily follows by estimating the binomial coefficients using Cauchy's formula for a disk of radius $2/3$, and we omit the details. Substituting $\xi = (\ln n)/\ln x$, and using the bound

$$\frac{\gamma_k(z)}{k!\,\Gamma(z-k)} \ll_A (c_7N+1)^N \qquad (|z| \leqslant A,\ 0 \leqslant k \leqslant N),$$

which follows from (5.7) and the reflection formula for the Γ function, we may therefore rewrite the main term of (5.40) as

$$\frac{x}{n}\Big\{\sum_{0\leqslant m\leqslant N}(\ln x)^{z-m-1}\sum_{h+k=m}\frac{\gamma_k(z)}{k!\,\Gamma(z-k)}\binom{z-k-1}{h}(-\ln n)^h$$
$$+ O\Big((c_8N+1)^N(\ln x)^{z-N-2}(\ln n)^{N+1}\Big)\Big\}.$$

Noting that, for $h+k=m$, we have

$$\frac{1}{\Gamma(z-k)}\binom{z-k-1}{h} = \frac{1}{h!\,\Gamma(z-m)},$$

we deduce from (5.40) that

$$\sum_{n\leqslant\sqrt{x}} g_z(n)T_z\Big(\frac{x}{n}\Big) = x\Big\{\sum_{0\leqslant m\leqslant N}\frac{(\ln x)^{z-m-1}}{\Gamma(z-m)}\sum_{k+h=m}\frac{\gamma_k(z)}{k!h!}\sum_{n\leqslant\sqrt{x}}\frac{g_z(n)(-\ln n)^h}{n} + O\Big(H_N(z)R_N(x;c_5,c_9)\Big)\Big\}.$$

For each h, the inner sum over n equals

$$G^{(h)}(1;z) + O_A\Big(2^N H_N(z)(\ln x)^{h-N-1-w}\Big)$$

by (5.38). The contribution of the error term to the previous expression is then

$$\ll_A 2^N H_N(z)x(\ln x)^{z-N-w-2}\sum_{k+h\leqslant N}\frac{1}{h!\,|\Gamma(z-k-h)|}\Big(\frac{2}{\ln x}\Big)^k$$

$$\ll_A H_N(z)x(\ln x)^{z-1}\Big(\frac{c_{10}N+1}{\ln x}\Big)^{N+1}.$$

We have therefore shown that

$$\sum_{n\leqslant\sqrt{x}} g_z(n)T_z\Big(\frac{x}{n}\Big) = x(\ln x)^{z-1}\Big\{\sum_{0\leqslant m\leqslant N}\frac{\lambda_m(z)}{(\ln x)^m} + O_A\Big(H_N(z)R_N(x;c_5,c_{11})\Big)\Big\}.$$

Taking (5.39) into account, this completes the proof of the theorem. □

Notes

§ 5.1. The complex integration method allows handling the case of Dirichlet series with singularities different in nature from those of type $(s-1)^{-z}$ as considered here. Delange (1954) gave an evaluation for a singularity of type

$$(s-1)^{-\omega}\Big(\log\frac{1}{s-1}\Big)^k$$

where $\omega \in \mathbb{R}$ and k is a non-negative integer—cf. Theorem 7.28. In Exercise 210 we sketch an argument adapted to a singularity of type

$$\mathrm{e}^{\lambda/(s-1)}.$$

In all cases, it appears of crucial importance to be able to assume analytic continuation in some suitable neighborhood (from which a point or a half-line may be taken out) of the singularities.

Except in the case of $\gamma_0(z) \equiv 1$, it seems difficult to give a simple expression for $\gamma_j(z)$. Writing the analytic continuation of $\zeta(s)$ for $\sigma > 0$

$$\zeta(s) = \frac{s}{s-1} - s\int_1^\infty \frac{\langle t\rangle}{t^{s+1}}\,\mathrm{d}t$$

readily gives, for $j \geqslant 1$,

$$\gamma_j(1) = (-1)^j \int_1^\infty \langle t\rangle\,(\ln t)^{j-1}\frac{\mathrm{d}t}{t^2}.$$

In particular, we have $\gamma_1(1) = \gamma - 1$, where γ is Euler's constant.

§ 5.3. The precise form obtained by Bateman for the error term in (5.18) is $\ll x\exp\big\{-c_{12}\sqrt{\ln x\ln_2 x}\big\}$ for any constant $c_{12} < \frac{1}{2}\sqrt{2}$. The method of the present chapter gives the same result by slightly altering the contour of integration. Balazard & Smati (1990) have shown that this estimate can be precisely recovered by an elementary method. Using complex analysis, Balazard & Tenenbaum (1998) established that the error term is $\ll x\exp\big\{-c(\ln x)^{3/5}/(\ln_2 x)^{1/5}\big\}$ for a suitable constant $c > 0$. This implies, in particular, that

$$a_n := \operatorname{card}\{m : \varphi(m) = n\} \ll n\,\mathrm{e}^{-c(\ln n)^{3/5}/(\ln_2 n)^{1/5}} \qquad (n \geqslant 3).$$

In 1935, Erdős showed the existence of a constant $\delta > 0$ such that $a_n > n^\delta$ for infinitely many integers n and conjectured that this result remained true for any $\delta < 1$. In the same direction, he ventured the hypothesis that the error term (5.18) is

$$\Omega\Big(x\mathrm{e}^{-c(\ln x)/\ln_2 x}\Big)$$

for a suitable positive constant c.

§ 5.4. The proof we give here is largely inspired by that of Delange (1971), in which we have made explicit the dependence of the error term on M and N.

Formula (5.31) provides a simpler approximation for $A(x)$ if the contribution to $\Phi'(x)$ from the circular part of the Hankel contour tends to 0 with the radius. For example, under the assumptions of Theorem 5.2 and additionally assuming that $\Re e\, z \leqslant 1 - \varepsilon$, we obtain in this way that

$$A(x) = \int_0^{c_{13}} x^{1-t}\alpha(t;z)\frac{\mathrm{d}t}{t^z} + O\Big(x\mathrm{e}^{-c_{14}\sqrt{\ln x}}\Big)$$

where c_{13} and c_{14} are positive constants depending only on A and ε, and where $\alpha(t;z)$ is the continuous function on $[0, c_{13}]$ defined by

$$\alpha(t;z) = \frac{\sin \pi z}{\pi} Z(1-t;z)G(1-t;z).$$

Of course, we could have obtained this result from (5.17) by choosing $N = N(x)$ tending to infinity suitably, and writing

$$\begin{aligned}\frac{1}{\Gamma(z-k)} &= \frac{\sin \pi z}{\pi}(-1)^k\Gamma(k+1-z)\\ &= \frac{\sin \pi z}{\pi}(-1)^k \int_0^X \mathrm{e}^{-t}t^{k-z}\,\mathrm{d}t + \text{ remainder}\end{aligned}$$

with $X := c_{13}\ln x$, then interchanging summation and integration in the main term of (5.17).

Exercises

200. Let $f(n)$ be a multiplicative arithmetic function such that $f(p) = \alpha$, $f(p^\nu) \ll p^{\delta\nu}$ $(\nu \geqslant 2)$ for some $\delta < \frac{1}{2}$. Evaluate the summatory function of $f(n)$.

201. Let $g(n)$ be the number of representations of n as a product of distinct numbers of the form $p-1$, where p is prime. Estimate $\sum_{n\leqslant x} g(n)$.

202. Evaluate, uniformly for $x \geqslant 2$, $1 \leqslant q \leqslant x$, the quantity

$$\sum_{n\leqslant x,\,(n,q)=1} 1/\tau(n).$$

203. Let $\lambda > 0$. Show that the asymptotic relation

$$\sum_{n\leqslant x}(-1)^n\lambda^{\omega(n)} = o\Big(\sum_{n\leqslant x}\lambda^{\omega(n)}\Big) \qquad (x\to\infty)$$

holds, if and only if, $\lambda = 1$.

204. Establish an asymptotic formula for $\sum_{1<n\leqslant x} t^{\omega(n)-1}$ valid uniformly for $0 \leqslant t \leqslant 1$. By integration, deduce an asymptotic formula for

$$\sum_{1<n\leqslant x} 1/\omega(n).$$

205. Use the method of Exercise 204 to evaluate $\sum_{1<n\leqslant x} \mu(n)^2/\omega(n)$.

206. Same question for $\sum_{1<n\leqslant x} \varphi(n)/\omega(n)$.

207. Define the arithmetic function $r(n)$ as the number of representations of the integer n in the form $n = [a,b]$ with $a \geqslant 1$, $b \geqslant 1$.

(a) Calculate $r(p^\nu)$. Show that r is multiplicative. Argue either directly from the definition, or by first computing $\sum_{d|n} r(d)$.

(b) Show that, for all $\beta > 0$, there exist two constants $G(\beta)$ and $\gamma(\beta)$, to be determined explicitly, such that

$$\sum_{1<n\leqslant x}\frac{1}{r(n)^\beta} = \frac{x}{(\ln x)^{\gamma(\beta)}}\Big\{G(\beta)+O\Big(\frac{1}{\ln x}\Big)\Big\}. \tag{5.41}$$

Make the dependence on β of the error term explicit.

(c) By integrating (5.41) with respect to β over $[0,1]$ show that

$$\sum_{1<n\leqslant x}\frac{1}{\ln r(n)} \sim \frac{Kx}{\ln_2 x} \qquad (x\to\infty),$$

where K is a constant to be made explicit.

208. (a) Let f and g be multiplicative arithmetic functions such that $f(p) = g(p)$ for all primes p and $|f(p^\nu)| + |g(p^\nu)| \ll_\varepsilon \mathrm{e}^{\varepsilon\nu}$, uniformly in p, for all integers $\nu \geqslant 1$ and every $\varepsilon > 0$. Suppose additionally that we have

$$\inf_{\substack{p\in\mathbb{P}\\ |z|\leqslant 3/4}} \Big|\sum_{\nu\geqslant 0} g(p^\nu)z^\nu\Big| > 0.$$

Show that there exists a constant M and a Dirichlet series $H(s)$ converging for $\sigma \geqslant 3/4$ such that

$$\sup_{\sigma\geqslant 3/4} |H(s)| \leqslant M, \qquad \sum_{n\geqslant 1}\frac{f(n)}{n^s} = H(s)\sum_{n\geqslant 1}\frac{g(n)}{n^s} \quad (\sigma > 1).$$

(b) For $\vartheta \in \mathbb{R}$, $n \in \mathbb{N}^*$, define $\tau(n,\vartheta) := \sum_{d|n} d^{i\vartheta}$. Show that, for all $p \in \mathbb{P}$, $\vartheta \in \mathbb{R}$, we have $|\tau(p,\vartheta)|^2 = 2 + 2\cos(\vartheta \ln p)$. By applying appropriately the result proved in (a), deduce that, for $\sigma > 1$ and $\vartheta \in \mathbb{R}$, we have

$$F_\vartheta(s) := \sum_{n\geqslant 1}\frac{\mu(n)^2|\tau(n,\vartheta)|^2}{\tau(n)n^s} = \zeta(s)\zeta(s+i\vartheta)^{1/2}\zeta(s-i\vartheta)^{1/2}H_\vartheta(s)$$

where $H_\vartheta(s)$ is a Dirichlet series converging and uniformly bounded in the half-plane $\sigma \geqslant \frac{3}{4}$. Deduce that, for $|\vartheta| \leqslant 1$, there exists a constant $a > 0$, such that, writing $\sigma_a(\tau) := a/\ln(3+|\tau|)$, the function $F_\vartheta(s)$ is meromorphically continuable to the domain

$$\mathcal{D} := \{s : \sigma \geqslant 1-\sigma_a(\tau)\} \setminus \Big([1-\sigma_a(\vartheta)+i\vartheta, 1+i\vartheta] \cup [1-\sigma_a(\vartheta)-i\vartheta, 1-i\vartheta]\Big).$$

(c) By noticing that

$$|1-p^{i\vartheta-1}| = \{1-c_p/p\}\sqrt{1+s_p^2/(p-c_p)^2}$$

with $c_p := \cos(\vartheta\ln p)$, $s_p := \sin(\vartheta\ln p)$, show that

$$H_\vartheta(1) = \prod_p \Big(1+\frac{s_p^2}{(p-c_p)^2}\Big)^{1/2}\Big(1-\frac{1+c_p+c_p^2}{p^2}+\frac{c_p+c_p^2}{p^3}\Big).$$

Deduce that $H_\vartheta(1) \gg 1$ for $|\vartheta| \leqslant 1$.

(d) Show that $\tau(n) \leqslant 2\sum_{d|n,\, d\leqslant\sqrt{n}} 1$ for any integer $n \geqslant 1$. Deduce that, uniformly for $x \geqslant 1$ and $\sqrt{x} \leqslant y \leqslant x$, we have

$$\sum_{|n-x|\leqslant y} \tau(n) \ll y\ln x.$$

(e) Show that, uniformly for $x \geqslant 3$ and $\vartheta \in \mathbb{R}$, we have

$$D(x;\vartheta) := \sum_{n\leqslant x}\frac{\mu(n)^2|\tau(n,\vartheta)|^2}{\tau(n)} = \frac{1}{2\pi i}\int_{\alpha-iT}^{\alpha+iT} F_\vartheta(s)x^s\frac{\mathrm{d}s}{s} + O\big(x\mathrm{e}^{-\sqrt{\ln x}}\big)$$

with $T := \mathrm{e}^{3\sqrt{\ln x}}$, $\alpha := 1 + 1/\ln x$.

(f) Assume from now on that $1/\ln x \leqslant |\vartheta| \leqslant 1$. Shift the integration segment to the left until $\sigma = 1 - \frac{1}{2}\sigma_a(T)$ producing truncated Hankel contours with radii $r := 1/(2\ln x)$ around the cuts $[1 - \sigma_a(\vartheta) \pm i\vartheta, 1 \pm i\vartheta]$. In particular, set

$$S_\pm := \{\sigma + i(\vartheta \pm 0) : 1 - \sigma_a(\vartheta) \leqslant \sigma \leqslant 1 - r\}$$
$$C := \{1 + i\vartheta + re^{i\varphi} : -\pi < \varphi < \pi\},$$

and define a positively oriented Hankel contour by $\mathcal{H} := S_- \cup C \cup S_+$. Show that there exists a constant $b > 0$ such that, for $x \geqslant x_0(b)$ and $1/\ln x \leqslant |\vartheta| \leqslant 1$, we have

$$D(x;\vartheta) = x|\zeta(1+i\vartheta)|H_\vartheta(1) + 2\Re e\, I(x;\vartheta) + O\big(xe^{-b\sqrt{\ln x}}\big),$$

with

$$I(x;\vartheta) := \frac{1}{2\pi i}\int_{\mathcal{H}} F_\vartheta(s)x^s\frac{ds}{s}.$$

(g) Show that for $\vartheta_0 \in \mathbb{R}$, $0 < |\vartheta_0| \leqslant 2$, $w \in \mathbb{C}$, $|w| \leqslant \frac{1}{2}|\vartheta_0|$, we have

$$\zeta(1 + i\vartheta_0 + w) = \zeta(1+i\vartheta_0)\{1 + O(|w|/|\vartheta_0|)\}.$$

Deduce that

$$I(x;\vartheta) = \frac{A(\vartheta)x^{1+i\vartheta}}{\sqrt{\pi\ln x}}\Big\{1 + O\Big(\frac{1}{|\vartheta|\ln x}\Big)\Big\},$$

where we have put $A(\vartheta) := \zeta(1+i\vartheta)\zeta(1+2i\vartheta)^{1/2}H_\vartheta(1+i\vartheta)/(1+i\vartheta)$.

(h) Show that

$$\int_0^1 D(x;\vartheta)\,d\vartheta \gg x\ln_2 x.$$

209. *Hooley's Δ-function: an average lower bound.*[1]

For $n \geqslant 1$ and $u \in \mathbb{R}$, define

$$\Delta(n,u) := \sum_{\substack{d\mid n\\ e^{u-1/2}<d\leqslant e^{u+1/2}}} 1, \qquad \Delta(n) := \sup_{u\in\mathbb{R}}\Delta(n,u).$$

(a) Let $w(\vartheta) := \frac{1}{2\pi}\Big(\frac{\sin\vartheta/2}{\vartheta/2}\Big)^2$, so that

$$\widehat{w}(u) := \int_{-\infty}^{\infty} w(\vartheta)e^{i\vartheta u}\,d\vartheta = \max(1-|u|,0) \qquad (u\in\mathbb{R}).$$

Show that

$$\int_{-\infty}^{\infty} w(\vartheta)|\tau(n,\vartheta)|^2\,d\vartheta \leqslant 2\Delta(n)\tau(n) \qquad (n\geqslant 1).$$

[1] See Hall & Tenenbaum (1982).

(b) From the above and the asymptotic lower bound established in Exercise 208(h), deduce that

$$\sum_{n\leqslant x}\Delta(n) \gg x\ln_2 x.$$

210. Let $F(s) := \sum_n a_n/n^s$ be a Dirichlet series with non-negative coefficients, converging for $\sigma > 1$. Assume that $F(s)$ can be extended by continuity to $\sigma = 1$, $s \neq 1$, and holomorphically to the punctured disk $0 < |s-1| < c$. Also assume that the extension satisfies the conditions

(i) $F(s) \ll (1+|\tau|)^{1-\delta}\sigma \geqslant 1,\ |\tau| \geqslant 1)$

(ii) $F(s) = \exp\left\{\dfrac{\lambda}{s-1}\right\}\sum_{n\geqslant 0}\alpha_n(s-1)^n \qquad (0 < |s-1| < c)$

where δ is a fixed number in $]0,1[$, $\lambda > 0$, and $\sum \alpha_n\xi^n$ is a power series with radius of convergence at least equal to c. Write $A(x) := \sum_{n\leqslant x} a_n$.

(a) Show that

$$\int_0^x A(t)\,\mathrm{d}t = \frac{1}{2\pi i}\int_{\mathcal{C}} F(s)x^{s+1}\frac{\mathrm{d}s}{s(s+1)} + O(x^2),$$

where $\mathcal{C}$ is the semi-circle $s = 1 + \frac{1}{2}ce^{i\vartheta}$, $-\frac{\pi}{2} \leqslant \vartheta \leqslant \frac{\pi}{2}$.

(b) In the above s-integral, use the change of variable $s = 1 + 1/w$ to establish the existence of a Laurent series $L(w) := \alpha_0 + \sum_{n\geqslant 1}\beta_n/w^n$, converging for sufficiently large $|w|$, and such that we have

$$\int_0^x A(t)\,\mathrm{d}t = \tfrac{1}{2}I(x) + O(x^2),$$

with

$$I(x) := \frac{x^2}{2\pi i}\int_{R-i\infty}^{R+i\infty}\exp\{w^{-1}\ln x + \lambda w\}L(w)\frac{\mathrm{d}w}{w^2} \qquad (R \geqslant R_0).$$

(c) Select $R := \sqrt{(\ln x)/\lambda}$ from now on. Show that $I(x)$ is infinitely differentiable, and the contribution to $I'(x)$ from the domain $|\tau| \geqslant (\ln x)^{1/3}$ is

$$\ll x\exp\left\{2\sqrt{\lambda\ln x} - \lambda^{3/2}(\ln x)^{1/6}\right\}.$$

(d) By expanding $(\ln x)/w + \lambda w$ as a power series up to order 4 for $|\tau| \leqslant (\ln x)^{1/3}$, show that the contribution to $I'(x)$ of the domain $|\tau| \leqslant (\ln x)^{1/3}$ equals

$$\frac{\lambda^{1/4}}{\sqrt{\pi}}\,x\,\frac{\mathrm{e}^{2\sqrt{\lambda\ln x}}}{(\ln x)^{3/4}}\left\{\alpha_0 + O\Big(\frac{1}{\sqrt{\ln x}}\Big)\right\}.$$

(e) Show that $I''(x) \ll I'(x)/x$.

(f) Using the monotonicity of the function $A(t)$, deduce from the previous results that

$$A(x) = \frac{\lambda^{1/4}}{2\sqrt{\pi}}\, x\, \frac{\mathrm{e}^{2\sqrt{\lambda \ln x}}}{(\ln x)^{3/4}} \Big\{ \alpha_0 + O\Big(\frac{1}{\sqrt{\ln x}}\Big) \Big\}.$$

211. *Oppenheim's problem on "factorisatio numerorum".*

Let $Q(n)$ be the number of ways of decomposing n as a product of factors > 1, regardless of order. Thus

$$Q(n) := \big| \{ (\nu_2, \nu_3, \ldots) : \prod_{j \geqslant 2} j^{\nu_j} = n \} \big|.$$

(a) Show that the Dirichlet series $F(s) := \sum_{n \geqslant 1} Q(n)/n^s$ admits the product representation

$$F(s) = \prod_{j \geqslant 2} \Big(1 - \frac{1}{j^s} \Big)^{-1}.$$

(b) Show that $F(s)$ satisfies the hypotheses of Exercise 210 with $\lambda = 1$.

(c) Establish Oppenheim's formula (1927)

$$\sum_{n \leqslant x} Q(n) = x \frac{\mathrm{e}^{2\sqrt{\ln x}}}{2\sqrt{\pi}(\ln x)^{3/4}} \Big\{ 1 + O\Big(\frac{1}{\sqrt{\ln x}}\Big) \Big\}.$$

212. *A theorem of Diamond (1984).*

Define $F_0(n) := \delta(n)$,[2] and, for $k \geqslant 1$,

$$F_k(n) := \sum_{\substack{\nu_1 \nu_2 \cdots \nu_k = n \\ \nu_j > 1 \ (j=1,\ldots,k)}} \prod_{j=1}^{k} \frac{1}{\ln \nu_j}.$$

Let F be the arithmetic function defined by

$$F(n) := \sum_{k \geqslant 0} F_k(n)/k! \quad (n \geqslant 1).$$

(a) Show that

$$\sum_{n \geqslant 1} \frac{F(n)}{n^s} = \zeta(s) \exp\Big\{ - \sum_{n \geqslant 2} \frac{\Lambda(n) - 1}{n^s \ln n} \Big\} \qquad (\sigma > 1).$$

(b) Establish the asymptotic formula

$$\sum_{n \leqslant x} F(n) = Kx + O\big(x \mathrm{e}^{-c\sqrt{\ln x}}\big) \qquad (x \to \infty)$$

[2] See p. 32.

with

$$K := \exp\Big\{\sum_{n\geqslant 2}\{1-\Lambda(n)\}/(n\ln n)\Big\} \approx 1.24292.$$

213. Show that, uniformly for $x \geqslant 2$ and $T > 0$, we have

$$\sum_{n\leqslant x,\,\omega(n)\leqslant T} 1 \ll x2^T/\sqrt{\ln x}.$$

Using the canonical decomposition of an integer n in the form $n = qs$ with $\mu(q)^2 = 1$, $(q,s) = 1$, $p|s \Rightarrow p^2|s$, show that, for each fixed integer $k \geqslant 1$, we have

$$\sum_{n\leqslant x,\,k|\tau(n)} 1 = c_k x + O\Big(x/\sqrt{\ln x}\Big).$$

For any prime number p, write $1 - c_p$ as an Eulerian product.

214. *Using the large sieve.* For $n \in \mathbb{Z}$ and $q \in \mathbb{N}^*$, denote by $r_q(n)$ the representative in $[0,q[$ of the residue class of n modulo q. Given $M \in \mathbb{N}$ and $N \in \mathbb{N}^*$, define

$$E(M,N) := \Big|\Big\{n \in]M, M+N] : (\forall p \in \mathbb{P})\ 0 \leqslant r_p(n) \leqslant \tfrac{2}{3}p\Big\}\Big|,$$

and write $E_N := \sup_{M\geqslant 0} E(M,N)$. We intend here to provide an upper bound for E_N as $N \to \infty$.

(a) Show that $E(0,10) = 4$.

(b) Let g be the multiplicative function defined by

$$(\forall p \in \mathbb{P}) \qquad g := \frac{p-1-\lfloor 2p/3\rfloor}{1+\lfloor 2p/3\rfloor}, \qquad g(p^\nu) := 0 \quad (\nu \geqslant 2).$$

Show that $g(p) = \frac{1}{2} + O(1/p)$ $(p \in \mathbb{P})$. Deduce that

$$\Big(1+\frac{g(p)}{p^s}\Big)\Big(1-\frac{1}{p^s}\Big)^{1/2} = 1+O\Big(\frac{1}{p^{\sigma+1}}+\frac{1}{p^{2\sigma}}\Big) \qquad (s=\sigma+i\tau, \sigma>0,\ p\in\mathbb{P}),$$

and then establish the existence of a function $H(s)$, holomorphic and bounded in the half-plane $\sigma = \Re e\, s \geqslant \frac{3}{4}$, such that the Dirichlet series $G(s) := \sum_{n\geqslant 1} g(n)/n^s$ has a holomorphic continuation of the type $G(s) = H(s)\zeta(s)^{1/2}$ in the domain

$$\mathcal{D} := \Big\{s = \sigma + i\tau \in \mathbb{C} : \sigma \geqslant 1 - \frac{c}{1+\ln^+|\tau|},\ s \notin [1-c,1]\Big\},$$

where c is a suitable positive constant.

(c) Estimate $L(Q) := \sum_{q\leqslant Q} g(q)$ as $Q \to \infty$.

(d) Using the large sieve inequality, show that

$$E_N \leqslant \Big\{1 + O\Big(\frac{1}{\ln N}\Big)\Big\}\frac{\sqrt{2\pi N \ln N}}{H}.$$

where $H := \prod_p \big(1 + g(p)/p\big)\big(1 - 1/p\big)^{1/2}$.

Two arithmetic applications

Here we intend to employ the results of the previous chapter, in particular Theorem 5.4, in order to investigate two concrete arithmetic problems.

6.1. Integers having k prime factors

The prime number theorem makes it easy to show by induction on the integer $k \geqslant 1$ that, for each fixed k, we have

$$N_k(x) := |\{n \leqslant x : \Omega(n) = k\}| \sim \frac{x}{\ln x} \frac{(\ln_2 x)^{k-1}}{(k-1)!} \quad (x \to \infty)$$

—see Landau (1909). The same result holds for the function

$$\pi_k(x) := |\{n \leqslant x : \omega(n) = k\}|.$$

Thus, the local laws on $[0, x]$ of the arithmetic functions Ω and ω are, in a certain sense, close to a Poisson law with parameter $\ln_2 x$. However, when k is allowed to tend to infinity with x, the study by induction of the asymptotic behavior of $\pi_k(x)$ or $N_k(x)$ becomes very technical—see Sathe (1953, 1954). Another line of attack, devised by Selberg (1954), consists, for example in the case of $\omega(n)$, in identifying $\pi_k(x)$ as the coefficient of z^k in the expression

$$\sum_{n \leqslant x} z^{\omega(n)} \tag{6.1}$$

and then applying Cauchy's integral formula. This program necessitates a good estimate for the sum (6.1), which will be provided by Theorem 5.2 or Theorem 5.4.

Indeed, let us consider the Dirichlet series

$$F_1(s; z) := \sum_{n \geqslant 1} \frac{z^{\omega(n)}}{n^s} = \prod_p \left(1 + \frac{z}{p^s - 1}\right) \qquad (\sigma > 1).$$

Then the function

$$G_1(s;z) := F_1(s;z)\zeta(s)^{-z} = \prod_p \Big(1 + \frac{z}{p^s - 1}\Big)\Big(1 - \frac{1}{p^s}\Big)^z$$

is expandable as a Dirichlet series $G_1(s;z) = \sum_{n\geqslant 1} b_{1z}(n)/n^s$ where b_{1z} is the multiplicative function whose values on prime powers are determined by the identity

$$1 + \sum_{\nu\geqslant 1} b_{1z}(p^\nu)\xi^\nu = \Big(1 + \frac{\xi z}{1-\xi}\Big)(1-\xi)^z \qquad (|\xi| < 1).$$

In particular, we have

$$b_{1z}(p) = 0 \tag{6.2}$$

and Cauchy's inequality implies that, for $|z| \leqslant A$,

$$\big|b_{1z}(p^\nu)\big| \leqslant M\cdot 2^{\nu/2} \qquad (\nu \geqslant 2) \tag{6.3}$$

with

$$M = M(A) := \sup_{|z|\leqslant A,\,|\xi|\leqslant 1/\sqrt{2}} \Big|\Big(1 + \frac{\xi z}{1-\xi}\Big)(1-\xi)^z\Big|.$$

Relations (6.2) and (6.3) show that we have, for $\sigma > \frac12$,

$$\sum_p \sum_{\nu\geqslant 1} \frac{\big|b_{1z}(p^\nu)\big|}{p^{\nu\sigma}} \leqslant 2M \sum_p \frac{1}{p^\sigma(p^\sigma - \sqrt{2})} \leqslant \frac{cM}{\sigma - 1/2}$$

where c is an absolute constant. By Theorem 1.3, we deduce that $G_1(s;z)$ is absolutely convergent for $\sigma > \frac12$ and that, for $\sigma \geqslant \frac34$, we have

$$G_1(s;z) \ll_A 1.$$

The assumptions of Theorem 5.2 are therefore satisfied, and we can formulate the following result.

Theorem 6.1. *For any positive constant A, there exist positive constants $c_1 = c_1(A)$ and $c_2 = c_2(A)$, such that, uniformly for $x \geqslant 3$, $N \geqslant 0$, $|z| \leqslant A$, we have*

$$\sum_{n\leqslant x} z^{\omega(n)} = x(\ln x)^{z-1}\Big\{\sum_{0\leqslant k\leqslant N} \frac{\lambda_k(z)}{(\ln x)^k} + O_A(R_N(x))\Big\} \tag{6.4}$$

with

$$R_N(x) := \mathrm{e}^{-c_1\sqrt{\ln x}} + \Big(\frac{c_2 N + 1}{\ln x}\Big)^{N+1} \tag{6.5}$$

and

$$\lambda_k(z) := \frac{1}{\Gamma(z-k)} \sum_{h+j=k} \frac{1}{h!\,j!} G_1^{(h)}(1;z)\gamma_j(z), \tag{6.6}$$

where the $\gamma_j(z)$ are the entire functions defined in Theorem 5.1.

It should be noted that $\lambda_k(0) = 0$ for all k. In particular, we write

$$\lambda_0(z) = z\lambda(z), \quad \text{with} \quad \lambda(z) := \frac{G_1(1;z)}{\Gamma(z+1)}. \tag{6.7}$$

Of course we can carry out a parallel study for

$$F_2(s;z) := \sum_{n\geqslant 1} \frac{z^{\Omega(n)}}{n^s} = \prod_p \Big(1 - \frac{z}{p^s}\Big)^{-1}.$$

The coefficient $b_{2z}(n)$ of $1/n^s$ in

$$G_2(s;z) := F_2(s;z)\zeta(s)^{-z} = \prod_p \Big(1 - \frac{z}{p^s}\Big)^{-1}\Big(1 - \frac{1}{p^s}\Big)^{z}$$

is again a multiplicative function of n, determined by the identity

$$1 + \sum_{\nu\geqslant 1} b_{2z}(p^\nu)\xi^\nu = \frac{(1-\xi)^z}{1-\xi z}.$$

As before we have $b_{2z}(p) = 0$, but in order to bound $|b_{2z}(p^\nu)|$, we must take account of the fact that the right-hand side is only holomorphic in ξ inside the disc $|\xi| < \min\big(1, 1/|z|\big)$. For all δ, $0 < \delta < 1$, Cauchy's integral formula gives

$$|b_{2z}(p^\nu)| \leqslant M(\delta)(2-\delta)^\nu \qquad (|z| \leqslant 2 - 2\delta)$$

with

$$M(\delta) := \sup_{\substack{|\xi|\leqslant 1/(2-\delta)\\ |z|\leqslant 2-2\delta}} \left|\frac{(1-\xi)^z}{1-\xi z}\right|.$$

This implies the absolute convergence of $G_2(s;z)$ for $\sigma \geqslant 1 - c_0(\delta)$ and Theorem 5.2 yields the result stated below.

Theorem 6.2. *For all δ, $0 < \delta < 1$, there exist positive constants $c_1 = c_1(\delta)$ and $c_2 = c_2(\delta)$ such that, uniformly for $x \geqslant 3$, $N \geqslant 0$, $|z| \leqslant 2 - \delta$, we have*

$$\sum_{n\leqslant x} z^{\Omega(n)} = x(\ln x)^{z-1}\Big\{\sum_{0\leqslant k\leqslant N} \frac{\nu_k(z)}{(\ln x)^k} + O_\delta(R_N(x))\Big\} \tag{6.8}$$

where $R_N(x)$ is defined by (6.5) *and where we have set*

$$\nu_k(z) := \frac{1}{\Gamma(z-k)} \sum_{h+j=k} \frac{1}{h!\,j!} G_2^{(h)}(1;z)\gamma_j(z).$$

As before, we have $\nu_k(0) = 0$ $(k \geqslant 0)$. We write

$$\nu_0(z) = z\nu(z), \text{ with } \nu(z) := \frac{G_2(1;z)}{\Gamma(z+1)}. \tag{6.9}$$

Starting from (6.4) and (6.8), the desired evaluations for $\pi_k(x)$ and $N_k(x)$ are direct consequences of the following general theorem.

Theorem 6.3. *Let $a_z(n)$ be an arithmetic function depending on a complex parameter z and with a power series expansion in the disk $|z| \leqslant A$*

$$a_z(n) = \sum_{k\geqslant 0} c_k(n) z^k \qquad (n = 1, 2, \ldots).$$

Let N be a non-negative integer. Suppose that there exist $N+1$ functions $h_j(z)$ $(0 \leqslant j \leqslant N)$, holomorphic for $|z| \leqslant A$, and a quantity $R_N(x)$, independent of z, such that, for $x \geqslant 3$ and $|z| \leqslant A$, we have

$$\sum_{n\leqslant x} a_z(n) = x(\ln x)^{z-1}\Big\{ \sum_{0\leqslant j\leqslant N} \frac{z h_j(z)}{(\ln x)^j} + O_A\big(R_N(x)\big)\Big\}. \tag{6.10}$$

Then, uniformly for $x \geqslant 3$ and $1 \leqslant k \leqslant A\ln_2 x$, we have

$$\begin{aligned} C_k(x) &:= \sum_{n\leqslant x} c_k(n) \\ &= \frac{x}{\ln x}\Big\{ \sum_{0\leqslant j\leqslant N} \frac{Q_{j,k}(\ln_2 x)}{(\ln x)^j} + O_A\Big(\frac{(\ln_2 x)^k}{k!} R_N(x)\Big)\Big\}, \end{aligned} \tag{6.11}$$

where we have put

$$Q_{j,k}(X) := \sum_{m+\ell=k-1} \frac{1}{m!\,\ell!} h_j^{(m)}(0) X^\ell \qquad (0 \leqslant j \leqslant N,\ k \geqslant 1) \tag{6.12}$$

If, in addition, we suppose that $|h_0''(z)| \leqslant B$ for $|z| \leqslant A$, we have, uniformly for $x \geqslant 3$ and $1 \leqslant k \leqslant A\ln_2 x$,

$$C_k(x) = \frac{x}{\ln x}\frac{(\ln_2 x)^{k-1}}{(k-1)!}\Big\{ h_0\Big(\frac{k-1}{\ln_2 x}\Big) + O_A\Big(\frac{B\cdot(k-1)}{(\ln_2 x)^2} + \frac{\ln_2 x}{k} R_0(x)\Big)\Big\}. \tag{6.13}$$

Proof. First of all we prove formula (6.11). Since the main term is the coefficient of z^k in the main term of (6.10), it suffices to evaluate the error term. By Cauchy's formula, this is

$$\ll_A \oint_{|z|=r} \frac{x R_N(x)}{\ln x} (\ln x)^{\Re e\, z} |z|^{-k-1} |\mathrm{d}z| \tag{6.14}$$

for all $r \leqslant A$. Select $r := k/\ln_2 x$. We have

$$\oint_{|z|=r} (\ln x)^{\Re e\, z} |z|^{-k-1} |\mathrm{d}z| = \Big(\frac{\ln_2 x}{k}\Big)^k \int_0^{2\pi} \mathrm{e}^{k\cos\vartheta}\,\mathrm{d}\vartheta$$

and

$$\begin{aligned} \int_0^{2\pi} \mathrm{e}^{k\cos\vartheta}\,\mathrm{d}\vartheta &\leqslant 2\int_0^{\pi/2} \mathrm{e}^{k\cos\vartheta}\,\mathrm{d}\vartheta + \pi = 2\int_0^1 \mathrm{e}^{kt}\frac{\mathrm{d}t}{\sqrt{1-t^2}} + \pi \\ &\leqslant 2\mathrm{e}^k \int_0^1 \mathrm{e}^{-k(1-t)}\frac{\mathrm{d}t}{\sqrt{1-t}} + \pi \leqslant 2\Gamma(\tfrac12)\mathrm{e}^k k^{-\frac12} + \pi. \end{aligned}$$

We thus obtain the stated result by substitution in (6.14).

When $k = 1$, formula (6.13) reduces to (6.11) for $N = 0$. Suppose therefore that $k \geqslant 2$, apply (6.11) with $N = 0$, and evaluate the main term by Cauchy's formula. For all $r \leqslant A$, we have

$$Q_{0,k}(X) = \frac{1}{2\pi i} \oint_{|z|=r} h_0(z) \mathrm{e}^{zX} z^{-k} \, \mathrm{d}z.$$

When $k \leqslant AX$, we can choose $r = (k-1)/X$. Noting that

$$\frac{1}{2\pi i} \oint_{|z|=r} (z-r) \mathrm{e}^{zX} z^{-k} \, \mathrm{d}z = \frac{X^{k-2}}{(k-2)!} - r\frac{X^{k-1}}{(k-1)!} = 0,$$

we see that

$$Q_{0,k}(X) = \frac{h_0(r)}{2\pi i} \oint_{|z|=r} \mathrm{e}^{zX} z^{-k} \, \mathrm{d}z + \frac{1}{2\pi i} \oint_{|z|=r} \{h_0(z) - h_0(r) - (z-r)h_0'(r)\} \mathrm{e}^{zX} z^{-k} \, \mathrm{d}z.$$

The first term on the right equals the stated main term $h_0(r)X^{k-1}/(k-1)!$. To evaluate the second, we note that the expression between curly brackets is equal to

$$(z-r)^2 \int_0^1 (1-t) h_0''(r + t(z-r)) \, \mathrm{d}t$$

and that we have $|r + t(z-r)| = |r(1-t) + tz| \leqslant r(1-t) + tr = r$ for each t in $[0,1]$. It follows that the modulus of the associated integral is

$$\leqslant \frac{B}{2\pi} \int_0^{2\pi} |\mathrm{e}^{i\vartheta} - 1|^2 \mathrm{e}^{rX\cos\vartheta} r^{3-k} \, \mathrm{d}\vartheta \leqslant \frac{B}{\pi} r^{3-k} \int_0^{2\pi} \mathrm{e}^{(k-1)\cos\vartheta} (1 - \cos\vartheta) \, \mathrm{d}\vartheta.$$

Bounding the integral over ϑ by

$$2\int_0^1 \mathrm{e}^{(k-1)t} \sqrt{1-t} \, \mathrm{d}t + 2\pi \leqslant 2\Gamma\left(\tfrac{3}{2}\right) \mathrm{e}^{k-1} (k-1)^{-3/2} + 2\pi$$

and applying Stirling's formula, we deduce that

$$Q_{0,k}(X) = h_0(r)\frac{X^{k-1}}{(k-1)!} + O\Big(B\frac{X^{k-1}}{(k-1)!}\frac{k-1}{X^2}\Big) \qquad (k \leqslant AX).$$

This finishes the proof of (6.13). □

Theorems 6.1, 6.2 and 6.3 now enable us to state explicit asymptotic formulae for $\pi_k(x)$ and $N_k(x)$. The role of the function $h_0(z)$ is played respectively by the functions

$$\lambda(z) := \frac{1}{\Gamma(z+1)} \prod_p \Big(1 + \frac{z}{p-1}\Big)\Big(1 - \frac{1}{p}\Big)^z \tag{6.15}$$

and

$$\nu(z) := \frac{1}{\Gamma(z+1)} \prod_p \Big(1 - \frac{z}{p}\Big)^{-1} \Big(1 - \frac{1}{p}\Big)^z. \tag{6.16}$$

Theorem 6.4. *Let $A > 0$. There exist positive constants $c_1 = c_1(A)$ and $c_2 = c_2(A)$ such that, uniformly for $x \geqslant 3$, $1 \leqslant k \leqslant A \ln_2 x$, $N \geqslant 0$, we have*

$$\pi_k(x) = \frac{x}{\ln x} \Big\{ \sum_{0 \leqslant j \leqslant N} \frac{P_{j,k}(\ln_2 x)}{(\ln x)^j} + O\Big(\frac{(\ln_2 x)^k}{k!} R_N(x)\Big) \Big\} \tag{6.17}$$

where $P_{j,k}(X)$ is a polynomial of degree at most $k-1$ and where $R_N(x)$ is defined by (6.5). *In particular, we have*

$$P_{0,k}(X) = \sum_{m+\ell=k-1} \frac{1}{m!\,\ell!} \lambda^{(m)}(0) X^\ell.$$

Moreover, under the same assumptions, we have

$$\pi_k(x) = \frac{x}{\ln x} \frac{(\ln_2 x)^{k-1}}{(k-1)!} \Big\{ \lambda\Big(\frac{k-1}{\ln_2 x}\Big) + O\Big(\frac{k}{(\ln_2 x)^2}\Big) \Big\}. \tag{6.18}$$

Theorem 6.5. *Let δ satisfy $0 < \delta < 1$. There exist strictly positive constants $c_1 = c_1(\delta)$ and $c_2 = c_2(\delta)$ such that, uniformly for $x \geqslant 3$, $1 \leqslant k \leqslant (2-\delta)\ln_2 x$, $N \geqslant 0$, we have*

$$N_k(x) = \frac{x}{\ln x} \Big\{ \sum_{0 \leqslant j \leqslant N} \frac{Q_{j,k}(\ln_2 x)}{(\ln x)^j} + O\Big(\frac{(\ln_2 x)^k}{k!} R_N(x)\Big) \Big\} \tag{6.19}$$

where $Q_{j,k}$ is a polynomial of degree at most $k-1$ and where $R_N(x)$ is defined by (6.5). *In particular, we have*

$$Q_{0,k}(X) = \sum_{m+\ell=k-1} \frac{1}{m!\,\ell!} \nu^{(m)}(0) X^\ell.$$

Moreover, under the same assumptions, we have

$$N_k(x) = \frac{x}{\ln x} \frac{(\ln_2 x)^{k-1}}{(k-1)!} \Big\{ \nu\Big(\frac{k-1}{\ln_2 x}\Big) + O\Big(\frac{k}{(\ln_2 x)^2}\Big) \Big\}. \tag{6.20}$$

The condition $k \leqslant (2-\delta)\ln_2 x$ is a natural restriction on the validity of (6.20) since $\nu(z)$ has a pole at $z = 2$. We can take advantage of this situation in order to evaluate $N_k(x)$ when $(2+\delta)\ln_2 x \leqslant k \leqslant A \ln_2 x$.

Theorem 6.6. *Assume $0 < \delta < 1$, and $A > 0$. Then, uniformly for $x \geqslant 3$ and $(2+\delta)\ln_2 x \leqslant k \leqslant A\ln_2 x$, we have*

$$N_k(x) = C\frac{x\ln x}{2^k}\Big\{1 + O\Big((\ln x)^{-\delta^2/5}\Big)\Big\}, \tag{6.21}$$

with

$$C := \tfrac{1}{4}\prod_{p>2}\Big(1 + \frac{1}{p(p-2)}\Big) \approx 0.378694.$$

Proof. Let $\varepsilon = \varepsilon(A)$ be a positive parameter to be specified later. First of all, we apply Cauchy's formula to (6.8) with $N = 0$, for the circle $|z| = 2-\varepsilon$. This gives

$$N_k(x) = \frac{x}{2\pi i}\oint_{|z|=2-\varepsilon} \nu(z)(\ln x)^{z-1}z^{-k}\,\mathrm{d}z + O_\varepsilon\Big(x(2-\varepsilon)^{-k}(\ln x)^{-\varepsilon}\Big), \tag{6.22}$$

where we have estimated trivially the contribution of the error term.

For $k \leqslant A\ln_2 x$, we have

$$(2-\varepsilon)^{-k}(\ln x)^{-\varepsilon} \leqslant 2^{-k}(\ln x)^{-A\ln(1-\varepsilon/2)-\varepsilon} \leqslant 2^{-k}(\ln x)^{1/2}$$

if $\varepsilon = \varepsilon(A)$ is sufficiently small.

It remains to evaluate the main term of (6.22). Since $\nu(z)$ has a pole at $z = 2$ with residue $-C$, we see that this quantity equals

$$C\frac{x\ln x}{2^k} - \frac{x}{2\pi i}\oint_{|z|=2+\delta} \nu(z)(\ln x)^{z-1}z^{-k}\,\mathrm{d}z.$$

The last integral is trivially

$$\ll_\delta (\ln x)^{1+\delta}(2+\delta)^{-k} \leqslant (\ln x)2^{-k}(\ln x)^{-\eta(\delta)}$$

with

$$\eta(\delta) := 2\Big\{\big(1+\tfrac{1}{2}\delta\big)\ln\big(1+\tfrac{1}{2}\delta\big) - \tfrac{1}{2}\delta\Big\} > \tfrac{1}{5}\delta^2.$$

This completes the proof. □

In Exercise 217 we propose an extension of this result to large values of k, i.e. such that $A\ln_2 x < k \leqslant (\ln x)/\ln 2$.

6.2. The average distribution of divisors: the arcsine law

In Part I, we obtained a certain amount of information on the average and extremal orders of the total number of divisors $\tau(n)$ of an integer n. A natural question to ask is whether these divisors are distributed in the in-

terval $[1, n]$ according to some definite law. Obviously, no result valid for all integers can be expected. Here we undertake a study on average.

For each integer n, let us define a random variable D_n taking the values $(\ln d)/\ln n$, as d runs through the set of the $\tau(n)$ divisors of n, with uniform probability $1/\tau(n)$. The distribution function F_n of D_n is then defined by

$$F_n(u) := \text{Prob}(D_n \leqslant u) = \frac{1}{\tau(n)} \sum_{d|n,\, d\leqslant n^u} 1 \qquad (0 \leqslant u \leqslant 1).$$

It is obvious that the sequence $\{F_n\}_{n=1}^{\infty}$ does not converge pointwise on $[0, 1]$. However, we shall see that the sequence of Cesàro means

$$G_N(u) := N^{-1} \sum_{n\leqslant N} F_n(u)$$

is *uniformly* convergent on $[0, 1]$. No less remarkably, this limit is the distribution function of a probability law well-known to specialists: the arcsine law, with density $1/\pi\sqrt{u(1-u)}$. Both large and small values have high probability: if D is a random variable with this distribution law, we have

$$\text{Prob}(D < 0.01 \text{ or } D > 0.99) \approx 0.128.$$

This indicates that, on average, an integer has many small (and correspondingly many large) divisors.

Theorem 6.7. *Uniformly for $x \geqslant 2$ and $0 \leqslant u \leqslant 1$, we have*

$$\frac{1}{x} \sum_{n\leqslant x} F_n(u) = \frac{2}{\pi} \arcsin\sqrt{u} + O\Big(\frac{1}{\sqrt{\ln x}}\Big). \tag{6.23}$$

This result is an easy consequence of the following theorem.

Theorem 6.8. *Set $h := \prod_{p\in\mathbb{P}} \sqrt{p(p-1)}\ln\{1/(1-1/p)\} \approx 0.969$. Then, uniformly for $x \geqslant 2$ and $d \geqslant 1$, we have*

$$\sum_{n\leqslant x} \frac{1}{\tau(nd)} = \frac{hx}{\sqrt{\pi \ln x}} \Big\{ g(d) + O\Big(\frac{(3/4)^{\omega(d)}}{\ln x}\Big)\Big\}, \tag{6.24}$$

where g is an arithmetic function satisfying

$$\sum_{d\leqslant x} g(d) = \frac{x}{h\sqrt{\pi \ln x}} \Big\{ 1 + O\Big(\frac{1}{\ln x}\Big)\Big\}. \tag{6.25}$$

Let us provisionally accept this theorem and see how it entails formula (6.23) above. The symmetry of the divisors of n about $\sqrt{n}$ allows us to write

$$\begin{aligned} F_n(u) &= \text{Prob}(D_n \geqslant 1-u) = 1 - \text{Prob}(D_n < 1-u) \\ &= 1 - F_n(1-u) + O\Big(\frac{1}{\tau(n)}\Big). \end{aligned}$$

Let $S(x,u)$ denote the left-hand side of (6.23). Summing the above equality for $n \leqslant x$ and evaluating the error term from (6.24) with $d = 1$, we obtain

$$S(x,u) + S(x,1-u) = 1 + O\Big(\frac{1}{\sqrt{\ln x}}\Big) \qquad (0 \leqslant u \leqslant 1).$$

Since we also have $(2/\pi)\arcsin\sqrt{u} + (2/\pi)\arcsin\sqrt{1-u} = 1$ $(0 \leqslant u \leqslant 1)$, we see that it suffices to establish (6.23) for $0 \leqslant u \leqslant \frac{1}{2}$.

With u in this range, we can write

$$S(x,u) = \frac{1}{x}\sum_{n\leqslant x}\frac{1}{\tau(n)}\sum_{d|n,\, d\leqslant n^u} 1 = \frac{1}{x}\sum_{n\leqslant x}\frac{1}{\tau(n)}\Big\{\sum_{d|n,\, d\leqslant x^u} 1 - \sum_{d|n,\, n^u<d\leqslant x^u} 1\Big\}$$

$$= \frac{1}{x}\sum_{d\leqslant x^u}\sum_{m\leqslant x/d}\frac{1}{\tau(md)} - R(x,u), \tag{6.26}$$

with

$$R(x,u) := \frac{1}{x}\sum_{d\leqslant x^u}\sum_{m\leqslant x/d,\,(md)^u<d}\frac{1}{\tau(md)} \leqslant \frac{1}{x}\sum_{d\leqslant x^u}\sum_{m\leqslant x/d,\, m\leqslant d^{(1-u)/u}}\frac{1}{\tau(m)}$$

$$\ll \frac{1}{x}\sum_{d\leqslant x^u}\frac{d^{(1-u)/u}}{\sqrt{1+\ln d^{(1-u)/u}}} \ll \frac{1}{\sqrt{1+\ln x^{1-u}}} \ll \frac{1}{\sqrt{\ln x}}.$$

Using (6.24) to estimate the inner sum in (6.26), it follows that

$$S(x,u) = \frac{h}{\sqrt{\pi}}\sum_{d\leqslant x^u}\frac{1}{d\sqrt{\ln x/d}}\Big\{g(d) + O\Big(\frac{(3/4)^{\omega(d)}}{\ln x}\Big)\Big\} + O\Big(\frac{1}{\sqrt{\ln x}}\Big)$$

where we have used the fact that $u \leqslant \frac{1}{2}$ to bound $1/(\ln x/d)$ by $O(1/\ln x)$.

Bounding $(\frac{3}{4})^{\omega(d)}$ by 1, we see that, in the summation over d, the contribution of the error terms is $\ll 1/\sqrt{\ln x}$. The main term is treated by partial integration. Let us set

$$\mathcal{G}(t) := \sum_{n\leqslant \mathrm{e}^t} g(n) = \frac{\mathrm{e}^t}{h\sqrt{\pi t}}\Big\{1 + O\Big(\frac{1}{t}\Big)\Big\} \qquad (t > 1).$$

We have

$$\begin{aligned}\frac{h}{\sqrt{\pi}}\sum_{d\leqslant x^u}\frac{g(d)}{d\sqrt{\ln x/d}} &= \frac{h}{\sqrt{\pi}}\int_{0-}^{u\ln x}\frac{\mathrm{e}^{-t}}{\sqrt{\ln x - t}}\,\mathrm{d}\mathcal{G}(t)\\ &= \frac{h}{\sqrt{\pi}}\int_0^{u\ln x}\mathcal{G}(t)\frac{\mathrm{e}^{-t}}{\sqrt{\ln x - t}}\Big\{1-\frac{1}{2(\ln x - t)}\Big\}\,\mathrm{d}t + O\Big(\frac{1}{\sqrt{\ln x}}\Big)\\ &= \frac{1}{\pi}\int_0^{u\ln x}\frac{1+O(1/(t+1))}{\sqrt{t(\ln x - t)}}\,\mathrm{d}t + O\Big(\frac{1}{\sqrt{\ln x}}\Big)\\ &= \frac{1}{\pi}\int_0^u\frac{dv}{\sqrt{v(1-v)}} + O\Big(\frac{1}{\sqrt{\ln x}}\Big) = \frac{2}{\pi}\arcsin\sqrt{u} + O\Big(\frac{1}{\sqrt{\ln x}}\Big).\end{aligned}$$

This establishes (6.23).

Proof of Theorem 6.8. Introduce the generating series

$$f_d(s) := \sum_{n\geqslant 1}\frac{1}{\tau(nd)n^s} \qquad (d=1,2,\ldots).$$

Although the function $n\mapsto\tau(nd)$ is not multiplicative, the formula

$$\tau(nd) = \prod_p (v_p(n)+v_p(d)+1)$$

allows expressing $f_d(s)$ as a product of Eulerian type, namely

$$f_d(s) = \prod_p\sum_{\nu\geqslant 0}\frac{p^{-\nu s}}{\nu+v_p(d)+1} = g_d(s)f_1(s),$$

with

$$g_d(s) := \prod_{p^\alpha\| d}\Big(\sum_{\nu\geqslant 0}\frac{p^{-\nu s}}{\nu+\alpha+1}\Big)\Big(\sum_{\nu\geqslant 0}\frac{p^{-\nu s}}{\nu+1}\Big)^{-1}.$$

For each d, $g_d(s)$ is a finite product of ratios of series which are each absolutely convergent for $\sigma>0$. For $\sigma\geqslant\frac{3}{4}$, the absolute values of the denominators are bounded from below by an absolute positive constant, indeed

$$\Big|\sum_{\nu\geqslant 0}\frac{p^{-\nu s}}{\nu+1}\Big| \geqslant 1-\sum_{\nu\geqslant 1}\frac{p^{-\nu\sigma}}{\nu} = 1+\ln\Big(1-\frac{1}{p^\sigma}\Big) \geqslant 1+\ln(1-2^{-3/4}) > 0.$$

That implies, for $\sigma\geqslant\frac{3}{4}$,

$$|g_d(s)| \leqslant \prod_{p^\alpha\| d}\Big\{\frac{1}{\alpha+1}+O\big(p^{-\sigma}\big)\Big\} \leqslant C\big(\tfrac{3}{4}\big)^{\omega(d)}, \tag{6.27}$$

where C is an absolute constant.

Now, we can also write $h(s) := f_1(s)\zeta(s)^{-1/2} = \sum_{n\geqslant 1} b(n)/n^s$ where $b(n)$ is the multiplicative function determined by the identity

$$\sum_{\nu\geqslant 0} b(p^\nu)\xi^\nu = \sum_{\nu\geqslant 0} \frac{\xi^\nu}{\nu+1}(1-\xi)^{1/2}.$$

In particular, $b(p) = 0$ for all p and

$$b(p^\nu) \ll \left(\tfrac{3}{2}\right)^\nu \qquad (\nu = 2, 3, \ldots).$$

This implies the absolute convergence of $h(s)$ for $\sigma > \frac{1}{2}$. The hypotheses of Theorem 5.2 are therefore satisfied for $f_d(s)$, with $z = \frac{1}{2}$ and $M \ll (3/4)^{\omega(d)}$. From this, noting that $h(1) = h$, we infer the validity of (6.24) with

$$g(d) = g_d(1).$$

It remains to show (6.25). To this end, we appeal once more to the Selberg–Delange Theorem 5.2 (or 5.4), noting that g is the multiplicative function defined by

$$g(p^\nu) := \Big(\sum_{j\geqslant 0} \frac{p^{-j}}{j+\nu+1}\Big)\Big(\sum_{j\geqslant 0} \frac{p^{-j}}{j+1}\Big)^{-1}.$$

For $\sigma > 0$, we can write

$$\sum_{n\geqslant 1} \frac{g(n)}{n^s} = \zeta(s)^{1/2} \sum_{n\geqslant 1} \frac{\beta(n)}{n^s}$$

where β is the multiplicative function determined by

$$\sum_{\nu\geqslant 1} \beta(p^\nu)\xi^\nu = (1-\xi)^{1/2} \sum_{\nu\geqslant 0} g(p^\nu)\xi^\nu \qquad (|\xi| < 1). \tag{6.28}$$

Since $|g(p^\nu)| \leqslant 1$, the right-hand side is holomorphic for $|\xi| < 1$ and we have

$$\beta(p^\nu) \ll (\tfrac{3}{2})^\nu \qquad (\nu = 1, 2, \ldots).$$

Moreover

$$\beta(p) = g(p) - \tfrac{1}{2} \ll 1/p.$$

This implies the absolute convergence of $\sum \beta(n)/n^s$ for $\sigma > \frac{1}{2}$.

Theorem 5.2 then provides the value

$$\sum_{n\leqslant x} g(n) = \frac{Bx}{\sqrt{\pi \ln x}}\Big\{1 + O\Big(\frac{1}{\ln x}\Big)\Big\}, \tag{6.29}$$

with

$$B := \prod_p \sum_{\nu\geqslant 0} \beta(p^\nu)/p^\nu.$$

By (6.28), we can rewrite the factor with index p in this product as

$$\Big(\sum_{\nu\geqslant 0} g(p^\nu)p^{-\nu}\Big)\Big(1-\frac{1}{p}\Big)\Big(1-\frac{1}{p}\Big)^{-1/2}$$
$$= \Big(\sum_{j\geqslant 0}\frac{p^{-j}}{j+1}\Big)^{-1}\Big(1-\frac{1}{p}\Big)^{-1/2}\sum_{\nu\geqslant 0}\Big(1-\frac{1}{p}\Big)\sum_{j\geqslant 0}\frac{p^{-\nu-j}}{\nu+j+1}$$
$$= \Big(\sum_{j\geqslant 0}\frac{p^{-j}}{j+1}\Big)^{-1}\Big(1-\frac{1}{p}\Big)^{-1/2} = \Big\{\sqrt{p(p-1)}\ln\Big(\frac{1}{1-1/p}\Big)\Big\}^{-1}.$$

Therefore we obtain

$$B = 1/h,$$

which concludes the proof of (6.25). □

Notes

§ 6.1. For an approach using induction on k to the study of $N_k(x)$ and $\pi_k(x)$ when k is bounded, see Hardy & Wright (1938), § 22.18. Theorems 6.1–6.5 are due to Selberg (1954) in the case $N = 0$. For arbitrary but fixed $N \geqslant 0$, they are special cases of results obtained by Delange (1971).

The asymptotic formula contained in the estimate of Theorem 6.6 was formulated by Selberg (1954). Using an elementary method, Nicolas (1984) extended this result by showing that, uniformly for $x \geqslant 3$, and

$$(2+\delta)\ln_2 x \leqslant k \leqslant \ln x/\ln 2,$$

we have

$$N_k(x) = Cy\ln y\Big\{1 + O\big((\ln y)^{-\eta}\big)\Big\} \tag{6.30}$$

with $y := x/2^k$ and $\eta = \eta(\delta) > 0$. Nicolas uses the fact that, for these large values of k, a dominating contribution to $N_k(x)$ arises from integers of the form $2^\ell m$ with $\ell \gg k$ and $\Omega(m) \sim 2\ln_2 y$—cf. Exercise 217, where we outline an analytic proof based on the Selberg–Delange method.

Balazard, Delange & Nicolas (1988) make precise the behavior of $N_k(x)$ when k is close to $2\ln_2 x$: for $|k - 2\ln_2 x| \leqslant A\sqrt{\ln_2 x}$, we have

$$N_k(x) = C\Phi\Big(\frac{k-2\ln_2 x}{\sqrt{2\ln_2 x}}\Big)\frac{x\ln x}{2^k}\Big\{1 + O_A\Big(\frac{1}{\sqrt{\ln_2 x}}\Big)\Big\}$$

where

$$\Phi(z) := \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{z} \mathrm{e}^{-t^2/2}\,\mathrm{d}t.$$

In his thesis, Balazard (1987) unified the study of $N_k(x)$ by giving an analytic approximation to $N_k(x)$ valid uniformly for $1 \leqslant k \leqslant (\ln x)/\ln 2$. See also Balazard, Delange & Nicolas (1988).

Under additional assumptions on higher-order derivatives of $h_0(z)$, it is possible to make formula (6.13) more precise. For example, if $h_0^{(4)}(z) \ll B$ for $|z| \leqslant A$, we obtain (with $r := (k-1)/\ln_2 x$)

$$C_k(x) = \frac{x}{\ln x}\frac{(\ln_2 x)^{k-1}}{(k-1)!}\left\{h_0(r) - \frac{rh_0''(r)}{2\ln_2 x} + O\left(\frac{B(k-1)^2}{(\ln_2 x)^4} + \frac{\ln_2 x}{k}R_0(x)\right)\right\}.$$

This immediately entails corresponding refinements for Theorems 6.4 and 6.5.

The problem of the asymptotic behavior of $\pi_k(x)$ for $k > A\ln_2 x$ is much more complicated than that of $N_k(x)$ since no single prime number plays

a special role. One had to wait many years before the limit of validity for Selberg's formula

$$\pi_k(x) \sim \lambda\Big(\frac{k-1}{\ln_2 x}\Big)\frac{x}{\ln x}\frac{(\ln_2 x)^{k-1}}{(k-1)!} \tag{6.31}$$

was determined. Hensley showed in 1987 that (6.31) holds if, and only if,

$$k = o\big((\ln_2 x/\ln_3 x)^2\big).$$

Using the saddle-point method in two variables,[1] Hildebrand & Tenenbaum (1988) showed that, uniformly for $k \ll (\ln_2 x)^2$, we have

$$\pi_k(x) = \lambda(r)\frac{x}{\ln x}\frac{(\ln_2 x)^{k-1}}{(k-1)!}\mathrm{e}^{-kh/2}\Big\{1+O\Big(\frac{1+r}{\ln_2 x}\Big)\Big\} \tag{6.32}$$

with $r := (k-1)/\ln_2 x$, and[2]

$$h := \left(\frac{\ln(2+\mathrm{e}^{\gamma} r\ln r)}{\ln_2 x}\right)^2.$$

The same article contains numerous other statements on the behavior of $\pi_k(x)$ for large values of k, notably the relation

$$\frac{\pi_{k+1}(x)}{\pi_k(x)} \sim \frac{L}{k} \qquad \Big(1 \leqslant k \ll \frac{\ln x}{(\ln_2 x)^2}\Big) \tag{6.33}$$

with $L := \ln\Big(\dfrac{\ln x}{k\ln(k+1)}\Big)$.

In a thesis supervised by the author, Kerner (2002), extended the domain of validity of the saddle-point method for this problem, obtaining an asymptotic formula for $\pi_k(x)$ which holds in the domain

$$k \leqslant K_{x,\varepsilon} := \Big\{1+\frac{1-\varepsilon}{\ln_2 x}\Big\}\frac{\ln x}{\ln_2 x} \qquad \text{for all } \varepsilon>0 \quad \text{and} \quad x\to\infty.$$

This allows an extension of (6.33), which implies, among other estimates,

$$\pi_{k+1}(x)/\pi_k(x) \asymp (L_1/k)^{\beta}$$

where $L_1 := \ln(K_{x,\varepsilon}/k)$, and $\beta = \beta(k,x)$ is a quantity precisely defined in the memoir and which satisfies in particular $\beta - 1 \asymp k/(L_1\ln x)$ throughout the whole domain under consideration.

For further information on this subject, and particularly for results concerning very large values of k, say $k \gg \ln x/\ln_2 x$, see Pomerance (1984).

Balazard (1990) showed that the sequence $k \mapsto \pi_k(x)$ is unimodal for sufficiently large x.

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[1] See Chap. III.5 for a description of the method in one variable in a typical special case, that of friable integers.

[2] This formula takes into account a remark of Norton—private communication.

As was shown by De Koninck & Tenenbaum (2002), Theorem 6.5 may be employed to evaluate the probability that an integer $n \leqslant x$ satisfies $\Omega(n) \leqslant \ln_2 x$, which constitutes a slightly biased approach to the problem of finding the median of the distribution of the values of $\Omega(n)$. We have

$$\sum_{\substack{n\leqslant x\\ \Omega(n)\leqslant \ln_2 x}} 1 = \tfrac{1}{2}x - x\frac{C + \langle \ln_2 x\rangle}{\sqrt{2\pi \ln_2 x}} + O\Big(\frac{x}{\ln_2 x}\Big)$$

with $C := \frac{1}{3} + \nu'(1) = \gamma - \frac{2}{3} + \sum_p \Big\{\frac{1}{p-1} - \ln\Big(\frac{1}{1-1/p}\Big)\Big\} \approx 0.36798.$

§ 6.2. Theorems 6.7 and 6.8 are due to Deshouillers, Dress & Tenenbaum (1979). As noted by these authors, the remainder term $O\big(1/\sqrt{\ln x}\big)$ is optimal in formula (6.23) if uniformity in u is required: for $u \in [0, (\ln 2)/\ln x[$ we have $F_n(u) = 1/\tau(n)$ $(n \leqslant x)$, so that

$$S(x,u) \sim \frac{hx}{\sqrt{\pi \ln x}}.$$

Exercises

215. *A formula of Delange (1959).* Put $\xi = \xi(x) := \ln_2 x$, and

$$T(x,u) := \sum_{n\leqslant x} u^{\omega(n)} \qquad (x \geqslant 1,\ 0 < u < \infty).$$

(a) Show that for all $\varepsilon > 0$, $u \leqslant 1 \leqslant v$, $x \geqslant 1$, we have

$$|\{n \leqslant x : |\omega(n) - \xi| > \varepsilon\xi\}| \leqslant u^{-(1-\varepsilon)\xi} T(x,u) + v^{-(1+\varepsilon)\xi} T(x,v).$$

Deduce that there exists an absolute constant c such that

$$|\{n \leqslant x : |\omega(n) - \xi| > \xi^{3/5}\}| \ll x \exp\{-c\xi^{1/5}\} \qquad (x \geqslant 3).$$

(b) Show that for $|k - \xi| \leqslant \xi^{3/5}$, we have

$$\mathrm{e}^{-\xi}\frac{\xi^k}{k!} \sim \frac{\mathrm{e}^{-\vartheta^2/2}}{\sqrt{2\pi\xi}} \qquad (\xi \to \infty)$$

with $\vartheta := (k-\xi)/\sqrt{\xi}$. Deduce the following asymptotic estimate, due to Delange and valid for any integer $m \geqslant 0$,

$$x^{-1}\sum_{n\leqslant x} \big(\omega(n) - \xi\big)^m = \big\{\mu_m + o(1)\big\}\xi^{m/2} \qquad (x \to \infty) \tag{6.34}$$

with

$$\mu_m := \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty} t^m \mathrm{e}^{-t^2/2}\,\mathrm{d}t = \begin{cases} 0 & (\text{if } m = 2k+1) \\ \dfrac{(2k)!}{2^k k!} & (\text{if } m = 2k). \end{cases}$$

216. Give an explicit value for the remainder term in the formula (6.34) in Exercise 215.

217. *Nicolas' theorem (1984).* Throughout this exercise, we write $y := x/2^k$. Let $N_k(x)$ (resp. $N'_k(x)$) denote the number of integers (resp. of odd integers) not exceeding x and such that $\Omega(n) = k$.

(a) Show that, for all $x \geqslant 1$, $k \geqslant 1$, we have $N_k(x) = \sum_{0\leqslant j\leqslant k} N'_j(2^j y)$.

(b) Show that, for all δ, $0 < \delta < 1$, and uniformly for $|z| \leqslant 3 - \delta$, $x \geqslant 3$, we have

$$\sum_{\substack{n\leqslant x \\ n\equiv 1 \,(\mathrm{mod}\, 2)}} z^{\Omega(n)} = x(\ln x)^{z-1}\Big\{zh(z) + O_\delta\Big(\frac{1}{\ln x}\Big)\Big\}$$

with $h(z) := \dfrac{2^{-z}}{\Gamma(z+1)}\displaystyle\prod_{p>2}\Big(1-\frac{z}{p}\Big)^{-1}\Big(1-\frac{1}{p}\Big)^z$.

(c) Using the previous result with $z = \frac{5}{2}$, establish the upper bound

$$N'_j(x) \ll x(\ln x)^{3/2}(\tfrac{2}{5})^j \qquad (x \geqslant 3,\ j \geqslant 1).$$

(d) Set $Y := \ln_2 3y$, and assume $y \geqslant 1$. Show that

$$\sum_{j>\frac{5}{2}Y} N_j'(2^j y) \ll y(\ln 3y)^{1-c_1}$$

with $c_1 = \frac{5}{2}\ln\frac{5}{4} - \frac{1}{2} > \frac{1}{20}$.

(e) Deduce from part (b) that, uniformly for $x \geqslant 3$ and $j \leqslant \frac{5}{2}\ln_2 x$, we have

$$N_j'(x) = \frac{x}{\ln x}\Big\{Q_j(\ln_2 x) + O\Big(\frac{(\ln_2 x)^j}{j!\ln x}\Big)\Big\}$$

with

$$Q_j(X) := \sum_{m+\ell=j-1} \frac{1}{m!\,\ell!} h^{(m)}(0) X^\ell.$$

(f) Show that $h^{(m)}(0)/m! \ll (\frac{3}{8})^m$ $(m \geqslant 0)$, and deduce that

$$Q_j(X) \ll (\tfrac{3}{8})^j \mathrm{e}^{8X/3} \quad (j > \tfrac{5}{2}X), \qquad Q_j'(X) \ll \frac{X^{j-2}}{(j-2)!} \quad (2 \leqslant j \leqslant \tfrac{5}{2}X).$$

(g) Deduce from parts (e) and (f) that, for $y \geqslant 1$, $j \leqslant \frac{5}{2}Y$, we have

$$N_j'(2^j y) = \frac{2^j y}{\ln 3y}\Big\{Q_j(Y) + O\Big(\frac{Y^{j+1}\mathrm{e}^{-Y}}{j!}\Big)\Big\}.$$

(h) Show that $\sum_{j\geqslant 1} 2^j Q_j(Y) = 2h(2)\mathrm{e}^{2Y}$ and that

$$\sum_{j>\frac{5}{2}Y} 2^j Q_j(Y) \ll \mathrm{e}^{(2-c_2)Y}$$

with $c_2 := \frac{5}{2}\ln\frac{4}{3} - \frac{2}{3} > \frac{1}{20}$.

(i) Show that, for all $k \in \big[\frac{5}{2}Y,\ (\ln x)/\ln 2\big]$, we have

$$N_k(x) = Cy\ln 3y\Big\{1 + O\big((\ln 3y)^{-1/20}\big)\Big\},$$

with

$$C := 2h(2) = \tfrac{1}{4}\prod_{p>2}\Big(1 + \frac{1}{p(p-2)}\Big).$$

218. *The Erdős–Kac theorem (1939).*

(a) Deduce from the central limit theorem in probability theory that for each real number λ we have

$$\lim_{x\to\infty} \sum_{k\leqslant x+\lambda\sqrt{x}} \mathrm{e}^{-x}\frac{x^k}{k!} = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\lambda} \mathrm{e}^{-t^2/2}\,\mathrm{d}t.$$

(b) Let $\varphi_x(t) := \mathrm{e}^{-x}x^t/\Gamma(t+1)$. Show that, for $x \geqslant 1$, $|t-x| \ll \sqrt{x}$, we have

$$\varphi_x(t) = \frac{\mathrm{e}^{-(t-x)^2/2x}}{\sqrt{2\pi x}}\Big\{1 + O\Big(\frac{1}{\sqrt{x}}\Big)\Big\}.$$

Estimate the Stieltjes integral $\displaystyle\int_{-\infty}^{x+\lambda\sqrt{x}} \varphi_x(t)\,\mathrm{d}\lfloor t\rfloor$ and thus recover the result of the previous question.

(c) Applying the estimates obtained by the Selberg–Delange method for $\pi_k(x) := \big|\{n \leqslant x : \omega(n) = k\}\big|$, establish the Erdős–Kac theorem:

$$(\forall \lambda \in \mathbb{R})\ \lim_{x\to\infty} \frac{1}{x}\big|\{n \leqslant x : \omega(n) \leqslant \ln_2 x + \lambda\sqrt{\ln_2 x}\}\big| = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\lambda} \mathrm{e}^{-t^2/2}\,\mathrm{d}t.$$

What explicit error term can you derive?

Tauberian Theorems

7.1. Introduction. Abelian/Tauberian theorems duality

In the course of the previous chapters we have developed a variety of methods for evaluating the summatory function of an arithmetic sequence, based on the analytic continuation of its Dirichlet series. In each case, the study of this continuation beyond the convergence domain revealed itself to be critical. In contrast, we aim here at results yielding the asymptotic behavior of summatory functions using the generating series exclusively at points of convergence. Theorems of this type have an appealing feature: ease of application, due to weakening of the hypotheses. This advantage, however, is balanced by the poor quality of the error terms in standard applications—although it can usually be shown that they are essentially optimal at the level of generality of the statements.

Let us begin by stating a classical result of Abel.

Theorem 7.1 (Abel). *Let $f(z) := \sum_{n\geqslant 0} a_n z^n$ be a power series, with radius of convergence 1, converging at $z = 1$. For any real number ϑ, $0 \leqslant \vartheta < \pi/2$, and any sector*

$$S(\vartheta) := \big\{z : |z| < 1,\, |\arg(1-z)| \leqslant \vartheta\big\},$$

we have

$$\lim_{z\to 1,\, z\in S(\vartheta)} f(z) = f(1).$$

Proof. Let $z = 1 - re^{i\varphi} \in S(\vartheta)$, with $r > 0$, $|\varphi| \leqslant \vartheta$. We have

$$|z|^2 = 1 - 2r\cos\varphi + r^2 < 1,$$

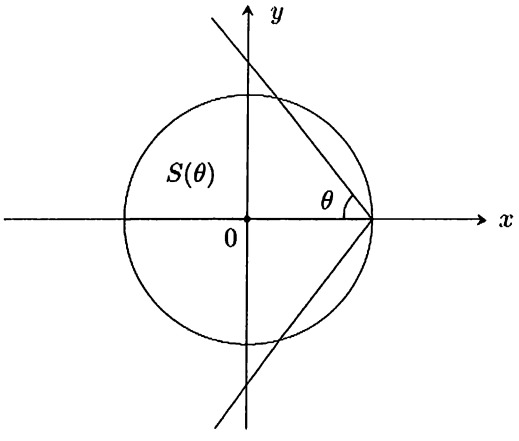

Figure 1. The sector $S(\vartheta)$.

and so $r < 2\cos\varphi$. Since $r \to 0$ as $z \to 1$, we can suppose $r \leqslant \cos\vartheta$. Let us denote by $S^*(\vartheta)$ the part of $S(\vartheta)$ corresponding to this supplementary condition.

If $z \in S^*(\vartheta)$, we then have

$$|1-z|\cos\vartheta = r\cos\vartheta \leqslant r(2\cos\varphi - r) = 1 - |z|^2 \leqslant 2(1-|z|). \tag{7.1}$$

It suffices to show that the convergence of the power series $f(z)$ is uniform in the sector $S^*(\vartheta)$, i.e. that

$$\sup_{z\in S^*(\vartheta)} \left| \sum_{n>N} a_n z^n \right| \to 0 \qquad (N \to \infty).$$

Set $A_n := \sum_{N<m\leqslant n} a_m$ $(n \geqslant N)$, $\varepsilon_N := \sup_{n>N} |A_n|$, so that our hypothesis of convergence of $f(1)$ is equivalent to $\lim_{N\to\infty} \varepsilon_N = 0$. Now partial summation readily yields, for $z \in S^*(\vartheta)$,

$$\left| \sum_{n>N} a_n z^n \right| = \left| (1-z) \sum_{n>N} A_n z^n \right| \leqslant \frac{\varepsilon_N |1-z||z|^N}{1-|z|} \leqslant \frac{2\varepsilon_N}{\cos\vartheta}.$$

□

Theorem 7.1 is the prototype of a class of statements called *Abelian* because they share with it the following characteristic: they establish that if a sequence (or a function) is sufficiently regular, then a certain average of its values also shows regular behavior. Thus, the well-known implication due to Cesàro

$$\lim_{n\to\infty} a_n = a \Rightarrow \lim_{n\to\infty} \frac{1}{n} \sum_{0\leqslant m\leqslant n} a_m = a$$

is an Abelian theorem.

Setting $b_n := \sum_{0\leqslant m\leqslant n} a_m$, so that

$$f(z) = \sum_{m\geqslant 0} a_m z^m = (1-z)\sum_{n\geqslant 0} b_n z^n,$$

Theorem 7.1 can be written equally well as

$$\lim_{n\to\infty} b_n = b \Rightarrow \lim_{z\to 1,\ z\in S(\vartheta)} (1-z)\sum_{n\geqslant 0} b_n z^n = b.$$

The converse of an Abelian theorem is in general false. For example, the sum of the power series

$$f(z) = \sum_{n\geqslant 0} (-1)^n z^n = \frac{1}{1+z}$$

tends to $\frac{1}{2}$ as $z \to 1$, but the series diverges at $z = 1$. A *Tauberian* theorem is one which provides a sufficient condition (called a *Tauberian condition*) for the truth of such a converse. The first result of this type is due to the Austrian mathematician Alfred Tauber (1897)—cf. § 7.2.

We conclude this section with an Abelian theorem concerning Dirichlet series.

Theorem 7.2. *Let $F(s) := \sum_{n\geqslant 1} a_n/n^s$ be a Dirichlet series converging for $\sigma > a \geqslant 0$. If there exist two constants c, ω, with $\omega > -1$, such that*

$$\sum_{n\leqslant x} a_n = \Big\{\frac{c}{\Gamma(\omega+1)} + o(1)\Big\} x^a (\ln x)^\omega \qquad (x\to\infty),$$

then we have

$$F(\sigma) = \frac{ca + o(1)}{(\sigma - a)^{\omega+1}} \qquad (\sigma \to a+).$$

Proof. Let us set $A(t) := \sum_{n\leqslant \mathrm{e}^t} a_n$ and

$$G(h) := \int_{0-}^{\infty} \mathrm{e}^{-(a+h)t}\,\mathrm{d}A(t) \qquad (h>0).$$

Since we have

$$\frac{c}{\Gamma(\omega+1)}\int_0^\infty \mathrm{e}^{-(a+h)t} t^\omega\,\mathrm{d}\{\mathrm{e}^{at}\} = \frac{ca}{\Gamma(\omega+1)}\int_0^\infty t^\omega \mathrm{e}^{-ht}\,\mathrm{d}t = \frac{ca}{h^{\omega+1}},$$

we can write

$$\begin{aligned} G(h) - \frac{ca}{h^{\omega+1}} &= (a+h)\int_0^\infty \mathrm{e}^{-(a+h)t} A(t)\,\mathrm{d}t - \frac{ca}{h^{\omega+1}} \\ &= \int_0^\infty \mathrm{e}^{-(a+h)t}\Big\{(a+h)A(t) - \frac{ac}{\Gamma(\omega+1)}\mathrm{e}^{at}t^\omega\Big\}\,\mathrm{d}t. \end{aligned}$$

By hypothesis there exists a function $\varepsilon(t)$, with $\lim_{t\to\infty}\varepsilon(t)=0$, such that the expression between curly brackets may be written as

$$\varepsilon(t)\mathrm{e}^{at}t^{\omega}+O(h\mathrm{e}^{at}t^{\omega}).$$

This implies, as required,

$$G(h)-\frac{ca}{h^{\omega+1}}=o\Big(\frac{1}{h^{\omega+1}}\Big)\qquad(h\to 0+).\qquad\square$$

7.2. Tauber's theorem

In its original form, Tauber's theorem (1897) is the exact converse of Abel's theorem.

Theorem 7.3 (Tauber). *Let $f(z):=\sum_{n\geqslant 0}a_nz^n$ be a power series with radius of convergence 1. Suppose that $f(z)$ tends to a limit ℓ as $z\to 1$ while remaining in the real interval $[0,1[$. Under the additional condition*

$$\text{(T)}\qquad \sum_{n\leqslant x}na_n=o(x)\qquad(x\to\infty),$$

the series $\sum a_n$ converges, and its sum is ℓ.

Proof. Let us set

$$\begin{aligned}
&A(x):=\sum_{n\leqslant x}a_n,\quad \alpha(x):=\int_0^x t\,\mathrm{d}A(t)\qquad(x>0),\\
&G(u):=\frac{\mathrm{e}^{-u}-1}{u},\quad g(u):=-G'(u)=\frac{(1+u)\mathrm{e}^{-u}-1}{u^2}\qquad(u>0),\\
&H(u):=\frac{\mathrm{e}^{-u}}{u},\quad h(u):=-H'(u)=\frac{(1+u)\mathrm{e}^{-u}}{u^2}\qquad(u>0).
\end{aligned}\tag{7.2}$$

We have

$$\begin{aligned}
f\big(\mathrm{e}^{-1/x}\big)-A(x)&=\frac{1}{x}\int_0^x G(t/x)t\,\mathrm{d}A(t)+\frac{1}{x}\int_x^\infty H(t/x)t\,\mathrm{d}A(t)\\
&=\frac{1}{x}\int_0^x\int_{t/x}^\infty g(u)\,\mathrm{d}u\,\mathrm{d}\alpha(t)+\frac{1}{x}\int_x^\infty\int_{t/x}^\infty h(u)\,\mathrm{d}u\,\mathrm{d}\alpha(t)\\
&=\frac{1}{x}\int_0^\infty g(u)\,\mathrm{d}u\int_0^{\min(1,u)x}\mathrm{d}\alpha(t)+\frac{1}{x}\int_1^\infty h(u)\,\mathrm{d}u\int_x^{ux}\mathrm{d}\alpha(t)\\
&=\int_0^1 g(u)\frac{\alpha(ux)}{x}\,\mathrm{d}u-\frac{\alpha(x)}{x}+\int_1^\infty h(u)\frac{\alpha(ux)}{x}\,\mathrm{d}u.
\end{aligned}\tag{7.3}$$

We have $\alpha(ux)/x\ll u$ uniformly in x since $\alpha(x)/x$ is bounded on $\mathbb{R}^+$. Since $\alpha(ux)/x$ tends to 0 as $x\to\infty$ for any fixed u, and since $ug(u)$ and

$uh(u)$ are, respectively, integrable on $[0,1]$ and $[1,\infty[$, we obtain, by dominated convergence, that

$$f(\mathrm{e}^{-1/x}) - A(x) = o(1) \quad (x \to \infty). \tag{7.4}$$

This completes the proof of Tauber's theorem. □

It should be noted that condition (T) is actually also *necessary* for the convergence of $\sum a_n$. Indeed, retaining the notation used in the course of the proof, we may write, by partial summation,

$$\frac{\alpha(x)}{x} = A(x) - \frac{1}{x}\int_0^x A(t)\,\mathrm{d}t = \frac{1}{x}\int_0^x \{A(x) - A(t)\}\,\mathrm{d}t. \tag{7.5}$$

Thus $A(x) \to \ell$ immediately entails $\alpha(x) = o(x)$.

We can hence finally formulate Tauber's theorem in integral, and more general, form as follows.

Theorem 7.4. *Let $\omega \in \mathbb{R}^+$ and $A : \mathbb{R}^+ \to \mathbb{R}$ be a function of bounded variation on any finite interval. Assume that the Laplace–Stieltjes integral*

$$F(\sigma) := \int_0^\infty \mathrm{e}^{-\sigma t}\,\mathrm{d}A(t)$$

converges for $\sigma > 0$ and satisfies $F(\sigma) = o(1/\sigma^\omega)$ $(\sigma \to 0+)$. Then the two following assertions are equivalent

$$\text{(i)}\ A(x) - A(0) = o(x^\omega) \qquad (x \to \infty)$$

$$\text{(ii)}\ \int_0^x t\,\mathrm{d}A(t) = o(x^{\omega+1}) \qquad (x \to \infty).$$

Proof. With appropriate changes in the notation, the calculation (7.3) allows us to express $F(\sigma) - A(1/\sigma) + A(0)$ as a function of the quantity $\alpha(x)$ defined in (7.2). The conclusion then results as before by appealing to Lebesgue's dominated convergence theorem. □

Remark. Theorem 7.3 corresponds to the case $\omega = 0$ of Theorem 7.4: the hypothesis $F(\sigma) = o(1)$ is then obtained by modifying the value of $A(0)$.

Considering the statement of Theorem 7.4, we can both extend and make more precise the notion of a Tauberian theorem. Given a real-valued function $\varphi(t,s)$, defined on $\mathbb{R}^+ \times S$ where $S \subseteq \mathbb{C}$, we define the φ-transform of a function A of bounded variation on any finite interval by

$$F(s) := \int_0^\infty \varphi(t,s)\,\mathrm{d}A(t) \tag{7.6}$$

whenever the integral converges. We next assume that the following Abelian theorem holds for some number s_0 in the closure of S: *If* $\lim_{t\to\infty} A(t) = \ell$,

then the integral (7.6) *converges for all* s *in* S *and we have*

$$\lim_{s\to s_0,\, s\in S} F(s) = \ell. \tag{7.7}$$

In these circumstances, we shall call *Tauberian* any theorem providing a sufficient condition to deduce from (7.7) that $\lim_{t\to\infty} A(t) = \ell$.

The scope of the adjective Tauberian can be further extended to results establishing that $A(t)$ has a definite behavior at infinity (for example, the existence of the limit for a certain φ-transform) under an assumption of type (7.7)—see in particular chapter 4 of Bingham, Goldie & Teugels (1987).

The reference work today for Tauberian theorems is the comprehensive book by Korevaar (2004).

7.3. The theorems of Hardy–Littlewood and Karamata

The Tauberian condition (T) in Theorem 7.3 readily follows from the assumption

$$a_n = o(1/n) \qquad (n \to \infty).$$

In 1913, Hardy & Littlewood showed that the one-sided condition

$$a_n \geqslant -K/n \qquad (n \geqslant 1),$$

where K is an arbitrary constant, suffices instead. The following theorem, due to Karamata (1931), allows a new, simpler proof of this result.

Theorem 7.5 (Karamata). *Let* $A(t)$ *be a non-decreasing function such that the integral*

$$F(\sigma) := \int_0^\infty \mathrm{e}^{-\sigma t}\, \mathrm{d}A(t) \tag{7.8}$$

converges for all $\sigma > 0$. *Assume there exist two real numbers* $c \geqslant 0$, $\omega > 0$, *such that*

$$F(\sigma) = \frac{c + o(1)}{\sigma^\omega} \qquad (\sigma \to 0+).$$

Then we have

$$A(x) = \frac{\{c + o(1)\}x^\omega}{\Gamma(\omega + 1)} \qquad (x \to +\infty).$$

Proof. We can assume without loss of generality that $A(0) = 0$. For any fixed integer $n \geqslant 0$, we have

$$\begin{aligned} F\big((n+1)\sigma\big) &= \int_0^\infty \mathrm{e}^{-n\sigma t}\mathrm{e}^{-\sigma t}\, \mathrm{d}A(t) = \frac{\{c + o(1)\}}{\Gamma(\omega)\sigma^\omega}\, \frac{\Gamma(\omega)}{(n+1)^\omega} \\ &= \frac{\{c + o(1)\}}{\Gamma(\omega)\sigma^\omega} \int_0^\infty \mathrm{e}^{-nt}\mathrm{e}^{-t} t^{\omega-1}\, \mathrm{d}t \qquad (\sigma \to 0+). \end{aligned}$$

It follows that, for each fixed polynomial P,

$$\int_0^\infty \frac{P(\mathrm{e}^{-\sigma t})}{\mathrm{e}^{\sigma t}}\,\mathrm{d}A(t) = \frac{\{c+o(1)\}}{\Gamma(\omega)\sigma^\omega}\int_0^\infty P(\mathrm{e}^{-t})\frac{t^{\omega-1}}{\mathrm{e}^t}\,\mathrm{d}t \quad (\sigma\to 0+). \tag{7.9}$$

We shall show that this relation persists when P is replaced by the function χ defined on $[0,1]$ by

$$\chi(\mathrm{e}^{-t}) = \begin{cases} \mathrm{e}^t & (0\leqslant t\leqslant 1)\\ 0 & (t>1)\end{cases}$$

and when σ tends to 0 in such a way that $1/\sigma$ avoids the points of discontinuity of A. Assuming this for the moment, it follows that

$$\begin{aligned} A\Big(\frac{1}{\sigma}\Big) &= \int_0^\infty \frac{\chi(\mathrm{e}^{-\sigma t})}{\mathrm{e}^{\sigma t}}\,\mathrm{d}A(t) = \frac{\{c+o(1)\}}{\Gamma(\omega)\sigma^\omega}\int_0^\infty \chi(\mathrm{e}^{-u})\frac{u^{\omega-1}}{\mathrm{e}^u}\,\mathrm{d}u\\ &= \frac{\{c+o(1)\}}{\Gamma(\omega)\sigma^\omega}\int_0^1 u^{\omega-1}\,\mathrm{d}u = \frac{c+o(1)}{\Gamma(\omega+1)\sigma^\omega}. \end{aligned}$$

From the monotonicity assumption on A, this asymptotic relation is actually valid without restriction on the values of σ, and the stated conclusion follows.

In order to show that (7.9) holds for χ, it is natural to approximate χ by polynomial functions. Let us consider the function $H(t)$ defined on $[0,1]$ by

$$H(t) := \frac{\chi(t)-t}{t(1-t)} \quad (0<t<1), \quad H(0)=-1, \quad H(1)=2. \tag{7.10}$$

It possesses a unique point of discontinuity, at $t = 1/\mathrm{e}$. For each $\varepsilon > 0$, there exists continuous functions f and g satisfying[1]

$$\begin{cases} f(t)\leqslant H(t)\leqslant g(t) & (0\leqslant t\leqslant 1)\\ g(t)-f(t)\leqslant \varepsilon & (|t-1/\mathrm{e}|>\varepsilon)\\ g(t)-f(t)\leqslant 12 & (|t-1/\mathrm{e}|\leqslant\varepsilon). \end{cases} \tag{7.11}$$

From the Weierstrass approximation theorem, there exist polynomials p and q such that

$$\max_{0\leqslant t\leqslant 1}|p(t)-f(t)|\leqslant\varepsilon, \qquad \max_{0\leqslant t\leqslant 1}|q(t)-g(t)|\leqslant\varepsilon. \tag{7.12}$$

Putting

$$P(t) := t+t(1-t)\big(p(t)-\varepsilon\big), \qquad Q(t) := t+t(1-t)\big(q(t)+\varepsilon\big),$$

we deduce from the above that

$$P(t)\leqslant\chi(t)\leqslant Q(t) \qquad (0\leqslant t\leqslant 1) \tag{7.13}$$

[1]The number 12 below serves here merely to exceed the value of the discontinuity of H at $t = 1/\mathrm{e}$.

and, for $0 < \varepsilon < \frac{1}{2}$,

$$\int_0^1 \frac{Q(t) - P(t)}{t(1-t)}\,\mathrm{d}t \leqslant \int_0^1 \{q(t) - p(t)\}\,\mathrm{d}t + 2\varepsilon.$$

It is easily deduced from (7.11) and (7.12) that the last integral does not exceed 27ε. By adjusting the value of ε in the preceding calculations, we therefore obtain the existence of two polynomials P and Q satisfying (7.13) and

$$\int_0^1 \frac{Q(t) - P(t)}{t(1-t)}\,\mathrm{d}t \leqslant \varepsilon. \tag{7.14}$$

It is now easy to establish the validity of (7.9) for the function χ when σ tends to $0+$ avoiding the discontinuities of $A(1/t)$. Under these conditions, we indeed have

$$\begin{aligned} A(1/\sigma) &= \int_0^\infty \frac{\chi(\mathrm{e}^{-\sigma t})}{\mathrm{e}^{-\sigma t}}\,\mathrm{d}A(t) \\ &\leqslant \int_0^\infty \frac{Q(\mathrm{e}^{-\sigma t})}{\mathrm{e}^{-\sigma t}}\,\mathrm{d}A(t) = \frac{\{c + o(1)\}}{\Gamma(\omega)\sigma^\omega} \int_0^\infty Q(\mathrm{e}^{-t})\frac{t^{\omega-1}}{\mathrm{e}^t}\,\mathrm{d}t. \end{aligned}$$

Similarly

$$A(1/\sigma) \geqslant \frac{\{c + o(1)\}}{\Gamma(\omega)\sigma^\omega} \int_0^\infty P(\mathrm{e}^{-t})\frac{t^{\omega-1}}{\mathrm{e}^t}\,\mathrm{d}t.$$

Now,

$$\int_0^\infty \frac{Q}{P}(\mathrm{e}^{-t})\mathrm{e}^{-t}t^{\omega-1}\,\mathrm{d}t \lessgtr \int_0^\infty \chi(\mathrm{e}^{-t})\mathrm{e}^{-t}t^{\omega-1}\,\mathrm{d}t \pm R$$

with

$$\begin{aligned} R &:= \int_0^\infty \{Q(\mathrm{e}^{-t}) - P(\mathrm{e}^{-t})\}\frac{t^{\omega-1}}{\mathrm{e}^t}\,\mathrm{d}t \\ &= \int_0^1 \Big\{\frac{Q(t) - P(t)}{t(1-t)}\Big\} t(1-t)\Big(\ln\frac{1}{t}\Big)^{\omega-1}\,\mathrm{d}t \leqslant \varepsilon M \end{aligned}$$

by (7.14), where $M := \sup_{0\leqslant t\leqslant 1} t(1-t)\{\ln(1/t)\}^{\omega-1}$ is independent of σ and ε. Letting successively σ and then ε tend to 0, we obtain the expected formula. □

Remark. If $\omega = 0$, the conclusion of Theorem 7.5 persists in the form

$$A(x) - A(0) \to c \qquad (x \to \infty).$$

This follows trivially from the partial integration formula

$$F(\sigma) = -A(0) + \sigma \int_0^\infty \mathrm{e}^{-\sigma t}A(t)\,\mathrm{d}t = \int_0^\infty \mathrm{e}^{-t}\big\{A(t/\sigma) - A(0)\big\}\,\mathrm{d}t.$$

Indeed, since A is non-decreasing, $A(t/\sigma)$ tends, for each t, to a limit $\alpha := \sup A(t)$, whether finite or not. By Lebesgue's dominated convergence theorem we hence have

$$c = \lim F(\sigma) = \int_0^\infty \mathrm{e}^{-t}\{\alpha - A(0)\}\,\mathrm{d}t = \alpha - A(0).$$

To deduce the Hardy–Littlewood theorem from Karamata's, we need a lemma of Landau's (1906, 1929) which is of independent interest.

Theorem 7.6 (Landau). *Let f be a twice differentiable function on $\mathbb{R}^+$, α an arbitrary real number, and M a positive constant. If we have*

(a) $f(x) = o(x^\alpha) \qquad (x \to +\infty, \text{resp. } 0+)$

(b) $f''(x) \leqslant Mx^{\alpha-2} \qquad (x > 0)$

then

$$f'(x) = o(x^{\alpha-1}) \qquad (x \to +\infty, \text{ resp. } 0+).$$

Proof. Let δ be a fixed real number with $0 < \delta \leqslant \frac{1}{2}$. Taylor's formula implies that, for all $x > 0$, there exist numbers $\vartheta_\pm = \vartheta_\pm(x, \delta) \in]0, 1[$ such that

$$f(x \pm \delta x) - f(x) = \pm\delta x f'(x) + \tfrac{1}{2}\delta^2 x^2 f''\big(x(1 \pm \vartheta_\pm\delta)\big).$$

The second derivative does not exceed

$$Mx^{\alpha-2}\{(1+\delta)^{\alpha-2} + (1-\delta)^{\alpha-2}\} \leqslant Kx^{\alpha-2}$$

where K is independent of δ. By (a) we can then write

$$o(x^\alpha) \leqslant \pm\delta x f'(x) + K\delta^2 x^\alpha \qquad (x \to +\infty, \text{ resp. } 0+).$$

Dividing the two sides by δx^α, and letting x tend to $+\infty$ or $0+$, we obtain

$$\lim \begin{matrix}\sup \\ \inf\end{matrix} x^{1-\alpha} f'(x) \begin{matrix}\leqslant \\ \geqslant\end{matrix} \pm K\delta.$$

The conclusion follows since δ is arbitrarily small. □

We are now in a position to establish a theorem which is more general than the Hardy–Littlewood theorem.

Given $\omega \geqslant 0$, we denote by $\mathcal{V}^*(\mathbb{R}^+)$ the class of functions $A : \mathbb{R}^+ \to \mathbb{R}$ of bounded variation on any bounded interval and such that the Laplace–Stieltjes integral (7.8) converges for $\sigma > 0$, and we define $\mathcal{K}(\omega)$ as the subclass of those A which satisfy, for a suitable (not necessarily positive) constant c, simultaneously

$$F(\sigma) := \int_0^\infty \mathrm{e}^{-\sigma t}\,\mathrm{d}A(t) = \frac{c + o(1)}{\sigma^\omega} \quad (\sigma \to 0+) \tag{7.15}$$

and

$$A(x) - A(0) = \frac{\{c + o(1)\}x^\omega}{\Gamma(\omega+1)} \quad (x \to \infty). \tag{7.16}$$

Note that the term $A(0)$ in (7.16) is significant only when $\omega = 0$.

Theorem 7.7 (Generalized Hardy–Littlewood–Karamata). *Assume $\omega \geqslant 0$ and $A \in \mathcal{V}^*(\mathbb{R}^+)$. If there exist $c \in \mathbb{R}$ such that* (7.15) *holds and $B \in \mathcal{K}(\omega+1)$ such that the measure $t\,\mathrm{d}A(t) + \mathrm{d}B(t)$ is positive, then $A \in \mathcal{K}(\omega)$.*

Proof. Up to altering the value of $A(0)$ when $\omega = 0$, or to replacing $A(x)$ by $A(x) - cx^\omega/\Gamma(\omega+1)$ when $\omega > 0$, we can assume $c = 0$. Moreover, for $0 < \sigma < 1$, we have

$$\begin{aligned} F''(\sigma) &= \int_0^\infty t^2 \mathrm{e}^{-\sigma t}\,\mathrm{d}A(t) \\ &\geqslant -\int_0^\infty t\mathrm{e}^{-\sigma t}\,\mathrm{d}B(t) = \int_0^\infty \{1-\sigma t\}B(t)\mathrm{e}^{-\sigma t}\,\mathrm{d}t \geqslant \frac{-K}{\sigma^{\omega+2}} \end{aligned}$$

with a suitable constant K, where the last inequality is a consequence of the fact that $B \in \mathcal{K}(\omega+1)$. By Theorem 7.6, we infer that

$$F'(\sigma) = -\int_0^\infty t\mathrm{e}^{-\sigma t}\,\mathrm{d}A(t) = o(1/\sigma^{\omega+1}) \qquad (\sigma \to 0+),$$

and hence, still using the fact that $B \in \mathcal{K}(\omega+1)$, for a suitable constant b,

$$\int_0^\infty \mathrm{e}^{-\sigma t}\{t\,\mathrm{d}A(t) + \mathrm{d}B(t)\} = \frac{b+o(1)}{\sigma^{\omega+1}}.$$

Karamata's Theorem 7.5 hence provides

$$\int_0^x \{t\,\mathrm{d}A(t) + \mathrm{d}B(t)\} = \frac{\{b+o(1)\}x^{\omega+1}}{\Gamma(\omega+2)} = \int_0^x \mathrm{d}B(t) + o\big(x^{\omega+1}\big).$$

The required conclusion then results from Theorem 7.4. □

Corollary 7.8 (Hardy–Littlewood). *Let $f(z) := \sum_{n\geqslant 0} a_n z^n$ be a power series with radius of convergence 1 and with real coefficients satisfying*

$$na_n \geqslant -K \qquad (n \geqslant 0) \tag{7.17}$$

for a suitable constant K. If

$$\lim_{z\to 1-} f(z) = \ell, \tag{7.18}$$

then we have $\sum_{n\geqslant 0} a_n = \ell$.

Proof. It suffices to apply Theorem 7.7 with

$$A(x) := \sum_{n\leqslant x} a_n, \qquad B(x) := K\lfloor x \rfloor, \qquad \omega := 0. \qquad \square$$

In the context of Dirichlet series we obtain the following result.

Corollary 7.9 ((Hardy–Littlewood–Karamata). *Let* $F(s) := \sum_{n\geqslant 1} a_n/n^s$ *be a Dirichlet series converging for* $\sigma > 1$. *Assume that there exist real numbers* c, K, ω, *with* $\omega \geqslant 0$, *such that we have*

$$a_n \geqslant -K(\ln n)^{\omega-1} \qquad (n \geqslant 2) \tag{7.19}$$

and

$$F(\sigma) = \frac{c+o(1)}{(\sigma-1)^\omega} \qquad (\sigma \to 1+). \tag{7.20}$$

Then we have

$$\sum_{n\leqslant x} \frac{a_n}{n} = \frac{\{c+o(1)\}}{\Gamma(\omega+1)}(\ln x)^\omega \qquad (x \to +\infty). \tag{7.21}$$

Remark. Condition (7.19) can be replaced more generally by $a_n \geqslant -b_n$ $(n \geqslant n_0)$ where the b_n are the coefficients of a Dirichlet series satisfying both (7.20) and (7.21) for some suitable constant c, but with the same ω. A condition of this type is, for example,

$$a_n \geqslant -K_1(\ln n)^{\omega-1} - K_2\tau(n)^\alpha \qquad (n \geqslant 2) \tag{7.22}$$

with $\alpha := \ln\omega/\ln 2$.

7.4. The remainder term in Karamata's theorem

In the course of the 1950s, Tauberian theory evolved in the direction of finding effective forms of the results, that is to say asymptotic formulae with explicit error terms. We give here an effective form of Karamata's Theorem 7.5 which is due to Freud (1952).

Theorem 7.10 (Karamata–Freud). *Let* $A(t)$ *be a non-decreasing function such that the integral*

$$F(\sigma) := \int_0^\infty \mathrm{e}^{-\sigma t}\,\mathrm{d}A(t)$$

converges for all $\sigma > 0$. *Suppose there exist two real numbers* $c \geqslant 0$, $\omega > 0$ *and a non-decreasing function* $\psi(t)$ *such that*

$$\psi(t) \to +\infty, \quad \psi(t)/t^\omega \text{ is non-increasing for large } t, \tag{7.23}$$

and

$$F(\sigma) = \Big\{c + O\Big(\frac{1}{\psi(1/\sigma)}\Big)\Big\}\sigma^{-\omega} \qquad (\sigma \to 0+). \tag{7.24}$$

Then we have

$$A(x) = \Big\{c + O\Big(\frac{1}{\ln\psi(x)}\Big)\Big\}\frac{x^\omega}{\Gamma(\omega+1)} \qquad (x \to \infty). \tag{7.25}$$

It is important to note right from the outset that the error term in (7.25) cannot be improved without additional assumptions, as shown by the following counter-example, due to Karamata (1952). Consider the increasing function

$$A(x) := \int_0^x \big(1 + \cos\{(\ln t)^2\}\big)\,\mathrm{d}t.$$

On the one hand, we have

$$A(x) = x + \Omega_{\pm}\Big(\frac{x}{\ln x}\Big). \tag{7.26}$$

Indeed, this results from simple partial integration:

$$\begin{aligned}\int_2^x \cos\{(\ln t)^2\}\,\mathrm{d}t &= \int_2^x \frac{t}{2\ln t}\,\mathrm{d}\sin\{(\ln t)^2\}\\ &= \Big[\frac{t}{2\ln t}\sin\{(\ln t)^2\}\Big]_2^x - \int_2^x \Big(1-\frac{1}{\ln t}\Big)\frac{\sin\{(\ln t)^2\}}{2\ln t}\,\mathrm{d}t\\ &= \frac{x}{\ln x}\Big\{\tfrac{1}{2}\sin\{(\ln x)^2\} + O\Big(\frac{1}{\ln x}\Big)\Big\}\end{aligned}$$

where the final integral has been estimated by a second integration by parts.

On the other hand, we shall see that

$$F(\sigma) = \frac{1}{\sigma} + O(1) \qquad (\sigma \to 0+). \tag{7.27}$$

It is clear that $F(\sigma) = \dfrac{1}{\sigma} + \Re e\, J(\sigma) + O(1)$, with

$$J(\sigma) := \int_1^\infty \exp\big\{-\sigma t + i(\ln t)^2\big\}\,\mathrm{d}t.$$

We estimate $J(\sigma)$ by replacing the integral over $[1,+\infty[$ by the complex integral over the contour made up of the circular arc

$$\Gamma := \big\{\mathrm{e}^{i\vartheta} : 0 \leqslant \vartheta \leqslant \pi/4\big\}$$

and the half-line $\Delta := \big\{r\mathrm{e}^{i\pi/4} : r \geqslant 1\big\}$. This transformation is justified for all $\sigma > 0$ by the bound

$$\big|\exp\big\{-\sigma z + i(\log z)^2\big\}\big| \leqslant R^{-2\vartheta}\mathrm{e}^{-\sigma R\cos\vartheta} \quad (z = R\mathrm{e}^{i\vartheta},\ 0 \leqslant \vartheta \leqslant \pi/4). \tag{7.28}$$

Moreover, the integral over Γ is clearly bounded, and the estimate (7.28) applied with $\vartheta = \pi/4 > \frac{1}{2}$ shows that the integral over Δ converges absolutely and is bounded independently of $\sigma > 0$. This implies (7.27).

The idea behind the proof of Theorem 7.10 consists in making explicit the polynomial approximations to the function χ arising in the proof of Karamata's theorem. Precision is measured in the L^1-norm, and it is necessary to control the size of the coefficients. We define the *length* $\ell(P)$ of a

polynomial $P(x) := \sum_{m=0}^{n} a_m x^m$ by

$$\ell(P) := \sum_{0 \leqslant m \leqslant n} |a_m|.$$

We shall establish the following one-sided approximation result.

Theorem 7.11. *Let $f(t)$ a function of bounded variation on $[0,1]$. There exist constants A_1, A_2, depending only on f, such that, for any integer $n \geqslant 1$, there exist two polynomials p and q, of degrees at most n, satisfying*

$$\text{(7.29)} \qquad \begin{cases} p(t) \leqslant f(t) \leqslant q(t) & (0 \leqslant t \leqslant 1) \\ \int_0^1 \big(q(t) - p(t)\big)\,\mathrm{d}t \leqslant A_1/n \\ \ell(p) + \ell(q) \leqslant A_2^n. \end{cases}$$

Initially, let us assume this result and see how Theorem 7.10 can be deduced from it.

By the second of the assumptions (7.23), relation (7.24) implies that, for any integer $m \geqslant 1$,

$$\int_0^\infty \mathrm{e}^{-m\sigma t}\mathrm{e}^{-\sigma t}\,\mathrm{d}A(t) = \frac{1}{\sigma^\omega}\Big\{\frac{c}{\Gamma(\omega)}\int_0^\infty \mathrm{e}^{-mt}\frac{t^{\omega-1}}{\mathrm{e}^t}\,\mathrm{d}t + O\Big(\frac{1}{\psi(1/\sigma)}\Big)\Big\}.$$

By linearity, we deduce that, for any polynomial P, we have

$$\text{(7.30)} \qquad \int_0^\infty \frac{P(\mathrm{e}^{-\sigma t})}{\mathrm{e}^{\sigma t}}\,\mathrm{d}A(t) = \frac{1}{\sigma^\omega}\Big\{\frac{c}{\Gamma(\omega)}\int_0^\infty \frac{P(\mathrm{e}^{-t})}{\mathrm{e}^t}t^{\omega-1}\,\mathrm{d}t + O\Big(\frac{\ell(P)}{\psi(1/\sigma)}\Big)\Big\}.$$

We now apply Theorem 7.11 to the function H defined by (7.10) and we set $P(t) := t + t(1-t)p(t)$, $Q(t) := t + t(1-t)q(t)$. We have $P(t) \leqslant \chi(t) \leqslant Q(t)$ for all t in $[0,1]$, and it follows from (7.30) that, if $1/\sigma$ is not a discontinuity of A, we also have

$$A(1/\sigma) \begin{matrix} \leqslant \\ \geqslant \end{matrix} \frac{1}{\sigma^\omega}\Big\{\frac{c}{\Gamma(\omega+1)} \pm R(\sigma)\Big\},$$

with

$$\begin{aligned} R(\sigma) &\ll \int_0^\infty \{Q(\mathrm{e}^{-t}) - P(\mathrm{e}^{-t})\}\mathrm{e}^{-t}t^{\omega-1}\,\mathrm{d}t + \frac{{A_2}^n}{\psi(1/\sigma)} \\ &= \int_0^1 \{q(t) - p(t)\}t(1-t)\Big(\ln\frac{1}{t}\Big)^{\omega-1}\,\mathrm{d}t + \frac{{A_2}^n}{\psi(1/\sigma)} \ll \frac{A_1}{n} + \frac{{A_2}^n}{\psi(1/\sigma)}. \end{aligned}$$

Selecting $n := \lfloor \ln\psi(1/\sigma)/(2\ln A_2)\rfloor$, we finally obtain the estimate (7.25).

In the course of the proof of Theorem 7.11, we shall appeal to the following classical computation concerning the Chebyshev polynomials $T_n(x)$ ($n \in \mathbb{N}^*$) defined on $[-1,1]$ by

$$T_n(\cos\vartheta) = \cos n\vartheta \quad (\vartheta \in \mathbb{R}).$$

Lemma 7.12. *For any integer $n \geqslant 1$, we have*

$$T_n(x) = \tfrac{1}{2} \sum_{0 \leqslant m \leqslant n/2} (-1)^m \frac{n}{n-m} \binom{n-m}{m} (2x)^{n-2m}. \tag{7.31}$$

In particular,

$$\ell(T_n) \leqslant n\big(1+\sqrt{2}\big)^n. \tag{7.32}$$

Proof. We start with the formula

$$\sin(n \pm 1)\vartheta = \sin n\vartheta \cos\vartheta \pm \cos n\vartheta \sin\vartheta,$$

from which we infer that

$$2\cos n\vartheta = \frac{\sin(n+1)\vartheta}{\sin\vartheta} - \frac{\sin(n-1)\vartheta}{\sin\vartheta}. \tag{7.33}$$

Set $f_\vartheta(r) := |1 - re^{i\vartheta}|^2$ $(0 \leqslant r < 1)$. On the one hand,

$$\begin{aligned} \frac{1}{f_\vartheta(r)} &= \frac{1}{(1-re^{i\vartheta})(1-re^{-i\vartheta})} = \frac{1}{2ir\sin\vartheta}\left\{\frac{1}{1-re^{i\vartheta}} - \frac{1}{1-re^{-i\vartheta}}\right\} \\ &= \sum_{n \geqslant 0} \frac{\sin(n+1)\vartheta}{\sin\vartheta} r^n, \end{aligned}$$

and, on the other,

$$\begin{aligned} \frac{1}{f_\vartheta(r)} &= \frac{1}{1-(2r\cos\vartheta - r^2)} = \sum_{j \geqslant 0} (2r\cos\vartheta - r^2)^j \\ &= \sum_{j \geqslant 0} \sum_{0 \leqslant k \leqslant j} \binom{j}{k} (2r\cos\vartheta)^k (-1)^{j-k} r^{2j-2k} \\ &= \sum_{n \geqslant 0} r^n \sum_{0 \leqslant \ell \leqslant n/2} (-1)^\ell \binom{n-\ell}{\ell} (2\cos\vartheta)^{n-2\ell}, \end{aligned}$$

where the last formula results from putting $\ell = j - k$. It follows that

$$\frac{\sin(n+1)\vartheta}{\sin\vartheta} = \sum_{0 \leqslant \ell \leqslant n/2} (-1)^\ell \binom{n-\ell}{\ell} (2\cos\vartheta)^{n-2\ell}. \tag{7.34}$$

Substituting in (7.33) and setting $x := \cos\vartheta$, we obtain

$$\begin{aligned} &2\cos n\vartheta \\ &= \sum_{0 \leqslant \ell \leqslant n/2} (-1)^\ell \binom{n-\ell}{\ell} (2x)^{n-2\ell} - \sum_{0 \leqslant h \leqslant n/2-1} (-1)^h \binom{n-2-h}{h} (2x)^{n-2-2h} \\ &= \sum_{0 \leqslant \ell \leqslant n/2} (-1)^\ell \binom{n-\ell}{\ell} (2x)^{n-2\ell} + \sum_{1 \leqslant \ell \leqslant n/2} (-1)^\ell \binom{n-1-\ell}{\ell-1} (2x)^{n-2\ell}, \end{aligned}$$

and hence finally (7.31).

From this we deduce (7.32) using the inequality

$$\binom{n-m}{m} y^m \leqslant (1+y)^{n-m}$$

with the choice $y := \frac{1}{2}\big(\sqrt{2}-1\big)$. □

Proof of Theorem 7.11. Let Y be the Heaviside function, defined by

$$Y(t) := \begin{cases} 0 & (t<0), \\ 1 & (t \geqslant 0). \end{cases}$$

We shall show the existence of absolute constants B_1, B_2 such that, for each $n \geqslant 1$, there exist polynomials R and S, of degrees $\leqslant n$, satisfying

$$\text{(7.35)} \qquad \begin{cases} R(t) \leqslant Y(t) \leqslant S(t) & (-1 \leqslant t \leqslant 1) \\ \displaystyle\int_{-1}^{1} \big\{S(t) - R(t)\big\}\,\mathrm{d}t \leqslant B_1/n \\ \ell(R) + \ell(S) \leqslant B_2^n. \end{cases}$$

The general statement follows from this special case. Indeed, if f_1 and f_2 are two non-decreasing functions such that $f = f_1 - f_2$, the polynomials defined on $[0,1]$ by

$$p(t) := f(0) + \int_0^1 R(t-\xi)\,\mathrm{d}f_1(\xi) - \int_0^1 S(t-\xi)\,\mathrm{d}f_2(\xi)$$

$$q(t) := f(0) + \int_0^1 S(t-\xi)\,\mathrm{d}f_1(\xi) - \int_0^1 R(t-\xi)\,\mathrm{d}f_2(\xi)$$

satisfy conditions (7.29) with

$$A_1 = B_1 \int_0^1 \mathrm{d}(f_1+f_2)(\xi), \qquad A_2 = 1 + 2|f(0)| + 2B_2 \int_0^1 \mathrm{d}(f_1+f_2)(\xi).$$

The construction of R and S uses the properties of Chebyshev polynomials. For $r \geqslant 1$, we denote by $x_\nu := \cos\big(\pi\big(\nu - \frac{1}{2}\big)/r\big)$ $(1 \leqslant \nu \leqslant r)$ the zeros of $T_r(x)$, and put $m := \big\lfloor \frac{1}{2}(r+1) \big\rfloor$, so that $x_{m+1} < 0 \leqslant x_m$. Then we define $R(t)$ as the unique polynomial of degree $n = 2r-2$ satisfying

$$R(x_\nu) = \begin{cases} 1 & (1 \leqslant \nu \leqslant m-1) \\ 0 & (m \leqslant \nu \leqslant r), \end{cases} \qquad R'(x_\nu) = 0 \quad (\nu \neq m)$$

Thus $R'(t)$ vanishes at least once on each of the $(r-2)$ intervals $]x_{\nu+1}, x_\nu[$ $(1 \leqslant \nu \leqslant r,\ \nu \neq m-1)$. This displays $r-1+r-2 = n-1$ zeros, hence $R'(t)$ does not vanish for $x_m \leqslant t < x_{m-1}$ and $R'(x_m) > 0$ since $R(x_{m-1}) = 1 > R(x_m) = 0$. This implies that $R(t)$ has local maxima at all the points x_ν $(\nu \neq m)$ and a minimum between each pair of consecutive maxima. In particular, we have $R(t) \leqslant Y(t)$ $(-1 \leqslant t \leqslant 1)$, as requested.

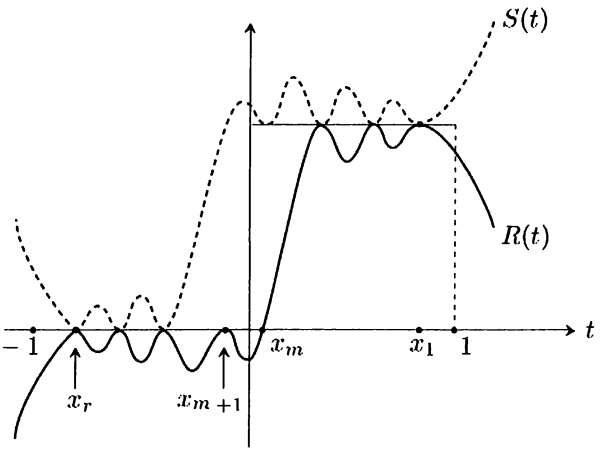

Figure 2. The polynomials R and S.

We define $S(t)$ symmetrically by the equations

$$S(x_\nu) = \begin{cases} 1 & (\nu \leqslant m+1) \\ 0 & (m+2 \leqslant \nu \leqslant r), \end{cases} \qquad S'(x_\nu) = 0 \quad (\nu \neq m+1).$$

and check that this implies $S(t) \geqslant Y(t) \quad (-1 \leqslant t \leqslant 1)$.

In order to obtain the second of the properties (7.35), we introduce the polynomials

$$V_\nu(x) := T_r'(x_\nu)^{-2}\Big(\frac{T_r(x)}{x - x_\nu}\Big)^2 \qquad (1 \leqslant \nu \leqslant r). \tag{7.36}$$

We have

$$V_\nu(x_j) = \begin{cases} 1 & (j = \nu) \\ 0 & (j \neq \nu) \end{cases}, \qquad V_\nu'(x_j) = 0 \quad (j \neq \nu).$$

The polynomial $W(x) := S(x) - R(x) - V_m(x) - V_{m+1}(x)$ has degree n, and vanishes at all the x_ν $(1 \leqslant \nu \leqslant r)$. Hence there exists a polynomial $Z(x)$, of degree $r-2$, such that $W(x) = T_r(x)Z(x)$. Whence

$$\int_{-1}^{1} W(x)\frac{\mathrm{d}x}{\sqrt{1-x^2}} = \int_0^\pi \cos ru \cdot Z(\cos u)\,\mathrm{d}u = 0$$

and so

$$\begin{aligned} \int_0^1 \{S(t) - R(t)\}\,\mathrm{d}t &\leqslant \int_{-1}^{1} \frac{S(x) - R(x)}{\sqrt{1-x^2}}\,\mathrm{d}x \\ &= \int_0^\pi \{V_m(\cos u) + V_{m+1}(\cos u)\}\,\mathrm{d}u. \end{aligned} \tag{7.37}$$

Set $\tau := \frac{\pi}{r}(m - \frac{1}{2})$, so that $x_m = \cos\tau$. We have

$$T_r'(\cos\tau) = r\frac{\sin r\tau}{\sin\tau} = \pm\frac{r}{\sin\tau}.$$

Substituting in (7.36) with $\nu = m$, we get

$$\begin{aligned}\int_0^\pi V_m(\cos u)\,du &= \Big(\frac{\sin\tau}{r}\Big)^2 \int_0^\pi \Big(\frac{\cos ru}{\cos u - \cos\tau}\Big)^2 du \\ &= \Big(\frac{\sin\tau}{r}\Big)^2 \int_0^\pi \Big\{\frac{\sin\big(r(u-\tau)/2\big)}{\sin\big((u-\tau)/2\big)} \cdot \frac{\sin\big(r(u+\tau)/2\big)}{\sin\big((u+\tau)/2\big)}\Big\}^2 du.\end{aligned}$$

We can assume $r \geqslant 3$. Then, for u in $[0,\pi]$, we have

$$\tfrac{1}{4}\pi\big(1 - 2/r\big) \leqslant \tfrac{1}{2}(u+\tau) \leqslant \tfrac{3}{4}\pi.$$

The integral over u is hence

$$\ll \int_{-\pi}^{\pi} \Big\{\frac{\sin\big(rv/2\big)}{\sin\big(v/2\big)}\Big\}^2 dv = 2\pi r,$$

from the classical form of the Fejér kernel

$$\Big(\frac{\sin(rv/2)}{\sin(v/2)}\Big)^2 = r + 2\sum_{1\leqslant j<r}(r-j)\cos(jv).$$

We have therefore shown that

$$\int_0^\pi V_m(\cos u)\,du \ll 1/r \ll 1/n.$$

A parallel computation provides the same estimate for the integral concerning V_{m+1}. Substituting this back into (7.37), we obtain the second property (7.35).

It remains to bound the lengths of R and S. Consider, for example, the case R. We have

$$R(x) = \sum_{0\leqslant j<m} M_j(x)$$

where $M_j(x)$ is the unique polynomial of degree n satisfying

$$M_j(x_\nu) = \begin{cases} 1 & (\nu = j) \\ 0 & (\nu \neq j)\end{cases}, \qquad M_j'(x_\nu) = 0 \quad (\nu \neq m).$$

It therefore suffices to evaluate the lengths of the M_j. We have

$$M_j(x) = \frac{1}{r^2}\Big(\frac{u_j x + v_j}{x - x_m}\Big)\Big(\frac{T_r(x)}{x - x_j}\Big)^2 \tag{7.38}$$

where u_j and v_j are determined by the equations

$$M_j(x_j) = 1, \qquad M_j'(x_j) = 0.$$

An easy calculation enables us to show that

$$u_j = 1 - x_j x_m, \qquad v_j = 2x_j^2 x_m - x_j^3 - x_m.$$

We hence have, in particular, that $\max(|u_j|, |v_j|) \leqslant 2$. From (7.32), we can write

$$T_r(x) = \vartheta_r \prod_{1 \leqslant \nu \leqslant r} (x - x_\nu)$$

with $|\vartheta_r| \leqslant r\big(1+\sqrt{2}\big)^r$. The required upper bound for $\ell(M_j)$ then follows easily from

$$M_j(x) = r^{-2}\vartheta_r(u_j x + v_j)(x - x_m) \prod_{\substack{1 \leqslant \nu \leqslant r \\ \nu \neq j,m}} (x - x_\nu)^2.$$

This completes the proof of Theorem 7.11. □

7.5. Ikehara's theorem

In terms of Dirichlet series, the Hardy–Littlewood–Karamata theorem may be described as a *limit Tauberian theorem*, in the sense that it relates the behavior of $\sum_{n \geqslant 1} a_n/n^\sigma$ for $\sigma > 1$ to that of $\sum_{n \leqslant x} a_n/n$. It makes no assumptions about the values of the series at non-real points, and thus does not yield a proof of the prime number theorem—see the notes for this chapter for more details concerning this last point.

This section is devoted to a Tauberian theorem of a different nature. It concerns obtaining asymptotic information about the summatory function $\sum_{n \leqslant x} a_n$ using assumptions on the values of the Dirichlet series at non-real points $s = \sigma + i\tau$ with abscissa $\sigma > 1$. Schematically, one passes from the half-plane $\sigma > 1$ to the point $s = 0$, and this discontinuity between the hypothesis and the conclusion gives us grounds for calling a result of this kind a *transcendental Tauberian theorem.*

Ikehara's theorem[2] (1931) belongs to this category. We give a formulation below which strengthens it in two directions. On the one hand, we generalize the assumption by considering series having a singularity of type $s^{-\omega-1}$ ($\omega > -1$) at $s = 0$—this form, which is today classical, is essentially due to Ingham (1935) and Delange (1954). On the other hand, we give an effective version of the conclusion, with an explicit error term. Such a result does not seem to exist as such in the literature, but follows rather easily from the method of Ganelius (1971).

[2]Often called in the literature the Wiener–Ikehara theorem, to account for the contribution of Wiener's Tauberian theory to the proof of the result.

Theorem 7.13 ("Effective" Ikehara–Ingham–Delange theorem). *Let $A(t)$ be a non-decreasing function such that the integral*

$$F(s) := \int_0^\infty \mathrm{e}^{-st}\,\mathrm{d}A(t) \tag{7.39}$$

converges for $\sigma > a > 0$. Suppose that there exist constants $c \geqslant 0$ and $\omega > -1$, such that the function

$$G(s) := \frac{F(s+a)}{s+a} - \frac{c}{s^{\omega+1}} \qquad (\sigma > 0) \tag{7.40}$$

satisfies

$$\eta(\sigma, T) := \sigma^\omega \int_{-T}^{T} |G(2\sigma + i\tau) - G(\sigma + i\tau)|\,\mathrm{d}\tau = o(1) \qquad (\sigma \to 0+) \tag{7.41}$$

for each fixed $T > 0$. Then we have

$$A(x) = \Big\{\frac{c}{\Gamma(\omega+1)} + O\big(\varrho(x)\big)\Big\}\mathrm{e}^{ax}x^\omega \qquad (x \geqslant 1), \tag{7.42}$$

with

$$\varrho(x) := \inf_{T \geqslant 32(a+1)} \Big\{\frac{1}{T} + \eta\Big(\frac{1}{x}, T\Big) + \frac{1}{(Tx)^{\omega+1}}\Big\}.$$

The implicit constant in (7.42) depends only on a, c and ω. An admissible choice for this constant is

$$52 + 1652c(a+1)(\omega+2) + 69c\big(1 + \mathrm{e}^{1-\omega}(\omega+1)^{\omega+2}\big)/\Gamma(\omega+1).$$

Ganelius' argument rests on a local version of Bohr's inequality (1935)

$$\|f\|_\infty \ll \|f'\|_\infty / T \tag{7.43}$$

which is valid, when both sides are finite, for any $\mathbb{R}$–integrable function whose Fourier transform $\widehat{f}(\tau)$ vanishes for $|\tau| \leqslant T$. Here we are content with a slightly weakened form of Ganelius' result, which is however sufficient for the applications envisaged. The proof depends on ideas similar to those in the classical Berry–Esseen inequality—cf. for instance, Feller (1971).

Theorem 7.14 (Ganelius). *Let g be an integrable and bounded function on $\mathbb{R}$. Suppose there is a positive real number T such that*

$$\sup_{x \leqslant y \leqslant x+1/T} \{g(y) - g(x)\} \leqslant K < \infty, \tag{7.44}$$

$$\widehat{g}(\tau) := \int_{-\infty}^{+\infty} \mathrm{e}^{-i\tau x} g(x)\,\mathrm{d}x = 0 \qquad (|\tau| \leqslant T). \tag{7.45}$$

Then we have

$$\|g\|_\infty := \sup_{x \in \mathbb{R}} |g(x)| \leqslant 16K. \tag{7.46}$$

Remark. We have made no attempt here to optimize the constant 16 appearing in (7.46).

Proof. We can suppose that $T = 1$: the general case will follow by considering $g(x/T)$. Assumption (7.45) implies that

$$\widehat{g}(\tau)\widehat{\chi}(\tau) = 0 \qquad (\tau \in \mathbb{R}) \tag{7.47}$$

for any integrable function χ whose Fourier transform has support contained in $[-1, 1]$. We choose

$$\chi(t) = \frac{1}{2\pi}\Big(\frac{\sin(t/2)}{t/2}\Big)^2$$

so that $\widehat{\chi}(\tau) = \max(1 - |\tau|, 0)$. Relation (7.47) implies that

$$\int_{-\infty}^{+\infty} g(x-t)\chi(t)\,\mathrm{d}t = 0 \qquad (x \in \mathbb{R}) \tag{7.48}$$

and the stated inequality will follow by using the fact that $\chi(t)$ "peaks" at $t = 0$. More precisely, we shall appeal to the inequality

$$I := \int_{|t|>5} \chi(t)\,\mathrm{d}t \leqslant \frac{4}{\pi}\int_5^\infty \frac{\mathrm{d}t}{t^2} = \frac{4}{5\pi}\cdot \tag{7.49}$$

Let ε, $0 < \varepsilon < 1$, be a fixed real number and let $\vartheta = \pm 1$ be so chosen that

$$\|g\|_\infty = \sup_{x\in\mathbb{R}}\{\vartheta g(x)\}.$$

Then there exists an $x_0 = x_0(\varepsilon)$ such that $\vartheta g(x_0) \geqslant (1-\varepsilon)\|g\|_\infty$. Applying (7.48) with $x = x_0 - 5\vartheta$ and taking account of (7.49) we can write

$$\begin{aligned}
0 &= \vartheta\int_{-\infty}^{+\infty} g(x_0 - t - 5\vartheta)\chi(t)\,\mathrm{d}t \\
&= \vartheta\int_{-5}^{5} g(x_0)\chi(t)\,\mathrm{d}t - \vartheta\int_{-5}^{5}\{g(x_0) - g(x_0 - t - 5\vartheta)\}\chi(t)\,\mathrm{d}t \\
&\qquad + \vartheta\int_{|t|>5} g(x_0 - t - 5\vartheta)\chi(t)\,\mathrm{d}t \\
&\geqslant (1-\varepsilon)(1-I)\|g\|_\infty - 10K(1-I) - I\|g\|_\infty,
\end{aligned}$$

where we have used the fact that, for all t such that $|t| \leqslant 5$, we have

$$\vartheta\{g(x_0) - g(x_0 - t - 5\vartheta)\} \leqslant \sup_{x<y\leqslant x+10}\{g(y) - g(x)\} \leqslant 10K.$$

We therefore obtain

$$\{(1-\varepsilon)(1-I) - I\}\|g\|_\infty \leqslant 10K(1-I)$$

and, letting ε tend to 0,

$$\|g\|_\infty \leqslant \frac{10K(1-I)}{1-2I} \leqslant 16K,$$

as stated. □

An application of Theorem 7.14 along the lines of Ganelius' work would lead to an effective Tauberian theorem with rather unusual conditions, and which would not entirely generalize the Ikehara–Ingham–Delange theorem. We proceed in a slightly different manner by noting that Theorem 7.14 easily implies a result which simultaneously contains the Berry–Esseen inequality (see § 7.6) and Theorem 7.13.

Theorem 7.15. *Let g be an integrable and bounded function on $\mathbb{R}$. Under assumption* (7.44)*, we have*

$$\|g\|_\infty \leqslant 16K + 6\int_{-T}^{T} |\widehat{g}(\tau)|\,\mathrm{d}\tau.$$

Proof. Assume $\varepsilon > 0$. We shall apply Theorem 7.14 to $g - f$ where f is the convolution of g with the integrable function

$$\alpha(t) := \frac{2}{\pi\varepsilon t^2}\sin\bigl(\varepsilon t/2\bigr)\sin\bigl((2T+\varepsilon)t/2\bigr)$$

whose Fourier transform is the trapezoidal function

$$\widehat{\alpha}(\tau) = \begin{cases} 1 & (|\tau| \leqslant T) \\ (T+\varepsilon-|\tau|)/\varepsilon & (T < |\tau| \leqslant T+\varepsilon) \\ 0 & (|\tau| > T+\varepsilon). \end{cases}$$

Since $\widehat{f} = \widehat{g}\,\widehat{\alpha}$ has compact support, we have

$$\|f\|_\infty \leqslant \frac{1}{2\pi}\|\widehat{f}\|_1 \leqslant \frac{1}{2\pi}\int_{-T-\varepsilon}^{T+\varepsilon} |\widehat{g}(\tau)|\,\mathrm{d}\tau. \tag{7.50}$$

It follows that

$$\sup_{x\leqslant y\leqslant x+1/T}\bigl\{g(y) - f(y) - \bigl(g(x) - f(x)\bigr)\bigr\} \leqslant K + 2\|f\|_\infty,$$

from which, by Theorem 7.14,

$$\|g\|_\infty \leqslant \|g-f\|_\infty + \|f\|_\infty \leqslant 16K + 33\|f\|_\infty.$$

Since $33/2\pi \leqslant 6$, the stated result follows by letting ε tend to 0. □

Proof of Theorem 7.13. Without loss of generality, we may assume that $A(0+) = 0$ and define $A(t)$ to be 0 for $t \leqslant 0$. As a first step, we apply Theorem 7.15 to the function

$$g_\sigma(t) := A(t)\mathrm{e}^{-(a+\sigma)t}(1 - \mathrm{e}^{-\sigma t})$$

whose Fourier transform is given, for any value of the positive parameter σ, by the formula

$$\widehat{g}_\sigma(\tau) = G(\sigma+i\tau) - G(2\sigma+i\tau) + c\bigl\{(\sigma+i\tau)^{-\omega-1} - (2\sigma+i\tau)^{-\omega-1}\bigr\}.$$

The modulus of the last term does not exceed

$$c(\omega+1)\left|\int_{\sigma+i\tau}^{2\sigma+i\tau}\frac{\mathrm{d}s}{s^{\omega+2}}\right| \leqslant \frac{c(\omega+1)\sigma}{|\sigma+i\tau|^{\omega+2}}.$$

In view of (7.41), we can thus write

$$\begin{aligned}\int_{-T}^{T}|\widehat{g}_\sigma(\tau)|\,\mathrm{d}\tau &\leqslant \frac{1}{\sigma^\omega}\left\{\eta(\sigma,T)+c(\omega+1)\sigma^{\omega+1}\int_{-T}^{T}\frac{\mathrm{d}\tau}{\max(\sigma,|\tau|)^{\omega+2}}\right\}\\ &\leqslant \frac{\eta(\sigma,T)+2c(\omega+2)}{\sigma^\omega}.\end{aligned}$$

Then the monotonicity and the positivity of A imply that, for $x>0$ and $y\geqslant 0$, we have

$$\begin{aligned}g_\sigma(x+y)-g_\sigma(x) &\geqslant A(x)\mathrm{e}^{-(a+\sigma)x}(1-\mathrm{e}^{-\sigma x})(\mathrm{e}^{-(a+\sigma)y}-1)\\ &\geqslant -(a+\sigma)\|g_\sigma\|_\infty y.\end{aligned}\tag{7.51}$$

This inequality is also satisfied when $x\leqslant 0$, since then $g_\sigma(x)=0$. Applying Theorem 7.15 to $-g_\sigma$, with $K=(a+\sigma)\|g_\sigma\|_\infty/T$, we deduce that, for $0<\sigma\leqslant 1$,

$$\|g_\sigma\|_\infty \leqslant \frac{6}{\sigma^\omega}\{\eta(\sigma,T)+2c(\omega+2)\}+\frac{16}{T}(a+1)\|g_\sigma\|_\infty.$$

Choosing $T=T_a:=32(a+1)$, we obtain

$$\|g_\sigma\|_\infty \leqslant M_1(\sigma)\sigma^{-\omega} \qquad (0<\sigma\leqslant 1),\tag{7.52}$$

with $M_1(\sigma):=12\{\eta(\sigma,T_a)+2c(\omega+2)\}$.

Let us set

$$B(t):=\begin{cases}\dfrac{c}{\Gamma(\omega+1)}\mathrm{e}^{-t}(1-\mathrm{e}^{-t})t^\omega & (t>0)\\ 0 & (t\leqslant 0).\end{cases}$$

The second and final step of the proof consists in a further application of Theorem 7.15, this time to the function

$$G_\sigma(t):=g_\sigma(t)-\sigma^{-\omega}B(\sigma t)=\left\{A(t)\mathrm{e}^{-at}-\frac{c}{\Gamma(\omega+1)}t^\omega\right\}\mathrm{e}^{-\sigma t}(1-\mathrm{e}^{-\sigma t})^+$$

with Fourier transform

$$\widehat{G}_\sigma(\tau)=G(\sigma+i\tau)-G(2\sigma+i\tau).$$

For all $t>0$, we have

$$\begin{aligned}B'(t)&=\frac{c}{\Gamma(\omega+1)}\mathrm{e}^{-t}t^{\omega-1}\{2t\mathrm{e}^{-t}-t+\omega(1-\mathrm{e}^{-t})\}\\ &\leqslant \frac{c(\omega+1)}{\Gamma(\omega+1)}\mathrm{e}^{-t}(1-\mathrm{e}^{-t})t^{\omega-1}\leqslant\frac{c(\omega+1)}{\Gamma(\omega+1)}\mathrm{e}^{-t}t^\omega.\end{aligned}$$

Furthermore, B is continuous on $\mathbb{R}$, and so equals the integral of its derivative, defined, for example, to be 0 at $t = 0$. Therefore we have, on the one hand, for $x < 0$, $x + y \geqslant 0$,

$$B(x+y) - B(x) \leqslant \int_0^y \frac{c(\omega+1)}{\Gamma(\omega+1)} \mathrm{e}^{-t} t^\omega \,\mathrm{d}t \leqslant \frac{c}{\Gamma(\omega+1)} y^{\omega+1},$$

and, on the other hand, for $x \geqslant 0$, $y \geqslant 0$,

$$\begin{aligned} B(x+y) - B(x) &\leqslant \frac{c(\omega+1)}{\Gamma(\omega+1)} \mathrm{e}^{-x} \int_x^{x+y} t^\omega \,\mathrm{d}t \\ &\leqslant \frac{c}{\Gamma(\omega+1)} \mathrm{e}^{-x} \Big\{ \vartheta(\omega)(\omega+1)(x+y)^\omega y + \big(1-\vartheta(\omega)\big) y^{\omega+1} \Big\} \end{aligned}$$

with $\vartheta(\omega) := 1$ if $\omega \geqslant 0$, and $\vartheta(\omega) := 0$ if $-1 < \omega < 0$. In the case $\omega < 0$, we have used the classical Minkowski inequality

$$(x+y)^{\omega+1} - x^{\omega+1} \leqslant y^{\omega+1} \qquad (-1 < \omega < 0).$$

By an easy calculation, we may deduce that, for $x \in \mathbb{R}$, $0 \leqslant y \leqslant 1/T \leqslant 1$, $0 < \sigma \leqslant 1$, we have

$$B(\sigma x + \sigma y) - B(\sigma x) \leqslant D\Big\{ \frac{\sigma}{T} + \Big(\frac{\sigma}{T}\Big)^{\omega+1} \Big\},$$

with

$$D := \frac{c}{\Gamma(\omega+1)} \big(1 + \mathrm{e}^{1-\omega} (\omega+1)^{\omega+2}\big).$$

Indeed, it can be checked that

$$D\Gamma(\omega+1) \geqslant c \max\{1, \vartheta(\omega)(\omega+1) \operatorname*{esup}_{t \geqslant 0} \mathrm{e}^{-t} t^\omega\}.$$

From (7.51) and (7.52), we obtain under the same conditions

$$\begin{aligned} G_\sigma(x+y) - G_\sigma(x) &\geqslant -\Big\{ \frac{(a+\sigma)\|g_\sigma\|_\infty}{T} + \frac{B(\sigma x + \sigma y) - B(\sigma x)}{\sigma^\omega} \Big\} \\ &\geqslant -\frac{1}{\sigma^\omega} \Big\{ \frac{M_2(\sigma)}{T} + D\Big(\frac{\sigma}{T}\Big)^{\omega+1} \Big\} \end{aligned}$$

with $M_2(\sigma) := (a+1)M_1(\sigma) + D = 12(a+1)\eta(\sigma, T_a) + 24c(a+1)(\omega+2) + D$. By then applying Theorem 7.15 to $-G_\sigma$, we infer that, for $T \geqslant T_a$,

$$\begin{aligned} |G_\sigma(x)| &\leqslant \frac{1}{\sigma^\omega} \Big\{ \frac{16 M_2(\sigma)}{T} + 6\eta(\sigma, T) + 16 D \Big(\frac{\sigma}{T}\Big)^{\omega+1} \Big\} \\ &\leqslant \frac{1}{\sigma^\omega} \Big\{ \frac{384c(a+1)(\omega+2) + 16D}{T} + 12\eta(\sigma, T) + 16D \Big(\frac{\sigma}{T}\Big)^{\omega+1} \Big\} \\ &\leqslant \frac{M}{\sigma^\omega} \Big\{ \frac{1}{T} + \eta(\sigma, T) + \Big(\frac{\sigma}{T}\Big)^{\omega+1} \Big\}, \end{aligned}$$

with

$$M := \max\{384c(a+1)(\omega+2)+16D,\ 12\}$$
$$\leqslant 12 + 384c(a+1)(\omega+2) + 16c\bigl(1+\mathrm{e}^{1-\omega}(\omega+1)^{\omega+2}\bigr)/\Gamma(\omega+1).$$

The conclusion follows by choosing $\sigma = 1/x$. The implicit constant in (7.42) does not exceed $\mathrm{e}M/(1-1/\mathrm{e})$. □

Ikehara's theorem implies the prime number theorem, assuming only that $\zeta(s) \neq 0$ for $\sigma = 1$, without requiring an upper bound for $1/\zeta(s)$—cf. Exercise 224. Of course, any explicit upper bound for $\zeta'(s)/\zeta(s)$ or for $1/\zeta(s)$ will provide, via Theorem 7.13, a corresponding effective version of the prime number theorem—see, in particular, Exercises 225 and 226.

7.6. The Berry–Esseen inequality

As mentioned in the previous section, we now show how Theorem 7.15 implies the probabilistic inequality of Berry–Esseen. According to usual convention, we say that a real function F of a real variable is a *distribution function*, if it is non-decreasing and satisfies

$$F(-\infty) = 0, \qquad F(+\infty) = 1. \tag{7.53}$$

The *characteristic function* of F is the Fourier–Stieltjes transform

$$f(\tau) := \int_{-\infty}^{+\infty} \mathrm{e}^{i\tau x}\,\mathrm{d}F(x).$$

Theorem 7.16 (Berry–Esseen). *Let F, G be two distribution functions with respective characteristic functions f, g. Suppose that G is differentiable and that G' is bounded on $\mathbb{R}$. Then we have for all $T > 0$*

$$\|F-G\|_\infty \leqslant 16\frac{\|G'\|_\infty}{T} + 6\int_{-T}^{T}\Bigl|\frac{f(\tau)-g(\tau)}{\tau}\Bigr|\,\mathrm{d}\tau. \tag{7.54}$$

Proof. Write $H := F - G$, and, for each value of the positive parameter ε, introduce the function

$$H_\varepsilon(x) := -\int_0^{+\infty} \mathrm{e}^{-\varepsilon t}\,\mathrm{d}H(x-t) = \mathrm{e}^{-\varepsilon x}\int_{-\infty}^{x} \mathrm{e}^{\varepsilon t}\,\mathrm{d}H(t). \tag{7.55}$$

It is easily verified that H_ε is integrable and that its Fourier transform has the value

$$\widehat{H}_\varepsilon(\tau) = \int_{-\infty}^{+\infty} \mathrm{e}^{-i\tau x}H_\varepsilon(x)\,\mathrm{d}x = \frac{f(-\tau)-g(-\tau)}{\varepsilon+i\tau}.$$

The second equality (7.55) gives, by partial integration,

$$H_\varepsilon(x) = H(x) - \varepsilon\mathrm{e}^{-\varepsilon x}\int_{-\infty}^{x} \mathrm{e}^{\varepsilon t}H(t)\,\mathrm{d}t. \tag{7.56}$$

Since $\|H\|_\infty \leqslant 1$ and $H(-\infty) = 0$, we easily deduce that we have, for any fixed x,

$$\lim_{\varepsilon \to 0+} H_\varepsilon(x) = H(x).$$

Write $\alpha := \|G'\|_\infty$. Clearly we have

$$H(x+y) - H(x) \geqslant -\alpha y \qquad (x \in \mathbb{R},\ y > 0)$$

and

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big\{\mathrm{e}^{-\varepsilon x}\int_{-\infty}^{x} \mathrm{e}^{\varepsilon t} H(t)\,\mathrm{d}t\Big\} = -\varepsilon \mathrm{e}^{-\varepsilon x}\int_{-\infty}^{x} \mathrm{e}^{\varepsilon t} H(t)\,\mathrm{d}t + H(x) \leqslant 2.$$

Substituting in (7.56), it follows that

$$H_\varepsilon(x+y) - H_\varepsilon(x) \geqslant -\frac{(\alpha + 2\varepsilon)}{T} \qquad \Big(x \in \mathbb{R},\ 0 \leqslant y \leqslant \frac{1}{T}\Big).$$

Applying Theorem 7.15 to $-H_\varepsilon$ with $K := \dfrac{\alpha + 2\varepsilon}{T}$, we obtain

$$|H_\varepsilon(x)| \leqslant 16\frac{(\alpha + 2\varepsilon)}{T} + 6\int_{-T}^{T} \Big|\frac{f(\tau) - g(\tau)}{\tau}\Big|\,\mathrm{d}\tau \qquad (x \in \mathbb{R}).$$

This yields (7.54) by letting ε tend to 0. □

Remarks. (i) The inequality (7.54) only provides a finite bound if $|f - g|$ is integrable with respect to the measure $\mathrm{d}\tau/\tau$. A sufficient condition for this is that f and g belong to the space E of functions φ such that $|1-\varphi(\tau)|/|\tau|$ is integrable at the origin. It is not difficult to show that a characteristic function φ is in E if, and only if, its original function H satisfies $\int_{\mathbb{R}} \ln^+ |z|\,\mathrm{d}H(z) < \infty$.

(ii) The numerical value of the constants in the Berry–Esseen inequality usually has little impact on the applications. Feller (1971) gives the values $24/\pi$ and $1/\pi$, both of which can be improved, respectively to π and 0.16318—cf. Vaaler (1985), th. 13.

7.7. Holomorphy as a Tauberian condition

Let us start with an extension of Theorem 7.14 which is essentially due to Ganelius (1971).

Theorem 7.17. *Let g be an integrable and bounded function on $\mathbb{R}$, C be a positive constant, and $\psi : \mathbb{R} \to [1, \infty[$ be a non-decreasing function such that $\psi(x) = 1$ for $x \leqslant 0$, $\psi(2x) \leqslant C\psi(x)$ for $x \geqslant 0$. Suppose that, for a suitable real number $T > 0$, we have*

(i) $\displaystyle\sup_{0 \leqslant y \leqslant 1/T} \{g(x+y) - g(x)\} \leqslant K/\psi(x) \qquad (x \in \mathbb{R})$,

(ii) $\widehat{g} \in \mathcal{C}^\infty([-T,T])$.

Then there exist an integer $N = N(C,K)$ and constant A, depending only on C, K, T and $M := \sup_{n \leqslant N} \|\widehat{g}^{(n)}\|_{[-T,T]}$, such that

$$|g(x)| \leqslant A/\psi(x) \qquad (x \in \mathbb{R}). \tag{7.57}$$

Proof. Suppose, without loss of generality, that $T = 1$. We may clearly restrict to establishing (7.57) for $x \geqslant x_0(C)$.

Let us first make the extra assumption that $\widehat{g}(\tau) = 0$ for $|\tau| \leqslant 1$. We then note that Theorem 7.14 allows us to bound $\|g\|_\infty$ from above by $16K$. Let χ be a rapidly decreasing non-negative function in $\mathcal{C}^\infty(\mathbb{R})$ with Fourier transform supported in $[-1,1]$, and such that $\widehat{\chi}(0) = 1$.[3] We have, in particular, $\chi(x) \ll_n 1/(1+|x|^n)$ for each integer n, and, consequently, $\chi(x) \leqslant B/\{(1+x^2)\psi(|x|)\}$ for a suitable constant B.

Besides, there exists an integer $R > 0$ such that

$$I_R := \int_{|t|>R} \chi(t)\,\mathrm{d}t < \frac{1}{2C+1}\cdot$$

Let $x \geqslant 4R$ and put $\vartheta := \operatorname{sgn} g(x)$. We have $g * \chi = 0$, whence

$$\begin{aligned}\vartheta g(x) &\int_{-R}^{R} \chi(t)\,\mathrm{d}t \\ &= \vartheta \int_{-R}^{R} \{g(x)-g(x-t-R\vartheta)\}\chi(t)\,\mathrm{d}t - \vartheta \int_{|t|>R} g(x-t-R\vartheta)\chi(t)\,\mathrm{d}t.\end{aligned}$$

It follows that

$$\begin{aligned}(1-I_R)|g(x)| &\leqslant \frac{2RK(1-I_R)}{\psi(x-2R)} + \int_{|t|>R} |g(x-t-R\vartheta)|\chi(t)\,\mathrm{d}t \\ &\leqslant \frac{A_1(1-I_R)}{\psi(x)} + \int_{|t|>R} |g(x-t-R\vartheta)|\chi(t)\,\mathrm{d}t,\end{aligned}$$

with $A_1 := 2CRK$. In the integral, we may bound $|g(x-t-R\vartheta)|$ by $\sup_{y>x/2}|g(y)|$ if $|t+R\vartheta| \leqslant x/2$ and by $16K$ otherwise. Noting that $I_R/(1-I_R) \leqslant 1/2C$, we get

$$\begin{aligned}|g(x)| &\leqslant \frac{A_1}{\psi(x)} + \frac{16K}{1-I_R}\int_{|t+R\vartheta|>x/2} \chi(t)\,\mathrm{d}t + \frac{1}{2C}\sup_{y>x/2}|g(y)| \\ &\leqslant \frac{A_1}{\psi(x)} + \frac{16C^2BK}{(1-I_R)\psi(x)}\int_{-\infty}^{\infty}\frac{\mathrm{d}t}{1+t^2} + \frac{1}{2C}\sup_{y>x/2}|g(y)| \\ &\leqslant \frac{A_2}{\psi(x)} + \frac{1}{2C}\sup_{y>x/2}|g(y)|,\end{aligned}$$

[3] A possible choice is $\chi(x) := \{\widehat{\sigma}(x)\}^2/\int_{\mathbb{R}}\{\widehat{\sigma}(t)\}^2\,\mathrm{d}t$, with

$$\sigma(t) := \exp\{-1/(1-4t^2)\}\mathbf{1}_{]-1/2,1/2[}(t) \qquad (t \in \mathbb{R}).$$

with $A_2 = A_2(C, K)$. Iterating, it follows that, for any integer $J \geqslant 0$,

$$|g(x)| \leqslant \sum_{0 \leqslant j \leqslant J} \frac{A_2}{(2C)^j \psi(x/2^j)} + \frac{1}{2^J C^J} \sup_{y > x/2^J} |g(y)|$$
$$\leqslant \sum_{0 \leqslant j \leqslant J} \frac{A_2 C^j}{2^j C^j \psi(x)} + \frac{16K}{2^J C^J}.$$

Letting J tend to infinity, we obtain, since $C \geqslant 1$,

$$|g(x)| \leqslant A_3/\psi(x)$$

with $A_3 := 2A_2$.

Now let us consider the general case when $\widehat{g}(\tau)$ does not necessarily vanish on $[-1, 1]$. We then introduce a function $\alpha \in \mathcal{C}^\infty(\mathbb{R}) \cap \mathcal{L}^1(\mathbb{R})$ such that $\widehat{\alpha}(\tau) = 1$ for $|\tau| \leqslant \frac{1}{2}$, $\widehat{\alpha}(\tau) = 0$ for $|\tau| \geqslant 1$, and we set

$$f(x) := \int_{\mathbb{R}} \mathrm{e}^{i\tau x} \widehat{\alpha}(\tau) \widehat{g}(\tau) \, \mathrm{d}\tau.$$

Successive integrations by parts show that $f(x) \ll_{M,N} 1/(1 + |x|^N)$ where $M = M_N$ is defined as in the statement of the theorem. We thus have

$$|f(x)| \leqslant A_4/\psi(x) \tag{7.58}$$

where A_4 only depends on C and M_N, with suitably chosen $N = N(C)$. Since we have $\widehat{f}(\tau) - \widehat{g}(\tau) = 0$ for $|\tau| \leqslant \frac{1}{2}$, we can apply the result established in the first part of the proof to obtain

$$|f(x) - g(x)| \leqslant A_5/\psi(x).$$

Taking (7.58) into account, this completes the proof. □

The above theorem provides an easy proof of the following result due to Korevaar (1954b), which is itself a refinement of a theorem of Fatou (1906).

Theorem 7.18 (Fatou-Korevaar). *Let $f(z) = \sum_{n \geqslant 0} a_n z^n$ be a power series converging for $|z| < 1$ and with a holomorphic continuation at $z = 1$. If there exists a function ψ satisfying the assumptions of Theorem 7.17 and such that*

$$a_n \geqslant -1/\psi(n) \qquad (n \geqslant 0), \tag{7.59}$$

then $\sum_{n \geqslant 0} a_n$ converges to $f(1)$ and we have

$$\sum_{n \leqslant x} a_n = f(1) + O\Big(\frac{1}{\psi(x)}\Big) \qquad (x \to \infty).$$

Proof. Let $r > 0$ be such that $f(z)$ can be holomorphically continued for $|z-1| \leqslant r$ and let ε satisfy $0 < \varepsilon < \ln\{1/(1-r)\}$. Let us set

$$g_\varepsilon(x) := \sum_{n \leqslant x} a_n \mathrm{e}^{-\varepsilon n} - f(\mathrm{e}^{-\varepsilon}) \mathbf{1}_{\mathbb{R}^+}(x),$$

so that

$$\widehat{g}_\varepsilon(\tau) = -\int_0^\infty \mathrm{e}^{-i\tau x} \sum_{n>x} a_n \mathrm{e}^{-\varepsilon n}\, \mathrm{d}x = \frac{f(\mathrm{e}^{-\varepsilon-i\tau}) - f(\mathrm{e}^{-\varepsilon})}{i\tau}.$$

For $\varepsilon > 0$ and $|\tau| \leqslant T$ with T sufficiently small, we then have $\widehat{g}_\varepsilon \in \mathcal{C}^\infty([-T,T])$ and the derivatives of $\widehat{g}_\varepsilon$ are bounded independently of ε. Since we also have

$$g_\varepsilon(x+y) - g_\varepsilon(x) = \sum_{x<n\leqslant x+y} \mathrm{e}^{-\varepsilon n} a_n - f(\mathrm{e}^{-\varepsilon}) \mathbf{1}_{]x,x+y]} \geqslant -\frac{1+y+M}{\psi(x)},$$

with $M := \sup_{|z-1|\leqslant r} |f(z)|$, we can apply Theorem 7.17 to obtain

$$g_\varepsilon(x) \ll 1/\psi(x) \qquad (x \geqslant 1)$$

where the constant involved is independent of ε. The stated result is deduced from this by letting ε tend to 0. □

Similarly we obtain an effective form of a theorem of Riesz (1909) concerning Dirichlet series.

Theorem 7.19. *Let $F(s) := \sum_{n\geqslant 1} a_n/n^s$ be a Dirichlet series converging for $\sigma > 1$ and with a holomorphic continuation at $s = 1$. If we have $a_n \geqslant -1/\psi(\ln n)$ $(n \geqslant 2)$ for a function ψ satisfying the assumptions of Theorem 7.17, then*

$$\sum_{n\leqslant x} \frac{a_n}{n} = F(1) + O\Big(\frac{1}{\psi(\ln x)}\Big) \qquad (x \to \infty). \tag{7.60}$$

Proof. It is sufficient to consider, for sufficiently small $\delta > 0$, the function

$$g_\delta(x) := \mathbf{1}_{\mathbb{R}^+}(x) \sum_{n>\mathrm{e}^x} \frac{a_n}{n^{1+\delta}}.$$

We omit the details, which are similar to those in the proof of Theorem 7.18. □

As noted by Landau (1910), the Tauberian condition can be replaced by an inequality on average, but bilateral. We derive here an effective version of Theorem 2.9.

Theorem 7.20. *In Theorem* 7.19, *the conclusion persists under the assumption*

$$\sum_{m\leqslant n} a_m \ll n/\psi(\ln n) \qquad (n \geqslant 2).$$

Proof. Write $\varphi_n(s) := n^{-s} - (n+1)^{-s} - sn^{-s-1}$ $(s \in \mathbb{C})$, so that

$$|\varphi_n(s)| \leqslant |s(s+1)|/n^{\sigma+2} \qquad (\sigma > 0).$$

Define $A_n := \sum_{m\leqslant n} a_m$. We then have $A_n \ll n/\psi(\ln n)$ $(n \geqslant 2)$. It is easily checked that, for any integer $N \geqslant 1$ and any $s \in \mathbb{C}$, we have

$$\sum_{1\leqslant n\leqslant N} \frac{a_n}{n^s} - s \sum_{1\leqslant n\leqslant N} \frac{A_n}{n^{s+1}} = \sum_{1\leqslant n\leqslant N} A_n\varphi_n(s) + \frac{A_N}{(N+1)^s}. \tag{7.61}$$

From this we deduce, taking $N \to \infty$, that

$$G(s) := \sum_{n\geqslant 1} \frac{A_n}{n^{s+1}} = \frac{F(s)}{s} - \frac{1}{s}\sum_{n\geqslant 1} A_n\varphi_n(s) \qquad (\sigma > 1).$$

The second series is holomorphic for $\sigma > 0$, therefore $G(s)$ has a holomorphic continuation at $\sigma = 1$. We can hence apply Theorem 7.19 to $G(s)$, which yields

$$\sum_{n\leqslant N} \frac{A_n}{n^2} = G(1) + O\Big(\frac{1}{\psi(\ln N)}\Big).$$

Substituting in (7.61) with $s = 1$, we obtain the stated result. □

7.8. Arithmetic Tauberian theorems

The attempts at elementary proofs of the prime number theorem led number theorists to consider a more general problem. Let f and g be arithmetic functions linked by the relation $f = g * \mathbf{1}$. Setting

$$F(x) := \sum_{n\leqslant x} f(n), \quad G(x) := \sum_{n\leqslant x} g(n),$$

one then aims at understanding what consequences for g can be derived from the assumption

$$F(x) = ax + o(x) \quad (x \to \infty). \tag{7.62}$$

The emblematical case is given by the choice $g = \mu$, $f = \delta$, and hence $a = 0$. The equation $G(x) = o(x)$ is then elementarily equivalent to the prime number theorem—see Theorem I.3.8. However, our present goal is not to find a new elementary proof of the prime number theorem, of which we will make free use in the sequel.

Theorem 7.21 (Ingham, 1945). *Under assumption* (7.62), *we have*

$$\sum_{n\leqslant x}\frac{g(n)}{n}=\frac{G(x)}{x}+a+o(1)\qquad(x\to\infty). \tag{7.63}$$

Proof. We have $F(x)=\sum_{n\leqslant x}g(n)\,\lfloor x/n\rfloor$. Up to altering the value of $g(1)$, we may hence assume $a=0$. Furthermore, $G(x)=\sum_{n\leqslant x}\mu(n)F(x/n)$. Therefore

$$\begin{aligned}\sum_{n\leqslant x}\frac{g(n)}{n}-\frac{G(x)}{x}&=\int_{1-}^{x}\Big(\frac{1}{u}-\frac{1}{x}\Big)\,\mathrm{d}G(u)=\int_1^x\frac{G(u)}{u^2}\,\mathrm{d}u\\&=\sum_{n\leqslant x}\mu(n)\int_n^x F(u/n)\frac{\mathrm{d}u}{u^2}=\sum_{n\leqslant x}\frac{\mu(n)}{n}\int_1^{x/n}F(v)\frac{\mathrm{d}v}{v^2}\\&=\int_1^x F(v)\sum_{n\leqslant x/v}\frac{\mu(n)}{n}\frac{\mathrm{d}v}{v^2}=o\Big(\int_1^x\frac{\mathrm{d}v}{v\{\ln(2x/v)\}^2}\Big)=o(1).\end{aligned}$$

□

Theorem 7.22 (Landau, 1910). *If g is real and satisfies $g(n)\geqslant -K$ $(n\geqslant 1)$ for some suitable real constant K, then* (7.63) *implies*

$$\sum_{n\geqslant 1}\frac{g(n)}{n}=a. \tag{7.64}$$

Proof. Up to altering $g(1)$, we can assume $a=0$. Then

$$\sum_{n\leqslant x}\frac{g(n)}{n}-\frac{G(x)}{x}=\int_1^x\frac{G(u)}{u^2}\,\mathrm{d}u=o(1). \tag{7.65}$$

If $G(N)>cN$ with $c>0$ for infinitely many integers N, then, for sufficiently small $N<u\leqslant(1+\varepsilon)N$ and ε, we have $G(u)>cN-K(u-N+1)>\frac{1}{2}cN$. This implies that

$$\int_N^{(1+\varepsilon)N}\frac{G(u)}{u^2}\,\mathrm{d}u>\tfrac{1}{2}c\varepsilon/(1+\varepsilon)^2,$$

and, by (7.65), we have a contradiction. Applying the same argument to $-G$, we deduce that $G(N)=o(N)$. □

The conjunction of Theorems 7.21 and 7.22 immediately provides a Tauberian condition on g for the implication (7.62) ⇒ (7.64).

Corollary 7.23 (Ingham, 1945). *If* (7.62) *is satisfied and we have*

$$\inf_n g(n)>-\infty,$$

then (7.64) *holds.*

Ingham's method also furnishes the following result.

Theorem 7.24 (Skałba, 1998). *If* (7.62) *is satisfied and if* $f(n) \ll 1$, *then we have* $G(x) = o(x)$ *and relation* (7.64) *holds.*

Proof. Assuming, without loss of generality, that $a = 0$, we have, by the hyperbola method (Theorem I.3.1), for each fixed $y \geqslant 2$,

$$G(x) = \sum_{m \leqslant y} \mu(m) F(x/m) + \sum_{n \leqslant x/y} f(n) M(x/n) - F(x/y) M(y). \tag{7.66}$$

Bounding $M(z)$ by $\ll z/(\ln z)^2$ and $f(n)$ by $O(1)$ in this sum, we readily see that $\limsup_{x\to\infty} |G(x)|/x \ll 1/\ln y$, and hence $G(x) = o(x)$. Substituting in (7.63), we obtain the required result. □

Corollary 7.25 (Skałba, 1998). *If* $f : \mathbb{R} \to \mathbb{C}$ *is* 2π*-periodic and Riemann integrable, then*

$$\frac{1}{2\pi} \int_0^{2\pi} f(u)\,\mathrm{d}u = \sum_{n \geqslant 1} \frac{g(n)}{n}.$$

Proof. Since $\langle n/2\pi \rangle$ is equidistributed modulo 1, the integral coincides with the mean value of $f(n) = f(2\pi \langle n/2\pi \rangle)$, by Weyl's criterion—Theorem I.6.13. □

As is mentioned in Skałba's article (1998), Drmota proved the converse of Theorem 7.24.

Theorem 7.26 (Drmota, 1998). *If* f *is real, satisfies* $\inf_n f(n) > -\infty$, *and if we have* (7.64), *then relation* (7.62) *holds.*

Proof. For each $y > 0$ we have

$$\sum_{n \geqslant 1} \frac{g(n)}{\mathrm{e}^{ny} - 1} = \sum_{n \geqslant 1} f(n) \mathrm{e}^{-ny}.$$

The left-hand side is the Lambert series associated with $g(n)/n$. Since the Lambert summation method is regular (see below), it equals $\{a + o(1)\}/y$ as $y \to 0+$. Karamata's Tauberian theorem (Theorem 7.5) thus implies (7.62). □

Let us now show the required property for the Lambert summation method. We set

$$\varphi(y) := y/(\mathrm{e}^y - 1).$$

Lemma 7.27 (Regularity of Lambert's summation method). *If* $\sum_n a_n = a$, *then*

$$\sum_n a_n \varphi(ny) = a + o(1) \qquad (y \to 0+).$$

Proof. We may clearly suppose $a = 0$. Let $A_n := \sum_{m \leqslant n} a_m$. Then $A_n = o(1)$ and

$$\sum_n a_n \varphi(ny) = \sum_n A_n \{\varphi(ny) - \varphi(ny + y)\}.$$

But φ is non-increasing on $\mathbb{R}^+$. Therefore, for each fixed N,

$$\Big|\sum_n A_n\{\varphi(ny)-\varphi(ny+y)\}\Big| \leqslant \sum_{n \leqslant N} |A_n|\{\varphi(ny)-\varphi(ny+y)\}+\varphi(Ny) \sup_{n>N} |A_n|.$$

It follows that

$$\limsup_{y \to 0+} \Big|\sum_n A_n\{\varphi(ny) - \varphi(ny + y)\}\Big| \leqslant \sup_{n>N} |A_n|$$

whence the result on letting N tend to infinity. □

Notes

§ 7.1. The restriction $z \in S(\vartheta)$ is necessary in Abel's theorem. The following counter-example is due to B. de Mathan. Let $\alpha_k := \exp(i/k)$ $(k = 1, 2, \ldots)$. The function

$$f(z) = \sum_{k \geqslant 1} \frac{\log(1 - z/\alpha_k)}{2^k}$$

is holomorphic for $|z| < 1$ because the series converges uniformly on any compact subset of the open disk. Since

$$\lim_{z \to \alpha_k} |f(z)| = \infty \qquad (k = 1, 2, \ldots),$$

we see that $f(z)$ fails to have a limit as $z \to 1$. But the Taylor series of $f(z)$ is

$$-\sum_{n \geqslant 1} \frac{z^n}{n} \sum_{k \geqslant 1} \frac{\alpha_k^{-n}}{2^k}.$$

By Abel's rule, this series converges for $z = 1$. Indeed, the partial sums are uniformly bounded:

$$\left| \sum_{1 \leqslant n \leqslant m} \sum_{k \geqslant 1} \frac{\overline{\alpha}_k^n}{2^k} \right| = \left| \sum_{k \geqslant 1} \frac{1}{2^k} \sum_{1 \leqslant n \leqslant m} \overline{\alpha}_k^n \right|$$

$$\leqslant \sum_{k \geqslant 1} \frac{1}{2^k} \left| \frac{1 - \overline{\alpha}_k^m}{1 - \overline{\alpha}_k} \right| \leqslant \sum_{k \geqslant 1} \frac{1}{2^{k-1} |1 - \mathrm{e}^{-i/k}|} < \infty.$$

§ 7.3. It is possible to extend Karamata's theorem by considering the more general assumption

$$F(\sigma) = \{c + o(1)\} \sigma^{-\omega} L(1/\sigma) \qquad (\sigma \to 0+)$$

where $L(t)$ is a *slowly varying* function, i.e. satisfies, for each fixed $k > 0$,

$$L(kt) \sim L(t) \qquad (t \to \infty).$$

The conclusion of Theorem 7.5 then holds on replacing x^ω by $x^\omega L(x)$—see Hardy (1949), th. 108.

§ 7.4. Theorem 7.11 is due to Freud (1952). See also Freud & Ganelius (1957). The self-contained proof given here essentially follows that of Korevaar (1954a). Ingham (1965) provides a direct proof of the Karamata–Freud theorem which does not explicitly appeal to the lemma on L^1–approximation by polynomials, but directly incorporates the use of "peak functions" in Karamata's proof.

§ 7.5. For developments relating to Bohr's inequality (7.43), see Hörmander (1954).

As indicated in Exercise 225, Theorem 7.13 yields, for all $\varepsilon > 0$, a remainder of order $\ll x/(\ln x)^{2-\varepsilon}$ in the prime number theorem. Rieger (1983), also using the Wiener-Ikehara method, obtained a remainder of order

$$\ll x \exp\{-c(\ln x)^{1/25}\}.$$

As Diamond remarked in his review (1988), it is not true *stricto sensu* that the Hardy–Littlewood–Karamata theorem cannot lead to a proof of the prime number theorem. Using a number of analytic properties of the Riemann zeta function, such as the functional equation, the product formula, the finiteness of the order, the absence of zeros on $\sigma = 1$, and a bound of type $N(T) \ll T^A$ for the number of zeros with ordinate not exceeding T, Littlewood indeed showed in 1971 that the Hardy–Littlewood–Karamata theorem can be used to obtain a "quick" proof of the prime number theorem. Since no known proof of the fact that $\zeta(s) \neq 0$ for $\sigma = 1$ furnishes an effective lower bound weaker than

$$|\zeta(s)| \gg \{\ln(3+|\tau|)\}^{-A} \qquad (\sigma \geqslant 1), \tag{7.67}$$

such a procedure would seem, however, to have a rather limited theoretical interest. Under assumption (7.67), Perron's formula on its own is sufficient to recover the prime number theorem and no Tauberian theorem is needed whatsoever—cf. § 4.2. The particular appeal of the Ikehara theorem is to yield the conclusion under the single minimal assumption $\zeta(s) \neq 0$ for $\sigma = 1$, without any additional information.

Delange (1954) generalized Ikehara's theorem to the case of a singularity of mixed type, involving both monomial and logarithmic terms in its principal part.

Theorem 7.28 (Delange). *Let $F(s) := \sum_{n\geqslant 1} a_n/n^s$ be a Dirichlet series with non-negative coefficients, converging for $\sigma > a > 0$. Suppose that $F(s)$ is holomorphic at all points on the line $\sigma = a$ other than $s = a$, and that, in the neighborhood of this point and for $\sigma > a$, we have*

$$F(s) = \frac{1}{(s-a)^{\omega+1}} \sum_{0\leqslant j\leqslant q} g_j(s)\Big\{\log\Big(\frac{1}{s-a}\Big)\Big\}^j + g(s),$$

where ω is some real number, and the $g_j(s)$ and $g(s)$ functions holomorphic at $s = a$, the number $g_q(a)$ being non-zero. Then:

(i) *if ω is not a negative integer, we have, as x tends to infinity,*

$$A(x) := \sum_{n\leqslant x} a_n \sim \frac{g_q(a)}{a\Gamma(\omega+1)} x^a (\ln x)^\omega (\ln_2 x)^q,$$

(ii) *if* $\omega = -m-1$ *for an integer* $m \geqslant 0$ *and if* $q \geqslant 1$, *we have, as* x *tends to infinity,*

$$A(x) \sim (-1)^m m! \frac{q g_q(a)}{a} \frac{x^a (\ln_2 x)^{q-1}}{(\ln x)^{m+1}}.$$

§ 7.6. We have given here the classical formulation of the Berry–Esseen inequality, in which absolute continuity of one of the two distribution functions is assumed. The same technique furnishes a fully general result, expressed in terms of the concentration function

$$Q_G(\ell) := \sup_z \{G(z+\ell) - G(z)\} \qquad (\ell > 0). \tag{7.68}$$

Elliott (1980, *lemma* 1.47) thus shows that the upper bound

$$\|F - G\|_\infty \ll Q_G\left(\frac{1}{T}\right) + \int_{-T}^{T} \left|\frac{f(\tau) - g(\tau)}{\tau}\right| \mathrm{d}\tau \tag{7.69}$$

holds, uniformly with respect to F and G, for any positive real number T.

Refining a result of Alladi (1987), Stef & Tenenbaum established (2001) an analogue of the Berry–Esseen inequality valid for the bilateral Laplace transform

$$\widehat{F}(u) := \int_{\mathbb{R}} \mathrm{e}^{-uz} \mathrm{d}F(z)$$

of distribution functions.

Theorem 7.29 (Stef & Tenenbaum, 2001). *Let* F *and* G *be two distribution functions, and let* $h : \mathbb{R}^+ \to \mathbb{R}^+$ *be a non-decreasing continuous function satisfying*

$$h(u) \gg u^4 \qquad (u \geqslant 0). \tag{7.70}$$

Let ε, κ, L *be real numbers such that* $0 < \varepsilon < 1/\{3 + h(2)\}$, $0 < \kappa \leqslant L$, *and verifying*

(i) $\big|\widehat{F}(u) - \widehat{G}(u)\big| \leqslant \varepsilon \qquad (0 \leqslant u \leqslant \kappa)$,

(ii) $\widehat{F}(u) + \widehat{G}(u) \ll h\big(|u|\big) < \infty \qquad (-L \leqslant u \leqslant L)$.

Then we have

$$\|F - G\|_\infty \ll Q_G\Big(\frac{\ln L}{L} + \frac{\ln W}{W}\Big), \tag{7.71}$$

where W *is any solution of the equation* $h(W) = 1/\varepsilon$. *The implicit constant in* (7.71) *depends at most on* κ *and on the implicit constants in* (7.70) *and (ii).*

In the particular case when G is absolutely continuous and has a bounded density, we derive the following corollary.

Corollary 7.30 (Stef & Tenenbaum). *Retain the hypotheses of Theorem 7.29 and suppose in addition that:*

(iii) *G is absolutely continuous with $G' \in L^\infty(\mathbb{R})$;*

(iv) $\exists \vartheta \geqslant 1 : \ln h(u) \ll 1 + u^\vartheta \qquad (0 \leqslant u \leqslant L)$.

Then

$$\|F - G\|_\infty \ll \left(\frac{\ln L}{L} + \frac{\ln|\ln\varepsilon|}{|\ln\varepsilon|^{1/\vartheta}}\right)\|G'\|_\infty \tag{7.72}$$

where the implicit constant depends at most on κ and on the implicit constants in (ii) and (iv).

As shown in the work of Stef & Tenenbaum, the exponent of $|\ln\varepsilon|$ appearing in (7.72) is optimal, at least when $\vartheta = 2k/(2k-1)$ $(k \in \mathbb{N}^*)$.

§ 7.7. Newman's theorem mentioned in the Notes for Chapter II.1 is another example of the use of holomorphy as a Tauberian condition. We give below an integral version due to Korevaar (1982) and Zagier (1997).

Theorem 7.31 (Newman, 1980; Korevaar, 1982; Zagier, 1997). *Let f be a locally integrable function, bounded on $\mathbb{R}^+$, and whose Laplace transform*

$$F(z) := \int_0^\infty f(t)\mathrm{e}^{-tz}\,\mathrm{d}t,$$

initially defined for $\Re\mathrm{e}\, z > 0$, can be analytically continued to the half-plane $\Re\mathrm{e}\, z \geqslant 0$. Then

$$\int_0^\infty f(t)\,\mathrm{d}t = F(0).$$

Proof. Put $F_T(z) := \int_0^T f(t)\mathrm{e}^{-tz}\,\mathrm{d}t$. This is clearly an entire function of z. We have to show that $F_T(0)$ tends to $F(0)$ as $T \to \infty$. Let $R > 0$ be a parameter that will later tend to infinity, and let L denote the boundary of the domain

$$\mathcal{D} := \{z \in \mathbb{C} : |z| \leqslant R,\ \Re\mathrm{e}\, z \geqslant -\delta\}$$

where $\delta > 0$ is so small (as a function of R) that F is holomorphic on $\mathcal{D}$. Then, by Cauchy's integral formula

$$F(0) - F_T(0) = \frac{1}{2\pi i}\oint_L \{F(z) - F_T(z)\}\mathrm{e}^{zT}\Big(1 + \frac{z^2}{R^2}\Big)\frac{\mathrm{d}z}{z}. \tag{7.73}$$

Let C be the boundary of the semicircle $L \cap \{z : \Re\mathrm{e}\, z > 0\}$ and put $M := \|f\|_\infty$. Since we have

$$|F(z) - F_T(z)| = \left|\int_T^\infty f(t)\mathrm{e}^{-zt}\,\mathrm{d}t\right| \leqslant M\mathrm{e}^{-T\Re\mathrm{e}\, z}/\Re\mathrm{e}\, z \qquad (\Re\mathrm{e}\, z > 0)$$

and

$$\left|e^{zT}\Big(1+\frac{z^2}{R^2}\Big)\frac{1}{z}\right| = e^{T\Re e\, z}\frac{2|\Re e\, z|}{R^2} \qquad (|z|=R), \tag{7.74}$$

we see that, on C, the integrand in (7.73) is bounded in absolute value by $2M/R^2$, which implies that the contribution of C to the integral in (7.73) is at most M/R. We shall bound the contribution of $B := L \setminus C$ by considering F and F_T separately. Since F_T is entire, we may replace B with the boundary of the semi-circle $C' := \{z \in \mathbb{C} : |z| = R,\ \Re e\, z < 0\}$. Since

$$|F_T(z)| = \left|\int_0^T f(t)e^{-zt}\,dt\right| \leqslant M\int_{-\infty}^T e^{-\Re e\, zt}\,dt = \frac{Me^{-\Re e\, zT}}{|\Re e\, z|} \qquad (\Re e\, z < 0),$$

we deduce by (7.74) that

$$\left|\frac{1}{2\pi i}\int_B F_T(z)e^{zT}\Big(1+\frac{z^2}{R^2}\Big)\frac{dz}{z}\right| \leqslant \frac{M}{R}.$$

Finally, for each fixed $R > 0$, we have

$$\lim_{T\to\infty}\frac{1}{2\pi i}\int_B F(z)e^{zT}\Big(1+\frac{z^2}{R^2}\Big)\frac{dz}{z} = 0$$

by Lebesgue's theorem. It follows that

$$\limsup_{T\to\infty}|F(0) - F_T(0)| \leqslant 2M/R.$$

We obtain the stated result by letting R tend to infinity. □

§ 7.8. Landau actually obtained a more general form of Theorem 7.22: if $\sum_{n\leqslant 1} g(n)/n = a$ according to the k-th order Cesàro mean summation method, and if $g(n) \geqslant -K$ $(n \geqslant 1)$, then (7.64) holds; see Hardy (1949), th. 64.

The proof of Theorem 7.24 presented here differs from that of Skałba, who mentions Corollary 7.23 but does not use it.

Deep extensions of arithmetic Tauberian theorems have been considered by Erdős & Ingham (1964).

Exercises

219. Let $\{a_n\}_{n=0}^{\infty} \in \mathbb{R}^{\mathbb{N}}$ and $b \in \mathbb{R}$. Set $s_n := \sum_{0 \leqslant m \leqslant n} a_m \quad (n \geqslant 0)$.

(a) Calculate $b_N := (1/N) \sum_{0 \leqslant n \leqslant N} s_n$ as a function of $a_0, \ldots, a_N$. Deduce that, assuming

$$\lim_{N \to \infty} b_N = b, \tag{7.75}$$

a necessary and sufficient condition for the series $\sum_{n \geqslant 0} a_n$ to converge to b is that $\sum_{0 \leqslant n \leqslant N} n a_n = o(N)$ $(N \to \infty)$.

(b) Assume condition (7.75) holds.

(i) Calculate s_n as a function of b_n and b_{n-1}. Show that the series $S(z) := \sum_{n \geqslant 0} s_n z^n$ and $A(z) := \sum_{n \geqslant 0} a_n z^n$ converge in the open unit disk.

(ii) Show that $S(x) = (1-x) \sum_{n \geqslant 0} n b_n x^n$ for $0 < x < 1$. Deduce that $\lim_{x \to 1-} (1-x) S(x) = b$. Writing $b_n = b + \varepsilon_n$ $(n \geqslant 0)$ where ε_n tends to 0, find an upper bound for $E(x) := |\sum_{n \geqslant 0} \varepsilon_n n x^n|$.

(iii) For $0 < x < 1$, write $(1-x)S(x)$ as a function of $A(x)$.

(c) Assume $K \in \mathbb{R}$. Is the condition $\inf_{n \in \mathbb{N}} n a_n > -K$ a sufficient Tauberian condition so that assumption (7.75) implies the convergence to b of the series $\sum_{n \geqslant 0} a_n$?

220. Deduce from Theorem 7.10 the estimate[4]

$$\sum_{n \leqslant x} \frac{\mu(n)}{n} \ll \frac{\ln x}{\ln_2 x} \qquad (x \to \infty).$$

221. Set $a_n := \sum_{d|n} (-1)^d 2^{\omega(n/d)}$, where $\omega(m)$ denotes the number of distinct prime factors of m. Find the asymptotic behavior of $\sum_{n \geqslant 1} a_n n^{-\sigma}$ when $\sigma \to 1+$. Appealing to the Karamata–Freud theorem, deduce that

$$\sum_{n \leqslant x} \frac{a_n}{n} \ll \frac{(\ln x)^3}{\ln_2 x} \qquad (x \to \infty).$$

By using the Selberg–Delange method, show that we have

$$\sum_{n \leqslant x} \frac{a_n}{n} \sim C (\ln x)^2$$

with $C := -(3 \ln 2)/\pi^2$.

222. Let $\{a_p\}$ be a bounded sequence indexed by the primes. Show that if $\sum a_p p^{-\sigma}$ tends to a limit as $\sigma \to 1+$ then the series $\sum a_p/p$ converges. Show that the result fails for a sequence indexed over the integers.

[4] This non-trivial estimate is, of course, weaker than (3.15), but only necessitates a very fragmentary piece of information on the Dirichlet series $1/\zeta(s)$ associated with the Möbius function.

223. Let A be a positive constant and $\{b_n\}_{n=1}^{\infty}$ a complex sequence verifying $|b_n| \leqslant A\tau(n)$ for any integer $n \geqslant 1$ and also

$$\sum_{n\geqslant 1} \frac{b_n}{n^\sigma} = o\Big(\frac{1}{(\sigma-1)^2}\Big) \qquad (\sigma \to 1+). \tag{7.76}$$

(a) Is it true that we necessarily have $\sum_{n\leqslant x} b_n/n = o\big((\ln x)^2\big)$ $(x \to \infty)$?

(b) Consider the same question for the asymptotic formula

$$\sum_{n\leqslant x} b_n = o(x \ln x) \qquad (x \to \infty).$$

(c) Let $\{a_n\}_{n=1}^{\infty}$ be a bounded complex sequence such that

$$\sum_{n\geqslant 1} \frac{a_n}{n^\sigma} = o\Big(\frac{1}{\sigma-1}\Big) \qquad (\sigma \to 1+).$$

Suppose now that $b_n := \sum_{d|n} a_d$. Show that (7.76) holds. Does the additional assumption on the structure of the sequence $n \mapsto b_n$ entail a modification of your conclusions on the validity of the relations considered in (a) and (b)?

224. Set $Z(s) := \zeta'(s)/\{s\zeta(s)\}$. Show that the assumption $\zeta(1+i\tau) \neq 0$ for $\tau \neq 0$ implies, for all fixed T, that

$$\lim_{\sigma\to 0+} \int_{-T}^{T} \Big| Z(1+2\sigma+i\tau) - Z(1+\sigma+i\tau) - \frac{1}{2\sigma+i\tau} + \frac{1}{\sigma+i\tau} \Big| \,\mathrm{d}\tau = 0.$$

Deduce the prime number theorem by applying Ikehara's theorem.

225. Using the elementary bounds proved in § 4.2

$$\zeta^{(k)}(s) \ll (\ln|\tau|)^{k+1}, \qquad \zeta(s)^{-1} \ll (\ln|\tau|)^7 \quad (|\tau| \geqslant 2,\ \sigma \geqslant 1),$$

show that the integral in Exercise 224 is $\ll \sigma(\ln T)^{19}$ for $T \geqslant 2$, $\sigma > 0$. From Theorem 7.13, deduce the effective estimate

$$\psi(x) := \sum_{n\leqslant x} \Lambda(n) = x + O\Big(x\frac{(\ln_2 x)^{19}}{\ln x}\Big).$$

226. Use the method in Exercise 225 to establish the bound

$$\sum_{n\leqslant x} \mu(n) \ll x\frac{(\ln_2 x)^{17}}{\ln x}.$$

[*Hint: Use the Dirichlet series* $\zeta(s) + 1/\zeta(s)$.]

227. *Berry–Esseen theorem.* Let X be a real random variable, with zero mean, variance equal to 1, and finite absolute moment of order 3 equal to ϱ. Set $F(x) := \mathrm{Prob}(X \leqslant x)$ $(x \in \mathbb{R})$ and

$$\varphi(\tau) := \int_{-\infty}^{+\infty} \mathrm{e}^{i\tau x}\, \mathrm{d}F(x) \qquad (\tau \in \mathbb{R}).$$

Finally, denote by $F_n(x)$ the distribution function of $(1/\sqrt{n}) \sum_{j=1}^{n} X_j$, where the X_j are independent variables with the same law as X.

(a) Show that $\varrho \geqslant 1$.

(b) Using the inequalities

$$\Big|\mathrm{e}^{iy} - \sum_{0 \leqslant j \leqslant m} \frac{1}{j!}(iy)^j\Big| \leqslant \frac{1}{(m+1)!}|y|^{m+1} \qquad (y \in \mathbb{R},\ m \geqslant 0)$$

and $|\log(1-y)+y| \leqslant |y|^2$ $(y \in \mathbb{C},\ |y| \leqslant \frac{1}{2})$, show that

$$|\varphi(\tau) - 1| \leqslant \tfrac{1}{2}\tau^2, \quad |\varphi(\tau) - 1 + \tfrac{1}{2}\tau^2| \leqslant \tfrac{1}{6}\varrho|\tau|^3 \quad (\tau \in \mathbb{R}),$$
$$\Big|\log\varphi(\tau) + \tfrac{1}{2}\tau^2\Big| \leqslant \tfrac{1}{4}\tau^4 + \tfrac{1}{6}\varrho|\tau|^3 \quad (|\tau| \leqslant 1).$$

(c) Show that, for all $n \geqslant 1$, we have

$$\Big|\varphi\Big(\frac{\tau}{\sqrt{n}}\Big)^n - \mathrm{e}^{-\tau^2/2}\Big| \ll \mathrm{e}^{-\tau^2/4}\Big\{\varrho\frac{|\tau|^3}{\sqrt{n}} + \frac{\tau^4}{n}\Big\} \quad (\varrho|\tau| \leqslant \sqrt{n})$$

(d) Deduce from the above that we have

$$\sup_{x\in\mathbb{R}} \Big|F_n(x) - \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{x} \mathrm{e}^{-t^2/2}\,\mathrm{d}t\Big| \leqslant C\frac{\varrho}{\sqrt{n}}$$

where C is a suitable absolute constant.[5]

228. *On an idea of Wirsing (1956).* Let $\mathcal{P}$ be a set of primes, and denote by ϑ the characteristic function of the set of integers all of whose prime factors belong to $\mathcal{P}$. Put $T(x) := \sum_{n\leqslant x}\vartheta(n)$. We assume there exist constants $\delta > 0$ and $K > 0$ such that, as $x \to \infty$,

$$\text{(i)} \sum_{p\leqslant x}\vartheta(p) \sim \frac{\delta x}{\ln x}, \qquad \text{(ii)} \prod_{p\leqslant x}\Big(1 + \frac{\vartheta(p)}{p-1}\Big) \sim K(\ln x)^{\delta}.$$

(a) Show that, when $x \to \infty$, we have

$$\text{(iii)} \sum_{n\leqslant x}\vartheta(n)\ln n \sim \delta x \sum_{m\leqslant x}\frac{\vartheta(m)}{m}, \qquad \text{(iv)}\ T(x) \sim \frac{\delta x}{\ln x}\sum_{m\leqslant x}\frac{\vartheta(m)}{m}.$$

[5] Feller (1970) shows the value $C = 3$ is admissible.

(b) Show that assumption (i) alone implies, as $\sigma \to 0+$,

$$\sum_{p\leqslant \exp(1/\sigma)} \frac{\vartheta(p)}{p}(1-p^{-\sigma}) - \sum_{p>\exp(1/\sigma)} \frac{\vartheta(p)}{p^{1+\sigma}} = \gamma\delta + o(1).$$

Deduce that

$$\sum_{m\geqslant 1} \frac{\vartheta(m)}{m^{1+\sigma}} \sim \mathrm{e}^{-\gamma\delta} \prod_{p\leqslant \exp(1/\sigma)} \Big(1-\frac{\vartheta(p)}{p}\Big)^{-1} \qquad (\sigma \to 0+).$$

(c) Show that, under hypotheses (i) and (ii), we have

$$T(x) \sim \frac{\delta K \mathrm{e}^{-\gamma\delta}}{\Gamma(\delta+1)} x(\log x)^{\delta-1} \qquad (x \to \infty).$$

229. *A converse to Ingham's theorem (Corollary 7.23).* Let $\{a_n\}_{n=1}^{\infty}$ be a bounded complex sequence such that $\sum_{n\geqslant 1} a_n/n = a$.

(a) Show that, for any fixed $y > 1$, we have

$$\sum_{N/y<n\leqslant N} a_n \Big\lfloor \frac{N}{n} \Big\rfloor = \sum_{m\leqslant y} \int_0^{N/m} \sum_{\max(t,N/y)<n\leqslant N/m} \frac{a_n}{n}\,\mathrm{d}t = o(N).$$

(b) Deduce that

$$\lim_{N\to\infty} \frac{1}{N} \sum_{1\leqslant n\leqslant N} a_n \Big\lfloor \frac{N}{n} \Big\rfloor = a. \tag{7.77}$$

(c) Application. Let $\vartheta \in \mathbb{R} \setminus \mathbb{Z}$. Define $\tau^*(n,\vartheta) := \sum_{d|n} \mathrm{e}(d\vartheta)$. Show that the quantity

$$\frac{1}{N} \sum_{n\leqslant N} \tau^*(n,\vartheta)$$

tends to a finite limit as N tends to infinity and calculate it.

230. (a) Show that $\sum_{n\geqslant 0} n\mathrm{e}^{-n^2\sigma} \sim 1/(2\sigma)$ as $\sigma \to 0+$.

(b) Deduce the following variant of the Hardy–Littlewood theorem: *let $K \in \mathbb{R}$ and let $\{a_n\}_{n=0}^{\infty}$ be a real sequence such that $na_n \geqslant -K$ $(n \geqslant 1)$ and*

$$F(\sigma) := \sum_{n\geqslant 0} a_n \mathrm{e}^{-\sigma n^2} = s + o(1) \qquad (\sigma \to 0+);$$

then the series $\sum_{n\geqslant 0} a_n$ converges and sums to s.

(c) Show that $\lim_{\sigma\to 0+} \sum_{n\geqslant 0} (-1)^n \mathrm{e}^{-\sigma n^2} = \frac{1}{2}$. What do you infer from this?

231. Let $g(n) := \sum_{d|n} \mu(d)\mathrm{e}^{in/d}$ $(n \geqslant 1)$. Show that the series $\sum_{n\geqslant 1} g(n)/n$ converges and calculate its sum.

Primes in arithmetic progressions

8.1. Introduction. Dirichlet characters

8.1.1. Definitions. Let a and q be coprime integers. There is, *a priori*, no obstacle to the existence of infinitely many primes p such that $p \equiv a \pmod{q}$. It is even reasonable, in the absence of any indication to the contrary, to imagine that the primes are distributed evenly amongst the $\varphi(q)$ classes which can actually contain infinitely many elements. Setting

$$\pi(x; a, q) := |\{p \leqslant x : p \equiv a \pmod{q}\}|,^{1}$$

we see, in view of the prime number theorem, that the assumption of equidistribution leads to the conjecture that

$$\pi(x; a, q) \sim \frac{x}{\varphi(q) \ln x} \qquad (x \to \infty).$$

This problem, which has its own captivating story parallel to that of the distribution of primes, was eventually settled in 1896 by La Vallée–Poussin, combining his approach to the prime number theorem (relying heavily, as seen in Chapter II.4, on Riemann's ideas) with specific tools forged by Dirichlet in the first half of the nineteenth century.

The first step in this direction consists in rigorously establishing the existence of infinitely many primes p in each admissible progression $a \pmod{q}$, with $(a, q) = 1$.

[1] This function is often denoted $\pi(x; q, a)$ in the literature. We prefer to leave the parameters a and q appearing in the same order as in writing the arithmetic progression $n \equiv a \pmod{q}$.

The cases $a = 3$, $q = 4$, and $a = 5$, $q = 6$, have been respectively handled in Exercises 34 and 35 (p. 26) via a fairly straightforward extension of Euclid's proof.

The case of the progression $p \equiv 1 \pmod 4$ is slightly more delicate. However, as seen in Exercise 38, p. 27, it can still be settled by a similar technique.

One might imagine that this type of approach, which calls for the existence of a polynomial with integer coefficients whose values at integer arguments are divisible by infinitely many primes in the selected progression, may be generalized to any admissible arithmetic progression. This is not so: Ram Murty established in 1988 that a "Euclidean" proof for the existence of infinitely many primes of the form $a + mq$ is possible if, and only if, $a^2 \equiv 1 \pmod q$.[2]

It was Dirichlet who, in 1837, provided the general solution to this problem.[3] His starting point was Euler's proof for the infinitude of the set of primes, which, as can be easily seen, also furnishes the divergence of the series $\sum_p 1/p$.

Dirichlet actually establishes that

$$\sum_{p \equiv a \pmod q} \frac{1}{p} = \infty$$

for all integers a, q such that $(a, q) = 1$. The salient point of his proof is the use of characters modulo q, that is, homomorphisms from the group $(\mathbb{Z}/q\mathbb{Z})^*$ of invertible residues $\pmod q$ into the multiplicative group of complex numbers of modulus 1.

We shall not develop here the classical general theory of characters of finite Abelian groups, and confine ourselves instead to briefly recalling the main properties that will be useful in the present context.

Let G be a finite Abelian group. Denote by $\widehat{G}$ the multiplicative group of homomorphisms $\chi : G \to \mathbb{C}^*$, known as the *characters* of G.

Let

$$G \cong \prod_{1 \leqslant j \leqslant k} G_j$$

be the decomposition of G as a direct product of cyclic groups. Put $n_j := |G_j|$ $(1 \leqslant j \leqslant k)$, so that $n := |G| = \prod_{1 \leqslant j \leqslant k} n_j$. If γ_j $(1 \leqslant j \leqslant k)$ denotes a generator of G_j, each element γ of G may be uniquely written as

[2] See also Murty & Thain (2006).

[3] Strictly speaking, Dirichlet's proof was only complete when the common difference q is prime. The general case derives from the "class number formula", which Dirichlet established in 1839/40.

a product

$$\gamma = \prod_{1\leqslant j\leqslant k} \gamma_j^{r_j} \qquad \big(1 \leqslant r_j \leqslant n_j\,(1 \leqslant j \leqslant k)\big). \tag{8.1}$$

For any character χ of G, we can therefore write

$$\chi(\gamma) = \prod_{1\leqslant j\leqslant k} \chi(\gamma_j)^{r_j}.$$

Thus, χ is entirely determined by the $\chi(\gamma_j)$ $(1 \leqslant j \leqslant k)$, and the fact that γ_j is of order n_j implies that $\chi(\gamma_j)$ is an n_j-th root of unity. Conversely, for every choice $\zeta_j = \mathrm{e}\big(h_j/n_j\big)$ $(1 \leqslant j \leqslant k)$ with $1 \leqslant h_j \leqslant n_j$, the mapping $\chi : G \to \mathbb{C}$ defined, with γ as given in (8.1), by

$$\chi(\gamma) = \prod_{1\leqslant j\leqslant k} \zeta_j^{r_j}$$

is a character of G. Moreover all characters produced in this way are obviously distinct. Thus $\widehat{G}$ is an Abelian group of order n with respect to the usual product of maps.

The following simple property constitutes one of the main attractions for the notion of a character of an Abelian group. We denote by χ_0 the trivial character defined by $\chi_0(\gamma) = 1$ for all $\gamma \in G$.

Theorem 8.1. *For any Abelian group G of order n, we have*

$$\sum_{\chi\in\widehat{G}} \chi(\gamma) = \begin{cases} n & \text{if } \gamma = 1 \\ 0 & \text{if } \gamma \neq 1 \end{cases} \qquad (\gamma \in G) \tag{8.2}$$

and

$$\sum_{\gamma\in G} \chi(\gamma) = \begin{cases} n & \text{if } \chi = \chi_0 \\ 0 & \text{if } \chi \neq \chi_0 \end{cases} \qquad (\chi \in \widehat{G}). \tag{8.3}$$

Proof. The proofs of these two dual relations being similar, we confine ourselves to establishing the first.

Let S denote the left-hand side of (8.2). If $\gamma = 1$, we have $\chi(\gamma) = 1$ for all χ, therefore $S = |\widehat{G}| = n$. If $\gamma \neq 1$, there exists a character, say χ_1, such that $\chi_1(\gamma) \neq 1$. Indeed, writing γ as in (8.1), up to reordering the factors, we can assume that $r_1 \neq n_1$. The map defined by $\chi_1(\gamma_1) = \mathrm{e}(1/n_1)$, $\chi_1(\gamma_j) = 1$ $(2 \leqslant j \leqslant k)$ has the required property. Since $\chi\chi_1$ runs over $\widehat{G}$ as χ runs over $\widehat{G}$, it follows that

$$\chi_1(\gamma)S = \sum_{\chi\in\widehat{G}} \chi_1(\gamma)\chi(\gamma) = \sum_{\chi\in\widehat{G}} (\chi\chi_1)(\gamma) = S,$$

whence $S = 0$. □

It follows from the above description of the character group that there are $\varphi(q)$ characters of $(\mathbb{Z}/q\mathbb{Z})^*$. They can be explicitly defined by decomposing $(\mathbb{Z}/q\mathbb{Z})^*$ as a direct product of cyclic groups. By the Chinese remainder theorem, we have, for any integer $q \geqslant 1$,

$$(\mathbb{Z}/q\mathbb{Z})^* \cong \prod_{p^\nu \| q} (\mathbb{Z}/p^\nu\mathbb{Z})^*. \tag{8.4}$$

Indeed, if we let q_j $(1 \leqslant j \leqslant \omega(q))$ denote the distinct prime powers dividing q and if, for each j, we let a_j denote the unique integer $(\mathrm{mod}\, q)$ such that

$$a_j \equiv 1 \,(\mathrm{mod}\, q_j), \qquad a_j \equiv 0 \,(\mathrm{mod}\, q_i) \quad (i \neq j),$$

then each integer m satisfies the congruence relation

$$m \equiv \sum_j a_j m_j \,(\mathrm{mod}\, q)$$

where m_j is the residue of $m \,(\mathrm{mod}\, q_j)$. Since $(m, q) = 1$ is equivalent to $(m_j, q_j) = 1$ for all j, we see that the canonical projection of the left-hand side of (8.4) on the right-hand side is well-defined and surjective, from which (8.4) follows.

When $p > 2$, the group $(\mathbb{Z}/p^\nu\mathbb{Z})^*$ is cyclic—see Exercises 232 and 233. When $p = 2$, there are three cases for the structure of $(\mathbb{Z}/2^\nu\mathbb{Z})^*$. If $\nu = 1$, the group is trivial. If $\nu = 2$, it is (cyclic) of order 2. If $\nu \geqslant 3$, it is the direct product of a group of order 2 with a cyclic group of order $2^{\nu-2}$. This last point can be checked by first noting that an easy induction yields

$$5^{2^{m-2}} = 1 + 2^m h_m \qquad (m \geqslant 2) \tag{8.5}$$

where h_m is odd. Applying this relation with $m = \nu$ we see that the order of 5 modulo 2^ν divides $2^{\nu-2}$. But, choosing $m = \nu - 1$, we also see that this order does not divide $2^{\nu-3}$. It is thus exactly equal to $2^{\nu-2}$, and we conclude by observing that 5 is not a square modulo 8— thereby excluding the possibility that $(\mathbb{Z}/2^\nu\mathbb{Z})^*$ is cyclic.

We can now make the characters of $(\mathbb{Z}/q\mathbb{Z})^*$ explicit. Write the prime power decomposition of q in the form

$$q = 2^\nu \prod_{1 \leqslant j \leqslant k} p_j^{\nu_j}, \quad 2 < p_1 < \ldots < p_k, \quad \nu \geqslant 0, \quad \nu_j > 0 \quad (1 \leqslant j \leqslant k).$$

For each index j, let g_j be a primitive root modulo $p_j^{\nu_j}$, i.e. a generator of $(\mathbb{Z}/p_j^{\nu_j}\mathbb{Z})^*$. Then, for each integer m such that $(m, q) = 1$, we can uniquely define $\varepsilon(m)$, $\eta(m)$, $\mu_j(m)$ $(1 \leqslant j \leqslant k)$ satisfying

$$m \equiv (-1)^{\varepsilon(m)} 5^{\eta(m)} \,(\mathrm{mod}\, 2^\nu), \quad \varepsilon(m) = 0 \text{ or } 1, \quad 0 \leqslant \eta(m) < 2^{\nu-2},$$

$$m \equiv g_j^{\mu_j(m)} \,(\mathrm{mod}\, p_j^{\nu_j}), \quad 0 \leqslant \mu_j(m) < \varphi(p_j^{\nu_j}) \quad (1 \leqslant j \leqslant k).$$

If $\nu = 0$ or 1, we make the convention that $\varepsilon(m) = 0$.

With this notation the characters of $(\mathbb{Z}/q\mathbb{Z})^*$ are the $\varphi(q)$ functions defined on this group by

$$\chi(m) = \mathrm{e}\left(\frac{\lambda\varepsilon(m)}{2} + \frac{\lambda'\eta(m)}{2^{\nu-2}} + \sum_{1\leqslant j\leqslant k} \frac{\lambda_j \mu_j(m)}{\varphi(p_j^{\nu_j})}\right) \quad (m \in (\mathbb{Z}/q\mathbb{Z})^*) \tag{8.6}$$

where the parameters λ, λ', λ_j are arbitrarily chosen so that

$$\lambda = 0 \text{ or } 1, \quad 0 \leqslant \lambda' < 2^{\nu-2}, \quad 0 \leqslant \lambda_j < \varphi(p_j^{\nu_j}) \quad (1 \leqslant j \leqslant k).$$

Definition 8.2. *A Dirichlet character to the modulus q (or: modulo q) is the arithmetic function extending a character χ of the group $(\mathbb{Z}/q\mathbb{Z})^*$ by means of the formula*

$$\chi(n) = \begin{cases} \chi(m), & \text{if } n \equiv m \,(\mathrm{mod}\, q),\ 1 \leqslant m \leqslant q,\ (m,q) = 1, \\ 0, & \text{if } (n,q) > 1. \end{cases} \tag{8.7}$$

Examples. (i) The unique non-trivial character to the modulus 3 is defined by

$$\chi_3(m) := \frac{\mathrm{e}(m/3) - \mathrm{e}(-m/3)}{i\sqrt{3}} = \begin{cases} 1 & \text{if } m \equiv 1 \,(\mathrm{mod}\, 3), \\ -1 & \text{if } m \equiv 2 \,(\mathrm{mod}\, 3). \end{cases}$$

(ii) Let χ_4 be in the unique non-trivial character to the modulus 4. We necessarily have $\chi_4(1) = 1$, $\chi_4(-1) = -1$, hence $\chi_4(2n+1) = (-1)^n$ for all $n \geqslant 0$.

(iii) Since 2 is a primitive root modulo 5, we have $m \equiv 2^{\mu_2(m)} \,(\mathrm{mod}\, 5)$ where μ_2 is defined on $(\mathbb{Z}/5\mathbb{Z})^*$ by

$$\mu_2(1) = 4, \quad \mu_2(2) = 1, \quad \mu_2(3) = 3, \quad \mu_2(4) = 2.$$

Selecting $\lambda_2 = 2$, we define a Dirichlet character to the modulus 5 by

$$\chi(n) = (-1)^{\mu_2(n)} = \begin{cases} 1 & \text{if } n \equiv \pm 1 \,(\mathrm{mod}\, 5), \\ -1 & \text{if } n \equiv \pm 2 \,(\mathrm{mod}\, 5) \end{cases}$$

and of course $\chi(n) = 0$ if $n \equiv 0 \,(\mathrm{mod}\, 5)$. This is the only real character modulo 5, and one sees easily that it is equal to the Legendre symbol. More generally, the *Jacobi symbol* modulo an odd number q is the character given by the formula

$$\chi(n) = \left(\frac{n}{q}\right) := \prod_{1\leqslant j\leqslant k} \left(\frac{n}{p_j}\right)^{\nu_j} = (-1)^{\sum_{1\leqslant j\leqslant k} \nu_j \mu_j(n)} \qquad (n \in (\mathbb{Z}/q\mathbb{Z})^*).$$

The homomorphism property of the characters of $(\mathbb{Z}/q\mathbb{Z})^*$ translates into the fact that *a Dirichlet character is a completely multiplicative arithmetic*

function. Theorem 8.1 then implies the following fundamental relations. As is customary, we call the extension of the unit character, viz.

$$\chi_0(n) = \begin{cases} 1 & \text{if } (n,q) = 1, \\ 0 & \text{if } (n,q) > 1, \end{cases} \tag{8.8}$$

the *principal character*, and denote it by χ_0,

Theorem 8.3 (Orthogonality relations).

(a) *For any integers $n, m \geqslant 1$, we have*

$$\frac{1}{\varphi(q)} \sum_{\chi \,(\mathrm{mod}\, q)} \chi(n)\overline{\chi(m)} = \begin{cases} 1 & \text{if } n \equiv m (\mathrm{mod}\, q) \text{ and } (m,q) = 1, \\ 0 & \text{otherwise.} \end{cases} \tag{8.9}$$

(b) *For all Dirichlet characters χ, χ', to the modulus q, we have*

$$\frac{1}{\varphi(q)} \sum_{1 \leqslant n \leqslant q} \chi(n)\overline{\chi'(n)} = \begin{cases} 1 & \text{if } \chi = \chi', \\ 0 & \text{otherwise.} \end{cases} \tag{8.10}$$

8.1.2. Primitive characters. Certain results of the theory take a simpler form for a special class of characters, called *primitive characters.*

Definition 8.4. *A Dirichlet character χ to the modulus q is said to be primitive if there exists no character χ_1 to the modulus $q_1 < q$, such that $\chi(n) = \chi_1(n)$ for all integers n coprime to q.*

If such a character χ_1 exists, we say that χ_1 induces χ. We then have, trivially,

$$(m,q) = (n,q) = 1 \text{ and } m \equiv n \,(\mathrm{mod}\, q_1) \Rightarrow \chi(m) = \chi(n). \tag{8.11}$$

Conversely, if q_1 satisfies (8.11), it is easy to show using the Chinese remainder theorem that, for any integer n satisfying $(n, q_1) = 1$, there exists an integer t such that $(n+tq_1, q) = 1$.[4] We can thus set $\chi_1(n) := \chi(n+tq_1)$, this definition being independent of the selected t. This does define a Dirichlet character χ_1 to the modulus q_1: by construction χ_1 is periodic modulo q_1 and, if $(mn, q_1) = 1$, we have, for suitable integers s and t,

$$\chi_1(mn) = \chi((m + sq_1)(n + tq_1)) = \chi(m + sq_1)\chi(n + tq_1) = \chi_1(n)\chi_1(m).$$

It is then clear that χ_1 induces χ.

We have thus established that *a necessary and sufficient condition for the character χ to be primitive is that q is the smallest of those integers q_1 satisfying* (8.11).

[4] Indeed, $n + tq_1$ is identically coprime to every prime factor p of q_1 since $(n, q_1) = 1$, and $n + tq_1$ may be made coprime to each prime factor p of q not dividing q_1 since q_1 is invertible modulo p.

If χ_1 to the modulus $q_1 < q$ induces χ and if q_1 is minimal, then $q_1|q$, since, if relation (8.11) is satisfied for q_1, then it also holds for (q, q_1).

Proposition 8.5. *Each imprimitive Dirichlet character χ to the modulus q is induced by a unique primitive character χ_1 modulo a proper divisor q_1 of q. Whenever $(n, q_1) = 1$, we have $\chi_1(n) := \chi(n + tq_1)$, where t is an arbitrary integer such that $(n + tq_1, q) = 1$.*

Proof. Let χ be an imprimitive character to the modulus q and let $q_1 < q$ be the smallest integer satisfying (8.11). We saw above that the map χ_1 defined as in the statement is a Dirichlet character inducing χ. We have $\chi_1(1) = \chi(1) = 1$, so χ_1 does not vanish identically. Furthermore, χ_1 is necessarily primitive since, if $(m, q_1) = (n, q_1) = 1$ and $m \equiv n \pmod{q_2}$ imply $\chi_1(m) = \chi_1(n)$, then q_2 satisfies (8.11) and hence $q_2 \geqslant q_1$, by the minimality of q_1.

Let us next establish uniqueness. If χ_2 is a primitive character to a modulus q_2 inducing χ, then q_2 satisfies (8.11) and thus $q_2 = q_1$ by minimality of q_1. Let n be such that $(n, q_1) = 1$. We then have $(n + tq_1, q) = 1$ for a suitable integer t. Therefore $\chi_2(n) = \chi_2(n + tq_1) = \chi(n + tq_1) = \chi_1(n)$. Thus $\chi_1 = \chi_2$. □

The modulus of the primitive character inducing a given Dirichlet character is usually referred to as its *conductor*: this is also the smallest modulus of a character inducing it.

8.1.3. Gaussian sums.

Definition 8.6. *Let χ be a Dirichlet character to the modulus q. The* Gaussian sum *associated to χ is the arithmetic function defined by*

$$G(n, \chi) := \sum_{1 \leqslant m \leqslant q} \chi(m)\mathrm{e}(mn/q) \qquad (n \geqslant 1).$$

Theorem 8.7. *Let χ be a primitive character to the modulus q. We have*

$$G(n, \chi) = \overline{\chi(n)}G(1, \chi) \qquad (n \geqslant 1). \tag{8.12}$$

Proof. If $\chi = \chi_0$, primitivity implies $q = 1$, and the result is trivial. We can thus suppose from now on that χ is not principal.

If $(n, q) = 1$, we have

$$\overline{\chi(n)}G(1, \chi) = \sum_{1 \leqslant m \leqslant q} \overline{\chi(n)}\chi(m)\mathrm{e}(m/q) = \sum_{1 \leqslant h \leqslant q} \overline{\chi(n)}\chi(nh)\mathrm{e}(nh/q) = G(n, \chi),$$

since nh runs over a complete system of residues modulo q as h runs over $[1, q]$. It may be noted that this formula actually holds even if χ is not primitive.

When $(n,q) > 1$, we have to show that $G(n,\chi) = 0$. Put $d = (n,q)$, $n = dn_1$, $q = dq_1$, so that $(n_1,q_1) = 1$. We may assume $q_1 > 1$: otherwise, we have $q|n$, whence

$$G(n,\chi) = \sum_{1\leqslant m\leqslant q} \chi(m) = 0.$$

Therefore we must establish the relation

$$\sum_{1\leqslant m\leqslant q} \chi(m)\mathrm{e}(mn_1/q_1) = 0. \tag{8.13}$$

Decompose each integer $m \in [1,q]$ as $m = uq_1+v$ with $0 \leqslant u < d$, $1 \leqslant v \leqslant q_1$. Then $\mathrm{e}(mn_1/q_1) = \mathrm{e}(vn_1/q_1)$ and the left-hand side of (8.13) equals

$$\sum_{1\leqslant v\leqslant q_1} \mathrm{e}(vn_1/q_1) \sum_{0\leqslant u<d} \chi(uq_1+v).$$

We shall show that the inner sum, call it $S(v)$, vanishes identically. We have

$$S(v+q_1) = \sum_{1\leqslant u\leqslant d} \chi(uq_1+v) = \sum_{0\leqslant u<d} \chi(uq_1+v) - \chi(v) + \chi(q+v) = S(v),$$

so S is q_1-periodic. Let c be such that $(c,q) = 1$, $c \equiv 1 \pmod{q_1}$. Then

$$\chi(c)S(v) = \sum_{0\leqslant u<d} \chi(cuq_1+cv) = \sum_{0\leqslant w<d} \chi(wq_1+cv)$$

since, as u runs over $\{0,1,\dots,d-1\}$, $w = cu$ runs through a complete system of residues modulo d. It follows that

$$\chi(c)S(v) = S(cv) = S(v), \tag{8.14}$$

where we have used the facts that $cv \equiv v \pmod{q_1}$ and that S is q_1-periodic. However, since χ is primitive, relation (8.11) is not satisfied for q_1. We can thus find integers c_1 and c_2 such that

$$(c_1c_2,q) = 1, \qquad \chi(c_1) \neq \chi(c_2), \qquad c_1 \equiv c_2 \pmod{q_1}.$$

Selecting $c \equiv c_1\overline{c_2} \pmod{q}$, we then have $\chi(c) \neq 1$, which, in view of (8.14), implies $S(v) = 0$. □

Theorem 8.8. *For every primitive Dirichlet character χ to the modulus q, we have*

$$|G(1,\chi)| = \sqrt{q}. \tag{8.15}$$

Proof. For any integer n, we have

$$|\chi(n)|^2\,|G(1,\chi)|^2 = |G(n,\chi)|^2 = \sum_{1\leqslant m,h\leqslant q} \chi(m)\overline{\chi(h)}\mathrm{e}((m-h)n/q).$$

Sum this relation over $1 \leqslant n \leqslant q$. The left-hand side equals $\varphi(q)|G(1,\chi)|^2$. The right-hand side can easily be computed by exchanging summations: the

inner sum over n vanishes unless $q|(m-h)$, in which case it equals q. It follows that

$$\varphi(q)|G(1,\chi)|^2 = q\sum_{1\leqslant m\leqslant q}|\chi(m)|^2 = q\varphi(q). \qquad \square$$

The following proposition, whose proof we leave as an exercise, allows calculating the modulus of Gaussian sums associated to imprimitive characters.

Proposition 8.9. *Let χ be an imprimitive Dirichlet character to the modulus q, and let χ_1 denote the primitive character, to the modulus q_1, which induces it. Put $r := q/q_1$. Then, we have*

$$G(1,\chi) = \begin{cases} \mu(r)\chi_1(r)G(1,\chi_1) & \text{if } (q_1,r)=1, \\ 0 & \text{if } (q_1,r)>1. \end{cases}$$

8.1.4. Bounds. For many questions related to the distribution of primes in arithmetic progressions, individual or average upper bounds for the quantity

$$K(x) := \sum_{n\leqslant x}\chi(n) \tag{8.16}$$

turn out to be crucial. The case $\chi = \chi_0$ is essentially relevant to sieve theory. For example, for any fixed $\varepsilon > 0$, we have

$$K(x) \ll x\varphi(q)/q$$

whenever $x \geqslant P^+(q)^\varepsilon$. If $\chi \neq \chi_0$, relation (8.10) applied with $\chi' = \chi_0$ implies by periodicity that

$$H(\chi) := \max_{x\geqslant 1}\big|K(x)\big| \leqslant \tfrac{1}{2}q. \tag{8.17}$$

The next theorem, discovered independently by G. Pólya and I. M. Vinogradov in 1918, allows considerable improvement of inequality (8.17).

Theorem 8.10 (Pólya–Vinogradov). *Let χ be a non-principal Dirichlet character to the modulus $q \geqslant 2$. We have*

$$H(\chi) \leqslant 2\sqrt{q}\ln q. \tag{8.18}$$

Proof. Suppose first χ is primitive. Using the classical notation

$$G(\chi) := G(1,\chi),$$

we have, by (8.12),

$$\chi(n) = \frac{1}{G(\overline{\chi})}\sum_{1\leqslant m\leqslant q}\overline{\chi(m)}\mathrm{e}(mn/q). \tag{8.19}$$

Whence

$$K(x) = \sum_{n\leqslant x} \chi(n) = \frac{1}{G(\overline{\chi})} \sum_{1\leqslant m\leqslant q} \overline{\chi(m)} \sum_{n\leqslant x} \mathrm{e}(mn/q).$$

Set $N := \lfloor x \rfloor$. The modulus of the inner sum equals

$$|\sin(\pi Nm/q)/\sin(\pi m/q)| \leqslant 1/|\sin(\pi m/q)|.$$

Since $\chi(q) = 0$, this yields

$$|K(x)| \leqslant \frac{1}{\sqrt{q}} \sum_{1\leqslant m<q} \frac{1}{|\sin(\pi m/q)|} \leqslant 2\sqrt{q} \sum_{1\leqslant m\leqslant (q-1)/2} \frac{1}{2m} < \sqrt{q}\ln q,$$

where we have used the bound

$$\sum_{1\leqslant m\leqslant t} \frac{1}{m} < \ln(2t+1) \qquad (t>0),$$

which is easily proved by induction on $k := \lfloor t \rfloor$.

If χ is imprimitive, it is induced by a primitive character χ_1 whose modulus q_1 divides q. We then have, denoting the summatory function of χ_1 by $K_1(y)$,

$$K(x) = \sum_{\substack{n\leqslant x\\(n,q)=1}} \chi(n) = \sum_{n\leqslant x} \chi(n) \sum_{d|(n,q)} \mu(d) = \sum_{d|q} \mu(d)\chi_1(d) K_1(x/d).$$

We thus obtain

$$H(\chi) \leqslant \sqrt{q_1}\ln q_1 \sum_{d|q} \Big|\mu(d)\chi_1(d)\Big|.$$

Since $\chi_1(d) = 0$ whenever $(d, q_1) > 1$, the modulus of the sum over d does not exceed

$$\sum_{t|q/q_1} \mu(t)^2 = 2^{\omega(q/q_1)} \leqslant \tau(q/q_1) \leqslant 2\sqrt{q/q_1}.$$

This yields (8.18). □

Remark. The above proof actually furnishes

$$\max_{x,y} |K(x) - K(y)| \leqslant 2\sqrt{q}\ln q_1 \leqslant 2\sqrt{q}\ln q. \tag{8.20}$$

It is easy to show that the Pólya–Vinogradov bound is not far from being optimal.

Theorem 8.11. *For any primitive Dirichlet character χ to the modulus q, we have*

$$H(\chi) \geqslant \tfrac{1}{4}\sqrt{q}. \tag{8.21}$$

Proof. Indeed, we have

$$G(\chi) = -\frac{2\pi i}{q}\int_0^q K(x)\mathrm{e}(x/q)\,\mathrm{d}x = -\frac{2\pi i}{q}\int_0^q K(q-x)\mathrm{e}(-x/q)\,\mathrm{d}x.$$

Now

$$\begin{aligned} K(q-x) &= \sum_{1\leqslant n\leqslant q-x}\chi(n) = \sum_{x\leqslant m<q}\chi(q-m) \\ &= \sum_{x\leqslant m<q}\chi(-m) = -\chi(-1)\sum_{1\leqslant m<x}\chi(m), \end{aligned}$$

and so $K(q-x) = -\chi(-1)K(x)$ except possibly for a finite number of values of x. It follows that

$$G(\chi) = -\frac{2\pi i}{q}\int_0^q K(x)\tfrac{1}{2}\{\mathrm{e}(x/q) - \chi(-1)\mathrm{e}(-x/q)\}\,\mathrm{d}x$$

and hence

$$\sqrt{q} = |G(\chi)| \leqslant H(\chi)\frac{2\pi}{q}\int_0^q \tfrac{1}{2}\,|\mathrm{e}(x/q) \pm \mathrm{e}(-x/q)|\,\mathrm{d}x = H(\chi)\int_0^{2\pi}|\varphi(u)|\,\mathrm{d}u$$

where $\varphi = \sin$ or $\cos$. This implies (8.21), as required. □

8.2. L-series. The prime number theorem for arithmetic progressions

8.2.1. L-series and primes in arithmetic progressions. For each character χ, the Dirichlet L-series is defined by the formula

$$L(s,\chi) := \sum_{n\geqslant 1}\frac{\chi(n)}{n^s} \qquad (\sigma > 1) \tag{8.22}$$

with Euler product

$$L(s,\chi) = \prod_p \Big(1 - \frac{\chi(p)}{p^s}\Big)^{-1} \qquad (\sigma > 1). \tag{8.23}$$

In particular, $L(s,\chi) \neq 0$ when $\sigma > 1$. When $\chi = \chi_0$, the above formula can also be written

$$L(s,\chi_0) = \prod_{p|q}(1-p^{-s})\zeta(s). \tag{8.24}$$

This defines a meromorphic extension of $L(s,\chi_0)$ to a function having a single simple pole in $\mathbb{C}$, at $s=1$, with residue $\varphi(q)/q$. When $\chi \neq \chi_0$, we get from (8.17) that the corresponding summatory function $K(x)$ is bounded. By Dirichlet's test we deduce that the series $L(s,\chi)$ converges for $\sigma > 0$, which actually determines the abscissa of convergence.

In conjunction with with orthogonality relations, the Euler product expansion (8.23) allows investigating prime numbers in arithmetic progressions. As in the case of the Riemann zeta function, we proceed via the logarithmic derivative.

Theorem 8.12. *For all integers a, q, $(a,q)=1$, we have*

$$\sum_{\substack{n\geqslant 1\\ n\equiv a \,(\mathrm{mod}\, q)}} \frac{\Lambda(n)}{n^s} = \frac{-1}{\varphi(q)} \sum_{\chi} \overline{\chi(a)} \frac{L'}{L}(s,\chi) \qquad (\sigma>1) \tag{8.25}$$

where the summation over χ extends to the $\varphi(q)$ Dirichlet characters to the modulus q.

Proof. By (8.23), for any χ, we have

$$\frac{-L'}{L}(s,\chi) = \sum_p \frac{\chi(p)\ln p}{p^s-\chi(p)} = \sum_p \sum_{\nu\geqslant 1} \frac{\chi(p^\nu)\ln p}{p^{\nu s}} = \sum_{n\geqslant 1} \frac{\chi(n)\Lambda(n)}{n^s}.$$

The result therefore follows from (8.9) with $m=a$, by interchanging summations on the right-hand side of (8.25). □

The Dirichlet series (8.25) is the Mellin–Stieltjes transform of the function

$$\psi(x;a,q) := \sum_{\substack{n\leqslant x\\ n\equiv a \,(\mathrm{mod}\, q)}} \Lambda(n). \tag{8.26}$$

As in the case of the classical Chebyshev function $\psi(x)=\psi(x;1,1)$, an integration by parts allows connecting the asymptotic behaviors of $\psi(x;a,q)$ and of the counting function of primes in the corresponding arithmetic progression, i.e.,

$$\pi(x;a,q) := \sum_{\substack{p\leqslant x\\ p\equiv a \,(\mathrm{mod}\, q)}} 1. \tag{8.27}$$

Dirichlet's theorem mentioned at the beginning of this chapter tells us that the expression (8.27) tends to infinity with x for all fixed a, q with $(a,q)=1$. Actually, Dirichlet established a more precise result, which, in a certain sense, implies that the primes are evenly distributed amongst the $\varphi(q)$ possible residue classes modulo q. He showed that we have

$$\sum_{\substack{p\leqslant x\\ p\equiv a \,(\mathrm{mod}\, q)}} \frac{1}{p} \sim \frac{\ln_2 x}{\varphi(q)} \qquad (x\to\infty). \tag{8.28}$$

This estimate readily follows from the assumption

$$L(1,\chi)\neq 0 \qquad (\chi\neq\chi_0) \tag{8.29}$$

by means of the Hardy–Littlewood–Karamata theorem (Proposition 7.9). Indeed, we deduce from (8.25) and (8.29) that, for fixed a and q, we have

$$\sum_{\substack{n\geqslant 1\\ n\equiv a\,(\mathrm{mod}\,q)}} \frac{\Lambda(n)}{n^\sigma} = \frac{-L'(\sigma,\chi_0)}{\varphi(q)L(\sigma,\chi_0)} + O_q(1) \qquad (\sigma>1), \tag{8.30}$$

and, by (8.24), we have under the same conditions

$$-\frac{L'}{L}(\sigma,\chi_0) = -\frac{\zeta'}{\zeta}(\sigma) - \sum_{p\mid q}\frac{\ln p}{p^\sigma-1} = \frac{1}{\sigma-1} + O_q(1). \tag{8.31}$$

Applying Theorem 7.10 of Karamata–Freud, we even obtain the remainder formula

$$\sum_{\substack{n\leqslant x\\ n\equiv a\,(\mathrm{mod}\,q)}} \frac{\Lambda(n)}{n} = \frac{\ln x}{\varphi(q)} + O_q\Big(\frac{\ln x}{\ln_2 x}\Big) \tag{8.32}$$

from which we infer by partial summation an effective version of (8.28), viz.

$$\sum_{\substack{p\leqslant x\\ p\equiv a\,(\mathrm{mod}\,q)}} \frac{1}{p} = \frac{\ln_2 x}{\varphi(q)} + O_q(\ln_3 x) \qquad (x\to\infty). \tag{8.33}$$

As shown by Mertens (1874), the fact that $L(1,\chi)\neq 0$ actually yields an improvement upon (8.33).

Theorem 8.13 (Mertens). *If $L(1,\chi)\neq 0$ for every non-principal character to the modulus q, then, for all a such that $(a,q)=1$, we have*

$$\sum_{\substack{p\leqslant x\\ p\equiv a\,(\mathrm{mod}\,q)}} \frac{1}{p} = \frac{\ln_2 x}{\varphi(q)} + c(a,q) + O\Big(\frac{1}{\ln x}\Big) \qquad (x\to\infty) \tag{8.34}$$

with

$$c(a,q) := \frac{1}{\varphi(q)}\Big\{\gamma - \sum_p\Big(\ln\Big(\frac{1}{1-1/p}\Big) - \frac{1}{p}\Big) + \sum_{\chi\neq\chi_0}\overline{\chi(a)}\sum_p\frac{\chi(p)}{p}\Big\}.$$

Proof. The convolution relation $\chi\ln = \chi(\Lambda * \mathbf{1}) = \chi\Lambda * \chi$ implies

$$\sum_{n\leqslant x}\frac{\chi(n)\ln n}{n} = \sum_{m\leqslant x}\frac{\chi(m)\Lambda(m)}{m}\sum_{d\leqslant x/m}\frac{\chi(d)}{d}.$$

By partial summation, we see the inner sum equals $L(1,\chi)+O(m/x)$, whence

$$\sum_{n\leqslant x}\frac{\chi(n)\ln n}{n} = L(1,\chi)\sum_{m\leqslant x}\frac{\chi(m)\Lambda(m)}{m} + O(1)$$
$$= L(1,\chi)\sum_{p\leqslant x}\frac{\chi(p)\ln p}{p} + O(1).$$

Since the left-hand side converges by Dirichlet's test, we deduce that the assumption $L(1,\chi)\neq 0$ implies

$$\sum_{p\leqslant x}\frac{\chi(p)\ln p}{p} \ll 1.$$

A further summation by parts then allows us to establish the convergence of $\sum_p \chi(p)/p$ and the asymptotic formula

$$\sum_{p\leqslant x}\frac{\chi(p)}{p} = \sum_{p}\frac{\chi(p)}{p} + O\Big(\frac{1}{\ln x}\Big) \qquad (x\to\infty).$$

This readily implies the stated result by linear combination of characters. □

Remark. If $R(x)$ denotes the error term in (8.34), we have

$$\pi(x;a,q) = \frac{\mathrm{li}(x)}{\varphi(q)} + \int_2^x \{R(x)-R(t)\}\,\mathrm{d}t + O(1).$$

Thus, we see that the prime number theorem for arithmetic progressions is equivalent to $R(x)=o(1/\ln x)$: indeed, the sufficiency part immediately follows from the above formula, and the converse may be derived, for example, via the elementary technique used for Corollary I.3.9(vi).

8.2.2. The numbers $L(1,\chi)$. We saw above that the assumption

$$\prod_{\chi\neq\chi_0} L(1,\chi)\neq 0$$

implies equidistribution of prime numbers in arithmetic progressions in the form (8.34).[5]

Conversely, we readily deduce from (8.34) that, for $\chi\neq\chi_0$,

$$\sum_p \frac{\chi(p)}{p} = \sum_{\substack{1\leqslant a\leqslant q\\(a,q)=1}}\chi(a)c(a,q).$$

In turn, this formula implies, by partial summation, that

$$F(\sigma,\chi) := \sum_p \frac{\chi(p)}{p^\sigma} = (\sigma-1)\int_1^\infty \sum_{p\leqslant x}\frac{\chi(p)}{p}\frac{\mathrm{d}x}{x^\sigma}$$

[5]This assertion also follows from the Hardy–Littlewood–Karamata Tauberian theorem as stated in Theorem 7.7. See, for example, Exercise 222, p. 354.

tends towards a finite limit as $\sigma \to 1+$. Since

$$L(\sigma,\chi) = \mathrm{e}^{F(\sigma,\chi)} \prod_p \frac{\mathrm{e}^{-\chi(p)/p^\sigma}}{1-\chi(p)/p^\sigma},$$

where, for all $\sigma \geqslant 1$, the infinite product converges absolutely and uniformly to a non-zero limit, we infer that $L(1,\chi) \neq 0$.

Thus, formally, it may be asserted that the validity of (8.34) for any a coprime to q is equivalent to $\prod_{\chi \neq \chi_0} L(1,\chi) \neq 0$.

As will be seen later, the value of $L(1,\chi)$ plays a special role in the theory when χ is a real primitive Dirichlet character. Relation (8.19) allows us to give a closed-form expression for this quantity.

We note at this point, that, for real χ, we have

$$G(\chi) = \sum_{1 \leqslant m \leqslant q} \chi(m)\mathrm{e}(m/q) = \chi(-1) \sum_{1 \leqslant m \leqslant q} \chi(m)\mathrm{e}(-m/q) = \chi(-1)\overline{G(\chi)},$$

therefore $G(\chi)$ is real if $\chi(-1) = 1$, and purely imaginary if $\chi(-1) = -1$.

Theorem 8.14. *Let χ be a real primitive Dirichlet character to the modulus $q > 1$. We have*

$$L(1,\chi) = \begin{cases} \dfrac{-i\pi}{qG(\chi)} \displaystyle\sum_{1 \leqslant m < q} m\chi(m) & \text{if } G(\chi) \in i\mathbb{R}, \\ \dfrac{-1}{G(\chi)} \displaystyle\sum_{1 \leqslant m < q} \chi(m) \ln\{2\sin(\pi m/q)\} & \text{if } G(\chi) \in \mathbb{R}. \end{cases} \tag{8.35}$$

Furthermore,

$$\begin{aligned} \sum_{1 \leqslant m < q} m\chi(m) &= -\frac{1-\chi(-1)}{2\pi i} qG(\chi)L(1,\chi) \\ &= \begin{cases} \dfrac{iqG(\chi)}{\pi} L(1,\chi) & \text{if } G(\chi) \in i\mathbb{R}, \\ 0 & \text{if } G(\chi) \in \mathbb{R}. \end{cases} \end{aligned} \tag{8.36}$$

Proof. Let us multiply (8.19) by $1/n$ and sum over $n \geqslant 1$. This yields

$$\begin{aligned} L(1,\chi) &= \frac{1}{G(\chi)} \sum_{1 \leqslant m < q} \chi(m) \log\Big(\frac{1}{1-\mathrm{e}(m/q)}\Big) \\ &= \frac{1}{G(\chi)} \sum_{1 \leqslant m < q} \chi(m) \log\Big(\frac{\mathrm{e}(-m/2q)}{-2i\sin(\pi m/q)}\Big) \\ &= \frac{1}{G(\chi)} \sum_{1 \leqslant m < q} \chi(m) \log\Big(\frac{\mathrm{e}^{i\pi(\frac{1}{2}-m/q)}}{2\sin(\pi m/q)}\Big), \end{aligned}$$

whence

$$G(\chi)L(1,\chi) = i\pi \sum_{1\leqslant m<q} \chi(m)\Big(\tfrac{1}{2} - \frac{m}{q}\Big) - \sum_{1\leqslant m<q} \chi(m)\ln\{2\sin(\pi m/q)\}.$$

This implies the stated formulae. □

The above theorem allows evaluating, for the same values of χ, the quadratic norm defined by

$$M_2(\chi)^2 := \frac{1}{q}\int_0^q |K(x)|^2\,\mathrm{d}x. \tag{8.37}$$

We have $M_2(\chi) \leqslant H(\chi)$, and we shall see that, whenever possible, it is indeed helpful to use average rather than pointwise bounds for $K(x)$.

Corollary 8.15. *Let χ be a real primitive Dirichlet character to the modulus $q > 1$. We have*

$$M_2(\chi)^2 \leqslant q\Big\{\frac{L(1,\chi)^2}{\pi^2} + \frac{1}{12}\Big\}. \tag{8.38}$$

Proof. The step function $K(x)$ is q-periodic and continuous on $\mathbb{R}\setminus\mathbb{Z}$. It hence has a convergent Fourier series expansion

$$K(x) = \sum_{\nu\in\mathbb{Z}} \widehat{K}(\nu)\mathrm{e}(\nu x/q) \qquad (x\in\mathbb{R}\setminus\mathbb{Z}).$$

We have

$$\begin{aligned}\widehat{K}(0) &:= \frac{1}{q}\int_0^q K(t)\,\mathrm{d}t = \frac{1}{q}\sum_{1\leqslant m<q}(q-m)\chi(m)\\ &= \frac{-1}{q}\sum_{1\leqslant m<q} m\chi(m) = \frac{1-\chi(-1)}{2\pi i}G(\chi)L(1,\chi),\end{aligned}$$

by (8.36). Similarly, for $\nu\neq 0$,

$$\widehat{K}(\nu) = \frac{1}{q}\int_0^q K(x)\mathrm{e}(-\nu x/q)\,\mathrm{d}x = \frac{1}{2i\pi\nu}\sum_{1\leqslant m\leqslant q}\chi(m)\mathrm{e}(-\nu m/q) = \frac{\chi(-\nu)G(\chi)}{2i\pi\nu}.$$

Whence

$$\begin{aligned}M_2(\chi)^2 &= \sum_{\nu\in\mathbb{Z}}|\widehat{K}(\nu)|^2 = \frac{\{1-\chi(-1)\}^2}{4\pi^2}qL(1,\chi)^2 + \frac{q}{2\pi^2}\sum_{\nu\geqslant 1}\frac{\chi(\nu)^2}{\nu^2}\\ &\leqslant q\Big\{\frac{L(1,\chi)^2}{\pi^2} + \frac{1}{12}\Big\}.\end{aligned} \tag{8.39}$$

□

8.2.3. The Siegel–Walfisz theorem. If assumption (8.29) is strengthened to

$$L(1+i\tau,\chi)\neq 0 \qquad (\tau\in\mathbb{R},\ \chi\neq\chi_0) \tag{8.40}$$

we obtain by Ikehara's theorem the more precise formula

$$\psi(x;a,q)\sim\frac{x}{\varphi(q)} \qquad (x\to\infty). \tag{8.41}$$

This result is commonly referred to as the prime number theorem for arithmetic progressions. It is one of the central questions in analytic number theory to determine effective versions of this asymptotic formula with respect to the three variables involved.

In order to deduce (8.41) from (8.40), it suffices to apply Theorem 7.13 to the Dirichlet series (8.25), i.e.,

$$F(s)=\int_0^\infty \mathrm{e}^{-st}\,\mathrm{d}\psi(\mathrm{e}^t;a,q).$$

By (8.40), each function $L'(s,\chi)/L(s,\chi)$, $\chi\neq\chi_0$, may be continued to a holomorphic function for $\sigma\geqslant 1$ (that is to say in an open set containing this half-plane), and it then follows from (8.24) and (8.25) that

$$G(s):=\frac{F(s+1)}{s+1}-\frac{1}{\varphi(q)s} \tag{8.42}$$

can be continued to a continuous function for $\sigma\geqslant 0$. Ikehara's theorem then allows us to deduce the validity of (8.41).

We shall see in the following section that (8.40) is indeed satisfied. We shall even obtain explicit lower bounds for $|L(1+i\tau,\chi)|$. This will enable us to employ the full strength of our effective version 7.13 of Ikehara's theorem—and hence to derive information on the dependence of the error term of (8.41) on x and q. We shall obtain the following result.

Theorem 8.16. *For any constant $A>0$, and uniformly for*

$$x\geqslant 3,\quad 1\leqslant q\leqslant(\ln x)^A,\quad (a,q)=1,$$

we have

$$\psi(x;a,q)=\frac{x}{\varphi(q)}+O\Big(x\frac{(\ln_2 x)^{19/2}}{\ln x}\Big). \tag{8.43}$$

In particular, the asymptotic formula (8.41) *holds, for each $\varepsilon>0$, uniformly in the domain*

$$x\geqslant 3,\qquad 1\leqslant q\leqslant(\ln x)^{1-\varepsilon}. \tag{8.44}$$

A finer study of the series $L(s,\chi)$ furnishes a zero-free region analogous to that for the zeta function. Uniformity in q is crucial and particularly difficult—incidentally raising an interesting open problem on the existence of the famous "Siegel zero", to which we will return later. In § 8.7, we shall establish the following assertion, which considerably strengthens Theorem 8.16—cf. Siegel (1936).

Theorem 8.17 (Siegel–Walfisz). *Under the assumptions of Theorem* 8.16, *we have*

$$\psi(x;a,q) = \frac{x}{\varphi(q)} + O\Big(x\mathrm{e}^{-c\sqrt{\ln x}}\Big) \tag{8.45}$$

where $c = c(A)$ *is a positive constant.*

A rather disappointing feature of Theorem 8.17 is its *ineffectiveness*: for $q \geqslant \ln x$, one cannot, in the present state of knowledge, provide numerical values for c or for the implicit constant in the Landau symbol. Actually, the best effective results known, based on the Pólya–Vinogradov inequality, will only yield an asymptotic formula for $\psi(x;a,q)$ if $q = o\big((\ln x)^2\big)$—see Corollary 8.31 for a precise formulation. In this respect, the effective estimate in Theorem 8.16 is hence of fairly good quality.

Before undertaking the proof of Theorem 8.16, a methodological remark is in order. For $q = 1$, the error term in (8.43) is significantly less precise than the one we obtained for the prime number theorem, even without using delicate properties of the function $\zeta(s)$ in the critical strip—cf. § 4.2. This suggests that we ought *a priori* to be able to improve Theorem 8.16 at least for sufficiently small, or even bounded, moduli q. It indeed turns out that the lower bounds for $|L(1+i\tau,\chi)|$ which we shall establish in the following section yield, under assumption (8.44), a zero-free region which in turn allows one to obtain an error term $\ll_A x/(\ln x)^A$ for all $A > 0$. The details are made precise in Exercise 237. Here, we have however chosen to apply the effective Ikehara theorem: on the one hand, it delivers very simply a domain of validity for (8.41) comparable to that obtained by contour integration, and, on the other hand, this gives us the opportunity to elaborate upon an arithmetic use of a Tauberian theorem.

8.3. Lower bounds for $|L(s,\chi)|$ when $\sigma \geqslant 1$. Proof of Theorem 8.16

For ease of notation, we define

$$\mathcal{L} = \mathcal{L}(q,\tau) := \ln(|\tau| + q + 1) \qquad (q \geqslant 1,\ \tau \in \mathbb{R}). \tag{8.46}$$

Theorem 8.18. *For* $k \geqslant 0$, $\chi \neq \chi_0$, *we have*

$$L^{(k)}(s,\chi) \ll_k \mathcal{L}^{k+1} \qquad (\sigma \geqslant 1). \tag{8.47}$$

Proof. We may assume without loss of generality that $1 \leqslant \sigma \leqslant 2$. Retaining notation (8.16), we have $|K(t)| \leqslant \frac{1}{2}q$ $(t \geqslant 0)$ by (8.17). Let $x \geqslant 2$, $T := |\tau| + 2$. We have

$$\begin{aligned} |L^{(k)}(s,\chi)| = \Big|\sum_{n\geqslant 1} \frac{\chi(n)(\ln n)^k}{n^s}\Big| &\leqslant \sum_{n\leqslant x} \frac{(\ln n)^k}{n} + \Big|\int_x^\infty \frac{(\ln t)^k}{t^s}\,\mathrm{d}K(t)\Big| \\ &\ll_k (\ln x)^{k+1} + \frac{(\ln x)^k|K(x)|}{x} + T\int_x^\infty \frac{|K(t)|(\ln t)^k}{t^2}\,\mathrm{d}t \\ &\ll_k (\ln x)^k\big\{\ln x + qT/x\big\}. \end{aligned}$$

The bound (8.47) follows from this estimate by selecting $x = qT$. □

Of course, the implicit constant in (8.47) could easily be specified. For the sake of future reference, we note that the above proof provides, for example,

$$|L'(\sigma,\chi)| \leqslant (1+\ln q)^2 \qquad (\chi \neq \chi_0,\ \sigma \geqslant 1). \tag{8.48}$$

Theorem 8.19. *For any Dirichlet character χ, we have*

$$L(\sigma,\chi_0)^3|L(\sigma+i\tau,\chi)|^4\,|L(\sigma+2i\tau,\chi^2)| \geqslant 1 \qquad (\sigma>1,\ \tau\in\mathbb{R}). \tag{8.49}$$

Proof. We proceed analogously to the proof of Theorem 3.11, using the inequality

$$V(\vartheta) = 3 + 4\cos\vartheta + \cos 2\vartheta \geqslant 0 \qquad (\vartheta\in\mathbb{R}) \tag{8.50}$$

which served to estimate $|\zeta(1+i\tau)|$ from below. We define for each character χ a function $\psi(n)$ with values in $[0, 2\pi[$ such that

$$\chi(n) = \chi_0(n)\mathrm{e}^{i\psi(n)}.$$

For $\sigma > 1$, we can write by (8.17)

$$\log L(s,\chi) = \sum_p\sum_{\nu\geqslant 1} \frac{\chi(p)^\nu}{\nu p^{\nu s}} = \sum_{p\nmid q}\sum_{\nu\geqslant 1} \frac{\exp\{i\big(\psi(p^\nu) - \nu\tau\ln p\big)\}}{\nu p^{\nu\sigma}}$$

whence, taking real parts,

$$\begin{aligned} 3\ln L(\sigma,\chi_0) + 4\ln|L(\sigma+i\tau,\chi)| + \ln|L(\sigma+2i\tau,\chi^2)| \\ = \sum_{p\nmid q}\sum_{\nu\geqslant 1} \frac{V\big(\psi(p^\nu) - \nu\tau\ln p\big)}{\nu p^{\nu\sigma}} \geqslant 0. \end{aligned}$$

The lower bound (8.49) now follows by exponentiation. □

The inequality (8.50)—as actually all classical methods for estimating $|L(1+i\tau)|$ from below—ascribes a special role to *real characters*, that is to characters χ such that

$$\chi^2 = \chi_0. \tag{8.51}$$

In this case, the factor $|L(\sigma+2i\tau,\chi^2)|$ becomes infinite as $\sigma\to 1+$, $\tau\to 0$, which has the consequence of reducing the effect of (8.49). The explicit description (8.6) shows that χ is real if and only if we have

$$\lambda' = \begin{cases} 0 & (\nu\leqslant 2) \\ 0,\ 2^{\nu-3} & (\nu\geqslant 3) \end{cases} \quad \text{and} \quad \lambda_j = 0 \text{ or } \tfrac{1}{2}\varphi(p_j^{\nu_j}) \quad (1\leqslant j\leqslant k).$$

Since $\varepsilon(m)=0$ when $\nu\leqslant 1$, we readily obtain that the number $r(q)$ of real characters modulo q is given by the formula

$$r(q) = \begin{cases} 2^{\omega(q)-1} & \text{if } q\equiv \pm 2 \pmod 8, \\ 2^{\omega(q)} & \text{if } q\equiv \pm 1, \pm 3, 4 \pmod 8, \\ 2^{\omega(q)+1} & \text{if } q\equiv 0 \pmod 8. \end{cases} \tag{8.52}$$

In all cases, we have $r(q)\leqslant \tau(q)$.

Theorem 8.20. *If $\chi^2\neq\chi_0$, we have*

$$L(\sigma+i\tau,\chi)^{-1} \ll \mathcal{L}^7 \qquad (\sigma\geqslant 1). \tag{8.53}$$

Proof. We apply the method of § 4.2. Set $s=\sigma+i\tau$, $s_0=s+\eta$, where η is a parameter satisfying $0<\eta\leqslant 1/\mathcal{L}$. It results from (8.47) that

$$|L(s,\chi)-L(s_0,\chi)| \leqslant C_0\eta\mathcal{L}^2 \tag{8.54}$$

for a suitable absolute constant C_0. On the other hand, Theorem 8.18 implies that

$$L(\sigma_0+2i\tau,\chi^2) \ll \mathcal{L} \qquad (\sigma_0:=\sigma+\eta), \tag{8.55}$$

since $\chi^2\neq\chi_0$. Substituting in (8.49), we obtain

$$|L(s_0,\chi)|^4 \gg \prod_{p\mid q}(1-p^{-\sigma_0})^{-3}\zeta(\sigma_0)^{-3}\mathcal{L}^{-1} \gg \eta^3\mathcal{L}^{-1}. \tag{8.56}$$

Taking account of (8.54), we see there is a positive constant C_1 such that

$$|L(s,\chi)| \geqslant C_1\eta^{3/4}\mathcal{L}^{-1/4} - C_0\eta\mathcal{L}^2 = C_2\{C_1\cdot C_2^{-1/4} - C_0\}\mathcal{L}^{-7}$$

for the choice $\eta := C_2/\mathcal{L}^9$. When C_2 is sufficiently small, the expression in curly brackets is positive, and we get (8.53). □

Theorem 8.21. *If $\chi^2=\chi_0$, $\chi\neq\chi_0$, we have*

$$L(\sigma,\chi) \geqslant \frac{1}{9\sqrt{q}} \qquad (\sigma\geqslant 1). \tag{8.57}$$

Proof. In the sequel, certain numerical details are left to the reader. Note that we necessarily have $q\geqslant 3$. Without loss of generality, we may assume that $1\leqslant\sigma\leqslant 1+3/(25\sqrt{q})$. Indeed, in the opposite circumstance we have

$$L(\sigma,\chi) = \prod_p \frac{1}{1-\chi(p)/p^\sigma} \geqslant \prod_p (1-p^{-\sigma}) = \frac{1}{\zeta(\sigma)} \geqslant \frac{\sigma-1}{\sigma} > \frac{1}{9\sqrt{q}}.$$

We may also suppose that $L(1,\chi) \leqslant \pi/6$: in the opposite case we deduce from (8.48) that

$$L(\sigma,\chi) \geqslant \tfrac{1}{6}\pi - (\sigma-1)(1+\ln q)^2 \geqslant \tfrac{1}{6}\pi - \frac{(1+\ln q)^2}{8\sqrt{q}} \geqslant \tfrac{1}{6}\pi - \tfrac{9}{20} > \frac{1}{9\sqrt{q}}\cdot$$

Let us assume, to begin with, that χ is primitive and consider the multiplicative function $f_\sigma(n) := \sum_{d|n} \chi(d) d^{1-\sigma}$. If $\sigma > 1$, we can write for each prime power p^ν

$$f_\sigma(p^\nu) = \frac{1-(\chi(p)p^{1-\sigma})^{\nu+1}}{1-\chi(p)p^{1-\sigma}} \geqslant \frac{1+(-1)^\nu p^{(1-\sigma)(\nu+1)}}{1+p^{1-\sigma}}$$

where the inequality follows from the fact that $\chi(p) = \pm 1$ or 0. In particular,

$$f_\sigma(p^\nu) \geqslant 0, \qquad f_\sigma(p^{2\nu}) \geqslant p^{2\nu(1-\sigma)} \qquad (\nu \geqslant 1).$$

By passage to the limit, these inequalities are still satisfied when $\sigma = 1$, so that

$$f_\sigma(n) \geqslant 0, \qquad f_\sigma(n^2) \geqslant n^{2(1-\sigma)} \qquad (n \geqslant 1,\ \sigma \geqslant 1).$$

We next give ourselves a parameter $\alpha > 0$, and consider the expression

$$F(\alpha,\sigma) := \sum_{n\geqslant 1} f_\sigma(n) \mathrm{e}^{-\alpha n}.$$

On the one hand, we have

$$\begin{aligned} F(\alpha,\sigma) &\geqslant \sum_{n\geqslant 1} n^{2(1-\sigma)} \mathrm{e}^{-\alpha n^2} \geqslant \int_1^\infty t^{2(1-\sigma)} \mathrm{e}^{-\alpha t^2}\,\mathrm{d}t \\ &\geqslant \tfrac{1}{2}\alpha^{\sigma-3/2}\Gamma\big(\tfrac{3}{2}-\sigma\big) - \frac{1}{3-2\sigma} \geqslant \tfrac{1}{2}\sqrt{\pi}\alpha^{\sigma-3/2} - \frac{1}{3-2\sigma}\cdot \end{aligned} \tag{8.58}$$

On the other hand, from the definition of f_σ, we can also write

$$F(\sigma,\alpha) = \sum_{m,\,d\geqslant 1} \chi(d) d^{1-\sigma} \mathrm{e}^{-\alpha m d} = \sum_{d\geqslant 1} \chi(d) \frac{d^{1-\sigma}}{\mathrm{e}^{\alpha d}-1} = \frac{L(\sigma,\chi)}{\alpha} - \sum_{d\geqslant 1} \chi(d) h(d)$$

with

$$h(t) := t^{1-\sigma} g(\alpha t), \quad g(t) := \frac{1}{t} - \frac{1}{\mathrm{e}^t - 1}\cdot$$

It is easily checked that h is positive and decreasing on $[1,+\infty[$. Furthermore, h' is negative and increasing on the same interval. We can therefore

write

$$F(\alpha,\sigma)-\frac{L(\sigma,\chi)}{\alpha}=-\int_{1-}^{\infty}h(t)\,\mathrm{d}K(t)=\int_1^{\infty}K(t)h'(t)\,\mathrm{d}t$$
$$=\int_1^{q+1}K(t)\sum_{j\geqslant 0}h'(t+jq)\,\mathrm{d}t\leqslant\int_1^{q+1}|K(t)|\Big\{|h'(t)|+\frac{1}{q}\int_t^{\infty}|h'(v)|\,\mathrm{d}v\Big\}\,\mathrm{d}t$$
$$=\int_1^{q+1}|K(t)|\Big\{-h'(t)+\frac{h(t)}{q}\Big\}\,\mathrm{d}t\leqslant\frac{M_2(\chi)}{\sqrt{q}}\sqrt{\int_1^{q+1}\big\{h(t)-qh'(t)\big\}^2\,\mathrm{d}t}$$
$$\leqslant\tfrac{1}{3}\sqrt{\int_1^{q+1}\big\{h(t)-qh'(t)\big\}^2\,\mathrm{d}t},$$

by the inequality $M_2(\chi)\leqslant\frac{1}{3}\sqrt{q}$ which follows from (8.38) in view of our initial reductions. At this stage, we observe that, for $t>0$,

$$g'(t)=\frac{t^2\mathrm{e}^t-(\mathrm{e}^t-1)^2}{t^2(\mathrm{e}^t-1)^2}=\frac{\mathrm{e}^t\{t^2-(2\sinh t/2)^2\}}{t^2(\mathrm{e}^t-1)^2}\leqslant 0,$$
$$g''(t)=\frac{2}{t^3}-\frac{\mathrm{e}^t(1+\mathrm{e}^t)}{(\mathrm{e}^t-1)^3}=\frac{2\{(\sinh t/2)^3-(t/2)^3\cosh t/2\}}{(t\sinh t/2)^3}\geqslant 0,$$

where the last inequality can be established by noting that

$$\varphi(u):=(\sinh u)^3-u^3\cosh u$$

satisfies $\varphi(0)=0$ and

$$\varphi'(u)=3\cosh u\{(\sinh u)^2-u^2\}-u^3\sinh u\geqslant u^3(u\cosh u-\sinh u)\geqslant 0.$$

Furthermore

$$g'(t)=-\sum_{k\geqslant 1}\frac{(2k-1)B_{2k}}{(2k)!}t^{2k-2}\qquad(|t|<2\pi),$$

where the B_{2k} denote the Bernoulli numbers. In particular, g' is continuous at 0 and $g'(0)=-\frac{1}{2}B_2=-\frac{1}{12}$. It follows, for $1\leqslant t\leqslant q+1$, that

$$h(t)\leqslant g(0)=\tfrac{1}{2},\quad |h'(t)|\leqslant\frac{(\sigma-1)g(0)}{t}+\alpha|g'(0)|=\frac{\sigma-1}{2t}+\frac{\alpha}{12}$$

and thus

$$\int_1^{q+1}\big\{h(t)-qh'(t)\big\}^2\,\mathrm{d}t\leqslant\int_1^{q+1}\Big\{\frac{q^2(\sigma-1)^2}{2t^2}+\frac{(\alpha q+6)^2}{72}\Big\}\,\mathrm{d}t$$
$$\leqslant\tfrac{1}{2}(\sigma-1)^2q^2+\tfrac{1}{72}(\alpha q+6)^2q$$
$$\leqslant\tfrac{1}{128}q+\tfrac{1}{72}q(\alpha q+6)^2.$$

Set $\alpha=\lambda/q$ and substitute in the initial inequality. Bounding $\frac{1}{128}$ by $\frac{1}{72}$, we obtain

$$L(\sigma,\chi)\geqslant\alpha F(\alpha,\sigma)-\frac{\lambda\sqrt{1+(\lambda+6)^2}}{18\sqrt{2q}}.$$

Inserting (8.58), it follows that

$$L(\sigma,\chi) \geqslant \tfrac{1}{2}\sqrt{\frac{\pi\lambda}{q}}\Big(\frac{\lambda}{q}\Big)^{\sigma-1} - \frac{\lambda}{q(3-2\sigma)} - \frac{\lambda\sqrt{1+(\lambda+6)^2}}{18\sqrt{2q}}.$$

Select $\lambda = \frac{1}{5}$ and observe that $(8q)^{1-\sigma} \geqslant (8q)^{-3/(25\sqrt{q})} > \frac{9}{10}$. This yields

$$L(\sigma,\chi)\sqrt{q} \geqslant \frac{9\sqrt{5\pi}}{100} - \frac{1}{5\sqrt{3}-6/5} - \frac{\sqrt{493}}{450} > \frac{1}{6}.$$

If χ is induced by a primitive character χ_1 to the modulus q_1, we have

$$L(\sigma,\chi) \geqslant L(\sigma,\chi_1) \prod_{p\mid q,\, p\nmid q_1} \Big(1 - \frac{\chi_1(p)}{p}\Big) \geqslant \frac{\sqrt{q_1}}{\sqrt{2q}} L(\sigma,\chi_1) \geqslant \frac{1}{6\sqrt{2q}} > \frac{1}{9\sqrt{q}}. \quad \square$$

Theorem 8.22. *There exists an absolute constant $c_0 > 0$ such that, for $\chi^2 = \chi_0$, $\chi \neq \chi_0$, $\sigma \geqslant 1$,*

$$L(s,\chi)^{-1} \ll \begin{cases} \mathcal{L}^6(\mathcal{L}+1/|\tau|) & \text{if } |\tau| > c_0 q^{-1/2}(\ln 2q)^{-2}, \\ \sqrt{q} & \text{if } |\tau| \leqslant c_0 q^{-1/2}(\ln 2q)^{-2}. \end{cases} \tag{8.59}$$

Proof. The second of these estimates follows immediately from Theorems 8.18 and 8.21. Indeed, a Taylor expansion to order 1 gives for all τ, $|\tau| \leqslant 1$,

$$|L(s,\chi)| \geqslant L(\sigma,\chi) + O\big(\tau(\ln 2q)^2\big),$$

from which, taking account of (8.57), the required estimate follows.

We can therefore assume that $|\tau| \gg q^{-1/2}(\ln 2q)^{-2}$. We then employ the method of Theorem 8.20. Let η be a positive parameter. Set $\sigma_0 = \sigma + \eta$, $s_0 = \sigma_0 + i\tau$, $s_1 = \sigma_0 + 2i\tau$. We have

$$L(s_1,\chi_0) = \prod_{p\mid q}(1-p^{-s_1})\zeta(s_1) \ll \prod_{p\mid q}(1-p^{-\sigma_0})^{-1}(\mathcal{L}+1/|\tau|).$$

It then follows from (8.49) that

$$L(s_0,\chi)^4 \gg \prod_{p\mid q}(1-p^{-\sigma_0})^{-2}\zeta(\sigma_0)^{-3}(\mathcal{L}+1/|\tau|)^{-1}.$$

By appealing to (8.47), we now deduce the existence of two positive absolute constants C_3 and C_4 such that

$$|L(s,\chi)| \geqslant C_3\eta^{3/4}(\mathcal{L}+1/|\tau|)^{-1/4} - C_4\eta\mathcal{L}^2.$$

Choosing $\eta = C_5(\mathcal{L}+1/|\tau|)^{-1}\mathcal{L}^{-8}$ for some suitable C_5, we obtain the required lower bound. $\square$

Proof of Theorem 8.16. We apply Theorem 7.13 to the series (8.25). If $G(s)$ is defined by (8.42), we obtain

$$\psi(x;a,q) = \frac{x}{\varphi(q)} + O\Big(x \min_{T\geqslant 64}\Big\{\frac{1}{T} + \eta\Big(\frac{1}{\ln x}, T\Big)\Big\}\Big) \tag{8.60}$$

with

$$\eta(\sigma,T) := \int_{-T}^{T} \big|G(2\sigma+i\tau) - G(\sigma+i\tau)\big|\,\mathrm{d}\tau \quad (\sigma>0). \tag{8.61}$$

We estimate this last quantity by bounding $|G'(s)|$. We have

$$G'(s) \ll \frac{1}{\varphi(q)} \sum_{\chi} |Z'(s,\chi)| \tag{8.62}$$

with

$$Z(s,\chi) := \begin{cases} \dfrac{-\zeta'(s+1)}{(s+1)\zeta(s+1)} - \dfrac{1}{s} + \dfrac{1}{(s+1)} \displaystyle\sum_{p\mid q} \frac{\ln p}{p^s} & (\chi=\chi_0), \\ \dfrac{-L'(s+1,\chi)}{(s+1)L(s+1,\chi)} & (\chi\neq\chi_0). \end{cases}$$

From the estimates in § 4.2 on $\zeta(s)$, and from Theorems 8.18, 8.20, 8.22 concerning $L(s,\chi)$, $\chi\neq\chi_0$, we obtain, after a routine calculation, for $\sigma\geqslant 0$

$$Z'(s,\chi) \ll \begin{cases} \mathcal{L}^{18}/|s+1| & \text{if } \chi^2\neq\chi_0 \text{ or } |\tau|>(\ln 2q)^{-1}, \\ \mathcal{L}^{16}/|\tau|^2 & \text{if } \chi^2=\chi_0 \text{ and } c_0(\ln 2q)^{-2}q^{-1/2} < |\tau| \leqslant (\ln 2q)^{-1}, \\ \mathcal{L}^4 q & \text{if } \chi^2=\chi_0 \text{ and } |\tau| < c_0(\ln 2q)^{-2}q^{-1/2}. \end{cases}$$

We substitute first in (8.62) and then in (8.61), taking (8.52) into account. We deduce

$$\eta(\sigma,T) \ll \sigma\mathcal{L}(q,T)^{19}.$$

Selecting $T = (\ln x)/(\ln_2 x)^{19/2}$, we obtain from (8.60) the stated formula for $\psi(x;a,q)$. □

8.4. The functional equation for the functions $L(s,\chi)$

Theorem 8.23. *Let χ be a primitive character to the modulus q. Set*

$$\alpha = \alpha(\chi) := \tfrac{1}{2}\{1-\chi(-1)\}.$$

The function $L(s,\chi)$ can be meromorphically continued to the entire complex plane. Its only singularity is a simple pole at $s=1$, with residue 1, when $\chi=\mathbf{1}$. Furthermore, $L(s,\chi)$ satisfies the functional equation

$$\xi(s,\chi) = E(\chi)\xi(1-s,\overline{\chi}) \qquad (s\in\mathbb{C})$$

with

$$\xi(s,\chi) := \Big(\frac{\pi}{q}\Big)^{-\frac{1}{2}(s+\alpha)}\Gamma\Big(\frac{s+\alpha}{2}\Big)L(s,\chi), \qquad E(\chi) := \frac{G(1,\chi)}{i^{\alpha}\sqrt{q}}.$$

Remark. If χ is not primitive, we have

$$L(s,\chi) = L(s,\chi_1) \prod_{p\mid q,\, p\nmid q_1} (1-\chi_1(p)/p^s) \qquad (\sigma>1) \tag{8.63}$$

where χ_1 is the primitive character inducing χ. An obvious extension of the continuation properties of $L(s,\chi)$ may be deduced from this formula.

Proof. Let us first show that $L(s,\chi)$ can be continued to a meromorphic function on the whole complex plane. The method used for the function $\zeta(s)$ applies with no extra difficulty. Thus, we confine ourselves to sketching the essential steps.

Defining

$$W(t,\chi) := \sum_{n\geqslant 1} \chi(n)\mathrm{e}^{-nt} = \bigl(1-\mathrm{e}^{-qt}\bigr)^{-1} \sum_{1\leqslant a<q} \chi(a)\mathrm{e}^{-at} \qquad (t>0),$$

we have

$$\Gamma(s)L(s,\chi) = \int_0^\infty t^{s-1}W(t,\chi)\,\mathrm{d}t \qquad (\sigma>1).$$

We obtain an analytic continuation of this integral by replacing the half-line of integration by the *Hankel contour* $\mathcal{C}_\varrho$ (see p. 233) where ϱ is a real parameter, $0<\varrho<2\pi/q$.

Since the function $z\mapsto z^{s-1}(1-\mathrm{e}^{-qz})^{-1}$ is holomorphic in the horizontal strip $|\Im m\, z|<2\pi$ punctured by the half-line $[0,+\infty[$, the integral

$$I(s,\chi) := \int_{\mathcal{C}_\varrho} z^{s-1}W(z,\chi)\,\mathrm{d}z,$$

is independent of ϱ in $]0,2\pi/q[$. It is absolutely convergent for each $s\in\mathbb{C}$ and uniformly convergent on any compact subset. It is therefore an entire function of s. We have

$$I(s,\chi) = \oint_{|z|=\varrho} W(z,\chi)z^{s-1}\,\mathrm{d}z + (\mathrm{e}^{2\pi i s}-1)\int_\varrho^\infty t^{s-1}W(t,\chi)\,\mathrm{d}t.$$

Taking into account the upper bound $|z^{s-1}(\mathrm{e}^z-1)^{-1}| \ll_s \varrho^{\sigma-2}$ $(|z|=\varrho\leqslant\pi)$, we obtain, by letting ϱ tend to 0,

$$I(s,\chi) = (\mathrm{e}^{2\pi i s}-1)\Gamma(s)L(s,\chi) \qquad (\sigma>1).$$

Using the reflection formula, we arrive at the expression

$$L(s,\chi) = \frac{\mathrm{e}^{-i\pi s}}{2\pi i}\Gamma(1-s)I(s,\chi), \tag{8.64}$$

which furnishes the analytic continuation to the entire complex plane. The only possible singularities of $L(s,\chi)$ are therefore at the positive integers. Since $L(s,\chi)$ converges for $\sigma>0$ if $\chi\neq\chi_0$, we see that $L(s,\chi)$ is in this case continuable to an entire function. Formula (8.24) implies that the continuation of $L(s,\chi_0)$ has a unique singularity, at $s=1$, which is a simple

pole with residue $\varphi(q)/q$. This residue hence equals 1 if χ is principal and primitive.

Let us now establish the functional equation. Let $\mathcal{H}_k := \mathcal{C}_{\varrho_k}$ be the Hankel contour with parameter $\varrho_k := (2k+1)\pi/q$, $k \geqslant 1$. Then, for $1 \leqslant a \leqslant q$, $z = \varrho_k e^{i\vartheta}$, the quantity

$$\frac{|e^{-az}|}{|1-e^{-qz}|} = \frac{e^{(2k+1)(1-a/q)\pi\cos\vartheta}}{|e^{(2k+1)\pi\cos\vartheta + i(2k+1)\pi\sin\vartheta} - 1|}$$

remains bounded as ϑ runs over $[0, 2\pi]$. Therefore

$$\Big| \int_{\mathcal{H}_k} W(z,s) z^{s-1}\, dz \Big| \ll_s k^{\sigma}. \tag{8.65}$$

Let ϱ be such that $0 < \varrho < 2\pi/q$. The contour $\mathcal{C}_\varrho - \mathcal{H}_k$ encloses, in the reverse sense, the poles $z_n = 2n\pi i/q$ ($n = \pm 1, \pm 2, \dots, \pm k$). The residue theorem then implies

$$\begin{aligned} I(s,\chi) &= \int_{\mathcal{C}_\varrho} z^{s-1} W(z,\chi)\, dz \\ &= \int_{\mathcal{H}_k} z^{s-1} W(z,\chi)\, dz - \frac{2\pi i}{q} \sum_{1\leqslant|n|\leqslant k} (z_n)^{s-1} \sum_{1\leqslant a\leqslant q} \chi(a) e^{-az_n} \\ &= \int_{\mathcal{H}_k} z^{s-1} W(z,\chi)\, dz + \Big(\frac{2\pi i}{q}\Big)^s \{e^{i\pi s} - \chi(-1)\} \sum_{1\leqslant n\leqslant k} G(n,\chi) n^{s-1}. \end{aligned}$$

We have $\chi(-1) = 1 - 2\alpha = e^{-i\pi\alpha}$. By (8.65), for each s in the half-plane $\sigma < 0$, we obtain, letting k tend to infinity,

$$I(s,\chi) = \big(e^{i\pi s} - e^{-i\pi\alpha}\big)\Big(\frac{2\pi i}{q}\Big)^s \sum_{n\geqslant 1} G(n,\chi) n^{s-1}.$$

From Theorem 8.8, we have $G(n,\chi) = \overline{\chi}(n) G(1,\chi)$ when χ is primitive. It follows that

$$I(s,\chi) = \big(e^{i\pi s} - e^{-i\pi\alpha}\big)\Big(\frac{2\pi i}{q}\Big)^s G(1,\chi) L(1-s,\overline{\chi}).$$

Substituting in (8.64), we obtain the asymmetric form of the functional equation

$$L(s,\chi) = E(\chi)\Big(\frac{2\pi}{q}\Big)^s \frac{\sqrt{q}}{\pi} \sin\big\{\tfrac{1}{2}\pi(s+\alpha)\big\}\Gamma(1-s) L(1-s,\overline{\chi}), \tag{8.66}$$

which therefore holds for all s by analytic continuation. We deduce the functional equation in the form of the statement by the reflection and duplication formulae for the Γ function. □

Remark. The trivial zeros of $L(s,\chi)$ are located at the points $-2k$ ($k \geqslant 0$) when $\alpha = 0$, i.e., $\chi(-1) = 1$, and at the points $-2k-1$ ($k \geqslant 0$) if $\alpha = 1$, i.e., $\chi(-1) = -1$.

8.5. Hadamard product formula and zero-free regions

Let us retain the notation $\alpha = \alpha(\chi)$ and $E(\chi)$ from Theorem 8.23. We shall now expand

$$\xi(s,\chi) := \Big(\frac{q}{\pi}\Big)^{(s+\alpha)/2}\Gamma\big(\tfrac{1}{2}(s+\alpha)\big)L(s,\chi)$$

as a Hadamard product using a technique close to that employed in Chap. II.3 for the zeta function.

Theorem 8.24. *Let χ be a non-principal primitive character to the modulus q. Then $L(s,\chi)$ has infinitely many zeros ϱ in the critical strip and we have*

$$\xi(s,\chi) = \mathrm{e}^{A+Bs}\prod_{\varrho}(1-s/\varrho)\mathrm{e}^{s/\varrho} \qquad (s\in\mathbb{C}), \tag{8.67}$$

the zeros ϱ being counted with multiplicity. The infinite product is absolutely convergent and we have

$$A = A(\chi) := \log\Big\{\Gamma\big(\tfrac{1}{2}(1+\alpha)\big)(q/\pi)^{(1+\alpha)/2}E(\chi)L(1,\overline{\chi})\Big\}$$

$$B = B(\chi) := \frac{\xi'}{\xi}(0,\chi) = -\frac{\xi'}{\xi}(1,\overline{\chi}).$$

Proof. The proof is similar to that carried out for the function $\zeta(s)$; we restrict to an outline. We first have, with notations (8.16) and (8.17),

$$|L(s,\chi)| = \Big|s\int_1^\infty \frac{K(x)}{x^{s+1}}\,\mathrm{d}x\Big| \leqslant |s|H(\chi)\int_1^\infty \frac{\mathrm{d}x}{x^{3/2}} \leqslant 2H(\chi)|s| \qquad (\sigma\geqslant\tfrac{1}{2}).$$

This implies that $\xi(s,\chi)$ is an entire function of order 1. Let us set

$$N(T,\chi) := |\{\varrho : L(\varrho,\chi)=0,\ |\Im m\,\varrho|\leqslant T\}|.$$

It should be noted that the zeros are not necessarily placed symmetrically with respect to the origin. It may be shown by complex integration that

$$\tfrac{1}{2}N(T,\chi) = \frac{T}{2\pi}\ln\frac{qT}{2\pi} - \frac{T}{2\pi} + O(\ln qT) \quad (T\geqslant 2). \tag{8.68}$$

The details are essentially identical with those in the case of $\zeta(s)$, but it is more convenient to consider the rectangle $\mathcal{R}$ with vertices $\frac{5}{2}\pm iT$, $-\frac{3}{2}\pm iT$, so as to avoid a possible trivial zero at $s=-1$. Thus, $\mathcal{R}$ always contains exactly one trivial zero of $L(s,\chi)$, either at 0, or at -1.

The estimate for $N(T,\chi)$ implies convergence of the infinite product, say $P(s,\chi)$, over the zeros and we deduce, as in the case of $\zeta(s)$, that $\log\{\xi(s,\chi)/P(s,\chi)\}$ is an affine function of s. □

The zero-free regions are essentially identical to those for the zeta function, with the possible exception of a real zero. The result is essentially due

to Landau (1918), its application to primitive characters having initially been derived by Page (1935).

Theorem 8.25 (Landau–Page). *There is an absolute constant $c > 0$ such that the region*

$$\mathcal{D}_q := \{s \in \mathbb{C} : \sigma > 1 - c/\ln qT\} \qquad (T := |\tau| + 2) \tag{8.69}$$

contains at most one zero of $\prod_{\chi(\mathrm{mod}\, q)} L(s,\chi)$ *and at most one zero of*

$$\prod_{q_1 \leqslant q} \prod_{\substack{\chi(\mathrm{mod}\, q_1) \\ \chi \text{ primitive}}} L(s,\chi).$$

In both cases, the possible exceptional zero is real and simple, and corresponds to a real character.

It is conjectured that the possible exceptional zero (commonly called "Siegel's zero") actually does not exist. The strongest conjecture in this direction is that all the zeros of L-functions are situated on the critical line $\sigma = \frac{1}{2}$—this is the *generalized Riemann hypothesis.*

The proof of Theorem 8.25 relies essentially on the following lemma, which is itself a consequence of the product formula and the functional equation for the functions $L(s,\chi)$. We retain the convention of denoting a generic zero of $L(s,\chi)$ by $\varrho = \beta + i\gamma$.

Lemma 8.26. *There is an absolute constant c_0 such that, for every non-principal character χ to the modulus q, we have*

$$-\Re e\, \frac{L'}{L}(s,\chi) \leqslant c_0 \ln qT - \sum_{\varrho} \Re e\, \frac{1}{s-\varrho} \qquad (\sigma > 1,\ T = |\tau| + 2) \tag{8.70}$$

where the series has non-negative terms.

Proof. We may assume without loss of generality that χ is primitive since, if χ_1 is the primitive character inducing χ, relation (8.63) implies that

$$\Big|\frac{L'}{L}(s,\chi) - \frac{L'}{L}(s,\chi_1)\Big| \leqslant \sum_{p|q} \frac{\ln p}{p^\sigma - 1} \leqslant \ln q \qquad (\sigma > 1). \tag{8.71}$$

Differentiating the product formula (8.67), we get

$$\begin{aligned} &-\Re e\, \frac{L'}{L}(s,\chi) \\ &\quad = \tfrac{1}{2} \ln \frac{q}{\pi} + \Re e\, \frac{\Gamma'}{2\Gamma}(\tfrac{1}{2}s + \tfrac{1}{2}\alpha) - \Re e\, B(\chi) - \Re e \sum_{\varrho} \Big(\frac{1}{s-\varrho} + \frac{1}{\varrho}\Big). \end{aligned} \tag{8.72}$$

Since χ is assumed primitive, the functional equation

$$\xi(s,\chi) = E(\chi)\xi(1-s,\overline{\chi})$$

yields, by logarithmic differentiation,

$$\frac{\xi'}{\xi}(s,\chi) = -\frac{\xi'}{\xi}(1-s,\overline{\chi}).$$

For $s=0$, we obtain from the product formula

$$B(\chi) = -B(\overline{\chi}) - \sum_{\varrho} \frac{1}{\overline{\varrho}} + \frac{1}{1-\overline{\varrho}}$$

because $\overline{\varrho}$ runs through all the zeros of $L(s,\overline{\chi})$ as ϱ ranges through those of $L(s,\chi)$. However, we have $B(\overline{\chi}) = \overline{B(\chi)}$, as can easily be deduced from the formula

$$B(\chi) = -\frac{\xi'}{\xi}(1,\overline{\chi}) = -\tfrac{1}{2}\ln(q/\pi) - \frac{\Gamma'}{2\Gamma}(\tfrac{1}{2}+\tfrac{1}{2}\alpha) - \frac{L'}{L}(1,\overline{\chi}).$$

It follows that

$$2\Re e\, B(\chi) = -\Re e \sum_{\varrho} \Big(\frac{1}{\overline{\varrho}} + \frac{1}{1-\overline{\varrho}}\Big).$$

In this series with non-negative, and hence permutable, terms, one may replace $1-\overline{\varrho}$ by ϱ. We thus obtain

$$\Re e\, B(\chi) = -\tfrac{1}{2}\sum_{\varrho} \Re e\, \Big(\frac{1}{\varrho} + \frac{1}{\overline{\varrho}}\Big) = -\sum_{\varrho} \Re e\, \frac{1}{\varrho}.$$

Substituting back into (8.72) yields

$$-\Re e\, \frac{L'}{L}(s,\chi) = \tfrac{1}{2}\ln\frac{q}{\pi} + \tfrac{1}{2}\Re e\, \frac{\Gamma'}{\Gamma}(\tfrac{1}{2}s+\tfrac{1}{2}\alpha) - \Re e \sum_{\varrho} \frac{1}{s-\varrho}.$$

The upper bound (8.70) follows from this in view of the estimate

$$\frac{\Gamma'(s)}{\Gamma(s)} \ll \ln T,$$

resulting from the complex Stirling formula. □

Proof of Theorem 8.25. We first observe that an argument analogous to that leading to (8.49) furnishes

$$-3\frac{L'}{L}(\sigma,\chi_0) - 4\Re e\, \frac{L'}{L}(s,\chi) - \Re e\, \frac{L'}{L}(s+i\tau,\chi^2) \geqslant 0. \qquad (\sigma > 1). \tag{8.73}$$

We now bound from above the three terms on the left-hand side. We first have

$$-3\frac{L'}{L}(\sigma,\chi_0) \leqslant -3\frac{\zeta'}{\zeta}(\sigma) \leqslant \frac{3}{\sigma-1} + c_1. \tag{8.74}$$

Here and in the sequel of this proof, we let c_j ($j \geqslant 0$) denote positive absolute constants.

Next, if $\varrho = \beta + i\gamma$ is a zero of $L(s,\chi)$ and if $\chi^2 \neq \chi_0$, we deduce from (8.70) that

$$\begin{aligned} -4\Re e\,\frac{L'}{L}(\sigma + i\gamma, \chi) &\leqslant 4c_0 \ln qT - \frac{4}{\sigma - \beta} \\ -\Re e\,\frac{L'}{L}(\sigma + 2i\gamma, \chi^2) &\leqslant 2c_0 \ln qT. \end{aligned} \qquad (\sigma > 1) \tag{8.75}$$

We may thus write

$$\frac{4}{\sigma - \beta} \leqslant \frac{3}{\sigma - 1} + c_2 \ln qT.$$

Choosing $\sigma = 1 + c_3/\ln qT$ with suitable c_3, we obtain

$$\beta \leqslant 1 - c_4/\ln qT.$$

It remains to consider the case when $\chi^2 = \chi_0$, that is when χ is real. To this end, we begin by sharpening the second inequality (8.75). By logarithmic differentiation of the product formula for $\zeta(s)$, we obtain

$$-\frac{\zeta'}{\zeta}(s) = \frac{1}{s-1} - b + \frac{\Gamma'}{2\Gamma}(\tfrac{1}{2}s + 1) - \sum_{\varrho} \Big(\frac{1}{\varrho} + \frac{1}{s - \varrho}\Big)$$

where the summation, now over the non-trivial zeros of $\zeta(s)$, has non-negative terms. We can therefore write

$$-\Re e\,\frac{\zeta'}{\zeta}(s) \leqslant \Re e\,\frac{1}{s-1} + c_5 \ln T.$$

Whence

$$-\Re e\,\frac{L'}{L}(s + i\tau, \chi^2) = -\Re e\,\frac{\zeta'}{\zeta}(s) + O(\ln q) \leqslant \Re e\,\frac{1}{s-1} + c_6 \ln qT.$$

Substituting back into (8.73) and using the first inequality (8.75), we obtain

$$\frac{4}{\sigma - \beta} \leqslant \frac{3}{\sigma - 1} + \Re e\,\frac{1}{\sigma - 1 + 2i\gamma} + c_7 \ln qT.$$

Let us select $\sigma = 1 + \delta/\ln qT$ and assume that $|\gamma| \geqslant \delta/\ln qT$. It follows, upon setting $\mathcal{L} := \ln qT$, that

$$\frac{4}{\sigma - \beta} \leqslant \frac{3\mathcal{L}}{\delta} + \frac{\mathcal{L}}{5\delta} + c_7\mathcal{L}$$

and thus

$$\beta \leqslant 1 - \frac{(4 - 5c_7\delta)\delta}{(16 + 5c_7\delta)\mathcal{L}}.$$

This implies, with a suitable choice of δ, that $\varrho \notin \mathcal{D}_q$ if the constant c_0 is sufficiently small.

Up to this point we have shown that, for each $\delta > 0$, there exists a constant $c(\delta) > 0$ such that, if we select $c = c(\delta)$ in the definition $\mathcal{D}_q$, then $\mathcal{D}_q$ contains no generic zero $\varrho = \beta + i\gamma$ of $L(s, \chi)$ provided one of the following conditions holds:

(i) $\chi^2 \neq \chi_0$.

(ii) $\chi^2 = \chi_0$ and $|\gamma| \geqslant \delta / \ln q$.

We will now embark upon showing that, when χ is real and δ is sufficiently small, $L(s, \chi)$ has at most a unique zero in the rectangle

$$1 - \frac{\delta}{\ln q} \leqslant \beta \leqslant 1, \qquad |\gamma| \leqslant \frac{\delta}{2 \ln q},$$

and that this zero is then necessarily real. Indeed, if $L(\varrho, \chi) = 0$ with $\gamma \neq 0$, we have, by (8.70),

$$-\Re e \frac{L'}{L}(\sigma, \chi) \leqslant c_0 \ln q - 2\Re e \frac{1}{\sigma - \varrho}$$

where the factor 2 arises from the fact that, since χ is real, the zeros are located symmetrically with respect to the real axis. However, since $\zeta(s)L(s, \chi) = \sum_{n \geqslant 1} f(n)/n^s$ with $f = \mathbf{1} * \chi \geqslant 0$,[6] we also have

$$-\Re e \frac{L'}{L}(\sigma, \chi) \geqslant \frac{\zeta'}{\zeta}(\sigma) \geqslant -\frac{1}{\sigma - 1} - c_1.$$

Therefore

$$\frac{2(\sigma - \beta)}{(\sigma - \beta)^2 + \gamma^2} \leqslant \frac{1}{\sigma - 1} + c_8 \ln q.$$

For $\sigma = 1 + 2\delta / \ln q$, so that $|\gamma| \leqslant \frac{1}{4}(\sigma - 1) \leqslant \frac{1}{4}(\sigma - \beta)$, we obtain

$$\frac{32}{17(\sigma - \beta)} \leqslant \left(\frac{1}{2\delta} + c_9\right) \ln q \leqslant \frac{9 \ln q}{17\delta}$$

for small enough δ. It follows that $\sigma - \beta \geqslant \frac{32}{9}\delta / \ln q$ and, in a contradiction, $1 - \beta \geqslant \frac{14}{9}\delta / \ln q$.

The same argument can clearly be repeated if $L(s, \chi)$ has two real zeros or a double zero on the segment $[1 - \delta / \ln q, 1]$.

In view of the properties already established, we can easily deduce from the next result, due to Landau (1918), the two assertions in the statement.

Lemma 8.27 (Landau). *Let χ_1 and χ_2 be two distinct primitive characters whose associated L-functions have real zeros β_1 and β_2. Then*

$$1 - \min(\beta_1, \beta_2) \geqslant c_{10} / \ln q_1 q_2.$$

Indeed, let us provisionally assume this lemma. Since $L(s, \chi)$ and $L(s, \chi_1)$ have the same zeros in the half-plane $\sigma > 0$ if χ_1 is the primitive character associated to χ, we see that $\prod_{\chi (\mathrm{mod}\, q)} L(s, \chi)$ has at most a

[6] See the proof of Theorem 8.21.

single real zero in $\mathcal{D}_q$ for a suitable constant $c > 0$. The second assertion (initially due to Page) is therefore immediate. □

Proof of Lemma 8.27. The character $\chi_1\chi_2$ is non-principal modulo q_1q_2. Indeed, otherwise we would have $\chi_1(n) = \chi_2(n)$ for every integer n prime to q_1q_2, and so these primitive characters would induce the same character modulo q_1q_2: this is impossible from the description in § 8.1.

Next, we deduce from Lemma 8.26 that, for $\sigma > 1$, we have

$$-\frac{L'}{L}(\sigma, \chi_1\chi_2) \leqslant c_0 \ln q_1q_2$$

and also

$$-\frac{L'}{L}(\sigma, \chi_j) \leqslant c_0 \ln q_j - \frac{1}{\sigma - \beta_j} \qquad (j = 1, 2).$$

However

$$-\frac{\zeta'}{\zeta}(\sigma) - \frac{L'}{L}(\sigma, \chi_1) - \frac{L'}{L}(\sigma, \chi_2) - \frac{L'}{L}(\sigma, \chi_1\chi_2)$$
$$= \sum_{n \geqslant 1} \frac{\Lambda(n)}{n^\sigma}\{1 + \chi_1(n)\}\{1 + \chi_2(n)\} \geqslant 0.$$

Therefore

$$\frac{1}{\sigma - \beta_1} + \frac{1}{\sigma - \beta_2} \leqslant \frac{1}{\sigma - 1} + c_{11} \ln q_1q_2.$$

Selecting $\sigma = 1 + \delta/\ln q_1q_2$ with sufficiently small δ, we reach the required conclusion. □

Remark. Landau's lemma immediately implies that the sequence of the q_j for which $\prod_{\chi \pmod{q_j}} L(s, \chi)$ has a real zero $\beta_j \geqslant 1 - c_{12}/\ln q_j$ satisfies $q_{j+1} > q_j^2$ provided the constant c_{12} is chosen sufficiently small. Indeed

$$c_{12}/\ln q_j \geqslant 1 - \min(\beta_j, \beta_{j+1}) \geqslant c_{11}/\ln q_jq_{j+1}$$

whence $q_jq_{j+1} \geqslant q_j^{c_{11}/c_{12}} \geqslant q_j^3$ if $c_{12} \leqslant \frac{1}{3}c_{11}$.

8.6. Explicit formulae for $\psi(x; \chi)$

Let us set

$$\psi(x; \chi) := \sum_{n \leqslant x} \Lambda(n)\chi(n), \qquad \psi^*(x; \chi) := \tfrac{1}{2}\{\psi(x-; \chi) + \psi(x; \chi)\},$$

so that

$$\psi(x; a, q) = \frac{1}{\varphi(q)} \sum_{\chi \pmod{q}} \overline{\chi(a)}\psi(x; \chi).$$

By the Landau–Page theorem (Theorem 8.25), the region $\mathcal{D}_q$ defined in (8.69) contains, provided the constant c is suitably chosen, at most one

zero of

$$\prod_{\chi(\mathrm{mod}\, q)} L(s,\chi)$$

and, if it exists, this zero is real, simple, and corresponds to a real character. Denote this character by χ_1, and put

$$\vartheta(\chi) := \begin{cases} 1 & \text{if } \chi = \chi_1, \\ 0 & \text{if } \chi \neq \chi_1. \end{cases}$$

In addition, designate by β_1 the putative exceptional zero of $L(s,\chi_1)$. By definition of $\mathcal{D}_q$, we have $1-\beta_1 < c/\ln 2q$. Throughout the remainder of this chapter, we adopt the convention that quantities where χ_1 or β_1 occur explicitly should be considered to vanish if there does not exist an exceptional zero in the sense described above.

Finally, we set

$$b(\chi) := \mathrm{Res}\left(\frac{L'(s,\chi)}{sL(s,\chi)}; 0\right). \tag{8.76}$$

Theorem 8.28. *Let χ be a non-principal Dirichlet character to the modulus q. For $x \geqslant 2$, we have*

$$\psi^*(x,\chi) = -\sum_{\varrho}{}'' \frac{x^\varrho}{\varrho} - (1-\alpha)\ln x - b(\chi) + \sum_{m\geqslant 1} \frac{x^{-2m+\alpha}}{2m-\alpha} \tag{8.77}$$

where the double apostrophe indicates that the sum over the non-trivial zeros ϱ of $L(s,\chi)$ is understood as the limit as $T\to\infty$ of $\sum_{|\Im m\,\varrho|\leqslant T}$.

Furthermore, for $q\geqslant 2$, $2\leqslant T\leqslant x$,

$$\psi(x;\chi) = -\vartheta(\chi)\frac{x^{\beta_1}}{\beta_1} - \sum_{|\gamma|\leqslant T}{}^* \frac{x^\varrho}{\varrho} + R_q(x,T) \tag{8.78}$$

with

$$R_q(x,T) \ll \frac{x(\ln qx)^2}{T} + x^{1/4},$$

and where the asterisk indicates that, if $\vartheta(\chi)=1$, the zeros β_1 and $1-\beta_1$ are omitted from the sum.

Proof. Let us initially consider the case when χ is primitive. Using the first effective Perron formula (Theorem 2.3) and arguing as in Lemma 4.7, we obtain

$$\psi(x;\chi) = \frac{-1}{2\pi i}\int_{\kappa-iT}^{\kappa+iT} \frac{L'}{L}(s,\chi)\frac{x^s}{s}\,\mathrm{d}s + O\left(\frac{x(\ln Tx)^2}{T} + \frac{x\ln x}{x+T\langle\langle x\rangle\rangle}\right) \tag{8.79}$$

with the notation

$$\langle\langle x\rangle\rangle := \min_{\substack{\nu\geqslant 1,\, p\in\mathbb{P}\\ p^\nu\neq x}} |x-p^\nu|.$$

We shall obtain the stated formula by moving the segment of integration and applying the residue theorem. The poles of the integrand are the zeros of $L(s,\chi)$ and the point $s=0$, which is hence a double pole if $\alpha(\chi)=0$, i.e., $\chi(-1)=1$. By the pigeonhole principle and in view of the asymptotic formula (8.68) for $N(T,\chi)$, in any interval of length $\gg \ln q$, we may select T such that $\min||\gamma|-T| \gg 1/\ln qT$.

By logarithmic differentiation of the product formula (8.67), we also have

$$-\frac{L'}{L}(s,\chi) = \tfrac{1}{2}\ln\frac{q}{\pi} + \frac{\Gamma'}{2\Gamma}\Big(\frac{s+\alpha}{2}\Big) - B(\chi) - \sum_{\varrho}\frac{s}{\varrho(s-\varrho)}, \tag{8.80}$$

so that logarithmically differentiating formula (0.4) for the Weierstrass product for Γ yields

$$\frac{\Gamma'}{2\Gamma}\Big(\frac{\alpha+s}{2}\Big) = -\frac{\gamma}{2} - \frac{1}{s+\alpha} - \sum_{n\geqslant 1}\Big(\frac{1}{2n+s+\alpha} - \frac{1}{2n}\Big).$$

Furthermore, by logarithmic differentiation of the functional equation in its asymmetric form (8.66), we see that

$$\begin{aligned}\frac{L'}{L}(s,\chi) &= \ln\Big(\frac{2\pi}{q}\Big) + \tfrac{1}{2}\pi\cot\{\tfrac{1}{2}\pi(s+\alpha)\} - \frac{\Gamma'}{\Gamma}(1-s) - \frac{L'}{L}(1-s,\overline{\chi}) \\ &\ll \ln(q|s|) \qquad (\sigma\leqslant -1).\end{aligned} \tag{8.81}$$

Indeed, the complex Stirling formula guarantees that

$$\frac{\Gamma'(1-s)}{\Gamma(1-s)} \ll \ln|s|$$

when $\sigma\leqslant -1$ and the cotangent formula, viz.

$$\pi\cot(\pi z) = \frac{1}{z} + \sum_{n\geqslant 1}\frac{2z}{z^2-n^2} \qquad (z\in\mathbb{C}\smallsetminus\mathbb{Z}),$$

implies $\cot\{\frac{1}{2}\pi(s+\alpha)\}\ll 1$ for $s=\sigma\pm iT$, $\sigma\leqslant -1$. We may therefore shift the line of integration to the left all the way to $-\infty$. We thus obtain, by the residue theorem,

$$\begin{aligned}\psi(x;\chi) = &-\sum_{|\gamma|\leqslant T}\frac{x^{\varrho}}{\varrho} + \sum_{n\geqslant 1}\frac{x^{\alpha-2n}}{2n-\alpha} \\ &+ \operatorname{Res}\Big(-\frac{L'}{L}(s,\chi)\frac{x^s}{s};0\Big) + O(R_1+R_2)\end{aligned} \tag{8.82}$$

with

$$R_1 := \frac{x\ln x}{x+T\langle\langle x\rangle\rangle} + \frac{x(\ln xT)^2}{T},$$
$$R_2 := \max_{\varepsilon=\pm1}\int_{-\infty}^{\kappa}\left|\frac{L'}{L}(\sigma+i\varepsilon T,\chi)\right|\frac{x^\sigma}{T}\,\mathrm{d}\sigma. \tag{8.83}$$

However we deduce easily from (8.68) that

$$\sum_{\varrho}\frac{1}{1+|t-\gamma|^2} \ll \ln\{q(2+|t|)\} \qquad (t\in\mathbb{R}), \tag{8.84}$$

hence, for $-1\leqslant\sigma\leqslant 2$, $s=\sigma+iT$, $s_0:=2+iT$,

$$\begin{aligned}\frac{L'}{L}(s,\chi) &= \frac{L'}{L}(s,\chi) - \frac{L'}{L}(s_0,\chi) + O(1)\\ &= \sum_{\varrho}\frac{2-\sigma}{(s-\varrho)(s_0-\varrho)} + O(\ln T) = \sum_{\substack{\varrho\\ |\tau-\gamma|\leqslant 1}}\frac{1}{s-\varrho} + O(\ln qT).\end{aligned}$$

There are $\ll \ln qT$ terms in the latter sum over ϱ and, by hypothesis, they all have modulus $\ll \ln qT$. This implies that, for $-1\leqslant\sigma\leqslant 2$, we have

$$\left|\frac{L'}{L}(s,\chi)\right| \ll (\ln qT)^2 \qquad (s=\sigma+iT,\ -1\leqslant\sigma\leqslant 2\leqslant T).$$

Taking (8.81) into account, we therefore obtain

$$\begin{aligned}R_2 &\ll \int_{-\infty}^{-1}\frac{x^\sigma}{T}\ln\{q(|\sigma|+T)\}\,\mathrm{d}\sigma + \int_{-1}^{\kappa}\frac{x^\sigma}{T}(\ln qT)^2\,\mathrm{d}\sigma\\ &\ll \frac{\ln qT}{Tx\ln x} + \frac{x(\ln qT)^2}{T\ln x} \ll \frac{x(\ln qT)^2}{T\ln x}.\end{aligned} \tag{8.85}$$

Now let us compute the residue at $s=0$ appearing in (8.82). Recall definition (8.76). We have $b(\chi)=L'(0,\chi)/L(0,\chi)$ if $\alpha=1$ and

$$\frac{L'}{L}(s,\chi) = \frac{1}{s} + b(\chi) + a_1 s + \cdots$$

if $\alpha=0$. It follows that, in all cases,

$$\operatorname{Res}\left(\frac{x^s L'(s,\chi)}{sL(s,\chi)};0\right) = (1-\alpha)\ln x + b(\chi).$$

Thus

$$\begin{aligned}\psi(x;\chi) = &-\sum_{|\gamma|\leqslant T}\frac{x^\varrho}{\varrho} + \sum_{m\geqslant 1}\frac{x^{-2m+\alpha}}{2m-\alpha} - (1-\alpha)\ln x - b(\chi)\\ &+ O\left(\frac{x\ln x}{x+T\langle\langle x\rangle\rangle} + \frac{x(\ln qxT)^2}{T}\right).\end{aligned} \tag{8.86}$$

We therefore get (8.77) by letting T tend to infinity.

It remains to establish (8.78). It plainly results from (8.86) that, when $2 \leqslant T \leqslant x$,

$$\psi(x;\chi) = -\sum_{|\gamma|\leqslant T} \frac{x^\varrho}{\varrho} - b(\chi) + O(R_3) \tag{8.87}$$

with

$$R_3 := \frac{x(\ln xq)^2}{T}.$$

We have yet to estimate $b(\chi)$ and include the contribution of the exceptional zero. Apply (8.80) at s and at 2 and subtract. We obtain

$$\frac{L'}{L}(s,\chi) = -\frac{\Gamma'}{2\Gamma}\left(\frac{s+\alpha}{2}\right) + \sum_\varrho \left(\frac{1}{s-\varrho} - \frac{1}{2-\varrho}\right) + O(1).$$

If $\alpha = 1$, we get, taking $s = 0$,

$$b(\chi) = \frac{L'}{L}(0,\chi) = -\sum_\varrho \left(\frac{1}{\varrho} + \frac{1}{2-\varrho}\right) + O(1) = -\sum_\varrho \frac{2}{\varrho(2-\varrho)} + O(1).$$

This formula still holds if $\alpha = 0$ since $-\Gamma'(s/2)/\{2\Gamma(s/2)\} = 1/s + O(1)$ as s tends to 0. Moreover, we deduce from (8.84) that

$$\sum_{|\gamma|\geqslant 1} \frac{2}{|\varrho(2-\varrho)|} \ll \sum_{|\gamma|\geqslant 1} \frac{1}{|2-\varrho|^2} \ll \ln q$$

while (8.68) implies

$$\sum_{|\gamma|\leqslant 1} \frac{1}{|2-\varrho|} \ll \ln q.$$

Therefore we conclude from the foregoing that

$$b(\chi) = -\sum_{|\gamma|\leqslant 1} \frac{1}{\varrho} + O(\ln q). \tag{8.88}$$

Substituting back into (8.87), we finally obtain

$$\psi(x;\chi) = -\sum_{|\gamma|\leqslant T} \frac{x^\varrho}{\varrho} + \sum_{|\gamma|\leqslant 1} \frac{1}{\varrho} + O(R_3).$$

By the Landau–Page theorem (Theorem 8.25), there is an absolute constant c such that $\prod_{\chi\neq\chi_0} L(s,\chi)$ has at most one zero $\varrho = \beta + i\gamma$ such that $|\gamma| \leqslant 1$ and $\beta \geqslant 1 - c/\ln q$, this zero being then the exceptional, real zero β_1. We can assume $c \leqslant \frac{1}{6}$, and thus $\beta_1 \geqslant \frac{3}{4}$. Then

$$\sum_{\substack{|\gamma|\leqslant 1\\ \varrho\neq\beta_1}} \frac{1}{\varrho} \ll (\ln q)^2,$$

whence

$$\psi(x;\chi) = -\sum_{|\gamma|\leqslant T}^{*} \frac{x^{\varrho}}{\varrho} - \left(\frac{x^{\beta_1}-1}{\beta_1} + \frac{x^{1-\beta_1}-1}{1-\beta_1}\right)\vartheta(\chi) + O(R_3)$$

$$= -\sum_{|\gamma|\leqslant T}^{*} \frac{x^{\varrho}}{\varrho} - \vartheta(\chi)\frac{x^{\beta_1}}{\beta_1} + O(R_3 + x^{1/4}),$$

since

$$\frac{x^{1-\beta_1}-1}{(1-\beta_1)} = (\ln x)\int_{\beta_1}^{1} x^{1-v}\,\mathrm{d}v \ll x^{1/4}, \qquad \frac{1}{\beta_1} \ll 1.$$

This completes the proof of (8.78) when χ is primitive.

When χ is induced by a primitive character χ^*, we have

$$|\psi(x;\chi) - \psi(x;\chi^*)| \leqslant \sum_{\substack{n\leqslant x\\(n,q)>1}} \Lambda(n) \leqslant \sum_{p|q}\sum_{\nu\leqslant \ln x/\ln p} \ln p \ll (\ln x)\ln q.$$

We obtain the stated formula for $\psi(x;\chi)$, but with β_1 being the exceptional zero associated to χ^*, and, consequently, not necessarily exceptional for χ. In this case, however, we would have $\beta_1 > 1 - c/\ln q^*$ and $\beta_1 < 1 - c/\ln q$. Formula (8.78) is then still valid since the term involving β_1 is now included in the sum over ϱ. □

8.7. Final form of the prime number theorem for arithmetic progressions

Theorem 8.29. *Let κ be an arbitrary positive constant. There exists a constant $c = c(\kappa) > 0$ such that, uniformly under the conditions $x \geqslant 2$, $1 \leqslant q \leqslant \mathrm{e}^{\kappa\sqrt{\ln x}}$, $(a,q) = 1$, we have*

$$\psi(x;a,q) = \frac{x}{\varphi(q)} - \frac{\chi_1(a)x^{\beta_1}}{\beta_1} + O\Big(x\mathrm{e}^{-c\sqrt{\ln x}}\Big). \tag{8.89}$$

Proof. The orthogonality relations for characters provide

$$\psi(x;a,q) = \frac{1}{\varphi(q)}\sum_{\chi}\overline{\chi(a)}\psi(x;\chi) = \frac{1}{\varphi(q)}\psi(x;\chi_0) + \frac{1}{\varphi(q)}\sum_{\chi\neq\chi_0}\overline{\chi(a)}\psi(x;\chi).$$

Moreover, we have

$$|\psi(x) - \psi(x;\chi_0)| \leqslant \sum_{\substack{n\leqslant x\\(n,q)>1}} \Lambda(n) \leqslant \sum_{p|q}\sum_{\nu\leqslant \ln x/\ln p} \ln p \ll (\ln x)\ln q$$

and, for a suitable constant $c_0 > 0$,

$$\psi(x) = x + O\Big(x\mathrm{e}^{-c_0\sqrt{\ln x}}\Big).$$

For $\chi \neq \chi_0$, we deduce from Theorem 8.28 that

$$\psi(x;\chi) = -\vartheta(\chi)\frac{x^{\beta_1}}{\beta_1} - \sideset{}{^*}\sum_{|\gamma|\leqslant T}\frac{x^\varrho}{\varrho} + O\Big(\frac{x(\ln qx)^2}{T} + x^{1/4}\Big) \qquad (1 \leqslant T \leqslant x).$$

In the sum over ϱ, each zero satisfies $\beta \leqslant 1 - c/\ln qT$, and so

$$x^\varrho \ll x\mathrm{e}^{-c_1 \ln x/\ln qT},$$

by Theorem 8.25. Furthermore, by (8.68)

$$\sum_{|\gamma|\leqslant 1}\frac{1}{|\varrho|} \ll (\ln q)^2, \qquad \sum_{1<|\gamma|\leqslant T}\frac{1}{|\varrho|} \ll (\ln qT)^2 \ll (\ln qx)^2.$$

The stated conclusion follows on choosing $T = \mathrm{e}^{\sqrt{\ln x}}$. □

In order to be able to consider the term involving the exceptional zero as a remainder term, we need a lower bound for $1 - \beta_1$.

Proposition 8.30. *Let $q \geqslant 3$. We have*

$$1 - \beta_1 \gg L(1,\chi_1)/(\ln q)^2 \gg 1/\{\sqrt{q}(\ln q)^2\}. \tag{8.90}$$

Proof. Up to changing the value of the implicit constant in the definition of $\mathcal{D}_q$, we can assume $\beta_1 > \frac{1}{2}$. For $\beta_1 \leqslant \sigma \leqslant 1$, $N > 1$, we then have

$$\begin{aligned} L'(\sigma,\chi_1) &= -\int_{1-}^{\infty}\frac{\ln t}{t^\sigma}\,\mathrm{d}K(t) \ll \sum_{n\leqslant N}\frac{\ln n}{n^\sigma} + \int_N^\infty \frac{|K(t)-K(N)|\ln t}{t^{\sigma+1}}\,\mathrm{d}t \\ &\ll \frac{N^{1-\sigma}-1}{1-\sigma}\ln N + H(\chi_1)\frac{\ln N}{N^\sigma}. \end{aligned}$$

Selecting $N := H(\chi_1)$, we obtain

$$L'(\sigma,\chi_1) \ll H(\chi_1)^{1-\beta_1}\big\{\ln H(\chi_1)\big\}^2 \ll (\ln q)^2,$$

since $H(\chi_1) \leqslant q$, $1 - \beta_1 \ll 1/\ln q$. Whence

$$L(1,\chi_1) = L(1,\chi_1) - L(\beta_1,\chi_1) \ll (1-\beta_1)(\ln q)^2.$$

In view of (8.57), this implies (8.90) as required. □

Corollary 8.31. *There exists a constant $c > 0$ such that, for any function $h(x)$ tending to infinity with x, and uniformly for $x \geqslant 3$, $(a,q) = 1$, $1 \leqslant q \leqslant (\ln x)^2/\{h(x)^2(\ln_2 x)^6\}$, we have*

$$\psi(x;a,q) = \frac{x}{\varphi(q)}\Big\{1 + O\Big(\frac{1}{(\ln x)^{c\,h(x)}}\Big)\Big\}. \tag{8.91}$$

Proof. It suffices to apply (8.89) and (8.90). □

Remark. All the constants occurring in (8.91) are effectively computable. In particular, one has

$$\psi(x;a,q) \sim \frac{x}{\varphi(q)} \tag{8.92}$$

as $x \to \infty$ and $q = o\big((\ln x)^2/(\ln_2 x)^6\big)$. This is essentially the best effective result known.

To widen the range of validity for the asymptotic formula (8.92) requires better lower bounds on $1-\beta_1$. Siegel's theorem provides such an improvement, although at the cost of ineffectiveness of the constants. We give below Estermann's proof (1948) of Siegel's theorem, however stating the result in the more general form of an effective version of the Phragmén–Landau theorem.

Theorem 8.32. *Let $F(s) := \sum_{n\geqslant 1} a_n/n^s$ be a Dirichlet series converging for $\sigma > 1$ and whose coefficients satisfy $a_n \geqslant \delta(n)$ for $n \geqslant 1$.*[7] *Assume there exist real numbers $r \in]0,1[$ and $M > 0$ such that:*

(i) *$f(s) := F(s)(s-1)$ can be analytically continued to the disk $|s-1| \leqslant r$;*

(ii) *$\sup_{|s-1|\leqslant r} |f(s)| \leqslant M$;*

(iii) *$\exists \beta \in [1-r/2, 1[\colon f(\beta) \geqslant 0$.*

Then we have

$$f(1) \geqslant \frac{c_1(1-\beta)}{M^{c_2(1-\beta)}} \tag{8.93}$$

where c_1 and c_2 are positive constants depending only on r.

Remark. We have $f(1) = \lim_{\sigma\to 1+} F(\sigma)(\sigma-1)$, hence $f(1) \geqslant 0$. If $f(1) = 0$, then F is analytic in the disk $|s-1| \leqslant r$ and so, by the Phragmén–Landau theorem, its abscissa of convergence is $< 1-r$. This implies $F(\beta) = \sum_{n\geqslant 1} a_n/n^\beta > 0$ and thus $f(\beta) = (\beta-1)F(\beta) < 0$. The assumptions of the theorem hence imply $f(1) > 0$, and (8.93) may be considered as an effective version of this lower bound.

Proof. Put $\beta = 1-\mu r$ with $0 < \mu \leqslant \frac{1}{2}$, and consider $\alpha := 1+\lambda r$, with $0 < \lambda \leqslant \frac{1}{2}$. Since $F(s)$ is holomorphic in the disk $|s-\alpha| < \lambda r$, we may write down, in this domain, the convergent Taylor expansion

$$F(s) = \sum_{n\geqslant 0} F_n(\alpha - s)^n.$$

[7] Recall that $\delta(n) := \delta_{1n}$ with Kronecker's notation.

We have $F_0 = F(\alpha) \geqslant 1$ and $F_n = (-1)^n F^{(n)}(\alpha)/n! \geqslant 0$ for all $n \geqslant 1$, hence $F_n \geqslant \delta(n)$. Moreover, for $|s-\alpha| < \lambda r$, we have

$$\frac{f(1)}{s-1} = f(1)\sum_{n\geqslant 0}\frac{(\alpha-s)^n}{(\alpha-1)^{n+1}} = \frac{f(1)}{\lambda r}\sum_{n\geqslant 0}\Big(\frac{\alpha-s}{\lambda r}\Big)^n.$$

However the Taylor expansion

$$F(s) - \frac{f(1)}{s-1} = \frac{f(s)-f(1)}{s-1} = \sum_{n\geqslant 0} b_n(\alpha-s)^n$$

converges in the disk $|s-\alpha| \leqslant (1-\lambda)r$ because it represents a holomorphic function. Cauchy's formula hence allows us to write

$$b_n = \frac{(-1)^n}{2\pi i}\oint_{|s-\alpha|=(1-\lambda)r} \frac{f(s)-f(1)}{s-1}\frac{\mathrm{d}s}{(s-\alpha)^{n+1}},$$

whence, since $|s-1| \geqslant |s-\alpha| - |\alpha-1| = (1-2\lambda)r$ for $|s-\alpha| = (1-\lambda)r$,

$$|b_n| \leqslant \frac{2M}{(1-2\lambda)(1-\lambda)^n r^{n+1}}.$$

From the formulae

$$b_n = F_n - \frac{f(1)}{(\lambda r)^{n+1}} \geqslant \delta(n) - \frac{f(1)}{(\lambda r)^{n+1}} \qquad (n \geqslant 0),$$

we deduce that, for any integer $N \geqslant 1$, and under the additional condition $\mu + 2\lambda < 1$, we have

$$\begin{aligned}\frac{f(1)}{1-\beta} &\geqslant \frac{f(\beta)-f(1)}{\beta-1}\\ &\geqslant 1 - \sum_{0\leqslant n<N}\frac{f(1)(\alpha-\beta)^n}{(\lambda r)^{n+1}} - \frac{2M}{(1-2\lambda)r}\sum_{n\geqslant N}\frac{(\alpha-\beta)^n}{(1-\lambda)^n r^n}\\ &\geqslant 1 - \frac{f(1)}{\lambda r}\frac{\{(\lambda+\mu)/\lambda\}^N - 1}{\{(\lambda+\mu)/\lambda\} - 1} - \frac{2M}{(1-2\lambda)r}\Big(\frac{\lambda+\mu}{1-\lambda}\Big)^N\frac{1}{1-(\lambda+\mu)/(1-\lambda)}.\end{aligned}$$

It follows furthermore that

$$\frac{f(1)}{\mu r}\Big(\frac{\lambda+\mu}{\lambda}\Big)^N \geqslant 1 - \frac{2M(1-\lambda)}{(1-2\lambda)(1-2\lambda-\mu)r}\Big(\frac{\lambda+\mu}{1-\lambda}\Big)^N. \tag{8.94}$$

Let us choose, for example, $\lambda = \frac{1}{8}$, so that $\lambda+\mu < \frac{5}{8}$, $(\lambda+\mu)/(1-\lambda) < \frac{5}{7}$. For $N = \lceil \ln(20M/r)/\ln\frac{7}{5}\rceil$, the right-hand side of (8.94) is hence $\geqslant \frac{1}{2}$. From this we infer

$$f(1) \geqslant \tfrac{1}{2}\mu r \mathrm{e}^{-8\mu N} \geqslant c_1 \mu r M^{-c_2\mu r} = c_1(1-\beta)M^{-c_2(1-\beta)}$$

with $c_1 \gg r^{8\mu} \gg r^4$ and $c_2 = 8/(r\ln\frac{7}{5})$. □

We are now in a position to establish Siegel's theorem.

Theorem 8.33 (Siegel). *Let $q \geqslant 2$. For any $\varepsilon > 0$, there exists a constant $c(\varepsilon) > 0$ such that $L(1,\chi) > c(\varepsilon)/q^{\varepsilon}$ for all non-principal real characters to the modulus q. In particular, we have*

$$1 - \beta_1 \gg_{\varepsilon} 1/q^{\varepsilon}.$$

Proof. Let χ_1 and χ_2 be real primitive characters with respective moduli q_1 and q_2. As seen in the proof of Lemma 8.27, $\chi_1\chi_2$ is non-principal (however not necessarily primitive) modulo q_1q_2. Consider the Dirichlet series

$$F(s) = \zeta(s)L(s,\chi_1)L(s,\chi_2)L(s,\chi_1\chi_2) = \sum_{n\geqslant 1} \frac{a_n}{n^s}. \tag{8.95}$$

We have $a_1 = 1$ and

$$\log F(s) = \sum_p \sum_{\nu\geqslant 1} \frac{\{1+\chi_1(p)^\nu\}\{1+\chi_2(p)^\nu\}}{\nu p^{\nu s}},$$

hence $a_n \geqslant 0$ for every integer $n \geqslant 2$. Moreover, F has a meromorphic continuation to the disk $|s-1| < 1$ of the form $f(s)/(s-1)$ with $f(1) = L(1,\chi_1)L(1,\chi_2)L(1,\chi_1\chi_2)$. Finally, it follows by partial summation that $L(\sigma,\chi) \ll q$ for any non-principal character to the modulus q and any σ in $]0,1[$. Therefore, $f(s) \ll (q_1q_2)^2$ for $0 < \sigma < 1$.

In order to fulfil the hypotheses of Theorem 8.32, we have to find a number β in, say, $[\frac{1}{2},1[$ such that $f(\beta) \geqslant 0$, or, equivalently, $F(\beta) \leqslant 0$.

Let $\varepsilon \in]0,\frac{1}{2}[$. If there is a non-principal real primitive character χ such that $L(s,\chi)$vanishes on $[1-\varepsilon,1[$, we choose $\chi_2 = \chi$ and $\beta = \beta_2$. Then we have $F(\beta_2) = 0$ for any χ_1. Otherwise, we have $L(\sigma,\chi) \neq 0$ for every non-principal real primitive character χ and all σ in $[1-\varepsilon,1[$. We then select $\beta = 1-\varepsilon/2$ and take χ_2 equal to some non-principal real primitive character. Since $\zeta(\beta) < 0$ and since the three L-functions in (8.95) are strictly positive on $[\beta,1]$, we must have $F(\beta) < 0$.

Thus, we have $F(\beta) \leqslant 0$ in all cases and (8.93) implies

$$L(1,\chi_1)L(1,\chi_2)L(1,\chi_1\chi_2) \geqslant c_1(1-\beta)/(q_1q_2)^{c_2(1-\beta)} \geqslant c_1(1-\beta)/(q_1q_2)^{c_2\varepsilon}.$$

When χ_2 is fixed, β only depends on ε. Furthermore, by Theorem 8.18, we have $L(1,\chi_1\chi_2) \ll \ln q_1q_2$ and $L(1,\chi_2) \ll \ln q_2$ where the constants involved are absolute. It hence follows that

$$L(1,\chi_1)(\ln q_1q_2)(\ln q_1) \gg_{\varepsilon} 1/q_1^{c_2\varepsilon}.$$

Since c_2 is absolute, this implies the stated result. □

Remark. The lower bound given for $L(1,\chi_1)$ in Siegel's theorem is ineffective: given ε, it does not provide a numerical value for $c(\varepsilon)$.

Proof of the Siegel–Walfisz theorem (Theorem 8.17). It suffices to substitute the lower bound for β_1 in the formula of Theorem 8.29 for $\varepsilon \leqslant 1/2A$. Note that this result is ineffective whenever $A \geqslant 1$: the Pólya–Vinogradov inequality is then useless. □

Notes

§ 8.1. For a more thorough study of the characters of finite Abelian groups the reader may refer, for example, to Ayoub (1963) or Ellison & Mendès France (1975).

The treatment of primitive characters and the proof of Theorem 8.7 follow Davenport (1980).

Kronecker's symbol is defined as a generalization of Legendre's: for $d \in \mathbb{Z}^*$, one puts

$$\left(\frac{d}{2}\right) := \begin{cases} 1 & \text{if } d \equiv \pm 1 \pmod 8, \\ 0 & \text{if } d \equiv 0 \pmod 2, \\ -1 & \text{if } d \equiv \pm 3 \pmod 8, \end{cases} \qquad \left(\frac{d}{n}\right) := \prod_{p^\nu \| n} \left(\frac{d}{p}\right)^\nu,$$

where the symbol $(d|p)$ retains Legendre's definition when $p > 2$. It is easy to verify that the mapping $n \mapsto (d|n)$ is a real character. It can be shown—see Davenport (1980)—that any real primitive character to the modulus q must be of the form

$$\chi(n) = \left(\frac{d}{n}\right) \tag{8.96}$$

where d is a *fundamental discriminant*, i.e., a product of pairwise coprime integers of the form -4, ± 8, $(-1)^{(p-1)/2}p$ where p is an odd prime and the right-hand side is defined in Kronecker's sense. The modulus of χ then equals $q = |d|$. Thus, there are two real primitive characters to the modulus q if $8 \| q$ and only one otherwise. We then have—see Landau (1966), th. 21—

$$G(n, \chi) = \chi(n)\sqrt{d}, \tag{8.97}$$

where the square root is taken as its principal branch.

For further improvements on the value of c such that $H(\chi) \geqslant c\sqrt{q}$, see Sárközy (1977) and Sokolovskii (1979). Besides, Montgomery & Vaughan (1979) showed that, for any $\varepsilon > 0$, we have $H(\chi) \ll_\varepsilon \sqrt{q}$ for all but at most $\varepsilon\varphi(q)$ non-principal characters to the modulus q.

More precisely, Paley showed in 1932 that, for infinitely many values of q, we actually have

$$\max_{\chi \neq \chi_0} H(\chi) \gg \sqrt{q} \ln_2 q.$$

Furthermore, Montgomery & Vaughan (1977) showed that, subject to the generalized Riemann hypothesis, we have

$$H(\chi) \ll \sqrt{q} \ln_2 q.$$

The inequality $|K(x)| \leqslant 2\sqrt{q}\ln q$ is clearly useless for small values of x, for instance $x \leqslant \sqrt{q}$. Burgess (1962, 1963) studied this question. He showed, among other estimates, that we have

$$\sum_{n\leqslant x}\chi(n) \ll x/q^{\delta} \qquad \big(x \geqslant q^{3/8+\varepsilon}\big) \tag{8.98}$$

where $\delta = \delta(\varepsilon) > 0$. Moreover, (8.98) holds for $x \geqslant q^{1/4+\varepsilon}$ if q has no cubic factor. See also Hildebrand (1986d).

§ 8.3. Our proof of Theorem 8.21 essentially follows that of Ellison & Mendès France (1975), chap. 7, A1, with an improvement derived from employing a bound for $M_2(\chi)$ instead of the trivial bound $H(\chi) \leqslant q/2$. The induced gain in Theorem 8.16 is not significant with the Tauberian method employed here, but the contour integration method actually provides in this way a better uniformity in q—see Exercise 237.

The theory of quadratic forms, and especially Dirichlet's class number formula—see, for example, Davenport (1980, chap. 6)—immediately delivers a bound comparable to that in Theorem 8.21. Ramaré (2001a) even notes that it implies the lower bound

$$L(1,\chi) \geqslant \pi 2^{\omega(q)-1}/\sqrt{q}$$

and he establishes a qualitatively equivalent inequality by purely analytic means.

For upper bounds on $L(1,\chi)$, see Ramaré (2001b, 2004).

§ 8.5. The presentation chosen here for the proof of Theorem 8.25 mainly follows that of Davenport (1980).

Miech (1969) showed that $c = 1/20$ is admissible in the statement of the Landau–Page theorem (Theorem 8.25) provided $q \geqslant q_0$. The best value of c presently known for sufficiently large q is $c = 0.10367$, due, according to Graham (1981b), to Schoenfeld in unpublished work—see Lemma 10 of Chen (1983) for a proof. Heath-Brown (1992) established that there is at most a single zero in the region $\sigma \geqslant 1 - 0.348/\ln q$, $|\tau| \leqslant 1$.

On the generalized Riemann hypothesis, the Siegel–Walfisz theorem may be improved to

$$\psi(x;a,q) = \frac{x}{\varphi(q)} + O\big(\sqrt{x}(\ln x)^2\big) \qquad (x \geqslant 2,\ q \geqslant 1). \tag{8.99}$$

The Bombieri–Vinogradov theorem—cf. Bombieri (1965), A. I. Vinogradov (1965, 1966)—establishes unconditionally that (8.99) is satisfied on average.

Theorem 8.34 (Bombieri–Vinogradov theorem). *Let A be a positive constant. Uniformly for $Q \geqslant 1$, $x \geqslant 1$, we have*

$$\sum_{q \leqslant Q} \max_{\substack{(a,q)=1 \\ y \leqslant x}} \Big| \psi(y; a, q) - \frac{y}{\varphi(q)} \Big| \ll \frac{x}{(\ln x)^A} + \sqrt{x} Q \big(\ln(Qx) \big)^4. \tag{8.100}$$

For an elementary proof of this result, based on a fruitful general method, see Vaughan (1980). The exponent 4 may be improved to $\frac{5}{2}+\varepsilon$, for any $\varepsilon > 0$, cf. Dress, Iwaniec & Tenenbaum (1983). The so-called Elliott–Halberstam conjecture asserts that, given any $\varepsilon > 0$, the left-hand side of (8.100) is $o(x)$ for $Q \leqslant x^{1-\varepsilon}$. This would indeed imply even sharper consequences than the generalized Riemann hypothesis in most arithmetical applications.

§ 8.7. Other analytic proofs of Siegel's theorem have been given by Chowla (1950) and Goldfeld (1974).

Exercises

232. (a) Let $p > 2$. For each $d \geqslant 1$, denote by $\psi(d)$ the number of elements of $(\mathbb{Z}/p\mathbb{Z})^*$ which are exactly of order d. Establish the formula $\sum_{d|(p-1)} \psi(d) = p - 1$.

(b) Let y be an element of order d in $(\mathbb{Z}/p\mathbb{Z})^*$. Show that the equation $x^d - 1 \equiv 0 \pmod{p}$ has exactly d solutions, and make them explicit. Deduce that each element of order d is one of the $\varphi(d)$ generators of the group generated by y.

(c) Show that $\psi(d) \leqslant \varphi(d)$ for all d, and hence that this inequality is an equality if $d|(p-1)$. Deduce that $(\mathbb{Z}/p\mathbb{Z})^*$ is cyclic.

233. Let $p > 2$, and g be a primitive root modulo p.

(a) Show that the order of $g \pmod{p^2}$ is either $p-1$ or $p(p-1)$. Show that, in the first case, $g+p$ does not have order $p-1$. Deduce that $(\mathbb{Z}/p^2\mathbb{Z})^*$ is cyclic.

(b) Let $\nu \geqslant 2$ and g be a primitive root mod p^2. Show that $g^{p-1} = 1+hp$ with $(h,p) = 1$. Deduce that

$$g^{\varphi(p^\nu)} \not\equiv 1 \pmod{p^{\nu+1}}$$

and hence that g generates $(\mathbb{Z}/p^{\nu+1}\mathbb{Z})^*$.

234. In this exercise, we do not assume the prime number theorem. However, Chebyshev's estimates may be freely employed.

Let q be an integer > 1 and let χ be a Dirichlet character to the modulus q. Denote by χ_0 the principal character modulo q and define

$$M(x;\chi) := \sum_{n\leqslant x} \mu(n)\chi(n),\ \psi(x;\chi) := \sum_{n\leqslant x} \chi(n)\Lambda(n),\ D(x;\chi) := \sum_{n\leqslant x} \chi(n)\ln n.$$

(a) Recall, without going into computational detail, the main steps in the elementary proof of the implication:

$$\lim_{x\to\infty} M(x)/x = 0 \ \Rightarrow\ \lim_{x\to\infty} \psi(x)/x = 1.$$

(b) Show that, for $x \geqslant 2$, $q \geqslant 1$, we have $\psi(x,\chi_0) = \psi(x) + O\big((\ln q)\ln x\big)$. Propose, and indeed establish, an improvement in the dependence of the error term upon q.

(c) Show that, for $\chi \neq \chi_0$, we have $D(x;\chi) \ll q\ln x$ $(x \geqslant 2)$. Propose and establish an improvement in the dependence upon q of this upper bound.

(d) Establish the convolution relation between arithmetic functions

$$\chi\Lambda = (\chi\mu) * (\chi\ln).$$

(e) Show that, if, for fixed $q > 1$ and every $\chi \neq \chi_0$, we have

$$M(x;\chi) = o(x) \qquad (x \to \infty),$$

then $\psi(x;a,q) \sim \psi(x)/\varphi(q)$ for all integers a such that $(a,q) = 1$.

(f) Is it true that the conclusion of the previous question also follows from the hypothesis

$$\sum_{\substack{n \leqslant x \\ n \equiv a \,(\mathrm{mod}\, q)}} \mu(n) = o(x) \qquad (x \to \infty)$$

for every integer a such that $(a,q) = 1$?

235. Using the method of Theorem 8.21, show that, for each non-principal, real Dirichlet character to the modulus q, we have $L(1,\chi) \geqslant 1/(4\sqrt{q})$.

236. Show that for $(a,q) = 1$, $\sigma > 1$, we have

$$\sum_{p \equiv a (\mathrm{mod}\, q)} \frac{1}{p^s} = \frac{1}{\varphi(q)} \sum_{\chi} \overline{\chi(a)} \log L(s,\chi) + h(s)$$

where $h(s)$ is a holomorphic function on the half-plane $\sigma > \frac{1}{2}$.

Assuming $L(s,\chi) \neq 0$ for $\sigma \geqslant 1$, deduce that

$$\sum_{p \equiv a (\mathrm{mod}\, q)} \frac{1}{p^s} = \frac{1}{\varphi(q)} \log\Big(\frac{1}{s-1}\Big) + h_1(s) \qquad (\sigma > 1)$$

where h_1 is holomorphic in the half-plane $\sigma \geqslant 1$. Then establish the prime number theorem for arithmetic progressions by applying the Delange Tauberian theorem—cf. Theorem 7.28.

237. Set $Q = \sqrt{q}(\ln 2q)^2$. Prove the existence of absolute positive constants c_1, c_2 such that the estimates of Theorems 8.18, 8.20 and 8.22 in fact hold for

$$1 - \sigma \leqslant \begin{cases} c_1 \mathcal{L}^{-9} & (\chi^2 \neq \chi_0) \\ c_1 \mathcal{L}^{-8} (\mathcal{L} + 1/|\tau|)^{-1} & (\chi^2 = \chi_0,\ |\tau| Q > c_2) \\ c_1 Q^{-1} & (\chi^2 = \chi_0,\ |\tau| Q \leqslant c_2). \end{cases}$$

By the method of § 4.2, deduce the existence of an absolute constant $c > 0$ and of a quantity $\delta(q) \gg q^{-1/2}(\ln q)^{-10}$ such that, uniformly for $q \leqslant \exp(\ln x)^{1/10}$,

$$\psi(x;a,q) = \frac{x}{\varphi(q)} + O\Big(x \exp\{-c(\ln x)^{1/10}\} + \frac{2^{\omega(q)}}{\varphi(q)} (\ln q)^9 x^{1-\delta(q)}\Big).$$

Show that, for each $\varepsilon > 0$, the asymptotic formula $\psi(x;a,q) \sim x/\varphi(q)$ persists uniformly for $q \leqslant (\ln x)^{2-\varepsilon}$.

238. *Squarefree integers in arithmetic progressions.* Set

$$Q(s,\chi) = \sum_{n\geqslant 1} \frac{\mu(n)^2\chi(n)}{n^s}.$$

Show that we have $Q(s,\chi) = L(s,\chi)H(s,\chi)$ where $H(s,\chi)$ is a bounded, holomorphic function for $\sigma \geqslant \sigma_0 > \frac{1}{2}$. Show that $L(s,\chi) \ll_\varepsilon (Tq)^{1-\sigma+\varepsilon}$ for $0 \leqslant \sigma \leqslant 1$, $|\tau|+1 \leqslant T$, and deduce that, for each $\varepsilon > 0$ and uniformly for $x \geqslant 2$, $q \geqslant 1$, $(a,q) = 1$, we have

$$\sum_{\substack{n\leqslant x\\ n\equiv a\,(\mathrm{mod}\,q)}} \mu(n)^2 = \frac{x}{q}\prod_{p\nmid q}(1-p^{-2}) + O_\varepsilon\Big(x^{2/3+\varepsilon}q^{1/3}\Big). \tag{8.101}$$

Show that the convolution argument used in the proof of Theorem I.3.10 actually yields a better upper bound on the error term, namely $O(\sqrt{x})$. Recover essentially this result (i.e. $\ll x^{1/2+\varepsilon}$) by the above analytic method[8] using Montgomery's upper bound (1971), *Theorem 10.1*,

$$\sum_{\chi(\mathrm{mod}\,q)} \int_{-T}^{T} |L(\tfrac{1}{2}+i\tau,\chi)|^4\,\mathrm{d}\tau \ll (qT)^{1+\varepsilon}.$$

239. Let χ_4 denote the unique non-principal Dirichlet character to the modulus 4. Show that $L(1,\chi_4) = \pi/4$. Show that the set of primes of the form $4m+3$ satisfies hypotheses (i) and (ii) of Exercise 228, with $\delta = \frac{1}{2}$ and $K = \sqrt{(2/\pi)}\,\mathrm{e}^{\gamma/2}A$, with $A := \prod_{p\equiv 3(\mathrm{mod}\,4)}\big(1-p^{-2}\big)^{-1/2}$. Deduce that the number $N_3(x)$ of those integers n not exceeding x all of whose prime factors are of the form $4m+3$ satisfies

$$N_3(x) \sim \frac{\sqrt{2}Ax}{\pi\sqrt{\ln x}}.$$

Give a remainder form of this result. Generalize.[9]

240. *Sums of two squares.* Retain the notation χ_4 of Exercise 239. Using the properties established in § I.4.8 for the characteristic function h of sums of two squares, establish the formula

$$\sum_{n\geqslant 1}\frac{h(n)}{n^s} = (1-2^{-s})^{-1/2}\zeta(s)^{1/2}L(s,\chi_4)^{1/2}\prod_{p\equiv 3\,(\mathrm{mod}\,4)}(1-p^{-2s})^{-1/2} \quad (\sigma>1)$$

where the square roots are taken as their principal branches. Employing the Selberg–Delange method, deduce that, for a suitable constant $c > 0$, we

[8] For refined, or average-type estimates, see Prachar (1958) and Warlimont (1969, 1980). In particular, Prachar shows by a very simple modification of the elementary method, that the error term in (8.101) is $\ll x^{1/2}q^{-1/4} + q^{1/2}$.

[9] See Landau (1909), vol. 2, pp. 641–669, and Wirsing (1956).

have

$$\sum_{n\leqslant x} h(n) = \int_0^{1/3} x^{1-t} g(t)\,\mathrm{d}t + O\Big(x\mathrm{e}^{-c\sqrt{\ln x}}\Big)$$

with

$$g(t) := \frac{\sqrt{-t\zeta(1-t)L(1-t,\chi_4)}}{\pi(1-t)\sqrt{t(1-2^{t-1})}} \prod_{p\equiv 3 \,(\mathrm{mod}\, 4)} \frac{1}{\sqrt{1-p^{2t-2}}}.$$

Deduce an asymptotic expansion of which formula (4.90) is the specialization to order 1.

241. *Distribution of divisors in residue classes.* For each Dirichlet character χ, set $\tau(n;\chi) := \sum_{d|n} \chi(d)$ $(n \in \mathbb{N}^*)$. For positive integers a, q, n, also define

$$\tau_q(n) := \sum_{\substack{d|n \\ (d,q)=1}} 1, \qquad \tau(n;a,q) := \sum_{\substack{d|n \\ d\equiv a \,(\mathrm{mod}\, q)}} 1.$$

(a) Show that, for all integers $n \geqslant 1$, $q \geqslant 1$, we have

$$\frac{1}{\varphi(q)} \sum_{\chi\neq\chi_0} |\tau(n;\chi)|^2 = \sum_{\substack{1\leqslant a\leqslant q \\ (a,q)=1}} \Big\{\tau(n;a,q) - \frac{\tau_q(n)}{\varphi(q)}\Big\}^2$$

where the sum over χ covers all non-principal characters to the modulus q.

(b) Let z, w be complex numbers such that $|z| < 1 = |w|$. Establish the identity

$$\sum_{\nu\geqslant 0} z^\nu \Big| \sum_{0\leqslant j\leqslant \nu} w^j \Big|^2 = \frac{1-z^2}{(1-z)^2(1-wz)(1-\overline{w}z)}.$$

Deduce that, for each non-principal character χ to the modulus q, we have

$$F(s,\chi) := \sum_{n\geqslant 1} \frac{|\tau(n;\chi)|^2}{n^s} = g_q(s) \frac{\zeta(s)^2 L(s,\chi) L(s,\overline{\chi})}{\zeta(2s)}$$

with $g_q(s) := \prod_{p|q}(1+1/p^s)^{-1}$.

(c) Show that, for a non-principal character χ to the modulus q, and for $s = \sigma + i\tau$, $0 < \varepsilon < \frac{1}{2}$, $\varepsilon \leqslant \sigma \leqslant 1$, $|\tau| + 2 \leqslant T$, we have

$$L(s,\chi) \ll_\varepsilon \big(q^{1/2}T\big)^{1-\sigma+\varepsilon}.$$

(d) Applying the second effective Perron formula to the segment $[\kappa - iT, \kappa + iT]$ with $\kappa := 1 + 1/\ln x$, $T := (x/q)^{3/14}$, and moving the abscissa of integration to $\sigma = \frac{1}{2} + \varepsilon$, show that, for each $\varepsilon > 0$ and uniformly for $1 \leqslant q \leqslant x$, we have

$$\sum_{n\leqslant x} |\tau(n;\chi)|^2 = xP(\ln x, \chi) + O(q^{3/14}x^{11/14+\varepsilon})$$

where $P(Y,\chi)$ is a polynomial of the first degree in Y, to be determined explicitly in terms of the quantities

$$a(\chi) := \frac{6}{\pi^2}|L(1,\chi)|^2 \prod_{p\mid q}(1+1/p)^{-1},$$

$$b(\chi) := \sum_{p\mid q}\frac{\ln p}{p+1} + 2\gamma - 1 - \frac{2\zeta'(2)}{\zeta(2)} + 2\Re e\,\frac{L'(1,\chi)}{L(1,\chi)}.$$

Recall that $\zeta(s) \ll T^{(1-\sigma)/3+\varepsilon}$ when $\frac{1}{2} \leqslant \sigma \leqslant 1$, $1 \leqslant |\tau| \leqslant T$.

(e) Deduce from above the validity, for $1 \leqslant q \leqslant x$, of the asymptotic formula[10]

$$\sum_{n\leqslant x}\sum_{\substack{1\leqslant a\leqslant q\\(a,q)=1}}\Big\{\tau(n;a,q) - \frac{\tau_q(n)}{\varphi(q)}\Big\}^2 = A_q x\ln x + B_q x + O_\varepsilon\big(x^{11/14+\varepsilon}q^{3/14}\big)$$

with $A_q := \dfrac{6g_q(1)}{\pi^2\varphi(q)}\displaystyle\sum_{\chi\neq\chi_0}|L(1,\chi)|^2$ and

$$B_q := A_q\Big\{\sum_{p\mid q}\frac{\ln p}{p+1} + 2\gamma - 1 - \frac{2\zeta'(2)}{\zeta(2)}\Big\} + \frac{12g_q(1)}{\pi^2\varphi(q)}\sum_{\chi\neq\chi_0}\Re e\,L'(1,\chi)L(1,\overline{\chi}).$$

242. Define $W(x) := \sum_{n\leqslant x}\Lambda(n)\mu(n+1)^2$ $(x \geqslant 1)$. Express $W(x)$ in terms of the quantities $\psi(x;-1,d^2)$ for $d \leqslant \sqrt{x+1}$. Using Theorem 8.34, deduce an asymptotic formula for $W(x)$, and then one for the number $\pi_0(x)$ of those primes $p \leqslant x$ such that $p+1$ is squarefree.

243. Denote by χ_4 the unique non-principal Dirichlet character modulo 4. Recall that $L(1,\chi_4) = \pi/4$.

(a) Show that, for all $n \geqslant 1$, we have $\tau(n) = 2\sum_{\substack{d\mid n\\ d\leqslant\sqrt{n}}} 1 + O(1)$.

(b) Let $\varrho(d)$ be the number of solutions of the congruence

$$n^2 + 1 \equiv 0 \,(\mathrm{mod}\, d).$$

Show that $\sum_{d\geqslant 1}\varrho(d)/d^s = \zeta(s)L(s,\chi_4)/\zeta(2s)$. Describe the domain of convergence.

(c) Applying a Tauberian theorem, deduce an asymptotic formula for

$$R(x) := \sum_{d\leqslant x}\varrho(d)/d.$$

[10] Hall (1970, 1971b) obtains the more precise error term $\ll (qx)^{1/2+\varepsilon}$.

(d) Set $S(x) := \sum_{d \leqslant x} \varrho(d)$. Show that

$$\sum_{n \leqslant x} \tau(n^2+1) = 2xR(x) + O\big(x + S(x)\big). \tag{8.102}$$

(e) Show that $\varrho \leqslant \lambda * \mathbf{1}$ where λ is the completely multiplicative function defined by $\lambda(p) = 1$ if $p = 2$ or $p \equiv 1 \pmod 4$ and $\lambda(p) = 0$ if $p \equiv 3 \pmod 4$.

(f) Show that, with $h := 1/\ln x$, we have

$$\sum_{p \leqslant x} c_p/p = \sum_{p} c_p/p^{1+h} + O(1)$$

for any bounded sequence $\{c_p\}_p$. Using the results obtained in parts (c) and (d), deduce an elementary proof of the formula $x + S(x) = o(xR(x))$.

(g) By employing a theorem of Selberg–Delange type, show that we have

$$S(x) = Cx + O(x/\ln x) \qquad (x \to \infty),$$

where C is a positive constant to be made explicit. Deduce an asymptotic formula for the left-hand side of (8.102).

244. *Number of representations of an integer as a sum of two squares.* Retain the notations χ_4 and ϱ from Exercise 243 and write $r(n)$ for the number of representations of $n \in \mathbb{N}^*$ as a sum $a^2 + b^2$ of two integral squares. Denote by $r^+(n)$ the number of representations satisfying the extra conditions $(a,b) = 1$, $a \in \mathbb{N}^*$, $b \in \mathbb{N}^*$, and by $r^*(n)$ the number of representations such that $(a,b) = 1$, $a \in \mathbb{Z}$, $b \in \mathbb{Z}$. Finally, for $n \in \mathbb{N}^*$ and $x \in (\mathbb{Z}/n\mathbb{Z})^*$, denote by $\overline{x}$ the inverse of x modulo n.

(a) Using the identity (4.88), show that, for all $n \in \mathbb{N}^*$, if the pairs (a,b) and (A,B) are counted by $r^+(n)$ and are such that $a\overline{b} \equiv A\overline{B} \pmod n$, then $a = A$ and $b = B$. Deduce that $r^+(n) \leqslant \varrho(n)$.

(b) Let $x \in [0, n[$ be an integer such that $x^2 + 1 \equiv 0 \pmod n$. Applying, with $Q := \lfloor \sqrt{n} \rfloor$, Dirichlet's approximation Theorem I.7.1 to

$$\vartheta := x/n,$$

construct a pair (a,b) counted by $r^+(n)$. Deduce that the formula $r^+(n) = \varrho(n)$ holds identically.

(c) Show that, for every $n \in \mathbb{N}^*$, we have $r^*(n) = 4\varrho(n)$.

(d) Show that, for every $n \in \mathbb{N}^*$, we have $r(n) = 4\sum_{d^2|n} \varrho(n/d^2)$. Appealing to the formula established in Exercise 243(b), deduce that

$$r(n) = 4\sum_{d|n} \chi_4(d) \qquad (n \geqslant 1). \tag{8.103}$$

Part III

Probabilistic Methods

Densities

1.1. Definitions. Natural density

Like other parts of mathematics (and maybe even more so!) number theory is confronted with the problem of rigorously formalizing intuitive notions. Foremost among these figures *the probability that an integer belongs to a given sequence.* The point here is to give a mathematical meaning to statements of the type: one integer in two is even, almost no integer is the sum of two squares, etc.

The first approach which comes to mind is naturally to have recourse to the established theory of probability. Defining a probability measure on $\mathbb{N}^*$ does indeed allow associating a probability with each subset $\mathcal{A}$ of the set of integers. However, the following result shows that such a framework fundamentally contradicts one of our strongest intuitions about numbers: that which suggests that the proportion of integers divisible by $a \geqslant 1$ is exactly $1/a$.

Theorem 1.1. *For $a \in \mathbb{N}^*$, denote by $a\mathbb{N}^*$ the set of positive multiples of a. There exists no probability measure P on $\mathbb{N}^*$ such that*

$$P(a\mathbb{N}^*) = 1/a \qquad (a = 1, 2, \ldots).$$

Proof. Let us argue by contradiction. Since

$$a\mathbb{N}^* \cap b\mathbb{N}^* = ab\mathbb{N}^*$$

whenever $(a, b) = 1$, we see that, under this assumption, the events $a\mathbb{N}^*$ and $b\mathbb{N}^*$ are independent. The same holds for their complements $\mathbb{N}_a$ and $\mathbb{N}_b$,

with the notation $\mathbb{N}_a := \mathbb{N}^* \setminus a\mathbb{N}^*$. Therefore

$$P(\mathbb{N}_a \cap \mathbb{N}_b) = \Big(1 - \frac{1}{a}\Big)\Big(1 - \frac{1}{b}\Big)$$

when $(a, b) = 1$. Inductively, we immediately obtain, for all integers m, n, $m < n$,

$$P(\{m\}) \leqslant P\Big(\bigcap_{m<p\leqslant n} \mathbb{N}_p\Big) = \prod_{m<p\leqslant n} \Big(1 - \frac{1}{p}\Big),$$

where, as usual, the letter p denotes a prime. Since n is arbitrarily large, we deduce from Mertens' theorem that $P(\{m\}) = 0$ for all $m \geqslant 1$, which yields the desired contradiction. □

To define a probability law on $\mathbb{N}^*$ amounts to specifying a convergent series with sum 1, viz.

$$\sum_{n\geqslant 1} \lambda_n = 1,$$

with $0 \leqslant \lambda_n \leqslant 1$ for all n. For any integer sequence[1] $\mathcal{A} \subseteq \mathbb{N}^*$, we then have

$$P(\mathcal{A}) = \sum_{a\in\mathcal{A}} \lambda_a$$

and we can further appreciate the discrepancy between such a model and intuition by noting that the probability of a sequence virtually depends only on its initial terms. Indeed, for each $\varepsilon > 0$ there is an $N = N_\varepsilon$ such that

$$P\big(\{1, 2, \ldots, N\}\big) \geqslant 1 - \varepsilon\ .$$

We thus find ourselves in a typical situation where the only theory at our disposal irrevocably invalidates two natural "theorems", namely

(i) $P(a\mathbb{N}^*) = 1/a \qquad (a = 1, 2, \ldots)$,

(ii) $P(\mathcal{A}) = P(\mathcal{B}) \qquad (|\mathcal{A}\Delta\mathcal{B}| < \infty)$,

where $\mathcal{A}\Delta\mathcal{B}$ denotes the symmetric difference of $\mathcal{A}$ and $\mathcal{B}$.

Making the choice (standard throughout the history of mathematics) of preferring intuitive theorems to established theories, it is not hard to circumvent these difficulties. We introduce a *divergent* series with non-negative terms

$$\sum_{n\geqslant 1} \lambda_n = \infty$$

and define the *density* $d(\mathcal{A})$ of a subset $\mathcal{A} \subseteq \mathbb{N}^*$ as the limit, when it exists, of the ratio

$$d(\mathcal{A}; x) := \sum_{a\leqslant x,\, a\in\mathcal{A}} \lambda_a \Big/ \sum_{n\leqslant x} \lambda_n \tag{1.1}$$

[1] In probabilistic number theory, it is customary to designate as *an integer sequence* a strictly increasing sequence of positive integers, which therefore can also be considered as a subset of $\mathbb{N}^*$.

as $x \to \infty$. It plainly follows that the density of any finite set is zero, but the concept thus introduced is not a measure on $\mathbb{N}^*$:

(iii) sequences possessing a density do not form a σ-algebra,

(iv) density is not countably additive.

The simplest choice consists in setting $\lambda_n = 1$ for all $n \geqslant 1$. We thus obtain the notion of *natural density*, or *asymptotic density*, both terms being commonly used. When it exists, the natural density of $\mathcal{A}$ is given by the formula

$$\mathbf{d}\mathcal{A} := \lim_{x\to\infty} x^{-1}\big|\{a \leqslant x : a \in \mathcal{A}\}\big|. \tag{1.2}$$

One denotes by upper (resp. lower) natural (or asymptotic) density the quantity $\overline{\mathbf{d}}\mathcal{A}$ (resp. $\underline{\mathbf{d}}\mathcal{A}$) obtained by replacing the symbol lim by lim sup (resp. lim inf) in (1.2).

Before proceeding further, we make four simple observations.

(a) Every arithmetic progression $n \equiv a \,(\mathrm{mod}\, q)$ has natural density, equal to $1/q$. This is immediate, since in this case we have

$$\big|\{a \leqslant x : a \in \mathcal{A}\}\big| = \lfloor x/q \rfloor + O(1).$$

The intuitive property (i) stated above is thus verified.

(b) A necessary and sufficient condition for the increasing sequence $a_1 < a_2 < a_3 < \dots$ to be of natural density α, $0 \leqslant \alpha \leqslant 1$, is that

$$\lim_{n\to\infty} n/a_n = \alpha. \tag{1.3}$$

The condition is clearly sufficient. We see that it is necessary by noting that, if $\{a_j\}_{j=1}^{\infty}$ has natural density α, then

$$n = \big|\{j : a_j \leqslant a_n\}\big| = \{\alpha + o(1)\}a_n \qquad (n \to \infty).$$

(c) There exist integer sequences failing to have natural density. Consider, for example, the sequence $\mathcal{A}$ of those integers n with leading digit equal to 1 in the expansion to base 10. We have

$$\mathcal{A} := \bigcup_{k\geqslant 0} \{n : 10^k \leqslant n < 2{\cdot}10^k\}. \tag{1.4}$$

Writing $A(x) := \big|\mathcal{A} \cap [1, x]\big|$, it follows that, for $m \geqslant 1$,

$$A\big(10^m - 1\big) = \sum_{0\leqslant k<m} 10^k = \tfrac{1}{9}\big(10^m - 1\big),$$

$$A\big(2{\cdot}10^m - 1\big) = \tfrac{1}{9}(10^m - 1) + 10^m = \tfrac{5}{9}\big(2{\cdot}10^m - 1\big) + \tfrac{4}{9}.$$

We easily infer that

$$\underline{\mathbf{d}}\mathcal{A} = \tfrac{1}{9}, \qquad \overline{\mathbf{d}}\mathcal{A} = \tfrac{5}{9}.$$

(d) A formal link between the notion of density and probability theory can be defined. In the case of natural density, it suffices to observe that, if ν_N denotes the probability measure on $\mathbb{N}^*$ obtained by assigning the uniform weight $1/N$ to each of the first N integers, then we have, subject to existence,

$$\mathbf{d}\mathcal{A} = \lim_{N\to\infty} \nu_N(\mathcal{A}).$$

This explains why the natural density echoes intuitive criteria: the density of a sequence is the limit of its frequency among the first N integers.

1.2. Logarithmic density

The density most used after the natural density is that which is obtained by choosing $\lambda_n = 1/n$ for $n \geqslant 1$. The concept thus defined is called *logarithmic density*. The traditional notation is $\boldsymbol{\delta}\mathcal{A}$, so, assuming existence,

$$\boldsymbol{\delta}\mathcal{A} := \lim_{x\to\infty} \frac{1}{\ln x} \sum_{a\leqslant x,\, a\in\mathcal{A}} \frac{1}{a}. \tag{1.5}$$

The upper $\overline{\boldsymbol{\delta}}\mathcal{A}$ and lower $\underline{\boldsymbol{\delta}}\mathcal{A}$ logarithmic densities are defined in an obvious way.

It is easy to construct examples of sequences failing to possess logarithmic density. The following theorem makes clear that they are to be sought amongst the sequences that do not have a natural density.

Theorem 1.2. *For any integer sequence $\mathcal{A}$, we have*

$$\underline{\mathbf{d}}\,\mathcal{A} \leqslant \underline{\boldsymbol{\delta}}\mathcal{A} \leqslant \overline{\boldsymbol{\delta}}\mathcal{A} \leqslant \overline{\mathbf{d}}\,\mathcal{A}. \tag{1.6}$$

In particular, if the sequence $\mathcal{A}$ has a natural density, then it also has a logarithmic density and the two are equal.

Proof. Define $A(x) := \sum_{a\leqslant x} 1$, and $L(x) := \sum_{a\leqslant x} 1/a$, where the sums are over elements a of $\mathcal{A}$. By partial summation, we have

$$L(x) = \frac{A(x)}{x} + \int_1^x \frac{A(t)}{t^2}\,\mathrm{d}t \qquad (x \geqslant 1). \tag{1.7}$$

Let $\varepsilon > 0$. There exists some $t_0 = t_0(\varepsilon)$ such that

$$\underline{\mathbf{d}}\,\mathcal{A} - \varepsilon \leqslant A(t)/t \leqslant \overline{\mathbf{d}}\mathcal{A} + \varepsilon \qquad (t > t_0).$$

Substituting in (1.7) for $x > t_0$, we infer that

$$\big(\underline{\mathbf{d}}\,\mathcal{A} - \varepsilon\big) \ln\big(x/t_0\big) \leqslant L(x) \leqslant 1 + \ln t_0 + \big(\overline{\mathbf{d}}\mathcal{A} + \varepsilon\big) \ln\big(x/t_0\big).$$

The stated result follows from these bounds upon letting x tend to ∞ and then ε to 0. □

The converse of Theorem 1.2 is false: the existence of logarithmic density in no way implies that of natural density. A counter-example is provided by the sequence (1.4). The following computation shows that it has a logarithmic density $\delta\mathcal{A} = (\ln 2)/\ln 10$. Indeed

$$L(x) = \sum_{a\leqslant x}\frac{1}{a} = \sum_{0\leqslant k\leqslant \ln x/\ln 10}\ \sum_{\substack{10^k\leqslant n<2\cdot 10^k\\ n\leqslant x}}\frac{1}{n}$$

$$= \sum_{0\leqslant k\leqslant \ln x/\ln 10}\Big\{\ln 2 + O\Big(\frac{1}{k+1}\Big)\Big\} + O(1) = \frac{\ln 2}{\ln 10}\ln x + O(\ln_2 x).$$

1.3. Analytic density

The method described in § 1.1 for defining the density of an integer sequence can be formally generalized. Instead of truncating a divergent series $\sum_{n\geqslant 1}\lambda_n$, we consider a continuous family $(P_\sigma)_{\sigma\in S}$ of probability laws on $\mathbb{N}^*$ and define

$$P_\sigma(\{n\}) =: \lambda_n(\sigma)\Big/\sum_{m\geqslant 1}\lambda_m(\sigma)$$

where each series $\sum_{m\geqslant 1}\lambda_m(\sigma)$ ($\sigma\in S$) is convergent. In addition, we consider a point σ_0, in the closure of S but not belonging to S, such that the series $\sum_{m\geqslant 1}\lambda_m(\sigma_0)$, where each term is defined by continuity, diverges. We then define $d(\mathcal{A})$ as the possible limit of the ratio

$$P_\sigma(\mathcal{A}) = \sum_{n\in\mathcal{A}}\lambda_n(\sigma)\Big/\sum_{m\geqslant 1}\lambda_m(\sigma) \tag{1.8}$$

as $\sigma\to\sigma_0$ while remaining in S. The case of Section 1.1 corresponds to $S :=]0,1]$, with $\sigma_0 := 0$, and

$$\lambda_n(\sigma) := \begin{cases}\lambda_n & (n\leqslant 1/\sigma)\\ 0 & (n>1/\sigma).\end{cases}$$

The convergence of the series $\sum_{m\geqslant 1}\lambda_m(\sigma)$ for $\sigma\in S$ implies, for each $\varepsilon>0$, the existence of an integer $N = N(\varepsilon,\sigma)$ such that

$$P_\sigma(\mathcal{A}) = \Big(\sum_{n\leqslant N,\, n\in\mathcal{A}}\lambda_n(\sigma)\Big/\sum_{m\leqslant N}\lambda_m(\sigma)\Big) + \vartheta\varepsilon,$$

with $|\vartheta|\leqslant 1$. By choosing $\varepsilon = \varepsilon(\sigma)\to 0$ as $\sigma\to\sigma_0$, we hence recover a definition analogous to that of Section 1.1, but in which the function λ_n also depends on x. In certain circumstances, it may be established by a Tauberian technique that there exists an equivalent procedure which is strictly of type (1.1). Even then, such a framework can be useful: when the series

$\sum \lambda_n(\sigma)$ are well chosen, considering (1.8) instead of (1.1) can notably simplify the calculations, or favor the use of certain analytic techniques. This stems from replacing truncation by a smoother procedure.

The basic example is obtained from the choice $S :=]1, \infty[$, $\sigma_0 := 1$ and $\lambda_n(\sigma) := 1/n^\sigma$ $(n \geqslant 1)$. This yields

$$P_\sigma(\mathcal{A}) = \frac{1}{\zeta(\sigma)} \sum_{n \in \mathcal{A}} \frac{1}{n^\sigma}. \tag{1.9}$$

The presence of the Dirichlet series associated to the indicator function of $\mathcal{A}$ opens the possibility of employing all the analytic and algebraic properties of this class of series—especially those connected to the convolution product—for the computation of density.

The *analytic density* of a sequence $\mathcal{A}$ is defined as the possible limit of the expression (1.9) as $\sigma \to 1+$. Instead of (1.9), we may, of course, equally well consider the quantity

$$(\sigma - 1) \sum_{n \in \mathcal{A}} \frac{1}{n^\sigma} \tag{1.10}$$

but it is often convenient to retain the factor $1/\zeta(\sigma)$ when Euler products are involved.

The expected connection between analytic density and a density of type (1.1) is made explicit in the following result.

Theorem 1.3. *Let $\mathcal{A}$ be an integer sequence. Then $\mathcal{A}$ has analytic density if, and only if, it has logarithmic density. In this case, the two densities are equal.*

Proof. Let us retain the notation $L(x) = \sum_{a \leqslant x} 1/a$ where summation is restricted to elements a of $\mathcal{A}$. By partial summation, we have

$$(\sigma - 1) \sum \frac{1}{a^\sigma} = (\sigma - 1)^2 \int_1^\infty \frac{L(x)}{x^\sigma} \, \mathrm{d}x \qquad (\sigma > 1). \tag{1.11}$$

Suppose first that $\mathcal{A}$ has logarithmic density $\delta = \delta\mathcal{A}$, i.e.,

$$L(x) = \{\delta + o(1)\} \ln x \qquad (x \to \infty).$$

Substituting in the right-hand side of (1.11), we infer that

$$(\sigma - 1) \sum \frac{1}{a^\sigma} = \delta + o(1) \qquad (\sigma \to 1+); \tag{1.12}$$

thus $\mathcal{A}$ has analytic density, equal to δ.

Conversely, if (1.12) holds, we can rewrite (1.11) in the following form, with $h := \sigma - 1$,

$$h \int_0^\infty \mathrm{e}^{-ht} \, \mathrm{d}L(\mathrm{e}^t) = h^2 \int_1^\infty L(x) \frac{\mathrm{d}x}{x^{1+h}} = \delta + o(1) \qquad (h \to 0+).$$

Karamata's Theorem II.7.5 then implies

$$L(e^t) = \{\delta + o(1)\}t \qquad (t \to \infty),$$

which is equivalent to the required result. □

1.4. Probabilistic number theory

The concepts introduced in the previous sections form the basis of a new branch of number theory. The key idea is that of natural density, which sheds new light on arithmetic functions.

Indeed, these functions usually have the specific property of varying in an intrinsically irregular and erratic manner, with the consequence that classical techniques of analysis are generally powerless to describe their behavior appropriately. Probabilistic number theory corresponds to the challenge, arising naturally in such circumstances, to undertake a statistical study of this behavior. Together with the notions of extremal and average orders, which allow a cursory classification, we shall also define (cf. Chapter III.3) the notion of a *normal order* of an arithmetic function, designed to reflect its "almost certain" behavior. In practice, this involves a process of neglecting a set of integers of zero density (obviously depending on the function considered) so as to eliminate aberrant values. Strikingly, this approach results in order and regularity suddenly emerging from apparent intrinsic chaos. The illuminating focus of the "almost everywhere" concept reveals a new field of investigation, requiring specific methods and yielding distinctive results.

Thus, as remarked by Delange (1982), one ought rather to speak of the "probabilistic theory of arithmetic functions". The typical viewpoint (see remark (d) in § 1.1) is that of considering an arithmetic function f as a random variable on the discrete space formed by the first N integers and equipped with the uniform law. The fundamental question then consists in determining in what sense(s) we can assert that the law of f tends toward a limit law as $N \to \infty$. We shall make this precise in the following chapters.

Notes

§§ 1.1–1.3. Many other types of densities for an integer sequence have been defined in the literature. Let us mention three.

(a) The *multiplicative density* of Davenport & Erdős (1951), connected with the distribution of $\mathcal{A}$ among the different subsequences of $\mathbb{N}^*$ obtained by removing those integers having a "large" prime factor—cf. Exercise 246.

(b) The *Schnirelmann* (1930) density, defined by $\sigma(\mathcal{A}) = \inf_{n \geqslant 1} A(n)/n$. It is connected with the *addition of sequences*; if we define

$$\mathcal{A} + \mathcal{B} := \{a + b : a \in \mathcal{A},\ b \in \mathcal{B}\},$$

an important theorem due to Mann (1942) states that

$$\sigma(\mathcal{A} + \mathcal{B}) \geqslant \min\{1, \sigma(\mathcal{A}) + \sigma(\mathcal{B})\}. \tag{1.13}$$

A fairly simple proof of (1.13) may be found in Halberstam & Roth (1983), chap. I, § 4.

(c) The *divisor density* of Hall (1978). When it exists, this is the unique number $\mathbf{D}\mathcal{A}$ satisfying

$$\sum_{d|n,\, d\in\mathcal{A}} 1 = \{\mathbf{D}\mathcal{A} + o(1)\}\tau(n) \qquad \text{pp},$$

where the symbol pp (for the French *presque partout*, i.e. almost everywhere) indicates that the relation thus marked occurs as $n \to \infty$ inside some suitable sequence of natural density 1. This definition is quite different from those in §§ 1.1–1.3. For example, for all α, β in $[0, 1]$ we can find a sequence $\mathcal{A}$ such that $\mathbf{d}\mathcal{A} = \alpha$, $\mathbf{D}\mathcal{A} = \beta$ (Hall, 1978). Other properties may be found in Hall (1981), Tenenbaum (1982), Dupain, Hall & Tenenbaum (1982), Hall & Tenenbaum (1986). See also Exercises 272 and 273.

§ 1.3. A deeper study of the properties of arithmetic functions in connection with the laws $P_\sigma(\mathcal{A})$ has been undertaken by Nanopoulos (1975, 1977, 1982).

Exercises

Recall that we denote by $P^+(n)$ the largest prime factor of an integer n, with the convention that $P^+(1) = 1$. Except in Exercise 252, we define, for $y \geqslant 2$, and $\mathcal{A} \subseteq \mathbb{N}^*$,

$$\mathcal{A}_y := \mathcal{A} \cap \{n : P^+(n) \leqslant y\}.$$

For each integer $j \geqslant 1$, we let $\mathcal{A}^{(j)}$ denote the finite sequence comprising the j smallest elements of $\mathcal{A}$ and we make the convention that $\mathcal{A}^{(0)} = \varnothing$.

245. For $n \geqslant 1$, write $n_y := \max\{d : d|n,\ P^+(d) \leqslant y\}$. Let $\mathcal{A} \subseteq \mathbb{N}^*$ be an integer sequence satisfying, for some $y \geqslant 2$, $\mathcal{A} = \{n : n_y \in \mathcal{A}_y\}$. Show that, for $s \in \mathbb{C}$, $\sigma > 1$, we have

$$\sum_{n\in\mathcal{A}} \frac{1}{n^s} = \zeta(s) \prod_{p\leqslant y} \Big(1 - \frac{1}{p^s}\Big) \sum_{a\in\mathcal{A}_y} \frac{1}{a^s}.$$

Deduce the existence of $\mathbf{d}\mathcal{A}$ and the formula

$$\mathbf{d}\mathcal{A} = \prod_{p\leqslant y} \big(1 - \frac{1}{p}\big) \sum_{a\in\mathcal{A}_y} \frac{1}{a}.$$

246. *Multiplicative density (Davenport & Erdős, 1951).*

For $y \geqslant 2$, and $\mathcal{A} \subseteq \mathbb{N}^*$, we define

$$\mathbf{m}_y\mathcal{A} := \prod_{p\leqslant y} \big(1 - \frac{1}{p}\big) \sum_{a\in\mathcal{A}_y} \frac{1}{a}.$$

If $\mathbf{m}_y\mathcal{A}$ tends to a limit $\mathbf{m}\mathcal{A}$ as $y \to \infty$, we say that $\mathbf{m}\mathcal{A}$ is the multiplicative density of $\mathcal{A}$. We also set

$$\overline{\mathbf{m}}\mathcal{A} := \limsup_{y\to\infty} \mathbf{m}_y\mathcal{A}, \qquad \underline{\mathbf{m}}\mathcal{A} := \liminf_{y\to\infty} \mathbf{m}_y\mathcal{A}.$$

(a) Show that, for all $\mathcal{A}$, we have $0 \leqslant \underline{\mathbf{m}}\mathcal{A} \leqslant \overline{\mathbf{m}}\mathcal{A} \leqslant 1$.

(b) Show that, for any integer sequence $\mathcal{A}$, we have $\overline{\boldsymbol{\delta}}\mathcal{A} \leqslant \mathrm{e}^{\gamma}\,\overline{\mathbf{m}}\mathcal{A}$.

(c) Let $\mathcal{A} := \{n \geqslant 1 : P^+(n) \leqslant \sqrt{n}\}$. Show that

$$\mathbf{d}\mathcal{A} = 1 - \ln 2, \qquad \mathbf{m}\mathcal{A} = 1 - \mathrm{e}^{-\gamma}.$$

247. *Sequential density of a set of multiples.*[2]

Let $\mathcal{A}$ be an integer sequence. Its *set of multiples* $\mathcal{M}(\mathcal{A})$ is the subset of $\mathbb{N}^*$ defined by $\mathcal{M}(\mathcal{A}) := \{am : a \in \mathcal{A},\ m \geqslant 1\}$.[3]

[2] Davenport & Erdős (1951).

[3] See Halberstam & Roth (1966), Erdős, Hall & Tenenbaum (1994).

(a) Show that, for each $j \geqslant 1$, $\mathcal{M}_j := \mathcal{M}(\mathcal{A}^{(j)}) \setminus \mathcal{M}(\mathcal{A}^{(j-1)})$ has natural density $\mathbf{d}\mathcal{M}_j = \Delta_j(\mathcal{A})$ given by

$$\Delta_j(\mathcal{A}) := \frac{1}{a_j} + \sum_{1 \leqslant k \leqslant j-1} (-1)^k \sum_{1 \leqslant i_1 < \ldots < i_k \leqslant j-1} \frac{1}{[a_{i_1}, \ldots, a_{i_k}, a_j]},$$

where a_j denotes the jth element of $\mathcal{A}$.

(b) Show that $\Delta(\mathcal{A}) := \sum_{j \geqslant 1} \Delta_j(\mathcal{A})$ is finite and satisfies $0 \leqslant \Delta(\mathcal{A}) \leqslant 1$. We say that $\Delta(\mathcal{A})$ is the *sequential density* of the set of multiples $\mathcal{M}(\mathcal{A})$.

(c) Show that, if $\sum_{j \geqslant 1} 1/a_j < \infty$, then $\mathcal{M}(\mathcal{A})$ has natural density and $\mathbf{d}\mathcal{M}(\mathcal{A}) = \Delta(\mathcal{A})$.

248. Let $\mathcal{A} \subseteq \mathbb{N}^*$, $y \geqslant 2$.

(a) Deduce from the results of Exercises 245 and 247 that

$$\mathbf{d}\mathcal{M}(\mathcal{A}_y) = \Delta(\mathcal{A}_y) = \mathbf{m}_y\mathcal{M}(\mathcal{A}).$$

(b) Show that $\mathbf{m}_y\mathcal{M}(\mathcal{A})$ is an increasing function of y.

(c) Let $j \geqslant 1$. Show that, for sufficiently large y, we have

$$\Delta_j(\mathcal{A}_y) = \Delta_j(\mathcal{A}).$$

Deduce that $\mathbf{m}\mathcal{M}(\mathcal{A}) \geqslant \Delta(\mathcal{A})$.

(d) Show that $\Delta(\mathcal{A}_y) \leqslant \Delta(\mathcal{A})$. Deduce that $\mathbf{m}\mathcal{M}(\mathcal{A}) = \Delta(\mathcal{A})$, in other words: *Any set of multiples has multiplicative density, equal to its sequential density* (Davenport–Erdős, 1951).

249. *The Davenport–Erdős theorem (1951).*

Let $\mathcal{A} \subseteq \mathbb{N}^*$, $y \geqslant 2$.

(a) Deduce from Exercise 245 that $\mathbf{d}\mathcal{M}(\mathcal{A}_y) = \mathbf{m}_y\mathcal{M}(\mathcal{A})$.

(b) Let $x > y$. Show that

$$\sum_{\substack{n \in \mathcal{M}(\mathcal{A}) \setminus \mathcal{M}(\mathcal{A}_y) \\ P^+(n) \leqslant x}} \frac{1}{n} = \prod_{p \leqslant x} \Big(1 - \frac{1}{p}\Big)^{-1} \{\mathbf{m}_x\mathcal{M}(\mathcal{A}) - \mathbf{m}_y\mathcal{M}(\mathcal{A})\}.$$

(c) By applying the result of Exercise 248, show that

$$\overline{\boldsymbol{\delta}}\mathcal{M}(\mathcal{A}) \leqslant \mathbf{m}\mathcal{M}(\mathcal{A}).$$

(d) Show that $\mathbf{m}\mathcal{M}(\mathcal{A}) = \boldsymbol{\delta}\mathcal{M}(\mathcal{A}) = \Delta(\mathcal{A}) = \underline{\mathbf{d}}\,\mathcal{M}(\mathcal{A})$, in other words: *Any set of multiples has a logarithmic density equal to its lower natural density, as well as to its multiplicative and sequential densities.*[4]

[4] A set of multiples does not necessarily have a natural density, cf. Exercise 271.

250. *The Davenport–Erdős theorem (proof of 1937).*

Let $\mathcal{A} \subseteq \mathbb{N}^*$. For $j \geqslant 1$, let $\vartheta_j(n)$ denote the set-theoretic indicator function of $\mathcal{M}_j := \mathcal{M}(\mathcal{A}^{(j)}) \setminus \mathcal{M}(\mathcal{A}^{(j-1)})$ and put $\Theta_j(n) := \sum_{1\leqslant i\leqslant j} \vartheta_i(n)$. We consider the Dirichlet series

$$F_j(s) := \sum_{n\geqslant 1} \frac{\vartheta_j(n)}{n^s}, \qquad F(s) := \sum_{j\geqslant 1} F_j(s) = \sum_{n\in\mathcal{M}(\mathcal{A})} \frac{1}{n^s},$$
$$G_j(s) := F_j(s)/\zeta(s), \qquad G(s) := F(s)/\zeta(s).$$

(a) Show that, for $n \geqslant 1$, $j \geqslant 1$, we have

$$\Theta_j(n) \ln n \geqslant \sum_{d|n} \Theta_j(d)\Lambda(n/d).$$

Deduce that $\sum_{1\leqslant i\leqslant j} G_i(\sigma)$ is a decreasing function of $\sigma > 1$.

(b) With the notation of Exercise 247, show that $G_j(1) = \Delta_j(\mathcal{A})$.

(c) Show that $\lim_{\sigma\to 1+} G(\sigma) = \Delta(\mathcal{A})$.

(d) Show that $\boldsymbol{\delta}\mathcal{M}(\mathcal{A}) = \Delta(\mathcal{A}) \geqslant \underline{\mathbf{d}}\,\mathcal{M}(\mathcal{A})$.

(e) Show that, for each $j \geqslant 1$, we have $\underline{\mathbf{d}}\,\mathcal{M}(\mathcal{A}) \geqslant \sum_{1\leqslant i\leqslant j} \Delta_i(\mathcal{A})$. Deduce that $\underline{\mathbf{d}}\,\mathcal{M}(\mathcal{A}) = \Delta(\mathcal{A})$.

251. *A theorem of Rényi (1955).*

Let $z \in \mathbb{C}$. Write $\varphi_z(n) := z^{\Omega(n)-\omega(n)}$, $\lambda_z(n) := (\varphi_z * \mu)(n)$.

(a) Show that, for $|z| < 2$, we have $\sum_{n\geqslant 1} |\lambda_z(n)|/n < +\infty$.

(b) Deduce an asymptotic formula for $\sum_{n\leqslant x} \varphi_z(n)$.

(c) Show that, for each integer $k \geqslant 0$, the sequence $\{n : \Omega(n)-\omega(n) = k\}$ has natural density d_k, given by the formula

$$\sum_{k\geqslant 0} d_k z^k = \frac{6}{\pi^2} \prod_p \Big(\frac{1 - z/(p+1)}{1 - z/p}\Big) \qquad (|z| < 2).$$

252. *Direct factors of* $\mathbb{N}^*$.

Two subsets $\mathcal{A}$ and $\mathcal{B}$ of $\mathbb{N}^*$ are said to form a pair of direct factors if $1 \in \mathcal{A}\cap\mathcal{B}$ and if each integer $n > 1$ decomposes uniquely as a product $n = ab$ with $a \in \mathcal{A}$, $b \in \mathcal{B}$. When this is so, we write $a = \pi_A(n)$, $b = \pi_B(n)$, with the convention that $\pi_A(1) = \pi_B(1) = 1$. For $y \geqslant 2$, $n \geqslant 1$, we denote by n_y the largest divisor of n free of prime factors $> y$, so that $n_y := \prod_{p^\nu \| n,\, p\leqslant y} p^\nu$, and set

$$\mathcal{A}_y := \{n : n \in \mathbb{N}^*,\, n_y \in \mathcal{A}\}, \quad A_y(x) := |\mathcal{A}_y \cap [1, x]|, \quad A(x) := |\mathcal{A} \cap [1, x]|.$$

Throughout the exercise, we let a (resp. b) denote generically an integer in $\mathcal{A}$ (resp. $\mathcal{B}$) where $(\mathcal{A}, \mathcal{B})$ is a given pair of direct factors of $\mathbb{N}^*$.

(a) Show that, for each $y \geqslant 2$, we have

$$\sum_{P^+(a)\leqslant y} \frac{1}{a} \sum_{P^+(b)\leqslant y} \frac{1}{b} = \prod_{p\leqslant y} \Big(1 - \frac{1}{p}\Big)^{-1}.$$

(b) Show the existence of $\mathbf{d}\mathcal{A}_y$ and the formula

$$\mathbf{d}\mathcal{A}_y = \Big(\sum_{P^+(b)\leqslant y} \frac{1}{b} \Big)^{-1}.$$

(c) Let $\varphi_y : \mathcal{A} \to \mathcal{A}_y$ be the map defined by $\varphi_y(a) = \pi_A(a_y)a/a_y$. Show that if $\varphi_y(a) = \varphi_y(a')$, then $a\pi_B(a'_y) = a'\pi_B(a_y)$. Deduce that φ_y is injective and establish that $A(x) \leqslant A_y(x)$ for $x \geqslant 1$, $y \geqslant 2$.

(d) Show that if $\sum 1/b = \infty$ then $\mathbf{d}\mathcal{A} = 0$.

(e) We now assume that $\sum 1/b < \infty$. Let α and β be the respective indicator functions of $\mathcal{A}$ and $\mathcal{B}$. Show that $\alpha = \mathbf{1} - \alpha * (\beta - \delta)$. Using the result of question (c) above, deduce that, for $y \geqslant 2$, we have

$$\underline{\mathbf{d}}\mathcal{A} \geqslant \Big(\sum_{P^+(b)\leqslant y} \frac{1}{b} \Big)^{-1} \Big(1 - \sum_{P^+(b)>y} \frac{1}{b} \Big).$$

(f) Show that $\mathcal{A}$ has a natural density, given by the formula

$$\mathbf{d}\mathcal{A} = \Big(\sum_{b\in\mathcal{B}} \frac{1}{b} \Big)^{-1}$$

where the right-hand side is interpreted as 0 when the series diverges.[5]

[5] This result is due to Saffari (1976) and Erdős, Saffari & Vaughan (1979). The proof presented here is essentially that of Daboussi (1979).

Limiting distributions of arithmetic functions

2.1. Definition—distribution functions

As seen in the previous chapter, probabilistic number theory may be regarded as the asymptotic study of the probability space

$$\Omega_N := \{n : 1 \leqslant n \leqslant N\}$$

equipped with the uniform law ν_N. In this context, an arithmetic function may be viewed as a sequence of random variables

$$f_N = (f, \nu_N) \qquad (N = 1, 2, \ldots)$$

taking the values $f(n)$, $1 \leqslant n \leqslant N$, with probability $1/N$. We intend to investigate from this perspective the classical probabilistic notion of a distribution function.

Recall (see Ch. II.7) that a distribution function[1] is a non-decreasing function $F : \mathbb{R} \to [0, 1]$, which is right-continuous and satisfies $F(-\infty) = 0$, $F(+\infty) = 1$.

The set $\mathcal{D}(F)$ of *discontinuity points of* F is thus at most countable, and only contains jump discontinuities. We denote by $\mathcal{C}(F)$ the complement of $\mathcal{D}(F)$, that is *the set of continuity points* of F. A *point of increase* for F is a real number z such that $F(z + \varepsilon) - F(z - \varepsilon) > 0$ for any $\varepsilon > 0$. Each discontinuity point is a point of increase, but the converse does not hold.

[1] Abbreviated as d.f.

Write $\mathcal{D}(F) = \{z_\nu\}_{\nu=1}^{\infty}$ and $s_\nu := F(z_\nu) - F(z_\nu -)$. The function

$$\Phi(z) = \sum_{z_\nu \leqslant z} s_\nu$$

increases exclusively by jumps and is constant on any closed interval not containing some z_ν. It is called a step-function. If $\mathcal{D}(F)$ is not empty, then Φ is, up to a multiplicative constant, a distribution function. Such a d.f. is said to be *purely discrete* or *atomic*. It is immediately checked that $F - \Phi$ is continuous.

The d.f. F is called *improper* (or is said to be the d.f. of an improper law) if it equals a one-point step-function, say

$$F(z) = \begin{cases} 0 & (z < z_0) \\ 1 & (z \geqslant z_0). \end{cases}$$

A simple example of a continuous d.f. is a function of the type

$$F(z) = \int_{-\infty}^{z} h(t)\,\mathrm{d}t$$

where $h \geqslant 0$ is Lebesgue-integrable and satisfies $\|h\|_1 = 1$. It is then said that F is *absolutely continuous*. The Radon–Nikodym theorem (cf. for example, Rudin (1970), th. 6.9) implies that each continuous d.f. may be written in the form

$$c_0 F_0 + c_1 F_1$$

where F_0 is absolutely continuous and F_1 is *purely singular*, that is to say continuous and such that

$$\int_{\mathcal{N}} \mathrm{d}F_1(z) = 1$$

where $\mathcal{N}$ is a subset of $\mathbb{R}$ with zero Lebesgue measure.

The above observations are collected into the following statement.

Theorem 2.1 (Lebesgue decomposition theorem). *Each d.f. F can be uniquely represented in the form*

$$F = \alpha_1 F_1 + \alpha_2 F_2 + \alpha_3 F_3$$

with $\alpha_1, \alpha_2, \alpha_3 \geqslant 0$, $\alpha_1 + \alpha_2 + \alpha_3 = 1$, and where the F_i are d.f.'s such that F_1 is absolutely continuous, F_2 is purely singular, and F_3 is atomic.

A sequence $\{F_n\}_{n=1}^{\infty}$ of d.f.'s is said to *converge weakly* to a function F if we have

$$\lim_{n\to\infty} F_n(z) = F(z) \qquad (z \in \mathcal{C}(F)). \tag{2.1}$$

It is to be emphasised that the weak limit F is necessarily bounded and non-decreasing, but need not be a d.f. We may always suppose that it is right-continuous for (2.1) imposes no constraint when $z \in \mathcal{D}(F)$.

Let us consider a real arithmetic function f. For each $N \geqslant 1$ the function

$$F_N(z) := \nu_N\{n : f(n) \leqslant z\} = \frac{1}{N}\big|\{n \leqslant N : f(n) \leqslant z\}\big| \tag{2.2}$$

is an atomic d.f.

Definition 2.2. *A real arithmetic function f is said to possess a (limiting) distribution function F (or: is said to have a limit law with d.f. F) if the sequence F_N defined by* (2.2) *converges weakly to F, and F is indeed a d.f.*

Thus the existence of a d.f. for f is equivalent to combining the two following assertions:

(i) The limit $F(z) := \lim_{N\to\infty} F_N(z)$ exists for all z belonging to a certain dense subset E of $\mathbb{R}$.

(ii) We have $\lim\limits_{\substack{z\to\pm\infty \\ z\in E}} F(z) = \begin{cases} 1 \\ 0 \end{cases}$.

Indeed, F can then be extended to a non-decreasing, right-continuous function. This implies the validity of (2.1).

The following result, which has often been implicitly used by Erdős, furnishes a practical sufficient condition for an arithmetic function to possess a limit law.

Theorem 2.3. *Let f be a real arithmetic function. Assume that for each $\varepsilon > 0$ there exists a function $n \mapsto a_\varepsilon(n)$, with values in $\mathbb{Z}^+$ and having the following properties:*

(i) $\lim\limits_{\varepsilon\to 0} \limsup\limits_{T\to+\infty} \overline{\mathbf{d}}\{n : a_\varepsilon(n) > T\} = 0$,

(ii) $\lim\limits_{\varepsilon\to 0} \overline{\mathbf{d}}\{n : |f(n) - f(a_\varepsilon(n))| > \varepsilon\} = 0$,

(iii) *for each $a \geqslant 1$, the density $\mathbf{d}\{n : a_\varepsilon(n) = a\}$ exists.*

Then f has a limit law.

Proof. Select, as $\eta \to 0+$, two functions

$$\varepsilon = \varepsilon(\eta) \to 0+ \quad \text{and} \quad T = T(\varepsilon(\eta)) \to +\infty$$

such that the upper density in (i) is $\leqslant \eta$. Denote by $\mathbf{d}(a, \varepsilon)$ the density mentioned in (iii) and write

$$F(z, \eta) := \sum_{a\leqslant T(\varepsilon),\, f(a)\leqslant z} \mathbf{d}(a, \varepsilon), \qquad F(z) := \limsup_{\eta\to 0} F(z, \eta).$$

We first show that $F_N(z)$, as defined by (2.2), converges weakly to F. For $z \in \mathcal{C}(F)$, we have

$$F_N(z) \leqslant \frac{1}{N} \sum_{\substack{n \leqslant N \\ a_\varepsilon(n) \leqslant T(\varepsilon) \\ f(a_\varepsilon(n)) \leqslant z+\varepsilon}} 1 + \frac{1}{N} \sum_{\substack{n \leqslant N \\ a_\varepsilon(n) > T(\varepsilon)}} 1 + \frac{1}{N} \sum_{\substack{n \leqslant N \\ |f(n)-f(a_\varepsilon(n))|>\varepsilon}} 1.$$

By (iii), the first term in this upper bound equals $F(z+\varepsilon,\eta) + o(1)$ as $N \to \infty$. The other two may be estimated using (i) and (ii). Letting successively N tend to $+\infty$ and η tend to 0, we obtain

$$\limsup_{N\to\infty} F_N(z) \leqslant \limsup_{\eta\to 0} F\big(z+\varepsilon(\eta),\eta\big) = F(z).$$

The last equality results from the facts that $F(w,\eta)$ is a non-decreasing function of w and that $z \in \mathcal{C}(F)$. We infer symmetrically

$$\liminf_{N\to\infty} F_N(z) \geqslant \limsup_{\eta\to 0} F\big(z-\varepsilon(\eta),\eta\big) = F(z).$$

Thus F_N converges weakly to F and we may normalize this function so that it becomes right-continuous.

It remains to establish that

$$F(-\infty) = 0, \qquad F(+\infty) = 1.$$

Since $F(z) = \lim F_N(z)$ for $z \in \mathcal{C}(F)$, we obviously have $0 \leqslant F \leqslant 1$. Now let $\varepsilon > 0$. Choose z in $\mathcal{C}(F)$ such that $z > \max\{f(a) : a \leqslant T(\varepsilon)\} + \varepsilon$. Then $f(n) > z$ implies either $a_\varepsilon(n) > T(\varepsilon)$, or $|f(n) - f(a_\varepsilon(n))| > \varepsilon$. By (i) and (ii) the corresponding density $1 - F(z)$ tends to 0 as $\eta \to 0+$. This implies $F(+\infty) = 1$. We obtain $F(-\infty) = 0$ analogously. □

In the probabilistic study of an arithmetic function, a natural normalization is obtained by introducing the expectation and variance of f relative to ν_N, viz.

$$\mathbb{E}_N(f) := \int_{-\infty}^{+\infty} z \, \mathrm{d}F_N(z) = \frac{1}{N} \sum_{1 \leqslant n \leqslant N} f(n), \tag{2.3}$$

and

$$\begin{aligned} \mathbb{V}_N(f) = \mathbb{D}_N(f)^2 &:= \int_{-\infty}^{+\infty} \big\{z - \mathbb{E}_N(f)\big\}^2 \, \mathrm{d}F_N(z) \\ &= \frac{1}{N} \sum_{1 \leqslant n \leqslant N} \{f(n) - \mathbb{E}_N(f)\}^2. \end{aligned} \tag{2.4}$$

This suggests a different approach to the problem of the distribution of values of an arithmetic function. Instead of studying the asymptotic

behavior of $F_N(z)$, we consider that of

$$G_N(z) := \nu_N\{n : f(n) \leqslant \mathbb{E}_N(f) + z\mathbb{D}_N(f)\}. \tag{2.5}$$

This perspective may be considered as central[2] in the probabilistic theory of numbers.

The sequence of d.f.'s $F_N(z)$ or $G_N(z)$ contains all the information concerning the arithmetic function f, and questions usually raised about f may be translated into equivalent ones regarding d.f.'s. We give two examples.

(a) It is equivalent to say that the integer sequence $\mathcal{A}$ has natural density α and to assert that its indicator function $\mathbf{1}_{\mathcal{A}}(n)$ has an atomic distribution function

$$F(z) = \begin{cases} 0 & (z < 0) \\ 1-\alpha & (0 \leqslant z < 1) \\ 1 & (z \geqslant 1). \end{cases}$$

(b) If there exist two real sequences A_N, B_N, such that

$$H_N(z) := F_N\big(A_N + zB_N\big)$$

converges weakly to a d.f. $H(z)$ and if the convergence of the expectations $\int_{\mathbb{R}} z\,\mathrm{d}H_N(z)$ is dominated in Lebesgue's sense, then we have

$$\mathbb{E}_N(f) = A_N + \Big\{\int_{-\infty}^{+\infty} z\,\mathrm{d}H(z) + o(1)\Big\}B_N \qquad (N \to \infty).$$

Thus a sufficiently precise knowledge of the d.f. of f provides an estimate for the mean value.

As a general rule, one should only consider the study of an arithmetic function to be complete when its d.f. has been adequately described.

2.2. Characteristic functions

As was mentioned in Ch. II.7, the characteristic function[3] of F is the Fourier transform of the Stieltjes measure $\mathrm{d}F(z)$, viz.

$$\varphi(\tau) := \int_{-\infty}^{+\infty} \mathrm{e}^{i\tau z}\,\mathrm{d}F(z) \qquad (\tau \in \mathbb{R}). \tag{2.6}$$

It is a uniformly continuous function on the real line, satisfying

$$|\varphi(\tau)| \leqslant 1 = \varphi(0) \qquad (\tau \in \mathbb{R}).$$

[2]No pun intended, although the central limit theorem and convergence to the Gaussian law are unavoidable in this branch of number theory: see, in particular, Theorem 4.15 and Exercise 287, p. 508.

[3]Abbreviated as c.f. in the sequel.

The correspondence $F \leftrightarrow \varphi$ is one-to-one as confirmed by the inversion formula

$$F(z+h) - F(z) = \lim_{T\to\infty} \frac{1}{2\pi} \int_{-T}^{T} \frac{1 - \mathrm{e}^{-i\tau h}}{i\tau} \mathrm{e}^{-i\tau z} \varphi(\tau)\, \mathrm{d}\tau \tag{2.7}$$

valid for z and $z+h$ in $\mathcal{C}(F)$. The proof of (2.7) is analogous to that of the classical Fourier inversion formula, and presents no new difficulty. If F and G have the same c.f., it follows from (2.7) that $F(z+h)-F(z) = G(z+h)-G(z)$ for almost all z, h. Letting h tend to $-\infty$, we derive that $F(z) = G(z)$ almost everywhere. Since F and G are non-decreasing and right-continuous, this does imply that $F = G$.

The famous *continuity theorem* of Paul Lévy connects weak convergence of d.f.'s to pointwise convergence of c.f.'s.

Theorem 2.4 (Continuity theorem, Lévy, 1925). *Let $\{F_n\}_{n=1}^{\infty}$ be a sequence of d.f.'s and $\{\varphi_n\}_{n=1}^{\infty}$ the sequence of their c.f.'s. Then F_n converges weakly to a d.f. F if, and only if, φ_n converges pointwise on $\mathbb{R}$ to a function φ which is continuous at 0. Furthermore, in this case, φ is the c.f. of F, and the convergence of φ_n to φ is uniform on any compact subset.*

This classical result is established in detail in most books on probability theory—see for example Cramér (1970), Feller (1971), Loève (1963) and Lukacs (1970). Here we limit ourselves to indicating the main steps. The first point is an identity which follows easily from (2.7).

Lemma 2.5. *Let F be a d.f., φ its c.f. For $z \in \mathbb{R}$, $h > 0$, we have*

$$\frac{1}{h}\int_{z}^{z+h} F(t)\,\mathrm{d}t - \frac{1}{h}\int_{z-h}^{z} F(t)\,\mathrm{d}t = \frac{1}{2\pi}\int_{-\infty}^{+\infty} \Big(\frac{\sin(\tau/2)}{\tau/2}\Big)^2 \mathrm{e}^{-i\tau z/h} \varphi\Big(\frac{\tau}{h}\Big)\,\mathrm{d}\tau. \tag{2.8}$$

Proof of lemma. It suffices to apply (2.7) to the d.f.

$$F_0(z) := \frac{1}{h}\int_{z}^{z+h} F(t)\,\mathrm{d}t.$$

Partial integration immediately yields that the c.f. of $F_0(z)$ is

$$\varphi(\tau)(1 - \mathrm{e}^{-i\tau h})/i\tau h.$$

An obvious change of variables then furnishes the result. □

Proof of the theorem. The necessity of the condition is easy. If F_n converges weakly to F, a standard argument provides the existence, for each $\varepsilon > 0$, of a real number $X = X(\varepsilon)$ such that

$$\sup_{n\geqslant 1} \sup_{\tau\in\mathbb{R}} \Big| \int_{|z|>X} \mathrm{e}^{i\tau z}\,\mathrm{d}F_n(z) \Big| \leqslant \sup_{n\geqslant 1} \int_{|z|>X} \mathrm{d}F_n(z) \leqslant \varepsilon.$$

Now X can be chosen in such a way that $\pm X \in \mathcal{C}(F)$, and we see, for example by partial integration, that

$$\int_{-X}^{X} e^{i\tau z}\, dF_n(z) \to \int_{-X}^{X} e^{i\tau z}\, dF(z)$$

uniformly on any compact set. Up to modifying the value of X, the last integral equals $\varphi(\tau) + O(\varepsilon)$. This implies that $\varphi_n \to \varphi$ uniformly on any compact subset.

In order to establish the converse, it suffices to show that if φ_n converges pointwise to φ and if φ is continuous at 0, then F_n converges weakly to a d.f. F. Indeed, it will then follow from the first part of the proof that φ is the c.f. of F and that the convergence $\varphi_n \to \varphi$ is uniform on any compact subset.

The first step consists in noting that we may extract from $\{F_n\}_{n=1}^{\infty}$ a subsequence $\{F_{n_j}\}_{j=1}^{\infty}$ converging weakly to a non–decreasing, right-continuous function F. This follows from a classical diagonal process, and we omit the details. We have trivially $0 \leqslant F \leqslant 1$, and it remains to show that F is a d.f., that is, $F(+\infty) - F(-\infty) = 1$. To this end, we apply (2.8) with $z = 0$, $F = F_{n_j}$, $\varphi = \varphi_{n_j}$, and let j tend to ∞. The conditions for applying Lebesgue's dominated convergence theorem are trivially fulfilled. We thus obtain (2.8) where F is the weak limit of the F_{n_j}, and φ is the pointwise limit of the φ_{n_j}. As $h \to +\infty$, the right-hand side tends to $F(+\infty) - F(-\infty)$ since F is non-decreasing, and the left-hand side tends to $\varphi(0)$ because φ is bounded and continuous at 0. Now, $\varphi(0) = 1$ since $\varphi_n(0) = 1$ for all n. The weak limit of the F_{n_j} is then certainly a d.f.

This also holds for any other weak limit, say F^*. And since F^* necessarily still has φ for its c.f. (as a result of the first part of the proof), we deduce that $F = F^*$. Therefore every weakly convergent subsequence of $\{F_n\}_{n=1}^{\infty}$ converges to F, which means that $\{F_n\}_{n=1}^{\infty}$ itself converges weakly to F, thereby completing the proof. □

When the limit law $F(z)$ is absolutely continuous with bounded density, say

$$F(z) = \int_{-\infty}^{z} h(t)\, dt,$$

the Berry–Esseen inequality (Theorem II.7.16) provides a quantitative estimate for the approximation of F by F_n. For each $T > 0$, we have

$$\sup_{z \in \mathbb{R}} |F_n(z) - F(z)| \ll \frac{1}{T} \sup_{t \in \mathbb{R}} |h(t)| + \int_{-T}^{T} \left| \frac{\varphi_n(\tau) - \varphi(\tau)}{\tau} \right| d\tau. \tag{2.9}$$

Let us assume that $\lim_{\varepsilon\to 0}\sup_n \int_0^\varepsilon |1-\varphi_n(\tau)|\,\mathrm{d}\tau/\tau = 0$.[4] Since φ_n tends to φ uniformly on any compact set, the right-hand side of (2.9) is therefore bounded, for a suitable choice $T = T_n$, by a quantity ε_n depending only on n and such that $\lim_{n\to\infty}\varepsilon_n = 0$. This bound is of major practical importance—cf. for example Theorem 4.15.

Bearing in mind the definition of the limiting d.f. of an arithmetical function, the continuity theorem immediately provides the following criterion.

Theorem 2.6. *Let f be a real arithmetic function. Then f possesses a d.f. F if, and only if, the sequence of functions*

$$\varphi_N(\tau) := \frac{1}{N}\sum_{n\leqslant N} \mathrm{e}^{i\tau f(n)} \tag{2.10}$$

converges pointwise on $\mathbb{R}$ to a function $\varphi(\tau)$ which is continuous at 0. In this case, φ is the c.f. of F.

If $F_N(z)$ is defined by (2.2), we indeed have $\varphi_N(\tau) = \int_{-\infty}^{+\infty} \mathrm{e}^{i\tau z}\,\mathrm{d}F_N(z)$. When f is additive, the function $n \mapsto \mathrm{e}^{i\tau f(n)}$ is, for each fixed τ, a multiplicative function of modulus 1. The problem of the existence of a limit law for f is therefore equivalent to the existence of the mean value of a multiplicative function with values in the unit disk. We shall see in Chapter III.4 how to exploit this duality.

[4] It may be shown that this condition is equivalent to the uniform $\mathrm{d}F_n$-integrability of $\ln^+|x|$, i.e., $\lim_{T\to\infty}\sup_n \int_{|x|>T} \ln|x|\,\mathrm{d}F_n(x) = 0$.

Notes

§ 2.1. For further details concerning the theory of d.f.'s, see Feller (1970, 1971), Loève (1963), or Lukacs (1970).

Theorem 2.3 is identical to *lemma* A2 of Hall & Tenenbaum (1988).

The normalisation (2.5) for d.f.'s of arithmetic functions has an obvious theoretical interest. In practice, it is often preferable to retain greater flexibility and to ask the more general question of determining those functions A_N, B_N for which the sequence of d.f.'s

$$H_N(z) := \nu_N\{n : f(n) \leqslant A_N + zB_N\} \qquad (N = 1, 2, \dots)$$

converges weakly to a d.f. $H(z)$. This is the point of view adopted by Elliott (1979). His book is today considered one of the essential references on probabilistic number theory.

§ 2.2. Our proof of Lévy's theorem essentially follows that of Cramér (1970).

The theory of c.f.'s is particularly suitable for the treatment of *convolutions* of d.f.'s. The *convolution* of two d.f. F and G is the d.f. H defined by

$$H(z) := \int_{-\infty}^{+\infty} F(z-y)\,\mathrm{d}G(y) = \int_{-\infty}^{+\infty} G(z-y)\,\mathrm{d}F(y).$$

If $\varphi(\tau)$ and $\gamma(\tau)$ are the c.f.'s of F and G respectively, then $\eta(\tau) = \varphi(\tau)\gamma(\tau)$ is the c.f. of H.

Parseval's formula reads

$$\int_{-\infty}^{+\infty} g(z)\,\mathrm{d}F(z) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} \widehat{g}(\tau)\varphi(\tau)\,\mathrm{d}\tau \tag{2.11}$$

for any function $g \in L^1(\mathbb{R})$ whose Fourier transform

$$\widehat{g}(\tau) = \int_{-\infty}^{+\infty} \mathrm{e}^{-i\tau z} g(z)\,\mathrm{d}z$$

is also in $L^1(\mathbb{R})$.

Let $\mathcal{C}_b(\mathbb{R})$ denote the space of all continuous and bounded functions on $\mathbb{R}$. Equation (2.11) remains valid in the form

$$\int_{-\infty}^{+\infty} g(z)\,\mathrm{d}F(z) = \lim_{\lambda\to\infty} \frac{1}{2\pi}\int_{-\lambda}^{\lambda} \Big(1 - \frac{|\tau|}{\lambda}\Big)\widehat{g}(\tau)\varphi(\tau)\,\mathrm{d}\tau \tag{2.12}$$

whenever $g \in L^2(\mathbb{R}) \cap \mathcal{C}_b(\mathbb{R})$. This can be quickly established using to the properties of the Fejér kernel

$$w_\lambda(z) := \frac{\lambda}{2\pi}\Big(\frac{\sin \lambda z/2}{\lambda z/2}\Big)^2$$

with Fourier transform $\widehat{w_\lambda}(\tau) = (1 - |\tau|/\lambda)^+$. Indeed, under the above assumptions for g, it is classically known that $g_\lambda := g * w_\lambda$ converges to g both pointwise and in quadratic mean. Furthermore, $\|g_\lambda\|_\infty \leqslant \|g\|_\infty$. Now, $\widehat{g_\lambda} = \widehat{g}\,\widehat{w_\lambda}$ lies in $L^1(\mathbb{R})$ as product of an element of $L^2(\mathbb{R})$ by a continuous function with compact support. Applying (2.11) to g_λ, and evaluating the limit of the left-hand side by Lebesgue's theorem of dominated convergence, we obtain (2.12).

Choosing, in (2.12), $g(z) = \sin\{T(z-y)\}/\{T(z-y)\}$, so that

$$\widehat{g}(\tau) = \pi \mathrm{e}^{-i\tau y}/T \quad (|\tau| \leqslant T), \qquad \widehat{g}(\tau) = 0 \quad (|\tau| > T),$$

we obtain

$$\int_{-\infty}^{+\infty} \frac{\sin\{T(z-y)\}}{T(z-y)}\,\mathrm{d}F(z) = \frac{1}{2T}\int_{-T}^{T} \mathrm{e}^{-i\tau y}\varphi(\tau)\,\mathrm{d}\tau.$$

Integrating over $\mathbb{R}$ with respect to $\mathrm{d}F(y)$, it follows that

$$\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} \frac{\sin\{T(z-y)\}}{T(z-y)}\,\mathrm{d}F(z)\,\mathrm{d}F(y) = \frac{1}{2T}\int_{-T}^{T} |\varphi(\tau)|^2\,\mathrm{d}\tau. \tag{2.13}$$

From which, by letting T tend to $+\infty$, we conclude that

$$\sum_{n\geqslant 0} s_n^2 = \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} |\varphi(\tau)|^2\,\mathrm{d}\tau \tag{2.14}$$

where $\{s_n\}_{n=1}^{\infty}$ is the sequence of jumps of F, ordered so as to be decreasing. In particular, *the relation*

$$\lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} |\varphi(\tau)|^2\,\mathrm{d}\tau = 0 \tag{2.15}$$

is a necessary and sufficient condition for F to be continuous.

As we shall see in Chapter III.4, the convergence of convolution products

$$F_1 * F_2 * \cdots * F_n \qquad (n \to \infty) \tag{2.16}$$

is of major importance for the study of limit laws of additive and multiplicative functions. In this context, we quote three fundamental results, which we assemble in a single statement. We denote by $\varphi_j(\tau)$ the c.f. of F_j and let σ_j be its maximal saltus, i.e.,

$$\sigma_j := \max_{z\in\mathbb{R}}\{F_j(z) - F_j(z-)\}.$$

Finally, we write $Y := \mathbf{1}_{[0,\infty[}$ for Heaviside's function.

Theorem 2.7. *The three following conditions are equivalent:*

(i) *The product* (2.16) *converges weakly, as $n \to \infty$, to a d.f. F,*

(ii) $\exists \delta > 0 : \lim_{m,n\to\infty} \prod_{m<j\leqslant n} \varphi_j(\tau) = 1 \quad (|\tau| \leqslant \delta)$,

(iii) $\lim_{m,n\to\infty} F_m * \cdots * F_n(z) = Y(z) \quad (z \neq 0)$.

Whenever any of these three conditions is fulfilled, we have:

(a) (Lévy, 1931) *F is continuous if, and only if,*

$$\exists n \geqslant 1 : \sigma_n = 0 \quad \textit{or} \quad \sum_{n\geqslant 1}(1-\sigma_n) = +\infty. \tag{2.17}$$

(b) (Jessen & Wintner, 1935) *If each F_n is atomic, then F is a pure law, that is F is either atomic, or purely singular continuous, or absolutely continuous.*

The equivalence of conditions (i), (ii) and (iii) easily follows from the continuity theorem. Put $F_{m,n} := F_m * \cdots * F_n$ and $\varphi_{m,n}(\tau) := \prod_{m<j\leqslant n} \varphi_j(\tau)$. If the product (2.16) converges weakly to F with c.f. φ, then φ is continuous and $\varphi(0) = 1$, hence φ does not vanish in a suitable neighborhood of the origin, say $|\tau| \leqslant \delta$. Lévy's theorem then immediately implies (ii). The implication (ii)$\Rightarrow$(iii) results from the following general inequality,

$$1 - \Re e\, \varphi(\tau) \geqslant \tfrac{1}{4}\big(1 - \Re e\, \varphi(2\tau)\big), \tag{2.18}$$

which holds for any c.f. φ and is derived, by integration with respect to $\mathrm{d}F(z)$, from the elementary inequality $1 - \cos(z\tau) \geqslant \frac{1}{4}\big(1 - \cos(2z\tau)\big)$. Applying (2.18) to $\varphi_{m,n}$, we see that $\lim_{m,n\to\infty} \varphi_{m,n}(\tau) = 1$ for $|\tau| \leqslant 2\delta$ and, finally, by iteration, for all τ. We deduce (iii) from this by the continuity theorem. It remains to show that (iii) implies (i). By a new application of the continuity theorem, we deduce from (iii) that $\varphi_{m,n}(\tau) \to 1$ uniformly on any compact set. This implies the uniform convergence, on any compact set, of the product $\prod_{j\geqslant 1} \varphi_j(\tau)$, whose limit is *ipso facto* continuous at the origin. A final application of Theorem 2.4 provides the required conclusion.

A sufficient condition for the existence of an absolutely continuous limiting distribution is given by Babu (1992)—see Exercise 259 for a special case.

A complete proof of statements (a) and (b) of Theorem 2.7 may be found in Elliott (1979), *lemma* 1.22. The proof of criterion (2.17) is based on the fruitful notion of *concentration function*, introduced by Paul Lévy in 1937. We give below a brief survey of some basic results. The book of Hengartner & Theodorescu (1973) provides a deeper account of this topic.

The concentration function[5] Q_F of a d.f. F is defined on $\mathbb{R}^+$ by the formula

$$Q_F(\ell) := \sup_{z\in\mathbb{R}} \big\{F(z+\ell) - F(z)\big\}. \tag{2.19}$$

We further define the *concentration* $Q(F) := Q_F(1)$ and observe that, for all $\ell > 0$, we have $Q_F(\ell) = Q(F_\ell)$, with $F_\ell(z) := F(\ell z)$. It is easy

[5] We have already encountered this notion: see p. 351.

to see that F is continuous if, and only if, $Q_F(\ell) \to 0$ as $\ell \to 0+$, so the concentration function may be regarded as a measure of the distance from F to the set of continuous d.f.'s. This notion is particularly well-suited to the study of convolutions of d.f.'s (or, what is the same, of sums of independent random variables), and indeed we have

$$Q(F * G) \leqslant Q(F) \tag{2.20}$$

for any pair (F, G) of d.f.'s. Inequality (2.20) follows immediately from the definition of the convolution product:

$$\begin{aligned} Q(F * G) &= \sup_{z \in \mathbb{R}} \int \int_{z < x+y \leqslant z+1} \mathrm{d}F(x)\, \mathrm{d}G(y) \\ &\leqslant \int_{\mathbb{R}} \mathrm{d}G(y) \sup_{z \in \mathbb{R}} \int_{]z-y, z-y+1]} \mathrm{d}F(x) = Q(F). \end{aligned}$$

A deeper inequality between the concentration of a convolution product and that of its factors has been given by Kolmogorov (1956, 1958), and then refined by Rogozin (1961).

Theorem 2.8 (Kolmogorov–Rogozin inequality). *Let $\{F_j\}_{j=1}^n$ be a finite sequence of d.f.'s and set $F := F_1 * \cdots * F_n$. For $\ell > 0$, we have*

$$Q_F(\ell) \leqslant \frac{C}{\sqrt{1 + \sum_{j=1}^n (1 - Q_{F_j}(\ell))}}, \tag{2.21}$$

where C is an absolute constant.

Prior to proving (2.21), let us see how this enables us to deduce assertion (a) of Theorem 2.7.

Assume first that $\sum_{j \geqslant 1} (1 - \sigma_j) < \infty$. We may suppose, without loss of generality, that $\sigma_j \neq 0$ for all j and hence that $\prod_{j \geqslant 1} \sigma_j > 0$. For each $j \geqslant 1$, we select a real number u_j such that $\sigma_j = F_j(u_j) - F_j(u_j-)$. The series $\sum_{j \geqslant 1} u_j$ is necessarily convergent: otherwise there exist a $\delta > 0$ and two integer sequences m_k, n_k tending to infinity such that $|\sum_{m_k < j \leqslant n_k} u_j| > \delta$ for all k, whence

$$\prod_{m_k < j \leqslant n_k} \sigma_j \leqslant \int_{|z| > \delta} \mathrm{d}F_{m_k, n_k}(z),$$

which yields the required contradiction, for the left-hand side approaches 1 as $k \to \infty$ whereas the right-hand side tends to 0. Setting $u := \sum_{j \geqslant 1} u_j$, we therefore obtain

$$F(u) - F(u-) \geqslant \int_{\prod_{j \geqslant 1} \{u_j\}} \prod_{j \geqslant 1} \mathrm{d}F_j(z_j) = \prod_{j \geqslant 1} \sigma_j > 0,$$

and F is not continuous at u.

Conversely, if condition (2.17) is realized and no σ_n vanishes, let $\varepsilon > 0$ be arbitrary and $m = m(\varepsilon)$ be such that $\sum_{1 \leqslant j \leqslant m}(1 - \sigma_j) \geqslant 1/\varepsilon^2$. For $\ell \leqslant \ell_0(\varepsilon)$ we have $Q_{F_j}(\ell) \leqslant \sigma_j + 1/m$ $(1 \leqslant j \leqslant m)$. Applying (2.20), and then (2.21), we obtain $Q_F(\ell) \ll \varepsilon$. This shows that $Q_F(\ell)$ tends to 0 with ℓ and hence that F is indeed continuous.

The following lemma provides upper and lower bounds for $Q(F)$ in terms of mean values of $\varphi(\tau)$. We simply denote by $w(z)$ the Fejér kernel $w_1(z)$.

Lemma 2.9. *Define* $K_1 := w(1) > 0.146$ *and* $K_2 := 1/\{2\pi w(\frac{1}{2})\} < 1.022$. *For any d.f.* F *with c.f.* φ, *we have*

$$K_1 \int_{-1}^{1} |\varphi(\tau)|^2 \, \mathrm{d}\tau \leqslant Q(F) \leqslant K_2 \int_{-1}^{1} |\varphi(\tau)| \, \mathrm{d}\tau. \tag{2.22}$$

Proof. For each $z \in \mathbb{R}$, we have

$$\begin{aligned} F(z + \tfrac{1}{2}) - F(z - \tfrac{1}{2}) &= \int_{z-1/2}^{z+1/2} \mathrm{d}F(v) \\ &\leqslant \frac{1}{w(1/2)} \int_{-\infty}^{+\infty} w(z - v) \, \mathrm{d}F(v) = K_2 \int_{-1}^{1} (1 - |\tau|) \mathrm{e}^{-i\tau z} \varphi(\tau) \, \mathrm{d}\tau, \end{aligned}$$

where the final equality follows from (2.11). This implies the upper bound in (2.22). For the lower bound, we note that

$$Q(F) = Q(F) \int_{-\infty}^{+\infty} \int_{z-1/2}^{z+1/2} \mathrm{d}F(v) \, \mathrm{d}z \geqslant \int_{-\infty}^{+\infty} \{F(z + \tfrac{1}{2}) - F(z - \tfrac{1}{2})\}^2 \, \mathrm{d}z.$$

However, writing the inversion formula (2.7) in the form

$$F(z + \tfrac{1}{2}) - F(z - \tfrac{1}{2}) = \frac{1}{2\pi} \lim_{T \to \infty} \int_{-T}^{T} \frac{\sin(\tau/2)}{\tau/2} \varphi(\tau) \mathrm{e}^{-i\tau z} \, \mathrm{d}\tau \, ,$$

and interpreting it as a Fourier transform in $L^2(\mathbb{R})$, enables us to write Plancherel's formula

$$\int_{-\infty}^{+\infty} \{F(z + \tfrac{1}{2}) - F(z - \tfrac{1}{2})\}^2 \, \mathrm{d}z = \int_{-\infty}^{+\infty} w(\tau) |\varphi(\tau)|^2 \, \mathrm{d}\tau \geqslant w(1) \int_{-1}^{1} |\varphi(\tau)|^2 \, \mathrm{d}\tau.$$

This provides the required inequality. □

We are now in a position to establish Theorem 2.8. The following proof is a variant of that of Esseen (1966), or Hengartner & Theodorescu (1973). We may assume without loss of generality that $\ell = 1$, and that $Q(F_j) < 1$

for $1 \leqslant j \leqslant n$. By (2.22), we may write

$$\begin{aligned} Q(F) &\leqslant K_2 \int_{-1}^{1} \prod_{j=1}^{n} |\varphi_j(\tau)| \, \mathrm{d}\tau \\ &\leqslant K_2 \int_{-1}^{1} \exp\Big\{ -\tfrac{1}{2} \sum_{1 \leqslant j \leqslant n} \big(1 - |\varphi_j(\tau)|^2\big) \Big\} \, \mathrm{d}\tau, \end{aligned} \tag{2.23}$$

as a consequence of the inequality $u \leqslant \mathrm{e}^{-(1-u^2)/2}$, valid for $0 \leqslant u \leqslant 1$.

Let $\check{F}_j(z) := 1 - F_j(-(z+))$ be the d.f. symmetric to $F_j(z)$, with c.f. $\overline{\varphi_j(\tau)}$. Define the symmetrized d.f. $G_j := F_j * \check{F}_j$, with c.f. hence equal to $|\varphi_j(\tau)|^2$. Thus

$$1 - |\varphi_j(\tau)|^2 = \int_{-\infty}^{+\infty} (1 - \cos(\tau z)) \, \mathrm{d}G_j(z). \tag{2.24}$$

Put $I := [-\frac{1}{2}, \frac{1}{2}[$ and

$$p_j := \int_{\mathbb{R} \smallsetminus I} \mathrm{d}G_j(z) = \int_{-\infty}^{+\infty} \mathrm{d}F_j(y) \int_{x \notin y - I} \mathrm{d}F_j(x) \geqslant 1 - Q(F_j) > 0.$$

Now introduce the d.f. H_j defined by

$$H_j(z) = \begin{cases} G_j(z)/p_j & \text{if } z \notin I, \\ G_j(-\frac{1}{2})/p_j & \text{if } z \in I. \end{cases}$$

We have $\mathrm{d}G_j \geqslant p_j \, \mathrm{d}H_j$ and insert this lower bound in (2.24). Put

$$T := \sum_{1 \leqslant j \leqslant n} p_j, \qquad \alpha_j := T/p_j,$$

so that $\sum_{1 \leqslant j \leqslant n} 1/\alpha_j = 1$. By Hölder's inequality with exponents α_j, we deduce from (2.23) and from the lower bound derived from (2.24) by introducing $\mathrm{d}H_j$, that

$$Q(F) \leqslant K_2 \prod_{j=1}^{n} \Big(\int_{-1}^{1} \exp\Big\{ -\tfrac{1}{2} \alpha_j p_j \int_{-\infty}^{+\infty} (1 - \cos(\tau z)) \, \mathrm{d}H_j(z) \Big\} \, \mathrm{d}\tau \Big)^{1/\alpha_j}.$$

We have $\alpha_j p_j = T$ for all j. By Jensen's inequality, the exponential in the above formula does not exceed

$$\int_{-\infty}^{+\infty} \mathrm{e}^{-T(1-\cos(\tau z))/2} \, \mathrm{d}H_j(z),$$

from which, permuting the order of integrations,

$$Q(F) \leqslant K_2 \prod_{j=1}^{n} \Big(\int_{-\infty}^{+\infty} \mathrm{d}H_j(z) \int_{-1}^{1} \mathrm{e}^{-T(1-\cos(\tau z))/2} \, \mathrm{d}\tau \Big)^{1/\alpha_j}.$$

To bound the inner integral, we may assume $|z| \geqslant \frac{1}{2}$ since I has zero measure for $\mathrm{d}H_j(z)$. It follows that

$$\begin{aligned}\int_{-1}^{1} \mathrm{e}^{-T(1-\cos(\tau z))/2}\,\mathrm{d}\tau &= \frac{2}{|z|}\int_0^{|z|} \mathrm{e}^{-T(1-\cos v)/2}\,\mathrm{d}v \\ &\leqslant \frac{2}{|z|}\Big(1+\frac{|z|}{\pi}\Big)\int_0^{\pi} \mathrm{e}^{-T(1-\cos v)/2}\,\mathrm{d}v \\ &\leqslant \Big(4+\frac{2}{\pi}\Big)\int_0^{+\infty} \mathrm{e}^{-Tv^2/\pi^2}\,\mathrm{d}v \leqslant \frac{19}{\sqrt{T}}.\end{aligned}$$

This yields the bound (2.21) when $T \geqslant 1$. Since the result is otherwise trivial, the proof of Theorem 2.8 is complete.

For additional information on c.f.'s, see the remarkable book of Lukacs (1970).

Exercises

253. *Limiting distribution of* $\varphi(n)/n$.

Let $\varepsilon > 0$, $y := 1/\varepsilon^2$, and $a_\varepsilon(n) := \prod_{p^\nu \| n,\, p \leqslant y} p^\nu$ $(n \geqslant 1)$.

(a) Show that

$$\mathbf{d}\{n : a_\varepsilon(n) = a\} = \frac{1}{a} \prod_{p \leqslant y} \Big(1 - \frac{1}{p}\Big)$$

for $a \geqslant 1$, $P^+(a) \leqslant y$. (Use, for example, Exercise 77, p. 105.)

(b) Establish the upper bound $\sum_{n \leqslant x} \ln a_\varepsilon(n) \ll x \ln\big(1/\varepsilon\big)$ $(x \geqslant 1)$.

(c) Write $f(n) = \varphi(n)/n$, where φ is Euler's function. Show that

$$\big|f(n) - f\big(a_\varepsilon(n)\big)\big| \leqslant \sum_{p \mid n,\, p > y} \frac{1}{p}.$$

(d) Show that $f(n)$ has a limit law by deducing from (b) and (c) that conditions (i) and (ii) of Theorem 2.3 are satisfied.

254. *A theorem of Schoenberg (1928).* For each $\tau \in \mathbb{R}$, determine an arithmetic function λ_τ satisfying $\big\{\varphi(n)/n\big\}^{i\tau} = \sum_{d \mid n} \lambda_\tau(d)$. Show that the series $\psi(\tau) := \sum_{d \geqslant 1} \lambda_\tau(d)/d$ is absolutely convergent, and thus recover the result of Exercise 253. Show that the limit law F is given by

$$F(\mathrm{e}^z) = \lim_{n \to \infty} F_{p_1} * F_{p_2} * \cdots * F_{p_n}(z)$$

where p_j denotes the jth prime number and $\mathrm{d}F_{p_j}$ is a linear combination of two Dirac measures. By applying Theorem 2.7(a), deduce that F is continuous.[6] Show that F is of pure type by using Theorem 2.7(b).

255. Recall that $\sigma(n) := \sum_{d \mid n} d$ $(n \geqslant 1)$. Reconsider Exercises 253 and 254 but with $\sigma(n)/n$ in place of $\varphi(n)/n$. Generalize.

256. *A sufficient condition for singularity* (Erdős, 1939). Throughout this exercise we assume as given a non-decreasing, integer-valued function $R(y)$, satisfying

(i) $\sum_p 1/pR(p) < \infty$,

(ii) For sufficiently large y, there exist $R = R(y)$ integers

$$m_1 < m_2 < \cdots < m_R$$

such that

(α) $P^+(m_j) \leqslant y$, $\mu(m_j)^2 = 1$ $(1 \leqslant j \leqslant R(y))$,

(β) $\sum_{1 \leqslant j \leqslant R(y)} 1/\varphi(m_j) \geqslant c \ln y$, where c is an absolute positive constant.

[6] This property can be established directly by appealing to criterion (2.15) of the Notes—see the end of the proof of Theorem 4.1.

Here, we investigate the following result due to Erdős: *Let f be a strongly additive arithmetic function such that $f(p) \ll 1/R(p)^3$. Then f has a d.f., which is either atomic or purely singular.*

(a) Compute $\lim_{x\to\infty} x^{-1}\sum_{n\leqslant x} e^{i\tau f(n)}$ by the method suggested in Exercise 254. Deduce the existence of the limit law, and show that it is of pure type.[7]

(b) Let $\lambda(I)$ denote the Lebesgue measure of a generic Riemann–integrable subset I of $\mathbb{R}$. Show that, if the limit law of f is absolutely continuous, then

$$\lim_{\lambda(I)\to 0} \overline{\mathbf{d}}\{n : f(n) \in I\} = 0.$$

(c) Let $\varepsilon > 0$, and $y = y(\varepsilon)$ be sufficiently large. Set $a_\varepsilon(n) := \prod_{p\leqslant y, p|n} p$. Show that, for $a \geqslant 1$, $\mu(a)^2 = 1$ and $P^+(a) \leqslant y$, we have

$$\mathbf{d}\{n : a_\varepsilon(n) = a\} = \frac{1}{\varphi(a)} \prod_{p\leqslant y} \Big(1 - \frac{1}{p}\Big)$$

and so that Theorem 2.3 may be applied to provide a second proof for the existence of the d.f. of f.

(d) The number $y(\varepsilon)$ being chosen sufficiently large, write

$$I := \bigcup_{1\leqslant j\leqslant R(y)} \Big(f(m_j) + \big[-1/R(y)^2,\, 1/R(y)^2\big]\Big).$$

Show that $\overline{\mathbf{d}}\{n : f(n) \in I\} \gg 1$. Conclude.

257. Show that, for all $\delta > 0$, the function $R(y) := y^\delta$ satisfies the hypotheses of Exercise 256. Deduce that the d.f. of $\varphi(n)/n$ and $\sigma(n)/n$ are purely singular. [*Assume here the results of Exercises* 254 *and* 255.]

258. Let $F(z)$ be a d.f. with c.f. $\varphi(\tau)$. Show that, if $\varphi \in L^2(\mathbb{R})$, then F is absolutely continuous. One may appeal to Plancherel's theorem and to Parseval's formula (2.11) for an arbitrary function g of class C^2 with compact support.

259. *A result of Saffari (1979).* Let f be the strongly additive function defined by $f(2) = 0$, $f(p) = (\ln p)^{-\alpha}$ $(p > 2)$, where $\alpha > 0$ is given.

(a) Show that f possesses a d.f. F, and compute its c.f. $\varphi(\tau)$.

(b) By calculating $|\varphi(\tau)|^2$, show that

$$\varphi(\tau) \asymp \exp\Big\{-2\sum_p \frac{1}{p}\sin^2\big(\tfrac{1}{2}\tau f(p)\big)\Big\}.$$

[7] See the Notes, Theorem 2.7.

(c) Evaluate the sum over p using the prime number theorem, and show that $|\varphi(\tau)| = |\tau|^{-1/\alpha+o(1)}$ $(|\tau| \to \infty)$. Appealing to the result of Exercise 258, deduce that F is absolutely continuous whenever $\alpha < 2$.

260. Let $h(n)$ be the number of prime factors p of n such that $n^{1/3} < p \leqslant n^{1/2}$. Show that $h(n)$ has a distribution function.

261. *Concentration of an additive function on divisors.* Given a real additive function f, we define, for each integer n, a d.f. F_n by

$$F_n(z) := \frac{1}{\tau(n)} \sum_{d|n,\, f(d)\leqslant z} 1.$$

(a) Compute the c.f. $\varphi_n(\tau)$ of F_n and deduce that $F_n = \mathop{\ast}_{p^\nu \| n} F_{p^\nu}$. Show that, if there exists an absolute positive constant c such that $Q(F_{p^\nu}) \leqslant 1 - c$, then we have $Q(F_n) \ll \{1+\omega(n)\}^{-1/2}$.

(b) For the case $f = \ln$, we define Hooley's Δ-function (1979)[8] by

$$\Delta(n) = Q(F_n)\tau(n) = \max_{u\in\mathbb{R}} \sum_{d|n,\, \mathrm{e}^u < d \leqslant \mathrm{e}^{u+1}} 1.$$

Show that $\Delta(n) \ll \tau(n)/\sqrt{1+\omega(n)}$, and establish that this upper bound is optimal up to the value of the implicit constant.

262. *Chains of divisors.* For any integer n, a subset $\{d_1, \dots, d_h\}$ of divisors of n is said to be an n-chain if

(i) $\Omega(d_1) + \Omega(d_h) = \Omega(n)$,

(ii) $d_{j+1}/d_j \in \mathbb{P}$ $\quad (1 \leqslant j < h)$.

(a) Let $n = mp^\nu$, with $p \nmid m$. Show that if $\{d_1, \dots, d_h\}$ is an m-chain, then $\mathcal{C}_r := \{d_r, d_r p, \dots, d_r p^{\nu+1-r}, d_{r+1}p^{\nu+1-r}, \dots, d_h p^{\nu+1-r}\}$ is an n-chain for all r, $1 \leqslant r \leqslant R := \min(h, \nu+1)$. Show that the $\mathcal{C}_r$ induce a partition of $\{d_j p^\mu : 1 \leqslant j \leqslant h,\, 0 \leqslant \mu \leqslant \nu\}$. Deduce, by induction on $k = \omega(n)$, that, for each n, the set of divisors of n is a disjoint union of n-chains. Denote by $\gamma(n)$ the maximum number of such n-chains.

(b) Show that each n-chain contains exactly one divisor d such that $\Omega(d) = \lfloor \Omega(n)/2 \rfloor$.

(c) Show that the function Ω fulfills the condition of Exercise 261 with $c = \frac{1}{2}$. Deduce an inequality for $\gamma(n)$.[9]

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[8] See Exercise 209, p. 294, and Exercise 279, p. 472.

[9] This result, as well as that of question (a) in Exercise 263, is due to de Bruijn, van Ebbenhorst Tengbergen & Kruyswijk (1949–51).

263. *Primitive sequences.* Retain the notation $\gamma(n)$ of Exercise 262. A sequence of integers $\mathcal{A}$ is said to be *primitive* if none of its elements divides any other. For each integer n, we put $\tau(n,\mathcal{A}) := \sum_{d|n,\, d\in\mathcal{A}} 1$.

(a) Show that, for every primitive sequence $\mathcal{A}$, we have $\tau(n,\mathcal{A}) \leqslant \gamma(n)$ $(n \geqslant 1)$.

(b) By evaluating $\sum_{n\leqslant x} \tau(n,\mathcal{A})$, prove Behrend's result (1935)

$$\sup_{\mathcal{A} \text{ primitive}} \sum_{a\leqslant x,\, a\in\mathcal{A}} \frac{1}{a} \ll \frac{\ln x}{\sqrt{\ln_2 x}}. \tag{2.25}$$

(c) By considering $\mathcal{A} = \mathcal{A}_x = \{n \geqslant 1 : \Omega(n) = \lfloor \ln_2 x \rfloor\}$ and appealing to the results obtained by the Selberg–Delange method, show that estimate (2.25) is optimal up to the value of the implicit constant.[10]

[10] Behrend's original proof, depending on a theorem of Sperner (1928), is reproduced in chap. 5 of Halberstam & Roth (1966). Erdős, Sárközy & Szemerédi (1967) have shown that the left-hand side of (2.25) is asymptotically $(\ln x)/\sqrt{2\pi \ln_2 x}$.

Normal order

3.1. Definition

The concept of normal order of an arithmetic function corresponds, in probabilistic number theory, to that of almost sure equality of random variables in probability theory. More precisely, *an arithmetic function f is said to have normal order g if g is an arithmetic function such that, for each $\varepsilon > 0$, we have*

$$|f(n) - g(n)| \leqslant \varepsilon |g(n)|$$

on a set of integers n of density 1. A convenient notation to express such a situation is[1]

$$f(n) = \{1 + o(1)\}g(n) \qquad \text{pp.} \tag{3.1}$$

Of course, a given function f may have several normal orders, which must all share the same asymptotic behavior. The notion is thus pertinent only for those functions g whose behavior is, in some sense, simpler than that of f. It is certainly with this idea in mind that, in an article which can be considered today as the birth of probabilistic number theory, Hardy & Ramanujan (1917) added the restriction that a normal order should be *elementary and monotone.*

It is always delicate to venture a definition of the word *elementary.* To fix ideas, we may give it here the relatively wide meaning of: *that which can be expressed by means of the symbols of real analysis.* Thus we consider as elementary a function such as $\mathrm{li}(x)$ but reject the label in the case of a function defined exclusively by a Cauchy integral. Although they indeed

[1] Recall the symbol pp (for the French *presque partout*, i.e., almost everywhere) means that the relation thus designated holds on a suitable subset of the positive integers with natural density 1.

hold in almost all known examples, we have not retained the limitations of Hardy & Ramanujan here, in order to allow as much flexibility as possible to the theoretical notion.

In terms of distribution functions (d.f.'s), the existence of a normal order can be interpreted, after suitable renormalization, as a convergence to an improper law. For instance, it is immediate, in the case of positive functions, that relation (3.1) is equivalent to the weak convergence of the d.f.'s

$$H_N(z) := \nu_N\{n : f(n)/g(n) \leqslant z\}$$

to $H(z) := \mathbf{1}_{[1,\infty[}(z)$.

In the course of this and the following chapters, we shall see that numerous arithmetic functions, with apparently chaotic variations, do indeed possess a normal order. Their behavior is thus satisfactorily described "almost everywhere" (pp), and the collection of information of this type contributes to elaborating our current model of a "normal" number—i.e., a number which is "random" in a sense suited to natural density. That such a concept lends itself to quantitative assertions, and can ultimately serve as the basis of a complete mathematical theory, is not the least of the attractions of the probabilistic viewpoint on the theory of numbers.

3.2. The Turán–Kubilius inequality

In the study of an arithmetic function f, it is often reasonable to consider the expectation (or some adequate approximation to it)

$$g(N) = \mathbb{E}_N(f) := \frac{1}{N} \sum_{1\leqslant n\leqslant N} f(n) \tag{3.2}$$

as a good candidate for a normal order of f. When g is monotone, a necessary condition for this is that its rate of growth be sufficiently slow so that

$$g(n) = \{1 + o(1)\}g(N) \tag{3.3}$$

holds for all but at most $o(N)$ integers $n \leqslant N$. It is then natural to expect that a dispersion computation, that is an evaluation of the empirical variance

$$\mathbb{V}_N(f) := \mathbb{E}_N\big(|f(n) - g(N)|^2\big), \tag{3.4}$$

will allow, by the Bienaymé–Chebyshev inequality, an effective proof that g is indeed a normal order for f.

This method is particularly efficient in the case of additive functions. The Turán–Kubilius inequality (Theorem 3.1) then provides an upper bound for (3.4) which is often sufficient to determine the normal order. Before stating the result, it is interesting to explain the underlying probabilistic ideas.

Let f be a complex additive function. We have

$$f(n) = \sum_{p^\nu \leqslant N} f(p^\nu)\xi_{p^\nu}(n) \qquad (n \leqslant N) \tag{3.5}$$

with

$$\xi_{p^\nu}(n) = \begin{cases} 1 & \text{if } p^\nu \| n, \\ 0 & \text{otherwise.} \end{cases} \tag{3.6}$$

In other words, in the probability space $\Omega_N := \{n : 1 \leqslant n \leqslant N\}$ equipped with the uniform law ν_N, we have an equality between random variables

$$f = \sum_{p^\nu \leqslant N} f(p^\nu)\xi_{p^\nu}.$$

The ξ_{p^ν} are Bernoulli random variables, determined by their expectations

$$\begin{aligned} w_N(p^\nu) &:= \mathbb{E}_N(\xi_{p^\nu}) = \nu_N\{n : \xi_{p^\nu}(n) = 1\} \\ &= \frac{1}{N}\Big(\Big\lfloor \frac{N}{p^\nu} \Big\rfloor - \Big\lfloor \frac{N}{p^{\nu+1}} \Big\rfloor \Big) = \frac{1 - 1/p}{p^\nu} + O\Big(\frac{1}{N}\Big). \end{aligned} \tag{3.7}$$

For $q \neq p$, we can further write

$$\begin{aligned} \mathbb{E}_N(\xi_{p^\nu}\xi_{q^\mu}) &= \frac{1}{N}\Big(\Big\lfloor \frac{N}{p^\nu q^\mu} \Big\rfloor - \Big\lfloor \frac{N}{p^{\nu+1}q^\mu} \Big\rfloor - \Big\lfloor \frac{N}{p^\nu q^{\mu+1}} \Big\rfloor + \Big\lfloor \frac{N}{p^{\nu+1}q^{\mu+1}} \Big\rfloor \Big) \\ &= \frac{(1-1/p)(1-1/q)}{p^\nu q^\mu} + O\Big(\frac{1}{N}\Big) = \mathbb{E}_N(\xi_{p^\nu})\mathbb{E}_N(\xi_{q^\mu}) + O\Big(\frac{1}{N}\Big). \end{aligned}$$

Thus ξ_{p^ν} and ξ_{q^μ} are asymptotically independent when $p \neq q$, and, for fixed p, the numbers $\mathbb{E}_N(\xi_{p^\nu})$ $(\nu = 0, 1, 2, \ldots)$ are close to the probabilities of a geometric random variable with parameter $(1 - 1/p)$. This leads to the heuristic assumption that the distribution law of f, as a random variable on $\Omega_N := \{n : 1 \leqslant n \leqslant N\}$, is close to that of

$$Z_N = Z_{f,N} := \sum_{p \leqslant N} \zeta_p \tag{3.8}$$

where the $\zeta_p = \zeta_p(f)$ are independent variables on an abstract probability space (Ω, P), with laws

$$P\big(\zeta_p = f(p^\nu)\big) = (1 - 1/p)p^{-\nu} \qquad (\nu = 0, 1, 2, \ldots). \tag{3.9}$$

[Note that $f(1) = 0$.] By a familiar abuse of notation, we interpret (3.9) by agreeing that, if several values (possibly infinite in number) $f(p^\nu)$ are equal, the corresponding probability is the sum of the probabilities occurring on the right-hand side. We may also assume, without loss of generality, that $f(p^\nu) = 0$ when $p^\nu > N$.

We have

$$\begin{aligned} \mathbb{E}(Z_{f,N}) &= \sum_{p^\nu \leqslant N} \frac{f(p^\nu)}{p^\nu}\Big(1-\frac{1}{p}\Big) \\ \mathbb{V}(Z_{f,N}) &= B_f(N)^2 - R_f(N)^2 \end{aligned} \tag{3.10}$$

where we have put

$$\begin{aligned} B_f(N)^2 &:= \sum_{p\leqslant N} \mathbb{E}(|\zeta_p|^2) = \sum_{p^\nu\leqslant N} \frac{|f(p^\nu)|^2}{p^\nu}\Big(1-\frac{1}{p}\Big) \\ R_f(N)^2 &:= \sum_{p\leqslant N} |\mathbb{E}(\zeta_p)|^2 = \sum_{p\leqslant N}\Big(1-\frac{1}{p}\Big)^2\Big|\sum_{1\leqslant\nu\leqslant(\ln N)/\ln p}\frac{f(p^\nu)}{p^\nu}\Big|^2 \end{aligned} \tag{3.11}$$

and where $\mathbb{E}$ and $\mathbb{V}$ respectively denote the expectation and the variance with respect to the abstract probability law P satisfying (3.9).

To fix ideas, we remark at the outset that, by the Cauchy–Schwarz inequality, we have

$$R_f(N)^2 \leqslant \sum_{p^\nu\leqslant N}\frac{|f(p^\nu)|^2}{p^{\nu+1}}\Big(1-\frac{1}{p}\Big) \leqslant \tfrac{1}{2}B_f(N)^2, \tag{3.12}$$

so that

$$\tfrac{1}{2}B_f(N)^2 \leqslant \sum_{p^\nu\leqslant N}\Big(1-\frac{1}{p}\Big)^2\frac{|f(p^\nu)|^2}{p^\nu} \leqslant \mathbb{V}(Z_{f,N}) \leqslant B_f(N)^2. \tag{3.13}$$

Moreover,

$$\begin{aligned} |\mathbb{E}_N(f) - \mathbb{E}(Z_{f,N})| &\leqslant \frac{1}{N}\sum_{p^\nu\leqslant N}|f(p^\nu)| \\ &\leqslant \frac{1}{N}\Big(\sum_{p^\nu\leqslant N}\frac{|f(p^\nu)|^2}{p^\nu}\sum_{p^\nu\leqslant N}p^\nu\Big)^{1/2} \ll \sqrt{\frac{\mathbb{V}(Z_{f,N})}{\ln 2N}}\,. \end{aligned} \tag{3.14}$$

The Turán–Kubilius inequality essentially asserts that

$$C_N := \sup_{f\neq 0}\mathbb{V}_N(f)/\mathbb{V}(Z_{f,N}) \ll 1 \qquad (N\geqslant 1). \tag{3.15}$$

Here $\mathbb{V}_N(f)$ denotes the empirical variance

$$\mathbb{V}_N(f) := \frac{1}{N}\sum_{1\leqslant n\leqslant N}|f(n)-\mathbb{E}_N(f)|^2\,, \tag{3.16}$$

which is often conveniently replaced with the *semi-empirical variance*

$$\mathbb{V}_N^*(f) := \frac{1}{N}\sum_{1\leqslant n\leqslant N}|f(n)-\mathbb{E}(Z_{f,N})|^2\,. \tag{3.17}$$

We will hence also be considering the quantity

$$C_N^* := \sup_{f\neq 0} \mathbb{V}_N^*(f)/\mathbb{V}(Z_{f,N}). \tag{3.18}$$

Temporarily anticipating (3.15), we easily deduce from (3.14) that

$$\mathbb{V}_N(f) = \mathbb{V}_N^*(f) + O\Big(\frac{\mathbb{V}(Z_{f,N})}{\ln 2N}\Big), \tag{3.19}$$

where the implicit constant is independent of f, and so

$$C_N^* = C_N + O\Big(\frac{1}{\ln N}\Big) \qquad (N \geqslant 2). \tag{3.20}$$

We further remark that (3.15) is qualitatively optimal: indeed

$$\min(C_N, C_N^*) \geqslant 2^{1-\langle (\ln N)/\ln 2\rangle} + O\Big(\frac{1}{N}\Big) \qquad (N \geqslant 1), \tag{3.21}$$

so that

$$\limsup_{N\to\infty} C_N = \limsup_{N\to\infty} C_N^* \geqslant 2. \tag{3.22}$$

Indeed, for $N \geqslant 2$, the choice

$$f(p^\nu) := \begin{cases} 1 & \text{if } p = 2 \text{ and } 2^\nu \leqslant N < 2^{\nu+1}, \\ 0 & \text{otherwise,} \end{cases}$$

yields, writing $v := \lfloor (\ln N)/\ln 2 \rfloor$,

$$\mathbb{E}_N(f) = \frac{1}{N}, \quad \mathbb{E}(Z_{f,N}) = \frac{1}{2^{v+1}} \leqslant \frac{1}{N},$$

$$\mathbb{V}_N(f) = \frac{1}{N}\Big(1 - \frac{1}{N}\Big), \quad \mathbb{V}_N^*(f) = \frac{1}{N} + O\Big(\frac{1}{N^2}\Big), \quad \mathbb{V}(Z_{f,N}) = \frac{1}{2^{v+1}}\Big(1 - \frac{1}{2^{v+1}}\Big).$$

We are now in a position to state the Turán–Kubilius inequality. We set

$$\varepsilon_N := \frac{8}{N}\Big(\sum_{\substack{p^\nu q^\mu \leqslant N \\ p\neq q}} p^\nu q^\mu\Big)^{1/2} + \frac{4}{N}\Big(2\sum_{p^\nu \leqslant N} p^\nu \sum_{p\leqslant N} \frac{1}{p}\Big)^{1/2}.$$

By standard partial summation, whose details we omit, the prime number theorem readily yields the estimate

$$\varepsilon_N \ll \sqrt{\frac{\ln_2 N}{\ln N}} \qquad (N \geqslant 3). \tag{3.23}$$

Theorem 3.1 (Turán–Kubilius inequality). *For each additive, complex-valued arithmetic function f, we have*

$$\mathbb{V}_N^*(f) \leqslant (4 + \varepsilon_N) B_f(N)^2 \leqslant (8 + 2\varepsilon_N)\mathbb{V}(Z_{f,N}) \qquad (N \geqslant 1). \tag{3.24}$$

In particular,

$$\max(C_N, C_N^*) \leqslant 8 + O\Big(\sqrt{\frac{\ln_2 N}{\ln N}}\Big) \qquad (N \geqslant 3). \tag{3.25}$$

Proof. It is sufficient to establish the first inequality in (3.24): indeed, the second results from (3.13).

Let us first consider the case when f is real and non-negative. We have

$$\mathbb{V}_N^*(f) = \mathbb{E}_N\big(\{f - \mathbb{E}(Z_{f,N}\}^2\big) = M_2 - 2\mathbb{E}(Z_{f,N})\mathbb{E}_N(f) + \mathbb{E}(Z_{f,N})^2 \tag{3.26}$$

with

$$\begin{aligned} M_2 &:= \frac{1}{N}\sum_{n\leqslant N} f(n)^2 = \frac{1}{N}\sum_{n\leqslant x}\sum_{p^\nu\|n,\, q^\mu\|n} f(p^\nu)f(q^\mu) \\ &= \sum_{p^\nu\leqslant N} f(p^\nu)^2\frac{1}{N}\sum_{\substack{n\leqslant N\\ p^\nu\|n}} 1 + \sum_{\substack{p^\nu q^\mu\leqslant N\\ p\neq q}} f(p^\nu)f(q^\mu)\frac{1}{N}\sum_{\substack{n\leqslant N\\ p^\nu\|n,\, q^\mu\|n}} 1. \end{aligned}$$

The first inner sum does not exceed N/p^ν, the second is at most equal to $Np^{-\nu}q^{-\mu}(1-1/p)(1-1/q)+2$. Therefore, with notation (3.11),

$$M_2 \leqslant 2B_f(N)^2 + \mathbb{E}(Z_{f,N})^2 + \frac{2}{N}\sum_{p^\nu q^\mu\leqslant N,\, p\neq q} f(p^\nu)f(q^\mu). \tag{3.27}$$

The last term may be estimated by the Cauchy–Schwarz inequality:

$$\sum_{\substack{p^\nu q^\mu\leqslant N\\ p\neq q}} \frac{f(p^\nu)}{p^{\nu/2}}\frac{f(q^\mu)}{q^{\mu/2}}\cdot p^{\nu/2}q^{\mu/2} \leqslant 2B_f(N)^2\sqrt{\sum_{\substack{p^\nu q^\mu\leqslant N\\ p\neq q}} p^\nu q^\mu}.$$

Substituting into (3.26) and taking (3.14) into account, we obtain

$$\mathbb{V}_N^*(f) \leqslant 2B_f(N)^2 + \frac{2\mathbb{E}(Z_{f,N})B_f(N)}{N}\sqrt{2\sum_{p^\nu\leqslant N} p^\nu} + \frac{4B_f(N)^2}{N}\sqrt{\sum_{\substack{p^\nu q^\mu\leqslant N\\ p\neq q}} p^\nu q^\mu}.$$

Another application of the Cauchy–Schwarz inequality, in the form

$$\mathbb{E}(Z_{f,N})^2 \leqslant \sum_{p^\nu\leqslant N}\frac{f(p^\nu)^2}{p^\nu}\Big(1-\frac{1}{p}\Big)\sum_{p^\nu\leqslant N}\frac{1-1/p}{p^\nu} \leqslant B_f(N)^2\sum_{p\leqslant N}\frac{1}{p},$$

thus implies, when f is real and non-negative,

$$\mathbb{V}_N^*(f) \leqslant \big(2+\tfrac{1}{2}\varepsilon_N\big)B_f(N)^2.$$

If f is real and takes values of both signs, we introduce the additive functions f^+ and f^-, defined by $f^\pm(p^\nu) = \max\big(\pm f(p^\nu), 0\big)$. We then have, for any integer $n \geqslant 1$,

$$\big\{f(n) - \mathbb{E}(Z_{f,N})\big\}^2 \leqslant 2\big\{f^+(n) - \mathbb{E}(Z_{f^+,N})\big\}^2 + 2\big\{f^-(n) - \mathbb{E}(Z_{f^-,N})\big\}^2$$

and $B_f(N)^2 = B_{f^+}(N)^2 + B_{f^-}(N)^2$ $(N \geqslant 1)$. This implies the validity of (3.24) in this case.

When f is complex-valued, we apply the above results to the real and imaginary parts of f. The inequality trivially continues to hold. □

The constants 4 and 8 appearing in (3.24) are not optimal. Kubilius showed in 1983 that, writing

$$B_f^+(N)^2 := \sum_{p^\nu \leqslant N} \frac{|f(p^\nu)|^2}{p^\nu},$$

we have

$$C_N^+ := \sup_{f \neq 0} \mathbb{V}_N^*(f)/B_f^+(N)^2 = \tfrac{3}{2} + O\Big(\frac{1}{\sqrt{\ln N}}\Big). \tag{3.28}$$

In the definition of C_N^+, the supremum may be indifferently taken over real or complex-valued additive functions. Two other, independent proofs of the validity and optimality of the constant $\frac{3}{2}$ are due to Hildebrand (1983) and Stein (1984).

As noted by La Bretèche & Tenenbaum (2005b), it is relatively easy to deduce from Hildebrand's proof and (3.22) that

$$\limsup_{N\to\infty} C_N^* = \limsup_{N\to\infty} C_N = 2. \tag{3.29}$$

This result provides a remarkable measure of the discrepancy between probabilistic number theory and the probability theory model: the fact that the constants C_N and C_N^* are bounded exhibits the influence of the (partial) independence of the ξ_{p^ν}; the fact that the common upper limit of these quantities is not equal to 1 underlines the limitations of this influence.

For a long time, classical statements of the Turán–Kubilius inequality were expressed in terms of C_N^+ or of similar quantities.[2] The fundamental character of the interpretation of C_N as the ratio of the empirical variance of an arithmetic function to that of its probabilistic model—the quantity C_N^* essentially reflecting the same phenomena but turning out to be more flexible—leads to favoring formulation (3.29). The same motivations are present, in an even more radical form, in the study of the friable extension of the Turán–Kubilius inequality, whose validity is conditional on the choice of an adequate model: see the Notes on Chapter III.5.

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[2] Hildebrand (1983) also compares $\mathbb{V}_N(f)$ to

$$B_f^*(N)^2 := \sum_{p^\nu \leqslant N} |f(p^\nu)|^2 w_N(p^\nu) - \sum_{p \leqslant N} \Big| \sum_{1 \leqslant \nu \leqslant (\ln N)/\ln p} f(p^\nu) w_N(p^\nu) \Big|^2,$$

where w_N is defined as in (3.7). The quantity $B_f^*(N)^2$ represents the sum of the variances, with respect to the measure ν_N, of the additive arithmetic functions f_p defined by

$$f_p(n) := \begin{cases} f(p^\nu) & \text{if } p^\nu \| n, \\ 0 & \text{if } p \nmid n. \end{cases}$$

3.3. Dual form of the Turán–Kubilius inequality

Consider an additive arithmetic function f and let $N \in \mathbb{N}^*$. For $1 \leqslant n \leqslant N$, we have

$$f(n) - \mathbb{E}(Z_{N,f}) = \sum_{r \leqslant N} c_{nr} y_r,$$

with

$$y_r := \begin{cases} f(p^\nu)\sqrt{\dfrac{1-1/p}{p^\nu}} & \text{if } r = p^\nu, \\ 0 & \text{if } r \neq p^\nu, \end{cases}$$

and, with the notation (3.6),

$$c_{nr} := \begin{cases} \xi_{p^\nu}(n)\sqrt{\dfrac{p^\nu}{1-1/p}} - \sqrt{\dfrac{1-1/p}{p^\nu}} & \text{if } r = p^\nu, \\ 0 & \text{if } r \neq p^\nu. \end{cases}$$

With the notation (3.18), the Turán–Kubilius inequality can therefore be written as

$$\sum_{n \leqslant N} \Big| \sum_{r \leqslant N} c_{nr} y_r \Big|^2 \leqslant N C_N^* \sum_{r \leqslant N} |y_r|^2.$$

Since the y_r are arbitrary complex numbers, we can apply the duality principle stated in Lemma I.4.8. It follows that

$$\sum_{r \leqslant N} \Big| \sum_{n \leqslant N} c_{nr} a_n \Big|^2 \leqslant N C_N^* \sum_{n \leqslant N} |a_n|^2$$

for any complex sequence $\{a_n\}_{n=1}^N$. Noting that, for $r = p^\nu$,

$$\sum_{n \leqslant N} c_{nr} a_n = \sqrt{\frac{p^\nu}{1-1/p}} \Big\{ \sum_{n \leqslant N,\, p^\nu \| n} a_n - \frac{1-1/p}{p^\nu} \sum_{n \leqslant N} a_n \Big\},$$

we therefore obtain the following result.

Theorem 3.2. *For any complex sequence $\{a_n\}_{n=1}^N$, we have*

$$\sum_{p^\nu \leqslant N} \frac{p^\nu}{1-1/p} \Big| \sum_{n \leqslant N,\, p^\nu \| n} a_n - \frac{1-1/p}{p^\nu} \sum_{n \leqslant N} a_n \Big|^2 \leqslant N C_N^* \sum_{n \leqslant N} |a_n|^2. \tag{3.30}$$

We recognize here a variant of Theorem I.4.14, obtained as a corollary to the large sieve inequality. Comparing the terms with index p, thus corresponding to $\nu = 1$, highlights two disparities with opposing influences: a single congruence class is considered for each prime number p, but the summation over p is significantly longer.

In practice this estimate is very useful owing to its complete generality and total uniformity. It expresses the fact that *on average* every sufficiently dense sequence a_n is well distributed amongst the congruence classes $n \equiv 0 \,(\mathrm{mod}\, p^\nu)$.

Elliott dedicated a whole book (1997) to the description and development of different methods elaborated to exploit the duality of which Theorem 3.2 is the prototype. Studying this work is strongly recommended to circumscribe the fundamental ideas of modern analytic and probabilistic number theory and explore them further.

3.4. The Hardy–Ramanujan theorem and other applications

As mentioned in Section 3.2, the Turán–Kubilius inequality is particularly well suited to determining the normal order of an additive function. We have the following general result, where the notation is that of (3.10) and (3.11).

Theorem 3.3. *Let f be a complex-valued additive function. Under the assumption that*

$$B_f(N) = o\big(\mathbb{E}(Z_{f,N})\big) \qquad (N \to \infty), \tag{3.31}$$

the function $n \mapsto g(n) := \mathbb{E}\big(Z_{f,n}\big)$ is a normal order for f.

Proof. Let us show first that $g(n) = \big\{1 + o(1)\big\} g(N)$ for all integers $n \leqslant N$ except at most $o(N)$. This is a consequence of the following upper bound, valid for $\sqrt{N} < n \leqslant N$,

$$\begin{aligned} |g(N) - g(n)| &= \Big| \sum_{n < p^\nu \leqslant N} f(p^\nu) \frac{1 - 1/p}{p^\nu} \Big| \\ &\leqslant \Big(\sum_{\sqrt{N} < p^\nu \leqslant N} \frac{1}{p^\nu} \sum_{p^\nu \leqslant N} \frac{|f(p^\nu)|^2}{p^\nu} \Big)^{1/2} \ll B_f(N) = o\big(g(N)\big). \end{aligned}$$

The assertion of the theorem hence results from the validity of

$$\nu_N\big\{n : |f(n) - g(N)| > \varepsilon |g(N)|\big\} = o(1), \tag{3.32}$$

for any fixed $\varepsilon > 0$. However, the left-hand side does not exceed

$$\mathbb{E}_N\Big(\Big|\frac{f(n) - g(N)}{\varepsilon g(N)}\Big|^2\Big) \ll \Big|\frac{B_f(N)}{\varepsilon g(N)}\Big|^2 = o(1), \tag{3.33}$$

by Theorem 3.1 and assumption (3.31). This concludes the proof. □

The classic application of Theorem 3.3 is to the functions $\omega(n)$ and $\Omega(n)$. In both cases, it is easily shown that

$$g(N) = \ln_2 N + O(1), \qquad B_f(N)^2 = \ln_2 N + O(1), \tag{3.34}$$

so that (3.31) is certainly satisfied. We thus obtain the remarkable result that *the number of prime factors of an integer n, counted with or without multiplicity, is normally asymptotic to* $\ln_2 n$. It is this statement which is usually referred to as the Hardy–Ramanujan theorem (1917). The following quantitative version is due to Turán (1934).

Theorem 3.4. *Let $\xi(N) \to \infty$. For $N \geqslant 3$, we have*

$$\left|\left\{n \leqslant N : |\omega(n) - \ln_2 N| > \xi(N)\sqrt{\ln_2 N}\right\}\right| \ll N/\xi(N)^2. \tag{3.35}$$

Moreover, the same upper bound holds with $\Omega(n)$ in place of $\omega(n)$.

Proof. Given (3.34), it is sufficient to select $\varepsilon = \xi(N)/\sqrt{\ln_2 N}$ in (3.33). □

The original proof by Hardy & Ramanujan rested on an upper bound, uniform for N and $k \geqslant 1$, for the *local laws* $\nu_N\{n : \omega(n) = k\}$ and, consequently, implied rather involved considerations—see Exercise 264, p. 467. It was with the intention of simplifying the argument that Turán established in 1934 the initial form of Theorem 3.1.

In view of the inequality $2^{\omega(n)} \leqslant \tau(n) \leqslant 2^{\Omega(n)}$, valid for any integer $n \geqslant 1$, we deduce immediately from Theorem 3.4 that we have

$$\tau(n) = (\ln n)^{\ln 2 + o(1)} \qquad \text{pp} \tag{3.36}$$

and even that

$$\tau(n) = (\ln n)^{\ln 2} \exp\left\{O\left(\xi(n)\sqrt{\ln_2 n}\right)\right\} \qquad \text{pp} \tag{3.37}$$

holds for any function $\xi(n) \to \infty$. *Stricto sensu*, these relations should be interpreted as the assertion that $\ln \tau(n)$ has a normal order. However (3.36) and (3.37) are often referred to by the rather loose statement that $\tau(n)$ has normal order $(\ln n)^{\ln 2}$.

For the first time, we meet here a common situation in probabilistic number theory: the average order—here $\ln n$—does not reflect the normal behavior—here $(\ln n)^{\ln 2}$. The source of this phenomenon lies in the fact that the sum

$$\sum_{n \leqslant x} \tau(n)$$

is dominated by a small number of abnormal integers, for which $\tau(n)$ is large. We shall see later—cf. Exercise 274(b)—that these are precisely those integers n such that $\omega(n) \sim 2\ln_2 n$. We actually have, for each $\varepsilon > 0$,

$$\sum_{\substack{n \leqslant x \\ |\omega(n) - 2\ln_2 n| \leqslant \varepsilon \ln_2 n}} \tau(n) \sim x \ln x \sim \sum_{n \leqslant x} \tau(n) \qquad (x \to \infty).$$

Such a situation leads to a vast field of practical applications. Indeed, it opens the possibility of considering, in certain summations, the arithmetic

function $\tau(n)$ as a weight coefficient which emphasizes those integers n for which $\omega(n) \sim 2\ln_2 n$.

We now give without proof some results which are straightforward applications of the Turán–Kubilius inequality. We let $\xi(n)$ denote an arbitrary quantity such that $\xi(n) \to \infty$.

(a) For $a \geqslant 1$, $q \geqslant 1$ and $(a,q) = 1$, let $\omega(n;a,q)$ be the number of distinct prime factors of n in the arithmetic progression $p \equiv a \,(\mathrm{mod}\, q)$. Then

$$\omega(n;a,q) = \frac{\ln_2 n}{\varphi(q)} + O\Big(\xi(n)\sqrt{\ln_2 n}\Big) \qquad \text{pp.} \tag{3.38}$$

More generally, if E denotes a set of prime numbers such that

$$E(x) := \sum_{p\leqslant x,\, p\in E} \frac{1}{p} \to \infty \qquad (x \to \infty),$$

then we have

$$\omega(n;E) := \sum_{p\mid n,\, p\in E} 1 = E(n) + O\Big(\xi(n)\sqrt{E(n)}\Big) \qquad \text{pp.} \tag{3.39}$$

(b) For fixed $k \geqslant 2$, consider the number $\tau_k(n)$ of representations of an integer n as a product of k positive integers. Then

$$\tau_k(n) = (\ln n)^{\ln k} \exp\Big\{O\Big(\xi(n)\sqrt{\ln_2 n}\Big)\Big\} \qquad \text{pp.} \tag{3.40}$$

Notice the disparity between the "normal" order $(\ln n)^{\ln k}$ and the average order $(\ln n)^{k-1}/(k-1)!$ of $\tau_k(n)$—cf. Theorem II.5.2.

(c) For $k \geqslant 1$, let $\delta_k(n)$ be the number of representations of n in the form $n = [m_1, \dots, m_k]$ with $m_1 \geqslant 1, \dots, m_k \geqslant 1$. Then

$$\delta_k(n) = (\ln n)^{\ln(2^k-1)} \exp\Big\{O\Big(\xi(n)\sqrt{\ln_2 n}\Big)\Big\} \qquad \text{pp.} \tag{3.41}$$

Here again the normal order is very far from the average order: using the convolution formula $\delta_k * \mathbf{1} = \tau^k$, it may be easily deduced from Theorem II.5.2 that we have

$$\sum_{n\leqslant x} \delta_k(n) \sim C_k x (\ln x)^{2^k-2} \tag{3.42}$$

with

$$C_k := \frac{1}{(2^k-2)!} \prod_p \Big(1-\frac{1}{p}\Big)^{2^k} \sum_{\nu\geqslant 0} \frac{(\nu+1)^k}{p^\nu}.$$

3.5. Effective mean value estimates for multiplicative functions

Here we describe another type of tool which furnishes an alternative proof of the Hardy–Ramanujan theorem. This concerns uniform estimates for mean values of multiplicative functions belonging to a certain class. It is important to note right away, and to bear in mind, that the applications are secondary to the fact that the bounds obtained are independent on the function chosen within this class.

Theorem 3.5. *Let f be a non-negative multiplicative function, which, for suitable constants A and B, satisfies*

$$\text{(i)} \quad \sum_{p \leqslant y} f(p) \ln p \leqslant Ay \qquad (y \geqslant 0),$$

$$\text{(ii)} \quad \sum_{p} \sum_{\nu \geqslant 2} \frac{f(p^\nu)}{p^\nu} \ln p^\nu \leqslant B.$$

Then, for $x > 1$, we have

$$\sum_{n \leqslant x} f(n) \leqslant (A + B + 1) \frac{x}{\ln x} \sum_{n \leqslant x} \frac{f(n)}{n}. \tag{3.43}$$

Proof. Denote the left-hand side of (3.43) by $S(x)$, and define

$$L(x) := \sum_{n \leqslant x} f(n)/n,$$

so that, trivially,

$$S(x) \leqslant xL(x) \qquad (x \geqslant 1). \tag{3.44}$$

Now we can write

$$\begin{aligned} S(x) \ln x &= \sum_{n \leqslant x} f(n) \ln \frac{x}{n} + \sum_{n \leqslant x} f(n) \sum_{p \| n} \ln p + \sum_{n \leqslant x} f(n) \sum_{\nu \geqslant 2, p^\nu \| n} \ln p^\nu \\ &= S_1 + S_2 + S_3 \qquad \text{(say)}. \end{aligned}$$

Clearly, $S_1 \leqslant xL(x)$. Making the change of variables $n = mp$ in S_2, we obtain

$$S_2 = \sum_{m \leqslant x} f(m) \sum_{p \leqslant x/m, p \nmid m} f(p) \ln p \leqslant AxL(x).$$

Finally, by interchanging summations, we get that

$$\begin{aligned} S_3 &= \sum_{p} \sum_{\nu \geqslant 2} f(p^\nu) \ln p^\nu \sum_{m \leqslant x/p^\nu, p \nmid m} f(m) \leqslant \sum_{p} \sum_{\nu \geqslant 2} f(p^\nu) \ln p^\nu S\Big(\frac{x}{p^\nu}\Big) \\ &\leqslant \sum_{p} \sum_{\nu \geqslant 2} f(p^\nu) \ln p^\nu \, \frac{x}{p^\nu} L\Big(\frac{x}{p^\nu}\Big) \leqslant BxL(x), \end{aligned}$$

where we have used (3.44) and the fact that L is non-decreasing.

This completes the proof. □

In practice, Theorem 3.5 is often used in the following form.

Corollary 3.6. *Let λ_1 and λ_2 be constants such that $\lambda_1 > 0$ and $0 \leqslant \lambda_2 < 2$. Uniformly for all multiplicative arithmetic functions f satisfying*

$$0 \leqslant f(p^\nu) \leqslant \lambda_1 \lambda_2^{\nu-1} \qquad (p \geqslant 2,\ \nu = 1, 2, \ldots), \tag{3.45}$$

we have

$$\sum_{n \leqslant x} f(n) \ll x \prod_{p \leqslant x} \Big(1 - \frac{1}{p}\Big) \sum_{\nu \geqslant 0} \frac{f(p^\nu)}{p^\nu} \qquad (x \geqslant 1). \tag{3.46}$$

The constant implied in (3.46) *does not exceed*

$$4\big(1 + 9\lambda_1 + \lambda_1\lambda_2/(2 - \lambda_2)^2\big). \tag{3.47}$$

Remark. Here we have not sought to optimize the numerical constants appearing in (3.47).

Proof. It is clear that (3.45) implies conditions (i) and (ii) of Theorem 3.5. By Theorem I.1.4, we have $A \leqslant \lambda_1 \ln 4$. In addition,

$$\begin{aligned} B &\leqslant \lambda_1 \sum_p \frac{\ln p}{p} \sum_{\nu \geqslant 2} \nu \Big(\frac{\lambda_2}{p}\Big)^{\nu-1} \leqslant 2\lambda_1\lambda_2 \sum_p \frac{\ln p}{(p - \lambda_2)^2} \\ &\leqslant \frac{2\lambda_1\lambda_2 \ln 2}{(2 - \lambda_2)^2} + 4\lambda_1 \sum_{n \geqslant 3} \frac{\ln n}{(n-2)^2}\,. \end{aligned}$$

It is easily shown that the series over n is $\leqslant \frac{5}{2}$. Observing that

$$\sum_{n \leqslant x} \frac{f(n)}{n} \leqslant \prod_{p \leqslant x} \sum_{\nu \geqslant 0} \frac{f(p^\nu)}{p^\nu},$$

we therefore obtain the stated estimate with a constant at most equal to

$$K \ln 4 \Big(\frac{1 + 10\lambda_1}{\ln 4} + \lambda_1 + \frac{\lambda_1\lambda_2}{(2 - \lambda_2)^2} \Big)$$

where $K := \sup_{x \geqslant 2} K(x)$ with $K(x) := (1/\ln x) \prod_{p \leqslant x} (1 - 1/p)^{-1}$.

We now prove that $K = K(2) = 2/\ln 2$. To see this, we first check numerically that $K(x) \leqslant K(2)$ for $x \leqslant 300$. Then we observe that for $x > y \geqslant 2$ we have

$$K(x) \leqslant \frac{\pi^2}{6 \ln x} \prod_{p \leqslant x} \Big(1 + \frac{1}{p}\Big) \leqslant \frac{\pi^2}{6 \ln x} \prod_{p \leqslant y} \Big(1 + \frac{1}{p}\Big) \exp\Big\{ \sum_{y < p \leqslant x} \frac{1}{p} \Big\}.$$

Using Mertens' first theorem (Theorem I.1.8) in the form

$$F(t) := \sum_{p \leqslant t} \frac{\ln p}{p} \leqslant \ln(4t) \qquad (t \geqslant 2)$$

and a partial summation, we readily get

$$\sum_{y<p\leqslant x}\frac{1}{p}\leqslant 1+\frac{\ln 4-F(y)}{\ln y}+\ln_2 x-\ln_2 y\qquad (x>y\geqslant 2),$$

from which it follows, taking $y=300$, that $K(x)<2.87<K(2)$ for $x>300$. This completes the proof. □

The following result, which is an immediate consequence of Corollary 3.6, will help to establish a strong form of Theorem 3.4. For $t>0$, we write

$$\omega(n,t):=\sum_{p\mid n,\,p\leqslant t}1,\qquad \Omega(n,t):=\sum_{p^\nu\|n,\,p\leqslant t}\nu\qquad (n\geqslant 1). \tag{3.48}$$

Theorem 3.7. *Let $y_0>0$. Uniformly for $0\leqslant y\leqslant y_0$, $x\geqslant t\geqslant 2$, we have*

$$\sum_{n\leqslant x}y^{\omega(n,t)}\ll x(\ln t)^{y-1}. \tag{3.49}$$

Under the additional hypothesis $y_0<2$, the same estimate holds for $\Omega(n,t)$.

Proof. For each t, the function $n\mapsto y^{\omega(n,t)}$ is multiplicative and satisfies (3.45) with $\lambda_1=1+y_0$, $\lambda_2=1$. This condition is still realized for $y^{\Omega(n,t)}$, with $\lambda_1=1+y_0$, $\lambda_2=\max(1,y_0)$. The upper bound (3.46) is then

$$x\prod_{p\leqslant t}\Big(1-\frac{1}{p}\Big)\Big\{1+\frac{y}{p}+O\Big(\frac{1}{p^2}\Big)\Big\}.$$

From this, we deduce the stated result by Mertens' formula. □

Theorem 3.8. *Uniformly for $x\geqslant t\geqslant 3$ and $0\leqslant\xi<\sqrt{\ln_2 t}$, we have*

$$\Big|\big\{n\leqslant x:|\omega(n,t)-\ln_2 t|>\xi\sqrt{\ln_2 t}\big\}\Big|\ll x\mathrm{e}^{-\xi^2/3}. \tag{3.50}$$

Furthermore, if $\xi\ll(\ln_2 t)^{1/6}$, we can replace $\mathrm{e}^{-\xi^2/3}$ by $\mathrm{e}^{-\xi^2/2}$. The same assertions hold for $\Omega(n,t)$ provided $\xi\leqslant c\sqrt{\ln_2 t}$ for some fixed constant $c<1$.

When $t=x$, we have $\omega(n,t)=\omega(n)$ for any integer $n\leqslant x$. In its range of validity, the upper bound (3.50) hence constitutes a remarkable improvement of (3.35).

Proof. Let us set $\delta:=\xi/\sqrt{\ln_2 t}$, so that $0\leqslant\delta<1$, and denote by $\chi(n)$ the indicator function of the set of integers n such that

$$|\omega(n,t)-\ln_2 t|>\delta\ln_2 t. \tag{3.51}$$

Plainly

$$\chi(n)\leqslant(1-\delta)^{\omega(n,t)-(1-\delta)\ln_2 t}+(1+\delta)^{\omega(n,t)-(1+\delta)\ln_2 t}.$$

Summing over $n \leqslant x$ and appealing to (3.49), we obtain

$$\sum_{n\leqslant x} \chi(n) \ll x(\ln t)^{-Q(1-\delta)} + x(\ln t)^{-Q(1+\delta)} \tag{3.52}$$

with

$$Q(y) := y\ln y - y + 1 = \int_0^{y-1} \ln(1+v)\,\mathrm{d}v \qquad (y>0). \tag{3.53}$$

For $0 \leqslant \delta \leqslant 1$, we have the estimates

$$Q(1\pm\delta) \geqslant \tfrac{1}{3}\delta^2 \quad \text{and} \quad Q(1\pm\delta) = \tfrac{1}{2}\delta^2 + O(\delta^3).$$

This implies the stated bounds. The proof is unchanged in the case of $\Omega(n,t)$. □

3.6. Normal structure of the sequence of prime factors of an integer

The factorization of an integer n is completely determined by the behavior of the step-function $\Omega(n,t)$. Here we deal with the technically more manageable function $\omega(n,t)$. For all intents and purposes, this choice is irrelevant since we have $0 \leqslant \Omega(n,t) - \omega(n,t) \leqslant \Omega(n) - \omega(n)$ for all n and t, and Theorems I.3.6 & I.3.7 imply that this last difference has bounded mean value: we deduce from this that the bound

$$\Omega(n,t) - \omega(n,t) \ll \xi(n) \qquad \text{pp} \tag{3.54}$$

holds uniformly in t for any given function $\xi(n) \to \infty$.

Theorem 3.8 describes the behavior of $\omega(n,t)$ for fixed t. However, only the variations as a function of t can yield information on the multiplicative structure of n. Thus we need a pp approximation to $t \mapsto \omega(n,t)$ with respect to *the norm of uniform convergence.* The following result, which is a simple consequence of Theorem 3.8, provides such an estimate.

Theorem 3.9. *Let $\varepsilon > 0$, and $\xi(n) \to \infty$. We have*

$$\sup_{\xi(n)\leqslant t\leqslant n} \left| \frac{\omega(n,t) - \ln_2 t}{\sqrt{2\ln_2 t \ln_3 t}} \right| \leqslant 1+\varepsilon \qquad \text{pp.} \tag{3.55}$$

Proof. Let ξ be an arbitrarily large real number. Plainly, $\xi(n) > \xi$ pp. It is hence sufficient to establish that the upper density of those integers n contradicting (3.55) with $\xi(n)$ replaced by ξ tends to 0 as $\xi \to \infty$. To this end, we introduce the *check-points*

$$t_j := \exp\exp j \qquad (j \geqslant \ln_2 \xi),$$

and note that, since $\omega(n,t)$ is a non-decreasing function of t, and since $\ln_2 t_{j+1} - \ln_2 t_j \leqslant 1$, we have, for sufficiently large ξ,

$$\sup_{\xi \leqslant t \leqslant n} \left| \frac{\omega(n,t) - \ln_2 t}{\sqrt{2 \ln_2 t \ln_3 t}} \right| > 1 + \varepsilon \;\Rightarrow\; \sup_{\xi \leqslant t_j \leqslant n} \left| \frac{\omega(n,t_j) - j}{\sqrt{2j \ln j}} \right| > \sqrt{1+\varepsilon}.$$

Choosing, in Theorem 3.8, $t = t_j$ and $\xi = \sqrt{(2+2\varepsilon)\ln j}$ for all integers j such that $\ln_2 \xi \leqslant j \leqslant \ln_2 x$, we see that the upper density of those integers n satisfying the last inequality is

$$\ll \sum_{j \geqslant \ln_2 \xi} \frac{1}{j^{1+\varepsilon}} \ll \frac{1}{(\ln_2 \xi)^{\varepsilon}}.$$

This completes the proof. □

Theorem 3.9 may be directly formulated in terms of the standard factorization of n. Let $p_j(n)$ denote the jth distinct prime factor of n, so that the canonical factorization may be written as

$$n = \prod_{1 \leqslant j \leqslant \omega(n)} p_j(n)^{\nu_j}$$

where the ν_j are strictly positive integers. Then $\omega(n, p_j(n)) = j$. With the change of variable $t \mapsto p_j(n)$ in (3.55), we obtain the following result.

Theorem 3.10. *Let $\varepsilon > 0$ and $\xi(n) \to \infty$. We have*

$$\sup_{\xi(n) \leqslant j \leqslant \omega(n)} \left| \frac{\ln_2 p_j(n) - j}{\sqrt{2j \ln j}} \right| \leqslant 1 + \varepsilon \qquad \text{pp.} \tag{3.56}$$

This result was announced in a more precise form by Erdős (1946), but without proof. Details were provided by Hall & Tenenbaum (1988), chap. 1. The main device to keep in mind is that, for a normal integer, the jth prime factor is roughly of size $\exp\exp j$. This generates a valuable heuristic model for the normal multiplicative structure of the integers—the scope of which, however, should not be overemphasized: taken too literally, this model leads to false conjectures; cf. Hall & Tenenbaum (1988) § 1.2, see also the Notes.

Notes

§ 3.2. The lower bound (3.22) was first observed by La Bretèche & Tenenbaum (2005b).

If f is restricted to the space of strongly additive functions, an elegant result of J. Lee (1989) specifies that

$$C_N^+ = \tfrac{3}{2} - \frac{\alpha}{\ln N} + O\left(\left(\frac{\ln_2 N}{\ln N}\right)^2\right) \tag{3.57}$$

where α is a positive absolute constant. This implies, in particular, the existence of an absolute N_0 such that any strongly additive f satisfies

$$\mathbb{V}_N^*(f) \leqslant \tfrac{3}{2} B_f^+(N)^2 \leqslant 6\mathbb{V}(Z_{f,N}) \qquad (N \geqslant N_0). \tag{3.58}$$

In general terms, it is expected that the distribution of f (with respect to ν_N) is close to that of $Z_{f,N}$. In this context, Ruzsa (1982) introduced an interesting methodological distinction by defining, for problems of this type:

(a) *the direct approach*, where the theory of probability appears only as a heuristic model and for which the proof is purely arithmetical;

(b) *the indirect approach*, characterized by a comparison of (f, ν_N) and $(Z_{f,N}, P)$ as random variables, and in which the arithmetical result is deduced as an additional step from a genuinely probabilistic result.

As for the Turán–Kubilius inequality, the direct classical statement (neglecting the numerical values of the constants) is

$$\mathbb{V}_N(f) \ll B_f^+(N)^2 \tag{3.59}$$

while the indirect statement is

$$\mathbb{V}_N(f) \ll \mathbb{V}(Z_{f,N}). \tag{3.60}$$

The first upper bound follows from the second by appealing to the probabilistic theorem on the variance of a sum of independent random variables: indeed,

$$V(\zeta_p) \leqslant E(\zeta_p^2) \leqslant \sum_{1 \leqslant \nu \leqslant (\ln N)/\ln p} \frac{|f(p^\nu)|^2}{p^\nu}.$$

Although deliberately presented in the form (3.60), appearing to be more fundamental, our proof of Theorem 3.1, which essentially follows that of Elliott (1979), is of direct type. The following remarkable result of Ruzsa provides an indirect approach, of which (3.60) is an immediate consequence.

Theorem 3.11 (Ruzsa, 1984). *Let f be a complex-valued additive function and let $F : \mathbb{R}^+ \to \mathbb{R}^+$ be non-decreasing. Uniformly for $A \in \mathbb{R}$ and $N \in \mathbb{Z}^+$, we have*

$$\mathbb{E}_N\big(F(|f-A|)\big) \ll \mathbb{E}\big(F(3|Z_{f,N}-A|)\big). \tag{3.61}$$

To determine the best constant in (3.61) is an interesting open problem. Disproving a conjecture made in his 1984 article, Ruzsa showed that it is not in general possible to replace the constant 3 by a function of N tending to 1. More precisely, he established (private communication) that the optimal constant is $\geqslant 1 + 2/\mathrm{e} \approx 1.73575$.

The Turán–Kubilius inequality does not always provide the exact order of magnitude of $\mathbb{V}_N(f)$ or $\mathbb{V}_N^*(f)$. In 1983, Ruzsa showed that, for real f,

$$\mathbb{V}_N(f) \asymp \min_{\lambda\in\mathbb{R}} \big\{\lambda^2 + B_N^+(f - \lambda \ln)^2\big\}. \tag{3.62}$$

Subsequently, Hildebrand obtained an asymptotic formula for $\mathbb{V}_N(f)$ as a function depending solely on the numbers $f(p)$, $p \leqslant N$. He derived several interesting corollaries from his result. One of these is a new proof that the factor $\frac{3}{2} + o(1)$ is admissible in (3.28). Another is a characterization of those additive functions such that

$$\mathbb{V}_N(f) \asymp B_f^+(N)^2. \tag{3.63}$$

Theorem 3.12 (Hildebrand, 1983). *Let f be an additive function. Then* (3.63) *holds if, and only if,*

$$\limsup_{N\to\infty} \frac{1}{B_f^+(N)\ln N}\Big|\sum_{p\leqslant N} \frac{f(p)\ln p}{p}\Big| < \frac{1}{\sqrt{2}}. \tag{3.64}$$

§ 3.3. Elliott (1979, chap. 4), gives several variants of the inequality (3.30) in Theorem 3.2. For instance, we have the following result.

Theorem 3.13 (Elliott, 1979). *For all real numbers $x \geqslant 2$ and for any complex sequence $\{a_n : 1 \leqslant n \leqslant x\}$, we have*

$$\sum_{p\leqslant\sqrt{x}} p\Big|\sum_{n\leqslant x,\, p\mid n} a_n - \frac{1}{p}\sum_{n\leqslant x} a_n\Big|^2 \leqslant \Big(1 + \frac{36}{\ln x}\Big)x\sum_{n\leqslant x}|a_n|^2.$$

Here again, the constant 36 is not optimal.

It is sometimes more useful to employ the dual form of (3.28) rather than that of (3.29). We then obtain, for any complex sequence $\{a_n\}_{n=1}^N$,

$$\sum_{p^\nu\leqslant N} p^\nu\Big|\sum_{p^\nu\parallel n,\, n\leqslant N} a_n - \frac{1-1/p}{p^\nu}\sum_{n\leqslant N} a_n\Big|^2 \leqslant NC_N^+\sum_{n\leqslant N}|a_n|^2.$$

In certain applications, the fact that $C_N^+ = \frac{3}{2} + o(1)$ may indeed reveal to be critical.

For a deeper study of the relations between the Turán–Kubilius inequality and the large sieve, see chapter 4 of Elliott (1979) and especially Elliott's book (1997).

§ 3.4. It was with the aim of establishing Theorem 3.4 that, in his 1934 article, Turán stated the inequality of Theorem 3.1 for the function $\omega(n)$, without giving a precise value to the constant. He soon (1936) generalized the results to the case when $0 \leqslant f(p) \leqslant C$. The method was later systematically exploited by Kubilius (1956, 1964).

§ 3.5. Theorem 3.5 appears in this form in Hall & Tenenbaum (1988), th. 01. It may be seen as a simplified, and slightly weaker, version of a result of Halberstam & Richert, itself generalizing a theorem of Hall (1974)—see Exercise 276—and which we now state.

Theorem 3.14 (Halberstam & Richert, 1979). *Let f be a non-negative multiplicative function which, for a suitable constant $\kappa > 0$, satisfies*

$$\text{(i)} \quad \sum_{p \leqslant y} f(p) \ln p \leqslant \kappa y + O\Big(\frac{y}{(\ln y)^2}\Big) \qquad (y \geqslant 2)$$

$$\text{(ii)} \quad \sum_{p \geqslant y} \sum_{\nu \geqslant 2} \frac{f(p^\nu)}{p^\nu} \ln p^\nu \ll \frac{1}{\ln y} \qquad (y \geqslant 2).$$

Then we have

$$\text{(3.65)} \qquad \sum_{n \leqslant x} f(n) \leqslant \Big\{1 + O\Big(\frac{1}{\ln x}\Big)\Big\} \frac{\kappa x}{\ln x} \sum_{n \leqslant x} \frac{f(n)}{n} \qquad (x \geqslant 2).$$

The implicit constant depends at most on the constants implied in (i) and (ii).

Analogous results for upper and lower bounds are given by Hildebrand (1984a, 1987b).

The constant κ in (3.65) is optimal, as shown by the choice

$$f(p^\nu) := \begin{cases} 0 & (p \leqslant x/\ln x), \\ \kappa & (x/\ln x < p \leqslant x). \end{cases}$$

The technique employed to prove Theorem 3.8 is an example of an extremely fruitful general elementary method— see Hall & Tenenbaum (1988), chap. 0, and also Section III.5.1 of the present work. In the case of Theorem 3.8, and for $t = x$, the idea goes back to Turán—cf. Elliott (1980), pp. 18–20.

For other applications of Theorem 3.5, see Exercises 268–274.

§ **3.6.** The more precise version of Theorem 3.10, announced by Erdős (1946) and established in detail in the book of Hall & Tenenbaum (1988), chap. 1, is the following.

Theorem 3.15 (Erdős, 1946). *Let $\varepsilon > 0$, and let $\xi(n)$ denote a function tending to ∞ with n, sufficiently slowly. We have*

$$\sup_{\xi(n)\leqslant j\leqslant\omega(n)} \left| \frac{\ln_2 p_j(n) - j}{\sqrt{2j \ln_2 j}} \right| \leqslant 1+\varepsilon \qquad \text{pp.} \tag{3.66}$$

Furthermore, if $1+\varepsilon$ is replaced by $1-\varepsilon$ on the right-hand side, the density of the set of those integers n satisfying (3.66) *is zero.*

The general idea in the proof of this result consists in showing that the law of the iterated logarithm is asymptotically applicable to the arithmetic function

$$\omega(n,t) = \sum_{p\leqslant t} \chi_p(n)$$

where $\chi_p(n)$ is the indicator function of the set of those integers n divisible by p. As previously seen, the χ_p are not completely independent, and this generates substantial difficulties.

For individual values of $j = j(n)$, it is possible to attain very precise information. Galambos has thus shown the following result.

Theorem 3.16 (Galambos, 1976).

(i) *Let $\varepsilon > 0$ and let $j = j(N) \to \infty$ in such a way that*

$$j(N) \leqslant \ln_2 N - (\ln_2 N)^{(1/2)+\varepsilon}.$$

Then we have

$$\lim_{N\to\infty} \nu_N\{n : \ln_2 p_j(n) \leqslant j + z\sqrt{j}\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} \mathrm{e}^{-t^2/2}\,\mathrm{d}t. \tag{3.67}$$

(ii) *Let $\varepsilon > 0$ and let $j = j(N) \to \infty$ in such a way that*

$$j(N) \leqslant (1-\varepsilon)\ln_2 N.$$

Then we have

$$\lim_{N\to\infty} \nu_N\{n : \ln_2 p_{j+1}(n) - \ln_2 p_j(n) \leqslant z\} = 1 - \mathrm{e}^{-z} \quad (z > 0). \tag{3.68}$$

Theorems 3.9, 3.10 or 3.15 are consistent with the prediction that the quantity $\omega(n;s,t) := \omega(n,t) - \omega(n,s)$ is normally of order $\ln(\ln t/\ln s)$. Erdős showed in 1969 that

$$\omega(n;s,t) \sim \ln(\ln t/\ln s) \qquad \text{pp}$$

holds uniformly for $3 \leqslant s \leqslant t \leqslant n$, provided that the right-hand side tends to infinity faster than $\ln_3 n$. This condition is indeed necessary. Other results

on the normal distribution of prime factors are available in the article of Bovey (1977).

We can apply Theorem 3.15 to determine the normal order of the jth divisor of an integer. Let $\{d_j(n) : 1 \leqslant j \leqslant \tau(n)\}$ denote the increasing sequence of divisors of n. We have the following result (a weaker form of which is described in Exercise 269).

Theorem 3.17 (Hall & Tenenbaum, 1988). *Let $\varepsilon > 0$, and let $\xi(n)$ be a function tending to ∞ with n, sufficiently slowly. We have*

$$\sup_{\xi(n)\leqslant j\leqslant\tau(n)} \left| \frac{\ln_2 d_j(n) - (\ln j)/\ln 2}{\sqrt{(2/\ln 2)\ln j \ln_3 j}} \right| \leqslant 1+\varepsilon \qquad \text{pp.} \tag{3.69}$$

Furthermore, if $1+\varepsilon$ is replaced by $1-\varepsilon$ in the right-hand side, the density of the set of those integers n satisfying (3.69) *is zero.*

Thus the heuristic model for the structure of the set of divisors of a normal integer is

$$d_j(n) \approx \exp\left\{j^{1/\ln 2}\right\} \qquad (\xi(n) \leqslant j \leqslant \tau(n)). \tag{3.70}$$

This agrees well with the normal order $(\ln n)^{\ln 2}$ of $\tau(n)$, but can lead to erroneous conclusions if considered as an asymptotic formula with an excessively local interpretation. Thus, an old conjecture of Erdős, established by Maier & Tenenbaum (1984), asserts that

$$\min_{1\leqslant j\leqslant\tau(n)} \frac{d_{j+1}(n)}{d_j(n)} \to 1, \qquad \text{pp.} \tag{3.71}$$

This implies that the above minimum is attained as $j \to \infty$, which contradicts (3.70) if we ascribe the symbol $\approx$ too precise a meaning.

Another model which also clashes with an excessively strict interpretation of (3.70) as a description of the structure of the set of divisors of a normal integer is that of a fractal object of dimension $\ln 2$. Such a model may be shown to be consistent with some rather strong arithmetical results concerning arithmetic functions closely linked to the local structure of the sequence of divisors—cf. Mendès France & Tenenbaum (1993).

Theorems 3.15–3.17 do not provide any information on the sequences $p_j(n)$ or $d_j(n)$ for small values of j. In 1979, Erdős introduced the densities $\Lambda_j(d)$, $\lambda_j(p)$, respectively corresponding to the sequences of those integers n satisfying $d_j(n) = d$ or $p_j(n) = p$. An asymptotic study of these densities was undertaken by Erdős & Tenenbaum (1989a), who prove, in particular, that $\Lambda_j(d) > 0$ if and only if $\tau(d) \leqslant j \leqslant d$. The study was continued by De Koninck & Tenenbaum (2002) and La Bretèche & Tenenbaum (2002).

Set

$$\lambda_k^*(y) := \mathbf{d}\{n : p_k(n) > y\} = \sum_{p>y} \lambda_k(p).$$

Strengthening a result of Erdős (1969), De Koninck & Tenenbaum show that, uniformly for $k \geqslant 1$ and $z \in \mathbb{R}$, we have

$$\lambda_k^*\big(\exp\exp\{k + z\sqrt{k}\}\big) = \Phi^*(z) - \frac{\Phi_0(z)}{\sqrt{2\pi k}} + O\Big(\frac{1}{k}\Big), \tag{3.72}$$

with

$$\Phi^*(z) := \int_z^\infty \mathrm{e}^{-u^2/2} \frac{\mathrm{d}u}{\sqrt{2\pi}}, \qquad \Phi_0(z) := \{\tfrac{1}{3} + A - \tfrac{1}{3}z^2\}\mathrm{e}^{-z^2/2},$$

$$A := \gamma - \sum_p \Big\{ \ln\Big(\frac{1}{1-1/p}\Big) - \frac{1}{p}\Big\} \approx 0.26150.$$

La Bretèche & Tenenbaum provide an asymptotic estimate for $\Lambda_k^* := \max_d \Lambda_k(d)$ as $k \to \infty$ and show in particular that this maximum is attained when

$$d = k^{\{1+o(1)\}(\ln_2 k)/\ln 4}.$$

Exercises

264. *Hardy–Ramanujan inequality (1917).*

(a) Show that for $x > 0$ and $0 < \alpha < 1 < \beta$ we have

$$\sum_{k \leqslant \alpha x} \frac{\mathrm{e}^{-x} x^k}{k!} < \frac{\mathrm{e}^{-Q(\alpha)x}}{(1-\alpha)\sqrt{\alpha x}} \quad \text{and} \quad \sum_{k \geqslant \beta x} \frac{\mathrm{e}^{-x} x^k}{k!} < \frac{\sqrt{\beta}\mathrm{e}^{-Q(\beta)x}}{(\beta-1)\sqrt{2\pi x}}$$

with $Q(y) := y \ln y - y + 1$.[3]

(b) Consider the function $\pi_k(x) := \big|\{n \leqslant x : \omega(n) = k\}\big|$. Show that, for $k \geqslant 1$, $x > 0$,

$$(k+1)\pi_{k+1}(x) \leqslant \sum_{p^\nu \leqslant x} \pi_k\big(x/p^\nu\big).$$

Deduce the existence of two absolute constants c_0 and c_1, such that

$$\pi_k(x) \leqslant \frac{c_0 x}{\ln x} \frac{(\ln_2 x + c_1)^{k-1}}{(k-1)!} \qquad (x \geqslant 3,\ k \geqslant 1).$$

(c) Recover, in this way, Theorem 3.8 for the function $\omega(n) = \omega(n, x)$.

(d) Adapt the method to handle the case of the function $\Omega(n)$.[4]

265. Let $A(n)$ be the additive arithmetic function defined by $A(p^\nu) = \nu p$— cf. Alladi & Erdős (1977, 1979).

(a) Show that $\displaystyle\sum_{n \leqslant x} A(n) = \frac{\pi^2 x^2}{12 \ln x}\Big\{1 + O\Big(\frac{1}{\ln x}\Big)\Big\}$.

(b) Show that, for all $n \geqslant 1$, we have $P^+(n) - 1 \leqslant A(n) \leqslant \Omega(n) P^+(n)$.

(c) Show that the sequence of those integers n such that $P^+(n) \geqslant \sqrt{n}$ has natural density, equal to $\ln 2$.

(d) Let $\chi(n)$ be the characteristic function of the set $\mathcal{A}$ of those integers n such that $P^+(n) \leqslant n^{2/5}$. Show that $\chi(n) \geqslant 1 - \sum_{p \mid n,\, p > n^{2/5}} 1$, and deduce that $\mathcal{A}$ has positive lower density.

(e) Show that $A(n)$ does not have a monotonic normal order.

266. Let $g(n)$ be the strongly additive function such that $g(p^\nu) = \ln p$.

(a) Show, in the notation of § 2, that we have $B_g(N) \sim \mathbb{E}_N(g)/\sqrt{2}$ as $N \to \infty$.

(b) Show that $\sum_{n \leqslant N} \big(\ln n - g(n)\big) \ll N$ $(N \geqslant 2)$.

(c) Deduce that g has a normal order, *not accessible by the Turán–Kubilius inequality.*

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[3] See Norton, 1976, and, for more precise estimates, 1978.

[4] For generalizations and lower bounds, see Norton (1976, 1979, 1982), Balazard (1987) and the references indicated in Exercise 286, p. 508.

267. Let $\delta > 1$ and $g_\delta(n) := \sum_{p|n}(\log p)^\delta$. Show that

$$\alpha_n^\delta(\log n)^\delta \leqslant g_\delta(n) \leqslant \alpha_n^{\delta-1}(\log n)^\delta \qquad (n \geqslant 1)$$

with $\alpha_n := \log P^+(n)/\log n$. Show that for each α, $\frac{1}{2} \leqslant \alpha \leqslant 1$, the sequence of those integers n such that $\alpha_n \geqslant \alpha$ has a natural density, and compute this density. Deduce that g_δ does not have a monotonic normal order.[5]

268. For $y \geqslant 2$, denote by $\chi(n;y)$ the indicator function of the set of those integers n such that $P^+(n) \leqslant y$ and let $\Psi(x,y)$ be its summatory function.

(a) Show that, for each $\alpha > 0$, we have

$$\Psi(x,y) \leqslant \sqrt{x} + \sum_{n\leqslant x} (n/\sqrt{x})^\alpha \chi(n;y).$$

(b) By applying Theorem 3.5 to the function $n \mapsto n^\alpha\chi(n;y)$ for some suitable choice of α, deduce that we have

$$\Psi(x,y) \leqslant c_1 x^{1-c_2/\log y} \qquad (x \geqslant y \geqslant 2)$$

where c_1 and c_2 are absolute positive constants.

269. Let $\xi(n)$ denote a function tending sufficiently slowly to ∞, and write $L(t) := \sqrt{2\log_2 t \log_3 t}$, $\omega(n,t) := \sum_{p|n,\, p\leqslant t} 1$, $\Omega(n,t) := \sum_{p^\nu\|n,\, p\leqslant t} \nu$, and $\tau(n,t) := \sum_{d|n,\, d\leqslant t} 1$.

(a) Show that, for each $\varepsilon > 0$, we have

$$\sup_{\xi(n)\leqslant t\leqslant n} \Big|\frac{\Omega(n,t) - \log_2 t}{L(t)}\Big| \leqslant 1+\varepsilon, \qquad \text{pp.}$$

(b) Deduce that

$$\tau(n,t) \leqslant (\log t)^{\log 2} 2^{(1+\varepsilon)L(t)} \quad \big(\xi(n) \leqslant t \leqslant n\big) \qquad \text{pp.}$$

(c) Let $\tau(n;t,u) := \big|\{d : d|n,\, d > t,\, P^+(d) \leqslant u\}\big|$. Show by using the result of Exercise 268 that we have

$$\sum_{n\leqslant x} \tau(n;t,u) \ll x(\log u)\exp\Big\{-c_2\frac{\log t}{\log u}\Big\} \qquad (2 \leqslant t,\, u \leqslant x).$$

(d) Set $\alpha := 1/\log 2$, $t_j := \exp j^\alpha$ $(j=1,2,\ldots)$, $u_j := \exp\{c_2 j^\alpha/(10\log j)\}$ $(j = 1,2,\ldots)$. Show that $\tau(n;t_j,u_j) = 0$ $\big(\xi(n) \leqslant j \leqslant (\log n)^{\log 2}\big)$ pp.

(e) Show that $2^{\omega(n,u)} \leqslant \tau(n,t) + \tau(n;t,u)$ and deduce that, for each $\varepsilon > 0$, we have

$$\tau(n,t_j) \geqslant j 2^{-(1+\varepsilon)L(t_j)} \quad \big(\xi(n) \leqslant j \leqslant (\log n)^{\log 2}\big), \quad \text{pp.}$$

[5] This also holds when $0 < \delta < 1$, see Exercise 291, p. 554.

(f) Establish that, for each $\varepsilon > 0$, we have

$$\sup_{\xi(n)\leqslant j\leqslant \tau(n)} \left| \frac{\ln_2 d_j(n) - \alpha \ln j}{\sqrt{2\alpha \ln j \ln_2 j}} \right| \leqslant 1+\varepsilon \qquad \text{pp.}$$

270. *Set of multiples of* $]y, 2y]$. Let $y \geqslant 2$, $\mathcal{A} := \mathbb{Z}^+ \cap]y, 2y]$, and $\mathcal{M}_y := \mathcal{M}(\mathcal{A})$ be the set of multiples of $\mathcal{A}$, that is the set of those integers n having at least one divisor d such that $y < d \leqslant 2y$.

(a) Show the existence of $\mathbf{d}\mathcal{M}_y =: \varepsilon_y$ and derive an expression for this quantity using the inclusion–exclusion principle.

(b) Let $\mathcal{B}_y := \{n : \Omega(n, y) \geqslant (\ln_2 y)/\ln 2\}$. Show the existence of $\mathbf{d}\mathcal{B}_y$ and establish the formula

$$\mathbf{d}\mathcal{B}_y = \prod_{p\leqslant y} (1 - \frac{1}{p}) \sum_{b\in\mathcal{B}_y,\, P^+(b)\leqslant y} \frac{1}{b}.$$

[*Cf. Exercise* 245, *page 421.*]

(c) Let $\chi_y(n)$ be the indicator function of $\mathcal{B}_y$. Show that, for all $z \geqslant 1$,

$$\chi_y(n) \leqslant (\ln y)^{-(\ln z)/\ln 2} z^{\Omega(n,y)} \qquad (n \geqslant 1).$$

By choosing the parameter z suitably, deduce that

$$\mathbf{d}\mathcal{B}_y \leqslant K(\ln y)^{-\delta}$$

where K is an absolute constant and $\delta := 1 - (1 + \ln_2 2)/\ln 2 \approx 0.08607$.

(d) Let $\mathcal{B}'_y := \mathcal{M}_y \setminus \mathcal{B}_y$. Show that $\mathcal{B}'_y$ has a natural density. Denote by $\chi'_y(n)$ the indicator function of $\mathcal{B}'_y$. Show that, for all z, $0 < z \leqslant 1$, we have

$$\chi'_y(n) \leqslant (\ln y)^{-(\ln z)/\ln 2} z^{\Omega(n,y)} \sum_{d|n,\, y<d\leqslant 2y} 1.$$

By choosing z suitably, deduce that $\mathbf{d}\mathcal{B}'_y \leqslant K'(\ln y)^{-\delta}$, where K' is absolute, and thus that $\varepsilon_y \leqslant K_0(\ln y)^{-\delta}$ with $K_0 = K + K'$.[6]

271. *A theorem of Besicovitch (1934).* With the notation $\mathcal{M}_y$ and ε_y from Exercise 270, let $H(x, y) := |\{n \leqslant x : n \in \mathcal{M}_y\}|$. Let $\varepsilon > 0$. Show the existence of a sequence $y_k \to \infty$ such that

(i) $\varepsilon_{y_k} \leqslant \varepsilon/2^{k+2}$ $\qquad (k \geqslant 0)$

(ii) $H(x, y_k) \leqslant 2\varepsilon_{y_k} x$ $\qquad (k \geqslant 0,\ x \geqslant y_{k+1})$.

Set $\mathcal{A} = \mathbb{Z}^+ \cap \bigcup_{k\geqslant 0}]y_k, 2y_k]$ and let $\mathcal{M} = \mathcal{M}(\mathcal{A})$ be the set of multiples of $\mathcal{A}$. Show that $\underline{\mathbf{d}}\mathcal{M} \leqslant \varepsilon$, $\overline{\mathbf{d}}\mathcal{M} \geqslant \frac{1}{2}$, and thus that *a set of multiples does not necessarily have a natural density.*

[6] Improving on estimates of Tenenbaum (1984)—see also Hall & Tenenbaum (1988), chapter 2—Ford (2008) showed that we actually have $\varepsilon_y \asymp (\ln y)^{-\delta}(\ln_2 y)^{-3/2}$.

Using the method of Exercise 270, find a uniform upper bound for $H(x,y)$ in the range $2 \leqslant y \leqslant \sqrt{x}$. Deduce that an admissible choice is $y_k = \exp\{c(2^k\varepsilon^{-1})^{1/\delta}\}$ with some sufficiently large absolute constant c.

272. *Divisor density (Hall, 1978).* Let $\mathcal{A}$ be an integer sequence with indicator function χ. We say that $\mathcal{A}$ has divisor density z, and then write $\mathbf{D}\mathcal{A} = z$, if we have

$$\tau(n,\mathcal{A}) := \sum_{d|n} \chi(d) = \{z + o(1)\}\tau(n) \qquad \text{pp.}$$

Throughout this exercise, we let the integers a and q, with $q \geqslant 2$, be given and put

$$\mathcal{A} := \{d : \omega(d) \equiv a \,(\mathrm{mod}\, q)\}.$$

(a) Show that $\chi(d) = (1/q)\sum_{j=0}^{q-1} \mathrm{e}^{2\pi i j(\omega(d)-a)/q}$ $(d \geqslant 1)$. Deduce the existence of $\mathbf{d}\mathcal{A}$ and indeed that $\mathbf{d}\mathcal{A} = 1/q$.

(b) Show that $\tau(n,\mathcal{A}) = (1/q)\sum_{j=0}^{q-1} \mathrm{e}^{-2\pi i j a/q} \prod_{p^\nu \| n}(1 + \nu \mathrm{e}^{2\pi i j/q})$.

(c) With the notation $m := \prod_{p\|n} p$, establish that

$$\big|\tau(n,\mathcal{A}) - \tau(n)/q\big| \leqslant \tau(n/m)\big\{2\cos \pi/q\big\}^{\omega(m)} = \tau(n)\big(\cos \pi/q\big)^{\omega(m)}.$$

(d) Show that the arithmetic function $n \mapsto \omega(m)$ is additive and has normal order $\ln_2 n$.

(e) Deduce that $\mathbf{D}\mathcal{A} = 1/q$.

273. *Another result on divisor density.*

Let δ be a fixed real number, $0 < \delta < \frac{1}{2}$. Set

$$\mathcal{A}(\delta) := \big\{m : m \geqslant 3,\, \omega(m) \leqslant (\tfrac{1}{2} + \delta)\ln_2 m\big\}.$$

(a) Show that $\mathbf{d}\mathcal{A}(\delta) = 0$.

(b) Show that for each ε, $0 < \varepsilon < 1$, we have

$$|\{p : p|n,\, \ln_2 p > (1-\varepsilon)\ln_2 n\}| \sim \varepsilon\omega(n) \quad \text{pp}$$

and deduce that $|\{d : d|n,\, \ln_2 d > (1-\varepsilon)\ln_2 n\}| \sim \tau(n)$ pp.

(c) Show that $\sum_{d|n} y^{\omega(d)} \leqslant (1+y)^{\Omega(n)}$ $(y \geqslant 0,\, n \geqslant 1)$ and use this inequality to establish that, for all η, $0 < \eta < \frac{1}{2}$, we have

$$|\{d : d|n, \omega(d) \leqslant (\tfrac{1}{2} + \eta)\Omega(n)\}| \sim \tau(n) \quad \text{pp.}$$

(d) Show that $\mathbf{D}\mathcal{A}(\delta) = 1$.

274. *Method of vanishing moments.*

(a) Show, for $-\frac{1}{2} \leqslant z \leqslant \frac{1}{2}$, that we have $\sum_{n\leqslant x} \tau(n)^z \ll x(\ln x)^{2^z-1}$. By choosing z sufficiently close to 0, deduce that, if $\alpha < \ln 2 < \beta$, then

$$\big|\{n \leqslant x : \tau(n) \notin [(\ln x)^\alpha, (\ln x)^\beta]\}\big| = o(x).$$

Sharpen this result by choosing $\alpha = \alpha(x) \to \ln 2-$ and $\beta = \beta(x) \to \ln 2+$.

(b) Establish the upper bound $\sum_{n\leqslant x} \tau(n)y^{\omega(n)} \ll x(\ln x)^{2y-1}$ uniformly for $x \geqslant 2$, $0 \leqslant y \leqslant y_0$. Deduce that, for $0 < \varepsilon < 1$, we have

$$\sum_{n\leqslant x,\, |\omega(n)-2\ln_2 x|>\varepsilon \ln_2 x} \tau(n) \ll x(\ln x)^{1-2\eta}$$

with $\eta := (1+\frac{1}{2}\varepsilon)\ln(1+\frac{1}{2}\varepsilon) - \frac{1}{2}\varepsilon > 0$.

275. *A theorem of Erdős & Hall (1974).*

Let $\alpha > \ln 2$, $\varepsilon_d \ll \big(\ln(d+1)\big)^{-\alpha}$ $(d \geqslant 1)$, and $f(n) := \sum_{d|n} \varepsilon_d$.

(a) Show that, for each $\varepsilon > 0$,

$$\lim_{z\to\infty} \overline{\mathbf{d}}\Big\{n : \sup_{d>z} \Omega(n,d)/\ln_2 d \geqslant 1+\varepsilon\Big\} = 0.$$

(b) Show that

$$\lim_{z\to\infty} \limsup_{x\to\infty} \frac{1}{x} \sum_{n\leqslant x} \sum_{d|n,\, d>z} |\varepsilon_d| 2^{-\Omega(n,d)} (\ln d)^{\ln 2} = 0.$$

(c) Establish, by an appropriate use of Theorem 2.3, that $f(n)$ has a limit law.[7]

276. *A theorem of Hall (1974).* Let f be a multiplicative function with $0 \leqslant f \leqslant 1$. Set $S(x) := \sum_{n\leqslant x} f(n)$.

(a) Let $k(n) := \prod_{p|n} p$. Show that

$$\sum_{n\leqslant x} f(n) \ln k(n) \leqslant \sum_{m\leqslant x} f(m)\psi\big(x/m\big)$$

and deduce that the left-hand side does not exceed $x\sum_{m\leqslant x} f(m)/m + O(x)$.

(b) Let $N(x,y)$ be the number of those integers $n \leqslant x$ such that $k(n) \leqslant y$. Show that, for all y with $1 \leqslant y \leqslant x$, we have

$$\sum_{n\leqslant x} f(n) \ln\big(x/k(n)\big) \leqslant N(x,y)\ln x + S(x)\ln(x/y).$$

(c) Appealing to Theorem II.1.16, deduce that we have

$$S(x) \leqslant \Big\{1 + O\Big(\frac{\ln_2 x}{\ln x}\Big)\Big\} \frac{x}{\ln x} \sum_{m\leqslant x} \frac{f(m)}{m},$$

where the implicit constant is absolute.[8]

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[7] Erdős & Hall actually prove that the corresponding d.f. is continuous.

[8] One may choose $y = x/\ln^3 x$ in (b)

277. Let f be an arithmetic function with a non-decreasing normal order. Show that, for all $\varepsilon > 0$, we have

$$\big|\{(m,n) : 1 \leqslant m \leqslant n \leqslant x,\ f(n) \leqslant (1-\varepsilon)f(m)\}\big| = o(x^2).$$

278. Let f be a multiplicative arithmetic function such that the set

$$\mathcal{A} := \{f(p) : p \in \mathbb{P}\}$$

has at least two elements. We assume furthermore that, for some fixed $\delta > 0$ and all primes p, we have $f(p) = 1 + O(p^{-\delta})$. Put $f_y(n) := \prod_{p^\nu \| n,\, p>y} f(p^\nu)$.

(a) For large y, find an upper bound for the mean value of $\mu(n)^2 |\ln f_y(n)|$ and deduce that, for all $\varepsilon > 0$ and any $a \in \mathcal{A}$, the sequence of those integers n such that $|f(n) - a| \leqslant \varepsilon$ has positive lower density.

(b) Using the result of Exercise 277, show that f fails to have an non-decreasing normal order.

(c) Applications: $f(n) := n/\varphi(n)$, $f(n) := \sigma(n)/n$.

279. *On the mean value of Hooley's Δ-function.*

Put $\Delta(n) := \max_{u\in\mathbb{R}} \sum_{d|n,\, \mathrm{e}^u < d \leqslant \mathrm{e}^{u+1}} 1$ and $\tau(n,\vartheta) := \sum_{d|n} d^{i\vartheta}$. By appealing to Lemma 2.9, show that

$$\tau(n)^{-1} \int_0^1 |\tau(n,\vartheta)|^2 \,\mathrm{d}\vartheta \ll \Delta(n) \ll \int_0^1 |\tau(n,\vartheta)| \,\mathrm{d}\vartheta.$$

Deduce the estimates[9]

$$x \ln_2 x \ll \sum_{n\leqslant x} \Delta(n) \ll x(\ln x)^{4/\pi - 1} \tag{3.73}$$

by using Theorem 3.5 for the upper bound and the Selberg–Delange method as suggested in Exercise 209, p. 294, for the lower bound.

280. *Erdős–Wintner theorem (1939): the sufficient condition via the Turán–Kubilius inequality.* Let f be a real additive arithmetic function such that the three series

$$\sum_{|f(p)|\leqslant R} \frac{f(p)}{p}, \quad \sum_{|f(p)|\leqslant R} \frac{f(p)^2}{p}, \quad \sum_{|f(p)|>R} \frac{1}{p}$$

converge for at least one value of $R > 0$.

(a) For $y \geqslant 2$, write $n_y := \prod_{\substack{p^\nu \| n \\ p^\nu \leqslant y}} p^\nu$. Show that $\mathbf{d}\{n : n_y = m\}$ exists for every integer $m \geqslant 1$ and that $\mathbf{d}\{n : n_y > T\}$ tends to 0 as $T \to \infty$.

(b) Let $\varepsilon > 0$. Observing that the inequality $|f(n) - f(n_y)| > \varepsilon$ implies that one of the three following conditions holds:

[9] The lower estimate in (3.73) is due to Hall & Tenenbaum (1982) and the upper estimate to Hooley (1979). Tenenbaum (1985) has shown that the exponent $4/\pi - 1$ may be replaced by $o(1)$; see also Hall & Tenenbaum (1988), chap. 7.

(i) $\exists p \| n \ : \quad |f(p)| > R,\ p > y,$

(ii) $\exists \nu > 1,\ p^\nu > y \ : \quad p^\nu \| n,$

(iii) $\Big| \sum_{\substack{p \| n,\ p > y \\ |f(p)| \leqslant R}} f(p) \Big| > \varepsilon,$

show that $\lim_{y \to \infty} \overline{\mathbf{d}}\{n : |f(n) - f(n_y)| > \varepsilon\} = 0$. The Turán–Kubilius inequality may be used to bound the upper density of the set of those integers satisfying (iii).

(c) Employing Theorem 2.3, prove that f has a limiting distribution.

Distribution of additive functions and mean values of multiplicative functions

4.1. The Erdős–Wintner theorem

The following result provides a complete solution to the problem of the existence of a limiting distribution for an additive function. Both its statement and its proof place it in the realm of comparison theorems between additive functions and sums of independent random variables. The underlying probabilistic result is Kolmogorov's three series theorem—cf., for example, Feller (1971), § IX.9.

Theorem 4.1 (Erdős–Wintner, 1939). *A necessary and sufficient condition for a real additive function $f(n)$ to have a limiting distribution is that the following three series converge simultaneously for at least one value of the positive real number R:*

$$\text{(a)} \quad \sum_{|f(p)|>R} \frac{1}{p}; \qquad \text{(b)} \quad \sum_{|f(p)|\leqslant R} \frac{f(p)^2}{p}; \qquad \text{(c)} \quad \sum_{|f(p)|\leqslant R} \frac{f(p)}{p}.$$

When these conditions are fulfilled, the characteristic function of the limit law is given by the convergent product

$$\varphi(\tau) = \prod_p \Big(1 - \frac{1}{p}\Big) \sum_{\nu\geqslant 0} \frac{e^{i\tau f(p^\nu)}}{p^\nu} \qquad (\tau \in \mathbb{R}). \tag{4.1}$$

The limit law is necessarily pure. It is continuous if, and only if,

$$\sum_{f(p)\neq 0} \frac{1}{p} = \infty. \tag{4.2}$$

Remarks. (i) Should the three series above converge for some value of $R > 0$, they do so for all values of $R > 0$. There is therefore no loss of generality, in practice, in taking $R = 1$.

(ii) It is to be emphasized that the existence of a limiting distribution for f does not depend on the values $f(p^\nu)$ when $\nu \geqslant 2$.

The original proof of this result, due solely to Erdős for the sufficiency, is direct, and rests upon delicate estimates for the frequencies $\nu_x\{n : f(n) \leqslant z\}$. We give here a simpler proof, due to Delange (1961), which depends in an essential way on Lévy's continuity theorem (Theorem 2.6). Delange establishes the following fundamental result, of which Theorem 4.1 is an easy consequence.

Theorem 4.2 (Delange). *Let g be a multiplicative function with values in the unit disk.*

(i) *If g has a non-zero mean value*

$$M(g) := \lim_{x\to\infty} \frac{1}{x} \sum_{n\leqslant x} g(n)$$

then we have:

(a) *the series $\sum_p \big(1 - g(p)\big)/p$ converges,*

(b) *there exists an integer $\nu \geqslant 1$ such that $g(2^\nu) \neq -1$.*

(ii) *If condition* (a) *above is fulfilled, then g has a mean value, given by the formula*

$$M(g) = \prod_p \Big(1 - \frac{1}{p}\Big) \sum_{\nu\geqslant 0} \frac{g(p^\nu)}{p^\nu}. \tag{4.3}$$

We provide a proof of this result in the next section. For the moment, let us show how Theorem 4.1 may be deduced from it—apart from the assertions concerning the purity and continuity of the limit law.

The following simple lemma will be useful.

Lemma 4.3. *Let $H \in \mathbb{R}^+$ and let $\{u_n\}_{n=1}^\infty$ and $\{v_n\}_{n=1}^\infty$ be two complex sequences such that $1 + u_n + v_n \neq 0$ $(n \geqslant 1)$ and*

$$\sum_{n\geqslant 1} (|u_n|^2 + |v_n|) \leqslant H. \tag{4.4}$$

Then the infinite product $\prod_{n\geqslant 1}(1+u_n+v_n)$ *converges if, and only if, the series* $\sum_{n\geqslant 1} u_n$ *converges. In this case, we have*

$$\Big|\prod_{n\geqslant 1}(1+u_n+v_n)\Big| \leqslant \exp\Big\{6H+\sum_{n\geqslant 1}\Re e\, u_n\Big\}. \tag{4.5}$$

Proof. If $|u_n|+|v_n|\geqslant \frac12$, then $|u_n|^2+|v_n|\geqslant |u_n|^2-|u_n|+\frac12\geqslant\frac14$. By (4.4), this implies that the set $\mathcal{E}$ of integers n for which $|u_n|+|v_n|\geqslant\frac12$ contains at most $4H$ elements.

When $n\notin\mathcal{E}$, we use the bound

$$|\log(1+z)-z|\leqslant |z|^2 \qquad (|z|\leqslant \tfrac12) \tag{4.6}$$

with $z=u_n+v_n$, where the logarithm is understood as its principal branch. For all m and M with $0\leqslant m<M$, we obtain

$$\begin{aligned}\Big|\sum_{\substack{m<n\leqslant M\\ n\notin\mathcal{E}}}(u_n+v_n)-\sum_{\substack{m<n\leqslant M\\ n\notin\mathcal{E}}}\log(1+u_n+v_n)\Big| &\leqslant \sum_{\substack{m<n\leqslant M\\ n\notin\mathcal{E}}}\big(|u_n|+|v_n|\big)^2\\ &\leqslant 2\sum_{\substack{m<n\leqslant M\\ n\notin\mathcal{E}}}\big(|u_n|^2+|v_n|^2\big)\leqslant \sum_{m<n\leqslant M}\big(2|u_n|^2+|v_n|\big).\end{aligned}$$

This shows that the convergence of the infinite product and that of the series $\sum u_n$ are equivalent. Moreover, it follows from this computation that

$$\begin{aligned}&\Big|\prod_{n\geqslant 1}(1+u_n+v_n)\Big|\\ &\quad\leqslant \prod_{n\in\mathcal{E}}\exp\{|u_n|+|v_n|\}\prod_{n\notin\mathcal{E}}\exp\{\Re e\,(u_n+v_n)+2|u_n|^2+|v_n|\}\\ &\quad\leqslant \exp\Big\{\sum_{n\geqslant 1}\Re e\, u_n+\sum_{n\in\mathcal{E}}(2|u_n|+|v_n|)+\sum_{n\notin\mathcal{E}}(2|u_n|^2+2|v_n|)\Big\}.\end{aligned}$$

We obtain (4.5) by observing that

$$\sum_{n\in\mathcal{E}}|u_n|\leqslant\Big(4H\sum_{n\geqslant 1}|u_n|^2\Big)^{1/2}\leqslant 2H. \qquad \square$$

We are now in a position to tackle the proof of Theorem 4.1.

For each real number τ, let us consider the multiplicative function of modulus 1 defined by

$$g_\tau(n):=\mathrm{e}^{i\tau f(n)} \qquad (n=1,2,\dots). \tag{4.7}$$

Suppose initially that f has a limit law. But by the continuity theorem (Theorem 2.6), it follows that g_τ has a mean value for each τ. Write $\varphi(\tau):=M(g_\tau)$. Then φ is the characteristic function of the limit law; in particular, $\varphi(0)=1$, and φ is continuous at $\tau=0$. Therefore there exists a

$T > 0$ such that $|\varphi(\tau)| \geqslant \frac{1}{2}$ for $|\tau| \leqslant T$. We shall establish the convergence of the three series in Theorem 4.1 when $R := 2/T$.

For $|\tau| \leqslant T$, assertion (i) of Theorem 4.2 implies the convergence of the series

$$\sum_p \frac{1 - g_\tau(p)}{p}. \tag{4.8}$$

Assertion (ii) then yields the representation

$$\varphi(\tau) = \prod_p \Big(1 - \frac{1}{p}\Big) \sum_{\nu \geqslant 0} \frac{g_\tau(p^\nu)}{p^\nu}. \tag{4.9}$$

The general term of this product may be written in the form

$$1 - \frac{1 - g_\tau(p)}{p} + \frac{h_\tau(p)}{p(p-1)},$$

with $|h_\tau(p)| \leqslant 2$. By Lemma 4.3, we then have

$$|\varphi(\tau)| \ll \exp\Big\{ - \sum_p \frac{1 - \cos\big(\tau f(p)\big)}{p} \Big\}, \tag{4.10}$$

and, since $|\varphi(\tau)| \geqslant \frac{1}{2}$ when $|\tau| \leqslant T$,

$$\sum_p \frac{1 - \cos\big(\tau f(p)\big)}{p} \ll 1 \qquad (|\tau| \leqslant T). \tag{4.11}$$

Given the inequality $1 - \cos\vartheta \geqslant 2\vartheta^2/\pi^2$ $(|\vartheta| \leqslant \pi)$, it follows that

$$\sum_{|f(p)| \leqslant 2/T} \frac{f(p)^2}{p} < \infty. \tag{4.12}$$

Averaging (4.11) over $[0, T]$, we further obtain

$$\sum_p \frac{1}{p} \Big\{ 1 - \frac{\sin\big(Tf(p)\big)}{Tf(p)} \Big\} \ll 1,$$

with an obvious convention when $f(p) = 0$. The expression in braces is non-negative for all p and exceeds $\frac{1}{2}$ when $|f(p)| > R = 2/T$. Therefore

$$\sum_{|f(p)| > R} \frac{1}{p} < \infty. \tag{4.13}$$

This allows us to deduce from the convergence of $\sum_p \sin\big(Tf(p)\big)/p$, which follows from that of (4.8), the convergence of

$$\sum_{|f(p)| \leqslant R} \frac{\sin\big(Tf(p)\big)}{p}.$$

Using (4.12) and the inequality $|\sin\vartheta - \vartheta| \leqslant \frac{1}{6}|\vartheta|^3 \leqslant \frac{1}{3}\vartheta^2$ ($|\vartheta| \leqslant 2$), we thus obtain the convergence of

$$\sum_{|f(p)|\leqslant R} \frac{f(p)}{p}. \tag{4.14}$$

This completes the proof of the necessity of the conditions in Theorem 4.1.

Conversely, let us assume that the three series in the statement converge for some suitable positive real number R.

The trivial bound $|1 - \mathrm{e}^{i\tau f(p)}| \leqslant 2$ then implies uniform convergence, for each $T > 0$, of the series

$$\sum_{|f(p)|>R} \frac{1 - \mathrm{e}^{i\tau f(p)}}{p} \qquad (|\tau| \leqslant T).$$

Furthermore, when $|f(p)| \leqslant R$, we can write

$$\begin{aligned} 1 - \mathrm{e}^{i\tau f(p)} &= -i\tau f(p) + \{1 - \cos(\tau f(p))\} + i\{\tau f(p) - \sin(\tau f(p))\} \\ &= -i\tau f(p) + O\Big(T^2 f(p)^2 + T^3 R f(p)^2\Big), \end{aligned}$$

uniformly for $|\tau| \leqslant T$. This yields uniform convergence on $[-T, T]$ for the series

$$\sum_{|f(p)|\leqslant R} \frac{1 - \mathrm{e}^{i\tau f(p)}}{p}.$$

Thus the series $\sum_p \{1 - g_\tau(p)\}/p$ is uniformly convergent on any compact subset. By assertion (ii) of Theorem 4.2, we deduce the existence, for all $\tau \in \mathbb{R}$, of the mean value $M(g_\tau) = \varphi(\tau)$, with $\varphi(\tau)$ given by (4.9). Since the general term of the product (4.9) equals $1 - \{1 - g_\tau(p)\}/p + O(1/p^2)$, Lemma 4.3 shows that the convergence of the infinite product is uniform on every compact set. Consequently, $\varphi(\tau)$ is continuous. By Theorem 2.6, f has a limiting distribution.

The assertions about the purity and the continuity of the limit law follow from Theorem 2.7, since this law is equal to the infinite convolution product

$$\mathop{*}_{p} F_p$$

where F_p is the atomic distribution law given by

$$F_p(z) = \sum_{f(p^\nu)\leqslant z} \frac{1}{p^\nu}\Big(1 - \frac{1}{p}\Big) \qquad (z \in \mathbb{R}).$$

In particular, the distance from 1 of the largest saltus of F_p equals $1/p + O(1/p^2)$ if $f(p) \neq 0$, and is $O(1/p^2)$ if $f(p) = 0$.

For convenience, we give here a direct proof of the fact that the condition

$$\sum_{f(p)\neq 0} \frac{1}{p} = \infty \tag{4.15}$$

is necessary and sufficient for the continuity of the limit law.

Let us first show that the condition is necessary. If the series (4.15) converges, then the sequence $\mathcal{A}$ of integers n which are squarefree and such that $p|n \Rightarrow f(p) = 0$, has positive density. This follows, for instance, from Theorem I.3.12, since the indicator function of $\mathcal{A}$ is multiplicative.[1] Since $f(n) = 0$ for all n in $\mathcal{A}$, the limit law is not continuous at the origin.

In order to prove the sufficiency of condition (4.15), we establish the continuity of the limit law in the equivalent form (cf. Chapter III.2, Notes)

$$\lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |\varphi(\tau)|^2 \, \mathrm{d}\tau = \lim_{T\to\infty} \tfrac{1}{2} \int_{-1}^{1} |\varphi(Ty)|^2 \, \mathrm{d}y = 0. \tag{4.16}$$

Let N be a sufficiently large integer, and let $a_1 < a_2 < \cdots < a_J$ denote the distinct non-zero values taken by the numbers $f(p)$, $p \leqslant N$. Set

$$\varepsilon_j := \sum_{p\leqslant N,\, f(p)=a_j} \frac{1}{p} \quad (1 \leqslant j \leqslant J), \qquad S_N := \sum_{1\leqslant j\leqslant J} \varepsilon_j = \sum_{p\leqslant N,\, f(p)\neq 0} \frac{1}{p}.$$

By Hölder's inequality, we have, for each integer $h \geqslant 1$ and any real number y with $|y| \leqslant 1$,

$$\Big(\sum_{1\leqslant j\leqslant J} \varepsilon_j \cos(a_j T y) \Big)^{2h} \leqslant S_N^{2h-1} \sum_{1\leqslant j\leqslant J} \varepsilon_j \cos^{2h}(a_j T y),$$

from which it follows that

$$\int_{-1}^{1} \Big(\sum_{1\leqslant j\leqslant J} \varepsilon_j \cos(a_j T y) \Big)^{2h} \mathrm{d}y \leqslant S_N^{2h-1} \sum_{1\leqslant j\leqslant J} \varepsilon_j \int_{-1}^{1} \cos^{2h}(a_j T y) \, \mathrm{d}y.$$

As $T \to \infty$, the last integral tends to

$$\frac{4}{\pi} \int_0^{\pi/2} (\cos w)^{2h} \, \mathrm{d}w \ll \frac{1}{\sqrt{h}},$$

where the bound follows from a classical evaluation of Wallis integrals—cf., for example, Exercise 3(b), p. 8. For each fixed N, and sufficiently large T, we can thus assert that the measure of the set of those y in $[-1, 1]$ such that

$$\Big| \sum_{1\leqslant j\leqslant J} \varepsilon_j \cos(a_j T y) \Big| \geqslant \Big(1 - \frac{1}{h}\Big) S_N$$

[1] We obtain, incidentally, that $\mathrm{d}\mathcal{A} = 6\pi^{-2} \prod_{f(p)\neq 0} (1 + 1/p)^{-1}$, but this explicit value is not needed here.

is $\ll 1/\sqrt{h}$. Using (4.10), it follows that

$$\int_{-1}^{1} |\varphi(Ty)|^2 \,\mathrm{d}y \ll \int_{-1}^{1} \exp\Big\{ -2 \sum_{1\leqslant j\leqslant J} \varepsilon_j \big(1-\cos(a_j Ty)\big)\Big\} \,\mathrm{d}y$$
$$\ll \frac{1}{\sqrt{h}} + \mathrm{e}^{-2S_N/h} \ll \sqrt{\frac{\ln S_N}{S_N}}$$

for the choice $h := \lfloor S_N/\ln S_N \rfloor$. Since (4.15) implies that $S_N \to \infty$ as $N \to \infty$, we obtain (4.16).

4.2. Delange's theorem

Here we give a proof of Theorem 4.2 also due to Delange, but differing from his original proof and, following an idea of Rényi (1965), depending on the Turán–Kubilius inequality.

The fundamental point is the following result.

Theorem 4.4 (Delange). *Let g be a multiplicative function with values in the unit disk. Under the assumption that*

$$\sum_p \frac{1-\Re\mathrm{e}\, g(p)}{p} < \infty, \tag{4.17}$$

we have

$$\frac{1}{x}\sum_{n\leqslant x} g(n) = \prod_{p\leqslant x}\Big(1-\frac{1}{p}\Big)\sum_{\nu\geqslant 0}\frac{g(p^\nu)}{p^\nu} + o(1) \qquad (x\to\infty). \tag{4.18}$$

Proof. For each real number $y \geqslant 2$, let us define a multiplicative function g_y by

$$g_y(p^\nu) := \begin{cases} g(p^\nu) & (p\leqslant y)\\ 1 & (p>y). \end{cases}$$

The behavior of g_y is simpler than that of g since g_y is only non-trivial on a finite set of primes. However, one can hope that g_y is a "good" approximation to g when y is large: this is the main thread of the proof.

Let us consider $h_y := \mu * g_y$. We have $h_y(p^\nu) = g(p^\nu) - g(p^{\nu-1})$ if $\nu \geqslant 1$, $p \leqslant y$, and $h_y(p^\nu) = 0$ if $\nu \geqslant 1$, $p > y$. This implies

$$H_y(x) := \sum_{m\leqslant x} |h_y(m)| + x\sum_{m>x}\frac{|h_y(m)|}{m}$$
$$\leqslant \sqrt{x}\sum_{m\geqslant 1}\frac{|h_y(m)|}{\sqrt{m}} \leqslant \sqrt{x}\prod_{p\leqslant y}\Big(1+\frac{2}{\sqrt{p}-1}\Big).$$

We easily deduce that, for each fixed y, g_y has a mean value: indeed

$$\sum_{n\leqslant x} g_y(n) = \sum_{m\leqslant x} h_y(m)\Big\lfloor \frac{x}{m}\Big\rfloor = x\sum_{m\geqslant 1}\frac{h_y(m)}{m} + O\big(H_y(x)\big)$$
$$= x\big\{M(g_y)+o(1)\big\},$$

with

$$M(g_y) := \prod_{p\leqslant y}\Big(1-\frac{1}{p}\Big)\sum_{\nu\geqslant 0}\frac{g(p^\nu)}{p^\nu}.$$

Now, let us define a multiplicative function r by $r(n) := |g(n)|$, and an additive function ϑ by

$$\vartheta(p^\nu) := \begin{cases} \arg\, g(p^\nu) & \text{if } g(p^\nu)\neq 0,\\ 0 & \text{if } g(p^\nu)=0,\end{cases}$$

where the argument is chosen within $]-\pi,\pi]$. Setting

$$A(x) := \sum_{p\leqslant x}\vartheta(p)/p,$$

we use Cauchy's criterion to show that the quantity

$$M(g_y)\mathrm{e}^{-iA(y)} \tag{4.19}$$

tends to a finite limit M as $y\to\infty$. For $2\leqslant y\leqslant z\leqslant x$, we have

$$S(x,y,z) := \frac{1}{x}\Big|\sum_{n\leqslant x} g_z(n)\mathrm{e}^{-iA(z)} - \sum_{n\leqslant x} g_y(n)\mathrm{e}^{-iA(y)}\Big|$$
$$\leqslant \frac{1}{x}\sum_{n\leqslant x}\Big|g_z(n)\mathrm{e}^{-iA(z)} - g_y(n)\mathrm{e}^{-iA(y)}\Big|$$
$$\leqslant \frac{1}{x}\sum_{n\leqslant x}\Big|\mathrm{e}^{-i\{A(z)-A(y)\}}\prod_{p^\nu\| n,\, y<p\leqslant z} g(p^\nu) - 1\Big|.$$

From the bound $|u_1\dots u_m - 1|\leqslant \sum_{j=1}^m |u_j-1|$, valid for all complex $u_1,\dots,u_m$ in the unit disk, we obtain

$$S(x,y,z)\leqslant \frac{1}{x}\sum_{n\leqslant x}\sum_{\substack{p^\nu\| n\\ y<p\leqslant z}}\big\{1-r(p^\nu)\big\} + \frac{1}{x}\sum_{n\leqslant x}\Big|\mathrm{e}^{i\{A(z)-A(y)-\vartheta_{y,z}(n)\}}-1\Big|$$

where $\vartheta_{y,z}$ is the additive function defined by $\vartheta_{y,z}(p^\nu) := \mathbf{1}_{]y,z]}(p)\vartheta(p^\nu)$.

We obtain an upper bound for the second sum over n by using the inequality

$$|\mathrm{e}^{iu}-1| = \Big|\int_0^u \mathrm{e}^{it}\,\mathrm{d}t\Big|\leqslant |u| \qquad (u\in\mathbb{R}).$$

After interchanging summations in the first sum, and applying the Cauchy–Schwarz inequality in the second, we can write

$$S(x,y,z) \leqslant \sum_{y<p\leqslant z} \frac{1-r(p)}{p} + 2\sum_{y<p\leqslant z} \frac{1}{p(p-1)} + \left\{\frac{1}{x}\sum_{n\leqslant x}\Big(\vartheta_{y,z}(n) - \big(A(z)-A(y)\big)\Big)^2\right\}^{1/2}$$

Estimating the last sum by the Turán–Kubilius inequality, we obtain

$$\begin{aligned} S(x,y,z) &\leqslant \sum_{y<p\leqslant z} \frac{1-r(p)}{p} + O\left(\frac{1}{y} + \sqrt{\sum_{y<p\leqslant z}\frac{\vartheta(p)^2}{p} + \frac{1}{y}}\right) \\ &\ll \sum_{y<p\leqslant z} \frac{1-\Re e\big(g(p)\big)}{p} + \frac{1}{\sqrt{y}} + \sqrt{\sum_{y<p\leqslant z}\frac{\vartheta(p)^2}{p}}. \end{aligned}$$

Now, we observe that, for each p, we have

$$\vartheta(p)^2 \leqslant \pi^2\{1-\Re e\, g(p)\}. \tag{4.20}$$

Indeed, either $|\vartheta(p)| > \pi/2$ and the inequality is satisfied since $\Re e\, g(p) \leqslant 0$, or $|\vartheta(p)| \leqslant \pi/2$ and we have

$$1-\Re e\, g(p) = 1 - r(p)\cos\vartheta(p) \geqslant 1-\cos\vartheta(p) \geqslant 2\{\vartheta(p)/\pi\}^2.$$

Substituting (4.20) into the above bound for $S(x,y,z)$, we derive

$$S(x,y,z) \ll \eta(y) \qquad (2\leqslant y\leqslant z\leqslant x) \tag{4.21}$$

with $\eta(y) := 1/\sqrt{y} + \sqrt{\sum_{p>y}\big(1-\Re e\, g(p)\big)/p} = o(1)$ $(y\to\infty)$. This shows that Cauchy's criterion is satisfied for (4.19), and hence there exists a complex number M such that

$$M(g_y)\mathrm{e}^{-iA(y)} = M + o(1) \qquad (y\to\infty). \tag{4.22}$$

Choosing $z = x$ in (4.21) and noting that $g_x(n) = g(n)$ for $n \leqslant x$, it follows that

$$\frac{1}{x}\sum_{n\leqslant x} g(n)\mathrm{e}^{-iA(x)} = \frac{1}{x}\sum_{n\leqslant x} g_y(n)\mathrm{e}^{-iA(y)} + O\big(\eta(y)\big).$$

Letting, successively, x and y tend to ∞, we obtain

$$\frac{\mathrm{e}^{-iA(x)}}{x}\sum_{n\leqslant x} g(n) = M + o(1) = M(g_x)\mathrm{e}^{-iA(x)} + o(1),$$

as required. □

We are now in a position to prove Theorem 4.2 of Delange. Assume first that g has a non-zero mean value $M(g)$. By partial summation, we then infer that

$$(\sigma-1)\sum_{n\geqslant 1}\frac{g(n)}{n^\sigma}\sim\frac{1}{\zeta(\sigma)}\sum_{n\geqslant 1}\frac{g(n)}{n^\sigma}\sim M(g)\qquad(\sigma\to 1+)$$

or, equivalently,

$$\lim_{\sigma\to 1+}\prod_p\Big(1-\frac{1}{p^\sigma}\Big)\sum_{\nu\geqslant 0}\frac{g(p^\nu)}{p^{\nu\sigma}}=M(g).$$

Noticing that the general term of this product has modulus $\leqslant 1$ and that

$$\Big|\sum_{\nu\geqslant 0}g(p^\nu)\Big(\frac{1}{p^\nu}-\frac{1}{p^{\nu\sigma}}\Big)\Big|\leqslant\frac{1-1/p^{\sigma-1}}{(p-1)(1-p^{-\sigma})}\ll\frac{(\sigma-1)\ln p}{p},$$

we can write

$$\begin{aligned}|M(g)|&\ll\Big|\prod_{p\leqslant\exp\{1/(\sigma-1)\}}(1-p^{-\sigma})\sum_{\nu\geqslant 0}\frac{g(p^\nu)}{p^{\nu\sigma}}\Big|\\&\ll\Big|\prod_{p\leqslant\exp\{1/(\sigma-1)\}}\Big\{(1-p^{-\sigma})\sum_{\nu\geqslant 0}\frac{g(p^\nu)}{p^\nu}+O\Big(\frac{(\sigma-1)\ln p}{p}\Big)\Big\}\Big|\\&\ll\Big|\prod_{p\leqslant\exp\{1/(\sigma-1)\}}\Big\{1-\frac{1-g(p)}{p}+O\Big(\frac{1}{p^2}+\frac{(\sigma-1)\ln p}{p}\Big)\Big\}\Big|\\&\ll\exp\Big\{-\sum_{p\leqslant\exp\{1/(\sigma-1)\}}\frac{1-\Re e\,g(p)}{p}\Big\},\end{aligned}$$

by Lemma 4.3.

This implies the convergence of the series $\sum_p\{1-\Re e\,g(p)\}/p$. By Theorem 4.4 it then follows that the product

$$\prod_p\Big(1-\frac{1}{p}\Big)\sum_{\nu\geqslant 0}\frac{g(p^\nu)}{p^\nu}\tag{4.23}$$

converges with value $M(g)$. Its general term equals

$$1-\frac{1-g(p)}{p}+O\Big(\frac{1}{p^2}\Big).$$

By a further application of Lemma 4.3, we then obtain the convergence of the series $\sum_p\{1-g(p)\}/p$. Finally, since $M(g)\neq 0$, no term in the infinite product (4.23) vanishes, and in particular $g(2^\nu)\neq -1$ must hold for at least one integer $\nu\geqslant 1$.

Let us now establish assertion (ii) of Theorem 4.2. If the series (a) converges, then, by Lemma 4.3, so does the product (4.23). Hence, the conclusion immediately follows from Theorem 4.4.

4.3. Halász's theorem

4.3.1. Statements. Delange's theorem (Theorem 4.2) furnishes a necessary and sufficient condition for a multiplicative function of modulus $\leqslant 1$ to have a non-zero mean value. Wirsing (1967) and Halász (1968) subsequently completed the study of the mean behavior of a multiplicative function of modulus $\leqslant 1$, the former for real, the latter for complex functions.

Theorem 4.5 (Halász). *Let g be a multiplicative function with values in the unit disk. If there exists some real number τ such that the series*

$$\sum_p \frac{1-\Re e\big(g(p)p^{-i\tau}\big)}{p} \tag{4.24}$$

converges, then we have

$$\frac{1}{x}\sum_{n\leqslant x} g(n) = \frac{x^{i\tau}}{1+i\tau}\prod_{p\leqslant x}\Big(1-\frac{1}{p}\Big)\sum_{\nu\geqslant 0}\frac{g(p^\nu)}{p^{\nu(1+i\tau)}} + o(1) \qquad (x\to\infty). \tag{4.25}$$

If there exists no real number τ such that the series (4.24) *converges, then we have*

$$\frac{1}{x}\sum_{n\leqslant x} g(n) = o(1). \tag{4.26}$$

The first assertion of this theorem follows easily from Theorem 4.4. We indeed deduce from the convergence of (4.24) that

$$\frac{1}{x}\sum_{n\leqslant x}\frac{g(n)}{n^{i\tau}} = \prod_{p\leqslant x}\Big(1-\frac{1}{p}\Big)\sum_{\nu\geqslant 0}\frac{g(p^\nu)}{p^{\nu(1+i\tau)}} + o(1) \qquad (x\to\infty). \tag{4.27}$$

Moreover, formula (4.22) applied to $g(n)n^{-i\tau}$ allows replacing the product on the right-hand side by $M_\tau \mathrm{e}^{iA_\tau(x)} + o(1)$, where M_τ is a suitable constant and

$$A_\tau(x) := \sum_{p\leqslant x} \vartheta_\tau(p)/p$$

with $\vartheta_\tau(p) := \arg\big\{g(p)p^{-i\tau}\big\} \in\,]-\pi,\pi]$. Inequality (4.20) then implies the convergence of the series $\sum_p \vartheta_\tau(p)^2/p$, whence, by applying the Cauchy-Schwarz inequality,

$$A_\tau(x) - A_\tau(y) = o\Big(1+\ln\Big(\frac{\ln x}{\ln y}\Big)\Big)$$

as $y\to\infty$ and $x\geqslant y$. In particular, the function L_τ defined by

$$L_\tau(\ln x) := \mathrm{e}^{iA_\tau(x)}$$

is *slowly varying*, i.e., satisfies

$$L_\tau(u) \sim L_\tau(v)$$

as $u, v \to \infty$, and $u \asymp v$. This property allows deducing (4.25) from (4.27) by a simple integration by parts. For convenience of later reference, we note that (4.25) may be rewritten in the form

$$\frac{1}{x}\sum_{n\leqslant x} g(n) = K_\tau x^{i\tau} L_\tau(\ln x) + o(1) \tag{4.28}$$

where K_τ is a suitable constant and L_τ is a slowly varying function of modulus 1.

We shall establish later the second assertion of Halász's theorem. Before this, we derive the following important corollary, initially proved by Wirsing (1967).

Theorem 4.6 (Wirsing). *Let g be a real multiplicative function with values in $[-1, 1]$. Then we have*

$$\lim_{x\to\infty} \frac{1}{x}\sum_{n\leqslant x} g(n) = \prod_p \Big(1 - \frac{1}{p}\Big)\sum_{\nu\geqslant 0} \frac{g(p^\nu)}{p^\nu} \tag{4.29}$$

where the infinite product is to be taken as zero when it diverges.

Proof. Under the assumptions of the above statement the series

$$\sum_p \big(1 - \Re e\{g(p)p^{-i\tau}\}\big)/p$$

diverges for all $\tau \neq 0$. This is an immediate consequence of the relation

$$\sum_{|\cos(\tau \ln p)|\leqslant \frac{1}{2}} \frac{1}{p} = \infty,$$

which is easily obtained by means of the prime number theorem.[2] Thus only the value $\tau = 0$ in Halász's theorem needs to be considered. □

Remark. Wirsing's theorem lies at least as deep as the prime number theorem since the function $g(n) = \mu(n)$ is within its range of application.

For the "divergent" case of Halász's theorem, we shall give an effective estimate essentially due to Montgomery (1978b). The argument is fundamentally identical to that employed by Halász in his 1971 article, partly devoted to making his 1968 results effective, but Montgomery also introduces certain refinements and numerous simplifications. We give a version that is slightly more precise than Montgomery's.

Set

$$G(x) := \sum_{n\leqslant x} g(n), \qquad F(s) := \sum_{n\geqslant 1} \frac{g(n)}{n^s} \qquad (\sigma > 1),$$

[2] Actually, an estimate of the type $\sum_{x<n\leqslant cx} \Lambda(n) \gg x$ $(x \to \infty)$ valid for any constant $c > 1$ would suffice.

and, for $T > 0$,

$$H_T(\alpha)^2 := \sum_{\substack{k\in\mathbb{Z}\\ |k|\leqslant T}} \frac{1}{k^2+1} \max_{\substack{\sigma=1+\alpha\\ |\tau-k|\leqslant\frac{1}{2}}} |F(s)|^2 \qquad (\alpha > 0). \tag{4.30}$$

Theorem 4.7. *With the above notation, and for each multiplicative function g with values in the unit disk, we have, uniformly in $T > 0$, $x \geqslant 2$,*

$$G(x) \ll \frac{x}{\ln x} \int_{1/\ln x}^{1} H_T(\alpha) \frac{\mathrm{d}\alpha}{\alpha} + \frac{x}{T}. \tag{4.31}$$

The result stated by Montgomery corresponds to the case $T = \infty$ of (4.31).

Let us first show how the second part of Halász's theorem may be deduced from this statement. To this end, we need the following lemma.

Lemma 4.8. *Let $\{\varphi_n(\tau)\}_{n=1}^{\infty}$ be a sequence of continuous functions such that, for each τ, $|\tau| \leqslant 1$, the sequence $\varphi_n(\tau)$ increases to ∞. Then, the convergence is uniform.*

Proof. We argue by contradiction. If the conclusion fails, then $\inf_{|\tau|\leqslant 1} \varphi_n(\tau)$ does not tend to infinity with n, and there exist some constant A and some subsequence of indices $\{n_j\}_{j=1}^{\infty}$, $n_j \to \infty$, such that $\inf_{|\tau|\leqslant 1} \varphi_{n_j}(\tau) \leqslant A$ for $j \geqslant 1$. Since the φ_{n_j} are continuous, for each j there exists some $\tau_j \in [-1, 1]$ such that $\varphi_{n_j}(\tau_j) = \inf_{|\tau|\leqslant 1} \varphi_{n_j}(\tau)$. By extracting a new subsequence if necessary, we may assume that $\tau_j \to \tau_0$ as $j \to \infty$. Now let n be some arbitrary fixed integer. For sufficiently large j, we have

$$\varphi_{n_j}(\tau_j) \geqslant \varphi_n(\tau_j) \geqslant \tfrac{1}{2}\varphi_n(\tau_0)$$

since $\varphi_n(\tau_j) \to \varphi_n(\tau_0)$. It follows that $\varphi_n(\tau_0) \leqslant 2A$, contradicting the hypothesis that $\varphi_n(\tau_0) \to \infty$ as $n \to \infty$. □

We can now complete the proof of Halász's theorem. By Lemma 4.3, we have

$$\begin{aligned} \frac{F(s)}{\zeta(\sigma)} &= \prod_p \Big(1 - \frac{1 - g(p)p^{-i\tau}}{p^{\sigma}} + \sum_{\nu\geqslant 2} \frac{g(p^\nu) - g(p^{\nu-1})p^{i\tau}}{p^{\nu s}}\Big) \\ &\ll \exp\Big\{-\sum_p \frac{1 - \Re\mathrm{e}\big(g(p)p^{-i\tau}\big)}{p^{\sigma}}\Big\} \end{aligned} \tag{4.32}$$

and, under the hypothesis that the series (4.24) diverges for any real number τ, Lemma 4.8 implies that the right-hand side tends to 0 uniformly in τ on any compact subset as $\sigma \to 1+$. We deduce that $H_T(\alpha) = o(1/\alpha)$ uniformly in T as $\alpha \to 0+$, from which, by substituting in (4.31), we get that $G(x) = o(x)$ as $x \to \infty$.

4.3.2. Lemmas. For the proof of Theorem 4.7, we need three auxiliary results. The first is a general inequality, due to Gallagher (1970), regarding mean values of trigonometric polynomials.

Lemma 4.9 (Gallagher). *Let $N \in \mathbb{N}^*$, and let $\{\lambda_n\}_{n=1}^N$ be a finite sequence of distinct real numbers. Put $\delta_n := \min_{m\neq n} |\lambda_m - \lambda_n|$. For all $\{a_n\}_{n=1}^N \in \mathbb{C}^N$ and all $T > 0$, we have*

$$\int_{-T}^{T} \Big| \sum_{1\leqslant n\leqslant N} a_n \mathrm{e}(\lambda_n t)\Big|^2 \mathrm{d}t \ll \sum_{1\leqslant n\leqslant N} |a_n|^2 \Big\{T + \frac{1}{\delta_n}\Big\}, \tag{4.33}$$

where the implicit constant is absolute. In particular, for any Dirichlet series $F(s) := \sum_{n\geqslant 1} a_n/n^s$ with abscissa of convergence $< \alpha$, and for all $T > 0$, we have, uniformly,

$$\int_{-T}^{T} |F(s)|^2 \,\mathrm{d}\tau \ll \sum_{n\geqslant 1} \frac{|a_n|^2}{n^{2\sigma}} (T+n) \qquad (\sigma \geqslant \alpha). \tag{4.34}$$

Proof. Inequality (4.34) follows immediately from the case $\lambda_n := \ln n$ $(n \geqslant 1)$ of (4.33) by passage to the limit in N.

To establish (4.33), we observe that, putting

$$A(x) := 2T \sum_{\substack{1\leqslant n\leqslant N\\ |x-\lambda_n|\leqslant 1/4T}} a_n, \qquad S(t) := \sum_{1\leqslant n\leqslant N} a_n \mathrm{e}(\lambda_n t),$$

we have

$$\widehat{A}(t) := \int_{\mathbb{R}} A(x)\mathrm{e}(-tx)\,\mathrm{d}x = S(-t)\frac{\sin(\pi t/2T)}{\pi t/2T}\cdot$$

By Plancherel's formula, we can therefore write

$$\int_{-T}^{T} |S(t)|^2 \,\mathrm{d}t \ll \int_{\mathbb{R}} \Big|S(-t)\frac{\sin(\pi t/2T)}{\pi t/2T}\Big|^2 \mathrm{d}t = \int_{\mathbb{R}} |A(x)|^2 \,\mathrm{d}x.$$

Write $N_k := \sum_{|2T\lambda_n - k|\leqslant 1} |a_n|$ $(k \in \mathbb{Z})$ and observe that $|A(x)| \leqslant 2TN_k$ when $|x - k/2T| \leqslant 1/4T$. It follows that

$$\int_{\mathbb{R}} |A(x)|^2 \,\mathrm{d}x \ll T \sum_{k\in\mathbb{Z}} N_k^2.$$

Denote by $\{\lambda_j^* : 1 \leqslant j \leqslant J_k\}$ the sequence of those λ_n counted by N_k, rearranged in increasing order, and define $b_j := |a_n|$ if $\lambda_j^* = \lambda_n$, $\delta_j^* := T(\lambda_{j+1}^* - \lambda_j^*)$ $(1 \leqslant j < J_k)$. If $J_k \leqslant 1$, we plainly have $N_k^2 \leqslant \sum_{|2T\lambda_n - k|\leqslant 1} |a_n|^2$. Otherwise, we may write, by the Cauchy–Schwarz inequality,

$$N_k^2 \leqslant T(\lambda_{J_k}^* - \lambda_1^*) \sum_{1\leqslant j<J_k} \frac{b_j^2}{\delta_j^*} + b_{J_k}^2 \leqslant \sum_{1\leqslant j<J_k} \frac{b_j^2}{\delta_j^*} + b_{J_k}^2 \leqslant \sum_{|2T\lambda_n - k|\leqslant 1} |a_n|^2 \Big(1 + \frac{1}{T\delta_n}\Big).$$

Summing over k, we derive the required inequality. □

Our second lemma is a general inequality[3] concerning quadratic means of almost periodic functions. We state it here in the context of Dirichlet series.

Lemma 4.10 (Montgomery–Wirsing). *Let* $A(s) := \sum_{n\geqslant 1} a_n/n^s$, *and* $B(s) := \sum_{n\geqslant 1} b_n/n^s$ *be two Dirichlet series which are convergent for* $\sigma > 1$ *and satisfy* $|a_n| \leqslant b_n$ *for* $n \geqslant 1$. *Then, for all* $T \geqslant 0$ *and all* $\sigma > 1$, *we have*

$$\int_{-T}^{T} |A(s)|^2 \,\mathrm{d}\tau \leqslant 3 \int_{-T}^{T} |B(s)|^2 \,\mathrm{d}\tau. \tag{4.35}$$

Proof. Consider the function $\chi(\tau) := \max(0, 1-|\tau|/T)$, with Fourier transform

$$\widehat{\chi}(t) = \int_{-\infty}^{+\infty} \mathrm{e}^{i\tau t}\chi(\tau)\,\mathrm{d}\tau = T\Big(\frac{\sin(tT/2)}{tT/2}\Big)^2.$$

For all $\tau_0 \in \mathbb{R}$, we have

$$\begin{aligned}\int_{-\infty}^{+\infty} \chi(\tau-\tau_0)|A(s)|^2\,\mathrm{d}\tau &= \sum_{m,n\geqslant 1} \frac{a_m\overline{a_n}}{(mn)^\sigma}\Big(\frac{n}{m}\Big)^{i\tau_0}\widehat{\chi}\Big(\ln\frac{m}{n}\Big)\\ &\leqslant \sum_{m,n\geqslant 1} \frac{b_m b_n}{(mn)^\sigma}\,\widehat{\chi}\Big(\ln\frac{n}{m}\Big) = \int_{-\infty}^{+\infty} \chi(\tau)|B(s)|^2\,\mathrm{d}\tau.\end{aligned}$$

Letting $h(\tau)$ denote the indicator function of $[-T,T]$, we have

$$h(\tau) \leqslant \chi(\tau-T) + \chi(\tau) + \chi(\tau+T),$$

whence

$$\int_{-T}^{T} |A(s)|^2\,\mathrm{d}\tau \leqslant 3\int_{-\infty}^{+\infty} \chi(\tau)|B(s)|^2\,\mathrm{d}\tau \leqslant 3\int_{-T}^{T} |B(s)|^2\,\mathrm{d}\tau. \qquad \square$$

The third auxiliary result is a convenient decomposition of the Euler product $F(s)$.

Lemma 4.11. *Let* g *be a multiplicative function of modulus* $\leqslant 1$. *We have*

$$\sum_{n\geqslant 1} \frac{g(n)}{n^s} = \big\{1 + D(s)\big\}F_1(s)J(s) \qquad (\sigma > 1) \tag{4.36}$$

with

$$D(s) := \sum_{\nu\geqslant 1} \frac{g(2^\nu)}{2^{\nu s}}, \qquad F_1(s) := \exp\sum_{p>2} \frac{g(p)}{p^s},$$

and where $J(s)$ *is holomorphic for* $\sigma > \frac{1}{2}$ *and satisfies*

$$1 \ll J(s) \ll 1, \qquad J'(s) \ll 1 \qquad (\sigma \geqslant 1). \tag{4.37}$$

[3] See Montgomery (1971), p.158, as well as Montgomery (1994), § 7.3, where it is mentioned that the argument employed here is due to Wirsing.

Proof. We have

$$J(s) = \prod_{p>2} \mathrm{e}^{-g(p)/p^s} \sum_{\nu\geqslant 0} \frac{g(p^\nu)}{p^{\nu s}}.$$

A typical factor in this product has value $1+O(p^{-2\sigma})$, and so $J(s)$ is certainly holomorphic for $\sigma > \frac{1}{2}$ and uniformly bounded for $\sigma \geqslant \frac{3}{4}$. In particular, this implies the upper bounds for $J(s)$ and $J'(s)$ in (4.37). Furthermore, since we have

$$\Big|\sum_{\nu\geqslant 1} g(p^\nu)p^{-\nu s}\Big| \leqslant \frac{1}{p-1} \leqslant \tfrac{1}{2} \qquad (p>2,\ \sigma\geqslant 1),$$

we infer from (4.6) that $|\log J(s)| \leqslant \sum_{p>2} 2/(p-1)^2$, from which follows the lower bound for $|J(s)|$ in (4.37). □

4.3.3. Proof of Theorem 4.7. Clearly, we may assume without loss of generality that T is arbitrarily large. It will be convenient to have at our disposal the lower bound

$$H_T(\alpha) \gg 1 \qquad (\alpha > 0). \tag{4.38}$$

For this, we first observe that, defining $\vartheta_\nu := \arg g(2^\nu)$ with $|\vartheta_\nu| \leqslant \pi$, we have

$$|1 + D(1+\alpha+i\tau)| \geqslant 1 - \sum_{\nu\geqslant 1} \frac{|\cos(\vartheta_\nu - \tau\nu\ln 2)|}{2^\nu} \geqslant \tfrac{1}{2}(1 - |\cos(\vartheta_1 - \tau\ln 2)|).$$

Let k_0 be one of the possibly two closest integers to $(\vartheta_1 - \frac{1}{2}\pi)/\ln 2$, so that $|1 + D(s)| \geqslant \frac{1}{2}(1 - \sin\ln 2) > 0$ for $\sigma \geqslant 1$, $|\tau - k_0| \leqslant \frac{1}{2}$. Plainly, $|k_0| \leqslant 8$. Furthermore

$$\int_{k_0-\frac{1}{2}}^{k_0+\frac{1}{2}} \log F_1(s)\,\mathrm{d}\tau = \sum_{p>2} \frac{g(p)}{p^\sigma} \int_{k_0-\frac{1}{2}}^{k_0+\frac{1}{2}} p^{-i\tau}\,\mathrm{d}\tau \ll \sum_p \frac{1}{p\ln p} \ll 1.$$

This implies that $\max_{|\tau-k_0|\leqslant\frac{1}{2}} |F_1(s)| \gg 1$, and hence

$$\max_{\substack{\sigma=1+\alpha\\ |\tau-k_0|\leqslant\frac{1}{2}}} |F(s)| \geqslant \min\big|\big(1+D(s)\big)J(s)\big| \max|F_1(s)| \gg 1,$$

from which (4.38) follows.

The first step in the proof consists in establishing an upper bound for $G(x)$ in terms of an average of itself, namely

$$|G(x)| \leqslant \frac{x}{\ln x}\int_1^x \frac{|G(t)|}{t^2}\,\mathrm{d}t + O\Big(\frac{x}{\ln x}\Big). \tag{4.39}$$

Write $K(x) := \sum_{n\leqslant x} g(n)\ln n$. Then

$$G(x)\ln x - K(x) = \sum_{n\leqslant x} g(n)\ln(x/n) \ll \sum_{n\leqslant x} \ln(x/n) \ll x. \tag{4.40}$$

Hence, in order to establish (4.39), it suffices to show that

$$|K(x)| \leqslant x \int_1^x \frac{|G(t)|}{t^2}\,\mathrm{d}t + O(x). \tag{4.41}$$

Now we can write

$$\begin{aligned} K(x) &= \sum_{n\leqslant x} g(n) \sum_{p^\nu \mid n} \ln p = \sum_{p^\nu \leqslant x} \ln p \sum_{m\leqslant x/p^\nu} g(mp^\nu) \\ &= \sum_{p^\nu \leqslant x} (\ln p)\Big\{ g(p^\nu) G\big(x/p^\nu\big) + O\big(x/p^{\nu+1}\big)\Big\} \\ &= \sum_{d\leqslant x} \Lambda(d) g(d) G\big(x/d\big) + O(x). \end{aligned} \tag{4.42}$$

Putting $R(v) := \psi(v) - v$, where ψ is the Chebyshev function, and assuming x is not an integer, we thus obtain,

$$\begin{aligned} |K(x)| &\leqslant \sum_{d\leqslant x} \Lambda(d) |G(x/d)| + O(x) = \int_1^x |G(x/t)|\,\mathrm{d}\psi(t) + O(x) \\ &= \int_1^x |G(x/t)|\,\mathrm{d}t + \int_1^x |G(x/t)|\,\mathrm{d}R(t) + O(x). \end{aligned}$$

The latter integral can be evaluated by partial summation using the fact that $\big|\mathrm{d}|G(v)|\big| \leqslant \mathrm{d}\,\lfloor v \rfloor$. We have

$$\begin{aligned} \int_1^x |G(x/t)|\,\mathrm{d}R(t) &= |G(x)| - \int_1^x R(x/t)\,\mathrm{d}|G(t)| \\ &\ll |G(x)| + \sum_{n\leqslant x} |R(x/n)| \ll x + \sum_{1\leqslant n\leqslant x} \frac{x}{n(\ln 2x/n)^2} \ll x, \end{aligned}$$

where we have used the prime number theorem in the form

$$R(v) \ll v/(\ln 2v)^2 \qquad (v \geqslant 1).$$

This indeed yields (4.41).

The second step consists in establishing the upper bound

$$\int_1^x \frac{|G(t)| \ln t}{t^2}\,\mathrm{d}t \ll H_T\Big(\frac{2}{\ln x}\Big) \ln x + \frac{(\ln x)^2}{T} \qquad (x \geqslant 2). \tag{4.43}$$

In (4.43), we may replace $|G(t)| \ln t$ by $|K(t)|$. Indeed, from (4.40), the implied error is $O(\ln x)$, which is acceptable in view of (4.38). Now, by the Cauchy–Schwarz inequality

$$\int_1^x \frac{|K(t)|}{t^2}\,\mathrm{d}t \leqslant \Big\{ \int_1^x \frac{|K(t)|^2}{t^3}\,\mathrm{d}t \int_1^x \frac{\mathrm{d}t}{t} \Big\}^{1/2},$$

we see that the proof of (4.43) may be reduced to that of

$$\int_1^\infty \frac{|K(t)|^2}{t^{3+2\alpha}}\,\mathrm{d}t \ll \frac{H_T(\alpha)^2}{\alpha} + \frac{1}{\alpha^3 T^2} \qquad (\alpha > 0). \tag{4.44}$$

Indeed, the required upper bound will then follow on selecting $\alpha = 2/\ln x$.

The equation

$$\int_0^\infty K(\mathrm{e}^u)\mathrm{e}^{-u\sigma}\mathrm{e}^{-iu\tau}\,\mathrm{d}u = \frac{-F'(s)}{s} \qquad (\sigma > 1)$$

allows us to write Plancherel's formula, viz.

$$\begin{aligned}\int_1^\infty \frac{|K(t)|^2}{t^{3+2\alpha}}\,\mathrm{d}t &= \int_0^\infty \frac{|K(\mathrm{e}^u)|^2}{\mathrm{e}^{2u(1+\alpha)}}\,\mathrm{d}u \\ &= \frac{1}{2\pi}\int_{-\infty}^{+\infty}\left|\frac{F'(1+\alpha+i\tau)}{1+\alpha+i\tau}\right|^2\mathrm{d}\tau.\end{aligned} \tag{4.45}$$

We estimate the contribution from the domain $|\tau| > T$ by Gallagher's lemma (Lemma 4.9): we have

$$\begin{aligned}\int_{|\tau|>T}\left|\frac{F'(1+\alpha+i\tau)}{1+\alpha+i\tau}\right|^2\mathrm{d}\tau &\leqslant \sum_{|k|\geqslant 1}\int_{-T}^{T}\frac{|F'(1+\alpha+2ikT+i\tau)|^2}{k^2T^2}\,\mathrm{d}\tau \\ &\ll \frac{1}{T^2}\sum_{n\geqslant 1}\frac{(T+n)(\ln n)^2}{n^{2+2\alpha}} \ll \frac{1}{T}+\frac{1}{\alpha^3T^2}.\end{aligned} \tag{4.46}$$

To estimate the contribution from the complementary range $|\tau| \leqslant T$, we write

$$\int_{|\tau|\leqslant T}\left|\frac{F'(1+\alpha+i\tau)}{1+\alpha+i\tau}\right|^2\mathrm{d}\tau \ll \sum_{|k|\leqslant T}\frac{1}{k^2+1}\int_{k-\frac12}^{k+\frac12}|F'(1+\alpha+i\tau)|^2\,\mathrm{d}\tau. \tag{4.47}$$

The last integral does not exceed

$$\max_{\substack{\sigma=1+\alpha\\ |\tau-k|\leqslant\frac12}}|F(s)|^2\int_{k-\frac12}^{k+\frac12}\left|\frac{F'}{F}(1+\alpha+i\tau)\right|^2\mathrm{d}\tau. \tag{4.48}$$

At this stage, we appeal to Lemma 4.11 in the form

$$\frac{F'}{F}(s) = \frac{D'(s)}{1+D(s)} + \frac{F_1'}{F_1}(s) + \frac{J'}{J}(s).$$

From (4.37), the last term is uniformly bounded. It therefore contributes $O(1)$ to the integral in (4.48). Moreover, we have

$$\frac{F_1'}{F_1}(s) = -\sum_{p>2}\frac{g(p)\ln p}{p^s}.$$

By Lemma 4.10, we thus have

$$\begin{aligned}\int_{k-\frac12}^{k+\frac12}\left|\frac{F_1'}{F_1}(1+\alpha+i\tau)\right|^2\mathrm{d}\tau &= \int_{-\frac12}^{\frac12}\left|\frac{F_1'}{F_1}(1+\alpha+ik+i\tau)\right|^2\mathrm{d}\tau \\ &\leqslant 3\int_{-\frac12}^{\frac12}\left|\frac{\zeta'}{\zeta}(1+\alpha+i\tau)\right|^2\mathrm{d}\tau \ll \int_{-\frac12}^{\frac12}\frac{\mathrm{d}\tau}{\alpha^2+\tau^2} \ll \frac{1}{\alpha}.\end{aligned}$$

Furthermore

$$\frac{1}{1+D(s)} = 1 + \sum_{\nu \geqslant 1} \big(-D(s)\big)^{\nu} \qquad (\sigma > 1).$$

As a Dirichlet series, this expression has coefficients which do not exceed, in absolute value, those of

$$1 + \sum_{\nu \geqslant 1} \Big(\sum_{m \geqslant 1} \frac{1}{2^{ms}} \Big)^{\nu} = \frac{2^s - 1}{2^s - 2} \ll \frac{1}{|s-1|}.$$

Given that $D'(s) \ll 1$ uniformly for $\sigma \geqslant 1$, we therefore deduce from Lemma 4.10 that

$$\begin{aligned} \int_{k-\frac{1}{2}}^{k+\frac{1}{2}} \Big| \frac{D'(1+\alpha+i\tau)}{1+D(1+\alpha+i\tau)} \Big|^2 \mathrm{d}\tau &\ll \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{\mathrm{d}\tau}{|1+D(1+\alpha+ik+i\tau)|^2} \\ &\ll \int_{-\frac{1}{2}}^{\frac{1}{2}} \Big| \frac{2^{1+\alpha+i\tau}-1}{2^{1+\alpha+i\tau}-2} \Big|^2 \mathrm{d}\tau \ll \frac{1}{\alpha}. \end{aligned}$$

It follows from the previous computations that the integral in (4.48) is $O(1/\alpha)$ uniformly in k. Substituting in (4.47) and taking (4.46) into account, this yields

$$\int_1^{\infty} \frac{|K(t)|^2}{t^{3+2\alpha}} \, \mathrm{d}t \ll \frac{H_T(\alpha)^2}{\alpha} + \frac{1}{\alpha^3 T^2} + \frac{1}{T}.$$

Since the latter term on the right-hand side is dominated by the sum of the two others, we obtain (4.44), and therefore (4.43).

We can now complete the argument. Since $H_T(\alpha) \gg 1$, the second term in the upper bound (4.39) is of the required order of magnitude. In order to evaluate the first, we use (4.43) in the form

$$\int_{\sqrt{y}}^{y} \frac{|G(t)|}{t^2} \, \mathrm{d}t \ll H_T\Big(\frac{2}{\ln y}\Big) \qquad (y \geqslant \mathrm{e}).$$

It follows that

$$\begin{aligned} \int_{\mathrm{e}^2}^{x} \frac{|G(t)|}{t^2} \, \mathrm{d}t &\ll \int_{\mathrm{e}^2}^{x} \frac{|G(t)|}{t^2} \int_t^{t^2} \frac{\mathrm{d}y}{y \ln y} \, \mathrm{d}t \ll \int_{\mathrm{e}^2}^{x^2} \frac{\mathrm{d}y}{y \ln y} \int_{\sqrt{y}}^{y} \frac{|G(t)|}{t^2} \, \mathrm{d}t \\ &\ll \int_{\mathrm{e}^2}^{x^2} H_T\Big(\frac{2}{\ln y}\Big) \frac{\mathrm{d}y}{y \ln y} = \int_{1/\ln x}^{1} \frac{H_T(\alpha)}{\alpha} \, \mathrm{d}\alpha. \end{aligned}$$

This concludes the proof of Theorem 4.7.

4.3.4. Applications. Our first application is an effective form of Halász' Theorem 4.5.

Corollary 4.12. *Let g be a multiplicative function modulus $\leqslant 1$. For $x \geqslant 2$, $T \geqslant 2$, set*

$$m(x,T) := \min_{|\tau|\leqslant T} \sum_{p\leqslant x} \frac{1-\Re e\big(g(p)p^{-i\tau}\big)}{p}, \quad R(x,T) := \frac{1+m(x,T)}{\mathrm{e}^{m(x,T)}} + \frac{1}{T}.$$

Then we have

$$\sum_{n\leqslant x} g(n) \ll x\, R(x,T). \tag{4.49}$$

Proof. Put

$$\mu(x,\tau) := \sum_{p\leqslant x} \Re e\left(g(p)/p^{1+i\tau}\right), \qquad \mu^*(x,T) := \max_{|\tau|\leqslant T} \mu(x,\tau),$$

so that $m(x,T) = \ln_2 x - \mu^*(x,T) + O(1)$.

We first establish a pseudo-monotonicity property for the function

$$x \mapsto \mu^*(x,T).$$

We plainly have

$$\left|\frac{\partial \mu(x,\tau)}{\partial \tau}\right| \leqslant \sum_{p\leqslant x} \frac{|g(p)|\ln p}{p} \ll \ln x,$$

so $\mu(x,\vartheta) = \mu(x,\tau) + O(1)$ for $|\vartheta - \tau| \leqslant 1/\ln x$ and

$$\mu(x,\tau) = \tfrac{1}{2}\ln x \int_{\tau-1/\ln x}^{\tau+1/\ln x} \mu(x,\vartheta)\,\mathrm{d}\vartheta + O(1).$$

Let τ_y, with $|\tau_y| \leqslant T - 1/\ln y$, be such that $\mu(y,\tau_y) = \mu^*(y, T-1/\ln y)$. We deduce from the above bounds that $\mu^*(y,T) \leqslant \mu(y,\tau_y) + O(1)$, hence

$$\begin{aligned}
\mu^*(y,T) &= \frac{\ln y}{2} \int_{\tau_y-1/\ln y}^{\tau_y+1/\ln y} \mu(y,\tau)\,\mathrm{d}\tau + O(1)\\
&= \frac{\ln y}{2} \int_{\tau_y-1/\ln y}^{\tau_y+1/\ln y} \Big(\mu(x,\tau) - \{\mu(x,\tau) - \mu(y,\tau)\}\Big)\,\mathrm{d}\tau + O(1)\\
&\leqslant \mu^*(x,T) - \frac{\ln y}{2} \int_{\tau_y-1/\ln y}^{\tau_y+1/\ln y} \sum_{y<p\leqslant x} \Re e\, \frac{g(p)}{p^{1+i\tau}}\,\mathrm{d}\tau + O(1)\\
&\leqslant \mu^*(x,T) + \ln y \sum_{y<p\leqslant x} \frac{|g(p)|}{p\ln p} + O(1) \leqslant \mu^*(x,T) + O(1).
\end{aligned} \tag{4.50}$$

The proof may now be completed. By (4.32), we have

$$\begin{aligned} \alpha F(1+\alpha+i\tau) &\ll \exp\Big\{-\sum_{p\leqslant \exp(1/\alpha)} \frac{1-\Re e\big(g(p)p^{-i\tau}\big)}{p^{1+\alpha}}\Big\} \\ &\ll \exp\Big\{-\sum_{p\leqslant \exp(1/\alpha)} \frac{1-\Re e\big(g(p)p^{-i\tau}\big)}{p}\Big\}, \end{aligned} \tag{4.51}$$

since $\sum_{p\leqslant \exp(1/\alpha)}(1/p - 1/p^{1+\alpha}) \ll 1$. Therefore, uniformly for $|\tau| \leqslant T$ and $1/\ln x \leqslant \alpha \leqslant 1$, we have

$$F(1+\alpha+i\tau) \ll \mathrm{e}^{-m(\exp(1/\alpha),T)}\alpha^{-1} \ll \mathrm{e}^{-m(x,T)}\ln x, \tag{4.52}$$

where the second evaluation results from (4.50). The same bound holds for the function $H_T(\alpha)$ defined by (4.30). Inserting this estimate into (4.31) when $\alpha \leqslant \alpha_0 := e^{m(x,T)}/\ln x$, and using the trivial bound $H_T(\alpha) \ll 1/\alpha$ otherwise, we get

$$\begin{aligned} \frac{1}{x}\sum_{n\leqslant x} g(n) &\ll \frac{1}{\ln x}\int_{1/\ln x}^{1} \frac{H_T(\alpha)}{\alpha}\,\mathrm{d}\alpha + \frac{1}{T} \\ &\ll e^{-m(x,T)}\ln(\alpha_0 \ln x) + \frac{1}{\alpha_0 \ln x} + \frac{1}{T} \ll R(x,T). \end{aligned}$$ □

We now derive a second consequence from Theorem 4.7, which concerns the mean value of real multiplicative functions. We shall make use of the following lemma.

Lemma 4.13. *Let h be a 2π-periodic function, with bounded variation on $[0, 2\pi]$, and with mean value*

$$\overline{h} := \frac{1}{2\pi}\int_0^{2\pi} h(t)\,\mathrm{d}t.$$

For all real numbers τ, w, z such that $\tau \neq 0$, $1 < w < z$, we have

$$\sum_{w<p\leqslant z} \frac{h(\tau \ln p)}{p} = \overline{h}\ln\Big(\frac{\ln z}{\ln w}\Big) + O\Big(\frac{V(h)}{|\tau|\ln w} + \frac{M(h)+(1+|\tau|)V(h)}{\mathrm{e}^{\sqrt{\ln w}}}\Big) \tag{4.53}$$

where $M(h) := \sup_t |h(t)|$, $V(h) := \int_0^{2\pi} |\,\mathrm{d}h(t)|$.

Proof. We may assume $\tau > 0$. We use the prime number theorem in the form $R(t) := \pi(t) - \mathrm{li}(t) \ll t\exp\{-2\sqrt{\ln t}\}$. By partial summation we can

write the left-hand side of (4.53) as

$$\int_w^z \frac{h(\tau\ln t)}{t\ln t}\,\mathrm{d}t + \Big[\frac{R(t)h(\tau\ln t)}{t}\Big]_w^z - \int_w^z R(t)\,\mathrm{d}\Big(\frac{h(\tau\ln t)}{t}\Big)$$
$$= \overline{h}\ln\Big(\frac{\ln z}{\ln w}\Big) + \int_{\tau\ln w}^{\tau\ln z}\big(h(t)-\overline{h}\big)\frac{\mathrm{d}t}{t} - \int_w^z \frac{R(t)}{t}\,\mathrm{d}h(\tau\ln t) + O\Big(\frac{M(h)}{\mathrm{e}^{\sqrt{\ln w}}}\Big).$$

Observing that, for all a, b, we have

$$\Big|\int_a^b \big(h(t)-\overline{h}\big)\,\mathrm{d}t\Big| \ll V(h)$$

and using the second mean value theorem, we see that the second term of the above sum is $\ll V(h)/(\tau\ln w)$. The third is

$$\ll \mathrm{e}^{-\sqrt{\ln w}}\int_{\tau\ln w}^{\tau\ln w+2\pi}\sum_{k=0}^{\infty}\mathrm{e}^{-\sqrt{(v+2\pi k)/\tau}}|\,\mathrm{d}h(v)| \ll (1+|\tau|)V(h)\mathrm{e}^{-\sqrt{\ln w}},$$

since the sum over k is $\ll(1+|\tau|)$. This implies (4.53). □

Theorem 4.14 (Hall & Tenenbaum, 1991). *Let φ_0 be the unique solution in $]0,2\pi[$ of the equation $\sin\varphi_0 + (\pi-\varphi_0)\cos\varphi_0 = \frac12\pi$. Put*

$$K := \cos\varphi_0 \approx 0.32867.$$

Then, for all real numbers $x \geqslant 2$, and uniformly for all multiplicative functions g with values in $[-1,1]$, we have

$$\sum_{n\leqslant x} g(n) \ll x\,\exp\Big\{-K\sum_{p\leqslant x}\frac{1-g(p)}{p}\Big\}. \tag{4.54}$$

Remark. Hall & Tenenbaum in fact show that the quoted constant K is optimal.

Proof. Set $h(\vartheta) := |\cos(\vartheta) - K|$. We first establish that, uniformly for $0<\alpha\leqslant 1$, $\tau\in\mathbb{R}$, we have

$$\sum_{p\leqslant\exp(1/\alpha)}\frac{h(\tau\ln p)}{p} \leqslant (1-K)\ln(1/\alpha) + O\big(\log_2(|\tau|+3)\big). \tag{4.55}$$

We may plainly assume $\tau>0$. For $\tau\leqslant\alpha$, relation (4.55) follows immediately, by summation over p, from the estimate

$$h(\tau\ln p) = 1 - K + O(\tau\ln p). \tag{4.56}$$

When $0<\alpha<\tau\leqslant 1$, write $w := \exp(1/\tau)$. We deduce from (4.56) that we have

$$\sum_{p\leqslant w}\frac{h(\tau\ln p)}{p} \leqslant (1-K)\log_2 w + O(1). \tag{4.57}$$

Then, putting $z := \exp(1/\alpha)$, and applying Lemma 4.13 to the function h with mean value

$$\overline{h} = \frac{1}{2\pi}\int_0^{2\pi} |\cos\vartheta - \cos\varphi_0|\,\mathrm{d}\vartheta = \frac{2}{\pi}\{\sin\varphi_0 - \varphi_0\cos\varphi_0\} + \cos\varphi_0$$
$$= 1 - \cos\varphi_0 = 1 - K,$$

we infer that

$$\sum_{w<p\leqslant z} \frac{h(\tau\ln p)}{p} \leqslant (1-K)\big(\ln(1/\alpha) - \ln_2 w\big) + O(1), \tag{4.58}$$

so that, given (4.57), relation (4.55) still holds in this case.

Suppose now that $1 < |\tau| \leqslant \exp(1/\sqrt{\alpha}) - 3$, and choose, in Lemma 4.13, $w := \exp\{\ln^2(3+|\tau|)\} \leqslant z = \exp(1/\alpha)$. Inequality (4.58) remains valid, and we deduce (4.55) by estimating the left-hand side of (4.57) trivially.

When $|\tau| > \exp(1/\sqrt{\alpha}) - 3$, we have $\ln(1/\alpha) \ll \ln_2(3+|\tau|)$, and the complete sum in (4.55) can be estimated trivially. We thus obtain that (4.55) holds in all cases.

Consider now the quantity $\lambda = \lambda(\alpha) \in [0,2]$, defined by

$$\sum_{p\leqslant\exp(1/\alpha)} \frac{1-g(p)}{p} = \lambda \sum_{p\leqslant\exp(1/\alpha)} \frac{1}{p}. \tag{4.59}$$

The inequality

$$\Re e\,\big(g(p)p^{i\tau}\big) = g(p)\big(\cos(\tau\ln p) - K\big) + Kg(p)$$
$$\leqslant |\cos(\tau\ln p) - K| + Kg(p) = h(\tau\ln p) + Kg(p)$$

yields, after dividing by p, summing over $p \leqslant \exp(1/\alpha)$ and employing (4.55) and (4.59), that

$$\Re e \sum_{p\leqslant\exp(1/\alpha)} \frac{g(p)}{p^{1+i\tau}} \leqslant (1-K\lambda)\ln(1/\alpha) + O\big(\ln_2(|\tau|+3)\big). \tag{4.60}$$

We can now embark on the last phase of the proof. If $F(s)$ denotes the Dirichlet series appearing in (4.30), then, for $s = 1+\alpha+i\tau$, we have that

$$F(s) \ll \exp\Big\{\Re e \sum_p \frac{g(p)}{p^s}\Big\} \ll \exp\Big\{\Re e \sum_{p\leqslant\exp(1/\alpha)} \frac{g(p)}{p^{1+i\tau}}\Big\}, \tag{4.61}$$

by virtue of the estimate

$$\sum_{p\leqslant\exp(1/\alpha)} \frac{1-p^{-\alpha}}{p} + \sum_{p>\exp(1/\alpha)} \frac{1}{p^{1+\alpha}} \ll 1,$$

which follows from the prime number theorem by partial summation.

We deduce from (4.60) and (4.61) that

$$F(s) \ll \alpha^{K\lambda-1} \ln^B(|\tau|+3)$$

where B is some suitable absolute constant. Montgomery's function

$$H(\alpha) := \lim_{T\to\infty} H_T(\alpha),$$

where H_T is defined by (4.30), thus satisfies

$$H(\alpha)^2 \ll \alpha^{2K\lambda-2} \sum_{k\in\mathbb{Z}} \frac{\ln^B(|k|+4)}{k^2+1}$$

whence

$$H(\alpha) \ll \alpha^{K\lambda-1}. \tag{4.62}$$

Then, set $S := \sum_{p\leqslant x} \{1-g(p)\}/p$. For $1/\ln x \leqslant \alpha \leqslant 1$, we have

$$\begin{aligned}\lambda \ln(1/\alpha) + O(1) &= \sum_{p\leqslant \exp(1/\alpha)} \frac{1-g(p)}{p} \geqslant \sum_{p\leqslant x} \frac{1-g(p)}{p} - \sum_{\exp(1/\alpha)<p\leqslant x} \frac{2}{p} \\ &\geqslant S - 2\ln_2 x + 2\ln(1/\alpha) + O(1)\end{aligned}$$

from which

$$\alpha^\lambda \ll \alpha^2 \mathrm{e}^{-S} \ln^2 x.$$

Substituting in (4.62), we obtain

$$H(\alpha) \ll \alpha^{2K-1} \mathrm{e}^{-KS} (\ln x)^{2K},$$

and thus, by Theorem 4.7,

$$G(x) \ll \frac{x}{\ln x} \mathrm{e}^{-KS} (\ln x)^{2K} \int_{1/\ln x}^1 \alpha^{2K-2}\, \mathrm{d}\alpha \ll x\mathrm{e}^{-KS},$$

as required. □

4.4. The Erdős–Kac theorem

The problem of the weak convergence of the frequencies

$$\nu_N\{n : f(n) \leqslant A_N + zB_N\} \qquad (N\to\infty) \tag{4.63}$$

is more delicate than that of the existence of a limit law. In the case of an additive function f, it is easy to appreciate the reason for this since the corresponding characteristic functions take the values

$$\frac{1}{N} \sum_{n\leqslant N} \mathrm{e}^{i\tau\{f(n)-A_N\}/B_N} \tag{4.64}$$

and so we need to estimate the average over the first N integers of a multiplicative function *depending on* N. The uniformity required for such a purpose does not follow from Theorems 4.2, 4.4 or 4.5, and the application

of Theorem 4.7 depends on sometimes delicate estimates for the function $H_T(\alpha)$—cf. the Notes on § 4.3 for references to the literature.

Here we shall treat the historically first example of a normalized distribution function. It concerns the arithmetic function $\omega(n)$, but an identical result also holds for $\Omega(n)$. The required effective mean value estimate is provided by Theorem II.5.2, of Selberg–Delange type. The link between characteristic function and distribution function is made explicit by the Berry–Esseen inequality—Theorem II.7.16. We let

$$\Phi(y) := \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{y} \mathrm{e}^{-t^2/2}\,\mathrm{d}t$$

denote the distribution function of the normal law, with characteristic function

$$\varphi(\tau) := \int_{-\infty}^{+\infty} \mathrm{e}^{i\tau y}\,\mathrm{d}\Phi(y) = \mathrm{e}^{-\tau^2/2}.$$

Theorem 4.15 (Erdős & Kac, 1939; Rényi & Turán, 1958). *Uniformly for $N \geqslant 2$, $y \in \mathbb{R}$, we have*

$$\nu_N\Big\{n : \omega(n) \leqslant \ln_2 N + y\sqrt{\ln_2 N}\Big\} = \Phi(y) + O\Big(\frac{1}{\sqrt{\ln_2 N}}\Big). \tag{4.65}$$

Proof. Let $F_N(y)$ denote the left-hand side of (4.65), and let $\varphi_N(\tau)$ be the corresponding characteristic function, namely

$$\varphi_N(\tau) := \frac{1}{N} \sum_{n \leqslant N} \exp\Big\{\frac{i\tau}{\sqrt{\ln_2 N}}\big(\omega(n) - \ln_2 N\big)\Big\} \qquad (\tau \in \mathbb{R}).$$

By Theorem II.6.1, we have, uniformly for $N \geqslant 2$, $t \in \mathbb{R}$,

$$\frac{1}{N} \sum_{n \leqslant N} \mathrm{e}^{it\omega(n)} = A(\mathrm{e}^{it})\big(\ln N\big)^{\mathrm{e}^{it}-1} + O\big((\ln N)^{\cos t - 2}\big) \tag{4.66}$$

where $A(z)$ is an entire function of z with $A(1) = 1$.

Write $T := \sqrt{\ln_2 N}$, and choose $t = \tau/T$. Since $\cos t - 1 \leqslant -2t^2/\pi^2$ for $|t| \leqslant 1$, we obtain, on the one hand,

$$\varphi_N(\tau) \ll \mathrm{e}^{-2\tau^2/\pi^2} \qquad (|\tau| \leqslant T). \tag{4.67}$$

We shall use this estimate for $T^{1/3} < |\tau| \leqslant T$.

Observing that

$$A(\mathrm{e}^{it}) = 1 + O(t), \qquad \mathrm{e}^{it} - 1 = it - \tfrac{1}{2}t^2 + O(t^3) \qquad (|t| \leqslant 1),$$

we may write, on the other hand, using (4.66),

$$\frac{1}{N} \sum_{n \leqslant N} \mathrm{e}^{i\tau\omega(n)/T} = \Big\{1 + O\Big(\frac{\tau}{T}\Big)\Big\} \exp\Big\{i\tau T - \tfrac{1}{2}\tau^2 + O\Big(\frac{\tau^3}{T}\Big)\Big\} + O\Big(\frac{1}{\ln N}\Big)$$

from which we deduce that

$$\varphi_N(\tau) = \mathrm{e}^{-\tau^2/2}\Big\{1 + O\Big(\frac{|\tau| + |\tau|^3}{T}\Big)\Big\} + O\Big(\frac{1}{\ln N}\Big) \qquad (|\tau| \leqslant T^{1/3}). \tag{4.68}$$

We shall use this estimate for $1/\ln N < |\tau| \leqslant T^{1/3}$.

When $|\tau| \leqslant 1/\ln N$, we simply insert the trivial estimate $\mathrm{e}^{iy} = 1 + O(y)$ ($y \in \mathbb{R}$) in the definition of $\varphi_N(\tau)$. This yields

$$\varphi_N(\tau) = 1 + O\Big(\frac{|\tau|}{TN}\sum_{n\leqslant N}|\omega(n) - \ln_2 N|\Big) = 1 + O(|\tau|), \tag{4.69}$$

bounding the sum over n by successive applications of the Cauchy–Schwarz and Turán–Kubilius inequalities.

Now consider the Berry–Esseen inequality

$$\sup_{y\in\mathbb{R}} |F_N(y) - \Phi(y)| \ll \frac{1}{T} + \int_{-T}^{T} \Big|\varphi_N(\tau) - \mathrm{e}^{-\tau^2/2}\Big| \frac{\mathrm{d}\tau}{|\tau|}.$$

We split the integral into three parts, I_1, I_2 and I_3, corresponding to the respective ranges of integration $T^{1/3} < |\tau| \leqslant T$, $1/\ln N < |\tau| \leqslant T^{1/3}$ and $0 \leqslant |\tau| \leqslant 1/\ln N$. By (4.67), (4.68) and (4.69), we obtain

$$I_1 \ll \int_{T^{1/3}}^{\infty} \mathrm{e}^{-2\tau^2/\pi^2}\frac{\mathrm{d}\tau}{\tau} \ll \frac{1}{T},$$

$$I_2 \ll \int_{-\infty}^{\infty} \Big(\frac{1+\tau^2}{T}\Big)\mathrm{e}^{-\tau^2/2}\,\mathrm{d}\tau + \frac{1}{\ln N}\int_{1/\ln N}^{T^{1/3}} \frac{\mathrm{d}\tau}{\tau} \ll \frac{1}{T},$$

and

$$I_3 \ll \int_{-1/\ln N}^{1/\ln N} \mathrm{d}\tau \ll \frac{1}{T}.$$

This completes the proof. □

Note that the error term in formula (4.65) is optimal. Indeed, we deduce from Theorem II.6.4 that, for $k := \lfloor \ln_2 N \rfloor$, we have

$$\nu_N\{n : \omega(n) = k\} \sim \frac{1}{\ln N}\frac{(\ln_2 N)^{k-1}}{(k-1)!} \sim \frac{1}{\sqrt{2\pi \ln_2 N}}. \tag{4.70}$$

Now, if $R(N)$ denotes the supremum over $y \in \mathbb{R}$ of the error term in (4.65), we obtain, expressing (4.70) as a difference of two frequencies of type (4.65),

$$\frac{1+o(1)}{\sqrt{2\pi \ln_2 N}} = \Phi\Big(\frac{\vartheta}{\sqrt{\ln_2 N}}\Big) - \Phi\Big(\frac{\vartheta - 1/2}{\sqrt{\ln_2 N}}\Big) + O\big(R(N)\big)$$

with $\vartheta := k - \ln_2 N$. Hence

$$R(N) \gg 1/\sqrt{\ln_2 N}.$$

Notes

§ 4.1. For a purely probabilistic proof (using Kolmogorov's theorem) of the Erdős–Wintner theorem, see Novoselov (1964) and Babu (1973). Another probabilistic interpretation is given in the excellent survey by Galambos (1970).

The original proof of Erdős for the sufficient condition in Theorem 4.1 was derived in three stages (1935/37/38).[4] The necessary condition was established by Erdős & Wintner in 1939.

The argument presented here to obtain directly—i.e., without invoking Theorem 2.7(a)—the continuity criterion for the limit law is a variant of a proof of Szüsz (1974), completed by himself in Elliott's book (1985), pp. 437–439. Another proof is given by Elliott (1979), pp. 220–224.

Elliott has proved deep extensions to Theorem 4.2. These include the following result, established in chapter 9 of his 1997 book. For $\alpha > 0$, we let $\mathcal{L}^\alpha(\mathbb{N}^*)$ denote the class of those arithmetic functions g such that

$$\|g\|_\alpha := \left\{ \limsup_{x\to\infty} \frac{1}{x} \sum_{n\leqslant x} |g(n)|^\alpha \right\}^{1/\alpha} < \infty.$$

Theorem 4.16 (Elliott). *Let $\alpha > 1$. A necessary and sufficient condition for a multiplicative arithmetic function g to belong to $\mathcal{L}^\alpha(\mathbb{N}^*)$ and to have a non-zero mean value $M(g)$ is that the series*

$$\sum_{p\in\mathbb{P}} \frac{g(p)-1}{p}, \quad \sum_{|g(p)|\leqslant 3/2} \frac{|g(p)-1|^2}{p}, \quad \sum_{|g(p)|>3/2} \frac{|g(p)|^\alpha}{p}, \quad \sum_{p\in\mathbb{P}} \sum_{\nu\geqslant 2} \frac{|g(p^\nu)|^\alpha}{p^\nu}$$

converge and that

$$\sum_{\nu\geqslant 0} \frac{g(p^\nu)}{p^\nu} \neq 0 \qquad (p\in\mathbb{P}).$$

When this is so, we have

$$M(g) = \prod_{p\in\mathbb{P}} \Big(1-\frac{1}{p}\Big) \sum_{\nu\geqslant 0} \frac{g(p^\nu)}{p^\nu}.$$

§ 4.2. Theorem 4.4 has not been published by Delange but has been the object of several oral expositions.

Another proof of part (i) of Theorem 4.2 is due to Daboussi (1982, 1989). In Exercise 282, we provide a third proof, inspired by Delange's original proof and based on the use of a Tauberian theorem.

A remarkable generalization of Theorem 4.4 is given by Elliott (1997), th. 11.1.

[4] See Exercise 280, p. 472, for a direct proof based on the Turán–Kubilius inequality.

§ 4.3. Numerous generalizations of Wirsing's theorem (Theorem 4.6) may be found in the literature. See, for example, Elliott (1997), chap. 17. Tenenbaum (2007) showed that, for any real g, if g^2 has a mean value, then so does g and, in these circumstances, we have $M(g) = 0$ whenever

$$\sum_{p\in\mathbb{P}} \frac{\{g(p)-1\}^2}{p} = \infty.$$

In view of the easy Theorem I.3.12, this immediately implies Theorem 4.6.

Montgomery (1994) deduced from Hilbert-type inequalities the following strengthening of (4.33)

$$\left| \int_{-T}^{T} \Big| \sum_{1\leqslant n\leqslant N} a_n \mathrm{e}(\lambda_n t) \Big|^2 \,\mathrm{d}t - T \sum_{1\leqslant n\leqslant N} |a_n|^2 \right| \leqslant \frac{3}{2} \sum_{1\leqslant n\leqslant N} \frac{|a_n|^2}{\delta_n}. \tag{4.71}$$

Theorem 4.7 is stated and proved by Montgomery (1978b) with $T = \infty$ and for a *completely* multiplicative function g. The extension to the general case is purely technical. In the same article, Montgomery remarks that his result enables one to establish estimates such as that in Theorem 4.14. See also Montgomery & Vaughan (2001). Evaluations of this type are referred to as *effective mean value estimates*, or *quantitative theorems* for mean values. They are, characteristically, uniform with respect to the summed function (in a certain class), which may, in particular, depend on x. Theorem 3.5 is a simple example of such a result. The estimate in Theorem 4.14 can be extended to complex-valued functions g such that $|g| \leqslant 1$ and $g(p)$ lies, for all p, in the ellipse

$$\Im m\,(\mathrm{e}^{-i\varphi}z)^2 \leqslant \delta^2\{1 - \Re e\,(\mathrm{e}^{-i\varphi}z)^2\}$$

where δ and φ are arbitrary parameters such that $0 \leqslant \delta < 1$, $0 \leqslant \varphi < \pi$. Hall & Tenenbaum (1991) give for each pair (δ, φ) the best constant $K(\delta, \varphi) > 0$ such that

$$G(x) \ll x\mathrm{e}^{-K(\delta,\varphi)\sum_{p\leqslant x}\{1-\Re e\, g(p)\}/p}.$$

The principle behind the upper bound (4.39) in the proof of Theorem 4.7 is due to Granville & Soundararajan (2003). The same work contains numerous results on effective upper bounds for sums of multiplicative functions, notably variants of Theorem 4.7 and a version of Corollary 4.12 whose operative numerical constants are made explicit. The authors also construct counter-examples supporting the qualitative optimality of their estimates.

For more general statements and the techniques necessary to prove them, see Elliott (1980), chap. 19, where the estimates of Halász (1971) are described and refined. Elliott proves the two following results; the first corresponds to Theorem 4.14 while the second is an effective version of Delange's theorem (Theorem 4.2).

Theorem 4.17. *Let g be a completely multiplicative function which, for some suitable constant $\lambda > 0$, satisfies $g(p) = 0$ or $\lambda \leqslant |g(p)| \leqslant 2 - \lambda$ for all p. Let ϑ_p be the argument of $g(p)$ when $g(p) \neq 0$. Assume there exist real numbers ϑ_0 and δ with $|\vartheta_0| \leqslant \pi$ and $\delta > 0$, such that $|e^{i\vartheta_p} - e^{i\vartheta_0}| \geqslant \delta$ $\big(g(p) \neq 0\big)$. Then we have*

$$\Big|\sum_{n\leqslant x} g(n)\Big| \ll x \exp\Big\{\sum_{p\leqslant x} \frac{|g(p)|-1}{p} - c\sum_{p\leqslant x} \frac{|g(p)| - \Re e\, g(p)}{p} + 2\lambda \sum_{\substack{p\leqslant x\\ g(p)=0}} \frac{1}{p}\Big\}$$

where c is a constant which depends only on δ and λ.

Theorem 4.18. *Let g be a completely multiplicative function which satisfies $|g(p) - 1| \leqslant \eta \leqslant \eta_0 < 1$ for all p. Then we have*

$$\Big|\sum_{n\leqslant x} g(n) - Ax\Big| \ll \eta x \exp\Big\{-\sum_{p\leqslant x} \frac{1 - \Re e\, g(p)}{p}\Big\} + x\Big(e^{-c_1/\eta} + (\ln x)^{-c_2}\Big) \exp\Big\{\sum_{p\leqslant x} \frac{|g(p)|-1}{p}\Big\}$$

where we have set $A := \exp\big\{\sum_{p\leqslant x} (g(p) - 1)/p\big\}$ and where c_1 and c_2 are positive constants depending only on η_0.

Theorem 4.5 yields a complete solution to the problem of the limiting distributions of shifted additive functions, namely the weak convergence of the d.f.'s

$$\nu_N\{n : f(n) \leqslant A_N + z\} \qquad (N \to \infty). \tag{4.72}$$

The following result, obtained independently by Elliott & Ryavec (1971), Levin & Timofeev (1971), Delange, and Kubilius (unpublished), is proved in Elliott (1979), chap. 7.

Theorem 4.19. *Let f be a real additive function. A necessary and sufficient condition for the existence of a function A_N such that the frequencies (4.72) converge weakly to a d.f. is that, for a suitable constant c, we have, on setting $h(n) := f(n) - c\ln n$ $(n \geqslant 1)$,*

$$\sum_p \frac{1}{p} \min(1, |h(p)|^2) < \infty.$$

In this case, we may choose $A_N := \sum_{p\leqslant N,\, |h(p)|\leqslant 1} h(p)/p + c\ln N$.

For this function A_N, the characteristic function of F is

$$\frac{1}{1 + ic\tau} \prod_{|h(p)|>1} w_p(\tau) \prod_{|h(p)|\leqslant 1} w_p(\tau) e^{-i\tau h(p)/p},$$

where we have put $w_p(\tau) := (1-1/p)\sum_{\nu\geqslant 0} \mathrm{e}^{-i\tau h(p^{-\nu})}p^{-\nu}$. *Moreover,* F *is of pure type. It is continuous if, and only if,*

$$\sum_{f(p)\neq 0} \frac{1}{p} = \infty.$$

§ 4.4. It is clear that the Erdős–Kac theorem can equally well be deduced from estimates for the frequencies $\nu_N\{n : \omega(n) = k\}$ by summing over appropriate values of k, as suggested in Exercise 218, p. 315. These estimates were obtained in Chap. II.6 by using the same basic tool as the one used here: the Selberg–Delange method.

The original proof by Erdős & Kac (1939) does not yield an explicit error term. LeVeque conjectured the bound $\ll 1/\sqrt{\ln_2 N}$ in 1949 and showed that $\ll (\ln_3 N)/(\ln_2 N)^{1/4}$ is admissible. This result was improved by Kubilius in 1956. LeVeque's conjecture was finally established by Rényi & Turán in 1958, employing the same method as in this chapter. Using the estimates for $\sum_{n\leqslant x} z^{\omega(n)}$ for complex z (cf. Theorem II.6.1), Delange (1959) obtained a more precise result: we have[5]

$$\nu_N\{n : \omega(n) \leqslant \ln_2 N + y\sqrt{\ln_2 N}\}$$
$$= \Phi(y) + \frac{\mathrm{e}^{-y^2/2}}{\sqrt{2\pi \ln_2 N}}\Big\{a - \tfrac{1}{6}y^2 + \big\langle \ln_2 N + y\sqrt{\ln_2 N}\big\rangle\Big\} + O\Big(\frac{1}{\ln_2 N}\Big)$$

where $a \approx 0.40516$.

[5] Recall the notation $\langle t\rangle$ for the fractional part of $t \in \mathbb{R}$.

Exercises

281. Show that each of the three types of possible distribution functions for an additive function can actually occur. [*Hint: see Exercises* 256–259, *pp. 440–441.*]

282. *Proof of the necessary condition in Delange's theorem via a Tauberian theorem.*[6] Let g be a complex multiplicative function taking values in the unit disk and having a non-zero mean value $M(g)$. For $\sigma > 0$ and $p \in \mathbb{P}$, set $G_p(\sigma) := \sum_{\nu \geqslant 0} g(p^\nu)/p^{\nu\sigma}$. Denote by log the principal branch of the complex logarithm.

(a) Show that $\lim_{\sigma \to 1+} \zeta(\sigma)^{-1} \prod_{p \in \mathbb{P}} G_p(\sigma) = M(g)$.

(b) Show that $G_p(\sigma) \in \mathbb{C} \setminus \mathbb{R}^-$ for all $\sigma > 1$, and that the limits

$$\lim_{\sigma \to 1+} \sum_{p \in \mathbb{P}} \log\Big\{(1 - p^{-\sigma})G_p(\sigma)\Big\}, \qquad \lim_{\sigma \to 1+} \Big\{\log G_2(\sigma) + \sum_{p > 3} \frac{g(p) - 1}{p^\sigma}\Big\}$$

are well defined in $\mathbb{C}$.

(c) Deduce from the above that $G_2(1) \neq 0$ and that $\sum_p \{g(p) - 1\}/p^\sigma$ tends to a finite limit as $\sigma \to 1+$.

(d) By employing Theorem II.7.7 with $\omega = 0$, establish the convergence of the series $\sum_p \{g(p) - 1\}/p$.

283. *On the distribution of multiplicative functions.*

Let g be a real multiplicative function.

(a) Show (using, for example, Theorem I.3.12) that we have

$$\mathbf{d}\{n : g(n) = 0\} = 1 - \prod_p \Big(1 - \frac{1}{p}\Big) \sum_{\nu \geqslant 0,\, g(p^\nu) \neq 0} \frac{1}{p^\nu},$$

where the infinite product is to be interpreted as 0 when it diverges. Deduce, in particular, that $g(n) = 0$ pp if, and only if, $\sum_{g(p)=0} 1/p = \infty$.

(b) Assume that g does not vanish pp, and set

$$g^*(n) := \prod_{p^\nu \| n,\, g(p^\nu) \neq 0} g(p^\nu).$$

[6] The proof offered here is based, like that of Delange (1961), on using the Hardy–Littlewood–Karamata Tauberian theorem (we employ the generalized form given in Theorem II.7.7), but the details are significantly different.

Show that g and g^* simultaneously possess limiting distributions. For this, consider first the case of a strongly multiplicative function and then invoke Lemma 1.45 of Elliott (1979).[7]

(c) Assume now that $g(n) \neq 0$ $(n \geqslant 1)$. Let E be a set of primes such that $\sum_{p\in E} 1/p < \infty$. Define

$$h(n) := \prod_{p^\nu \| n,\, p\notin E} g(p^\nu) \quad \text{and} \quad A := \{a : p|a \Rightarrow p \in E\},$$

reserving the letter a to denote a generic integer belonging to A. Show that $\sum_{a\in A} 1/a < \infty$. Show that, for all $N \geqslant 1$, $z \in \mathbb{R}$,

$$\nu_N\{n : g(n) \leqslant z\} = \frac{1}{N} \sum_{a\leqslant N} \sum_{a'\leqslant N/a} \mu(a') \sum_{m\leqslant N/aa',\, g(a)h(m)\leqslant z} 1.$$

Deduce that, if h has a distribution function H, then g has a distribution function G, given almost everywhere by

$$G(z) = \prod_{p\in E} \Big(1 - \frac{1}{p}\Big) \sum_{a\in A} \frac{H_a\big(z/g(a)\big)}{a}$$

where we have set $H_a(z) := H(z)$ if $g(a) > 0$, and $H_a(z) := 1 - H(z)$ if $g(a) < 0$.

(d) Establish the following extension of a theorem of Galambos & Szüsz (1986): *Let g be a multiplicative function such that $g(n) \neq 0$ for $n \geqslant 1$, and such that $\sum_{g(p)<0} 1/p < \infty$. Then g has a limit law which is continuous at the origin if, and only if, the additive function $\ln|g(n)|$ satisfies the conditions of the Erdős–Wintner theorem.*[8]

284. *Daboussi's theorem (1974).* An arithmetic function f is said to have property (D) if $n \mapsto f(n)\mathrm{e}(\alpha n)$ has zero mean value for all $\alpha \in \mathbb{R} \setminus \mathbb{Q}$.

(a) Show that, if g has property (D) and satisfies $\sum_{n\leqslant x} |g(n)| \ll x$ and $\sum_{n\geqslant 1} |h(n)|/n < \infty$, then $h * g$ has property (D).

(b) Let f a multiplicative function with modulus at most 1. Show that if $g(n) = \mu(n)^2 f(n)$ has property (D), the same holds for f.

(c) In the sequel of this exercise, define, for $y \geqslant 2$, the completely multiplicative functions u_y and v_y by

$$u_y(p) := 1 \quad (p > y), \qquad u_y(p) := 0 \quad (p \leqslant y), \qquad v_y(p) := 1 - u_y(p).$$

Show that $u_y := \mathbf{1} * v_y\mu$, and deduce that u_y has property (D).

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[7] This is a variant of the continuity theorem, stipulating that an arithmetic function g has a distribution function if, and only if, for all fixed $\tau \in \mathbb{R}$ and $j \in \{0, 1\}$, the arithmetic function

$$G_{j\tau}(n) := \begin{cases} |g(n)|^{i\tau} \operatorname{sgn} g(n)^j & \text{if } g(n) \neq 0, \\ 0 & \text{if } g(n) = 0, \end{cases}$$

has a mean value $\varphi_j(\tau)$ continuous at $\tau = 0$.

[8] A necessary and sufficient condition for the existence of a d.f. for strongly multiplicative real functions is given by Galambos (1971). For the general case, see Elliott (1979), th. 7.11.

(d) Let f be a multiplicative function of modulus $\leqslant 1$ satisfying $f(p^\nu)=0$ whenever $\nu \geqslant 2$. Show that $f = fu_y * fv_y$. Deduce by a suitable application of the Cauchy–Schwarz inequality that, for $x \geqslant 1$, we have

$$\Big|\frac{1}{x}\sum_{n\leqslant x} f(n)\mathrm{e}(\alpha n)\Big|^2 \leqslant A(x)\{B_1(x)+B_2(x)\}$$

with $A(x) := (1/x)\sum_{n\leqslant x} u_y(n)$ and

$$B_1(x) := \frac{1}{x}\sum_{d\leqslant x}\big|f(d)v_y(d)\big|^2 \sum_{n\leqslant x/d} u_y(n)$$

$$B_2(x) := \sum_{1\leqslant d,d'\leqslant x,\, d\neq d'} fv_y(d)\overline{fv_y(d')}\,\frac{1}{x}\sum_{n\leqslant x/d',\, n\leqslant x/d} u_y(n)\mathrm{e}\big(\alpha n(d-d')\big).$$

(e) Show that $\lim_{x\to\infty} A(x)=\prod_{p\leqslant y}(1-1/p)$. Show that $B_1(x)\leqslant 1$. Using the fact that $fv_y(d)$ is non-zero for at most a finite number of integers d, show that $B_2(x)\to 0$ as $x\to+\infty$.

(f) Letting $y\to\infty$, prove Daboussi's theorem: *Every multiplicative function of modulus* $\leqslant 1$ *has property* (D).[9]

285. *An estimate of Montgomery & Vaughan (2001).*

Define $M(n,x) = \sum_{d|n,\, d\leqslant x}\mu(d)$.

(a) Using Lemma 4.13, show that, uniformly for $0<\alpha\leqslant 1$ and $\tau\neq 0$, we have

$$-\sum_{p>\exp(1/|\tau|)}\frac{\cos(\tau\ln p)}{p^{1+\alpha}} \leqslant 2\ln_2(3+|\tau|)+O(1),$$

$$\sum_{p>\exp(1/|\tau|)}\frac{|\cos(\tau\ln p)|}{p^{1+\alpha}} \leqslant \frac{2}{\pi}\ln\frac{1}{\alpha}+(2-\frac{4}{\pi})\ln_2(3+|\tau|)+O(1).$$

(b) Show that, for every integer $n\geqslant 1$, we have

$$\Big|\prod_{p|n}\big(1-p^{-1-\alpha-i\tau}\big)\Big| \ll \alpha^{-1/\pi}\Big(\ln(3+|\tau|)\Big)^{2-2/\pi} \qquad (0<\alpha\leqslant 1,\ \tau\neq 0).$$

(c) Using Theorem 4.7, deduce that we have

$$\max_{n\geqslant 1}|M(n,x)| \ll x\big(\ln x\big)^{(1/\pi)-1} \qquad (x\geqslant 2). \tag{4.73}$$

[9]The proof described here, though different from Daboussi's original proof (cf. Daboussi & Delange, 1982), is also due to Daboussi (1989).

(d) Show that the left-hand side of (4.73) is $\gg x/\ln x$. Evaluate the integral $\int_1^\infty M(n_x,t)\,\mathrm{d}t/t^{2+i}$ for $n_x := \prod_{p\leqslant x,\,\cos\ln p\leqslant 0} p$, and deduce that the exponent $1/\pi$ in (4.73) is optimal.[10]

286. *A theorem of Halász (1971).* Let E be a set of prime numbers. Write $E(x) := \sum_{p\leqslant x,\,p\in E} 1/p$ and consider the arithmetic function defined by the formula $\Omega(n;E) := \sum_{p^\nu\|n,\,p\in E}\nu$.

(a) Show, using Theorem 4.17, that if $\delta>0$, $\delta\leqslant|z|\leqslant 2-\delta$, then, for some constant $c_1>0$ depending only on δ, we have

$$\sum_{n\leqslant x} z^{\Omega(n;E)} \ll x\exp\big\{\big(|z|-1-c_1(|z|-\Re ez)\big)E(x)\big\}.$$

(b) Using Theorem 4.18, show that, if $|z-1|\leqslant\frac12$, then, for absolute constants $c_2>0$, $c_3>0$, we have that

$$\Big|\frac1x\sum_{n\leqslant x} z^{\Omega(n;E)} - \mathrm{e}^{(z-1)E(x)}\Big| \ll |z-1|\mathrm{e}^{(\Re ez-1)E(x)} + \mathrm{e}^{(|z|-1)E(x)}\Big\{\mathrm{e}^{-c_2/|z-1|}+(\ln x)^{-c_3}\Big\}.$$

(c) Employing Cauchy's formula in a manner similar to that in the proof of Theorem II.6.3, infer from (a) and (b) that we have, uniformly for $E(x)\geqslant 2$ and $\delta E(x)\leqslant m\leqslant(2-\delta)E(x)$,[11]

$$\sum_{n\leqslant x,\,\Omega(n;E)=m} 1 = x\frac{E(x)^m}{m!}\mathrm{e}^{-E(x)}\Big\{1+O\Big(\frac{1}{\sqrt{E(x)}}+\frac{|m-E(x)|}{E(x)}\Big)\Big\}.$$

287. *Convergence to the Gaussian law.*

Throughout the exercise, we set $y=y(x)=x^{1/\ln_2 x}$.

(a) Let $\mathcal{A}(x)$ be the class of those multiplicative functions G which take values in the unit disk and which satisfy $G(p^\nu)=1$ for $p>y$ and all $\nu\geqslant 1$. Show that, uniformly for $x\geqslant 2$ and $G\in\mathcal{A}(x)$, we have

$$\sum_{n\leqslant x} G(n) = x\prod_{p\leqslant y}\Big(1-\frac1p\Big)\sum_{\nu\geqslant 0}\frac{G(p^\nu)}{p^\nu}+O\Big(\frac{x}{\ln x}\Big).$$

(b) Let $\{g_x : x\geqslant 2\}$ be a family of multiplicative functions of modulus of most 1 such that $\lim_{x\to+\infty}\sum_{y<p\leqslant x}|g_x(p)-1|/p=0$. Introducing

$$G_x(n) := \prod_{p\leqslant y,\,p^\nu\|n} g_x(p^\nu)$$

[10] By the method of Hall & Tenenbaum (1991), it can actually be shown that $\ll$ in (4.73) may be replaced by $\asymp$.

[11] For more precise results, see Halász (1972) and Sárközy (1977b).

and using part (a), show that we have

$$\sum_{n\leqslant x} g_x(n) = x\prod_{p\leqslant x}\Big(1-\frac{1}{p}\Big)\sum_{\nu\geqslant 0}\frac{g_x(p^\nu)}{p^\nu} + o(x).$$

(c) Let f a real additive function satisfying the following properties as x tends to infinity:

(i) $D(x)^2 := \sum_{p\leqslant x} f(p)^2/p \to +\infty$, (ii) $\sum_{p\leqslant x} f(p)/p = o\big(D(x)\big)$,

(iii) $\sum_{y<p\leqslant x} |f(p)|/p = o\big(D(x)\big)$, (iv) $\max_{p\leqslant x} |f(p)| = o\big(D(x)\big)$.

For $\tau\in\mathbb{R}$, put $g_x(n,\tau) := \exp\{i\tau f(n)/D(x)\}$. Show that, for fixed τ and all $p\leqslant x$, we have

$$1+\frac{g_x(p,\tau)-1}{p} = \exp\Big\{\frac{i\tau f(p)}{pD(x)} - \frac{\tau^2 f(p)^2}{2pD(x)^2} + O\Big(\frac{|f(p)|^3}{pD(x)^3}+\frac{|f(p)|^2}{p^2D(x)^2}\Big)\Big\}.$$

Deduce the existence, and find the value, of $\lim_{x\to+\infty} x^{-1}\sum_{n\leqslant x} g_x(n,\tau)$. Show that, for all $z\in\mathbb{R}$, we have

$$\lim_{x\to\infty}\frac{1}{x}\big|\{n\leqslant x : f(n)\leqslant zD(x)\}\big| = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{z} \mathrm{e}^{-t^2/2}\,\mathrm{d}t.$$

(d) Application: Does the strongly additive arithmetic function defined by $f(n) := \sum_{p|n}(-1)^{\lfloor \ln_2 p\rfloor}$ have a normal order?

Friable integers. The saddle-point method

5.1. Introduction. Rankin's method

An integer n is called y–friable if its largest prime factor $P^+(n)$ does not exceed y.[1] Denote by

$$S(x,y) := \{n \leqslant x : P^+(n) \leqslant y\} \tag{5.1}$$

the set of y-friable integers not exceeding x. In this chapter, we shall describe the various methods which have been developed in the literature for the evaluation of the quantity

$$\Psi(x,y) := \big|S(x,y)\big|. \tag{5.2}$$

This question is indeed crucial in numerous arithmetical problems, and the techniques that have been implemented to tackle it are instructive from several points of view.

[1] Friable integers are called *smooth* in part of the literature on this topic, although there seems to be a wide consensus that the word smooth is overused and that the word friable is more appropriate: (a) it is only employed in this context; (b) it is evocative; (c) it is the same in English and in French; (d) it is easy to decline in expressions such as friability parameter, friable summation; (e) the integers referred to are anything but smooth, since their distribution is rather violently irregular. With the belief that the words of science are part of science, we follow a steadily increasing terminological usage by employing exclusively the word friable in this book for this meaning.

For each fixed $y \geqslant 1$, $\Psi(x,y)$ is the summatory function of the multiplicative function

$$\chi(n,y) := \begin{cases} 1 & \text{if } P^+(n) \leqslant y, \\ 0 & \text{if } P^+(n) > y. \end{cases} \tag{5.3}$$

The associated Dirichlet series is the leading factor of the Euler product for the Riemann zeta function, i.e.

$$\zeta(s,y) := \prod_{p \leqslant y} \Big(1 - \frac{1}{p^s}\Big)^{-1} = \sum_{n \geqslant 1} \frac{\chi(n,y)}{n^s}. \tag{5.4}$$

In 1938, in an article devoted to large gaps between consecutive primes,[2] Rankin devised a technique for finding an upper bound for $\Psi(x,y)$ which rests essentially upon the fact that the abscissa of convergence for $\zeta(s,y)$ is $\sigma_c = 0$. Indeed, for any $\sigma > 0$, we have

$$\Psi(x,y) \leqslant \sum_{n \geqslant 1,\, P^+(n) \leqslant y} \Big(\frac{x}{n}\Big)^{\sigma} = x^{\sigma}\zeta(\sigma,y). \tag{5.5}$$

We then obtain an explicit upper bound by choosing σ optimally. This procedure, simple but remarkably efficient, can be generalized in various ways. It is now known as *Rankin's method.* We apply it, in a rudimentary form, in the proof of Theorem 5.1 below.

The quantity

$$u := \frac{\ln x}{\ln y} \qquad (x \geqslant y \geqslant 2) \tag{5.6}$$

plays a pivotal role in the study of the asymptotic behavior of $\Psi(x,y)$. In the course of this chapter, we make systematic use of the notation (5.6).

Theorem 5.1. *We have*

$$\Psi(x,y) \ll x\mathrm{e}^{-u/2} \qquad (x \geqslant y \geqslant 2). \tag{5.7}$$

Proof. We can assume $y \geqslant 11$: otherwise, we have

$$\Psi(x,y) \leqslant \Psi(x,7) \ll (\ln x)^4$$

whereas the upper bound in (5.7) is of the order of a power of x. This being so, we have for all $\alpha \geqslant 0$ that

$$\Psi(x,y) \leqslant x^{3/4} + \sum_{n \leqslant x} \Big(\frac{n}{x^{3/4}}\Big)^{\alpha} \chi(n,y). \tag{5.8}$$

[2] See Theorem 5.26 *infra.*

With the selection $\alpha := 2/(3\ln y)$, the multiplicative function $n \mapsto n^\alpha \chi(n,y)$ satisfies the hypotheses of Corollary 3.6. We obtain

$$\sum_{n\leqslant x} n^\alpha \chi(n,y) \ll x \prod_{p\leqslant y}\Big(1+\frac{p^\alpha-1}{p}\Big) \leqslant x\exp\Big\{O\Big(\sum_{p\leqslant y}\frac{\alpha\ln p}{p}\Big)\Big\} \ll x.$$

Substituting in (5.8) and noting that $\frac14 \ln x > \frac12 u$ for $y \geqslant 11$, we obtain the stated result. □

The above proof is clearly not optimal. A more thorough study of Rankin's bound (5.5) enabled de Bruijn (1966) to obtain a uniform estimate providing an asymptotic formula for $\ln\Psi(x,y)$ as u and y tend to infinity. We now give a slightly more precise version with relative error term tending to 0 as soon as $y \to \infty$. We set

$$Z := \frac{\ln x}{\ln y}\ln\Big(1+\frac{y}{\ln x}\Big)+\frac{y}{\ln y}\ln\Big(1+\frac{\ln x}{y}\Big) = u\int_0^1 \ln\Big(1+\frac{y}{v\ln x}\Big)\,\mathrm{d}v. \tag{5.9}$$

Theorem 5.2. *Uniformly for $x \geqslant y \geqslant 2$, we have*

$$\ln\Psi(x,y) = Z\Big\{1+O\Big(\frac{1}{\ln y}+\frac{1}{\ln_2 2x}\Big)\Big\}. \tag{5.10}$$

Proof. We may plainly assume x to be sufficiently large. Set

$$\varphi_y(\sigma) := \frac{-\zeta'(\sigma,y)}{\zeta(\sigma,y)} = \sum_{p\leqslant y}\frac{\ln p}{p^\sigma-1} \qquad (0<\sigma\leqslant 1). \tag{5.11}$$

The optimal value of the parameter σ in (5.5) is the (unique) solution $\alpha = \alpha(x,y)$ of the equation

$$\varphi_y(\alpha) = \ln x.$$

The first stage of the proof consists in finding an explicit approximation for $\alpha(x,y)$. To this end, we appeal to the prime number theorem so as to evaluate $\varphi_y(\sigma)$ uniformly for $y \geqslant 2$, $0 < \sigma \leqslant 1$. We obtain

$$\varphi_y(\sigma) = \frac{y}{y^\sigma-1}\cdot\frac{1-y^{-(1-\sigma)}}{1-\sigma}\Big\{1+O\Big(\frac{1}{\ln y}\Big)\Big\}. \tag{5.12}$$

We omit the proof, somewhat technical, of this estimate, but indicate the main steps of the argument in Exercise 288.

We can deduce from (5.12) the estimate

$$\alpha(x,y) = \beta(x,y)\Big\{1+O\Big(\frac{\ln_2 y}{\ln y}\Big)\Big\} \qquad (x\geqslant y\geqslant 2) \tag{5.13}$$

where $\beta = \beta(x,y)$ is the solution of the equation $y/(y^\beta-1) = \ln x$, namely

$$\beta(x,y) := \frac{\ln(1+y/\ln x)}{\ln y}. \tag{5.14}$$

To show (5.13), we first observe that

$$\tfrac{1}{2} \leqslant (1 - y^{-(1-\sigma)})/(1-\sigma) \leqslant \ln y \quad (0 \leqslant \sigma \leqslant 1),$$

from which we infer, inserting in (5.12), that $\alpha = \beta + O(\ln_2 y/\ln y)$. This yields (5.13) if $\beta > \frac{1}{2}$. If $\beta \leqslant \frac{1}{2}$, we substitute the above estimate back into (5.12) to obtain

$$\ln x = \frac{y}{y^\beta - 1} = \frac{y}{(1-\beta)(y^\alpha - 1)}\Big\{1 + O\Big(\frac{\ln_2 y}{\ln y}\Big)\Big\},$$

whence $y^\alpha = y^\beta\{1 + O(\beta \ln_2 y)\}$, so that (5.13) also holds in this case.

The upper bound implied in Theorem 5.2 will follow by choosing $\sigma = \beta$ in (5.5). Note that

$$Z = \beta \ln x + \frac{y}{\ln y} \ln\Big(\frac{1}{1-y^{-\beta}}\Big). \tag{5.15}$$

By (5.5), we have

$$\Psi(x,y) \leqslant x^\beta \zeta(\beta, y) = \zeta(1,y) \exp\Big\{\beta \ln x + \int_\beta^1 \varphi_y(\sigma)\,\mathrm{d}\sigma\Big\}. \tag{5.16}$$

If $y \leqslant (\ln x)^2$, we certainly have $\beta \leqslant \frac{2}{3}$. Using (5.12), we can then write

$$\int_\beta^1 \varphi_y(\sigma)\,\mathrm{d}\sigma = \Big\{1 + O\Big(\frac{1}{\ln y}\Big)\Big\}\frac{y}{\ln y}\int_\beta^{2/3} \frac{(\ln y)\mathrm{d}\sigma}{(1-\sigma)(y^\sigma - 1)} + O\Big(\int_{2/3}^1 y^{1-\sigma} \ln y\,\mathrm{d}\sigma\Big).$$

The second error term is $\ll y^{1/3}$. The integral over σ equals

$$\int_\beta^{2/3} \frac{\ln y}{y^\sigma - 1}\,\mathrm{d}\sigma + O\Big(\int_\beta^{2/3} \frac{\sigma \ln y}{y^\sigma - 1}\,\mathrm{d}\sigma\Big) = \ln\Big(\frac{1-y^{-2/3}}{1-y^{-\beta}}\Big) + O\Big(\frac{1}{\ln y}\int_{\beta \ln y}^\infty \frac{t\,\mathrm{d}t}{\mathrm{e}^t - 1}\Big)$$
$$= \ln\Big(\frac{1}{1-y^{-\beta}}\Big) + O\Big(\frac{1}{y^{2/3}} + \frac{1+\beta\ln y}{y^\beta \ln y}\Big).$$

It follows that

$$\int_\beta^1 \varphi_y(\sigma)\,\mathrm{d}\sigma = \Big\{1 + O\Big(\frac{1}{\ln y}\Big)\Big\}\frac{y}{\ln y}\ln\Big(\frac{1}{1-y^{-\beta}}\Big) + O\Big(\frac{y(1+\beta\ln y)}{y^\beta (\ln y)^2}\Big).$$

We claim that the second error term is $\ll Z/\ln y$. Indeed, when $y \leqslant \ln x$, we have $\beta \ln y \ll 1$, from which $Z \gg y/\ln y$, and when $\ln x < y \leqslant (\ln x)^2$ we have $\beta \ln y \gg 1$, from which

$$\frac{y(1+\beta\ln y)}{y^\beta(\ln y)^2} \asymp \frac{\beta y^{1-\beta}}{\ln y} \asymp \beta u \ll \frac{Z}{\ln y}.$$

Substituting in (5.16), we obtain that the bound

$$\ln \Psi(x,y) \leqslant Z\Big\{1+O\Big(\frac{1}{\ln y}\Big)\Big\} \tag{5.17}$$

holds for $y \leqslant (\ln x)^2$.

When $(\ln x)^2 < y \leqslant x$, we have $\beta \gg 1$. It hence follows from (5.12) that

$$\varphi_y(\sigma) = \frac{y^{1-\sigma}-1}{1-\sigma}\Big\{1+O\Big(\frac{1}{\ln y}\Big)\Big\} \qquad (\beta \leqslant \sigma \leqslant 1).$$

This immediately implies

$$\int_\beta^1 \varphi_y(\sigma)\,\mathrm{d}\sigma = \Big\{1+O\Big(\frac{1}{\ln y}\Big)\Big\}\int_0^{(1-\beta)\ln y} \frac{\mathrm{e}^v-1}{v}\,\mathrm{d}v \ll \frac{y^{1-\beta}}{(1-\beta)\ln y}.$$

Observing that $y^{1-\beta} = \Big(\dfrac{1}{y}+\dfrac{1}{\ln x}\Big)^{-1} \asymp \ln x$, we see that

$$\int_\beta^1 \varphi_y(\sigma)\,\mathrm{d}\sigma \ll \frac{\ln x}{\ln_2 x},$$

from which

$$\ln \Psi(x,y) \leqslant \beta \ln x + O\Big(\frac{\ln x}{\ln_2 x}\Big) \leqslant Z\Big\{1+O\Big(\frac{1}{\ln_2 x}\Big)\Big\}.$$

We have thus established, in all cases, the required upper bound implicit in (5.10). For the corresponding lower bound, we set $v := \lceil u \rceil$, $z := x^{1/v}$, and $k := \pi(z)$. Then $\Psi(x,y) \geqslant \Psi(x,z)$ and it is clear that each k-tuple $(\nu_1,\dots,\nu_k)$ of integers $\geqslant 0$ such that $\sum_{j=1}^k \nu_j \leqslant v$ provides at least one integer counted in $\Psi(x,z)$, namely $\prod_{j=1}^k p_j^{\nu_j}$ (where p_j denotes the j-th prime number). It follows that

$$\Psi(x,y) \geqslant \binom{k+v}{v},$$

from which, applying Stirling's formula,

$$\ln \Psi(x,y) \geqslant (k+v)\ln(k+v) - k\ln k - v\ln v + O\big(1+\ln\min(k,v)\big).$$

The main term in this lower bound can be rewritten as $\int_0^k \ln(1+v/t)\,\mathrm{d}t$. We hence see that the error term implied in replacing k by $z/\ln z$ is

$$\ll \frac{z}{(\ln z)^2}\ln\Big(1+\frac{v}{k}\Big) \ll \frac{z}{(\ln z)^2}\ln\Big(1+\frac{\ln x}{z}\Big).$$

Thus we can write

$$\ln \Psi(x,y) \geqslant W(z)\Big\{1+O\Big(\frac{1}{\ln z}\Big)\Big\} \tag{5.18}$$

with

$$W(z) := \frac{\ln x}{\ln z}\ln\Big(1+\frac{z}{\ln x}\Big) + \frac{z}{\ln z}\ln\Big(1+\frac{\ln x}{z}\Big).$$

It may be easily checked that

$$W'(t) = -\frac{W(t)}{t\ln t} + \frac{1}{\ln t}\ln\Big(1+\frac{\ln x}{t}\Big).$$

This implies, after a small calculation, that we have for $\sqrt{y} \leqslant t \leqslant y^2$,

$$W'(t) \ll \begin{cases} \dfrac{\ln x}{t\ln t} & \big(y \leqslant (\ln x)^4\big), \\ \dfrac{(\ln x)\ln_2 x}{t(\ln t)^2} & \big((\ln x)^4 < y \leqslant x\big). \end{cases}$$

Bearing in mind that $(\ln y)/\ln z = 1 + O(1/u)$, it follows in a few lines that

$$Z - W(z) = W(y) - W(z) \ll \min(\ln y, \ln_2 x) \ll Z\Big(\frac{1}{\ln y} + \frac{1}{\ln_2 x}\Big).$$

Substituting in (5.18), we obtain

$$\ln\Psi(x,y) \geqslant Z\Big\{1 + O\Big(\frac{1}{\ln y} + \frac{1}{\ln_2 x}\Big)\Big\}.$$

This completes the proof of Theorem 5.2. □

Theorem 5.2 demonstrates a change in behavior for $\Psi(x,y)$ when y passes through the threshold $\ln x$. This phenomenon will be more fully described in the next sections. It may be explained, at least partially, by the fact that the relation

$$\prod_{p\leqslant y} p = o(x),$$

valid, as $x \to \infty$, whenever $y \leqslant (1-\varepsilon)\ln x$, imposes a special structure on the integers counted in $\Psi(x,y)$ when y is "small", namely that some exponents in their canonical prime decomposition *must* then be "large".

5.2. The geometric method

We have seen in the course of the proof of the lower bound in Theorem 5.2 that the problem of $\Psi(x,y)$ can be interpreted as counting points with integral coordinates inside some polyhedron of $\mathbb{R}^k$, with $k := \pi(y)$. When k is not too large, a satisfactory estimate can quite easily be obtained.

Let $\{a_j\}_{j=1}^{\infty}$ be a sequence of positive real numbers. We set

$$N_k(z) := \Big|\Big\{(\nu_1,\dots,\nu_k) \in \mathbb{N}^k : \textstyle\sum_{1\leqslant j\leqslant k} \nu_j a_j \leqslant z\Big\}\Big|.$$

Theorem 5.3. *For $k \geqslant 1$, $z \geqslant 0$, we have*

$$\frac{z^k}{k!}\prod_{1\leqslant j\leqslant k}\frac{1}{a_j} < N_k(z) \leqslant \frac{\big(z+\sum_{1\leqslant j\leqslant k} a_j\big)^k}{k!}\prod_{1\leqslant j\leqslant k}\frac{1}{a_j}. \tag{5.19}$$

For the choice $k := \pi(y)$, $z := \ln x$, $a_j := \ln p_j$ (where p_j denotes the j-th prime number), we have $N_k(z) = \Psi(x,y)$ and $\sum_{1\leqslant j\leqslant k} a_j = \vartheta(y) \ll y$. This hence immediately yields the following result.

Corollary 5.4 (Ennola, 1969). *Uniformly for $2 \leqslant y \leqslant \sqrt{\ln x \ln_2 x}$, we have*

$$\Psi(x,y) = \frac{1}{\pi(y)!} \prod_{p\leqslant y} \Big(\frac{\ln x}{\ln p}\Big)\Big\{1 + O\Big(\frac{y^2}{\ln x \ln y}\Big)\Big\}. \tag{5.20}$$

Proof of Theorem 5.3. We argue by induction on k. The result holds for $k = 1$, since

$$\frac{z}{a_1} < N_1(z) = 1 + \Big\lfloor \frac{z}{a_1} \Big\rfloor \leqslant \frac{z + a_1}{a_1}.$$

Suppose the double inequality (5.19) is satisfied for $k-1$ with $k \geqslant 2$. We then have

$$N_k(z) = \sum_{0\leqslant \nu\leqslant z/a_k} N_{k-1}(z - \nu a_k),$$

from which

$$\frac{1}{(k-1)!} \prod_{1\leqslant j<k} \frac{1}{a_j}\, S_k(z) < N_k(z) \leqslant \frac{1}{(k-1)!} \prod_{1\leqslant j<k} \frac{1}{a_j}\, S_k(w)$$

where we have set $S_k(t) := \sum_{0\leqslant\nu\leqslant z/a_k} (t - \nu a_k)^{k-1}$ and $w := z + \sum_{1\leqslant j<k} a_j$.

Let $n := \lfloor z/a_k \rfloor$. The Euler–Maclaurin formula of order 0 (cf. Theorem I.0.7) implies

$$\begin{aligned} S_k(z) = \sum_{0\leqslant\nu\leqslant n} (z - \nu a_k)^{k-1} &= \int_0^n (z - t a_k)^{k-1}\,\mathrm{d}t + \tfrac{1}{2}\big\{(z - n a_k)^{k-1} + z^{k-1}\big\} \\ &\quad - (k-1)a_k \int_0^n B_1(t)(z - t a_k)^{k-2}\,\mathrm{d}t, \end{aligned}$$

with $B_1(t) := \langle t\rangle - \frac{1}{2}$. In absolute value, the last integral is bounded above by

$$-\tfrac{1}{2}\int_0^n \frac{\mathrm{d}}{\mathrm{d}t}\big\{(z - t a_k)^{k-1}\big\}\,\mathrm{d}t = \tfrac{1}{2}\big\{z^{k-1} - (z - n a_k)^{k-1}\big\}.$$

Therefore

$$S_k(z) \geqslant \frac{1}{a_k}\Big\{\frac{z^k}{k} - \frac{(z - n a_k)^k}{k}\Big\} + (z - n a_k)^{k-1} > \frac{z^k}{k a_k},$$

since $(z - n a_k)/k a_k \leqslant 1/k < 1$. This establishes the desired lower bound.

In order to obtain the upper bound, we again apply the Euler–Maclaurin formula, but with z replaced by w. This yields

$$S_k(w) \leqslant \int_0^n (w - t a_k)^{k-1}\,\mathrm{d}t + w^{k-1} \leqslant \frac{w^k + k a_k w^{k-1}}{k a_k} < \frac{(w + a_k)^k}{k a_k}.$$

This completes the proof. □

In addition to the information it provides on the global behavior of $\Psi(x,y)$, Ennola's theorem clarifies the local nature of this function's variations. When y is small, it can thus be seen that $\Psi(x,y)$ is very sensitive to irregularities in the distribution of primes. For example, it is easy to deduce from (5.20) that no continuous function of x and y is asymptotic to $\Psi(x,y)$ for $y \leqslant \sqrt{\ln x}$. Indeed, if y is a prime number, we then have, as x and y tend to infinity,

$$\frac{\Psi(x,y)}{\Psi(x,y-)} \sim \frac{\ln x}{\pi(y)\ln y} \sim \frac{\ln x}{y} \to \infty.$$

We shall return to this point in § 5.5.

5.3. Functional equations

Like most of the quantities arising in sieve problems, $\Psi(x,y)$ satisfies a functional equation.

Theorem 5.5. *For $x \geqslant 1$, $y \geqslant 1$, we have*

$$\Psi(x,y) = 1 + \sum_{p \leqslant y} \Psi\big(x/p, p\big). \tag{5.21}$$

Proof. If $n > 1$ is counted by $\Psi(x,y)$, then we can write $n = mp$ with $p = P^+(n)$. This last condition is equivalent to $P^+(m) \leqslant p$. Therefore

$$\Psi(x,y) = 1 + \sum_{p \leqslant y} \sum_{n \leqslant x,\, P^+(n)=p} 1 = 1 + \sum_{p \leqslant y} \sum_{m \leqslant x/p,\, P^+(m) \leqslant p} 1,$$

which is (5.21). □

In practice, this equation is often used to express the differences

$$\Psi(x,z) - \Psi(x,y).$$

Corollary 5.6 (Buchstab's identity). *For $x \geqslant 1$, $z \geqslant y \geqslant 1$, we have*

$$\Psi(x,y) = \Psi(x,z) - \sum_{y < p \leqslant z} \Psi(x/p, p). \tag{5.22}$$

This identity allows evaluating $\Psi(x,y)$ by induction on $\lfloor u \rfloor = \lfloor \ln x / \ln y \rfloor$. Indeed, we have $\Psi(x,y) = \lfloor x \rfloor$ if $y \geqslant x$, that is $u \leqslant 1$. Noticing then that we have $x/p \leqslant p$ in (5.22) whenever $y \geqslant \sqrt{x}$, we can write (with $z = x$), for $\sqrt{x} \leqslant y \leqslant x$,

$$\Psi(x,y) = \lfloor x \rfloor - \sum_{y < p \leqslant x} \lfloor x/p \rfloor \sim x(1 - \ln u)$$

by a straightforward application of the prime number theorem. Of course, it is then possible to feed this information back into (5.22). Selecting $z = \sqrt{x}$, we obtain, for $x^{1/3} \leqslant y \leqslant x^{1/2}$,

$$\Psi(x,y) \sim x(1-\ln 2) - \sum_{y<p\leqslant\sqrt{x}} \frac{x}{p}\Big\{1 - \ln\Big(\frac{\ln(x/p)}{\ln p}\Big)\Big\} \sim x\varrho(u),$$

with

$$\varrho(u) := 1 - \ln u + \int_2^u \ln(v-1)\frac{\mathrm{d}v}{v} \qquad (2 \leqslant u \leqslant 3).$$

It is clear that this procedure yields, for each $\varepsilon > 0$, the asymptotic formula

$$\Psi(x,y) \sim x\varrho(u) \qquad (x^\varepsilon < y \leqslant x) \tag{5.23}$$

where the function ϱ is defined by the initial condition $\varrho(u) = 1$ $(0 \leqslant u \leqslant 1)$ and the smooth version of equation (5.22), namely

$$\varrho(u) = \varrho(k) - \int_k^u \varrho(v-1)\frac{\mathrm{d}v}{v} \qquad (k < u \leqslant k+1). \tag{5.24}$$

This function was discovered by Dickman in 1930 and today bears his name. It is continuous at $u = 1$, differentiable for $u > 1$, and satisfies the *delay differential equation*

$$u\varrho'(u) + \varrho(u-1) = 0 \qquad (u > 1) \tag{5.25}$$

obtained by differentiating (5.24). The occurrence of an equation of this type, which reflects Buchstab's identity, is characteristic of sieve problems.

Theorem 5.7. *Dickman's function $\varrho(u)$ satisfies the following properties*

(i) $\varrho(u) = \displaystyle\int_{u-1}^u \varrho(v)\,\mathrm{d}v$ $\qquad (u \geqslant 1)$

(ii) $\varrho(u) > 0$ $\qquad (u \geqslant 0)$

(iii) $\varrho'(u) < 0$ $\qquad (u > 1)$

(iv) $\varrho(u) \leqslant 1/\Gamma(u+1)$ $\qquad (u \geqslant 0).$

Proof. Part (i) follows immediately from (5.25) and the initial conditions for ϱ. The two sides of (i) have the same derivative for $u > 1$ and take the same value at $u = 1$.

Let us prove (ii). Set $\tau := \inf\{u : \varrho(u) = 0\}$. If τ is finite, then $\tau > 1$ since ϱ is continuous and satisfies $\varrho(u) = 1$ for $0 \leqslant u \leqslant 1$. By (i), we can then write

$$0 = \tau\varrho(\tau) = \int_{\tau-1}^{\tau} \varrho(v)\,\mathrm{d}v.$$

The continuity of ϱ then implies, by the very definition of τ, that the right-hand side is strictly positive. This shows τ is not finite, and (ii) follows.

Part (iii) is an immediate consequence of (ii) and the functional equation (5.25).

We obtain (iv) by induction on $k := \lfloor u \rfloor$. The property holds for $k = 0$ because $\varrho(u) = 1$ $(0 \leqslant u \leqslant 1)$. If $k \geqslant 1$, we deduce from (i), (ii), (iii) and the inductive assumption that we have

$$\varrho(u) = \frac{1}{u}\int_{u-1}^{u} \varrho(v)\,\mathrm{d}v \leqslant \frac{\varrho(u-1)}{u} \leqslant \frac{1}{u\Gamma(u)} = \frac{1}{\Gamma(u+1)}. \qquad \square$$

We shall see (Corollary 5.19) that the asymptotic formula (5.23) remains true in a very large (x,y)-domain. A first attempt at an inductive use of Buchstab's identity is the object of the following theorem, which represents partial success in this direction.

Theorem 5.8. *Uniformly for $x \geqslant y \geqslant 2$, we have*

$$\Psi(x,y) = x\varrho(u) + O\Big(\frac{x}{\ln y}\Big). \tag{5.26}$$

Proof. Given the rapid decrease of $\varrho(u)$ as $u \to \infty$, it is enough to give a proof for $u \leqslant 2\ln_2 y$. Otherwise, the error term in (5.26) is of larger order than the main term, and the result follows, for example, from Theorem 5.1.

So let $\Delta(x,y)$ be the quantity implicitly defined by the relation

$$\Psi(x,y) = x\varrho(u) + \frac{x\Delta(x,y)}{\ln y}.$$

As previously noted, we have $\Delta(x,y) \ll 1$ for $y \geqslant 2$, $1 \leqslant u \leqslant 2$. Indeed, relation (5.22) with $z = x$ yields

$$\Psi(x,y) = \lfloor x \rfloor - \sum_{y<p\leqslant x} \lfloor x/p \rfloor = x(1-\ln u) + O\big(\pi(x)\big) = x\varrho(u) + O\Big(\frac{x}{\ln x}\Big).$$

We use induction on $k \leqslant 1 + 2\ln_2 y$ to show that the quantity

$$\Delta_k(y) := 1 + \sup\big\{|\Delta(X,Y)| : y \leqslant Y \leqslant X \leqslant Y^k\big\}$$

is finite and uniformly bounded. Let us hence assume that

$$\Delta_k(y) < \infty \qquad (k \geqslant 2),$$

and consider those X, Y such that $Y \geqslant y$ and $Y^2 < X \leqslant Y^{k+1}$. Applying (5.22) for (X,Y) with $z = \sqrt{X}$ and using the notation $U := (\ln X)/\ln Y$,

we obtain

$$\begin{aligned}\Psi(X,Y) &= \Psi(X,\sqrt{X}) - \sum_{Y<p\leqslant\sqrt{X}} \Psi(X/p,p)\\ &= X\Big\{\varrho(2) - \int_2^U \varrho(v-1)\,\mathrm{d}L(v)\Big\}\\ &\quad + \frac{X}{\ln Y}\Big\{\frac{2}{U}\Delta(X,\sqrt{X}) - \sum_{Y<p\leqslant\sqrt{X}} \frac{\Delta(X/p,p)\ln Y}{p\ln p}\Big\},\end{aligned} \tag{5.27}$$

with

$$L(v) := -\sum_{Y<p\leqslant X^{1/v}} \frac{1}{p} = \ln(v/U) + O\big(\mathrm{e}^{-\sqrt{\ln Y}}\big).$$

Property (iii) of Theorem 5.7 shows that ϱ is non-increasing. By partial summation, we immediately derive

$$\int_2^U \varrho(v-1)\,\mathrm{d}L(v) = \int_2^U \varrho(v-1)\frac{\mathrm{d}v}{v} + O\Big(\mathrm{e}^{-\sqrt{\ln Y}}\Big) = \varrho(2) - \varrho(U) + O\Big(\mathrm{e}^{-\sqrt{\ln Y}}\Big).$$

Furthermore, for $Y < p \leqslant \sqrt{X}$, we have

$$1 \leqslant \frac{\ln(X/p)}{\ln p} \leqslant U - 1 \leqslant k.$$

For these values of p, we can thus bound $\Delta(X/p,p)$ from above by $\Delta_k(y)-1$. Substituting in (5.27), it then follows that

$$\begin{aligned}1 + |\Delta(X,Y)| &\leqslant 1 + C_0\,\mathrm{e}^{-\frac{1}{2}\sqrt{\ln Y}} + \big(\Delta_k(y)-1\big)\Big\{\frac{2}{U} + \sum_{Y<p\leqslant\sqrt{X}} \frac{\ln Y}{p\ln p}\Big\}\\ &\leqslant \Delta_k(y)\Big(1 + C_1\,\mathrm{e}^{-\frac{1}{2}\sqrt{\ln Y}}\Big),\end{aligned}$$

by the prime number theorem. Taking the supremum over X, Y such that $Y \geqslant y$, $k < U \leqslant k+1$ and iterating, it follows that

$$\Delta_{\lfloor 2\ln_2 y\rfloor}(y) \ll \Big(1 + C_1\,\mathrm{e}^{-\frac{1}{2}\sqrt{\ln y}}\Big)^{2\ln_2 y} \ll 1,$$

which is the desired conclusion. □

The proof of Theorem 5.8 may be refined by using a smooth main term whose variations mimic Buchstab's identity, satisfied by $\Psi(x,y)$, more closely. In this way we improve the error term $O(x/\ln y)$, but we cannot evade the inconvenience, inherent to the method, of a remainder term independent of u—and thus of little worth when u is large. In this direction,

de Bruijn (1951b) chose the main term

$$\Lambda(x,y) = \begin{cases} x\displaystyle\int_{-\infty}^{\infty} \varrho(u-v)\,\mathrm{d}\big(\lfloor y^v\rfloor / y^v\big) & (x \notin \mathbb{N}^*), \\ \Lambda(x+,y) & (x \in \mathbb{N}^*). \end{cases} \tag{5.28}$$

In § 5.5, we shall meet a second heuristic justification of this definition and in the Notes we give an asymptotic expansion which describes the behavior of this function in a large (x,y)-domain. For the moment, let us be satisfied with the observation that

$$\Lambda(x,y) = \lfloor x\rfloor \qquad (x \leqslant y) \tag{5.29}$$

and

$$\Lambda(x,y) = \Lambda(x,z) - \int_y^z \Lambda\Big(\frac{x}{t},t\Big)\frac{\mathrm{d}t}{\ln t} \qquad (y \leqslant z). \tag{5.30}$$

Thus $\Lambda(x,y)$ satisfies the same initial condition as $\Psi(x,y)$ and obeys a functional equation which is the smooth version of Buchstab's identity.

Employing a method analogous to that of Theorem 5.8, de Bruijn obtained the formula

$$\Psi(x,y) = \Lambda(x,y) + O\Big(\frac{xu^2}{L_\varepsilon(y)}\Big) \qquad (x \geqslant y \geqslant 2), \tag{5.31}$$

with

$$L_\varepsilon(y) := \exp\big\{(\ln y)^{(3/5)-\varepsilon}\big\}. \tag{5.32}$$

We shall show in § 5.5 that the main term in (5.31) has order $x\varrho(u)$ whenever $y \geqslant (\ln x)^2$. From the upper bound (iv) of Theorem 5.7, we hence see that (5.31) can only yield an asymptotic formula for $\Psi(x,y)$ in the domain

$$\exp\big\{(\ln x)^{(5/8)+\varepsilon}\big\} \leqslant y \leqslant x. \tag{5.33}$$

This region may be considered as the natural limit of Buchstab's iterative method.

However, de Bruijn's result has been considerably improved by Hildebrand (1986a) using another functional equation, namely, with notation (5.1),

$$\Psi(x,y)\ln x = \int_1^x \Psi(t,y)\frac{\mathrm{d}t}{t} + \sum_{d\in S(x,y)} \Lambda(d)\Psi\Big(\frac{x}{d},y\Big). \tag{5.34}$$

Equation (5.34) has two advantages over (5.22): on the one hand, since it only involves a fixed value of the parameter y, it takes the form of an equation in a single variable, and so is easier to iterate; on the other hand, by expressing $\Psi(x,y)$ as an average of itself, it generates by iteration a

powerful regulating effect. This latter feature was actually already present, but dampened, in Buchstab's identity. The cause of this phenomenon can be traced back to the very source of the method, which consists in calculating $\Psi(x,y)$ starting from $\Psi(x,z)$ and deleting positive terms. As one might foresee, iterating Buchstab's identity produces alternating signs—and hence potential cancellations which have to be regarded as unexploited gains in the error terms.

Hildebrand established the validity of the formula

$$\Psi(x,y) = x\varrho(u)\Big\{1 + O\Big(\frac{\ln(u+1)}{\ln y}\Big)\Big\} \tag{5.35}$$

uniformly, for each $\varepsilon > 0$, in the domain

$$\exp\big\{(\ln_2 x)^{(5/3)+\varepsilon}\big\} \leqslant y \leqslant x. \tag{5.36}$$

We shall obtain this result in § 5.5 as a consequence of a theorem generalizing both (5.31) and (5.35), and based on a purely analytical method. We will thus not undertake here to deduce (5.35) from (5.34), but merely indicate how (5.34) can be established.

It actually suffices to compute in two ways the quantity

$$S := \sum_{n \in S(x,y)} \ln n.$$

On the one hand, we have, by partial summation,

$$S = \Psi(x,y)\ln x - \int_1^x \Psi(t,y)\frac{\mathrm{d}t}{t},$$

and on the other, by the convolution identity for Λ,

$$S = \sum_{n \in S(x,y)} \sum_{d \mid n} \Lambda(d) = \sum_{d \in S(x,y)} \Lambda(d)\Psi\big(x/d, y\big).$$

For another application of (5.34), see Theorem 5.27 in the Notes.

5.4. Dickman's function

Here, we intend to complete the analytic study of the Dickman function $\varrho(u)$ introduced in the previous section, and, in particular, to clarify its asymptotic behavior.

Recall that we defined $\varrho(u)$ as the solution, continuous at $u = 1$ and differentiable for $u > 1$, of the delay differential equation

$$u\varrho'(u) + \varrho(u-1) = 0 \qquad (u > 1) \tag{5.37}$$

with the initial condition $\varrho(u) = 1$ $(0 \leqslant u \leqslant 1)$. It will be convenient in the sequel to extend $\varrho(u)$ to all of $\mathbb{R}$ in such a way that (5.37) is satisfied everywhere. We hence let

$$\varrho(u) := 0 \qquad (u < 0). \tag{5.38}$$

Thus defined, $\varrho(u)$ is right-continuous and has a single discontinuity, which is a jump, at $u = 0$. It follows immediately that the k-th derivative, $\varrho^{(k)}(u)$, is defined on $\mathbb{R} \setminus \{0, 1, \dots, k\}$ and also has discontinuities of the first kind at the exceptional points $u = j$ $(0 \leqslant j \leqslant k)$. We extend it to $\mathbb{R}$ by right-continuity.

Our method for studying the Dickman function rests on the computation of the Laplace transform

$$\widehat{\varrho}(s) := \int_0^\infty \mathrm{e}^{-st} \varrho(t)\, \mathrm{d}t \tag{5.39}$$

and evaluating $\varrho(u)$ by Laplace inversion. By Theorem 5.7(iv), the integral (5.39) is absolutely convergent for all complex numbers s, and thus defines an entire function of s.

By change of variable $v = ts$ we can write, for $s \in \mathbb{R}^{+*}$,

$$s\widehat{\varrho}(s) = \int_0^\infty \mathrm{e}^{-v} \varrho(v/s)\, \mathrm{d}v, \tag{5.40}$$

from which, by (5.37), we infer that

$$\frac{\mathrm{d}}{\mathrm{d}s}\{s\widehat{\varrho}(s)\} = \frac{1}{s}\int_0^\infty \mathrm{e}^{-v} \varrho\Big(\frac{v}{s} - 1\Big)\, \mathrm{d}v = \int_0^\infty \mathrm{e}^{-ts} \varrho(t-1)\, \mathrm{d}t = \mathrm{e}^{-s}\widehat{\varrho}(s).$$

Solving this first-order differential equation yields, for a suitable constant C,

$$s\widehat{\varrho}(s) = C\, \mathrm{e}^{-J(s)}, \tag{5.41}$$

with

$$J(s) := \int_0^\infty \frac{\mathrm{e}^{-s-t}}{s+t}\, \mathrm{d}t. \tag{5.42}$$

The integral $J(s)$ is evidently continuable as a holomorphic function on $\mathbb{C} \setminus]-\infty, 0]$. The following lemma, of crucial use here, gives further information about this continuation. We set

$$I(s) := \int_0^s \frac{\mathrm{e}^t - 1}{t}\, \mathrm{d}t \qquad (s \in \mathbb{C}). \tag{5.43}$$

Lemma 5.9. *For $s \in \mathbb{C} \setminus]-\infty, 0]$, we have*

$$I(-s) + J(s) + \gamma + \log s = 0 \tag{5.44}$$

where γ denotes Euler's constant.

Proof. The classical formula $\gamma = -\Gamma'(1)$ immediately yields, after partial integration,

$$-\gamma = \int_0^{-1} \frac{e^t - 1}{t}\,dt + \int_{-1}^{-\infty} e^t \frac{dt}{t}.$$

Without altering the value of this expression, we can replace the real path $[-1, -\infty[$ in the second integral by the polygonal line made up of the segment $[-1, -s]$ and the half-line $\{-s - t : t \geqslant 0\}$. It follows that

$$-\gamma = \int_0^{-1} \frac{e^t - 1}{t}\,dt + \int_{-1}^{-s} \frac{e^t - 1}{t}\,dt + \int_{-1}^{-s} \frac{dt}{t} + \int_0^{\infty} \frac{e^{-s-t}}{s+t}\,dt.$$

Rearranging the two first integrals, we obtain (5.44). □

This result enables us to complete the calculation of $\widehat{\varrho}(s)$.

Theorem 5.10. *For all complex numbers s, we have*

$$\widehat{\varrho}(s) = e^{\gamma + I(-s)}. \tag{5.45}$$

Proof. By (5.41) and (5.44) we may write, when s is not real and negative,

$$s\widehat{\varrho}(s) = C\,e^{-J(s)} = Cse^{\gamma + I(-s)},$$

from which

$$\widehat{\varrho}(s) = C\,e^{\gamma + I(-s)}.$$

Now, on the one hand, it follows from (5.40) that

$$\lim_{s \to +\infty} s\widehat{\varrho}(s) = \varrho(0+) = 1,$$

while, on the other, (5.41) implies that

$$\lim_{s \to +\infty} s\widehat{\varrho}(s) = C.$$

Therefore, we must have $C = 1$, and the proof of (5.45) is complete. □

In Exercise 289, we indicate an arithmetic proof, resting on Theorems 5.1 and 5.8, of the formula $\widehat{\varrho}(0) = e^{\gamma}$. The value is then provided by Mertens' formula.

The regularity and the rapid decrease of $\varrho(u)$ at infinity show that the inverse Laplace integral

$$\varrho(u) = \frac{1}{2\pi i} \int_{\alpha - i\infty}^{\alpha + i\infty} \widehat{\varrho}(s) e^{us}\,ds \tag{5.46}$$

converges for all $\alpha \in \mathbb{R}$ and all $u \neq 0$.[3] In accordance with the principles of the saddle-point method,[4] it can be expected, by choosing α to be a zero of the derivative of the integrand, that the integral (5.46) as a whole will

[3] See, for example, Widder (1946), theorems II.7.3 or II.7.5

[4] See de Bruijn (1970), chap. 5.

be dominated by the contribution from a small neighborhood of the real point α. This will then allow the determination of an asymptotic formula for $\varrho(u)$ by carrying out a Taylor expansion of $\widehat{\varrho}(s)\mathrm{e}^{us}$ in the neighborhood of $s = \alpha$.

The explicit formula (5.45) immediately yields the value of α. We must choose $\alpha = -\xi(u)$ where $\xi = \xi(u)$ is the unique real, non-zero root of the equation

$$\mathrm{e}^{\xi} = 1 + u\xi \qquad (u > 0,\ u \neq 1). \tag{5.47}$$

By convention we set $\xi(1) = 0$. The following two technical lemmas will allow us to carry out the evaluation of the integral (5.46).

Lemma 5.11. *For $u \geqslant 3$, we have*

$$\xi(u) = \ln(u \ln u) + O\Big(\frac{\ln_2 u}{\ln u}\Big). \tag{5.48}$$

Proof. We trivially have $1 \ll \xi(u) \ll \ln u$. This simple first estimate yields the result by iterating (5.47). Indeed, we have

$$\begin{aligned}\xi &= \ln u + \ln(\xi + 1/u) = \ln u + \ln\big(\ln u + \ln(\xi + 1/u) + 1/u\big)\\ &= \ln(u \ln u) + O\Big(\frac{\ln(\xi + 1/u)}{\ln u}\Big).\end{aligned}$$

□

Lemma 5.12. *For $u > 1$, $s = -\xi(u) + i\tau$, $\tau \in \mathbb{R}$, we have*

$$\widehat{\varrho}(s) \ll \begin{cases} \exp\{I(\xi) - \tau^2 u/2\pi^2\} & (|\tau| \leqslant \pi)\\ \exp\{I(\xi) - u/(\pi^2 + \xi^2)\} & (|\tau| > \pi)\end{cases} \tag{5.49}$$

and

$$\widehat{\varrho}(s) = \frac{1}{s}\Big\{1 + O\Big(\frac{1 + u\xi}{s}\Big)\Big\} \qquad (|\tau| > 1 + u\xi). \tag{5.50}$$

Proof. Note, first of all, that (5.50) follows immediately from (5.45) and (5.44) in the form

$$s\widehat{\varrho}(s) = \mathrm{e}^{-J(s)}. \tag{5.51}$$

Indeed, it suffices to use the trivial bound $J(s) \ll \mathrm{e}^{-\sigma}|\tau|^{-1}$ with $\sigma = -\xi(u)$.

In order to establish (5.49), we introduce the quantity

$$H(\tau) := I(\xi) - \Re\mathrm{e}\, I(-s) = \int_0^1 \mathrm{e}^{h\xi}\,\frac{1 - \cos(h\tau)}{h}\,\mathrm{d}h.$$

When $|\tau| \leqslant \pi$, we have $1 - \cos(h\tau) \geqslant 2\tau^2 h^2/\pi^2$ for $0 \leqslant h \leqslant 1$, from which

$$H(\tau) \geqslant \frac{2\tau^2}{\pi^2}\int_0^1 h\,\mathrm{e}^{\xi h}\,\mathrm{d}h.$$

The desired conclusion then follows from the lower bound

$$\int_0^1 h\,\mathrm{e}^{h\xi}\,\mathrm{d}h \geqslant \tfrac{1}{2}\int_{1/2}^1 \mathrm{e}^{h\xi}\,\mathrm{d}h \geqslant \tfrac{1}{4}\int_0^1 \mathrm{e}^{h\xi}\,\mathrm{d}h = \tfrac{1}{4}u.$$

When $|\tau| > \pi$, we note that we can assume u to be sufficiently large since the conclusion is otherwise trivial. Taking (5.48) into account, we may hence suppose that

$$u > \pi^2 + \xi^2.$$

We can then write

$$\begin{aligned} H(\tau) &\geqslant \int_0^1 \mathrm{e}^{h\xi}\bigl\{1-\cos(h\tau)\bigr\}\,\mathrm{d}h = u - \Re\mathrm{e}\Bigl(\frac{\mathrm{e}^{\xi+i\tau}-1}{\xi+i\tau}\Bigr) \\ &= u - \Re\mathrm{e}\Bigl(\frac{u\xi\mathrm{e}^{i\tau}+\mathrm{e}^{i\tau}-1}{\xi+i\tau}\Bigr) \geqslant u\Bigl(1-\frac{\xi\cos\vartheta}{\sqrt{\pi^2+\xi^2}}\Bigr) - \frac{2}{\sqrt{\tau^2+\xi^2}}, \end{aligned}$$

with $\vartheta := \tau - \arctan(\tau/\xi)$. The factor of u in the last expression is at least equal to $\pi^2/(2\pi^2+2\xi^2)$. It follows that

$$H(\tau) \geqslant \frac{u\pi^2}{2(\pi^2+\xi^2)} - \frac{2u}{\pi^2+\xi^2} > \frac{u}{\pi^2+\xi^2}. \qquad \square$$

We are now in a position to establish the following result which provides an asymptotic formula with remainder for $\varrho(u)$ as $u \to \infty$.

Theorem 5.13 (de Bruijn; Alladi). *For $u \geqslant 1$, we have*

$$\varrho(u) = \sqrt{\frac{\xi'(u)}{2\pi}}\,\mathrm{e}^{\gamma-u\xi+I(\xi)}\Bigl\{1+O\Bigl(\frac{1}{u}\Bigr)\Bigr\}. \tag{5.52}$$

Remarks. Taking the derivative of (5.47) with respect to u, we immediately obtain $\xi'(u) = \xi/\bigl(1+u(\xi-1)\bigr)$. In particular, we have $\xi'(u) \sim 1/u$ as $u \to \infty$. Furthermore, it is useful to note that the main term in (5.52) can be transformed by means of the identity

$$u\xi - I(\xi) = \int_1^u \xi(t)\,\mathrm{d}t \qquad (u > 0). \tag{5.53}$$

Proof. Let $\delta = \delta(u) := \pi\sqrt{2\ln(u+1)/u}$ and

$$K(u) := \frac{1}{2\pi}\int_{-\delta}^{\delta} \widehat{\varrho}(s)\mathrm{e}^{us}\,\mathrm{d}\tau,$$

where $s = -\xi(u) + i\tau$. As a first step, we show that the difference

$$\varrho(u) - K(u) = \frac{1}{2\pi i}\int_{|\tau|>\delta} \widehat{\varrho}(s)\mathrm{e}^{us}\,\mathrm{d}s \tag{5.54}$$

is dominated by the error term in (5.52).

To this end, we use Lemma 5.12. The contribution to the integral (5.54) from the domain $\delta < |\tau| \leqslant \pi$ is

$$\ll \mathrm{e}^{-u\xi+I(\xi)} \int_{\delta}^{\infty} \mathrm{e}^{-\tau^2 u/2\pi^2}\,\mathrm{d}\tau \ll \frac{\mathrm{e}^{-u\xi+I(\xi)}}{\sqrt{u}} \int_{\ln(u+1)}^{\infty} \mathrm{e}^{-t}\frac{\mathrm{d}t}{\sqrt{t}} \ll \frac{\mathrm{e}^{-u\xi+I(\xi)}}{u^{3/2}}.$$

Similarly, the contribution of the domain $\pi < |\tau| \leqslant 1+u\xi$ is

$$\ll (1+u\xi)\exp\big\{-u\xi+I(\xi)-u/(\pi^2+\xi^2)\big\}.$$

These bounds are clearly acceptable.

Finally, using (5.50), we can evaluate the contribution from the range $|\tau| > 1+u\xi$. We have

$$\int_{1+u\xi}^{\infty} \widehat{\varrho}(s)\mathrm{e}^{us}\,\mathrm{d}\tau = \int_{1+u\xi}^{\infty} \frac{\mathrm{e}^{-u\xi+i\tau u}}{\tau}\Big\{1+O\Big(\frac{u\xi}{\tau}\Big)\Big\}\,\mathrm{d}\tau \ll \mathrm{e}^{-u\xi}$$

where we appealed to the second mean value theorem in order to handle the term involving $1/\tau$. Since $I(\xi) \sim u$, this upper bound is also of the required order of magnitude.

It remains to evaluate $K(u)$. For this, we consider the Taylor expansion of $\widehat{\varrho}(s)$ in the neighborhood of $\tau = 0$. Noting that, for $\Re\mathrm{e}\, s = -\xi$ and $k \geqslant 1$, we have

$$\big|I^{(k)}(s)\big| = \Big|\int_0^1 h^{k-1}\mathrm{e}^{hs}\,\mathrm{d}h\Big| \leqslant I^{(k)}(\xi) \leqslant I'(\xi) = u,$$

this yields

$$I(\xi-i\tau) = I(\xi) - i\tau u - \tfrac{1}{2}\tau^2 I''(\xi) - \tfrac{1}{6}i\tau^3 I'''(\xi) + O(u\tau^4).$$

We can thus write, for $\tau \ll u^{-1/4}$,

$$\exp\big\{I(\xi-i\tau)+us\big\} = \exp\big\{I(\xi)-u\xi-\tfrac{1}{2}\tau^2 I''(\xi)\big\}\big(1+h(u)\big),$$

with

$$h(u) = \exp\big\{-\tfrac{1}{6}i\tau^3 I'''(\xi)+O(u\tau^4)\big\} - 1 = -\tfrac{1}{6}i\tau^3 I'''(\xi) + O(u\tau^4+u^2\tau^6).$$

Substituting in the integral defining $K(u)$ and noting that the contribution of the term in τ^3 vanishes by symmetry, we obtain

$$K(u) = \frac{\mathrm{e}^{-u\xi+I(\xi)}}{2\pi}\int_{-\delta}^{\delta} \mathrm{e}^{-\tau^2 I''(\xi)/2}\big\{1+O(u\tau^4+u^2\tau^6)\big\}\,\mathrm{d}\tau. \tag{5.55}$$

The contribution of the error terms is

$$\ll \frac{\mathrm{e}^{-u\xi+I(\xi)}}{u\sqrt{I''(\xi)}}.$$

That of the main term is evaluated by extending the integral to infinity and bounding above the contribution of the range $|\tau| > \delta$. Since

$$I''(\xi) = 1/\xi'(u) = u - (u-1)/\xi, \tag{5.56}$$

this yields

$$\int_{-\delta}^{\delta} \mathrm{e}^{-\tau^2 I''(\xi)/2}\,\mathrm{d}\tau = \sqrt{\frac{2\pi}{I''(\xi)}}\Big\{1+O\Big(\frac{1}{u}\Big)\Big\}.$$

Substituting this estimate in (5.55) and taking account of (5.56), we finally obtain the stated formula (5.52). □

Corollary 5.14. *For any integer $k \geqslant 0$ and any real number $u_0 > 1$, we have*

$$\varrho^{(k)}(u) = (-1)^k \xi(u)^k \varrho(u)\big\{1+O(1/u)\big\} \qquad (u \geqslant u_0). \tag{5.57}$$

Proof. Differentiating the functional equation (5.37) k times, we obtain

$$u\varrho^{(k+1)}(u) = -\varrho^{(k)}(u-1) - k\varrho^{(k)}(u) \qquad (u>1). \tag{5.58}$$

This immediately implies by induction on k that

$$(-1)^k \varrho^{(k)}(u) > 0 \qquad (u>1). \tag{5.59}$$

Therefore, (5.57) holds when u is bounded.

Relation (5.56) shows that

$$\xi'(u) \sim 1/u, \qquad \xi''(u) \sim -1/u^2 \qquad (u\to\infty). \tag{5.60}$$

We deduce from this that, for sufficiently large u,

$$\xi'(u-1) = \xi'(u)\big\{1+O(1/u)\big\}, \qquad \int_{u-1}^{u} \xi(t)\,\mathrm{d}t = \xi(u)+O(1/u). \tag{5.61}$$

Substituting in the asymptotic formula (5.52) written with (5.53) taken into account, we obtain

$$\varrho(u-1) = \varrho(u)\mathrm{e}^{\xi(u)}\big\{1+O(1/u)\big\} = u\xi(u)\varrho(u)\big\{1+O(1/u)\big\}. \tag{5.62}$$

This allows us to prove (5.57) by induction on k. Assuming that the formula is satisfied up to order k, we can write, by (5.58), (5.60)—in the form $\xi(u-1) = \xi(u)\big(1+O(1/u)\big)$—and (5.62),

$$\begin{aligned} u\varrho^{(k+1)}(u) &= (-1)^k \xi(u)^k\big\{-u\xi(u)\varrho(u) + O\big(\xi(u)\varrho(u)\big)\big\}\big\{1+O(1/u)\big\} \\ &= (-1)^{k+1}\xi(u)^{k+1} u\varrho(u)\big\{1+O(1/u)\big\}. \end{aligned}$$

The proof of (5.57) is thus complete. □

Corollary 5.15. *Uniformly for $0 \leqslant v \leqslant u$, we have*

$$\varrho(u-v) \ll \varrho(u)\mathrm{e}^{v\xi(u)}. \tag{5.63}$$

Proof. If $u-1 < v \leqslant u$, we have $\varrho(u-v) = 1$. In this case, the bound (5.63) follows from (5.52), since $I(\xi) \sim u$. If $0 \leqslant v \leqslant u-1$, we deduce from (5.52) and (5.53) that

$$\frac{\varrho(u-v)}{\varrho(u)} \ll \frac{u}{u-v} \exp\Big\{\int_0^v \xi(u-t)\,\mathrm{d}t\Big\}.$$

By (5.60), we have $\xi(u-t) \leqslant \xi(u) - ct/u$ for a suitable positive absolute constant c. The required result then follows from the elementary inequality

$$\frac{cv^2}{2u} \geqslant \ln\left(\frac{u}{u-v}\right) + O(1) \qquad (0 \leqslant v \leqslant u-1). \tag{5.64}$$

□

5.5. Approximation to $\Psi(x,y)$ by the saddle-point method

For any $\alpha > 0$, Perron's formula yields the expression

$$\Psi(x,y) = \frac{1}{2\pi i}\int_{\alpha-i\infty}^{\alpha+i\infty} \zeta(s,y)x^s \frac{\mathrm{d}s}{s} \qquad (x \notin \mathbb{N}^*). \tag{5.65}$$

We shall see that the saddle-point method, employed in the previous section to evaluate Dickman's function, also works for the integral (5.65).

The optimal choice for the parameter α is the (unique) solution $\alpha(x,y)$ of the transcendental equation

$$\frac{-\zeta'}{\zeta}(\alpha,y) = \sum_{p\leqslant y} \frac{\ln p}{p^\alpha - 1} = \ln x. \tag{5.66}$$

As we have seen in the course of the proof of Theorem 5.2, the quantity $\alpha(x,y)$ can equally well be regarded as the optimal value of the parameter in Rankin's method, and formula (5.13) provides an asymptotic formula for $\alpha(x,y)$ as $y \to \infty$. However, the implicit nature of $\alpha(x,y)$ leads one to expect a certain lack of flexibility in the asymptotic formula resulting from such a treatment. The main aim of this section is to establish that, without loss of precision, $\alpha(x,y)$ can be replaced by an explicit approximation in a suitable subregion for (x,y). The result is an extension of both formulae (5.31) and (5.35), due to de Bruijn and Hildebrand respectively.

The explicit approximation to $\alpha(x,y)$ is suggested in a natural way by the following lemma. We set

$$Y_\varepsilon := \exp\big\{(\ln y)^{(3/2)-\varepsilon}\big\}$$

and recall the notation

$$u := \frac{\ln x}{\ln y}, \qquad L_\varepsilon(y) := \exp\big\{(\ln y)^{(3/5)-\varepsilon}\big\},$$

which is used throughout this section. For the purpose of later applications, we give a formulation more general than is presently necessary.

Lemma 5.16. *For any $\varepsilon \in]0, \frac{1}{2}[$, there exists some real number $y_0 = y_0(\varepsilon)$ such that, uniformly under the conditions*

$$y \geqslant y_0(\varepsilon), \qquad \sigma \geqslant 1 - \frac{1}{(\ln y)^{2/5+\varepsilon} + \mathcal{L}_\tau^{2/3+\varepsilon}}, \qquad |\tau| \leqslant Y_{3\varepsilon}, \tag{5.67}$$

where we have put $\mathcal{L}_\tau := \ln(2+|\tau|)$, we have

$$\zeta(s,y) = \zeta(s)(s-1)\ln y\, \widehat{\varrho}\big((s-1)\ln y\big)\Big\{1+O\Big(\frac{1}{L_\varepsilon(y)} + y^{-1/\mathcal{L}_\tau^{2/3+\varepsilon}}\Big)\Big\}. \tag{5.68}$$

Proof. Observe first that, under the above hypotheses, we have

$$-\frac{\zeta'}{\zeta}(s,y) = \sum_{n\leqslant y} \frac{\Lambda(n)}{n^s} + O\big(y^{\frac{1}{2}-\sigma}\big). \tag{5.69}$$

Indeed, the error term has absolute value at most equal to

$$\sum_{\substack{n>y\\ P^+(n)\leqslant y}} \frac{\Lambda(n)}{n^\sigma} = \sum_{p\leqslant y} \sum_{\nu > \frac{\ln y}{\ln p}} \frac{\ln p}{p^{\nu\sigma}} \ll \sum_{p\leqslant \sqrt{y}} \frac{\ln p}{y^\sigma} + \sum_{\sqrt{y}<p\leqslant y} \frac{\ln p}{p^{2\sigma}} \ll y^{\frac{1}{2}-\sigma}.$$

For $\sigma \leqslant 1$, the effective Perron formula (Corollary II.2.4) allows us to write the main term of (5.69) in the form

$$\frac{-1}{2\pi i}\int_{\kappa-iT}^{\kappa+iT} \frac{\zeta'}{\zeta}(s+w)\frac{y^w}{w}\,\mathrm{d}w + O\Big(\frac{y^{1-\sigma}\ln^3 y}{T} + \frac{\ln y}{y^\sigma}\Big) \tag{5.70}$$

with $\kappa := 1-\sigma+1/\ln y$, $T := (|\tau|+1)L_\varepsilon(y)^3$. In order to estimate the w-integral, we move the segment of integration towards the left as far as

$$-\eta := 1-\sigma-1/(\ln T)^{2/3+\varepsilon}.$$

The real part of the argument of $-\zeta'/\zeta$ remains constantly greater than or equal to $\sigma-\eta$, hence the shifted segment stays in the interior of the Korobov–Vinogradov zero-free region for $\zeta(s)$.[5] Thus, the segment of integration has crossed exactly two poles of the integrand, namely $w=0$ and $w=1-s$. By the residue theorem, the integral in (5.70) equals

$$-\frac{\zeta'}{\zeta}(s) + \frac{y^{1-s}}{1-s} - \frac{1}{2\pi i}\int_{\mathcal{W}} \frac{\zeta'}{\zeta}(s+w)\frac{y^w}{w}\,\mathrm{d}w \tag{5.71}$$

where $\mathcal{W}$ denotes the polygonal line joining the points $\kappa \pm iT$, passing through $-\eta \pm iT$, and traced out clockwise. When $w \in \mathcal{W}$, we have[6]

$$-\frac{\zeta'}{\zeta}(s+w) \ll \ln T \ll (\ln y)^2.$$

[5] See the Notes for Chapter II.3.

[6] See the Notes on § II.3.10.

Hence the contribution from the vertical segment of $\mathcal{W}$ to the integral in (5.71) is

$$\ll y^{-\eta}\int_0^T \frac{(\ln y)^2}{\eta+t}\,\mathrm{d}t \ll y^{1-\sigma}(\ln y)^2\ln(T/\eta)\,y^{-1/(\ln T)^{2/3+\varepsilon}}$$
$$\ll (\ln y)^4\{L_\varepsilon(y)^{-2}+y^{-1/\mathcal{L}_\tau^{2/3+2\varepsilon/3}}\}.$$

That of the horizontal segments can be bounded by

$$\ll y^{1-\sigma}T^{-1}(\ln y)^2 \ll y^{1-\sigma}L_\varepsilon(y)^{-2}.$$

Taking account of (5.69), we finally obtain, for $\sigma \leqslant 1$,

$$\frac{\zeta'}{\zeta}(s,y)=\frac{\zeta'}{\zeta}(s)+\frac{y^{1-s}}{s-1}+O\Big(y^{1-\sigma}\Big\{\frac{1}{L_\varepsilon(y)^2}+y^{-1/\mathcal{L}_\tau^{2/3+2\varepsilon/3}}\Big\}\Big). \tag{5.72}$$

When $\sigma > 1$, an integration by parts shows, in view of (5.69), that this relation continues to be valid up to multiplying the error term by σ—we omit the details which are standard. Assuming temporarily that $\tau \neq 0$ and integrating (5.72) thus modified along the half-line $\{s+t : t \geqslant 0\}$, we obtain

$$\zeta(s,y)=\zeta(s)\mathrm{e}^{-J((s-1)\ln y)}\Big\{1+O\Big(\frac{1}{L_\varepsilon(y)}+y^{-1/\mathcal{L}_\tau^{2/3+\varepsilon}}\Big)\Big\}$$

with the notation (5.42) for $J(s)$. The stated result therefore follows from the formula $s\widehat{\varrho}(s)=\mathrm{e}^{-J(s)}$—the case $\tau=0$ being dealt with by passing to the limit. □

The fact that the inverse Laplace integral for $\widehat{\varrho}(s)$ can be satisfactorily evaluated on the line $\sigma=-\xi(u)$ and the occurrence of this same function in the approximation to $\zeta(s,y)$ are two strong heuristic reasons for choosing the abscissa α in the Perron integral (5.65) in such a way that

$$(\alpha-1)\ln y=-\xi(u),$$

that is $\alpha=\alpha_0$, with

$$\alpha_0 := 1-\frac{\xi(u)}{\ln y}. \tag{5.73}$$

It is actually easy to verify, using estimates for the function[7] $\varphi_y(\sigma)$ resulting from the prime number theorem—cf. Exercise 288—that α_0 is an excellent approximation to $\alpha(x,y)$ in a large domain for x,y: we have

$$\alpha(x,y)=\alpha_0+O\Big(\frac{1}{L_\varepsilon(y)}+\frac{1}{(\ln x)\ln y}\Big) \tag{5.74}$$

uniformly for $x \geqslant x_0(\varepsilon)$, $(\ln x)^{1+\varepsilon} \leqslant y \leqslant x$.

In order that Lemma 5.16 be applicable with $\sigma=\alpha_0$ as integration abscissa and with relative error term $1+O(1/L_\varepsilon(y))$, it is necessary that

[7] Defined in (5.11).

$\xi(u) \leqslant (\ln y)^{3/5-\varepsilon}$ for some suitable $\varepsilon > 0$, that is, taking account of Lemma 5.11, that x and y belong to the region of the plane defined for some fixed real number $\varepsilon > 0$ by

$$(H_\varepsilon) \qquad x > x_0(\varepsilon), \qquad \exp\{(\ln_2 x)^{(5/3)+\varepsilon}\} \leqslant y \leqslant x.$$

Under this assumption, Lemma 5.16 hence suggests that the quantity

$$(5.75) \qquad \frac{1}{2\pi i}\int_{\alpha_0-i\infty}^{\alpha_0+i\infty} \zeta(s)(s-1)\ln y\,\widehat{\varrho}\big((s-1)\ln y\big)x^s\frac{\mathrm{d}s}{s}$$

is a good approximation to $\Psi(x,y)$. The change of variables $(s-1)\ln y \mapsto s$ allows us to rewrite expression (5.75) as

$$(5.76) \qquad \frac{x}{2\pi i}\int_{-\xi(u)-i\infty}^{-\xi(u)+i\infty} \widehat{\varrho}(s)\zeta\Big(1+\frac{s}{\ln y}\Big)\frac{s}{s+\ln y}\mathrm{e}^{us}\,\mathrm{d}s.$$

Now, it is easily verified that, for all complex numbers s, we have

$$(5.77) \qquad \frac{s}{s+\ln y}\zeta\Big(1+\frac{s}{\ln y}\Big) = \int_{-\infty}^{\infty}\mathrm{e}^{-st}\,\mathrm{d}\Big(\frac{\lfloor y^t\rfloor}{y^t}\Big).$$

The convolution theorem then shows that the factor multiplying x in (5.76) is the inverse Laplace integral for

$$\int_{-\infty}^{\infty}\varrho(u-v)\,\mathrm{d}\big(\lfloor y^v\rfloor/y^v\big) = \Lambda(x,y)/x.$$

Thus, in the context of a saddle-point method analysis, de Bruijn's function $\Lambda(x,y)$ appears as the natural approximation to $\Psi(x,y)$. Actually, the above heuristic considerations can be regarded as a the outline of a genuine argument.

We are now in position to state the main result of this section.

Theorem 5.17 (Saias, 1989). *Let $\varepsilon > 0$. For (x,y) in the domain (H_ε), we have*

$$(5.78) \qquad \Psi(x,y) = \Lambda(x,y)\Big\{1+O\Big(\frac{1}{L_\varepsilon(y)}\Big)\Big\}.$$

Prior to embarking on the proof, let us show how this theorem implies Hildebrand's asymptotic formula (5.35).

Lemma 5.18. *Let $\varepsilon > 0$. Under the conditions*

$$(5.79) \qquad x > x_0(\varepsilon), \qquad (\ln x)^{1+\varepsilon} \leqslant y \leqslant x,$$

we have

$$(5.80) \qquad \Lambda(x,y) = x\varrho(u)\Big\{1+O\Big(\frac{\ln(u+1)}{\ln y}\Big)\Big\}.$$

Proof. By partial summation, we can write

$$\Lambda(x,y) = x\varrho(u) - \langle x\rangle - x\int_0^u \varrho'(u-v)\frac{\langle y^v\rangle}{y^v}\,\mathrm{d}v. \tag{5.81}$$

By Corollaries 5.14 and 5.15, we have, for all v with $0 \leqslant v \leqslant u$,

$$\varrho'(u-v) \ll \xi(u)\varrho(u-v) \ll \ln(u+1)\varrho(u)\mathrm{e}^{\xi(u)v}.$$

Now, in the domain (5.79), we have

$$\xi(u) \leqslant \ln_2 x + O(1) \leqslant \big(1 - \tfrac{1}{2}\varepsilon\big)\ln y.$$

This implies

$$\begin{aligned}\int_0^u \frac{\varrho'(u-v)}{y^v}\,\mathrm{d}v &\ll \ln(u+1)\varrho(u)\int_0^u \Big(\frac{\mathrm{e}^{\xi(u)}}{y}\Big)^v\,\mathrm{d}v\\ &\ll \frac{\varrho(u)\ln(u+1)}{\ln y},\end{aligned} \tag{5.82}$$

and therefore furnishes the desired conclusion. □

Corollary 5.19 (Hildebrand). *Let $\varepsilon > 0$. Uniformly under the conditions*

$$x \geqslant 3, \qquad \exp\big\{(\ln_2 x)^{(5/3)+\varepsilon}\big\} \leqslant y \leqslant x,$$

we have

$$\Psi(x,y) = x\varrho(u)\Big\{1 + O\Big(\frac{\ln(u+1)}{\ln y}\Big)\Big\}. \tag{5.83}$$

Proof. When $x > x_0(\varepsilon)$, this follows immediately from (5.78) and (5.80). Since $\varrho(u) > 0$ for $u \geqslant 1$, the conclusion is trivial for $3 \leqslant x \leqslant x_0(\varepsilon)$. □

Of course, the information contained in estimate (5.78) is much more precise than that of formula (5.83). In particular, an asymptotic expansion for $\Psi(x,y)$ in terms of powers of $\ln(u+1)/\ln y$ can be deduced from it—see the Notes.

The first step in the proof of Theorem 5.17 consists in truncating the Perron integral in such a way that Lemma 5.16 can be used to approximate the integrand. We state the corresponding result below.

Lemma 5.20. *Let $\varepsilon > 0$. Set $T := L_{\varepsilon/2}(y)$. For x, y in (H_ε), we have*

$$\Psi(x,y) = \frac{1}{2\pi i}\int_{\alpha_0 - iT^2}^{\alpha_0 + iT^2} \zeta(s,y)\frac{x^s}{s}\,\mathrm{d}s + O\Big(\frac{x\varrho(u)}{L_\varepsilon(y)}\Big). \tag{5.84}$$

Proof. We appeal to Theorem II.2.3. Denoting by R the error term in (5.84), we have

$$R \ll x^{\alpha_0}\sum_{P^+(n)\leqslant y}\frac{n^{-\alpha_0}}{1+T^2|\ln(x/n)|} \ll \frac{x^{\alpha_0}\zeta(\alpha_0,y)}{T} + R_T$$

with $R_T := \Psi(x+x/T,y) - \Psi(x-x/T,y)$. When u is not too large, we use the trivial upper bound

$$R_T \ll x/T.$$

Otherwise, we proceed as in Exercise 171, p. 228, introducing the weight function

$$w(t) := \frac{1}{2\pi}\Big(\frac{\sin\big(tT/2\big)}{tT/2}\Big)^2 \qquad (t \in \mathbb{R})$$

with Fourier transform

$$\widehat{w}(\tau) := \int_{-\infty}^{\infty} w(t)\mathrm{e}^{-it\tau}\,\mathrm{d}t = \frac{1}{T}\Big(1-\Big|\frac{\tau}{T}\Big|\Big)^+.$$

Then we have

$$R_T \ll \sum_{P^+(n)\leqslant y}\Big(\frac{x}{n}\Big)^{\alpha_0} w\Big(\ln\frac{x}{n}\Big) = \frac{1}{2\pi i}\int_{\alpha_0-iT}^{\alpha_0+iT}\zeta(s,y)x^s\widehat{w}(\tau)\,\mathrm{d}s.$$

Hence we can finally write

$$R \ll \frac{x^{\alpha_0}\zeta(\alpha_0,y)}{\sqrt{T}} + \min\Big\{\frac{x}{T},\, x^{\alpha_0}\max_{\sqrt{T}\leqslant|\tau|\leqslant T}\big|\zeta(\alpha_0+i\tau,y)\big|\Big\}. \tag{5.85}$$

By Lemma 5.16 and Theorem 5.13, we have

$$\begin{aligned} x^{\alpha_0}\zeta(\alpha_0,y) &\sim x\,\mathrm{e}^{-u\xi(u)}\zeta\Big(1-\frac{\xi(u)}{\ln y}\Big)\big(-\xi(u)\big)\widehat{\varrho}\big(-\xi(u)\big) \\ &\ll x\ln y\sqrt{u}\,\varrho(u). \end{aligned} \tag{5.86}$$

The first term on the right-hand side of (5.85) is therefore acceptable. The same holds for the second if $u \leqslant 2\ln L_\varepsilon(y)$. Indeed, we then have

$$\frac{1}{T} \ll \frac{u^{-2u}}{\sqrt{T}} \ll \frac{\varrho(u)}{L_\varepsilon(y)}.$$

If $u > 2\ln L_\varepsilon(y)$, we can use Lemma 5.16 (with $\frac{1}{2}\varepsilon$ in place of ε) and Lemma 5.12. For $\sqrt{T} < |\tau| \leqslant T$, $s = \alpha_0 + i\tau$, we have

$$\zeta(s,y) \ll \zeta(s)(s-1)\ln y\,\widehat{\varrho}\big((s-1)\ln y\big) \ll \zeta(s) \ll \ln T$$

since $|\tau|\ln y \geqslant \sqrt{T} \geqslant 1+u\xi(u)$. By Theorem 5.13, we therefore deduce from the above that the second term on the right-hand side of (5.85) is

$$\ll x^{\alpha_0}\ln T \ll x\mathrm{e}^{-u\xi(u)}\ln y \ll x\varrho(u)/L_\varepsilon(y)$$

where we have used the fact that $I(\xi) \sim u$ as $u \to \infty$. This finishes the proof of Lemma 5.20. □

We are now in a position to complete the proof of Theorem 5.17. The first step consists in replacing $\zeta(s,y)$ by its smooth approximation in the integral of (5.84). Applying Lemma 5.16 with $\frac{1}{2}\varepsilon$ instead of ε, we see that the error introduced by this process is at most

$$\ll \frac{x^{\alpha_0}\zeta(\alpha_0,y)}{T}\int_0^{T^2}\frac{\mathrm{d}\tau}{|\alpha_0+i\tau|}$$

with $T = L_{\varepsilon/2}(y)$. From (5.86), we see that this upper bound is of the required order of magnitude.

By carrying out the change of variables $(s-1)\ln y \mapsto s$, we can rewrite the new main term in the form

$$\begin{aligned} P &:= \frac{x}{2\pi i}\int_{-\xi(u)-iT^2\ln y}^{-\xi(u)+iT^2\ln y}\widehat{\varrho}(s)\zeta\Big(1+\frac{s}{\ln y}\Big)\frac{s}{s+\ln y}\mathrm{e}^{us}\,\mathrm{d}s \\ &= \frac{x}{2\pi i}\int_{-\xi(u)-iT^2\ln y}^{-\xi(u)+iT^2\ln y}\widehat{\lambda}_y(s)\mathrm{e}^{us}\,\mathrm{d}s \end{aligned}$$

where we have set

$$\lambda_y(u) := \int_{-\infty}^{\infty}\varrho(u-v)\,\mathrm{d}\big(\lfloor y^v\rfloor/y^v\big) \qquad (y^u\notin\mathbb{N}^*) \tag{5.87}$$

with the convention that $\lambda_y(u)=\lambda_y(u+)$ if $y^u\in\mathbb{N}^*$, so that $\Lambda(x,y)=x\lambda_y(u)$ holds identically.

Splitting the integral of (5.87) according to whether $v\leqslant u/2$ or $v>u/2$, we immediately see that the bound

$$\lambda_y(u) \ll \varrho(u/2)+y^{-u/2} \tag{5.88}$$

holds uniformly for $u\geqslant 0$, $y\geqslant 2$. This implies that the inverse Laplace integral

$$\lambda_y(u) = \frac{1}{2\pi i}\int_{-\xi(u)-i\infty}^{-\xi(u)+i\infty}\widehat{\lambda}_y(s)\mathrm{e}^{us}\,\mathrm{d}s \tag{5.89}$$

converges whenever $\xi(u)<\frac{1}{2}\ln y$ and $y^u\notin\mathbb{N}^*$, and therefore certainly in the domain (H_ε) excluding pairs (u,y) such that $y^u\in\mathbb{N}^*$. When $x=y^u\in\mathbb{N}^*$, the integral (5.89) converges in principal value to

$$\tfrac{1}{2}\big\{\lambda_y(u)+\lambda_y(u-)\big\} = \lambda_y(u)+\frac{1}{2x}.$$

We may therefore assume in the sequel that x is a half-integer. It follows that

$$P = \Lambda(x,y)+O\Big(x\int_{\substack{\sigma=-\xi(u)\\ |\tau|>T^2\ln y}}\widehat{\lambda}_y(s)\mathrm{e}^{us}\,\mathrm{d}s\Big). \tag{5.90}$$

When $s=-\xi(u)+i\tau$, $|\tau|>T^2$, we have by Lemma 5.12 that

$$s\widehat{\varrho}(s)=1+O\Big(\frac{1+u\xi}{s}\Big) \tag{5.91}$$

and it results from the usual estimates for the zeta function that

$$\frac{1}{s+\ln y}\zeta\Big(1+\frac{s}{\ln y}\Big)\ll|s|^{-1/2}. \tag{5.92}$$

The error term in (5.90) is thus

$$\begin{aligned}&\ll x\int_{\substack{\sigma=-\xi(u)\\ |\tau|>T^2\ln y}}\zeta\Big(1+\frac{s}{\ln y}\Big)\frac{\mathrm{e}^{us}}{s+\ln y}\,\mathrm{d}s+O\Big(x\frac{\mathrm{e}^{-u\xi(u)}(1+u\xi(u))}{T\sqrt{\ln y}}\Big)\\ &=\int_{\substack{\sigma=\alpha_0\\ |\tau|>T^2}}\zeta(s)\frac{x^s}{s}\,\mathrm{d}s+O\Big(\frac{x\varrho(u)}{L_\varepsilon(y)}\Big).\end{aligned}$$

In order to estimate the last integral, we use the approximation to $\zeta(s)$ by the partial sum of the series in the form of Corollary II.3.5, namely

$$\zeta(s)=\sum_{n\leqslant|\tau|}\frac{1}{n^s}-\frac{|\tau|^{1-s}}{1-s}+O\Big(\frac{1}{|\tau|^\sigma}\Big)=\sum_{n\leqslant|\tau|}\frac{1}{n^s}+O\Big(\frac{1}{|\tau|^\sigma}\Big).$$

It follows that

$$\int_{\substack{\sigma=\alpha_0\\ |\tau|>T^2}}\zeta(s)x^s\frac{\mathrm{d}s}{s}=\sum_{n\geqslant1}\int_{|\tau|\geqslant\max(n,T^2)}\Big(\frac{x}{n}\Big)^s\frac{\mathrm{d}s}{s}+O\Big(\frac{x^{\alpha_0}}{T}\Big).$$

This last term is clearly acceptable. Formula (2.7) allows us to estimate the general term of the sum over n by

$$\ll\Big(\frac{x}{n}\Big)^{\alpha_0}\frac{1}{1+(n+T^2)|\ln(x/n)|}.$$

Considering separately the two cases obtained in comparing $|x-n|$ to $x^{3/4}$, we therefore obtain the result that the sum over n is

$$\ll\sum_{n\geqslant1}\frac{x^{\alpha_0}}{n^{3/2}+T^2}+x^{3/4}\ll\frac{x\varrho(u)}{T}.$$

This shows that the error term in (5.90) is certainly of the required order, and thus completes the proof of Theorem 5.17.

When condition (H_ε) is not fulfilled, the saddle-point method is still efficient, provided that one chooses the theoretical abscissa of integration $\sigma=\alpha(x,y)$ as defined by (5.66). This development was carried out by

Hildebrand and the author (1986), who obtain the following result valid for $x \geqslant y \geqslant 2$. We recall the notation

$$\varphi_y(\sigma) = \sum_{p \leqslant y} \frac{\ln p}{p^\sigma - 1}, \qquad \varphi_y'(\sigma) = \frac{\mathrm{d}\varphi_y(\sigma)}{\mathrm{d}\sigma} = -\sum_{p \leqslant y} \frac{p^\sigma (\ln p)^2}{(p^\sigma - 1)^2}.$$

Theorem 5.21 (Hildebrand–Tenenbaum). *Uniformly for $x \geqslant y \geqslant 2$, we have*

$$\Psi(x,y) = \frac{x^\alpha \zeta(\alpha, y)}{\alpha \sqrt{2\pi |\varphi_y'(\alpha)|}} \Big\{ 1 + O\Big(\frac{1}{u} + \frac{\ln y}{y} \Big) \Big\}, \tag{5.93}$$

$$|\varphi_y'(\alpha)| = \Big(1 + \frac{\ln x}{y} \Big) \ln x \ln y \Big\{ 1 + O\Big(\frac{1}{\ln(u+1)} + \frac{1}{\ln y} \Big) \Big\}. \tag{5.94}$$

Furthermore, if $0 < \varepsilon < \frac{1}{2}$ and $y \geqslant (\ln x)^{1+\varepsilon}$, we have

$$\Psi(x,y) = x\varrho(u) \exp\Big\{ O\Big(\frac{\ln(u+1)}{\ln y} + \frac{u}{L_\varepsilon(y)} \Big) \Big\}. \tag{5.95}$$

We shall not prove this result here, and restrict ourselves to some comments.

First of all, let us observe that estimate (5.13) for α and formula (5.94) imply

$$\ln y \ll \alpha \sqrt{2\pi |\varphi_y'(\alpha)|} \sim \ln\Big(1 + \frac{y}{\ln x} \Big) \sqrt{2\pi u \Big(1 + \frac{\ln x}{y} \Big)} \ll \sqrt{\frac{y}{\ln y}}. \tag{5.96}$$

This allows us to compare Rankin's upper bound with the true order of magnitude of $\Psi(x,y)$. Although it never provides the exact order, Rankin's method is remarkably efficient: when, for instance, $y = \ln x$, every estimate available in the literature implies an error factor $\gg \exp\{y^{1+o(1)}\}$.

Next, we note that (5.95), which is actually proved using a slight strengthening of Theorem 5.8 for small values of u, strictly contains Corollary 5.19.

Finally, we draw the reader's attention to the interest of a formula like (5.93), depending on an implicit parameter such as $\alpha(x,y)$. As we have already seen, inserting an approximation for α yields a genuine asymptotic formula. Moreover, there exists an application of a different type, resting on the fact that small variations of $\alpha(x,y)$ are relatively easy to study. This allows an investigation of the *local behavior* of $\Psi(x,y)$, even in regions where the global behavior is not completely understood. The following result, characteristic of the method, can be obtained by an immediate extension of theorem 3 of Hildebrand & Tenenbaum (1986).

Theorem 5.22. *Uniformly for $x \geqslant y \geqslant 2$, $c \geqslant 1$ and with $t := (\ln c)/\ln y$, we have*

$$\Psi(cx,y) = \Psi(x,y)c^{\alpha(x,y)}\Big\{1 + O\Big((t^2+1)\Big(\frac{1}{u} + \frac{\ln y}{y}\Big)\Big)\Big\}. \tag{5.97}$$

Thus, for example, we have

$$\Psi(2x,y) \sim \Psi(x,y) \Leftrightarrow y \leqslant (\ln x)^{1+o(1)}$$

and

$$\Psi(2x,y) \sim 2\Psi(x,y) \Leftrightarrow (\ln y)/\ln_2 x \to \infty.$$

For another application of Theorem 5.22 see Exercise 296.

Estimate (5.93) also provides information on the local behavior with respect to the variable y. It is, for example, easy to deduce from (5.93) and the formula (5.13) for $\alpha(x,y)$ that

$$\Psi(x,y)/\Psi(x,y-) \sim (\ln x)/y \tag{5.98}$$

when y is a prime number with $y \leqslant (\ln x)^{1-\varepsilon}$. Therefore, there does not exist a continuous function asymptotic to $\Psi(x,y)$ in this region. Hildebrand (1986f) has shown that a continuous approximation cannot be "too" precise in the range $y \leqslant (\ln x)^{2-\varepsilon}$.

The following theorem gives a uniform one-sided bound for the local behavior of $\Psi(x,y)$. It is an easy consequence of (5.93).

Theorem 5.23. *Uniformly for $x \geqslant y \geqslant 2$, $c \geqslant 1$, we have*

$$\Psi(cx,y) \leqslant c^{\alpha}\Psi(x,y)\Big\{1 + O\Big(\frac{1}{u} + \frac{\ln y}{y}\Big)\Big\}. \tag{5.99}$$

Proof. Set $\alpha_1 := \alpha(cx,y)$. Then $\alpha_1 \leqslant \alpha$. By (5.93), we can write

$$\Psi(cx,y) = \frac{(cx)^{\alpha_1}\zeta(\alpha_1,y)}{\alpha_1\sqrt{2\pi|\varphi_y''(\alpha_1)|}}\Big\{1 + O\Big(\frac{1}{u} + \frac{\ln y}{y}\Big)\Big\}.$$

From the definition of α_1, we have $(cx)^{\alpha_1}\zeta(\alpha_1,y) \leqslant (cx)^{\alpha}\zeta(\alpha,y)$. Furthermore, a routine calculation allows us to verify that $\sigma \mapsto \sigma\sqrt{|\varphi_y''(\sigma)|}$ is a decreasing function of σ. Applying (5.93) a second time, after having replaced α_1 by α on the right-hand side, we obtain the stated inequality. □

5.6. Jacobsthal's function and Rankin's theorem

In this section, our purpose is to use the upper bounds previously obtained for $\Psi(x,y)$ in order to establish Rankin's theorem on large differences between consecutive primes.

To this end, we infer[8] from Theorems 5.2 and 5.21 that we have

$$\Psi(x,y) \ll xu^{-u} + \sqrt{x} \qquad (x \geqslant y \geqslant 2). \tag{5.100}$$

We shall also employ the following result which can be easily deduced from the large sieve or from Selberg's sieve.[9]

Lemma 5.24. *Let M, $N \in \mathbb{N}^*$ and let $\mathcal{A}$ be a set of integers included in the interval $]M, M+N]$. Assume that, for each prime p, $\mathcal{A}$ is excluded from $w(p)$ residue classes modulo p and that $w(p) \ll 1$. Then we have*

$$|\mathcal{A}| \ll N \prod_{p \leqslant N} \Big(1 - \frac{w(p)}{p}\Big).$$

Jacobsthal's function, denoted by $n \mapsto j(n)$, is defined as the arithmetic function which associates to an integer n the largest gap $j(n)$ between two integers coprime to n.

Maier and Pomerance (1990) introduce the maximal number $j^*(n)$ of consecutive integers which can be sieved by the prime factors of n, that is

$$j^*(n) := \sup\big\{x \in \mathbb{N} : \exists t \in \mathbb{N}, \exists \{a_p\}_{p|n} : \forall m \in]t, x+t]\ \exists p|n : m \equiv (\mathrm{mod}\, a_p)p\big\}.$$

It is easy to see that

$$j^*(n) = j(n) - 1 \qquad (n \geqslant 1). \tag{5.101}$$

Indeed, on the one hand, if $\{a_k\}_{k=1}^{\varphi(n)}$ denotes the increasing sequence of integers in $[1,n]$ which are coprime to n and if $j(n) = a_{m+1} - a_m$, then the interval $]a_m, a_{m+1} - 1]$ is sieved out by the prime factors of n, therefore $j(n) - 1 \leqslant j^*(n)$. On the other hand, if $]t, t + j^*(n)]$ is an interval sieved by the prime factors of n and if $\{a_p\}_{p|n}$ is a system of associated residues, the Chinese remainder theorem guarantees the existence of an integer N such that $N \equiv -a_p \,(\mathrm{mod}\, p)$ for all $p|n$. It follows that each integer of the form $N + m$ with $t + 1 \leqslant m \leqslant t + j^*(n)$ is not coprime to n. Therefore $j^*(n) \leqslant j(n) - 1$.

Remark. We also have

$$j^*(n) = \sup\big\{x \in \mathbb{N} : \exists \{a_p\}_{p|n} : \forall m \in]0, x]\ \exists p|n : m \equiv a_p \,(\mathrm{mod}\, p)\big\}.$$

Indeed, this amounts to replacing a_p by $a_p - t$ in the definition of $j^*(n)$.

Denote by $\mathbb{P} := \{p_n\}_{n \geqslant 1}$ the increasing sequence of primes, put $d_n := p_{n+1} - p_n$ $(n \geqslant 1)$ and

$$P(z) := \prod_{p \leqslant z} p.$$

[8] The easy details are left to the reader.

[9] It is also possible to give a proof, however more complicated, using Brun's sieve.

All lower bounds available in the literature for d_n result from lower bounds on $j\big(P(z)\big)$. Indeed, if $j\big(P(z)\big) > x$, there exists an interval $]t, t+x]$ entirely sieved by the primes $\leqslant z$. From above, we infer that there exists some integer $N \leqslant P(z)$ such that the interval $]N+t, N+t+x]$ contains solely integers m such that $P^-(m) \leqslant z$, and, in particular, does not contain any prime number. Since we can always choose $t \leqslant P(z)$, it follows that

$$j(P(z)) > x \Rightarrow \exists p_k \leqslant 2P(z) : p_{k+1} - p_k \geqslant x.$$

Recall the notation $\Phi(x,y)$, introduced in § I.4.2, for the number of integers $n \leqslant x$ such that $P^-(n) > y$. The following lemma expresses quantitatively the idea that an interval of length x can be sieved in two stages, first by eliminating those integers having a small prime factor, and then by considering congruence classes relative to larger prime numbers.

Lemma 5.25. *Let t, x, z, and Z be integers such that*

$$\pi(Z) - \pi(z) \geqslant \Phi(x+t, z) - \Phi(t, z). \tag{5.102}$$

Then we have

$$j\big(P(Z)\big) > x. \tag{5.103}$$

Proof. To each prime p in $]z, Z]$, we may associate, surjectively, an integer m counted in $\Phi(x+t,z) - \Phi(t,z)$. Let us thus then denote by k_p the residue of m modulo p. Put

$$a_p := \begin{cases} 0 & \text{if } p \leqslant z, \\ -k_p & \text{if } z < p \leqslant Z. \end{cases}$$

By the Chinese remainder theorem, there exists an integer N such that $N \equiv a_p \,(\mathrm{mod}\, p)$ for all $p \leqslant Z$. The interval $]N+t, N+t+x]$ contains no integer n such that $P^-(n) > Z$. Indeed, if $n = N+m$ with $t < m \leqslant t+x$, either $P^-(m) \leqslant z$ and so $P^-(m) | N$, whence $P^-(m) | n$, or $P^-(m) > z$ and there exists a prime $p \in]z, Z]$ such that $N + m \equiv 0 \,(\mathrm{mod}\, p)$, hence $P^-(n) \leqslant Z$.

Thus the interval $]N+t, N+t+x]$ is entirely made up of integers sieved by the prime factors of $P(Z)$. Therefore

$$j^*\big(P(Z)\big) \geqslant x$$

and (5.103) follows. □

Theorem 5.26 (Rankin, 1938). *We have*

$$j\big(P(Z)\big) \gg \frac{Z \ln Z \ln_3 Z}{(\ln_2 Z)^2} \qquad (Z \geqslant 100). \tag{5.104}$$

In particular, there exists a constant $c > 0$ such that, for infinitely many values of $n \in \mathbb{N}$, we have

$$d_n > c \frac{\ln p_n \ln_2 p_n \ln_4 p_n}{(\ln_3 p_n)^2}. \tag{5.105}$$

Proof. Let $z \geqslant 2$, $M := P(z)$, and v, w satisfy $2 \leqslant v \leqslant w \leqslant z$. Put

$$M_1 := P(v), \qquad M_2 := \prod_{v<p\leqslant w} p, \qquad M_3 := \prod_{w<p\leqslant z} p,$$

so that $M = M_1M_2M_3$, and select a $t \leqslant M$ such that

$$t \equiv 0 \,(\mathrm{mod}\, M_1M_3), \qquad t \equiv 1 \,(\mathrm{mod}\, M_2).$$

Then each integer n in $]t, t+x]$ such that $(M,n) = 1$, i.e., each integer n counted by $\Phi(t+x,z) - \Phi(t,z)$ can be written as $n = t+m$ with

$$1 \leqslant m \leqslant x, \qquad (m, M_1M_3) = 1, \qquad (m+1, M_2) = 1. \tag{5.106}$$

Write $m = ab$ with $a \mid M_2^\infty$ and $P^-(b) > z$. If $b \neq 1$, then $b > z$ and so $a \leqslant x/z$. We choose $v := x/z$, so that

$$b \neq 1 \Rightarrow a = 1.$$

We also require that $z > \sqrt{x}$, which entails

$$b \neq 1 \Rightarrow b \in \mathbb{P}.$$

We therefore see that m satisfies (5.106) if, and only if, one of the two following conditions holds:

(i) $m \mid M_2^\infty$, and so $P^+(m) \leqslant w$,

(ii) $m \in \mathbb{P}$, $m > z$, $(m+1, M_2) = 1$.

The number H of solutions of (5.106) hence satisfies

$$\Phi(t+x,z) - \Phi(t,z) \leqslant H \ll \Psi(x,w) + \frac{x}{\ln x}\frac{\ln v}{\ln w},$$

where the last term corresponds to an upper bound for the number of solutions of (ii) by Lemma 5.24.

Select $x = z(\ln z)(\ln_3 z)/(\ln_2 z)^2$ and $w = z^{A \ln_3 z/\ln_2 z}$, where A is some suitable constant. Then $u := \ln x/\ln w \sim \ln_2 z/(A \ln_3 z)$, hence

$$u \ln u \sim (1/A) \ln_2 z \sim (1/A) \ln_2 x.$$

It follows that

$$\Psi(x,w) \ll xu^{-u} \ll x/(\ln x)^{1/A} \ll z/(\ln z)^2$$

provided A is sufficiently small. From this we deduce that

$$\Phi(t+x,z) - \Phi(t,z) \ll \frac{z}{(\ln z)^2} + \frac{x}{\ln x}\frac{\ln v}{\ln w} \ll \frac{z}{\ln z}.$$

With the notation from Lemma 5.25, we may thus choose $Z \ll z$.

Since $M = P(Z) \leqslant \mathrm{e}^{BZ}$ by Chebyshev's upper bound, it follows that $d_n \geqslant x$ whenever $p_n \leqslant \mathrm{e}^{Cz}$. This implies the required lower bound. □

Notes

§ 5.1. It is often of importance in number theory to decompose an integer into two or more factors determined by the size of their prime divisors. This partially explains the frequent occurrence of the function $\Psi(x,y)$ in the analytic theory of numbers. Daboussi (1984) showed that the prime number theorem can be proved by analyzing the limiting case in the model consisting of y-friable integers for a fixed y. This result is only one example of application of a fertile general method resting on the properties of friable integers—cf. Daboussi (1989).

For historical surveys of the numerous works dedicated to the asymptotic behavior of $\Psi(x,y)$ and the role of friable numbers in number theory, see, in particular, Norton (1971), Hildebrand & Tenenbaum (1993b), and Pomerance (1995).

§ 5.2. Ennola (1969) also provides a more complicated asymptotic formula, valid when $y \leqslant (\ln x)^{3/4}$.

§ 5.3. The iterative method of Hildebrand is simply illustrated in the proof of the following result.

Theorem 5.27 (Hildebrand, 1986). *Let $\varepsilon > 0$. There exists a constant $C_3 > 0$ such that, uniformly for $x \geqslant y \geqslant 2$, we have*

$$\Psi(x,y) \geqslant x\varrho(u)\exp\big\{-C_3u/L_\varepsilon(u)\big\}. \tag{5.107}$$

Proof. We may suppose $u \leqslant y^2$. Otherwise Theorem 5.13 easily shows that the right-hand side of (5.107) does not exceed 1.

When $u \leqslant y^2$, we can assume y to be sufficiently large, modifying C_3 if need be. Thus let us fix $y \geqslant y_0$ and set

$$\delta(u) := \inf_{0\leqslant v\leqslant u} \Psi(y^v,y)/y^v\varrho(v).$$

The functional equation (5.34) allows us to write

$$\Psi(x,y)\ln x \geqslant \sum_{d\leqslant y}\Lambda(d)\Psi(x/d,y) \geqslant x\delta(u)S_{\frac12} + x\delta\big(u-\tfrac12\big)\big(S_1 - S_{\frac12}\big), \tag{5.108}$$

where

$$S_\vartheta := \sum_{d\leqslant y^\vartheta}\frac{\Lambda(d)}{d}\varrho\Big(u-\frac{\ln d}{\ln y}\Big) \qquad (0\leqslant\vartheta\leqslant 1).$$

Assume temporarily the estimate

$$S_\vartheta = \ln y\int_0^\vartheta \varrho(u-v)\,\mathrm{d}v + O\Big(\varrho(u)\Big\{1+\frac{u\ln(u+1)}{L_\varepsilon(y)}\Big\}\Big). \tag{5.109}$$

Since $\int_0^1 \varrho(u-v)\,\mathrm{d}v = u\varrho(u)$, by substituting in (5.108) and then dividing by $x\varrho(u)\ln x = xu\varrho(u)\ln y$, we obtain

$$\frac{\Psi(x,y)}{x\varrho(u)} \geqslant \delta(u)r(u) + \delta\big(u-\tfrac12\big)\big\{1 - r(u) + O\big(R_\varepsilon(u)\big)\big\}$$

with

$$r(u) := \frac{1}{u\varrho(u)}\int_0^{\frac12} \varrho(u-v)\,\mathrm{d}v, \qquad R_\varepsilon(u) := \frac{1}{u\ln y} + \frac{\ln(u+1)}{L_\varepsilon(y)}.$$

Since ϱ is non-increasing, we have $r(u) \leqslant \frac12$. Similarly, since δ is also non-increasing, we may write

$$\delta(u)r(u) + \delta\big(u-\tfrac12\big)\big(1-r(u)\big) \geqslant \tfrac12\big\{\delta(u) + \delta\big(u-\tfrac12\big)\big\}$$

and hence, for $y \geqslant y_0$, $u \leqslant y^2$,

$$\delta(u) \geqslant \delta\big(u-\tfrac12\big)\Big\{1 + O\big(R_\varepsilon(u)\big)\Big\} \geqslant \delta\big(u-\tfrac12\big)\exp\Big\{O\big(R_\varepsilon(u)\big)\Big\}.$$

Iterating, it follows that

$$\begin{aligned}\delta(u) &\geqslant \exp\Big\{O\Big(\sum_{k\leqslant u} R_\varepsilon\big(\tfrac12 k\big)\Big)\Big\} \geqslant \exp\Big\{O\Big(\frac{\ln(u+1)}{\ln y} + \frac{u\ln(u+1)}{L_\varepsilon(y)}\Big)\Big\}\\ &\geqslant \exp\big\{O\big(u/L_{2\varepsilon}(u)\big)\big\}.\end{aligned}$$

It remains to show (5.109). The prime number theorem implies that

$$\sum_{d\leqslant y^v} \frac{\Lambda(d)}{d} = v\ln y - \gamma + O\big(1/L_\varepsilon(y^v)\big).$$

Therefore

$$S_\vartheta = \int_0^\vartheta \varrho(u-v)\,\mathrm{d}\{v\ln y\} + \Big[O\Big(\frac{\varrho(u-v)}{L_\varepsilon(y^v)}\Big)\Big]_0^\vartheta + O\Big(\int_0^\vartheta \frac{\varrho'(u-v)}{L_\varepsilon(y^v)}\,\mathrm{d}v\Big).$$

The first term is indeed the main term of (5.109). Set $\kappa := \frac35 - \varepsilon$. Corollary 5.15 allows us to bound the second term by

$$\ll \varrho(u) + \varrho(u)\exp\big\{\vartheta\xi(u) - (\vartheta\ln y)^\kappa\big\} \ll \varrho(u)\Big\{1 + \frac{u\ln(u+1)}{L_\varepsilon(y)}\Big\}.$$

By Corollaries 5.14 and 5.15, we see that the third term is

$$\ll \varrho(u)\ln(u+1)\int_0^1 \exp\big\{v\xi(u) - (v\ln y)^\kappa\big\}\,\mathrm{d}v.$$

If $\xi(u) \leqslant \frac12(\ln y)^\kappa$, the last integral is at most

$$\int_0^1 \exp\Big\{-\tfrac12(v\ln y)^\kappa\Big\}\,\mathrm{d}v \ll \frac{1}{\ln y} \ll \frac{1}{\ln(u+1)}.$$

If $\xi(u) > \frac{1}{2}(\ln y)^\kappa$, it does not exceed

$$\int_0^{1/2} \mathrm{e}^{v\xi(u)}\,\mathrm{d}v + \int_{1/2}^1 \exp\bigl(v\xi(u) - (\tfrac{1}{2}\ln y)^\kappa\bigr)\,\mathrm{d}v \ll \frac{\mathrm{e}^{\xi(u)/2}}{\ln(u+1)} + \frac{\mathrm{e}^{\xi(u)}}{L_{2\varepsilon}(y)\ln(u+1)} \ll \frac{u}{L_{2\varepsilon}(y)}.$$

That establishes (5.109) and completes the proof. □

§ 5.4. As noted by Hildebrand & Tenenbaum (1993), standard techniques from asymptotic analysis[10] may be applied to prove that, for large u, we have

$$\xi(u) = \ln u + \ln_2 u + \sum_{m\geqslant 0}\sum_{k\geqslant 1} c_{mk}\Bigl(\frac{1}{\ln u}\Bigr)^m\Bigl(\frac{1+u\ln_2 u}{u\ln u}\Bigr)^k \tag{5.110}$$

with

$$c_{mk} := \binom{m+k}{m}\operatorname{Res}\Bigl(\frac{z^m}{(\mathrm{e}^z-1)^{m+k}}\Bigl(\frac{z\mathrm{e}^z}{\mathrm{e}^z-1} - \frac{m}{m+k}\Bigr); 0\Bigr) \quad (m\geqslant 0,\ k\geqslant 0).$$

In the form presented here, Theorem 5.13 was proved by Alladi (1982), improving on an asymptotic formula due to de Bruijn (1951a). The method employed by these authors is different from ours and, amongst other things, depends on the following result.

Theorem 5.28 (de Bruijn; Alladi). *Let $f(u)$ be a continuous function for $u > 0$ which satisfies*

$$uf(u) = \int_{u-1}^u f(t)\,\mathrm{d}t \qquad (u > 1). \tag{5.111}$$

Then, for some suitable constant C and all λ with $0 < \lambda < \frac{1}{4}$, we have

$$f(u) = \bigl\{C + O\bigl(\mathrm{e}^{-u^\lambda}\bigr)\bigr\}\varrho(u) \qquad (u\to\infty).$$

The original proof uses the theory of Volterra equations. An alternative route consists in tackling $f(t)$ directly by inverse Laplace transform using the saddle-point method. Hildebrand and the author (1993a) show in this way that the general solution of a delay differential equation of the type $uf'(u) + af(u) + bf(u-1) = 0$ $(u > 1)$ with $a, b \in \mathbb{C}$ may be represented as a convergent series

$$f(u) = \alpha F(u) + \sum_{n\in\mathbb{Z}} \alpha_n F_n(u) \qquad (u > 2)$$

[10] See especially de Bruijn (1970).

where the constants α, α_n can be calculated explicitly from the initial conditions (i.e. the value of $f(u)$ on $[0,1]$) and the functions F, F_n are fundamental solutions of the equation, only depending on a and b, whose asymptotic behavior can be described. Furthermore, this representation is also an asymptotic expansion. In the case of equation (5.111), the first order approximation yields the following result.

Theorem 5.29 (Hildebrand–Tenenbaum). *Let $0 < a < 2\pi^2$. Under the hypotheses of Theorem 5.28, we have*

$$f(u) = \Big\{C + O\big(\mathrm{e}^{-au/(\ln u)^2}\big)\Big\}\varrho(u) \qquad (u \to \infty)$$

with $C := \int_{-\infty}^{\infty}\int_0^1 f(v)(1-\mathrm{e}^{-vt})\mathrm{e}^{-t}\,\mathrm{d}v\,\mathrm{d}t/t\widehat{\varrho}(t)$.

The method employed here to establish Theorem 5.13 easily yields an asymptotic expansion of $\varrho(u)$ in powers of $1/u$. More precisely, there exists a sequence $\{h_j\}_{j\geqslant 1}$ of real analytic functions such that, for $u \geqslant 1$, $N \geqslant 0$, $\xi = \xi(u)$, we have

$$\varrho(u) = \frac{\exp\{\gamma - u\xi + I(\xi)\}}{\sqrt{2\pi I''(\xi)}}\Big\{1 + \sum_{1\leqslant j\leqslant N}\frac{h_j(\xi)}{I''(\xi)^j} + O_N\Big(\frac{1}{u^{N+1}}\Big)\Big\}. \tag{5.112}$$

Furthermore, we have $u - (u-1)/\xi \leqslant I''(\xi) \leqslant u$ $(u \geqslant 1)$ and for each fixed $j \geqslant 1$, there exists a constant c_j such that

$$h_j(\xi) = c_j + O(1/\xi) \qquad (u \to \infty). \tag{5.113}$$

For a detailed proof of such a result and its generalization to (positive real) convolution powers of $\varrho(u)$, see Smida (1991), who extends a theorem of Hensley (1986a). The case of complex convolution powers of $\varrho(u)$ has been treated by Hildebrand (1990).

The local behavior of the Dickman function turns out to be an essential tool for analyzing de Bruijn's approximation $\Lambda(x,y)$. In particular, Corollary 5.15 can be sharpened by the formula

$$\varrho(u-v) = \varrho(u)\mathrm{e}^{v\xi(u)+O(v^2/(u+v^2))} \quad (0 \leqslant v \leqslant u) \tag{5.114}$$

which follows from (5.63) and lemma 6.1 of Fouvry & Tenenbaum (1996).

In this context, the logarithmic derivative $r(u) := -\varrho'(u)/\varrho(u)$ often plays an essential role. Hildebrand (1986) showed that r is non-decreasing on $[1,\infty[$, and Evertse, Moree, Stewart & Tijdeman (2003) observed that we have

$$r(u-1) \leqslant \xi(u) \leqslant r(u) \qquad (u > 1).$$

La Bretèche & Tenenbaum (2005b) proved that

$$r(v) - \xi(v) \ll 1/v, \quad r'(v) - \xi'(v) \ll 1/v^2, \quad \text{and} \quad r''(v) \ll 1/v^2 \qquad (v \geqslant 1).$$

Martin & Tenenbaum (2010) provide further estimates for this function, in particular

$$r'''(v) \ll 1/v^3 \qquad (v \geqslant 1).$$

§ 5.5. The estimate (5.74) for $\alpha(x,y)$ is established in Hildebrand & Tenenbaum (1986), § 7. See also Lemma 6.1.

As shown by Saias (1989), an asymptotic expansion for $\Lambda(x,y)$ can be obtained. The most convenient way to arrive at this result consists in writing down the Taylor–Lagrange formula for $\varrho(u)$—cf. Fouvry & Tenenbaum (1991), lemma 4.2. If we put

$$r_{mj} := \varrho^{(j)}(m) - \varrho^{(j)}(m-) \qquad (0 \leqslant m \leqslant j),$$

then for $u \geqslant 0$, $v \geqslant 0$, $k \geqslant 0$, we have,

$$\varrho(u-v) = \sum_{0\leqslant j\leqslant k} \frac{(-1)^j}{j!}\varrho^{(j)}(u)v^j + R_k(u,v) + J_k(u,v), \tag{5.115}$$

with

$$R_k(u,v) := \sum_{0\leqslant j\leqslant k} \frac{(-1)^{j+1}}{j!} \sum_{\substack{u-v<m\leqslant j\\ m\leqslant u}} r_{mj}(v+m-u)^j,$$

and

$$J_k(u,v) := \frac{(-1)^{k+1}}{k!}\int_0^v (v-w)^k \varrho^{(k+1)}(u-w)\,\mathrm{d}w.$$

Integrating this formula with respect to the measure $\mathrm{d}\big(\lfloor y^v\rfloor/y^v\big)$, we obtain, for all fixed $\varepsilon > 0$, $k \geqslant 0$,

$$\Lambda(x,y) = x\sum_{0\leqslant j\leqslant k} a_j \frac{\varrho^{(j)}(u)}{(\ln y)^j} + O\Big(x\frac{\varrho^{(k+1)}(u)}{(\ln y)^{k+1}}\Big) \tag{5.116}$$

uniformly under the conditions

$$x \geqslant 2, \qquad (\ln x)^{1+\varepsilon} \leqslant y \leqslant x, \qquad \min_{0\leqslant j\leqslant k,\, j\leqslant u}\Big(\frac{u-j}{k+1-j}\Big) \geqslant \frac{\ln_2 y}{\ln y}, \tag{5.117}$$

where the a_j are the Taylor coefficients at $s=0$ of $s\zeta(s+1)/(s+1)$. The final condition in (5.117) is indeed necessary—cf. Saias (1989).

Although Theorem 5.17 was obtained in a thesis supervised by the author, the proof given here is somewhat simpler than the original proof.

Hildebrand (1984a) showed that extending the validity range of the formula

$$\Psi(x,y) = x\varrho(u)\big\{1 + O\big(\ln(u+1)/\ln y\big)\big\} \tag{5.118}$$

to $y \geqslant (\ln x)^{2+\varepsilon}$ is equivalent to the Riemann hypothesis.

The method of proof for Lemma 5.20 provides an estimate for the number of friable integers in short intervals. See Lemma 6.13 and Theorem 4 of Hildebrand & Tenenbaum (1986).

A more complete account of the saddle-point method and its arithmetic applications may be found in the author's survey (1988).

Knowledge of the local behavior of $\Psi(x,y)$ and its generalization

$$\Psi_m(x,y) := \sum_{\substack{n\in S(x,y)\\(n,m)=1}} 1 \tag{5.119}$$

is critical in most of the arithmetic problems involving friable integers. This explains why the study of local fluctuations of $\Psi_m(x,y)$ has given rise to a considerable number of works in the last decades: see especially Hensley (1986b), Friedlander & Granville (1993), Hildebrand (1985, 1986f), Granville (1993), Hildebrand & Tenenbaum (1986), and, for the special case $m = \prod_{p\leqslant z} p$, Saias (1995). The work of La Bretèche & Tenenbaum (2005a) provides, for the local behavior of the quantity (5.119), a synthetic study with formulae as precise and general as present-day techniques allow. Among others, we mention the following result.

Recall the definition of the saddle-point $\alpha(x,y)$ in (5.66). We use the notation $\overline{u} := \min\{u, y/\log y\}$, $u_y := \overline{u} + (\log y)/\log(u+1)$, and set $\delta := \frac12$ if $y \leqslant (\log x)^2$, $\delta := 0$ otherwise. Furthermore, we introduce the quantity

$$W_m := \log p_{\omega(m)} \asymp \log\{\omega(m)+2\} \qquad (m \geqslant 1)$$

where p_k denotes the kth prime number and where, for simplicity, we make the convention that $p_0 = 2$. We then write

$$\vartheta_m = \vartheta_m(y) := \frac{W_m}{\log y}, \quad \overline{\alpha} = \overline{\alpha}(x,y) := \frac{\log(1+y/\log x)}{\log y},$$

$$E_m := E_m(x,y) = \begin{cases} \dfrac{\vartheta_m\{u\log(u+2)\}^{\vartheta_m}}{1+\vartheta_m\log(u+2)} & \text{if } y > (\log x)^2, \\[2ex] \dfrac{\overline{u}\,(\overline{u}\log y)^{\vartheta_m}}{u\log y}\Big(1+\dfrac{1}{\vartheta_m\overline{\alpha}\log y}\Big) & \text{if } y \leqslant (\log x)^2. \end{cases}$$

Theorem 5.30 (La Bretèche & Tenenbaum). *There exist strictly positive absolute constants b_1, b_2, and a function $b = b(x,y;d,m)$ satisfying $b_1 \leqslant b \leqslant b_2$ $(x \geqslant y \geqslant 2,\, d \geqslant 1,\, m \geqslant 1)$ such that, under the conditions $1 \leqslant d \leqslant x/y$ and*

$$x \geqslant y \geqslant 2, \quad P(m) \leqslant y, \quad \omega(m) \ll \sqrt{y}/(\log y)^{\delta}, \tag{5.120}$$

we have, uniformly,

$$\Psi_m\Big(\frac{x}{d},y\Big)=\{1+O(h_m)\}\Big(1-\frac{t^2}{u^2+\overline{u}^2}\Big)^{b\overline{u}}\prod_{p\mid m}(1-p^{-\alpha})\frac{\Psi(x,y)}{d^\alpha} \tag{5.121}$$

with $\alpha:=\alpha(x,y)$, $t:=(\ln d)/\ln y$, *and*

$$h_m=h_m(x,y;d):=\frac{1}{u_y}+(1+E_m)\Big(\frac{t}{u}+\frac{E_m}{\overline{u}}\Big).$$

When $\omega(m)\ll 1$, *formula* (5.121) *holds uniformly under the conditions* $1\leqslant d\leqslant x$ *and* (5.120), *and we have*

$$h_m\asymp h_1\asymp\frac{1}{u_y}+\frac{t}{u}\cdot \tag{5.122}$$

Another result in the same article provides, in the domain (H_ε) and for $P(m)\leqslant y$, $\omega(m)\ll y^{1/\ln(u+2)}$, the estimate

$$\Psi_m(x,y)=x\varrho(u)Z(\beta)\prod_{p\mid m}(1-p^{-\beta})\Big\{1+O\Big(\frac{\vartheta_m^2}{u}+\vartheta_m\ln(u+1)\Big(\frac{y}{x}\Big)^{1/(W_m+6)}\Big)\Big\}$$

where $\beta:=1+\varrho'(u)/\{\varrho(u)\ln y\}$, $Z(s):=(s-1)\zeta(s)/s$. This yields, in the same region of validity, a strengthening of Hildebrand's formula (5.83).

For the study of friable mean values of arithmetic functions, which has given rise to an abundant literature, see especially Tenenbaum & Wu (2003, 2008a, 2008b), Hanrot, Tenenbaum & Wu (2008) and the references indicated there.

Theorem 5.30 suggests that the approximation

$$\frac{\Psi_p(x/p^\nu,y)}{\Psi(x,y)}\approx\frac{1}{p^{\nu\alpha}}\Big(1-\frac{1}{p^\alpha}\Big), \tag{5.123}$$

with $\alpha=\alpha(x,y)$, which has the advantage of being a simple function of p when x and y are fixed, is pertinent in a large domain of the four variables and without any restriction on the relative sizes of x and y. This leads to modelling an additive arithmetic function f defined on the set $S(x,y)$ of y-friable integers not exceeding x by the random variable

$$Z_{f,x,y}:=\sum_{p\leqslant y}\xi_p$$

where the ξ_p are independent variables on an abstract probability space (Ω,P) with geometric laws

$$P(\xi_p=f(p^\nu))=\frac{1}{p^{\nu\alpha}}\Big(1-\frac{1}{p^\alpha}\Big)\qquad(\nu=0,1,2,\ldots).$$

Here, we make the convention that if several (possibly infinite in number) values $f(p^\nu)$ are equal, the corresponding probability is obtained by summing the probabilities on the right-hand side. We have

$$\mathbb{E}(Z_{f,x,y}) = \sum_{p^\nu \in S(x,y)} \frac{f(p^\nu)}{p^{\nu\alpha}}\Big(1-\frac{1}{p^\alpha}\Big),$$

$$\mathbb{V}(Z_{f,x,y}) = \sum_{p^\nu \in S(x,y)} \frac{|f(p^\nu)|^2}{p^{\nu\alpha}}\Big(1-\frac{1}{p^\alpha}\Big) - \sum_{p\leqslant y}\Big(1-\frac{1}{p^\alpha}\Big)^2\Big|\sum_{p^\nu\leqslant x}\frac{f(p^\nu)}{p^{\nu\alpha}}\Big|^2.$$

With this notation, the analogue of the semi-empirical variance (3.17) is the quantity

$$\mathbb{V}^*_{x,y}(f) := \frac{1}{\Psi(x,y)}\sum_{n\in S(x,y)} |f(n)-\mathbb{E}(Z_{f,x,y})|^2.$$

La Bretèche & Tenenbaum (2005b, 2014) have shown that the friable version of the Turán–Kubilius inequality holds uniformly for $x \geqslant y \geqslant 2$: we have

$$C^*(x,y) := \sup_{f\neq 0}\frac{\mathbb{V}^*_{x,y}(f)}{\mathbb{V}(Z_{f,x,y})} \ll 1 \qquad (x\geqslant y\geqslant 2). \tag{5.124}$$

Furthermore

$$C^*(x,y) = 1+o(1) \qquad \Big(\frac{1}{u}+\frac{\ln x}{y}\to 0\Big). \tag{5.125}$$

The same estimates still hold after replacing the semi-empirical by the empirical variance.

The approach described in Exercise 280, p. 472, can be generalized to deduce from (5.124) a friable Erdős–Wintner theorem: see La Bretèche & Tenenbaum (2005b). Contrary to the classical case, for an arithmetic function f on $S(x,y)$, the existence of a limit law, as x and y tend to infinity, involves in this context the values $f(p^\nu)$ for $1\leqslant \nu \leqslant 1/\alpha(x,y)+o(1)$.

All these results can naturally be interpreted in the framework of the Kubilius model for the probabilistic theory of numbers: see the Notes for Chapter III.6.

§ 5.6. It is natural to suspect that the construction in Lemma 5.25 can be improved upon. Indeed, on probabilistic grounds, one would expect that for each $p\in\,]z,Z]$ there exists a congruence class k_p containing $\gg \ln z/\ln_2 z$ integers m in $]t,t+x]$ such that $P^-(m) > z$. This is consistent with the conjecture that condition (5.102) may be replaced by

$$\frac{(Z-z)\ln_2 z}{(\ln z)^2} \gg \Phi(x+t,z)-\Phi(t,z). \tag{5.126}$$

This leads to multiplying Rankin's lower bound by $(\ln z)/\ln_2 z$ and therefore to conjecture that

$$j\big(P(z)\big) \asymp \frac{z(\ln z)^2 \ln_3 z}{(\ln_2 z)^3}. \tag{5.127}$$

The constant c in (5.105) can be specified. For many years all known results were of the type $c > be^\gamma - \varepsilon$ for arbitrary $\varepsilon > 0$ and $n > n_0(\varepsilon)$. Rankin (1938) obtained $b = 1/3$, Maier & Pomerance (1990) showed we can choose $b \approx 1.31256$. The best result of this shape, namely $b = 2$, is due to Pintz (1997). However, Ford, Green, Konyagin, Maynard & Tao announced very recently (2014) that $(\ln_3 p_n)^2$ can be replaced by $\ln_3 p_n$ in the denominator of the right-hand side of (5.105).

Iwaniec (1978) proved that Jacobsthal's function satisfies

$$j(n) \ll \omega(n)^2 (\ln\{2\omega(n)\})^2 \qquad (n \geqslant 2).$$

Exercises

288. For $\varepsilon > 0$, $y \geqslant 2$, define $L_\varepsilon(y) := \exp\{(\log y)^{(3/5)-\varepsilon}\}$.

(a) Applying a strong form of the prime number theorem, show that, for each $\varepsilon > 0$ and uniformly for $y \geqslant 2$, $\sigma > 0$, we have

$$\sum_{p \leqslant y} \frac{\log p}{p^\sigma - 1} = \Big\{1 + O\Big(\frac{1}{L_\varepsilon(y)}\Big)\Big\} \int_{3/2}^{y} \frac{\mathrm{d}t}{t^\sigma - 1} + O(1).$$

[*Hint: Integrate by parts the contribution to the p-sum from the remainder* $R(t) := \vartheta(t) - t$, *and then consider separately the three cases obtained according to the position of* σ *relative to the points* $\frac{1}{2}$ *and* $1 - 2\log L_\varepsilon(y)/\log y$.]

(b) Let $\delta > 0$. Show that, uniformly for $\sigma \geqslant \delta$, we have

$$\int_{3/2}^{y} \frac{\mathrm{d}t}{t^\sigma - 1} = \frac{1 + O\big(1/L_\varepsilon(y)\big)}{1 - y^{-\sigma}} \int_1^y \frac{\mathrm{d}t}{t^\sigma} + O(1).$$

Show that the conclusion still holds for $\sigma \geqslant 2(\log_2 2y)/\log y$, up to replacing $L_\varepsilon(y)$ by $\log y$ in the error term. [*Hint: Use the relation*

$$1/(t^\sigma - 1) = t^{-\sigma} + O\big(t^{-2\sigma}/(1 - 2^{-\sigma})\big),$$

valid for $t \geqslant \frac{3}{2}$.]

(c) Assume now that $0 < \sigma \leqslant 2/3$. Establish the relations

$$\int_1^y \frac{\mathrm{d}t}{t^\sigma} = \Big\{1 + O\big(y^{-1/6}\big)\Big\} \int_{\sqrt{y}}^{y} \frac{\mathrm{d}t}{t^\sigma}, \quad \int_{3/2}^{y} \frac{\mathrm{d}t}{t^\sigma - 1} = \Big\{1 + O\Big(\frac{\log y}{y^{1/6}}\Big)\Big\} \int_{\sqrt{y}}^{y} \frac{\mathrm{d}t}{t^\sigma - 1}.$$

Show that

$$\int_{\sqrt{y}}^{y} \Big(\frac{1}{1 - t^{-\sigma}} - \frac{1}{1 - y^{-\sigma}}\Big) \frac{\mathrm{d}t}{t^\sigma} \ll \frac{y}{(y^\sigma - 1)\log y}.$$

[*Hint: Use the inequality* $1 - (y/t)^{-\sigma} \leqslant \sigma \log(y/t)$.]

(d) Establish the following formula, uniformly for $y \geqslant 2$ and $\sigma > 0$,

$$\sum_{p \leqslant y} \frac{\log p}{p^\sigma - 1} = \frac{1 + O(E)}{1 - y^{-\sigma}} \int_1^y \frac{\mathrm{d}t}{t^\sigma} + O(1)$$

with $E \ll_\delta 1/L_\varepsilon(y)$ if $\sigma \geqslant \delta$, and $E \ll 1/\log y$ for all $\sigma > 0$. Show that $(1 - \sigma)(1 - y^{-\sigma})\log y \ll y^{1-\sigma} - 1$ for $0 < \sigma \leqslant 1$ and deduce formula (5.12) of § 5.1.

289. *Arithmetic proof of the formula* $\widehat{\varrho}(0) = \mathrm{e}^{\gamma}$.

(a) Deduce from Theorem 5.1 that, for $x \geqslant y \geqslant 2$, we have

$$\sum_{n>x,\, P^+(n)\leqslant y} \frac{1}{n} \ll \mathrm{e}^{-u/2} \ln y.$$

(b) Deduce from Theorem 5.8 that, for $x \geqslant y \geqslant 2$, we have

$$\sum_{n\in S(x,y)} \frac{1}{n} = \ln y \int_0^u \varrho(v)\,\mathrm{d}v + O(u).$$

(c) Using Mertens' formula, show that, for $u \geqslant 1$, $y \geqslant 2$, we have

$$\int_0^u \varrho(v)\,\mathrm{d}v = \mathrm{e}^{\gamma} + O\Big(\frac{u}{\ln y} + \mathrm{e}^{-u/2}\Big)$$

and infer from this, with a suitable selection of $u = u(y)$, that $\widehat{\varrho}(0) = \mathrm{e}^{\gamma}$.

290. *Mean value of* $\ln P^+(n)$. Define $S(x) := \sum_{n\leqslant x} \ln P^+(n)$.

(a) Show that $S(x) = x \ln x - \displaystyle\int_1^x \big(\Psi(x,y)/y\big)\,\mathrm{d}y + O(\ln x)$.

(b) Applying Theorem 5.8, deduce that we have

$$S(x) = \alpha x \ln x + O(x \ln_2 x)$$

with $\alpha := 1 - \displaystyle\int_1^{\infty} \frac{\varrho(v)}{v^2}\,\mathrm{d}v \approx 0.62433$.

(c) Deduce from part (a) and from Theorem 5.17 that, for all $\varepsilon > 0$, we have

$$S(x) = x \ln x - \int_1^x \Lambda(x,y)\frac{\mathrm{d}y}{y} + O\Big(x \exp\big\{-(\ln x)^{(3/8)-\varepsilon}\big\}\Big).$$

(d) Show that, for $|s| < 1$, we have

$$\int_{1-}^{\infty} t^{-s}\,\mathrm{d}\Big(\frac{\lfloor t\rfloor}{t}\Big) = \frac{s\zeta(s+1)}{s+1}$$

where the integral is uniformly convergent on any compact subset. Deduce that

$$\int_1^x \ln t\,\mathrm{d}\Big(\frac{\lfloor t\rfloor}{t}\Big) = 1 - \gamma + O\Big(\frac{\ln x}{x}\Big),$$

where γ is Euler's constant.

(e) Show that

$$\begin{aligned}\int_1^x \Lambda(x,y)\frac{\mathrm{d}y}{y} &= x\int_{1-}^x \ln\Big(\frac{x}{t}\Big)\int_{1-(\ln t)/\ln x}^{\infty} \frac{\varrho(v)}{v^2}\,\mathrm{d}v\,\mathrm{d}\Big(\frac{\lfloor t\rfloor}{t}\Big)\\ &= (1-\alpha)x\ln x + \alpha(1-\gamma)x + O(\ln x)\end{aligned}$$

and deduce the asymptotic formula

$$S(x) = \alpha x \ln x - \alpha(1-\gamma)x + O\Big(x\exp\big\{-(\ln x)^{(3/8)-\varepsilon}\big\}\Big).$$

291. Let δ be some fixed real number with $0<\delta<1$. Set $g_\delta(n):=\sum_{p|n}(\ln p)^\delta$.

(a) Show that $\sum_{n\leqslant x} g_\delta(n) \sim (x/\delta)(\ln x)^\delta \quad (x\to\infty)$.

(b) Define $N(x;\lambda) := \big|\{n\leqslant x : g_\delta(n)\leqslant(\lambda/\delta)(\ln x)^\delta\}\big| \quad (\lambda>1)$. Show that $N(x;\lambda)\geqslant\{(\lambda-1)/\lambda+o(1)\}x \quad (x\to\infty)$.

(c) For $n>1$, set $\alpha_n := \big(\ln P^+(n)\big)/\ln n$. Show that α_n has a distribution function and compute it.

(d) Show that for all $n>1$, we have $g_\delta(n)\geqslant(\alpha_n)^{\delta-1}(\ln n)^{\delta-1}g_1(n)$. Using Exercise 266 (p. 467) and the previous question, show that, for each $\lambda>0$, there exists a quantity $c(\lambda,\delta)>0$ such that

$$N(x;\lambda)\leqslant\{1-c(\lambda,\delta)+o(1)\}x \quad (x\to\infty).$$

(e) Show that g_δ fails to have a non-decreasing normal order.

292. Let $N_k(z)$ be the quantity defined in § 5.2.

(a) Calculate $F(\sigma) := \int_{0-}^{\infty} \mathrm{e}^{-\sigma t}\,\mathrm{d}N_k(t)$.

(b) Show that $N_k(z)\leqslant \mathrm{e}^k F(k/z)$.

(c) Deduce the estimate

$$N_k(z)\leqslant\Big(\frac{\mathrm{e}z}{k}\Big)^k \exp\Big\{\frac{k}{2z}\sum_{1\leqslant i\leqslant k} a_i\Big\}\prod_{i=1}^{k}\frac{1}{a_i}.$$

For which values of z is this bound more precise than that of Theorem 5.3?

293. *Integers for which the product of small prime factors is large.*

(a) Using the large sieve (Corollary I.4.13), show that, for $x\geqslant y\geqslant 2$, we have

$$\Phi(x,y) := \big|\{n\leqslant x : P^-(n)>y\}\big| \ll \frac{x}{\ln y}.$$

(b) Set $\Theta(x,y,z) := \big|\{n\leqslant x : \prod_{p^\nu\|n,\,p\leqslant y} p^\nu > z\}\big|$. Show that

$$\Theta(x,y,z)\leqslant \sum_{z<a\leqslant x/y,\ P^+(a)\leqslant y} \Phi(x/a,y)+\Psi(x,y).$$

Deduce that, uniformly for $x\geqslant z\geqslant y\geqslant 2$ and with $v := (\ln z)/\ln y$, we have

$$\Theta(x,y,z)\ll x\mathrm{e}^{-v/2}.$$

(c) Appealing to Theorems 5.2 and 5.21, show that, for each $\varepsilon>0$ and uniformly for $x\geqslant y\geqslant 2$, we have

$$\Psi(x,y)\ll xu^{-u}+x^\varepsilon.$$

Deduce that, for any $\varepsilon > 0$ and uniformly for $x \geqslant z \geqslant y \geqslant 2$, $v := (\ln z)/\ln y$,[11] we have

$$\Theta(x,y,z) \ll_\varepsilon xv^{-v} + xz^{-1+\varepsilon}.$$

294. Put $\overline{u} := \min(u, y/\ln y)$, $u_y := \overline{u} + (\ln y)/\ln(u+1)$. Employing Theorem 5.30, show that, uniformly for $x \geqslant y \geqslant 2$,[12] we have

$$\sum_{n\in S(x,y)} \ln n = \Big\{\ln x - \frac{1}{\alpha} + O\Big(\frac{1}{\alpha u_y}\Big)\Big\}\Psi(x,y)$$
$$= \Big\{\ln x - \frac{\ln y + O(\ln_2 y)}{\ln(1+y/\ln x)}\Big\}\Psi(x,y).$$

295. Retain the notation u_y from Exercise 294. Using Theorem 5.30, show that, uniformly for $x \geqslant y \geqslant 2$,[13] we have

$$\Psi(x,y;\Omega-\omega) := \sum_{n\in S(x,y)} \{\Omega(n)-\omega(n)\} = \Big\{1+O\Big(\frac{1}{u_y}\Big)\Big\}\Psi(x,y)\sum_{p\leqslant y}\frac{1}{p^\alpha(p^\alpha-1)}.$$

Deduce that

$$\inf_{2\leqslant y\leqslant z(x)} \frac{\Psi(x,y;\Omega-\omega)}{\Psi(x,y)} \to \infty \quad (x\to\infty)$$

if, and only if, $z(x) \leqslant (\ln x)^{2+o(1)}$.

296. *Squarefree friable integers.* Using Corollary 5.19, Theorem 5.22 and the bound (5.13), show that, uniformly for $x \geqslant y \geqslant 2$, $x \to \infty$, we have

$$\sum_{n\in S(x,y)} \mu(n)^2 = \{1/\zeta(\beta_1) + o(1)\}\Psi(x,y)$$

with $\beta_1 := \max(1, 2\beta(x,y))$ where $\beta(x,y)$ is defined by (5.14).[14]

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[11] An asymptotic formula for $\Theta(x,y,z)$ has been established by Tenenbaum (2006): see Theorem 6.23.

[12] See La Bretèche & Tenenbaum (2005a), cor. 2.8.

[13] See La Bretèche & Tenenbaum (2005a), cor. 2.11.

[14] More precise results have been established by Ivić & Tenenbaum (1986), and Naïmi (1988). La Bretèche & Tenenbaum (2005a) prove the uniform formula

$$\sum_{n\in S(x,y)} \mu(n)^2 = \Psi(x,y)\{1/\zeta(2\alpha,y) + O(R)\} \qquad (x\geqslant y\geqslant 2)$$

with $R := \min\{\Theta_y(\ln\Theta_y)^2/u_y, \ln(\overline{u}+1)/\sqrt{\overline{u}}\} \ll (\ln_2 y)/\ln y$ where $\overline{u}$ and u_y are defined as in Exercise 294 and $\Theta_y := \prod_{p\leqslant y}(1+1/p^{2\alpha})$. See also La Bretèche & Tenenbaum (2002).

Integers free of small prime factors

6.1. Introduction

Here we undertake a study dual to that of the previous chapter, namely the evaluation of the quantity

$$\Phi(x,y) := \big|\{n \leqslant x : P^-(n) > y\}\big| \qquad (x \geqslant y \geqslant 2).$$

As is the case for $\Psi(x,y)$, this function occurs constantly in analytic and probabilistic number theory. In particular, it is of fundamental use in sieve problems.

Retain the systematic notation

$$u := \frac{\ln x}{\ln y} \qquad (x \geqslant y \geqslant 2)$$

introduced in Chapter III.5. We have seen (Theorem I.4.3) that Brun's pure sieve provides the asymptotic formula

$$\Phi(x,y) = \frac{x}{\zeta(1,y)}\Big\{1 + O\Big(\frac{1}{(\ln y)^2}\Big)\Big\} \qquad (2 \leqslant y \leqslant x^{1/10\ln_2 x}) \tag{6.1}$$

while the fundamental lemma of the combinatorial sieve (Theorem I.4.4) implies the following estimate, valid uniformly for $x \geqslant y \geqslant 2$,

$$\Phi(x,y) = \frac{x}{\zeta(1,y)}\big\{1 + O\big(u^{-u/2}\big)\big\} + O\big(\Psi(x,y)\big). \tag{6.2}$$

The appearance of $\Psi(x,y)$ as an error term in the evaluation of $\Phi(x,y)$ is not surprising. The characteristic function $\eta(n;y)$ of the set of those

integers n such that $P^-(n) > y$ is indeed multiplicative, and, by the Möbius inversion formula, it can be written as

$$\eta(n;y) = \sum_{d|n,\, P^+(d)\leqslant y} \mu(d) \qquad (n \geqslant 1). \tag{6.3}$$

Upon summing over $n \leqslant x$, we get

$$\Phi(x,y) = \sum_{d\leqslant x,\, P^+(d)\leqslant y} \mu(d)\, \lfloor x/d \rfloor \qquad (x \geqslant y \geqslant 2). \tag{6.4}$$

The main term in (6.2) corresponds to the quantity obtained from (6.4) by suppressing the integer parts and extending the summation to infinity. It is clear that the error implied in this process involves $\Psi(x,y)$.

In order to obtain, by this approach, our first result, we use the following technical estimate for the saddle-point $\alpha(x,y)$ defined in (5.66) as the unique solution of the equation

$$\sum_{p\leqslant y} \frac{\ln p}{p^\alpha - 1} = \ln x. \tag{6.5}$$

For this, we introduce the further notation

$$\overline{u} := \min\big(u, y/\ln y\big) = \min(\ln x, y)/\ln y. \tag{6.6}$$

Lemma 6.1. *Uniformly for $x \geqslant y \geqslant 2$, we have*

$$\frac{y^{1-\alpha} - 1}{(1-\alpha)\ln y} \asymp \overline{u}. \tag{6.7}$$

Proof. Evaluating the left-hand side of (6.5) by formula (5.12), we readily obtain

$$u(1 - y^{-\alpha}) = \frac{y^{1-\alpha} - 1}{(1-\alpha)\ln y}\Big\{1 + O\Big(\frac{1}{\ln y}\Big)\Big\}. \tag{6.8}$$

Now the estimate (5.13), namely

$$\alpha = \frac{\ln(1 + y/\ln x)}{\ln y}\Big\{1 + O\Big(\frac{\ln_2 y}{\ln y}\Big)\Big\}, \tag{6.9}$$

implies that

$$1 - y^{-\alpha} \asymp \min\big(1, y/\ln x\big)$$

provided y is sufficiently large. Actually, this estimate remains true for bounded y: on the one hand, by (6.5), we have $(\ln 2)/(2^\alpha - 1) \leqslant \ln x$, hence $1 - y^{-\alpha} \geqslant 1 - 2^{-\alpha} \gg 1/\ln x$, and, on the other hand, we infer from (6.9) that $1 - y^{-\alpha} \leqslant \alpha \ln y \ll 1/\ln x$.

We immediately deduce the required result by substituting in (6.8). □

As in Chapter III.5, let $\xi(u)$ denote the unique real, non-zero root of the equation

$$\mathrm{e}^{\xi} = 1 + u\xi \tag{6.10}$$

for $u > 0$, $u \neq 1$, and define $\xi(1) := 0$. We can rewrite (6.7) as

$$(1-\alpha)\log y = \xi(b\overline{u}) \qquad (b \asymp 1).$$

Estimates (5.48) and (5.60) for $\xi(u)$ and $\xi'(u)$ therefore allow us to deduce immediately from Lemma 6.1 that

$$\alpha = 1 - \frac{\log\big(\overline{u}\log(\overline{u}+1)\big) + O(1)}{\log y} \qquad (x \geqslant y \geqslant 2). \tag{6.11}$$

Theorem 6.2. *Uniformly for $x \geqslant y \geqslant 2$, we have*

$$\Phi(x,y) = \frac{x}{\zeta(1,y)} + O\big(\Psi(x,y)\big). \tag{6.12}$$

Remark. This estimate is only non-trivial for

$$y \leqslant x^{c\log_3 x/\log_2 x},$$

where $c = c(x) \to 1$. Outside such a range, formula (6.12) is weaker than the sieve upper bound (cf., for example, Corollary I.4.13)

$$\Phi(x,y) \ll \frac{x}{\log y} \qquad (x \geqslant y \geqslant 2).$$

Proof. As we just observed, we can assume that u is sufficiently large, so that (6.9) and (6.11) imply

$$(1-\alpha)\log y \geqslant c_0 > 0. \tag{6.13}$$

By (6.4), the error term in (6.12) equals

$$-\sum_{\substack{d\leqslant x\\ P^+(d)\leqslant y}} \mu(d)\Big\langle\frac{x}{d}\Big\rangle - x\sum_{\substack{d>x\\ P^+(d)\leqslant y}}\frac{\mu(d)}{d} \ll \Psi(x,y) + x\int_x^\infty |M(t,y)|\frac{\mathrm{d}t}{t^2},$$

with

$$M(x,y) := \sum_{d\leqslant x,\, P^+(d)\leqslant y} \mu(d).$$

Bounding $|M(t,y)|$ trivially by $\Psi(t,y)$, we obtain, by Theorem 5.23, a result slightly inferior to the required estimate. Indeed

$$x\int_x^\infty \Psi(t,y)\frac{\mathrm{d}t}{t^2} \ll x^{1-\alpha}\Psi(x,y)\int_x^\infty t^{\alpha-2}\,\mathrm{d}t \ll \frac{\Psi(x,y)}{1-\alpha}.$$

However, a result of the author (1990), also proved by the saddle-point method, states that we have

$$M(x,y) \ll \Psi(x,y)\Big(\frac{\exp\{-c_1 u/\log^2(u+1)\}}{\log y} + \exp\big\{-(\log y)^{(3/2)-\varepsilon}\big\}\Big) \tag{6.14}$$

uniformly for $x \geqslant y \geqslant 2$. Thus, we can certainly divide the above upper bound for the t-integral by $\ln y$, and the required conclusion then follows from (6.13). □

6.2. Functional equations

In common with $\Psi(x,y)$ and most functions arising in sieve problems for primes, $\Phi(x,y)$ too has a functional equation.

Theorem 6.3. *For $x \geqslant 1$, $y \geqslant 1$, we have*

$$\Phi(x,y) = 1 + \sum_{y<p\leqslant x} \sum_{\nu\geqslant 1} \Phi\big(x/p^\nu, p\big). \tag{6.15}$$

Remark. As in Chapter III.5, this equation is normally used in Buchstab's form

$$\Phi(x,y) = \Phi(x,z) + \sum_{y<p\leqslant z} \sum_{\nu\geqslant 1} \Phi\big(x/p^\nu, p\big) \qquad (x \geqslant z \geqslant y \geqslant 1). \tag{6.16}$$

Proof. If $n > 1$ is counted by $\Phi(x,y)$, then n can be uniquely written as $n = p^\nu m$, with $p = P^-(n)$ and $p \nmid m$. These two conditions are equivalent to $P^-(m) > p$. It follows that

$$\Phi(x,y) = 1 + \sum_{y<p\leqslant x} \sum_{\substack{n\leqslant x\\ P^-(n)=p}} 1 = 1 + \sum_{y<p\leqslant x} \sum_{\nu\geqslant 1} \sum_{\substack{m\leqslant x/p^\nu\\ P^-(m)>p}} 1,$$

which is (6.15). □

Using the trivial upper bound x/p^ν of $\Phi(x/p^\nu, p)$ for $\nu \geqslant 2$, we deduce from (6.16) the approximate equation

$$\Phi(x,y) = \Phi(x,z) + \sum_{y<p\leqslant z} \Phi\big(x/p, p\big) + O\big(x/y\big) \qquad (x \geqslant z \geqslant y \geqslant 1). \tag{6.17}$$

Had we defined $\Phi(x,y)$ using the weaker inequality $P^-(n) \geqslant y$ instead of strict inequality, the functional equation (6.17) would hold without an error term. However, we shall retain here the definition as previously stated, for it is more convenient when decompositions of integers into products with terms defined according to the size of their prime factors are involved.

As in § 5.3, the functional equation may be used to estimate the sieve function. One argues by induction on $\lfloor u \rfloor$ starting from the trivial case $u \leqslant 1$. Considering just the first two steps leads to the general form of the approximation.

When $\sqrt{x} < y \leqslant x$, the inner sum in (6.15) is identically 1 (since $P^-(1) = \infty$) and we have $\Phi(x,y) = \pi(x) - \pi(y) + 1$. Substituting back

into (6.17) with $z = \sqrt{x}$, we deduce that, for $x^{1/3} < y \leqslant x^{1/2}$,

$$\Phi(x,y) = \Phi(x,\sqrt{x}) + \sum_{y<p\leqslant\sqrt{x}} \Phi\big(x/p,p\big) + O(x^{2/3})$$
$$= \frac{x}{\ln x} + \sum_{y<p\leqslant\sqrt{x}} \frac{x}{p\ln(x/p)} + O\Big(\frac{x}{\ln^2 x}\Big).$$

For $2 \leqslant v \leqslant u$, let us set

$$G(v) := \sum_{x^{1/v}<p\leqslant\sqrt{x}} \frac{1}{p} = \ln(v/2) + O\Big(\mathrm{e}^{-\sqrt{\ln y}}\Big) \tag{6.18}$$

where the estimate follows in a standard way from the prime number theorem. The sum over p in the last evaluation of $\Phi(x,y)$ equals

$$\frac{x}{\ln x}\int_{2-}^{x} \frac{v}{v-1}\,\mathrm{d}G(v) = \frac{x}{\ln y}\Big\{\frac{1+\ln(u-1)}{u}\Big\} + O\Big(\frac{x}{(\ln x)^2}\Big),$$

by partial summation using (6.18).

If, for $1 \leqslant u \leqslant 3$, we define a function $\omega(u)$ by $u\omega(u) := 1$ $(1 \leqslant u \leqslant 2)$ and $u\omega(u) := 1 + \ln(u-1)$ $(2 < u \leqslant 3)$, we have thus proved that

$$\Phi(x,y) = \frac{x\omega(u) - y}{\ln y} + O\Big(\frac{x}{\ln^2 y}\Big) \qquad (x^{1/3} \leqslant y \leqslant x). \tag{6.19}$$

When $y \leqslant x/\ln x$, the second main term $-y/\ln y$ is absorbed by the error term. If $x^{1/4} < y \leqslant x^{1/3}$, we can use (6.19) to evaluate $\Phi(x/p,p)$ in (6.17). On the assumption that (6.19) continues to hold, we can iterate this procedure; the function $\omega(u)$ is then defined inductively on intervals of length 1 by the relation stemming from (6.17) and (6.19)

$$\frac{x\omega(u)}{\ln y} \sim \frac{x}{\ln x} + \sum_{y<p\leqslant\sqrt{x}} \frac{x}{p\ln p}\,\omega\Big(\frac{\ln x}{\ln p} - 1\Big) \sim \frac{x}{\ln x}\Big\{1 + \int_2^u \omega(v-1)\,\mathrm{d}v\Big\},$$

that is

$$u\omega(u) = 1 + \int_1^{u-1} \omega(v)\,\mathrm{d}v \qquad (u > 2). \tag{6.20}$$

This function was discovered (in exactly this way) by Buchstab in 1937, and today bears his name. It is the unique continuous solution for $u > 1$ to the delay differential equation

$$\big\{u\omega(u)\big\}' = \omega(u-1) \qquad (u > 2) \tag{6.21}$$

with initial condition

$$u\omega(u) = 1 \qquad (1 \leqslant u \leqslant 2). \tag{6.22}$$

We extend the definition of $\omega(u)$ to 0 for $u < 1$, so that (6.21) is satisfied for $u \in \mathbb{R} \setminus \{1, 2\}$. For purposes of later reference, we now observe that, by induction on $k := \lfloor u \rfloor$, (6.20) and (6.22) easily imply the bounds

$$\tfrac{1}{2} \leqslant \omega(u) \leqslant 1 \qquad (u \geqslant 1). \tag{6.23}$$

The iterative approach presented above for the evaluation of $\Phi(x, y)$ leads in a natural way to the following result, which is the counterpart of Theorem 5.8 for $\Psi(x, y)$. However, the similarity of the methods is in contrast with a remarkable discrepancy in the quality of the approximations they provide: while Theorem 5.8 is only useful in a restricted region, we now obtain a genuine asymptotic formula with remainder, uniform for $y \to \infty$, $y \leqslant x/2$.

Theorem 6.4. *Uniformly for $x \geqslant y \geqslant 2$, we have*

$$\Phi(x, y) = \frac{x\omega(u) - y}{\ln y} + O\Big(\frac{x}{(\ln y)^2}\Big). \tag{6.24}$$

Proof. Since the result is trivial for bounded y, we assume $y \geqslant y_0$ where y_0 is a sufficiently large absolute constant. Furthermore, we make the assumption that $u > 3$ since we have already established the result for $1 \leqslant u \leqslant 3$.

Let $\Delta(x, y)$ be the function implicitly defined by the formula

$$\Phi(x, y) = \frac{x}{\ln y}\Big\{\omega(u) + \frac{\Delta(x, y)}{\ln y}\Big\}. \tag{6.25}$$

We shall establish, by induction on the integer $k \geqslant 3$, that the quantity

$$\Delta_k := \sup\big\{|\Delta(x, y)| : y \geqslant y_0,\ 2 < u \leqslant k\big\}$$

is finite and bounded independently of k.

We have $\Delta_3 < \infty$. Let $k \geqslant 3$ be such that $\Delta_k < \infty$. If x, y satisfy $y \geqslant y_0$ and $k < u \leqslant k + 1$, we obtain, by inserting (6.25) in (6.17) with $z = x^{1/3}$,

$$\Phi(x, y) = \Phi\big(x, x^{1/3}\big) + \sum_{y < p \leqslant x^{1/3}} \frac{x}{p \ln p}\Big\{\omega\Big(\frac{\ln x}{\ln p} - 1\Big) + \frac{\vartheta_p \Delta_k}{\ln p}\Big\} + O\Big(\frac{x}{y}\Big) \tag{6.26}$$

with $\vartheta_p = \vartheta_p(x) \in [-1, 1]$. We estimate $\Phi(x, x^{1/3})$ using (6.19) with $y = x^{1/3}$. Furthermore, partial summation allows us to write, for sufficiently large y_0,

$$\sum_{p > y} \frac{1}{p(\ln p)^2} = \frac{\frac{1}{2} + O\big(\mathrm{e}^{-\sqrt{\ln y}}\big)}{(\ln y)^2} \leqslant \frac{3}{4(\ln y)^2}. \tag{6.27}$$

Finally, we note that

$$\sum_{y < p \leqslant x^{1/3}} \frac{1}{p \ln p}\,\omega\Big(\frac{\ln x}{\ln p} - 1\Big) = \frac{1}{\ln x} \int_{3-}^{u} v\,\omega(v - 1)\,\mathrm{d}G(v)$$

where $G(v)$ is defined as in (6.18). Now it follows from (6.21) and (6.23) that the function $v \mapsto v\omega(v-1)$ is continuous for $v \neq 2$, has a jump discontinuity at $v = 2$, and is differentiable for $v \neq 2, 3$, with a uniformly bounded derivative.

In view of (6.18), we can therefore write

$$\begin{aligned}\int_{3-}^{u} v\omega(v-1)\,\mathrm{d}G(v) &= \int_{3}^{u} \omega(v-1)\,\mathrm{d}v + \int_{3-}^{u} v\omega(v-1)\,\mathrm{d}\Big\{O\Big(\mathrm{e}^{-\sqrt{\ln y}}\Big)\Big\}\\ &= u\,\omega(u) - 3\omega(3) + \int_{3-}^{u} O\Big(\mathrm{e}^{-\sqrt{\ln y}}\Big)\,\mathrm{d}\{v\,\omega(v-1)\}\\ &= u\,\omega(u) - 3\omega(3) + O\Big(u\mathrm{e}^{-\sqrt{\ln y}}\Big).\end{aligned}$$

Substituting in (6.26) and taking (6.27) into account, we obtain

$$\Phi(x,y) = \frac{x}{\ln y}\Big\{\omega(u) + O\Big(\mathrm{e}^{-\sqrt{\ln y}}\Big)\Big\} + \frac{x\{\vartheta\Delta_k + O(1)\}}{(\ln y)^2}$$

with $|\vartheta| \leqslant \frac{3}{4}$. We hence deduce the existence of an absolute constant C such that

$$\Delta_{k+1} \leqslant \max\Big(\Delta_k, \tfrac{3}{4}\Delta_k + C\Big).$$

Since $\Delta_k \leqslant \Delta_{k+1}$ it finally follows that $\Delta_{k+1} \leqslant \max(\Delta_k, 4C)$. This completes the proof. □

Corollary 6.5. *We have*

$$\omega(u) = \mathrm{e}^{-\gamma} + O\big(u^{-u/2}\big) \qquad (u \geqslant 1). \tag{6.28}$$

Proof. The real number $u \geqslant 1$ being given, we set $y := \exp\{u^{u/2}\}$ and $x := y^u$. By Theorem 5.13 and Corollary 5.19, we then have

$$\Psi(x,y) \ll xu^{-u}.$$

Comparing (6.12) and (6.24) and appealing to Mertens' formula for estimating $\zeta(1,y)$, we readily obtain (6.28). □

It is not difficult to improve on (6.28) by working directly with the delay differential equation (6.21)—insofar as the limit at infinity of $\omega(u)$ is already known. The result obtained is made precise in the following statement. We define $\omega'(u)$ by right-continuity at $u = 1$ and $u = 2$.

Theorem 6.6. *We have*

$$|\omega'(u)| \leqslant \varrho(u) \qquad (u \in \mathbb{R}), \tag{6.29}$$

$$\omega(u) = \mathrm{e}^{-\gamma} + O\Big(\frac{\varrho(u)}{\ln(u+1)}\Big) \qquad (u \geqslant 1). \tag{6.30}$$

Proof. For $1 < u < 2$, we have $\omega'(u) = -1/u^2$ and $\varrho(u) = 1 - \ln u$. An elementary calculus computation then shows that $|\omega'(u)| < \varrho(u)$. This inequality is also true for $u = 2$ since $\omega'(2) = \omega'(2+) = 1/4$. If we define

$$\tau := \inf\{u > 1 : |\omega'(u)| \geqslant \varrho(u)\},$$

we therefore have $\tau > 2$. Now, since $\omega(u)$ is continuous for $u > 1$, we infer from (6.21) that, for $u > 2$, we have

$$u\omega'(u) = \omega(u-1) - \omega(u) = -\int_{u-1}^{u} \omega'(t)\,\mathrm{d}t. \tag{6.31}$$

If τ is finite, we obtain for $u = \tau$ that

$$\tau|\omega'(\tau)| \leqslant \int_{\tau-1}^{\tau} |\omega'(t)|\,\mathrm{d}t < \int_{\tau-1}^{\tau} \varrho(t)\,\mathrm{d}t = \tau\varrho(\tau)$$

from which we get $|\omega'(\tau)| < \varrho(\tau)$, a contradiction. Therefore τ is not finite, and (6.29) follows.

The rapid decrease of $\varrho(u)$ at infinity allows us to write, for $u > 1$,

$$\omega(u) - \mathrm{e}^{-\gamma} = -\int_{u}^{\infty} \omega'(t)\,\mathrm{d}t \ll \int_{u}^{\infty} \varrho(t)\,\mathrm{d}t.$$

We deduce (6.30) by employing Corollary 5.14 in the form

$$\int_{u}^{\infty} \varrho(t)\,\mathrm{d}t \ll \int_{u}^{\infty} \frac{-\varrho'(t)}{\ln(t+1)}\,\mathrm{d}t \ll \frac{\varrho(u)}{\ln(u+1)}\cdot \tag{6.32}$$

□

6.3. Buchstab's function

The object of this section is to undertake the asymptotic study of $\omega(u)$ by the saddle-point method. Apart from the explicit calculation of the Laplace transform

$$\widehat{\omega}(s) := \int_{0}^{\infty} \mathrm{e}^{-su}\omega(u)\,\mathrm{d}u, \tag{6.33}$$

this also leads to a remarkable sharpening of Theorem 6.6.

By differentiation under the integral sign, we may write, for $\sigma > 0$,

$$\frac{\mathrm{d}}{\mathrm{d}s}\widehat{\omega}(s) = -\int_{0}^{\infty} \mathrm{e}^{-su}u\omega(u)\,\mathrm{d}u = \left[\frac{\mathrm{e}^{-su}u\omega(u)}{s}\right]_{0}^{\infty} - \frac{1}{s}\int_{0}^{\infty} \mathrm{e}^{-su}\,\mathrm{d}\{u\omega(u)\}.$$

The term in the square brackets vanishes. Denoting by δ_1 the Dirac measure at $u = 1$, we deduce from (6.21) and (6.22) the following equality between measures:

$$\mathrm{d}\{u\omega(u)\} = \delta_1 + \{u\omega(u)\}'\,\mathrm{d}u = \delta_1 + \omega(u-1)\,\mathrm{d}u.$$

After the change of variable $(u-1) \mapsto u$, we obtain

$$\frac{\mathrm{d}}{\mathrm{d}s}\widehat{\omega}(s) = -\frac{\mathrm{e}^{-s}}{s}\{1 + \widehat{\omega}(s)\}$$

from which it follows that $1+\widehat{\omega}(s) = Ce^{J(s)}$ for some suitable constant C with

$$J(s) := \int_0^\infty \frac{e^{-s-t}}{s+t}\,dt.$$

Lemma 5.9 then shows that

$$1+\widehat{\omega}(s) = \frac{C}{s\widehat{\varrho}(s)} \tag{6.34}$$

and in particular that

$$\lim_{s\to+\infty} \widehat{\omega}(s) = C-1,$$

since $s\widehat{\varrho}(s) = \int_0^\infty \varrho(u/s)e^{-u}\,du \to \varrho(0) = 1$ as $s\to+\infty$. Now, it clearly follows from (6.23) that $\widehat{\omega}(s)\to 0$ as $s\to+\infty$. We hence have $C=1$, and we can state the following result.

Theorem 6.7. *The function $\widehat{\omega}(s)$, defined by* (6.33) *for $\sigma>0$, extends to a meromorphic function on $\mathbb{C}$, given explicitly by the formula*

$$1+\widehat{\omega}(s) = \frac{1}{s\widehat{\varrho}(s)} \qquad (s\neq 0). \tag{6.35}$$

When s is not a negative real number, we have

$$1+\widehat{\omega}(s) = e^{J(s)}. \tag{6.36}$$

Remarks. (i) Formula (6.35) can be established in a purely arithmetical way by appealing to the elementary estimates for $\Phi(x,y)$ and $\Psi(x,y)$ deduced from the functional equations (Theorems 6.4 and 5.8) and using the natural duality between these quantities: see Exercise 297.

(ii) Conversely, one may use (6.35) to recover analytically that

$$\lim_{u\to\infty} \omega(u) = e^{-\gamma},$$

which was proved in Corollary 6.5 using Mertens' formula. For this, observe that (6.35) implies

$$\lim_{s\to 0+} s\widehat{\omega}(s) = 1/\widehat{\varrho}(0) = e^{-\gamma}.$$

Bearing in mind the positivity of ω, Karamata's Tauberian theorem (Theorem II.7.5) implies

$$\int_0^u \omega(t)\,dt = \{e^{-\gamma}+o(1)\}u \qquad (u\to\infty).$$

However, it follows from (6.21) and (6.23) that $u\omega'(u)\ll 1$. For a suitable function $\varepsilon(u)\to 0$ and for all h, $0<h\leqslant u$, we thus obtain, on the one hand, that

$$\frac{1}{h}\int_u^{u+h} \omega(t)\,dt = e^{-\gamma} + O\Big(\varepsilon(u)\frac{u}{h}\Big)$$

and, on the other, that

$$\frac{1}{h}\int_u^{u+h} \omega(t)\,\mathrm{d}t = \omega(u) + \frac{1}{h}\int_u^{u+h} O\Big(\frac{h}{u}\Big)\,\mathrm{d}t = \omega(u) + O\Big(\frac{h}{u}\Big).$$

It thus suffices to choose $h := u\sqrt{\varepsilon(u)}$.

It is easy to deduce from the delay differential equation (6.21)–(6.22) satisfied by the Buchstab function that it is of class $\mathcal{C}^j$ on $\mathbb{R}\setminus\{1,2,\dots,j+1\}$ for each integer $j \geqslant 0$. At the exceptional points, successive derivatives have jump discontinuities, and, by convention, we extend the definition of $\omega^{(j)}(u)$ to the entire real line by right-continuity.

In particular, $\omega'(u)$ is of bounded variation on every bounded interval. Combined with the rapid decrease at infinity (Theorem 6.6), this property is sufficient to imply the convergence (to the value $\omega'(u)$) of the inverse Laplace integral for $u \neq 1$ or 2, for any abscissa of integration. We can therefore use the saddle-point method to evaluate $\omega'(u)$ at infinity. With the notation

$$H(u) := \exp\big\{u/\ln^2(u+2)\big\} \qquad (u \geqslant 0),$$

we obtain the following result.

Theorem 6.8. *For a suitable positive absolute constant a, we have*

$$\begin{matrix}\omega(u) - \mathrm{e}^{-\gamma} \\ \omega'(u)\end{matrix} \ll \varrho(u)H(u)^{-a} \qquad (u \geqslant 0). \tag{6.37}$$

Proof. We can suppose $u > 2$. First, observe that the second stated upper bound easily implies the first since

$$\omega(u) - \mathrm{e}^{-\gamma} = -\int_u^\infty \omega'(t)\,\mathrm{d}t.$$

The conclusion then follows by using the fact that $H(u)^{-a}$ is decreasing and appealing to the estimate (6.32).

Let $h(s)$ be the Laplace transform of $\omega'(u)$. As we already noted, we have

$$\omega'(u) = \frac{1}{2\pi i}\int_{\kappa-i\infty}^{\kappa+i\infty} h(s)\mathrm{e}^{us}\,\mathrm{d}s \qquad (u > 2) \tag{6.38}$$

for any real number κ. Now

$$h(s) = s\widehat{\omega}(s) - \mathrm{e}^{-s} \tag{6.39}$$

where the second term arises from the discontinuity of $\omega(u)$ at $u = 1$. Taking account of (6.35), it follows that

$$h(s) = \widehat{\varrho}(s)^{-1} - \mathrm{e}^{-s} - s. \tag{6.40}$$

Recall that

$$\widehat{\varrho}(s) = \mathrm{e}^{\gamma + I(-s)} \qquad \Big(I(s) := \int_0^s \frac{\mathrm{e}^v - 1}{v}\,\mathrm{d}v\Big).$$

In accordance with the underlying principles of the saddle-point method, and temporarily neglecting the influence of the term $-\mathrm{e}^{-s} - s$ in this computation, we need to choose the abscissa of integration κ in such a way that at least one of the complex solutions to the equation

$$\mathrm{e}^{-s} = 1 + su \tag{6.41}$$

has abscissa κ. Contrary to the situation we met in the study of Dickman's function, the saddle-point equation (6.41) does not have a non-trivial real solution. However, we shall see that, for sufficiently large u, it has a solution with real part close to $-\xi(u)$.

Indeed, let us make the change of variables $s = -\xi(u) + i\pi - z$ in (6.41). We obtain

$$z\lambda(z) = w \tag{6.42}$$

with

$$\lambda(z) := (1 + u\xi)\Big(\frac{\mathrm{e}^z - 1}{z}\Big) - u, \qquad w := -i\pi u - 2.$$

Since $\xi(u) \sim \ln u\,(u \to \infty)$—cf. Lemma 5.11—it follows that there exists some absolute positive constant u_0 such that

$$|z\lambda(z)| > |w| \qquad \big(|z| = 2\pi/\ln(u+1),\ u > u_0\big).$$

Since $\lambda(z)$ does not vanish for $|z| < 2\pi/\ln(u+1)$, $u > u_0$, this implies by Rouché's theorem that equation (6.42) has in this disk a unique solution. Lagrange's theorem actually states that this solution is an analytic function of w and provides an explicit formula for coefficients of its Taylor expansion—see for example Whittaker & Watson (1927), § 7.32. We make no use of this last piece of information in this proof.

The integral (6.38) could be evaluated by choosing precisely $\kappa = -\xi(u) - \Re\mathrm{e}\, z$. This would lead to a result slightly sharper than (6.37)—see the Notes. However, we will be satisfied here with working with the simpler choice $\kappa = -\xi(u)$ which is heuristically justified by the fact that (6.42) has a solution approaching 0 as $u \to \infty$. This will allow us to use the estimates obtained in Chapter III.5 for $\widehat{\varrho}(s)$ on the line $\sigma = -\xi(u)$.

For the remainder of this proof, we hence let $\kappa = -\xi(u)$. We first consider the contribution to (6.38) from the domain $|\tau| > \mathrm{e}^{\xi}$. From the estimate

$$J(s) = \frac{\mathrm{e}^{-s}}{s}\Big(1 - \frac{1}{s}\Big) + O\Big(\frac{\mathrm{e}^{-s}}{|\tau|^3}\Big) \qquad (\tau \neq 0),$$

it follows, using (6.39) and (6.36), that, for $s = -\xi(u)+i\tau$, $|\tau| > \mathrm{e}^{\xi}$, we have

$$h(s) = s\Big\{1+\frac{\mathrm{e}^{-s}}{s}\Big(1-\frac{1}{s}\Big)+\frac{\mathrm{e}^{-2s}}{2s^2}\Big(1-\frac{1}{s}\Big)^2+O\Big(\frac{\mathrm{e}^{3\xi}}{|\tau|^3}\Big)\Big\}-s-\mathrm{e}^{-s}$$
$$= -\frac{\mathrm{e}^{-s}}{i\tau}+\frac{\mathrm{e}^{-2s}}{2i\tau}+O\Big(\frac{\mathrm{e}^{3\xi}}{\tau^2}\Big).$$

We can estimate the contribution of the main terms to the integral by the second mean value theorem; bounding that of the remainder term in a straightforward manner, we obtain that the contribution to (6.38) of the domain $|\tau| > \mathrm{e}^{\xi}$ is

$$\ll \mathrm{e}^{-u\xi+2\xi} \ll \varrho(u)\mathrm{e}^{-u/2}$$

where the second estimate follows from Theorem 5.13.

In order to evaluate the complementary contribution, we use $h(s)$ in the form (6.40). We shall show that, for some suitable positive constant a, we have

$$\widehat{\varrho}(s)^{-1} \ll \widehat{\varrho}(-\xi)H(u)^{-2a} \qquad (\sigma = -\xi,\ |\tau| \leqslant \mathrm{e}^{\xi}). \tag{6.43}$$

Temporarily admitting this upper bound, it follows that

$$\int_{|\tau|\leqslant \mathrm{e}^{\xi}} h(s)\mathrm{e}^{su}\,\mathrm{d}s \ll \mathrm{e}^{-u\xi+\xi}\big\{\widehat{\varrho}(-\xi)H(u)^{-2a}+\mathrm{e}^{\xi}\big\} \ll \varrho(u)H(u)^{-a},$$

by Theorem 5.13.

It remains to establish (6.43), that is (setting $T := u/\ln^2(u+1)$) to show that

$$\Re e\big\{I(\xi+i\tau)+I(\xi)\big\} \geqslant 2aT+O(1) \qquad (|\tau| \leqslant \mathrm{e}^{\xi}). \tag{6.44}$$

The left-hand side equals

$$\int_0^1 \Big\{\mathrm{e}^{v\xi}\big(1+\cos(\tau v)\big)-2\Big\}\frac{\mathrm{d}v}{v}$$
$$\geqslant -\int_0^{\frac12}\big(1-\cos(\tau v)\big)\frac{\mathrm{d}v}{v}+\int_{\frac12}^1\Big\{\mathrm{e}^{v\xi}\big(1+\cos(\tau v)\big)-\frac{2}{v}\Big\}\mathrm{d}v$$
$$= -2\int_0^{\frac14\tau}(\sin v)^2\frac{\mathrm{d}v}{v}+\Big[\frac{\mathrm{e}^{v\xi}}{\xi}+\Re e\Big(\frac{\mathrm{e}^{v(\xi+i\tau)}}{\xi+i\tau}\Big)\Big]_{\frac12}^1-\ln 4$$
$$\geqslant O\big(\ln(1+\tau^2)\big)+\mathrm{e}^{\xi}\Big\{\frac{1}{\xi}+\frac{\cos\lambda}{\sqrt{\xi^2+\tau^2}}\Big\}+O\Big(\sqrt{\frac{u}{\xi}}\Big)$$

with $\lambda := \tau-\arctan(\tau/\xi)$. There exists some absolute constant τ_0 such that $\cos\lambda \geqslant 0$ for $|\tau| \leqslant \tau_0$. For such τ, the above lower bound is $\geqslant u+O(\sqrt{u/\xi})$. For $|\tau| > \tau_0$, it is

$$\geqslant \mathrm{e}^{\xi}\Big(\frac{1}{\xi}-\frac{1}{\sqrt{\xi^2+\tau_0^2}}\Big)+O\Big(\sqrt{\frac{u}{\xi}}\Big) \gg \frac{\mathrm{e}^{\xi}}{\xi^3}+O\Big(\sqrt{\frac{u}{\xi}}\Big) \gg T+O(1).$$

This establishes (6.44) and hence completes the proof of Theorem 6.8. □

Corollary 6.9. *Let j be an integer $\geqslant 1$. We have*

$$\omega^{(j)}(u) \ll \varrho^{(j)}(u)H(u)^{-a} \qquad (u \to \infty). \tag{6.45}$$

Proof. Equation (6.21) and our convention of extending ω and all its derivatives by right-continuity allows writing, after j differentiations,

$$u\omega^{(j+1)}(u) = \omega^{(j)}(u-1) - (j+1)\omega^{(j)}(u) \qquad (u \in \mathbb{R}). \tag{6.46}$$

By induction on j and appeal to Corollary 5.14 in the form

$$\begin{aligned} \varrho^{(j)}(u-1) &\ll \varrho(u-1)\log^j(u+1) = -u\varrho'(u)\log^j(u+1) \\ &\ll u\varrho(u)\log^{j+1}(u+1) \ll u\varrho^{(j+1)}(u), \end{aligned} \qquad (u \geqslant 1)$$

this implies (6.45). □

6.4. Approximations to $\Phi(x,y)$ by the saddle-point method

As with its dual $\Psi(x,y)$, the function $\Phi(x,y)$ can be estimated by the saddle-point method starting from its Perron integral. Nevertheless, two differences stand out. On the one hand the associated Dirichlet series $\zeta(s)/\zeta(s,y)$ now has a pole at $s=1$; the possibility of selecting the abscissa of integration optimally will thus depend on a preliminary use of the residue theorem. On the other hand, it is no longer a question of determining the asymptotic behavior of the function under study, but of estimating the error term relative to an explicit main term—cf. Theorem 6.4.

Throughout this section we retain some of the notation from Chapter III.5, such as the domain

$$x > x_0(\varepsilon), \quad \exp\left\{(\log_2 x)^{(5/3)+\varepsilon}\right\} \leqslant y \leqslant x, \tag{H_ε}$$

and the function

$$L_\varepsilon(y) := \exp\left\{(\log y)^{(3/5)-\varepsilon}\right\}. \tag{6.47}$$

In addition, we introduce the following notation:

$$Y_\varepsilon := \exp\left\{(\log y)^{(3/2)-\varepsilon}\right\}, \quad E(x,y) := H(u)^{-c}L_\varepsilon(y)^{-1} + Y_\varepsilon^{-1}, \tag{6.48}$$

$$\mu_y(u) := \int_0^\infty \omega(u-v)y^{-v}\,\mathrm{d}v, \quad W(x,y) := x\mu_y(u)\frac{\mathrm{e}^\gamma \log y}{\zeta(1,y)}. \tag{6.49}$$

The letter c, with or without index, stands for some absolute positive constant.

The main purpose of this section consists in establishing the following result.

Theorem 6.10. *Suppose $\varepsilon > 0$. For $x \geqslant y \geqslant 2$, we have*

$$\Phi(x,y) - W(x,y) \ll \begin{cases} \Psi(x,y)\,E(x,y) & \text{(in the domain } H_\varepsilon\text{)} \\ \Psi(x,y) & \text{(outside } H_\varepsilon\text{)}. \end{cases} \tag{6.50}$$

We begin by proving the second bound. It is an easy consequence of the following lemma.

Lemma 6.11. *Let $\delta > 1$. Uniformly for $x \geqslant y \geqslant 2$, we have*

$$\big|\mu_y(u)\mathrm{e}^\gamma \ln y - 1\big| \ll \varrho(u)H(u)^{-c_1} + x^{-1}(\mathrm{e}^{\delta\overline{u}} + y). \tag{6.51}$$

Temporarily assuming this estimate, we argue as follows. By Theorem 5.27, we have

$$x\varrho(u)H(u)^{-c_1} \ll \Psi(x,y)H(u)^{-c_1/2}.$$

Furthermore, it is easy to see that the approximation Z for $\ln\Psi(x,y)$ from Theorem 5.2 satisfies

$$Z := u\ln\Big(1+\frac{y}{\ln x}\Big) + \frac{y}{\ln y}\ln\Big(1+\frac{\ln x}{y}\Big) \geqslant \overline{u}\ln 4$$

and also

$$Z \geqslant \tfrac{1}{2}\overline{u} + \ln y + O(1) \qquad (u \geqslant 2).$$

For $1 < \delta < \ln 4 - \frac{1}{3}$, we therefore have

$$y + \mathrm{e}^{\delta\overline{u}} \ll \Psi(x,y)\mathrm{e}^{-\overline{u}/3}.$$

Substituting back in (6.51), we thus obtain

$$W(x,y) - \frac{x}{\zeta(1,y)} \ll \frac{\Psi(x,y)}{\ln y}H(\overline{u})^{-c_1/2}. \tag{6.52}$$

The second bound in (6.50) follows from this estimate and Theorem 6.2.

Proof of Lemma 6.11. A simple integration by parts enables us to write

$$\mu_y(u)\ln y = \omega(u) - \frac{y}{x} - \int_0^\infty \omega'(u-v)y^{-v}\,\mathrm{d}v \tag{6.53}$$

where the second term of the right-hand side arises from the discontinuity of $\omega(u)$ at $u = 1$.

Taking into consideration the bound for $\omega(u) - \mathrm{e}^{-\gamma}$ in Theorem 6.8, we see that the proof of (6.51) reduces to showing that, for $x \geqslant y \geqslant 2$, we have

$$\int_0^\infty \big|\omega'(u-v)\big|y^{-v}\,\mathrm{d}v \ll \varrho(u)H(u)^{-c_1} + \mathrm{e}^{\delta\overline{u}}x^{-1}. \tag{6.54}$$

Observe first that, for any $\varepsilon > 0$, we have

$$H(u-v)^{-a} \ll_\varepsilon H(u)^{-a}\mathrm{e}^{\varepsilon v} \qquad (0 \leqslant v \leqslant u)$$

where a is the absolute constant of Theorem 6.8. Put $y_\varepsilon := ye^{-\varepsilon}$. By Theorem 6.8, we then obtain

$$\int_0^\infty |\omega'(u-v)|y^{-v}\,dv \ll_\varepsilon H(u)^{-a}\int_0^\infty \varrho(u-v)y_\varepsilon^{-v}\,dv. \tag{6.55}$$

Let λ be a parameter $\geqslant 1$. We may write the last integral as

$$y_\varepsilon^{-u}\int_0^u \varrho(t)y_\varepsilon^t\,dt \leqslant y_\varepsilon^{-u}\int_0^\infty \varrho(t)y_\varepsilon^t\lambda^{u-t}\,dt = \big(y_\varepsilon/\lambda\big)^{-u}\widehat{\varrho}\big(-\ln(y_\varepsilon/\lambda)\big).$$

If $y_\varepsilon \geqslant e^{\xi(u)}$, we may choose $\lambda = y_\varepsilon e^{-\xi(u)}$; taking Theorem 5.13 into account, we find that the left-hand side of (6.55) is

$$\ll_\varepsilon H(u)^{-a}e^{-u\xi(u)}\widehat{\varrho}\big(-\xi(u)\big) \ll H(u)^{-a/2}\varrho(u).$$

The bound (6.54) is therefore satisfied in this case.

If $y_\varepsilon < e^{\xi(u)}$, we appeal to the upper bound (6.29) for $|\omega'(u)|$ in Theorem 6.6. The left-hand side of (6.55) does not exceed

$$\int_0^\infty \varrho(u-v)y^{-v}\,dv = \int_0^u \varrho(t)y^{t-u}\,dt \leqslant y^{-u}\widehat{\varrho}(-\ln y) = y^{-u}e^{\gamma+I(\ln y)}.$$

Since $I(\ln y) \sim y/\ln y$ as $y \to \infty$ and $I\big(\varepsilon+\xi(u)\big) \sim e^\varepsilon u$ as $u \to \infty$, we certainly have, in the region under consideration,

$$e^{\gamma+I(\ln y)} \ll e^{\delta\overline{u}}$$

provided that $\varepsilon = \varepsilon(\delta)$ is sufficiently small.

This completes the proof of (6.54) and hence that of Lemma 6.11. □

We now turn our attention to the first upper bound in Theorem 6.10, concerning the domain (H_ε). By Perron's formula, for any real number $\kappa > 1$, we have

$$\Phi(x,y) = \frac{1}{2\pi i}\int_{\kappa-i\infty}^{\kappa+i\infty}\frac{\zeta(s)x^s}{\zeta(s,y)s}\,ds \qquad (x \notin \mathbb{Z}^+). \tag{6.56}$$

The residue at $s=1$ has value $x/\zeta(1,y)$. The integrand is a holomorphic function of s for $s \neq 0$ or 1, and tends to 0 as $|\tau| \to \infty$ in every vertical strip $0 < \sigma_0 \leqslant \sigma \leqslant 1$. We can hence move the abscissa of integration to the left until

$$\sigma = \alpha_0 := 1 - \frac{\xi(u)}{\ln y}.$$

We obtain

$$\Phi(x,y) = \frac{x}{\zeta(1,y)} + \frac{1}{2\pi i}\int_{\alpha_0-i\infty}^{\alpha_0+i\infty}\frac{\zeta(s)x^s}{\zeta(s,y)s}\,ds. \tag{6.57}$$

Lemma 5.16 and Theorem 6.7 allow us to approximate, when $|\tau| \leqslant L_\varepsilon(y)$, the integrand of (6.57) by

$$\frac{x\mathrm{e}^{uz}\{1+\widehat{\omega}(z)\}}{1+z/\ln y} \qquad (z := (s-1)\ln y).$$

Neglecting for the time being the influence of the domain $|\tau| > L_\varepsilon(y)$ on the integral in (6.57), we obtain the heuristic approximation to $\Phi(x,y)$

$$W_1(x,y) := \frac{x}{\zeta(1,y)} + \frac{x}{2\pi i}\int_{-\xi(u)-i\infty}^{-\xi(u)+i\infty} \frac{1+\widehat{\omega}(s)}{s+\ln y}\mathrm{e}^{us}\,\mathrm{d}s. \tag{6.58}$$

We shall see below that this integral is indeed convergent. By Theorem 6.7, $\widehat{\omega}(s)$ has a simple pole at $s=0$, with residue $\mathrm{e}^{-\gamma}$. Moving now the line of integration in (6.58) to $\sigma = \kappa > 0$, we obtain

$$W_1(x,y) = \frac{x}{\zeta(1,y)} - \frac{x\mathrm{e}^{-\gamma}}{\ln y} + \frac{x}{2\pi i}\int_{\kappa-i\infty}^{\kappa+i\infty} \frac{1+\widehat{\omega}(s)}{s+\ln y}\,\mathrm{e}^{us}\,\mathrm{d}s. \tag{6.59}$$

However, $\big(1+\widehat{\omega}(s)\big)/(s+\ln y)$ is the Laplace transform of the function

$$t \mapsto y^{-t} + \int_0^\infty \omega(t-v)y^{-v}\,\mathrm{d}v$$

which is continuous and of bounded variation on each bounded interval. The Laplace inversion theorem—cf. Widder (1946), th. II.7.3—then yields the explicit formula

$$W_1(x,y) = 1 + x\Big\{\frac{1}{\zeta(1,y)} - \frac{\mathrm{e}^{-\gamma}}{\ln y} + \mu_y(u)\Big\}. \tag{6.60}$$

Now we observe that, in the domain (H_ε), Lemma 6.11 implies

$$\begin{aligned} W_1(x,y) - W(x,y) &= 1 + x\Big(\frac{1}{\zeta(1,y)} - \frac{\mathrm{e}^{-\gamma}}{\ln y}\Big)\big(1-\mu_y(u)\mathrm{e}^{\gamma}\ln y\big) \\ &\ll x\varrho(u)H(u)^{-c_1}L_\varepsilon(y)^{-1} \ll \Psi(x,y)E(x,y). \end{aligned} \tag{6.61}$$

This difference is of the same order as the error term stated in (6.50). We can therefore regard $W(x,y)$ as the natural approximation for $\Phi(x,y)$ arising from the saddle-point method.

It remains to make rigorous the argument sketched above. On several occasions, we shall appeal to the following estimate which is a consequence of Theorem 5.13, Lemma 5.16, and Corollary 5.19. We have, uniformly in (H_ε),

$$x^{\alpha_0}\zeta(\alpha_0,y) \asymp \Psi(x,y)\sqrt{u}\ln y. \tag{6.62}$$

Set $L := L_{\varepsilon/2}(y)$. We shall establish the following three estimates:

$$\int_{\substack{\sigma=\alpha_0\\ |\tau|>L}} \frac{\zeta(s)x^s}{\zeta(s,y)s}\,\mathrm{d}s \ll \Psi(x,y)E(x,y), \tag{6.63}$$

$$\frac{1}{L}\int_{\substack{\sigma=\alpha_0\\ |\tau|\leqslant L}} \Big|\frac{\zeta(s)x^s}{\zeta(s,y)s}\Big|\,\mathrm{d}s \ll \Psi(x,y)E(x,y), \tag{6.64}$$

$$x\int_{\substack{\sigma=-\xi\\ |\tau|>L}} \frac{1+\widehat{\omega}(s)}{s+\ln y}\mathrm{e}^{us}\,\mathrm{d}s \ll \Psi(x,y)E(x,y). \tag{6.65}$$

This clearly suffices to establish Theorem 6.10: the bound (6.63) allows truncating the integral (6.56), the estimate (6.64) permits neglecting the error term created by using Lemma 5.16 (smooth approximation to $\zeta(s,y)$), and the bound (6.65) enables us to extend to infinity the integral obtained by replacing $\zeta(s,y)$ by its smooth approximation.

The most difficult of these is the estimate (6.63). We shall need the following two auxiliary results.

Lemma 6.12. *For (x,y) in (H_ε), $s=\alpha_0+i\tau$, we have*

$$\zeta(s,y) \ll \zeta(\alpha_0,y)\exp\Big\{-\frac{c_2\tau^2 u}{(1-\alpha_0)^2+\tau^2}\Big\} \qquad \Big(\frac{1}{\ln y}\leqslant|\tau|\leqslant Y_\varepsilon\Big), \tag{6.66}$$

$$\zeta(s,y)^{-1} \ll \zeta(\alpha_0,y)H(u)^{-c_2} \qquad (|\tau|\leqslant Y_\varepsilon). \tag{6.67}$$

Proof. Let us first prove (6.66). An easy calculation gives

$$\begin{aligned}\Big|\frac{1-p^{-\alpha_0}}{1-p^{-s}}\Big| &= \Big(1+\frac{2\big(1-\cos(\tau\ln p)\big)}{p^{\alpha_0}(1-p^{-\alpha_0})^2}\Big)^{-1/2}\\ &\leqslant \exp\Big\{-\frac{1-\cos(\tau\ln p)}{p^{\alpha_0}}\Big\}\end{aligned} \tag{6.68}$$

from which we get

$$|\zeta(s,y)|\zeta(\alpha_0,y)^{-1} \leqslant \mathrm{e}^{-X}, \quad \text{with} \quad X := \sum_{p\leqslant y}\frac{1-\cos(\tau\ln p)}{p^{\alpha_0}}.$$

We need a lower bound for X. We have

$$X\ln y \geqslant A(\alpha_0) - \Re\mathrm{e}\,A(\alpha_0+i\tau) + O(1) \tag{6.69}$$

with

$$A(s) := \sum_{n\leqslant y}\frac{\Lambda(n)}{n^s}.$$

The error term in (6.69) accounts for the contribution of the composite integers n in $A(s)$.

The effective Perron formula (Corollary II.2.4) allows us to write

$$A(s) = \frac{-1}{2\pi i}\int_{\kappa-iT}^{\kappa+iT} \frac{\zeta'}{\zeta}(s+w)\frac{y^w}{w}\,dw + O\Big(\frac{\ln y}{y^{\alpha_0}}\Big)$$

with $\kappa := 1-\alpha_0+1/\ln y$, $T := Y_\varepsilon^2$. We move the segment of integration as far as $\Re e\, w = 1-\alpha_0-(\ln T)^{-(2/3)-\varepsilon/10}$, so that the point $s+w$ remains constantly in Vinogradov's zero-free region. The only pole encountered is the point $w = 1-s$. The contribution of the vertical segment is

$$\ll (\ln T)^2 y^{1-\alpha_0}\exp\Big\{-\frac{\ln y}{(\ln T)^{(2/3)+\varepsilon/10}}\Big\} \ll y^{1-\alpha_0}\exp\big\{-(\ln y)^{\varepsilon/2}\big\}$$

and it is checked without difficulty that the same bound also holds for the contribution of the horizontal segments. We thus obtain

$$A(s) = \frac{y^{1-s}}{1-s} + O\Big(y^{1-\alpha_0}\exp\big\{-(\ln y)^{\varepsilon/2}\big\}\Big).$$

Substituting in (6.69), we infer that

$$X\ln y \geqslant \frac{y^{1-\alpha_0}}{1-\alpha_0}\,\Re e\Big(1-\frac{1-\alpha_0}{1-s}\Big) + O\Big(y^{1-\alpha_0}\exp\big\{-(\ln y)^{\varepsilon/2}\big\}\Big)$$

and therefore

$$X \geqslant \frac{e^{\xi(u)}\tau^2}{\xi(u)\big((1-\alpha_0)^2+\tau^2\big)} + O\Big(\exp\big\{\xi(u)-(\ln y)^{\varepsilon/2}\big\}\Big) \gg \frac{u\tau^2}{(1-\alpha_0)^2+\tau^2},$$

since, for $|\tau| \geqslant 1/\ln y$, we have

$$\frac{\tau^2}{(1-\alpha_0)^2+\tau^2} \geqslant \frac{1}{\xi(u)^2+1} \gg \frac{1}{(\ln y)^2}. \tag{6.70}$$

This establishes (6.66).

The proof of (6.67) is analogous. The starting point is

$$\begin{aligned}\big|(1-p^{-s})(1-p^{-\alpha_0})\big| &= \big(1-p^{-2\alpha_0}\big)\Big(1-\frac{2\big(1+\cos(\tau\ln p)\big)}{p^{\alpha_0}\big(1+p^{-\alpha_0}\big)^2}\Big)^{1/2}\\ &\leqslant \exp\Big\{-\frac{1+\cos(\tau\ln p)}{4p^{\alpha_0}}\Big\},\end{aligned}$$

from which it follows that

$$\big|\zeta(s,y)\zeta(\alpha_0,y)\big|^{-1} \leqslant e^{-V/4}, \quad \text{with} \quad V := \sum_{p\leqslant y}\frac{1+\cos(\tau\ln p)}{p^{\alpha_0}}.$$

If $|\tau| > 1/\ln y$, the lower bound established for X is also valid for V: it suffices to consider $A(\alpha_0) + \Re e\, A(\alpha_0+i\tau)$. Taking (6.70) into account, we

certainly have (6.67) in this case. For $|\tau| \leqslant 1/\ln y$, we can write

$$V \geqslant \frac{1+\cos 1}{\ln y} \sum_{p \leqslant y} \frac{\ln p}{p^{\alpha_0}} \gg \frac{y^{1-\alpha_0}-1}{(1-\alpha_0)\ln y} = u.$$

This completes the proof of Lemma 6.12. □

Lemma 6.13. *Let $\varepsilon > 0$. Uniformly for (x,y) in (H_ε) and $1 \leqslant z \leqslant Y_\varepsilon$, we have*

$$\Psi\big(x + x/z, y\big) - \Psi(x,y) \ll \Psi(x,y)\sqrt{u}\ln y\big\{1/z + \mathrm{e}^{-c_3 u}\big\}. \tag{6.71}$$

Proof. We proceed as in Lemma 5.20. Setting

$$w(t) := \frac{1}{2\pi}\Big(\frac{\sin tz}{tz}\Big)^2, \quad \widehat{w}(\tau) := \int_{-\infty}^{+\infty} w(t)\mathrm{e}^{i\tau t}\,\mathrm{d}t = \frac{1}{2z}\Big(1 - \Big|\frac{\tau}{2z}\Big|\Big)^+,$$

we see the left-hand side of (6.71) is

$$\begin{aligned} &\ll \sum_{P^+(n)\leqslant y} \Big(\frac{x}{n}\Big)^{\alpha_0} w\Big(\ln\frac{x}{n}\Big) = \frac{1}{2\pi i}\int_{\alpha_0-2iz}^{\alpha_0+2iz} \zeta(s,y)x^s\widehat{w}(\tau)\,\mathrm{d}s \\ &\ll \frac{x^{\alpha_0}\zeta(\alpha_0,y)}{z} + x^{\alpha_0}\max_{1\leqslant|\tau|\leqslant 2z}\big|\zeta(\alpha_0+i\tau,y)\big|. \end{aligned}$$

The stated result then follows from (6.66) and (6.62). □

Proof of (6.63). Set

$$T := \begin{cases} L & \text{if} \quad 1 \leqslant u \leqslant (\ln y)^{3(1-\varepsilon)/5}, \\ L\mathrm{e}^{c_4 u} & \text{if} \quad (\ln y)^{3(1-\varepsilon)/5} < u \leqslant (\ln y)^{(3-\varepsilon)/2}, \\ Y_{\varepsilon/3} & \text{if} \quad (\ln y)^{(3-\varepsilon)/2} < u \leqslant L_\varepsilon(y), \end{cases} \tag{6.72}$$

where c_4 is a suitable absolute constant. When $T \neq L$—that is when $u > \big(\ln y\big)^{3(1-\varepsilon)/5}$—we have by (6.67), estimating $\zeta(s)$ by Theorem II.3.9,

$$\int_{\substack{\sigma=\alpha_0 \\ L<|\tau|\leqslant T}} \frac{\zeta(s)x^s}{\zeta(s,y)s}\,\mathrm{d}s \ll x^{\alpha_0}\zeta(\alpha_0,y)H(u)^{-c_2}\frac{T^{1-\alpha_0}}{(1-\alpha_0)^2} \ll \Psi(x,y)E(x,y)$$

for some suitable choice of the constant c appearing in the definition of $E(x,y)$, formula (6.48). Thus, it remains to show that the integral

$$R := \int_{\substack{\sigma=\alpha_0 \\ |\tau|>T}} \frac{\zeta(s)x^s}{\zeta(s,y)s}\,\mathrm{d}s$$

also has the same order of magnitude as that of the previous upper bound.

The crude form of the approximate functional equation for the zeta function given in Corollary II.3.5 allows writing, for $s = \alpha_0 + i\tau$,

$$\zeta(s) = \sum_{n \leqslant |\tau|} \frac{1}{n^s} + O\Big(\frac{1}{|\tau|^{\alpha_0}}\Big). \tag{6.73}$$

It ensues that

$$\frac{\zeta(s)}{\zeta(s,y)} = \sum_{n \leqslant |\tau|} \sum_{P(m) \leqslant y} \frac{\mu(m)}{(mn)^s} + O\Big(\frac{\zeta(\alpha_0, y)}{|\tau|^{\alpha_0}}\Big). \tag{6.74}$$

The contribution to R of the error term in (6.74) is

$$\ll x^{\alpha_0} \zeta(\alpha_0, y) \int_T^\infty \frac{\mathrm{d}\tau}{\tau^{1+\alpha_0}} \ll \frac{x^{\alpha_0} \zeta(\alpha_0, y)}{\sqrt{T}} \ll \Psi(x,y) E(x,y),$$

where we have again appealed to (6.62).

The first effective Perron formula in the form of the bound (2.7) allows us to write, uniformly for $z > 0$,

$$\int_{\substack{\sigma = \alpha_0 \\ |\tau| > T}} \frac{z^s}{s} \, \mathrm{d}s \ll \frac{z^{\alpha_0}}{1 + T|\ln z|}.$$

The contribution to R of the main term of (6.74) can therefore be estimated as follows:

$$\begin{aligned} &\sum_{n \geqslant 1} \sum_{P^+(m) \leqslant y} \mu(m) \int_{\substack{\sigma = \alpha_0 \\ |\tau| \geqslant \max(T,n)}} \Big(\frac{x}{mn}\Big)^s \frac{\mathrm{d}s}{s} \\ &\qquad \ll \sum_{n \geqslant 1} \sum_{P^+(m) \leqslant y} \frac{(x/mn)^{\alpha_0}}{1 + (T+n)|\ln(x/mn)|}. \end{aligned} \tag{6.75}$$

The contribution to (6.75) arising from the pairs (m,n) such that $|\ln(x/mn)| > 1$ is

$$\ll x^{\alpha_0} \zeta(\alpha_0, y) \sum_{n \geqslant 1} \frac{1}{n^{\alpha_0}(T+n)} \ll \frac{x^{\alpha_0} \zeta(\alpha_0, y)}{\sqrt{T}} \ll \Psi(x,y) E(x,y).$$

For the remaining pairs (m,n), such that $|\ln(x/mn)| \leqslant 1$, it is useful to bear in mind that $T \leqslant x^\varepsilon$ whenever x, and so y, is sufficiently large. Let S_1 and S_2 denote the respective contributions corresponding to the additional conditions $m \leqslant x/T$ and $m > x/T$. On the one hand, with the notation (5.1),

$$\begin{aligned} S_1 &\leqslant \sum_{m \in S(x/T,y)} \sum_{x/em < n \leqslant ex/m} \frac{1}{1 + |n - x/m|} \ll \sum_{m \in S(x/T,y)} \ln(\mathrm{e}x/m) \\ &\ll \Psi\big(x/T, y\big) \ln x \leqslant \big(x/T\big)^{\alpha_0} \zeta(\alpha_0, y) \ln x \ll \Psi(x,y) E(x,y), \end{aligned}$$

where the penultimate inequality is just Rankin's upper bound for the quantity $\Psi(x/T, y)$.

When (m,n) is counted in S_2, we certainly have $n \leqslant \mathrm{e}T$. From this, we obtain that

$$S_2 \ll \sum_{n\leqslant \mathrm{e}T} \sum_{\substack{x/\mathrm{e}n<m\leqslant \mathrm{e}x/n \\ P^+(m)\leqslant y}} \frac{1}{1+T|x/mn-1|}$$
$$\ll \sum_{n\leqslant \mathrm{e}T} \Big\{ \frac{\Psi(x/n,y)}{\sqrt{T}} + \sum_{\substack{|m-x/n|\leqslant x/(n\sqrt{T}) \\ P^+(m)\leqslant y}} 1 \Big\}.$$

The last sum over m can be estimated using Lemma 6.13, which furnishes the bound

$$\ll \Psi(x/n,y)E(x,y)^2$$

provided that $u > (\ln y)^{3(1-\varepsilon)/5}$. In the opposite case, we appeal to the trivial bound

$$\ll x/\big(n\sqrt{T}\big) \ll (x/n)u^{-2u}T^{-1/3} \ll \Psi(x/n,y)E(x,y)^2.$$

Hence, in any case, we have that

$$S_2 \ll \sum_{n\leqslant T} \Psi(x/n,y)E(x,y)^2 \leqslant x^{\alpha_0}\zeta(\alpha_0,y)E(x,y)^2 \sum_{n\leqslant \mathrm{e}T} \frac{1}{n^{\alpha_0}}$$
$$\ll x^{\alpha_0}\zeta(\alpha_0,y)E(x,y)^2 T^{1-\alpha_0}/(1-\alpha_0) \ll \Psi(x,y)E(x,y).$$

This completes the proof of the bound (6.63). □

Proof of (6.64). Employing (6.67) and then (6.62), we can bound the left-hand side of (6.64) by

$$\ll \frac{x^{\alpha_0}\zeta(\alpha_0,y)}{L} H\big(u\big)^{-c_2} \ln L \ll \Psi(x,y)E(x,y). \qquad \square$$

Proof of (6.65). Formula (6.36) for $1+\widehat{\omega}(s)$ and the trivial bound $J(s) \ll \mathrm{e}^{-\sigma}|\tau|^{-1}$ imply

$$1+\widehat{\omega}(s) = 1 + O\Big(\frac{1+u\xi(u)}{s}\Big) \qquad (|\tau| > L).$$

This shows that the integral in (6.65) is conditionally convergent. The second mean value theorem allows us to write

$$\int_{\substack{\sigma=-\xi \\ |\tau|>L}} \frac{1+\widehat{\omega}(s)}{s+\ln y} \mathrm{e}^{us}\,\mathrm{d}s = \int_{\substack{\sigma=-\xi \\ |\tau|>L}} \frac{\mathrm{e}^{us}}{s} \Big\{1 + O\Big(\frac{u\xi(u)+\ln y}{s}\Big)\Big\}\,\mathrm{d}s \ll \mathrm{e}^{-u\xi}L^{-1}.$$

Since, by Theorem 5.13 and Corollary 5.19, we have

$$\Psi(x,y) \gg x\mathrm{e}^{-u\xi}\mathrm{e}^{u/2}$$

in the domain (H_ε), we obtain the required estimate (6.65). □

This completes the proof of Theorem 6.10.

Corollary 6.14. *Let $\varepsilon > 0$. For (x,y) in (H_ε), we have*

$$\Phi(x,y) - \frac{x}{\zeta(1,y)} \ll \frac{x\varrho(u)}{\ln y}\Big(H\big(u\big)^{-c_5} + Y_\varepsilon^{-1}\Big) \tag{6.76}$$

Proof. This results immediately from (6.50) and (6.52). □

Corollary 6.15. *Let $\varepsilon > 0$. For (x,y) in (H_ε), we have*

$$\Phi(x,y) = \big(x\omega(u) - y\big)\frac{\mathrm{e}^\gamma}{\zeta(1,y)} + O\Big(\frac{x\varrho(u)}{\ln^2 y}\Big\{H(u)^{-c_6} + Y_\varepsilon^{-1}\Big\}\Big). \tag{6.77}$$

Proof. The error term in (6.77) is compatible with the first bound of (6.50). It thus suffices to show that (6.77) is valid with $\Phi(x,y)$ replaced by $W(x,y)$. According to (6.53), we have

$$W(x,y) = \big(x\omega(u) - y\big)\frac{\mathrm{e}^\gamma}{\zeta(1,y)} - \frac{x\mathrm{e}^\gamma}{\zeta(1,y)}\int_0^\infty \omega'(u-v)y^{-v}\,\mathrm{d}v.$$

By (6.55), with $\varepsilon = 1$, and Corollary 5.15, the last integral is

$$\ll \frac{\varrho(u)}{H(u)^a}\int_0^\infty \Big(\frac{\mathrm{e}^{1+\xi(u)}}{y}\Big)^v \mathrm{d}v \ll \frac{\varrho(u)}{H(u)^a \ln y}$$

since, in (H_ε), we have, for example, that $\mathrm{e}^{1+\xi(u)} \leqslant \sqrt{y}$ whenever x, and hence y, is sufficiently large. This implies (6.77). □

Corollary 6.16. *Uniformly for $x \geqslant 2y \geqslant 5$, we have*

$$\Phi(x,y) = \frac{\mathrm{e}^\gamma\{x\omega(u) - y\}}{\zeta(1,y)}\Big\{1 + O\Big(\frac{\mathrm{e}^{-u/3}}{\ln y}\Big)\Big\}. \tag{6.78}$$

Proof. It is easily checked that $x\omega(u) - y \gg x$ under the stated conditions. The result is therefore (amply) implied by (6.77) for (x,y) in (H_ε). Otherwise, we deduce from (6.53) and (6.54) that

$$W(x,y) - \big(x\omega(u) - y\big)\frac{\mathrm{e}^\gamma}{\zeta(1,y)} \ll \Psi(x,y) \ll \frac{x\mathrm{e}^{-u/3}}{(\ln y)^2}$$

where the final estimate follows from Theorem 5.1. This yields the stated result. □

6.5. The Kubilius model

In recent years, the importance, in number theory, of the duality between friable and sifted integers has been steadily increasing. The reason for this stems as much from the development of new theoretical methods where the associated canonical decomposition plays an essential role, as from fundamental applications to factorization problems.

The algorithmic applications are linked to the fact that friable integers are precisely those which can be easily factorized: one only needs to try small primes.

The incidence of friable integers in arithmetic problems originates in particular from the development of sieve techniques. We previously mentioned[1] Daboussi's (1984) elementary proof of the prime number theorem where the decomposition $n = ab$ with

$$P^+(a) \leqslant y < P^-(b)$$

for fixed y is non-trivially exploited.

Numerous developments of the circle method, notably those due to Vaughan & Wooley,[2] depend strongly on properties of friable integers, which arise as summation sets for certain exponential sums. The good distribution properties of divisors of friable integers (for example, $d_{j+1} \leqslant y d_j$ where d_j and d_{j+1} are consecutive divisors of a y-friable integer n) allow controlling the number of solutions of certain Diophantine equations, which are in turn interpreted as L^p-norms for some exponential sums.

The results obtained through the saddle-point method for the sieve functions $\Psi(x,y)$ and $\Phi(x,y)$ also have applications in probabilistic number theory. We intend here to describe one of these, fundamental in nature.

Let $n \in \mathbb{N}^*$, write $\Omega_n := \{1, 2, \dots, n\}$ and let $\mathcal{T}_y$ be the σ-algebra comprising those events in Ω_n that can be defined by divisibility conditions involving only prime powers p^j with $p \leqslant y$. Thus, $\mathcal{T}_y$-measurable events in Ω_n coincide with unions of sets of the type

$$E_a := \{m \leqslant n : m = ab,\ P^-(b) > y\} \qquad (P^+(a) \leqslant y).$$

For the uniform empirical measure ν_n on Ω_n, we therefore have

$$\nu_n(E_a) = \frac{1}{n}\Phi\Big(\frac{n}{a}, y\Big) \qquad (P^+(a) \leqslant y).$$

[1] See the Notes for § 5.1.

[2] See, for example, chap. 12 of Vaughan (1997).

A probabilistic model for $(\Omega_n, \nu_n, \mathcal{T}_y)$ is obtained by considering, on some abstract set Ω, $\pi(y)$ partitions

$$\Omega = \bigcup_{j\geqslant 0} \omega_{p,j} \qquad (p \leqslant y)$$

and the σ-algebra $\mathcal{T}_y^*$ generated by finite intersections of sets $\omega_{p,j}$. Then, for $P^+(a) \leqslant y$, the set E_a may be mimicked by $E_a^* := \cap_{p^j \| a} \omega_{p,j}$ and ν_n is compared to the probability measure P_y defined on Ω by

$$P_y(\omega_{p,j}) = (1 - 1/p)p^{-j} \qquad (p \leqslant y,\ j \in \mathbb{N}) \tag{6.79}$$

with the condition that $\omega_{p,j}$ and $\omega_{q,k}$ are independent events if $p \neq q$. To any $\mathcal{T}_y$-measurable subset E of Ω_n, we can hence associate a $\mathcal{T}_y^*$-measurable subset E^* of Ω, defined by the formula

$$E^* := \bigcup_{E_a \subset E} E_a^*.$$

The Kubilius gauge $K(n,y)$ is a measure of the distance between the probability spaces $(\Omega_n, \nu_n, \mathcal{T}_y)$ and $(\Omega, P_y, \mathcal{T}_y^*)$. It is defined by

$$K(n,y) := \sup_{E\in\mathcal{T}_y} |\nu_n(E) - P_y(E^*)|, \tag{6.80}$$

whence, in view of (6.79), writing $E = \cup_{a\in A} E_a$,

$$K(n,y) = \sup_{A\subset S(\infty,y)} \left| \sum_{a\in A} \left\{ \frac{1}{n}\Phi\Big(\frac{n}{a}, y\Big) - \frac{1}{a\zeta(1,y)} \right\} \right|.$$

It is sufficient to consider the supremum over those subsets A such that $\nu_n(E) - P_y(E^*) \geqslant 0$: the complementary set is obtained by changing A to

$$\overline{A} := S(\infty, y) \setminus A,$$

which only changes the sign of the quantity involved. Thus, $K(n,y)$ may be approximated to arbitrary precision by a quantity of the type

$$\sum_{a\in A} \left\{ \frac{1}{n}\Phi\Big(\frac{n}{a}, y\Big) - \frac{1}{a\zeta(1,y)} \right\}.$$

This expression is clearly maximal when

$$A := \big\{ a \in S(\infty, y) : \Phi(n/a, y) \geqslant n/(a\zeta(1,y)) \big\},$$

and this incidentally shows that the supremum in (6.80) is attained. It follows that

$$K(n,y) = \sum_{P^+(a)\leqslant y} \Big\{ \frac{1}{n}\Phi\Big(\frac{n}{a},y\Big) - \frac{1}{a\zeta(1,y)} \Big\}^+$$
$$= \tfrac12 \sum_{P^+(a)\leqslant y} \Big\{ \Big| \frac{1}{n}\Phi\Big(\frac{n}{a},y\Big) - \frac{1}{a\zeta(1,y)} \Big| + \frac{1}{n}\Phi\Big(\frac{n}{a},y\Big) - \frac{1}{a\zeta(1,y)} \Big\},$$

and therefore for all $n \in \mathbb{N}$, $y \geqslant 1$,

$$K(n,y) = \tfrac12 \sum_{P^+(a)\leqslant y} \Big| \frac{1}{n}\Phi\Big(\frac{n}{a},y\Big) - \frac{1}{a\zeta(1,y)} \Big|. \tag{6.81}$$

The classical version of the "fundamental lemma" of Kubilius' model, due to Kubilius (1956), Barban & Vinogradov (1964) and Elliott (1979) states that, with the notation $u := (\log n)/\log y$, we have

$$K(n,y) \ll u^{-u/8} + n^{-1/15}. \tag{6.82}$$

It is worth noticing that the threshold from which $K(n,y)$ tends to 0, namely $y = n^{o(1)}$, coincides with that from which the friable Turán–Kubilius constant tends to 1—see (5.125). Each of these two phenomena expresses a specific form of the asymptotic independence of divisibility conditions involving powers of 'small' primes.

The approximations to $\Phi(x,y)$ and $\Psi(x,y)$ arising from the saddle-point method yield a better estimate for $K(n,y)$. The result may be partly stated in terms of the function

$$\Xi(u) := \tfrac12 \int_{\mathbb{R}} |\omega(v) - \mathrm{e}^{-\gamma}| \varrho(u-v)\,\mathrm{d}v + \tfrac12 \varrho(u). \tag{6.83}$$

This quantity was introduced by Arratia & Stark (1997), who proved that, for each fixed u, we have $\lim_{y\to\infty} K(y^u,y) = \Xi(u)$. It can be shown that

$$\Xi(u) = \varrho(u) 2^{u+o(u)} \qquad (u \to \infty), \tag{6.84}$$

[3]

and so $\Xi(u)$ tends to 0 very quickly: in particular,

$$\Xi(u) \ll u^{-u} \qquad (u \geqslant 1).$$

The following statement has been obtained by the author (1999).

Theorem 6.17. *Let $n \in \mathbb{N}$ and $y \in \mathbb{R}$ satisfy $n \geqslant y \geqslant 2$. For each $\varepsilon > 0$, we have*

$$K(n,y) \ll_\varepsilon \varrho(u) 2^{(1+\varepsilon)u} + n^{-1+\varepsilon} \tag{6.85}$$

[3] A more precise estimate is provided by Tenenbaum (1999).

where we have put $u := (\ln n)/\ln y$. *Furthermore, for each* $\varepsilon > 0$, *the approximation*

(6.86)
$$K(n,y) = \Xi(u)\Big\{1 + O\Big(\frac{\ln(u+1) + 2^{-u}\ln_2 y}{\ln y}\Big)\Big\} + O\Big(\frac{\varrho(u/2)^2}{\exp\{(\ln y)^{3/2-\varepsilon}\}}\Big)$$

holds uniformly in the domain

(H_ε) $$n \geqslant 3, \qquad \exp\{(\ln_2 n)^{5/3+\varepsilon}\} \leqslant y \leqslant n.$$

As established in the same work, the parameter ε can be replaced with 0 in neither of its two occurrences on the right-hand side of (6.85). In particular, we deduce from (6.85) the quasi-optimal improvement of (6.82)

(6.87) $$K(n,y) \ll_\varepsilon u^{-u} + n^{-1+\varepsilon} \qquad (n \geqslant y \geqslant 2),$$

while (6.86) implies the validity, for all $\varepsilon > 0$, of the asymptotic formula

(6.88) $$K(n,y) = \Xi(u)\Big\{1 + O\Big(\frac{\ln_2 y}{\ln y}\Big)\Big\}$$

in the domain

(G_ε) $$n \geqslant 3, \qquad \exp\{(\ln n)^{2/5+\varepsilon}\} \leqslant y \leqslant n.$$

Notes

§ 6.3. As brought up in the proof of Theorem 6.8, the finer asymptotic behavior of the Buchstab function $\omega(u)$ is connected with the equation

$$e^{\zeta} = 1 - u\zeta. \tag{6.89}$$

Just as (6.10), this equation has infinitely many complex solutions, close to the line $\Re e\,\zeta = \xi(u)$. However, while (6.10) has a unique non-zero real solution for $u > 1$, playing *ipso facto* a fundamental role, (6.89) fails to have a non-trivial real solution. Therefore the two conjugate solutions closest to the real axis are those describing more precisely the asymptotic behavior of $\omega(u)$, applying the saddle-point method with the 'right' choice for the line of integration. Let $\zeta_0(u)$ and $\zeta_{-1}(u) = \overline{\zeta_0(u)}$ denote[4] these solutions. Hildebrand & Tenenbaum (1993a) established that $\zeta_0(u)$ is, for sufficiently large real u, the unique solution of (6.89) satisfying

$$|\zeta_0(u) - \xi(u) + i\pi| \leqslant \pi$$

and that we actually have

$$\zeta_0(u) = \xi(u) + \frac{\pi^2}{2\xi(u)^2} - i\frac{\pi\xi(u)}{\xi(u)-1} + O\Big(\frac{1}{\xi(u)^3}\Big), \tag{6.90}$$

$$u\zeta_0'(u) = \frac{-\zeta_0(u)}{\zeta_0(u)-1} + O\Big(\frac{1}{u\xi(u)}\Big) = \frac{-\xi(u)+i\pi}{\xi(u)-1-i\pi} + O\Big(\frac{1}{\xi(u)^3}\Big). \tag{6.91}$$

Let us next define

$$\Phi(u) := \frac{\exp\{-u\zeta_0(u) - I(\zeta_0(u))\}}{\zeta_0(u)\sqrt{2\pi u\{1-1/\zeta_0(u)\}}}. \tag{6.92}$$

Hildebrand (1990) showed that

$$\omega(u) - e^{-\gamma} = -2e^{-\gamma}R(u)\{\cos\vartheta(u) + O(1/u)\} \tag{6.93}$$

with $R(u) := |\Phi(u)|$, $\vartheta(u) := \arg\Phi(u)$. As shown by the author (1999), the above asymptotic formula may now be readily derived from the general theory of Hildebrand & Tenenbaum (1993a), which in this case yields additional information, in particular the formulae

$$R(u) = \varrho(u)\exp\Big\{\frac{-\pi^2 u}{2\xi(u)^2} + O\Big(\frac{u}{\xi(u)^3}\Big)\Big\} \tag{6.94}$$

and

$$\vartheta'(u) = \frac{\pi\xi(u)}{\xi(u)-1} + O\Big(\frac{1}{\xi(u)^3}\Big) = 2\pi\Big(1 + \frac{1}{\ln u}\Big) + O\Big(\frac{\ln_2 u}{(\ln u)^2}\Big). \tag{6.95}$$

It is also worthwhile noticing that R is strictly decreasing: this easily follows from (6.92).

[4] In accordance with the notation employed in Hildebrand & Tenenbaum (1993a).

Formulae (6.93) and (6.95) imply in particular that $\omega(u) - \mathrm{e}^{-\gamma}$ changes sign infinitely often. As shown by Hildebrand (1990), if λ_n denotes the nth zero of this quantity, we may deduce from (6.95) that

$$\lambda_{n+1} - \lambda_n = 1 - \frac{1}{\ln n} + O\Big(\frac{\ln_2 n}{(\ln n)^2}\Big).$$

For an extension of these results to the general solution of a delay differential equation of the type

$$uf'(u) + af(u) + bf(u-1) = 0$$

with arbitrary complex numbers, a, b, see Hildebrand & Tenenbaum (1993a).

§ 6.4. Writing out the kth order Taylor–Lagrange formula for $\omega(u)$,[5] we can generalize formula (6.53) in the following way.

Lemma 6.18. *Let* $\omega_{mj} := \omega^{(j)}(m) - \omega^{(j)}(m-)$ $(j \geqslant 0,\ 1 \leqslant m \leqslant j+1)$. *For any integer* $k \geqslant 0$, *we have*

(6.96)

$$\mu_y(u) \ln y = \sum_{0\leqslant j\leqslant k} \frac{(-1)^j \omega^{(j)}(u)}{(\ln y)^j} + \sum_{\substack{1\leqslant m\leqslant k+1\\ m\leqslant u}} y^{m-u} \sum_{m-1\leqslant j\leqslant k} \frac{(-1)^{j+1}\omega_{mj}}{(\ln y)^j}$$
$$+ \frac{(-1)^{k+1}}{(\ln y)^k} \int_0^\infty \omega^{(k+1)}(u-v) y^{-v}\, \mathrm{d}v.$$

This allows sharpening Corollary 6.15 by providing, in the domain (H_ε), an asymptotic expansion of $\Phi(x,y)$ in powers of $1/\ln y$. It is indeed easy to show that in this region the integral in (6.96) is

$$\ll_k \frac{\varrho(u)}{\ln y} H(u)^{-c}.$$

In the same way as (6.93) and (6.94) show that Theorem 6.8 is optimal up to the value of the exponent of $H(u)$, it can be established, in an analogous sense, that Corollary 6.14 is essentially optimal—see Friedlander, Granville, Hildebrand & Maier (1991). In this article, the authors study sign changes of $\Phi(x,y) - x/\zeta(1,y)$ in order to deduce oscillation theorems for the distribution of prime numbers. The original idea underlying this somewhat surprising connection is due to Maier (1985)—see also Hildebrand & Maier (1989) and Friedlander & Granville (1989).

Theorem 6.10 is significantly more precise than the classical result of de Bruijn (1950)

$$\Phi(x,y) = W(x,y) + O\Big(\frac{x}{L_\varepsilon(y)}\Big) \qquad (x \geqslant y \geqslant 2). \tag{6.97}$$

[5] See Tenenbaum (1990), lemma 6.

This result is clear when (x,y) belongs to (H_ε). In the contrary case, the error term $O\big(\Psi(x,y)\big)$ of Theorem 6.10 is certainly

$$\ll x\mathrm{e}^{-L_\varepsilon(y)}.$$

It can be shown that Lemma 6.12 holds for $x \geqslant y \geqslant \ln x$, subject to replacing α_0 by the actual solution $\alpha(x,y)$ of equation (6.5)—cf. Hildebrand & Tenenbaum (1986), lemma 8, and Tenenbaum (1990), corollary to lemma 1.

The following result provides a refinement of Theorem 6.10. We recall the notation in (6.47), (6.48) and (6.49).

Theorem 6.19 (Tenenbaum, 1999). *Let $\varepsilon > 0$. Uniformly for $x \geqslant y \geqslant 2$, we have*

$$\Phi(x,y) = W(x,y) + O(E(x,y)), \tag{6.98}$$

with

$$E(x,y) := \begin{cases} xR(u)/L_\varepsilon(y) + x\varrho(u)/Y_\varepsilon & \text{if } (x,y) \in (H_\varepsilon), \\ \Psi(x,y) & \text{otherwise.} \end{cases} \tag{6.99}$$

Corollary 6.20 (Tenenbaum, 1999).

Let $\varepsilon > 0$. Uniformly for $(x,y) \in (H_\varepsilon)$, we have

$$\Phi(x,y) = \frac{x}{\zeta(1,y)} + O\Big(\frac{xR(u)}{\ln y} + \frac{x\varrho(u)}{Y_\varepsilon}\Big), \tag{6.100}$$

$$\Phi(x,y) = \frac{\mathrm{e}^\gamma\{x\omega(u) - y\}}{\zeta(1,y)} + O\Big(\frac{xR(u)\ln(u+1)}{(\ln y)^2} + \frac{x\varrho(u)}{Y_\varepsilon}\Big). \tag{6.101}$$

Established in the same article, a second corollary of Theorem 6.19 concerns the mean value of the function $\ln P^-(n)$. The dual result, relative to $\ln P^+(n)$ and refining an asymptotic formula due to de Bruijn (1951b), is established in Exercise 290, p. 553. The main interest of this corollary is that it illustrates, on a practical example, the various possibilities for exploiting the specific form of the main term $W(x,y)$ in (6.98).[6] The technique can be easily adapted to obtain precise mean value estimates for a wide class of arithmetic functions of the form $f(P^-(n))$. We set

$$A := \frac{1+\gamma}{\mathrm{e}^\gamma} + \int_1^\infty \{\omega(t) - \mathrm{e}^{-\gamma}\}\frac{\mathrm{d}t}{t} + \sum_p \Big\{\frac{\ln p}{(p-1)\zeta(1,p)} + \mathrm{e}^{-\gamma}\ln\Big(1-\frac{1}{p}\Big)\Big\}.$$

Corollary 6.21. *Let $\varepsilon > 0$. Uniformly for $x \geqslant 2$, we have*

$$\sum_{1<n\leqslant x} \ln P^-(n) = \mathrm{e}^{-\gamma}x\ln_2 x + Ax + O\Big(x\mathrm{e}^{-(\ln x)^{3/8-\varepsilon}}\Big).$$

[6] For a dual study, concerning friable integers, of this approach, see Tenenbaum & Wu (2008b).

§ 6.5. The estimates for $\omega(u) - \mathrm{e}^{\gamma}$ described in the Notes for § 6.3 can in turn be used to evaluate asymptotically the function $\Xi(u)$ from(6.83), and in particular to prove (6.84).

The estimates given in Theorem 6.17 for the Kubilius gauge $K(n,y)$ stem essentially from Hildebrand's formula (5.83) for $\Psi(x,y)$ and from Corollary 6.20 regarding $\Phi(x,y)$.

Theorem 6.17 may be applied in a number of ways. We shall describe two of them.

The first application consists in a strengthening of theorem 12.5 of Elliott (1980). Recall that the Lévy distance $L(F,G)$ of two distribution functions F and G is defined by the formula

$$L(F,G) := \inf\{\delta > 0 : F(z-\delta) - \delta \leqslant G(z) \leqslant F(z+\delta) + \delta \ (\forall z \in \mathbb{R})\}.$$

Corollary 6.22 (Tenenbaum, 1999). *Let f be a real additive function and let A_N and B_N be real quantities defined for each integer $N \geqslant 1$, such that $B_N > 0$. Set*

$$F_N(z) := \nu_N(f(n) \leqslant A_N + zB_N), \qquad G_N(z) := P(Z_{f,N} \leqslant A_N + zB_N),$$

where $Z_{f,N}$ denotes the random variable defined in (3.8). *Then, for any $\varepsilon > 0$ and uniformly for $0 < h < 1$ and $2 \leqslant y \leqslant N$, we have*

$$L(F_N, G_N) \ll_{\varepsilon} \sum_{\substack{y<p\leqslant N \\ |f(p^\nu)|>hB_N/u}} \sum_{\nu\geqslant 1} \frac{1}{p^\nu} + h + \varrho(u)2^{(1+\varepsilon)u} + N^{-1+\varepsilon}. \tag{6.102}$$

Our second application concerns the function

$$\Theta(x,y,z) := |\{n \leqslant x : n_y > z\}|$$

where n_y is the largest y-friable divisor of n. Erdős' principle of the double exponential growth for the sequence of prime factors of a normal integer (cf., for example, Theorem 3.10) suggests that $\Theta(x,y,z)$ should be small compared to x whenever $\ln z$ is large compared to $\ln y$. Indeed, we have actually already obtained a uniform bound for this quantity in Exercise 293, p. 554. We now describe a sharper result.

Put

$$\begin{aligned}
\lambda_1(w) &:= \int_w^\infty \varrho(t)\,\mathrm{d}t \qquad (w \geqslant 0),\\
\sigma(u,v) &:= \int_v^\infty \varrho(t)\omega(u-t)\,\mathrm{d}t \quad (u \geqslant 1,\ v \geqslant 0),\\
\vartheta(u,v) &:= \varrho(u) + \sigma(u,v) \quad (u \geqslant 1,\ v \geqslant 0),\\
\kappa(u,v) &:= (1-\gamma)\varrho(u-1) + \gamma\varrho(v)\omega(u-v) \quad (u \geqslant 1,\ v \geqslant 0).
\end{aligned}$$

It easily follows from the estimates for Dickman's function given in Chapter III.5 that

$$\lambda_1(w) = \frac{\varrho(w)}{\ln(w+1)}\Big\{1 + O\Big(\frac{1}{\ln(w+1)}\Big)\Big\} \quad (w \geqslant 1). \tag{6.103}$$

Since we have $\frac{1}{2} \leqslant \omega(t) \leqslant 1$ $(t \geqslant 1)$ and $\varrho(v+h) \gg \varrho(v)\{v\ln(v+1)\}^{-h}$ for $0 \leqslant h \leqslant 1 \leqslant v$, this implies in particular that $\vartheta(u,v) \asymp \lambda_1(v)$ whenever $u - 1 - v \gg 1/\ln(v+1)$.

Theorem 6.23 (Tenenbaum, 1999, 2006). *Let $\varepsilon > 0$. Under the conditions*

$$x \geqslant 2, \quad y \geqslant 2, \qquad 1 \leqslant z \leqslant \exp\exp\big\{(\ln y)^{3/5-\varepsilon}\big\} \tag{6.104}$$

we have

$$\Theta(x,y,z) = \Big\{\mathrm{e}^{-\gamma} + O\Big(\frac{\ln(v+2)}{\ln y}\Big)\Big\}x\lambda_1(v) + O\Big(x\varrho(u)2^{(1+\varepsilon)u} + x^{\varepsilon}\Big), \tag{6.105}$$

where $u := (\ln x)/\ln y$, $v := (\ln z)/\ln y$. If, in addition, $1 \leqslant z \leqslant x/y$, we have uniformly

$$\Theta(x,y,z) = x\Big\{\vartheta(u,v) - \frac{\kappa(u,v)}{\ln y} + O\Big(\frac{\lambda_1(v)}{\ln y} + \frac{\varrho(v)\ln(v+2)}{(\ln y)^2} + \frac{1}{z}\Big)\Big\}. \tag{6.106}$$

In particular, if

$$(\ln y)^2 \leqslant z \leqslant \min\Big(x/y^{1+\varepsilon/\ln(u+1)}, \exp\exp\big\{(\ln y)^{3/5-\varepsilon}\big\}\Big),$$

we have

$$\Theta(x,y,z) = \Big\{1 + O\Big(\frac{\ln(v+2)}{\ln y}\Big)\Big\}\vartheta(u,v)x. \tag{6.107}$$

We note that the condition $z \leqslant x/y$ is not restrictive since we have $\Theta(x,y,z) = \Psi(x,y) - \Psi(z,y)$ when $x/y < z \leqslant x$.

Exercises

297. (a) With the notation (5.1), establish the following identity for $x \geqslant y \geqslant 2$:

$$\lfloor x \rfloor = \sum_{n \in S(x/y,y)} \Phi(x/n, y) + \Psi(x, y) - \Psi(x/y, y).$$

(b) Using Theorems 5.8 and 6.4, deduce from (a) the convolution identity

$$\varrho(u) + \int_0^u \varrho(v)\omega(u-v)\,\mathrm{d}v = 1 \qquad (u \geqslant 0).$$

Hence recover (6.35).

In the following exercises, we let $\Psi_k(x, y, z)$ denote the number of those integers $n \leqslant x$ satisfying

$$\Omega(n; y, z) := \sum_{z < p \leqslant y,\, p^\nu \| n} \nu = k.$$

We also consistently set $x = y^u$, $z = y^\lambda$.

298. For $0 \leqslant \lambda \leqslant 1$ and $u \geqslant 0$, set

$$\vartheta_0(\lambda, u) := \varrho(u/\lambda) + \int_0^u \varrho(v/\lambda)\,\omega(u-v)\,\mathrm{d}v$$

with the convention that $\vartheta_0(0, u) := 0$.

(a) Show that, for $u \geqslant 1$, we have

$$\vartheta_0(\lambda, u) \leqslant \varrho(u/\lambda) + \lambda \int_0^{(u-1)/\lambda} \varrho(v)\,\mathrm{d}v$$

and deduce that $\vartheta_0(\lambda, u) \leqslant \mathrm{e}^\gamma \lambda$ for $u \geqslant 1$.

(b) Establish the bounds

$$\tfrac{1}{2}\lambda \leqslant \vartheta_0(\lambda, u) \leqslant \mathrm{e}^\gamma \lambda \qquad (0 \leqslant \lambda \leqslant 1,\ u \geqslant 1 + \lambda).$$

(c) Show that

$$\lim_{u \to +\infty} \vartheta_0(\lambda, u) = \lambda.$$

299. (a) Establish Buchstab's identity

$$\Psi_0(x, y, z) = \Psi(x, z) + \sum_{y < p \leqslant x} \Psi_0(x/p, p-1, z).$$

(b) Using, in addition to other results, that of Exercise 216, show that, uniformly for $y \geqslant z \geqslant 2$ and $x \geqslant yz$, we have

$$\Psi_0(x, y, z) = x\vartheta_0(\lambda, u)\{1 + O(1/\ln z)\}.$$

300. (a) Show that

$$\lambda\frac{\partial\vartheta_0}{\partial\lambda}(\lambda,u)=\vartheta_0(\lambda,u-\lambda)\qquad(0\leqslant\lambda\leqslant1,\ u\geqslant1).$$

(b) By using the results (a) and (b) of Exercise 299, show that, for $0\leqslant\lambda\leqslant1$, $u\geqslant1$, we have

$$u\frac{\partial\vartheta_0}{\partial u}(\lambda,u)=\vartheta_0(\lambda,u-1)-\vartheta_0(\lambda,u-\lambda).$$

(c) Show the existence of an absolute constant A such that we have

$$\Big|\frac{\partial\vartheta_0}{\partial u}(\lambda,u)\Big|\leqslant A\lambda\varrho(u)\qquad(3\leqslant u\leqslant4)$$

and deduce that

$$\vartheta_0(\lambda,u)=\lambda\big\{1+O\big(\varrho(u)\big)\big\}\qquad(u\geqslant1).$$

301. *An approximate Poisson law.* Define inductively a sequence of functions $\{\vartheta_k(\lambda,u)\}_{k=0}^{\infty}$ by the formula

$$k\,\vartheta_k(\lambda,u)=\int_\lambda^1\vartheta_{k-1}(\lambda,u-v)\frac{\mathrm{d}v}{v}\qquad(k\geqslant1).$$

Using the results of Exercises 298 and 300, show that, under the conditions $k\geqslant0$, $0\leqslant\lambda\leqslant1$, $u\geqslant k+1+\lambda$, we have

(a) $\frac12\dfrac{\lambda}{k!}\Big(\ln\dfrac1\lambda\Big)^k\leqslant\vartheta_k(\lambda,u)\leqslant\mathrm{e}^\gamma\dfrac{\lambda}{k!}\Big(\ln\dfrac1\lambda\Big)^k$,

(b) $\vartheta_k(\lambda,u)=\dfrac{\lambda}{k!}\Big(\ln\dfrac1\lambda\Big)^k\big\{1+O\big(\varrho(u-k)\big)\big\}$.

302. Under the triple condition $y\geqslant z\big(1+1/L_\varepsilon(z)\big)$, $zy^{k+1}\leqslant x$ and $k\leqslant\ln(1/\lambda)L_\varepsilon(z)$, show that we have, uniformly,

$$\Psi_k(x,y,z)=x\vartheta_k(\lambda,u)\big\{1+O(1/\ln z)\big\}.$$

303. (a) Show that $\vartheta_k(\lambda,u)=0$ if $u<k\lambda$.

(b) Deduce, for $0<\lambda\leqslant1$, $u>0$, that $\sum_{k=0}^{\infty}\vartheta_k(\lambda,u)=1$.

(c) Define $\kappa(\lambda,u):=1/u$ $(\lambda\leqslant u\leqslant1)$, $\kappa(\lambda,u):=0$ (otherwise). Show that

$$k!\vartheta_k(\lambda,u)=\kappa^{*k}*\vartheta_0(\lambda,u)$$

where the convolutions are carried out with respect to the variable u.

(d) Calculate the Laplace transform $\widehat{\kappa}_\lambda(s):=\int_0^\infty\mathrm{e}^{-us}\kappa(\lambda,u)\,\mathrm{d}u$ and deduce, by using the result in part (b), that we have

$$\widehat{\vartheta}_0(\lambda,s):=\int_0^\infty\mathrm{e}^{-us}\vartheta_0(\lambda,u)\,\mathrm{d}u=\frac1s\exp\Big\{-\int_{\lambda s}^s\mathrm{e}^{-v}\frac{\mathrm{d}v}{v}\Big\}.$$

(e) Recover the above result by using the definition of $\vartheta_0(\lambda,u)$ from Exercise 298.

304. Let Y denote the Heaviside function, viz. $Y(u) := \mathbf{1}_{[0,\infty[}(u)$ $(u \in \mathbb{R})$.

(a) Show that, for $0 \leqslant \lambda \leqslant 1$, $u \geqslant 0$, we have

$$\vartheta_0(\lambda, u) = \sum_{k \geqslant 0} \frac{(-1)^k}{k!} (\kappa^{*k} * Y)(\lambda, u)$$

and deduce that, for all $k \geqslant 0$, we have

$$\vartheta_k(\lambda, u) = \frac{1}{k!} \sum_{j \geqslant 0} \frac{(-1)^j}{j!} F_{k+j}(\lambda, u)$$

with

$$F_0(\lambda, u) = 1, \quad F_k(\lambda, u) = \int_{\substack{[\lambda,1]^k \\ \sum v_i \leqslant u}} \frac{\mathrm{d}v_1}{v_1} \cdots \frac{\mathrm{d}v_k}{v_k}.$$

(b) Recover the differential equation in Exercise 300(a).

305. Let $\widehat{\varrho}_\lambda(s) := \int_0^\infty \mathrm{e}^{-us} \varrho(u/\lambda)\, \mathrm{d}u$. Show that, for $s \neq 0$, we have

$$s\widehat{\varrho}(s)\widehat{\vartheta}_0(\lambda, s) = \widehat{\varrho}_\lambda(s) = \lambda \widehat{\varrho}(\lambda s)$$

and deduce the validity of the convolution formula

$$\int_0^u \vartheta_0(\lambda, v) \varrho(u - v)\, \mathrm{d}v = \lambda \int_0^{u/\lambda} \varrho(v)\, \mathrm{d}v.$$

306. (a) Show that for all $\xi \in \mathbb{C}$, we have

$$\sum_{k \geqslant 0} \vartheta_k(\lambda, u) \xi^k = \sum_{k \geqslant 0} \frac{1}{k!} F_k(\lambda, u)(\xi - 1)^k$$

where the functions $F_k(\lambda, u)$ are as defined in Exercise 304.

(b) Deduce, for all $\xi \geqslant 1$, that

$$\sum_{k \geqslant \xi \ln(1/\lambda)} \vartheta_k(\lambda, u) \leqslant \lambda^{Q(\xi)} \qquad \big(Q(\xi) := \xi \ln \xi - \xi + 1\big).$$

307. Define $\widehat{\vartheta}_k(\lambda, s) := \int_0^\infty \mathrm{e}^{-us} \vartheta_k(\lambda, u)\, \mathrm{d}u$.

(a) Using the formula proved in Exercise 303, write down the expansion to order 1 at $s = 0$ of $\widehat{\vartheta}_0(\lambda, s)$.

(b) Show that $\int_0^\infty \{\vartheta_0(\lambda, u) - \lambda\}\, \mathrm{d}u = \lambda(1 - \lambda)$.

(c) Compute $\widehat{\vartheta}_k(\lambda, s)$ and write down its expansion to order 1 at $s = 0$.

(d) Show, for $k \geqslant 1$, that

$$\int_0^\infty \left\{ \vartheta_k(\lambda, u) - \frac{\lambda}{k!} \Big(\ln \frac{1}{\lambda} \Big)^k \right\} \mathrm{d}u = \frac{\lambda(1 - \lambda)}{k!} \Big(\ln \frac{1}{\lambda} \Big)^{k-1} \Big\{ \ln \frac{1}{\lambda} - k \Big\}.$$

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